



Intel® 64 and IA-32 Architectures Software Developer's Manual

Volume 4: Model-Specific Registers

NOTE: The *Intel® 64 and IA-32 Architectures Software Developer's Manual* consists of ten volumes: *Basic Architecture*, Order Number 253665; *Instruction Set Reference, A-L*, Order Number 253666; *Instruction Set Reference, M-U*, Order Number 253667; *Instruction Set Reference, V*, Order Number 326018; *Instruction Set Reference, W-Z*, Order Number 334569; *System Programming Guide, Part 1*, Order Number 253668; *System Programming Guide, Part 2*, Order Number 253669; *System Programming Guide, Part 3*, Order Number 326019; *System Programming Guide, Part 4*, Order Number 332831; *Model-Specific Registers*, Order Number 335592. Refer to all ten volumes when evaluating your design needs.

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The Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 4: Model-Specific Registers (order number 335592) is part of a set that describes the architecture and programming environment of Intel® 64 and IA-32 architecture processors. Other volumes in this set are:

- Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 1: Basic Architecture (order number 253665).
- Intel® 64 and IA-32 Architectures Software Developer's Manual, Volumes 2A, 2B, 2C, & 2D: Instruction Set Reference (order numbers 253666, 253667, 326018, and 334569).
- The Intel® 64 and IA-32 Architectures Software Developer's Manual, Volumes 3A, 3B, 3C, & 3D: System Programming Guide (order numbers 253668, 253669, 326019, and 332831).

The Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 1, describes the basic architecture and programming environment of Intel 64 and IA-32 processors. The Intel® 64 and IA-32 Architectures Software Developer's Manual, Volumes 2A, 2B, 2C, & 2D, describe the instruction set of the processor and the opcode structure. These volumes apply to application programmers and to programmers who write operating systems or executives. The Intel® 64 and IA-32 Architectures Software Developer's Manual, Volumes 3A, 3B, 3C, & 3D, describe the operating-system support environment of Intel 64 and IA-32 processors. These volumes target operating-system and BIOS designers. In addition, the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 3B, and the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 3C, address the programming environment for classes of software that host operating systems. The Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 4, describes the model-specific registers of Intel 64 and IA-32 processors.

1.1 OVERVIEW OF THE MODEL-SPECIFIC REGISTERS

A description of this manual's content follows:

Chapter 1 — About This Manual. Gives an overview of all volumes of the Intel® 64 and IA-32 Architectures Software Developer's Manual, with chapter-specific details for the current volume.

Chapter 2 — Model-Specific Registers (MSRs). Lists the MSRs available in Intel processors, and describes their functions.

CHAPTER 2 MODEL-SPECIFIC REGISTERS (MSRS)

This chapter lists MSRs across Intel processor families. All MSRs listed can be read with the RDMSR and written with the WRMSR instructions. The scope of an MSR defines the set of processors that access the same MSR with RDMSR and WRMSR. Thread-scope MSRs are unique to every logical processor. Core-scope MSRs are shared by the threads in the same core; similarly for module-scope, die-scope, and package-scope.

When a processor package contains a single die, die-scope and package-scope are synonymous. When a package contains multiple die, they are distinct.

NOTE

For information on hierarchical level types supported, refer to the CPUID Leaf 1FH definition for the actual level type numbers: "V2 Extended Topology Enumeration Leaf" in the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 2A. Also see Section 10.9.1, "Hierarchical Mapping of Shared Resources," in the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 3A.

Register addresses are given in both hexadecimal and decimal. The register name is the mnemonic register name and the bit description describes individual bits in registers.

Model specific registers and its bit-fields may be supported for a finite range of processor families/models. To distinguish between different processor family and/or models, software must use CPUID.01H leaf function to query the combination of DisplayFamily and DisplayModel to determine model-specific availability of MSRs (see CPUID instruction in Chapter 3, "Instruction Set Reference, A-L," in the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 2A). Table 2-1 lists the signature values of DisplayFamily and DisplayModel for various processor families or processor number series.

Table 2-1. CPUID Signature Values of DisplayFamily_DisplayModel

DisplayFamily_DisplayModel	Processor Families/Processor Number Series
06_BDH	Intel® Series 2 Core™ Ultra processors supporting Lunar Lake performance hybrid architecture
06_ADH, 06_AEH	Intel® Xeon® 6 P-core processors based on Granite Rapids microarchitecture
06_AFH	Intel® Xeon® 6 E-core processors based on Sierra Forest microarchitecture
06_AAH	Intel® Core™ Ultra 7 processors supporting Meteor Lake performance hybrid architecture
06_CFH	5th generation Intel® Xeon® Scalable Processor Family based on Emerald Rapids microarchitecture
06_8FH	4th generation Intel® Xeon® Scalable Processor Family based on Sapphire Rapids microarchitecture
06_BAH, 06_B7H, 06_BFH	13th generation Intel® Core™ processors supporting Raptor Lake performance hybrid architecture
06_97H, 06_9AH	12th generation Intel® Core™ processors supporting Alder Lake performance hybrid architecture
06_8CH, 06_8DH	11th generation Intel® Core™ processors based on Tiger Lake microarchitecture
06_A7H	11th generation Intel® Core™ processors based on Rocket Lake microarchitecture
06_7DH, 06_7EH	10th generation Intel® Core™ processors based on Ice Lake microarchitecture
06_A5H, 06_A6H	10th generation Intel® Core™ processors based on Comet Lake microarchitecture
06_66H	Intel® Core™ processors based on Cannon Lake microarchitecture
06_8EH, 06_9EH	7th generation Intel® Core™ processors based on Kaby Lake microarchitecture, 8th and 9th generation Intel® Core™ processors based on Coffee Lake microarchitecture, Intel® Xeon® E processors based on Coffee Lake microarchitecture
06_6AH, 06_6CH	3rd generation Intel® Xeon® Scalable Processor Family based on Ice Lake microarchitecture

Table 2-1. CPUID Signature Values of DisplayFamily_DisplayModel (Contd.)

DisplayFamily_DisplayModel	Processor Families/Processor Number Series
06_55H	Intel® Xeon® Scalable Processor Family based on Skylake microarchitecture, 2nd generation Intel® Xeon® Scalable Processor Family based on Cascade Lake product, and 3rd generation Intel® Xeon® Scalable Processor Family based on Cooper Lake product
06_4EH, 06_5EH	6th generation Intel Core processors and Intel Xeon processor E3-1500m v5 product family and E3-1200 v5 product family based on Skylake microarchitecture
06_85H	Intel® Xeon Phi™ Processor 7215, 7285, 7295 Series based on Knights Mill microarchitecture
06_57H	Intel® Xeon Phi™ Processor 3200, 5200, 7200 Series based on Knights Landing microarchitecture
06_56H	Intel Xeon processor D-1500 product family based on Broadwell microarchitecture
06_4FH	Intel Xeon processor E5 v4 Family based on Broadwell microarchitecture, Intel Xeon processor E7 v4 Family, Intel Core i7-69xx Processor Extreme Edition
06_47H	5th generation Intel Core processors, Intel Xeon processor E3-1200 v4 product family based on Broadwell microarchitecture
06_3DH	Intel Core M-5xxx Processor, 5th generation Intel Core processors based on Broadwell microarchitecture
06_3FH	Intel Xeon processor E5-4600/2600/1600 v3 product families, Intel Xeon processor E7 v3 product families based on Haswell-E microarchitecture, Intel Core i7-59xx Processor Extreme Edition
06_3CH, 06_45H, 06_46H	4th Generation Intel Core processor and Intel Xeon processor E3-1200 v3 product family based on Haswell microarchitecture
06_3EH	Intel Xeon processor E7-8800/4800/2800 v2 product families based on Ivy Bridge-E microarchitecture
06_3EH	Intel Xeon processor E5-2600/1600 v2 product families and Intel Xeon processor E5-2400 v2 product family based on Ivy Bridge-E microarchitecture, Intel Core i7-49xx Processor Extreme Edition
06_3AH	3rd Generation Intel Core Processor and Intel Xeon processor E3-1200 v2 product family based on Ivy Bridge microarchitecture
06_2DH	Intel Xeon processor E5 Family based on Sandy Bridge microarchitecture, Intel Core i7-39xx Processor Extreme Edition
06_2FH	Intel Xeon Processor E7 Family
06_2AH	Intel Xeon processor E3-1200 product family; 2nd Generation Intel Core i7, i5, i3 Processors 2xxx Series
06_2EH	Intel Xeon processor 7500, 6500 series
06_25H, 06_2CH	Intel Xeon processors 3600, 5600 series, Intel Core i7, i5, and i3 Processors
06_1EH, 06_1FH	Intel Core i7 and i5 Processors
06_1AH	Intel Core i7 Processor, Intel Xeon processor 3400, 3500, 5500 series
06_1DH	Intel Xeon processor MP 7400 series
06_17H	Intel Xeon processor 3100, 3300, 5200, 5400 series, Intel Core 2 Quad processors 8000, 9000 series
06_0FH	Intel Xeon processor 3000, 3200, 5100, 5300, 7300 series, Intel Core 2 Quad processor 6000 series, Intel Core 2 Extreme 6000 series, Intel Core 2 Duo 4000, 5000, 6000, 7000 series processors, Intel Pentium dual-core processors
06_0EH	Intel Core Duo, Intel Core Solo processors
06_0DH	Intel Pentium M processor
06_86H, 06_96H, 06_9CH	Intel Atom® processors, Intel® Celeron® processors, Intel® Pentium® processors, and Intel® Pentium® Silver processors based on Tremont Microarchitecture
06_7AH	Intel Atom processors based on Goldmont Plus microarchitecture
06_5FH	Intel Atom processors based on Goldmont microarchitecture (Denverton)
06_5CH	Intel Atom processors based on Goldmont microarchitecture

Table 2-1. CPUID Signature Values of DisplayFamily_DisplayModel (Contd.)

DisplayFamily_DisplayModel	Processor Families/Processor Number Series
06_4CH	Intel Atom processor X7-Z8000 and X5-Z8000 series based on Airmont microarchitecture
06_5DH	Intel Atom processor X3-C3000 based on Silvermont microarchitecture
06_5AH	Intel Atom processor Z3500 series
06_4AH	Intel Atom processor Z3400 series
06_37H	Intel Atom processor E3000 series, Z3600 series, Z3700 series
06_4DH	Intel Atom processor C2000 series
06_36H	Intel Atom processor S1000 Series
06_1CH, 06_26H, 06_27H, 06_35H, 06_36H	Intel Atom processor family, Intel Atom processor D2000, N2000, E2000, Z2000, C1000 series
0F_06H	Intel Xeon processor 7100, 5000 Series, Intel Xeon Processor MP, Intel Pentium 4, Pentium D processors
0F_03H, 0F_04H	Intel Xeon processor, Intel Xeon processor MP, Intel Pentium 4, Pentium D processors
06_09H	Intel Pentium M processor
0F_02H	Intel Xeon Processor, Intel Xeon processor MP, Intel Pentium 4 processors
0F_0H, 0F_01H	Intel Xeon Processor, Intel Xeon processor MP, Intel Pentium 4 processors
06_7H, 06_08H, 06_0AH, 06_0BH	Intel Pentium III Xeon processor, Intel Pentium III processor
06_03H, 06_05H	Intel Pentium II Xeon processor, Intel Pentium II processor
06_01H	Intel Pentium Pro processor
05_01H, 05_02H, 05_04H	Intel Pentium processor, Intel Pentium processor with MMX Technology
06_BEH	Intel® Processor and Intel® Core™ i3 and Intel® Core™ 3 N-Series Processors and Intel Atom® x7000 Processor Series based on Gracemont microarchitecture

The Intel® Quark™ SoC X1000 processor can be identified by the signature of DisplayFamily_DisplayModel = 05_09H and SteppingID = 0

2.1 ARCHITECTURAL MSRS

Many MSRs have carried over from one generation of IA-32 processors to the next and to Intel 64 processors. A subset of MSRs and associated bit fields, which do not change on future processor generations, are now considered architectural MSRs. For historical reasons (beginning with the Pentium 4 processor), these “architectural MSRs” were given the prefix “IA32_”. Table 2-2 lists the architectural MSRs, their addresses, their current names, their names in previous IA-32 processors, and bit fields that are considered architectural. MSR addresses outside Table 2-2 and certain bit fields in an MSR address that may overlap with architectural MSR addresses are model-specific. Code that accesses a model-specific MSR and that is executed on a processor that does not support that MSR will generate an exception.

Architectural MSR or individual bit fields in an architectural MSR may be introduced or transitioned at the granularity of certain processor family/model or the presence of certain CPUID feature flags. The right-most column of Table 2-2 provides information on the introduction of each architectural MSR or its individual fields. This information is expressed either as signature values of “DF_DM” (see Table 2-1) or via CPUID flags.

Certain bit field position may be related to the maximum physical address width, the value of which is expressed as “MAXPHYADDR” in Table 2-2. “MAXPHYADDR” is reported by CPUID.8000_0008H leaf.

MSR address range between 40000000H - 4000FFFFH is marked as a specially reserved range. All existing and future processors will not implement any features using any MSR in this range.

Table 2-2. IA-32 Architectural MSRs

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 0H, 0		IA32_P5_MC_ADDR (P5_MC_ADDR)
See Section 2.23, "MSRs in Pentium Processors."		Pentium Processor (05_01H)
Register Address: 1H, 1		IA32_P5_MC_TYPE (P5_MC_TYPE)
See Section 2.23, "MSRs in Pentium Processors."		DF_DM = 05_01H
Register Address: 6H, 6		IA32_MONITOR_FILTER_SIZE
See Section 10.10.5, "Monitor/Mwait Address Range Determination."		0F_03H
Register Address: 10H, 16		IA32_TIME_STAMP_COUNTER (TSC)
See Section 19.17, "Time-Stamp Counter."		05_01H
Register Address: 17H, 23		IA32_PLATFORM_ID (MSR_PLATFORM_ID)
Platform ID (R/O) The operating system can use this MSR to determine "slot" information for the processor and the proper microcode update to load.		06_01H
49:0	Reserved.	
52:50	Platform ID (R/O) Contains information concerning the intended platform for the processor. 52 51 50 0 0 0 Processor Flag 0 0 0 1 Processor Flag 1 0 1 0 Processor Flag 2 0 1 1 Processor Flag 3 1 0 0 Processor Flag 4 1 0 1 Processor Flag 5 1 1 0 Processor Flag 6 1 1 1 Processor Flag 7	
63:53	Reserved.	
Register Address: 1BH, 27		IA32_APIC_BASE (APIC_BASE)
This register holds the APIC base address, permitting the relocation of the APIC memory map. See Section 12.4.4, "Local APIC Status and Location," and Section 12.4.5, "Relocating the Local APIC Registers."		06_01H
7:0	Reserved.	
8	BSP Flag (R/W)	
9	Reserved.	
10	Enable x2APIC mode.	06_1AH
11	APIC Global Enable (R/W)	
(MAXPHYADDR - 1):12	APIC Base (R/W)	
63: MAXPHYADDR	Reserved.	
Register Address: 2FH, 47		IA32_BARRIER
IA32_BARRIER (R/O) The IA32_BARRIER MSR ensures ordered execution by acting like LFENCE, controlling the sequencing of subsequent MSR reads after prior MSR reads and instructions.		CPUID.07H.01H:EAX[27]=1

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
31:0	DATA Reserved. Always 0.	
63:32	Reserved.	
Register Address: 3AH, 58		IA32_FEATURE_CONTROL
Control Features in Intel 64 Processor (R/W)		If any one enumeration condition for defined bit field holds.
0	Lock bit (R/WO): (1 = locked). When set, locks this MSR from being written; writes to this bit will result in GP(0). Note: Once the Lock bit is set, the contents of this register cannot be modified. Therefore the lock bit must be set after configuring support for Intel Virtualization Technology and prior to transferring control to an option ROM or the OS. Hence, once the Lock bit is set, the entire IA32_FEATURE_CONTROL contents are preserved across RESET when PWRGOOD is not deasserted.	If any one enumeration condition for defined bit field position greater than bit 0 holds.
1	Enable VMX inside SMX operation (R/WL) This bit enables a system executive to use VMX in conjunction with SMX to support Intel® Trusted Execution Technology. BIOS must set this bit only when the CPUID function 1 returns VMX feature flag and SMX feature flag set (ECX bits 5 and 6 respectively).	If CPUID.01H:ECX[5] = 1 && CPUID.01H:ECX[6] = 1
2	Enable VMX outside SMX operation (R/WL) This bit enables VMX for a system executive that does not require SMX. BIOS must set this bit only when the CPUID function 1 returns the VMX feature flag set (ECX bit 5).	If CPUID.01H:ECX[5] = 1
7:3	Reserved.	
14:8	SENTER Local Function Enables (R/WL) When set, each bit in the field represents an enable control for a corresponding SENTER function. This field is supported only if CPUID.1:ECX.[bit 6] is set.	If CPUID.01H:ECX[6] = 1
15	SENTER Global Enable (R/WL) This bit must be set to enable SENTER leaf functions. This bit is supported only if CPUID.1:ECX.[bit 6] is set.	If CPUID.01H:ECX[6] = 1
16	Reserved.	
17	SGX Launch Control Enable (R/WL) This bit must be set to enable runtime re-configuration of SGX Launch Control via the IA32_SGXLEPUBKEYHASHn MSR.	If CPUID.(EAX=07H, ECX=0H): ECX[30] = 1
18	SGX Global Enable (R/WL) This bit must be set to enable SGX leaf functions.	If CPUID.(EAX=07H, ECX=0H): EBX[2] = 1
19	Reserved.	
20	LMCE On (R/WL) When set, system software can program the MSRs associated with LMCE to configure delivery of some machine check exceptions to a single logical processor.	If IA32_MCG_CAP[27] = 1
63:21	Reserved.	
Register Address: 3BH, 59		IA32_TSC_ADJUST

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Per Logical Processor TSC Adjust (R/Write to clear)		If CPUID.(EAX=07H, ECX=0H): EBX[1] = 1
63:0	THREAD_ADJUST Local offset value of the IA32_TSC for a logical processor. Reset value is zero. A write to IA32_TSC will modify the local offset in IA32_TSC_ADJUST and the content of IA32_TSC, but does not affect the internal invariant TSC hardware.	
Register Address: 48H, 72		IA32_SPEC_CTRL
Speculation Control (R/W) The MSR bits are defined as logical processor scope. On some core implementations, the bits may impact sibling logical processors on the same core. This MSR has a value of 0 after reset and is unaffected by INIT# or SIPI#.		If any one of the enumeration conditions for defined bit field positions holds.
0	Indirect Branch Restricted Speculation (IBRS). Restricts speculation of indirect branch.	If CPUID.(EAX=07H, ECX=0):EDX[26]=1
1	Single Thread Indirect Branch Predictors (STIBP). Prevents indirect branch predictions on all logical processors on the core from being controlled by any sibling logical processor in the same core.	If CPUID.(EAX=07H, ECX=0):EDX[27]=1
2	Speculative Store Bypass Disable (SSBD) delays speculative execution of a load until the addresses for all older stores are known.	If CPUID.(EAX=07H, ECX=0):EDX[31]=1
3	IPRED_DIS_U If 1, enables IPRED_DIS control for CPL3.	If CPUID.(EAX=07H, ECX=2):EDX[1]=1
4	IPRED_DIS_S If 1, enables IPRED_DIS control for CPL0/1/2.	If CPUID.(EAX=07H, ECX=2):EDX[1]=1
5	RRSBA_DIS_U If 1, disables RRSBA behavior for CPL3.	If CPUID.(EAX=07H, ECX=2):EDX[2]=1
6	RRSBA_DIS_S If 1, disables RRSBA behavior for CPL0/1/2.	If CPUID.(EAX=07H, ECX=2):EDX[2]=1
7	PSFD If 1, disables Fast Store Forwarding Predictor. Note that setting bit 2 (SSBD) also disables this.	If CPUID.(EAX=07H, ECX=2):EDX[0]=1
8	DDPD_U If 1, disables the Data Dependent Prefetcher that examines data values in memory while CPL = 3. Note that setting bit 2 (SSBD) also disables this.	If CPUID.(EAX=07H, ECX=2):EDX[3]=1
9	Reserved.	
10	BHI_DIS_S When '1, enables BHI_DIS_S behavior.	If CPUID.(EAX=07H, ECX=2):EDX[4]=1
63:11	Reserved.	
Register Address: 49H, 73		IA32_PRED_CMD
Prediction Command (WO) Gives software a way to issue commands that affect the state of predictors.		If any one of the enumeration conditions for defined bit field positions holds.

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
0	Indirect Branch Prediction Barrier (IBPB)	If CPUID.(EAX=07H, ECX=0):EDX[26]=1
63:1	Reserved.	
Register Address: 4EH, 78		IA32_PPIN_CTL
Protected Processor Inventory Number Enable Control (R/W)		If CPUID.(EAX=07H, ECX=01H):EBX[0]=1 ¹
0	LockOut (R/W/O) If 0, indicates that further writes to IA32_PPIN_CTL is allowed. If 1, indicates that further writes to IA32_PPIN_CTL is disallowed. Writing 1 to this bit is only permitted if the Enable_PPIN bit is clear. The Privileged System Software Inventory Agent should read IA32_PPIN_CTL[bit 1] to determine if IA32_PPIN is accessible. The Privileged System Software Inventory Agent is not expected to write to this MSR.	
1	Enable_PPIN (R/W) If 1, indicates that IA32_PPIN is accessible using RDMSR. If 0, indicates that IA32_PPIN is inaccessible using RDMSR. Any attempt to read IA32_PPIN will cause #GP.	
63:2	Reserved.	
Register Address: 4FH, 79		IA32_PPIN
Protected Processor Inventory Number (R/O)		If CPUID.(EAX=07H, ECX=01H):EBX[0]=1 ¹
63:0	Protected Processor Inventory Number (R/O) A unique value within a given CPUID family/model/stepping signature that a privileged inventory initialization agent can access to identify each physical processor, when access to IA32_PPIN is enabled. Access to IA32_PPIN is permitted only if IA32_PPIN_CTL[bits 1:0] = '10b'.	
Register Address: 79H, 121		IA32_BIOS_UPDT_TRIG (BIOS_UPDT_TRIG)
BIOS Update Trigger (W) Executing a WRMSR instruction to this MSR causes a microcode update to be loaded into the processor. See Section 11.11.6, "Microcode Update Loader." A processor may prevent writing to this MSR when loading guest states on VM entries or saving guest states on VM exits.		06_01H
Register Address: 7AH, 122		IA32_FEATURE_ACTIVATION
Feature Activation (R/W) Implements Feature Activation command. WRMSR to this address activates all 'activatable' features on this thread.		
0	SE Secure Enclaves feature activation.	
1	KL Keylocker feature activation.	
63:2	Reserved.	
Register Address: 7BH, 123		IA32_MCU_ENUMERATION

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA32_MCU_ENUMERATION (R/O) Enumeration of architectural features.		
0	UNIFORM_MCU_AVAIL When set to 1, uniform microcode update is available, and UNIFORM_MCU_SCOPE (bits [10:8]) indicates the scope of writes to IA32_BIOS_UPDT_TRIG. When set to 0, uniform microcode update is not available, and writes to IA32_BIOS_UPDT_TRIG are core scoped.	
1	UNIFORM_MCU_CONFIG_REQD When set to 1, indicates that configuration is required to ensure that all MCU components are updated on WRMSR 79H, and UNIFORM_MCU_CONFIG_COMPLETE (bit 2) should be checked to determine whether the necessary configuration has been completed. When set to 0, indicates that no configuration is required, and UNIFORM_MCU_CONFIG_COMPLETE should be ignored.	
2	UNIFORM_MCU_CONFIG_COMPLETE If UNIFORM_MCU_CONFIG_REQD (bit 1) is 0, then this bit should be ignored. If UNIFORM_MCU_CONFIG_REQD is 1, then this bit indicates whether all necessary configurations have been completed to ensure that all MCU components will be updated on WRMSR 79H.	
3	ARCH_ROLLBACK_SVN_COMMIT When set to 1, indicates support for the MCU deferred SVN architecture, SVN reporting architecture, and MCU rollback architecture.	
4	MCU_STAGING When set to 1, indicates that the microcode update staging capability is supported by the processor. When supported, the use of the MCU staging capability is recommended to reduce the latency of the IA32_BIOS_UPDT_TRIG operation.	
7:5	Reserved for future use.	
15:8	UNIFORM_MCU_SCOPE Indicates the current* uniform microcode update scope: ▪ 0x02: Core Scoped ▪ 0x03: Module Scoped** ▪ 0x04: Tile Scoped** ▪ 0x05: Die Scoped** ▪ 0x80: Package Scoped ▪ 0xC0: Platform Scoped All others: Reserved for future use * The value of this field reflects the state of platform configuration and may change as the configuration changes during the boot process. Once configuration is complete, it is not expected to change during runtime. ** If these domains are enumerated by CPUID.1F, then this field may also report them as appropriate.	
63:16	Reserved for future use.	
Register Address: 7CH, 124		IA32_MCU_STATUS

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
MCU Status (R/O) Communicates results from the previous patch loads.		
0	MCU_PARTIAL_UPDATE When set to 1, indicates that the most recent write to IA32_BIOS_UPDT_TRIG resulted in a partial update. This means that microcode update components were only partially updated after some portion of the MCU had already been committed and the Revision ID had been updated.	
1	AUTH_FAIL_ON_MCU_COMPONENT When set to 1, indicates that an authentication failure occurred on some portion of the MCU after another portion of the MCU had already been committed and the Revision ID had already been updated on the most recent write to IA32_BIOS_UPDT_TRIG.	
2	Reserved for future use.	
3	POST_BIOS_MCU When set to 1, indicates that an update was successfully loaded via IA32_BIOS_UPDT_TRIG after bit 0 of MSR_BIOS_DONE (address 151H) was set to 1.	
63:4	Reserved for future use.	
Register Address: 82H, 130		IA32_FZM_RANGE_INDEX
IA32_FZM_RANGE_INDEX (R/W) Index and Domain handle for a valid FZM region. Programmed by software and used by other FRM MSRs FZM Range Index register to R/W Domain Index.		
3:0	REGION_INDEX Holds the Index of domain.	
7:4	Reserved.	
12:8	DOMAIN_HANDLE Holds the Domain Handle.	
63:13	Reserved.	
Register Address: 83H, 131		IA32_FZM_DOMAIN_CONFIG
IA32_FZM_DOMAIN_CONFIG (R/O) Bit mask of valid regions within the domain identified by FZM_RANGE_INDEX.		
63:0	REGION_BITMAP Bitmap of valid regions for a given domain.	
Register Address: 84H, 132		IA32_FZM_RANGE_STARTADDR
IA32_FZM_RANGE_STARTADDR (R/O) Start address of the FZM range pointed to by FZM_RANGE_INDEX.		
51:0	START_ADDR Start address of the specified domain in FZM_RANGE_INDEX.	
63:52	Reserved.	
Register Address: 85H, 133		IA32_FZM_RANGE_ENDADDR

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA32_FZM_RANGE_ENDADDR (R/O) End address of the specified domain in FZM_RANGE_INDEX.		
51:0	END_ADDR End address of the specified domain in FZM_RANGE_INDEX.	
63:52	Reserved.	
Register Address: 86H, 134		IA32_FZM_RANGE_WRITESTATUS
IA32_FZM_RANGE_WRITESTATUS (R/O) Write status of the FZM range pointed to by FZM_RANGE_INDEX.		
0	WRITE_STATUS Write status of the specified domain in FZM_RANGE_INDEX.	
1	READ_STATUS Read status of the specified domain in FZM_RANGE_INDEX.	
63:2	Reserved.	
Register Address: 87H, 135		IA32_MKTME_KEYID_PARTITIONING
MKTME KEY ID Partitioning (R/O) Enumerates the number of activated KeyIDs for Intel TME-MK and Intel TDX.		
31:0	NUM_MKTME_KIDS Number of activated Intel TME-MK KeyIDs. This field is supported on all parts that enumerate support for Intel Total Memory Encryption - Multi-Key (Intel TME-MK). If IA32_TME_ACTIVATE.LOCK is 1, this field reports MAX_ACTIVATE_MKTME_HKIDS (KMK-1) else report 0. Intel TME-MK KIDs will always span the KID range [1 ... NUM_MKTME_KIDS].	
63:32	NUM_TDX_PRIV_KIDS Number of activated TDX private KeyIDs. This field is supported on all parts that enumerate support for SEAM mode. If IA32_TME_ACTIVATE.LOCK is 1, This field reports MAX_ACTIVATE_TDX_HKIDS (KTD) else report 0. TDX private KIDs will always span the range [NUM_MKTME_KIDS+1... (NUM_MKTME_KIDS + NUM_TDX_PRIV_KIDS)].	
Register Address: 8BH, 139		IA32_BIOS_SIGN_ID (BIOS_SIGN/BBL_CR_D3)
BIOS Update Signature (R/W) Returns the microcode update signature following the execution of CPUID.01H. A processor may prevent writing to this MSR when loading guest states on VM entries or saving guest states on VM exits.		06_01H
31:0	Reserved.	
63:32	PATCH_SIGN_ID It is recommended that this field be preloaded with zero prior to executing CPUID. If the field remains zero following the execution of CPUID, this indicates that no microcode update is loaded. Any non-zero value is the microcode update signature patch signature ID.	
Register Address: 8CH, 140		IA32_SGXLEPUBKEYHASH0

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA32_SGXLEPUBKEYHASH[63:0] (R/W) Bits 63:0 of the SHA256 digest of the SIGSTRUCT.MODULUS for SGX Launch Enclave. On reset, the default value is the digest of Intel's signing key.		Read permitted if CPUID.(EAX=12H,ECX=0H): EAX[0]=1 && CPUID.(EAX=07H, ECX=0H):ECX[30]=1. Write permitted if CPUID.(EAX=12H,ECX=0H): EAX[0]=1 && IA32_FEATURE_CONTROL[17]=1 && IA32_FEATURE_CONTROL[0] = 1.
Register Address: 8DH, 141		IA32_SGXLEPUBKEYHASH1
IA32_SGXLEPUBKEYHASH[127:64] (R/W) Bits 127:64 of the SHA256 digest of the SIGSTRUCT.MODULUS for SGX Launch Enclave. On reset, the default value is the digest of Intel's signing key.		Same comment in MSR listing for IA32_SGXLEPUBKEYHASH0 (MSR address 8CH, 140) applies here.
Register Address: 8EH, 142		IA32_SGXLEPUBKEYHASH2
IA32_SGXLEPUBKEYHASH[191:128] (R/W) Bits 191:128 of the SHA256 digest of the SIGSTRUCT.MODULUS for SGX Launch Enclave. On reset, the default value is the digest of Intel's signing key.		Same comment in MSR listing for IA32_SGXLEPUBKEYHASH0 (MSR address 8CH, 140) applies here.
Register Address: 8FH, 143		IA32_SGXLEPUBKEYHASH3
IA32_SGXLEPUBKEYHASH[255:192] (R/W) Bits 255:192 of the SHA256 digest of the SIGSTRUCT.MODULUS for SGX Launch Enclave. On reset, the default value is the digest of Intel's signing key.		Same comment in MSR listing for IA32_SGXLEPUBKEYHASH0 (MSR address 8CH, 140) applies here.
Register Address: 90H, 144		IA32_SGXLEPUBKEYHASH4
IA32_SGXLEPUBKEYHASH[319:256] (R/W) Bits 319:256 of the SHA256 digest of the SIGSTRUCT.MODULUS for SGX Launch Enclave. On reset, the default value is the digest of Intel's signing key.		Same comment in MSR listing for IA32_SGXLEPUBKEYHASH0 (MSR address 8CH, 140) applies here.
Register Address: 91H, 145		IA32_SGXLEPUBKEYHASH5
IA32_SGXLEPUBKEYHASH[383:320] (R/W) Bits 383:320 of the SHA256 digest of the SIGSTRUCT.MODULUS for SGX Launch Enclave. On reset, the default value is the digest of Intel's signing key.		Same comment in MSR listing for IA32_SGXLEPUBKEYHASH0 (MSR address 8CH, 140) applies here.
Register Address: 9BH, 155		IA32_SMM_MONITOR_CTL
SMM Monitor Configuration (R/W)		If CPUID.01H: ECX[5]=1 CPUID.01H: ECX[6] = 1
0	Valid (R/W)	
1	Reserved.	
2	Controls SMI unblocking by VMXOFF (see Section 33.14.4).	If IA32_VMX_MISC[28]
11:3	Reserved.	
31:12	MSEG Base (R/W)	
63:32	Reserved.	
Register Address: 9EH, 158		IA32_SMBASE
Base address of the logical processor's SMRAM image (R/O, SMM only).		If IA32_VMX_MISC[15]
Register Address: BCH, 188		IA32_MISC_PACKAGE_CTLs

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Power Filtering Control (R/W) This MSR has a value of 0 after reset and is unaffected by INIT# or SIPI#.		If IA32_ARCH_CAPABILITIES [10] = 1
0	ENERGY_FILTERING_ENABLE (R/W) If set, RAPL MSRs report filtered processor power consumption data. This bit can be changed from 0 to 1, but cannot be changed from 1 to 0. After setting, all attempts to clear it are ignored until the next processor reset.	If IA32_ARCH_CAPABILITIES [11] = 1
63:1	Reserved.	
Register Address: BDH, 189		IA32_XAPIC_DISABLE_STATUS
xAPIC Disable Status (R/O)		If CPUID.(EAX=07H, ECX=0):EDX[29]=1 and IA32_ARCH_CAPABILITIES [21] = 1
0	LEGACY_XAPIC_DISABLED When set, indicates that the local APIC is in x2APIC mode (IA32_APIC_BASE.EXTD = 1) and that attempts to clear IA32_APIC_BASE.EXTD will fail (e.g., WRMSR will #GP).	
63:1	Reserved.	
Register Address: C1H, 193		IA32_PMC0 (PERFCTR0)
General Performance Counter 0 (R/W)		If CPUID.0AH: EAX[15:8] > 0
Register Address: C2H, 194		IA32_PMC1 (PERFCTR1)
General Performance Counter 1 (R/W)		If CPUID.0AH: EAX[15:8] > 1
Register Address: C3H, 195		IA32_PMC2
General Performance Counter 2 (R/W)		If CPUID.0AH: EAX[15:8] > 2
Register Address: C4H, 196		IA32_PMC3
General Performance Counter 3 (R/W)		If CPUID.0AH: EAX[15:8] > 3
Register Address: C5H, 197		IA32_PMC4
General Performance Counter 4 (R/W)		If CPUID.0AH: EAX[15:8] > 4
Register Address: C6H, 198		IA32_PMC5
General Performance Counter 5 (R/W)		If CPUID.0AH: EAX[15:8] > 5
Register Address: C7H, 199		IA32_PMC6
General Performance Counter 6 (R/W)		If CPUID.0AH: EAX[15:8] > 6
Register Address: C8H, 200		IA32_PMC7
General Performance Counter 7 (R/W)		If CPUID.0AH: EAX[15:8] > 7
Register Address: C9H, 201		IA32_PMC8
General Performance Counter 8 (R/W)		If CPUID.0AH: EAX[15:8] > 8
Register Address: CAH, 202		IA32_PMC9
General Performance Counter 9 (R/W)		If CPUID.0AH: EAX[15:8] > 9
Register Address: CFH, 207		IA32_CORE_CAPABILITIES

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA32 Core Capabilities Register		If CPUID.(EAX=07H, ECX=0):EDX[30] = 1
63:0	Reserved.	No architecturally defined bits.
Register Address: E1H, 225		IA32_UMWAIT_CONTROL
UMWAIT Control (R/W)		
0	C0.2 is not allowed by the OS. Value of "1" means all C0.2 requests revert to C0.1.	
1	Reserved.	
31:2	Determines the maximum time in TSC-quanta that the processor can reside in either C0.1 or C0.2. A zero value indicates no maximum time. The maximum time value is a 32-bit value where the upper 30 bits come from this field and the lower two bits are zero.	
Register Address: E7H, 231		IA32_MPERF
TSC Frequency Clock Counter (R/Write to clear)		If CPUID.06H: ECX[0] = 1
63:0	C0_MCNT: C0 TSC Frequency Clock Count Increments at fixed interval (relative to TSC freq.) when the logical processor is in C0. Cleared upon overflow / wrap-around of IA32_APERF.	
Register Address: E8H, 232		IA32_APERF
Actual Performance Clock Counter (R/Write to clear)		If CPUID.06H: ECX[0] = 1
63:0	C0_ACNT: C0 Actual Frequency Clock Count Accumulates core clock counts at the coordinated clock frequency, when the logical processor is in C0. Cleared upon overflow / wrap-around of IA32_MPERF.	
Register Address: FEH, 254		IA32_MTRRCAP (MTRRcap)
MTRR Capability (R/O) See Section 13.11.2.1, "IA32_MTRR_DEF_TYPE MSR."		06_01H
7:0	VCNT: The number of variable memory type ranges in the processor.	
8	Fixed range MTRRs are supported when set.	
9	Reserved.	
10	WC Supported when set.	
11	SMRR Supported when set.	
12	PRMRR supported when set.	
63:13	Reserved.	
Register Address: 10AH, 266		IA32_ARCH_CAPABILITIES
Enumeration of Architectural Features (R/O)		If CPUID.(EAX=07H, ECX=0):EDX[29]=1
0	RDCL_NO: The processor is not susceptible to Rogue Data Cache Load (RDCL).	
1	IBRS_ALL: The processor supports enhanced IBRS.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
2	RSBA: The processor supports RSB Alternate. Alternative branch predictors may be used by RET instructions when the RSB is empty. SW using retpoline may be affected by this behavior.	
3	SKIP_L1DFL_VMENTRY: A value of 1 indicates the hypervisor need not flush the L1D on VM entry.	
4	SSB_NO: Processor is not susceptible to Speculative Store Bypass.	
5	MDS_NO: Processor is not susceptible to Microarchitectural Data Sampling (MDS).	
6	IF_PSCCHANGE_MC_NO: The processor is not susceptible to a machine check error due to modifying the size of a code page without TLB invalidation.	
7	TSX_CTRL: If 1, indicates presence of IA32_TSX_CTRL MSR.	
8	TAA_NO: If 1, processor is not affected by TAA.	
9	MCU_CONTROL: If 1, the processor supports the IA32_MCU_CONTROL MSR.	
10	MISC_PACKAGE_CTL: The processor supports IA32_MISC_PACKAGE_CTL MSR.	
11	ENERGY_FILTERING_CTL: The processor supports setting and reading the IA32_MISC_PACKAGE_CTL[0] (ENERGY_FILTERING_ENABLE) bit.	
12	DOITM: If 1, the processor supports Data Operand Independent Timing Mode.	
13	SBDR_SSDP_NO: The processor is not affected by either the Shared Buffers Data Read (SBDR) vulnerability or the Sideband Stale Data Propagator (SSDP).	
14	FBSDP_NO: The processor is not affected by the Fill Buffer Stale Data Propagator (FBSDP).	
15	PSDP_NO: The processor is not affected by vulnerabilities involving the Primary Stale Data Propagator (PSDP).	
16	MCU_ENUMERATION: If 1, the processor supports the IA32_MCU_ENUMERATION and IA32_MCU_STATUS MSRs.	
17	FB_CLEAR: If 1, the processor supports overwrite of fill buffer values as part of MD_CLEAR operations with the VERW instruction.	
18	FB_CLEAR_CTRL: If 1, the processor supports the IA32_MCU_OPT_CTRL MSR and allows software to set bit 3 of that MSR (FB_CLEAR_DIS).	
19	RRSBA: A value of 1 indicates the processor may have the RRSBA alternate prediction behavior, if not disabled by RRSBA_DIS_U or RRSBA_DIS_S.	
20	BHI_NO: A value of 1 indicates BHI_NO branch prediction behavior, regardless of the value of IA32_SPEC_CTRL[BHI_DIS_S] MSR bit.	
21	XAPIC_DISABLE_STATUS: Enumerates that the IA32_XAPIC_DISABLE_STATUS MSR exists, and that bit 0 specifies whether the legacy xAPIC is disabled and APIC state is locked to x2APIC.	
22	MCU_EXTENDED_SERVICE: If 1, the processor supports MCU Extended servicing - IA32_MCU_EXT_SERVICE MSR.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
23	OVERCLOCKING_STATUS: If set, the IA32_OVERCLOCKING_STATUS MSR exists.	
24	PBRBSB_NO: If 1, the processor is not affected by issues related to Post-Barrier Return Stack Buffer Predictions.	
25	GDS_CTRL: If 1, the processor supports the GDS_MITG_DIS and GDS_MITG_LOCK bits of the IA32_MCU_OPT_CTRL MSR.	
26	GDS_NO: If 1, the processor is not affected by Gather Data Sampling.	
27	RFDS_NO: If 1, the processor is not affected by Register File Data Sampling.	
28	RFDS_CLEAR: If 1, when VERW is executed the processor will clear stale data from register files affected by Register File Data Sampling.	
29	IGN_UMONITOR_SUPPORT If 0, IA32_MCU_OPT_CTRL bit 6 (IGN_UMONITOR) is not supported. If 1, it indicates support of IA32_MCU_OPT_CTRL bit 6 (IGN_UMONITOR).	
30	MON_UMON_MITG_SUPPORT If 0, IA32_MCU_OPT_CTRL bit 7 (MON_UMON_MITG) is not supported. If 1, it indicates support of IA32_MCU_OPT_CTRL bit 7 (MON_UMON_MITG).	
31	Reserved.	
32	PBOPT_SUPPORT If 0, IA32_PBOPT_CTRL bit 0 (Prediction Barrier Option [PBOPT]) is not supported. If 1, IA32_PBOPT_CTRL bit 0 (Prediction Barrier Option [PBOPT]) is supported.	
61:33	Reserved.	
62	ITS_NO If 0, the hypervisor indicates that the system is not affected by Indirect Target Selection. If 1, the hypervisor indicates that the system may be affected by Indirect Target Selection.	
63	Reserved.	
Register Address: 10BH, 267		IA32_FLUSH_CMD
Flush Command (W0) Gives software a way to invalidate structures with finer granularity than other architectural methods.		If any one of the enumeration conditions for defined bit field positions holds.
0	L1D_FLUSH Writeback and invalidate the L1 data cache.	If CPUID.(EAX=07H, ECX=0):EDX[28]=1
63:1	Reserved.	
Register Address: 10FH, 271		IA32_TSX_FORCE_ABORT
TSX Force Abort		If CPUID.(EAX=07H, ECX=0):EDX[13]=1

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
0	RTM_FORCE_ABORT If 1, all RTM transactions abort with EAX code 0.	R/W, Default: 0 If CPUID.(EAX=07H,ECX=0):EDX[11]=1, bit 0 is always 1 and writes to change it are ignored. If SDV_ENABLE_RTM is 1, bit 0 is always 0 and writes to change it are ignored.
1	TSX_CPUID_CLEAR When set, CPUID.(EAX=07H,ECX=0):EBX[11]=0 and CPUID.(EAX=07H,ECX=0):EBX[4]=0.	R/W, Default: 0 Can be set only if CPUID.(EAX=07H,ECX=0):EDX[11]=1 or if SDV_ENABLE_RTM is 1.
2	SDV_ENABLE_RTM When set, CPUID.(EAX=07H,ECX=0):EDX[11]=0 and the processor may not force abort RTM. This unsupported mode should only be used for software development and not for production usage.	R/W, Default: 0 If 0, can be set only if CPUID.(EAX=07H,ECX=0):EDX[11]=1.
63:3	Reserved.	
Register Address: 122H, 290		IA32_TSX_CTRL
IA32_TSX_CTRL (R/W)		Thread scope. Not architecturally serializing. Available when CPUID.ARCH_CAP(EAX=7H, ECX = 0):EDX[29] = 1 and IA32_ARCH_CAPABILITIES.bit 7 = 1.
0	RTM_DISABLE When set to 1, XBEGIN will always abort with EAX code 0.	
1	TSX_CPUID_CLEAR When set to 1, CPUID.07H.EBX.RTM [bit 11] and CPUID.07H.EBX.HLE [bit 4] report 0. When set to 0 and the SKU supports TSX, these bits will return 1.	
63:2	Reserved.	
Register Address: 123H, 291		IA32_MCU_OPT_CTRL
Microcode Update Option Control (R/W)		If CPUID.(EAX=07H, ECX=0):EDX[9]=1 or CPUID.(EAX=07H, ECX=0H):EDX[11]=1 IA32_ARCH_CAPABILITIES [18] = 1 or IA32_ARCH_CAPABILITIES [25]=1 or IA32_ARCH_CAPABILITIES [29]=1 or IA32_ARCH_CAPABILITIES [30]=1

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
0	RNGDS_MITG_DIS (R/W) If 0 (default), SRBDS mitigation is enabled for RDRAND and RDSEED. If 1, SRBDS mitigation is disabled for RDRAND and RDSEED executed outside of Intel SGX enclaves.	If CPUID.(EAX=07H, ECX=0):EDX[9]=1
1	RTM_ALLOW If 0, XBEGIN will always abort with EAX code 0. If 1, XBEGIN behavior depends on the value of IA32_TSX_CTRL[RTM_DISABLE].	If CPUID.(EAX=07H, ECX=0H):EDX[11]=1 Read/Write Setting RTM_LOCKED prevents writes to this bit.
2	RTM_LOCKED When 1, RTM_ALLOW is locked at zero, writes to RTM_ALLOW will be ignored.	If CPUID.(EAX=07H, ECX=0H):EDX[11]=1 Read-Only status bit.
3	FB_CLEAR_DIS If 1, prevents the VERW instruction from performing an FB_CLEAR action.	If IA32_ARCH_CAPABILITIES [18]=1
4	GDS_MITG_DIS If 0, the Gather Data Sampling mitigation is enabled (patch load time default). If 1 on all threads for a given core, the Gather Data Sampling mitigation is disabled.	If IA32_ARCH_CAPABILITIES [25]=1
5	GDS_MITG_LOCK If 0, not locked, and GDS_MITG_DIS is under OS control. If 1, locked and GDS_MITG_DIS is forced to 0 (writes are ignored).	If IA32_ARCH_CAPABILITIES [25]=1
6	IGN_UMONITOR If 0, enable CPL0-3 software to use the UMONITOR/UMWAIT instructions. If 1 (default), disable UMONITOR functionality. CPL0-3 software will be able to call the UMONITOR instruction without causing a fault, however the address monitoring hardware will not be armed. When UMWAIT is called, it will not enter an implementation-dependent optimized state.	If IA32_ARCH_CAPABILITIES [29]=1
7	MON_UMON_MITG If 0 (default), disabled. If 1, enable: Flush the thread's previously monitored address from the CPU caches as part of the (U)MONITOR instruction. Additionally, for every 4th (U)MONITOR instruction within a core, flush the peer hyperthread's monitored address from the CPU caches as well. This will increase the latency of the instruction. This may have a minor impact on workloads using the (U)MONITOR instruction.	If IA32_ARCH_CAPABILITIES [30]=1
63:8	Reserved.	
Register Address: 174H, 372		IA32_SYSENTER_CS
SYSENTER_CS_MSR (R/W)		06_01H
15:0	CS Selector.	
31:16	Not used.	Can be read and written.

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
63:32	Not used.	Writes ignored; reads return zero.
Register Address: 175H, 373		IA32_SYSENTER_ESP
SYSENTER_ESP_MSR (R/W)		06_01H
Register Address: 176H, 374		IA32_SYSENTER_EIP
SYSENTER_EIP_MSR (R/W)		06_01H
Register Address: 179H, 377		IA32_MCG_CAP (MCG_CAP)
Global Machine Check Capability (R/O)		06_01H
7:0	Count: Number of reporting banks.	
8	MCG_CTL_P: IA32_MCG_CTL is present if this bit is set.	
9	MCG_EXT_P: Extended machine check state registers are present if this bit is set.	
10	MCP_CMCL_P: Support for corrected MC error event is present.	06_01H
11	MCG_TES_P: Threshold-based error status register are present if this bit is set.	
15:12	Reserved.	
23:16	MCG_EXT_CNT: Number of extended machine check state registers present.	
24	MCG_SER_P: The processor supports software error recovery if this bit is set.	
25	Reserved.	
26	MCG_ELOG_P: Indicates that the processor allows platform firmware to be invoked when an error is detected so that it may provide additional platform specific information in an ACPI format "Generic Error Data Entry" that augments the data included in machine check bank registers.	06_3EH
27	MCG_LMCE_P: Indicates that the processor supports extended state in IA32_MCG_STATUS and associated MSR necessary to configure Local Machine Check Exception (LMCE).	06_3EH
63:28	Reserved.	
Register Address: 17AH, 378		IA32_MCG_STATUS (MCG_STATUS)
Global Machine Check Status (R/W)		06_01H
0	RIPV. Restart IP valid.	06_01H
1	EIPV. Error IP valid.	06_01H
2	MCIP. Machine check in progress.	06_01H
3	LMCE_S.	If IA32_MCG_CAP.LMCE_P[27] =1
63:4	Reserved.	
Register Address: 17BH, 379		IA32_MCG_CTL (MCG_CTL)
Global Machine Check Control (R/W)		If IA32_MCG_CAP.CTL_P[8] =1
Register Address: 180H—185H, 384—389		N/A
Reserved		06_0EH ²

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 186H, 390		IA32_PERFEVTSEL0 (PERFEVTSEL0)
Performance Event Select Register 0 (R/W)		If CPUID.0AH: EAX[15:8] > 0
7:0	Event Select: Selects a performance event logic unit.	
15:8	UMask: Qualifies the microarchitectural condition to detect on the selected event logic.	
16	USR: Counts while in privilege level is not ring 0.	
17	OS: Counts while in privilege level is ring 0.	
18	Edge: Enables edge detection if set.	
19	PC: Enables pin control.	
20	INT: Enables interrupt on counter overflow.	
21	AnyThread: When set to 1, it enables counting the associated event conditions occurring across all logical processors sharing a processor core. When set to 0, the counter only increments the associated event conditions occurring in the logical processor which programmed the MSR.	
22	EN: Enables the corresponding performance counter to commence counting when this bit is set.	
23	INV: Invert the CMASK.	
31:24	CMASK: When CMASK is not zero, the corresponding performance counter increments each cycle if the event count is greater than or equal to the CMASK.	
63:32	Reserved.	
Register Address: 187H, 391		IA32_PERFEVTSEL1 (PERFEVTSEL1)
Performance Event Select Register 1 (R/W)		If CPUID.0AH: EAX[15:8] > 1
Register Address: 188H, 392		IA32_PERFEVTSEL2
Performance Event Select Register 2 (R/W)		If CPUID.0AH: EAX[15:8] > 2
Register Address: 189H, 393		IA32_PERFEVTSEL3
Performance Event Select Register 3 (R/W)		If CPUID.0AH: EAX[15:8] > 3
Register Address: 18AH, 394		IA32_PERFEVTSEL4
Performance Event Select Register 4 (R/W)		If CPUID.0AH: EAX[15:8] > 4
Register Address: 18BH, 395		IA32_PERFEVTSEL5
Performance Event Select Register 5 (R/W)		If CPUID.0AH: EAX[15:8] > 5
Register Address: 18CH, 396		IA32_PERFEVTSEL6
Performance Event Select Register 6 (R/W)		If CPUID.0AH: EAX[15:8] > 6
Register Address: 18DH, 397		IA32_PERFEVTSEL7
Performance Event Select Register 7 (R/W)		If CPUID.0AH: EAX[15:8] > 7
Register Address: 18EH, 398		IA32_PERFEVTSEL8
Performance Event Select Register 8 (R/W)		If CPUID.0AH: EAX[15:8] > 8
Register Address: 18FH, 399		IA32_PERFEVTSEL9
Performance Event Select Register 9 (R/W)		If CPUID.0AH: EAX[15:8] > 9
Register Address: 18AH–194H, 394–404		N/A

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Reserved.		06_0EH ³
Register Address: 195H, 405		IA32_OVERCLOCKING_STATUS
Overclocking Status (R/O) IA32_ARCH_CAPABILITIES[bit 23] enumerates support for this MSR.		
0	Overclocking Utilized Indicates if specific forms of overclocking have been enabled on this boot or reset cycle: 0 indicates no, 1 indicates yes.	
1	Undervolt Protection Indicates if the “Dynamic OC Undervolt Protection” security feature is active: 0 indicates disabled, 1 indicates enabled.	
2	Overclocking Secure Status Indicates that overclocking capabilities have been unlocked by BIOS, with or without overclocking: 0 indicates Not Secured, 1 indicates Secure.	
63:3	Reserved.	
Register Address: 196H–197H, 406–407		N/A
Reserved.		06_0EH ³
Register Address: 198H, 408		IA32_PERF_STATUS
Current Performance Status (R/O) See Section 16.1.1, “Software Interface For Initiating Performance State Transitions.”		0F_03H
15:0	Current Performance State Value.	
63:16	Reserved.	
Register Address: 199H, 409		IA32_PERF_CTL
Performance Control MSR (R/W) Software makes a request for a new Performance state (P-State) by writing this MSR. See Section 16.1.1, “Software Interface For Initiating Performance State Transitions.”		0F_03H
15:0	Target performance State Value.	
31:16	Reserved.	
32	Intel® Dynamic Acceleration Technology Engage (R/W) When set to 1: Disengages Intel Dynamic Acceleration Technology.	06_0FH (Mobile only)
63:33	Reserved.	
Register Address: 19AH, 410		IA32_CLOCK_MODULATION
Clock Modulation Control (R/W) See Section 16.8.3, “Software Controlled Clock Modulation.”		If CPUID.01H:EDX[22] = 1
0	Extended On-Demand Clock Modulation Duty Cycle.	If CPUID.06H:EAX[5] = 1
3:1	On-Demand Clock Modulation Duty Cycle: Specific encoded values for target duty cycle modulation.	If CPUID.01H:EDX[22] = 1
4	On-Demand Clock Modulation Enable: Set 1 to enable modulation.	If CPUID.01H:EDX[22] = 1
63:5	Reserved.	
Register Address: 19BH, 411		IA32_THERM_INTERRUPT

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Thermal Interrupt Control (R/W) Enables and disables the generation of an interrupt on temperature transitions detected with the processor's thermal sensors and thermal monitor. See Section 16.8.2, "Thermal Monitor."		If CPUID.01H:EDX[22] = 1
0	High-Temperature Interrupt Enable	If CPUID.01H:EDX[22] = 1
1	Low-Temperature Interrupt Enable	If CPUID.01H:EDX[22] = 1
2	PROCHOT# Interrupt Enable	If CPUID.01H:EDX[22] = 1
3	FORCEPR# Interrupt Enable	If CPUID.01H:EDX[22] = 1
4	Critical Temperature Interrupt Enable	If CPUID.01H:EDX[22] = 1
7:5	Reserved.	
14:8	Threshold #1 Value	If CPUID.01H:EDX[22] = 1
15	Threshold #1 Interrupt Enable	If CPUID.01H:EDX[22] = 1
22:16	Threshold #2 Value	If CPUID.01H:EDX[22] = 1
23	Threshold #2 Interrupt Enable	If CPUID.01H:EDX[22] = 1
24	Power Limit Notification Enable	If CPUID.06H:EAX[4] = 1
63:25	Reserved.	
Register Address: 19CH, 412		IA32_THERM_STATUS
Thermal Status Information (R/O) Contains status information about the processor's thermal sensor and automatic thermal monitoring facilities. See Section 16.8.2, "Thermal Monitor."		If CPUID.01H:EDX[22] = 1
0	Thermal Status (R/O)	If CPUID.01H:EDX[22] = 1
1	Thermal Status Log (R/W)	If CPUID.01H:EDX[22] = 1
2	PROCHOT # or FORCEPR# event (R/O)	If CPUID.01H:EDX[22] = 1
3	PROCHOT # or FORCEPR# log (R/WCO)	If CPUID.01H:EDX[22] = 1
4	Critical Temperature Status (R/O)	If CPUID.01H:EDX[22] = 1
5	Critical Temperature Status log (R/WCO)	If CPUID.01H:EDX[22] = 1
6	Thermal Threshold #1 Status (R/O)	If CPUID.01H:ECX[8] = 1
7	Thermal Threshold #1 log (R/WCO)	If CPUID.01H:ECX[8] = 1
8	Thermal Threshold #2 Status (R/O)	If CPUID.01H:ECX[8] = 1
9	Thermal Threshold #2 log (R/WCO)	If CPUID.01H:ECX[8] = 1
10	Power Limitation Status (R/O)	If CPUID.06H:EAX[4] = 1
11	Power Limitation log (R/WCO)	If CPUID.06H:EAX[4] = 1
12	Current Limit Status (R/O)	If CPUID.06H:EAX[7] = 1
13	Current Limit log (R/WCO)	If CPUID.06H:EAX[7] = 1
14	Cross Domain Limit Status (R/O)	If CPUID.06H:EAX[7] = 1
15	Cross Domain Limit log (R/WCO)	If CPUID.06H:EAX[7] = 1
22:16	Digital Readout (R/O)	If CPUID.06H:EAX[0] = 1
26:23	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
30:27	Resolution in Degrees Celsius (R/O)	If CPUID.06H:EAX[0] = 1
31	Reading Valid (R/O)	If CPUID.06H:EAX[0] = 1
63:32	Reserved.	
Register Address: 1A0H, 416		IA32_MISC_ENABLE
Enable Misc. Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		
0	Fast-Strings Enable When set, the fast-strings feature (for REP MOVSB and REP STOSB) is enabled (default). When clear, fast-strings are disabled.	OF_OH
2:1	Reserved.	
3	Automatic Thermal Control Circuit Enable (R/W) 1 = Setting this bit enables the thermal control circuit (TCC) portion of the Intel Thermal Monitor feature. This allows the processor to automatically reduce power consumption in response to TCC activation. 0 = Disabled. Note: In some products clearing this bit might be ignored in critical thermal conditions, and TM1, TM2, and adaptive thermal throttling will still be activated. The default value of this field varies with product. See respective tables where default value is listed.	OF_OH
6:4	Reserved.	
7	Performance Monitoring Available (R) 1 = Performance monitoring enabled. 0 = Performance monitoring disabled.	OF_OH
10:8	Reserved.	
11	Branch Trace Storage Unavailable (R/O) 1 = Processor doesn't support branch trace storage (BTS). 0 = BTS is supported.	OF_OH
12	Processor Event Based Sampling (PEBS) Unavailable (R/O) 1 = PEBS is not supported. 0 = PEBS is supported.	06_0FH
15:13	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W) 0 = Enhanced Intel SpeedStep Technology disabled. 1 = Enhanced Intel SpeedStep Technology enabled.	If CPUID.01H: ECX[7] = 1
17	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
18	ENABLE MONITOR FSM (R/W) When this bit is set to 0, the MONITOR feature flag is not set (CPUID.01H:ECX[bit 3] = 0). This indicates that MONITOR/MWAIT are not supported. Software attempts to execute MONITOR/MWAIT will cause #UD when this bit is 0. When this bit is set to 1 (default), MONITOR/MWAIT are supported (CPUID.01H:ECX[bit 3] = 1). If the SSE3 feature flag ECX[0] is not set (CPUID.01H:ECX[bit 0] = 0), the OS must not attempt to alter this bit. BIOS must leave it in the default state. Writing this bit when the SSE3 feature flag is set to 0 may generate a #GP exception.	0F_03H
21:19	Reserved.	
22	Limit CPUID Maxval (R/W) When this bit is set to 1, CPUID.00H returns a maximum value in EAX[7:0] of 2.	CPUID.0H:EAX > 2 and CPUID.(EAX = 07H, ECX = 1):EBX. CPUIDMAXVAL_LIM_RMV [bit 3] = 0
23	xTPR Message Disable (R/W) When set to 1, xTPR messages are disabled. xTPR messages are optional messages that allow the processor to inform the chipset of its priority.	If CPUID.01H:ECX[14] = 1
63:24	Reserved. Note: Some older processors defined one of these bits as a disable for the execute-disable feature of paging. If a processor supports this bit, this information is provided in the model-specific tables. See Table 2-3 for the definition of this bit.	
Register Address: 1B0H, 432		IA32_ENERGY_PERF_BIAS
Performance Energy Bias Hint (R/W)		If CPUID.6H:ECX[3] = 1
3:0	Power Policy Preference: 0 indicates preference to highest performance. 15 indicates preference to maximize energy saving.	
63:4	Reserved.	
Register Address: 1B1H, 433		IA32_PACKAGE_THERM_STATUS
Package Thermal Status Information (R/O) Contains status information about the package's thermal sensor. See Section 16.9, "Package Level Thermal Management."		If CPUID.06H: EAX[6] = 1
0	Pkg Thermal Status (R/O)	
1	Pkg Thermal Status Log (R/W)	
2	Pkg PROCHOT # event. (R/O)	
3	Pkg PROCHOT # log. (R/WCO)	
4	Pkg Critical Temperature Status. (R/O)	
5	Pkg Critical Temperature Status Log. (R/WCO)	
6	Pkg Thermal Threshold #1 Status. (R/O)	
7	Pkg Thermal Threshold #1 Log. (R/WCO)	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
8	Pkg Thermal Threshold #2 Status. (R/O)	
9	Pkg Thermal Threshold #1 Log. (R/WC0)	
10	Pkg Power Limitation Status. (R/O)	
11	Pkg Power Limitation Log. (R/WC0)	
15:12	Reserved.	
22:16	Pkg Digital Readout. (R/O)	
25:23	Reserved.	
26	Hardware Feedback Interface Structure Change Status.	If CPUID.06H:EAX.[19] = 1
63:27	Reserved.	
Register Address: 1B2H, 434		IA32_PACKAGE_THERM_INTERRUPT
Pkg Thermal Interrupt Control (R/W) Enables and disables the generation of an interrupt on temperature transitions detected with the package's thermal sensor. See Section 16.9, "Package Level Thermal Management."		If CPUID.06H: EAX[6] = 1
0	Pkg High-Temperature Interrupt Enable.	
1	Pkg Low-Temperature Interrupt Enable.	
2	Pkg PROCHOT# Interrupt Enable.	
3	Reserved.	
4	Pkg Overheat Interrupt Enable.	
7:5	Reserved.	
14:8	Pkg Threshold #1 Value.	
15	Pkg Threshold #1 Interrupt Enable.	
22:16	Pkg Threshold #2 Value.	
23	Pkg Threshold #2 Interrupt Enable.	
24	Pkg Power Limit Notification Enable.	
25	Hardware Feedback Interrupt Enable.	If CPUID.06H:EAX.[19] = 1
63:26	Reserved.	
Register Address: 1C4H, 452		IA32_XFD
Extended Feature Disable Control (R/W) Controls which XSAVE-enabled features are temporarily disabled. See Section 13.14 of the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 1.		If CPUID.(EAX=0DH,ECX=1): EAX[4] = 1
Register Address: 1C5H, 453		IA32_XFD_ERR
Extended Feature Disable Error Code (R/W) Reports which XSAVE-enabled features caused a fault due to being disabled. See Section 13.14 of the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 1.		If CPUID.(EAX=0DH,ECX=1): EAX[4] = 1
Register Address: 1D9H, 473		IA32_DEBUGCTL (MSR_DEBUGCTLA, MSR_DEBUGCTLB)
Trace/Profile Resource Control (R/W)		06_0EH

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
0	LBR: Setting this bit to 1 enables the processor to record a running trace of the most recent branches taken by the processor in the LBR stack.	06_01H
1	BTF: Setting this bit to 1 enables the processor to treat EFLAGS.TF as single-step on branches instead of single-step on instructions.	06_01H
2	BLD: Enable OS bus-lock detection. See Section 19.3.1.6 of the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 3B.	If (CPUID.(EAX=07H, ECX=0):ECX[24] = 1)
5:3	Reserved.	
6	TR: Setting this bit to 1 enables branch trace messages to be sent.	06_0EH
7	BTS: Setting this bit enables branch trace messages (BTMs) to be logged in a BTS buffer.	06_0EH
8	BTINT: When clear, BTMs are logged in a BTS buffer in circular fashion. When this bit is set, an interrupt is generated by the BTS facility when the BTS buffer is full.	06_0EH
9	1: BTS_OFF_OS: When set, BTS or BTM is skipped if CPL = 0.	06_0FH
10	BTS_OFF_USR: When set, BTS or BTM is skipped if CPL > 0.	06_0FH
11	FREEZE_LBRS_ON_PMI: When set, the LBR stack is frozen on a PMI request.	If CPUID.01H: ECX[15] = 1 && CPUID.0AH: EAX[7:0] > 1
12	FREEZE_PERFMON_ON_PMI: When set, each ENABLE bit of the global counter control MSR are frozen (address 38FH) on a PMI request.	If CPUID.01H: ECX[15] = 1 && CPUID.0AH: EAX[7:0] > 1
13	ENABLE_UNCORE_PMI: When set, enables the logical processor to receive and generate PMI on behalf of the uncore.	06_1AH
14	FREEZE_WHILE_SMM: When set, freezes PerfMon and trace messages while in SMM.	If IA32_PERF_CAPABILITIES[12] = 1
15	RTM_DEBUG: When set, enables DR7 debug bit on XBEGIN.	If (CPUID.(EAX=07H, ECX=0):EBX[11] = 1)
63:16	Reserved.	
Register Address: 1DDH, 477		IA32_LER_FROM_IP
Last Event Record Source IP Register (R/W)		
63:0	FROM_IP The source IP of the recorded branch or event, in canonical form.	Reset Value: 0
Register Address: 1DEH, 478		IA32_LER_TO_IP
Last Event Record Destination IP Register (R/W)		
63:0	TO_IP The destination IP of the recorded branch or event, in canonical form.	Reset Value: 0
Register Address: 1E0H, 480		IA32_LER_INFO
Last Event Record Info Register (R/W)		
55:0	Undefined, may be zero or non-zero. Writes of non-zero values do not fault, but reads may return a different value.	Reset Value: 0
59:56	BR_TYPE The branch type recorded by this LBR. Encodings match those of IA32_LBR_x_INFO.	Reset Value: 0

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
60	Undefined, may be zero or non-zero. Writes of non-zero values do not fault, but reads may return a different value.	Reset Value: 0
61	TSX_ABORT This LBR record is a TSX abort. On processors that do not support Intel® TSX (CPUID.07H.EBX.HLE[bit 4]=0 and CPUID.07H.EBX.RTM[bit 11]=0), this bit is undefined.	Reset Value: 0
62	IN_TSX This LBR record records a branch that retired during a TSX transaction. On processors that do not support Intel® TSX (CPUID.07H.EBX.HLE[bit 4]=0 and CPUID.07H.EBX.RTM[bit 11]=0), this bit is undefined.	Reset Value: 0
63	MISPRED The recorded branch taken/not-taken resolution (for conditional branches) or target (for any indirect branch, including RETs) was mispredicted.	Reset Value: 0
Register Address: 1F2H, 498		IA32_SMRR_PHYSBASE
SMRR Base Address (Writeable only in SMM) Base address of SMM memory range.		If IA32_MTRRCAP.SMRR[11] = 1
7:0	Type. Specifies memory type of the range.	
11:8	Reserved.	
31:12	PhysBase SMRR physical Base Address.	
63:32	Reserved.	
Register Address: 1F3H, 499		IA32_SMRR_PHYSMASK
SMRR Range Mask (Writeable only in SMM) Range Mask of SMM memory range.		If IA32_MTRRCAP[SMRR] = 1
10:0	Reserved.	
11	Valid Enable range mask.	
31:12	PhysMask SMRR address range mask.	
63:32	Reserved.	
Register Address: 1F8H, 504		IA32_PLATFORM_DCA_CAP
DCA Capability (R)		If CPUID.01H: ECX[18] = 1
Register Address: 1F9H, 505		IA32_CPU_DCA_CAP
If set, CPU supports Prefetch-Hint type.		If CPUID.01H: ECX[18] = 1
Register Address: 1FAH, 506		IA32_DCA_0_CAP
DCA type 0 Status and Control register.		If CPUID.01H: ECX[18] = 1
0	DCA_ACTIVE: Set by HW when DCA is fuse-enabled and no defeatures are set.	
2:1	TRANSACTION	
6:3	DCA_TYPE	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
10:7	DCA_QUEUE_SIZE	
12:11	Reserved.	
16:13	DCA_DELAY: Writes will update the register but have no HW side-effect.	
23:17	Reserved.	
24	SW_BLOCK: SW can request DCA block by setting this bit.	
25	Reserved.	
26	HW_BLOCK: Set when DCA is blocked by HW (e.g., CRO.CD = 1).	
31:27	Reserved.	
Register Address: 200H, 512		IA32_MTRR_PHYSBASE0 (MTRRphysBase0)
See Section 13.11.2.3, "Variable Range MTRRs."		If IA32_MTRRCAP[7:0] > 0
Register Address: 201H, 513		IA32_MTRR_PHYSMASK0
MTRRphysMask0		If IA32_MTRRCAP[7:0] > 0
Register Address: 202H, 514		IA32_MTRR_PHYSBASE1
MTRRphysBase1		If IA32_MTRRCAP[7:0] > 1
Register Address: 203H, 515		IA32_MTRR_PHYSMASK1
MTRRphysMask1		If IA32_MTRRCAP[7:0] > 1
Register Address: 204H, 516		IA32_MTRR_PHYSBASE2
MTRRphysBase2		If IA32_MTRRCAP[7:0] > 2
Register Address: 205H, 517		IA32_MTRR_PHYSMASK2
MTRRphysMask2		If IA32_MTRRCAP[7:0] > 2
Register Address: 206H, 518		IA32_MTRR_PHYSBASE3
MTRRphysBase3		If IA32_MTRRCAP[7:0] > 3
Register Address: 207H, 519		IA32_MTRR_PHYSMASK3
MTRRphysMask3		If IA32_MTRRCAP[7:0] > 3
Register Address: 208H, 520		IA32_MTRR_PHYSBASE4
MTRRphysBase4		If IA32_MTRRCAP[7:0] > 4
Register Address: 209H, 521		IA32_MTRR_PHYSMASK4
MTRRphysMask4		If IA32_MTRRCAP[7:0] > 4
Register Address: 20AH, 522		IA32_MTRR_PHYSBASE5
MTRRphysBase5		If IA32_MTRRCAP[7:0] > 5
Register Address: 20BH, 523		IA32_MTRR_PHYSMASK5
MTRRphysMask5		If IA32_MTRRCAP[7:0] > 5
Register Address: 20CH, 524		IA32_MTRR_PHYSBASE6
MTRRphysBase6		If IA32_MTRRCAP[7:0] > 6
Register Address: 20DH, 525		IA32_MTRR_PHYSMASK6
MTRRphysMask6		If IA32_MTRRCAP[7:0] > 6
Register Address: 20EH, 526		IA32_MTRR_PHYSBASE7

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
MTRRphysBase7		If IA32_MTRRCAP[7:0] > 7
Register Address: 20FH, 527		IA32_MTRR_PHYSMASK7
MTRRphysMask7		If IA32_MTRRCAP[7:0] > 7
Register Address: 210H, 528		IA32_MTRR_PHYSBASE8
MTRRphysBase8		If IA32_MTRRCAP[7:0] > 8
Register Address: 211H, 529		IA32_MTRR_PHYSMASK8
MTRRphysMask8		If IA32_MTRRCAP[7:0] > 8
Register Address: 212H, 530		IA32_MTRR_PHYSBASE9
MTRRphysBase9		If IA32_MTRRCAP[7:0] > 9
Register Address: 213H, 531		IA32_MTRR_PHYSMASK9
MTRRphysMask9		If IA32_MTRRCAP[7:0] > 9
Register Address: 250H, 592		IA32_MTRR_FIX64K_00000
MTRRfix64K_00000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 258H, 600		IA32_MTRR_FIX16K_80000
MTRRfix16K_80000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 259H, 601		IA32_MTRR_FIX16K_A0000
MTRRfix16K_A0000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 268H, 616		IA32_MTRR_FIX4K_C0000 (MTRRfix4K_C0000)
See Section 13.11.2.2, "Fixed Range MTRRs."		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 269H, 617		IA32_MTRR_FIX4K_C8000
MTRRfix4K_C8000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 26AH, 618		IA32_MTRR_FIX4K_D0000
MTRRfix4K_D0000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 26BH, 619		IA32_MTRR_FIX4K_D8000
MTRRfix4K_D8000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 26CH, 620		IA32_MTRR_FIX4K_E0000
MTRRfix4K_E0000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 26DH, 621		IA32_MTRR_FIX4K_E8000
MTRRfix4K_E8000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 26EH, 622		IA32_MTRR_FIX4K_F0000
MTRRfix4K_F0000		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 26FH, 623		IA32_MTRR_FIX4K_F8000
MTRRfix4K_F8000.		If CPUID.01H: EDX.MTRR[12] = 1
Register Address: 277H, 631		IA32_PAT
IA32_PAT (R/W)		If CPUID.01H: EDX.MTRR[16] = 1
2:0	PA0	
7:3	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
10:8	PA1	
15:11	Reserved.	
18:16	PA2	
23:19	Reserved.	
26:24	PA3	
31:27	Reserved.	
34:32	PA4	
39:35	Reserved.	
42:40	PA5	
47:43	Reserved.	
50:48	PA6	
55:51	Reserved.	
58:56	PA7	
63:59	Reserved.	
Register Address: 280H, 640		IA32_MC0_CTL2
MSR to enable/disable CMCI capability for bank 0. (R/W) See Section 17.3.2.5, "IA32_MCI_CTL2 MSRs."		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 0
14:0	Corrected error count threshold.	
29:15	Reserved.	
30	CMCI_EN	
63:31	Reserved.	
Register Address: 281H, 641		IA32_MC1_CTL2
Same fields as IA32_MC0_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 1
Register Address: 282H, 642		IA32_MC2_CTL2
Same fields as IA32_MC0_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 2
Register Address: 283H, 643		IA32_MC3_CTL2
Same fields as IA32_MC0_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 3
Register Address: 284H, 644		IA32_MC4_CTL2
Same fields as IA32_MC0_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 4
Register Address: 285H, 645		IA32_MC5_CTL2
Same fields as IA32_MC0_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 5
Register Address: 286H, 646		IA32_MC6_CTL2
Same fields as IA32_MC0_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 6

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 287H, 647	IA32_MC7_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 7
Register Address: 288H, 648	IA32_MC8_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 8
Register Address: 289H, 649	IA32_MC9_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 9
Register Address: 28AH, 650	IA32_MC10_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 10
Register Address: 28BH, 651	IA32_MC11_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 11
Register Address: 28CH, 652	IA32_MC12_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 12
Register Address: 28DH, 653	IA32_MC13_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 13
Register Address: 28EH, 654	IA32_MC14_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 14
Register Address: 28FH, 655	IA32_MC15_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 15
Register Address: 290H, 656	IA32_MC16_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 16
Register Address: 291H, 657	IA32_MC17_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 17
Register Address: 292H, 658	IA32_MC18_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 18
Register Address: 293H, 659	IA32_MC19_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 19
Register Address: 294H, 660	IA32_MC20_CTL2	
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 20

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 295H, 661		IA32_MC21_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 21
Register Address: 296H, 662		IA32_MC22_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 22
Register Address: 297H, 663		IA32_MC23_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 23
Register Address: 298H, 664		IA32_MC24_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 24
Register Address: 299H, 665		IA32_MC25_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 25
Register Address: 29AH, 666		IA32_MC26_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 26
Register Address: 29BH, 667		IA32_MC27_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 27
Register Address: 29CH, 668		IA32_MC28_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 28
Register Address: 29DH, 669		IA32_MC29_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 29
Register Address: 29EH, 670		IA32_MC30_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 30
Register Address: 29FH, 671		IA32_MC31_CTL2
Same fields as IA32_MCO_CTL2. (R/W)		If IA32_MCG_CAP[10] = 1 && IA32_MCG_CAP[7:0] > 31
Register Address: 2DCH, 732		IA32_INTEGRITY_STATUS
IA32_INTEGRITY_STATUS (R/O) Provides status information for integrity features.		If any one enumeration condition for defined bit field holds.
0	STATIC_LOCKSTEP 0: Static Lockstep Mode is not active on this logical processor. 1: Static Lockstep Mode is active on this logical processor.	If CPUID(EAX=07H, ECX=1H).EDX[24]=1
63:1	Reserved.	
Register Address: 2FFH, 767		IA32_MTRR_DEF_TYPE

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
MTRRdefType (R/W)		If CPUID.01H: EDX.MTRR[12] = 1
2:0	Default Memory Type	
9:3	Reserved.	
10	Fixed Range MTRR Enable	
11	MTRR Enable	
63:12	Reserved.	
Register Address: 309H, 777		IA32_FIXED_CTR0
Fixed-Function Performance Counter 0 (R/W): Counts Instr_Retired.Any.		If CPUID.0AH:EDX[4:0] > 0 CPUID.0AH:ECX[0] = 1 CPUID.23H.1H:EBX[0] = 1
Register Address: 30AH, 778		IA32_FIXED_CTR1
Fixed-Function Performance Counter 1 (R/W): Counts CPU_CLK_Unhalted.Core.		If CPUID.0AH:EDX[4:0] > 1 CPUID.0AH:ECX[1] = 1 CPUID.23H.1H:EBX[1] = 1
Register Address: 30BH, 779		IA32_FIXED_CTR2
Fixed-Function Performance Counter 2 (R/W): Counts CPU_CLK_Unhalted.Ref.		If CPUID.0AH:EDX[4:0] > 2 CPUID.0AH:ECX[2] = 1 CPUID.23H.1H:EBX[2] = 1
Register Address: 30CH, 780		IA32_FIXED_CTR3
Fixed-Function Performance Counter 3 (R/W): Top-down Microarchitecture Analysis unhalted number of available slots.		If CPUID.0AH:EDX[4:0] > 3 CPUID.0AH:ECX[3] = 1 CPUID.23H.1H:EBX[3] = 1
Register Address: 30DH, 781		IA32_FIXED_CTR4
Fixed-Function Performance Counter 4 (R/W): Top-down bad speculation.		If CPUID.0AH:EDX[4:0] > 4 CPUID.0AH:ECX[4] = 1 CPUID.23H.1H:EBX[4] = 1
47:0	FIXED_COUNTER Top-down bad speculation counter.	
63:46	Reserved.	
Register Address: 30EH, 782		IA32_FIXED_CTR5
Fixed-Function Performance Counter 5 (R/W): Top-down Frontend Bound.		If CPUID.0AH:EDX[4:0] > 5 CPUID.0AH:ECX[5] = 1 CPUID.23H.1H:EBX[5] = 1
47:0	FIXED_COUNTER Top-down Frontend Bound counter.	
63:46	Reserved.	
Register Address: 30FH, 783		IA32_FIXED_CTR6
Fixed-Function Performance Counter 6 (R/W): Top-down retiring.		If CPUID.0AH:EDX[4:0] > 6 CPUID.0AH:ECX[6] = 1 CPUID.23H.1H:EBX[6] = 1
47:0	FIXED_COUNTER Top-down Retiring counter.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
63:46	Reserved.	
Register Address: 345H, 837		IA32_PERF_CAPABILITIES
Read Only MSR that enumerates the existence of performance monitoring features. (R/O)		If CPUID.01H: ECX[15] = 1
5:0	LBR format	
6	PEBS Trap	
7	PEBSSaveArchRegs	
11:8	PEBS Record Format	
12	1: Freeze while SMM is supported.	
13	1: Full width of counter writable via IA32_A_PMCx.	
14	PEBS_BASELINE	
15	1: Performance metrics available.	
16	1: PEBS output will be written into the Intel PT trace stream.	If CPUID.0x7.0.EBX[25]=1
17	1: Indicates support for PEBS Retire Latency output.	
18	TSX_ADDRESS	
19	RDPMC_METRICS_CLEAR	
63:20	Reserved.	
Register Address: 38DH, 909		IA32_FIXED_CTR_CTRL
Fixed-Function Performance Counter Control (R/W) Counter increments while the results of ANDing respective enable bit in IA32_PERF_GLOBAL_CTRL with the corresponding OS or USR bits in this MSR is true.		If CPUID.0AH: EAX[7:0] > 1
0	EN0_OS: Enable Fixed Counter 0 to count while CPL = 0.	
1	EN0_Usr: Enable Fixed Counter 0 to count while CPL > 0.	
2	AnyThr0: When set to 1, it enables counting the associated event conditions occurring across all logical processors sharing a processor core. When set to 0, the counter only increments the associated event conditions occurring in the logical processor which programmed the MSR.	If CPUID.0AH:EAX[7:0] > 2 && CPUID.0AH:EDX[15]=0
3	EN0_PMI: Enable PMI when fixed counter 0 overflows.	
4	EN1_OS: Enable Fixed Counter 1 to count while CPL = 0.	
5	EN1_Usr: Enable Fixed Counter 1 to count while CPL > 0.	
6	AnyThr1: When set to 1, it enables counting the associated event conditions occurring across all logical processors sharing a processor core. When set to 0, the counter only increments the associated event conditions occurring in the logical processor which programmed the MSR.	If CPUID.0AH:EAX[7:0] > 2 && CPUID.0AH:EDX[15]=0
7	EN1_PMI: Enable PMI when fixed counter 1 overflows.	
8	EN2_OS: Enable Fixed Counter 2 to count while CPL = 0.	
9	EN2_Usr: Enable Fixed Counter 2 to count while CPL > 0.	
10	AnyThr2: When set to 1, it enables counting the associated event conditions occurring across all logical processors sharing a processor core. When set to 0, the counter only increments the associated event conditions occurring in the logical processor which programmed the MSR.	If CPUID.0AH:EAX[7:0] > 2 && CPUID.0AH:EDX[15]=0
11	EN2_PMI: Enable PMI when fixed counter 2 overflows.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
12	EN3_OS: Enable Fixed Counter 3 to count while CPL = 0.	
13	EN3_Usr: Enable Fixed Counter 3 to count while CPL > 0.	
14	Reserved.	
15	EN3_PMI: Enable PMI when fixed counter 3 overflows.	
63:16	Reserved.	
Register Address: 38EH, 910		IA32_PERF_GLOBAL_STATUS
Global Performance Counter Status (R/O)		If CPUID.0AH: EAX[7:0] > 0 (CPUID.(EAX=07H, ECX=0):EBX[25] = 1 && CPUID.(EAX=014H, ECX=0):ECX[0] = 1)
0	Ovf_PMC0: Overflow status of IA32_PMC0.	If CPUID.0AH: EAX[15:8] > 0
1	Ovf_PMC1: Overflow status of IA32_PMC1.	If CPUID.0AH: EAX[15:8] > 1
2	Ovf_PMC2: Overflow status of IA32_PMC2.	If CPUID.0AH: EAX[15:8] > 2
3	Ovf_PMC3: Overflow status of IA32_PMC3.	If CPUID.0AH: EAX[15:8] > 3
n	Ovf_PMCn: Overflow status of IA32_PMCn.	If CPUID.0AH: EAX[15:8] > n
31:n+1	Reserved.	
32	Ovf_FixedCtr0: Overflow status of IA32_FIXED_CTR0.	If CPUID.0AH: EAX[7:0] > 1
33	Ovf_FixedCtr1: Overflow status of IA32_FIXED_CTR1.	If CPUID.0AH: EAX[7:0] > 1
34	Ovf_FixedCtr2: Overflow status of IA32_FIXED_CTR2.	If CPUID.0AH: EAX[7:0] > 1
32+m	Ovf_FixedCtrm: Overflow status of IA32_FIXED_CTRm.	If CPUID.0AH:ECX[m] == 1 CPUID.0AH:EDX[4:0] > m
47:33+m	Reserved.	
48	OVF_PERF_METRICS: If this bit is set, it indicates that PERF_METRIC counter has overflowed and a PMI is triggered; however, an overflow of fixed counter 3 should normally happen first. If this bit is clear no overflow occurred.	
54:49	Reserved.	
55	Trace_ToPA_PMI: A PMI occurred due to a ToPA entry memory buffer that was completely filled.	If CPUID.(EAX=07H, ECX=0):EBX[25] = 1 && CPUID.(EAX=014H, ECX=0):ECX[0] = 1
57:56	Reserved.	
58	LBR_Frz. LBRs are frozen due to: - IA32_DEBUGCTL.FREEZE_LBR_ON_PMI=1. - The LBR stack overflowed.	If CPUID.0AH: EAX[7:0] > 3
59	CTR_Frz. Performance counters in the core PMU are frozen due to: - IA32_DEBUGCTL.FREEZE_PERFMON_ON_PMI=1. - One or more core PMU counters overflowed.	If CPUID.0AH: EAX[7:0] > 3
60	ASCI: Data in the performance counters in the core PMU may include contributions from the direct or indirect operation Intel SGX to protect an enclave.	If the processor supports Intel® SGX.
61	Ovf_Uncore: Uncore counter overflow status.	If CPUID.0AH: EAX[7:0] > 2

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
62	OvfBuf: DS SAVE area Buffer overflow status.	If CPUID.0AH: EAX[7:0] > 0
63	CondChgd: Status bits of this register have changed.	If CPUID.0AH: EAX[7:0] > 0
Register Address: 38FH, 911		IA32_PERF_GLOBAL_CTRL
Global Performance Counter Control (R/W) Counter increments while the result of ANDing the respective enable bit in this MSR with the corresponding OS or USR bits in the general-purpose or fixed counter control MSR is true.		If CPUID.0AH: EAX[7:0] > 0
0	EN_PMC0	If CPUID.0AH: EAX[15:8] > 0
1	EN_PMC1	If CPUID.0AH: EAX[15:8] > 1
2	EN_PMC2	If CPUID.0AH: EAX[15:8] > 2
n	EN_PMCn	If CPUID.0AH: EAX[15:8] > n
31:n+1	Reserved.	
32	EN_FIXED_CTR0	If CPUID.0AH: EDX[4:0] > 0
33	EN_FIXED_CTR1	If CPUID.0AH: EDX[4:0] > 1
34	EN_FIXED_CTR2	If CPUID.0AH: EDX[4:0] > 2
32+m	EN_FIXED_CTRm	If CPUID.0AH: ECX[m] == 1 CPUID.0AH: EDX[4:0] > m
47:33+m	Reserved.	
48	EN_PERF_METRICS: If this bit is set and fixed counter 3 is effectively enabled, built-in performance metrics are enabled.	
63:49	Reserved.	
Register Address: 390H, 912		IA32_PERF_GLOBAL_STATUS_RESET
Global Performance Counter Overflow Reset Control (R/W)		If CPUID.0AH: EAX[7:0] > 3 (CPUID.(EAX=07H, ECX=0):EBX[25] = 1 && CPUID.(EAX=14H, ECX=0):ECX[0] = 1)
0	Set 1 to Clear Ovf_PMC0 bit.	If CPUID.0AH: EAX[15:8] > 0
1	Set 1 to Clear Ovf_PMC1 bit.	If CPUID.0AH: EAX[15:8] > 1
2	Set 1 to Clear Ovf_PMC2 bit.	If CPUID.0AH: EAX[15:8] > 2
n	Set 1 to Clear Ovf_PMCn bit.	If CPUID.0AH: EAX[15:8] > n
31:n	Reserved.	
32	Set 1 to Clear Ovf_FIXED_CTR0 bit.	If CPUID.0AH: EDX[4:0] > 0
33	Set 1 to Clear Ovf_FIXED_CTR1 bit.	If CPUID.0AH: EDX[4:0] > 1
34	Set 1 to Clear Ovf_FIXED_CTR2 bit.	If CPUID.0AH: EDX[4:0] > 2
32+m	Set 1 to Clear Ovf_FIXED_CTRm bit.	If CPUID.0AH: ECX[m] == 1 CPUID.0AH: EDX[4:0] > m
47:33+m	Reserved.	
48	RESET_OVF_PERF_METRICS: If this bit is set, it will clear the status bit in the IA32_PERF_GLOBAL_STATUS register for the PERF_METRICS counters.	
54:49	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
55	Set 1 to Clear Trace_ToPA_PMI bit.	If CPUID.(EAX=07H, ECX=0):EBX[25] = 1 && CPUID.(EAX=014H, ECX=0):ECX[0] = 1
57:56	Reserved.	
58	Set 1 to Clear LBR_Frz bit.	If CPUID.0AH: EAX[7:0] > 3
59	Set 1 to Clear CTR_Frz bit.	If CPUID.0AH: EAX[7:0] > 3
60	Set 1 to Clear ASCI bit.	If the processor supports Intel® SGX.
61	Set 1 to Clear Ovf_Uncore bit.	06_2EH
62	Set 1 to Clear OvfBuf bit.	If CPUID.0AH: EAX[7:0] > 0
63	Set 1 to clear CondChgd bit.	If CPUID.0AH: EAX[7:0] > 0
Register Address: 391H, 913		IA32_PERF_GLOBAL_STATUS_SET
Global Performance Counter Overflow Set Control (R/W)		If CPUID.0AH: EAX[7:0] > 3 (CPUID.(EAX=07H, ECX=0):EBX[25] = 1 && CPUID.(EAX=014H, ECX=0):ECX[0] = 1)
0	Set 1 to cause Ovf_PMC0 = 1.	If CPUID.0AH: EAX[7:0] > 3
1	Set 1 to cause Ovf_PMC1 = 1.	If CPUID.0AH: EAX[15:8] > 1
2	Set 1 to cause Ovf_PMC2 = 1.	If CPUID.0AH: EAX[15:8] > 2
n	Set 1 to cause Ovf_PMCn = 1.	If CPUID.0AH: EAX[15:8] > n
31:n	Reserved.	
32	Set 1 to cause Ovf_FIXED_CTR0 = 1.	If CPUID.0AH: EAX[7:0] > 3
33	Set 1 to cause Ovf_FIXED_CTR1 = 1.	If CPUID.0AH: EAX[7:0] > 3
34	Set 1 to cause Ovf_FIXED_CTR2 = 1.	If CPUID.0AH: EAX[7:0] > 3
32+m	Set 1 to cause Ovf_FIXED_CTRm = 1.	If CPUID.0AH:ECX[m] == 1 CPUID.0AH:EDX[4:0] > m
47:33+m	Reserved.	
48	SET_OVF_PERF_METRICS: If this bit is set, it will set the status bit in the IA32_PERF_GLOBAL_STATUS register for the PERF_METRICS counters.	
54:49	Reserved.	
55	Set 1 to cause Trace_ToPA_PMI = 1.	If CPUID.(EAX=07H, ECX=0):EBX[25] = 1 && CPUID.(EAX=014H, ECX=0):ECX[0] = 1
57:56	Reserved.	
58	Set 1 to cause LBR_Frz = 1.	If CPUID.0AH: EAX[7:0] > 3
59	Set 1 to cause CTR_Frz = 1.	If CPUID.0AH: EAX[7:0] > 3
60	Set 1 to cause ASCI = 1.	If the processor supports Intel® SGX.
61	Set 1 to cause Ovf_Uncore = 1.	If CPUID.0AH: EAX[7:0] > 3

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
62	Set 1 to cause OvfBuf = 1.	If CPUID.0AH: EAX[7:0] > 3
63	Reserved.	
Register Address: 392H, 914		IA32_PERF_GLOBAL_INUSE
Indicator that core PerfMon interface is in use. (R/O)		If CPUID.0AH: EAX[7:0] > 3
0	IA32_PERFEVTSEL0 in use.	
1	IA32_PERFEVTSEL1 in use.	If CPUID.0AH: EAX[15:8] > 1
2	IA32_PERFEVTSEL2 in use.	If CPUID.0AH: EAX[15:8] > 2
n	IA32_PERFEVTSELn in use.	If CPUID.0AH: EAX[15:8] > n
31:n+1	Reserved.	
32	IA32_FIXED_CTR0 in use.	
33	IA32_FIXED_CTR1 in use.	
34	IA32_FIXED_CTR2 in use.	
32+m	IA32_FIXED_CTRm in use.	
62:33+m	Reserved or model specific.	
63	PMI in use.	
Register Address: 3F1H, 1009		IA32_PEBS_ENABLE
PEBS Control (R/W)		
0	Enable PEBS on IA32_PMC0.	06_0FH
3:1	Reserved or model specific.	
31:4	Reserved.	
35:32	Reserved or model specific.	
63:36	Reserved.	
Register Address: 400H, 1024		IA32_MCO_CTL
MCO_CTL		If IA32_MCG_CAP.CNT > 0
Register Address: 401H, 1025		IA32_MCO_STATUS
MCO_STATUS		If IA32_MCG_CAP.CNT > 0
Register Address: 402H, 1026		IA32_MCO_ADDR ¹
MCO_ADDR		If IA32_MCG_CAP.CNT > 0
Register Address: 403H, 1027		IA32_MCO_MISC
MCO_MISC		If IA32_MCG_CAP.CNT > 0
Register Address: 404H, 1028		IA32_MC1_CTL
MC1_CTL		If IA32_MCG_CAP.CNT > 1
Register Address: 405H, 1029		IA32_MC1_STATUS
MC1_STATUS		If IA32_MCG_CAP.CNT > 1
Register Address: 406H, 1030		IA32_MC1_ADDR ²
MC1_ADDR		If IA32_MCG_CAP.CNT > 1
Register Address: 407H, 1031		IA32_MC1_MISC

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
MC1_MISC		If IA32_MCG_CAP.CNT >1
Register Address: 408H, 1032	IA32_MC2_CTL	
MC2_CTL		If IA32_MCG_CAP.CNT >2
Register Address: 409H, 1033	IA32_MC2_STATUS	
MC2_STATUS		If IA32_MCG_CAP.CNT >2
Register Address: 40AH, 1034	IA32_MC2_ADDR ¹	
MC2_ADDR		If IA32_MCG_CAP.CNT >2
Register Address: 40BH, 1035	IA32_MC2_MISC	
MC2_MISC		If IA32_MCG_CAP.CNT >2
Register Address: 40CH, 1036	IA32_MC3_CTL	
MC3_CTL		If IA32_MCG_CAP.CNT >3
Register Address: 40DH, 1037	IA32_MC3_STATUS	
MC3_STATUS		If IA32_MCG_CAP.CNT >3
Register Address: 40EH, 1038	IA32_MC3_ADDR ¹	
MC3_ADDR		If IA32_MCG_CAP.CNT >3
Register Address: 40FH, 1039	IA32_MC3_MISC	
MC3_MISC		If IA32_MCG_CAP.CNT >3
Register Address: 410H, 1040	IA32_MC4_CTL	
MC4_CTL		If IA32_MCG_CAP.CNT >4
Register Address: 411H, 1041	IA32_MC4_STATUS	
MC4_STATUS		If IA32_MCG_CAP.CNT >4
Register Address: 412H, 1042	IA32_MC4_ADDR ¹	
MC4_ADDR		If IA32_MCG_CAP.CNT >4
Register Address: 413H, 1043	IA32_MC4_MISC	
MC4_MISC		If IA32_MCG_CAP.CNT >4
Register Address: 414H, 1044	IA32_MC5_CTL	
MC5_CTL		If IA32_MCG_CAP.CNT >5
Register Address: 415H, 1045	IA32_MC5_STATUS	
MC5_STATUS		If IA32_MCG_CAP.CNT >5
Register Address: 416H, 1046	IA32_MC5_ADDR ¹	
MC5_ADDR		If IA32_MCG_CAP.CNT >5
Register Address: 417H, 1047	IA32_MC5_MISC	
MC5_MISC		If IA32_MCG_CAP.CNT >5
Register Address: 418H, 1048	IA32_MC6_CTL	
MC6_CTL		If IA32_MCG_CAP.CNT >6
Register Address: 419H, 1049	IA32_MC6_STATUS	
MC6_STATUS		If IA32_MCG_CAP.CNT >6

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 41AH, 1050	IA32_MC6_ADDR ¹	
MC6_ADDR		If IA32_MCG_CAP.CNT >6
Register Address: 41BH, 1051	IA32_MC6_MISC	
MC6_MISC		If IA32_MCG_CAP.CNT >6
Register Address: 41CH, 1052	IA32_MC7_CTL	
MC7_CTL		If IA32_MCG_CAP.CNT >7
Register Address: 41DH, 1053	IA32_MC7_STATUS	
MC7_STATUS		If IA32_MCG_CAP.CNT >7
Register Address: 41EH, 1054	IA32_MC7_ADDR ¹	
MC7_ADDR		If IA32_MCG_CAP.CNT >7
Register Address: 41FH, 1055	IA32_MC7_MISC	
MC7_MISC		If IA32_MCG_CAP.CNT >7
Register Address: 420H, 1056	IA32_MC8_CTL	
MC8_CTL		If IA32_MCG_CAP.CNT >8
Register Address: 421H, 1057	IA32_MC8_STATUS	
MC8_STATUS		If IA32_MCG_CAP.CNT >8
Register Address: 422H, 1058	IA32_MC8_ADDR ¹	
MC8_ADDR		If IA32_MCG_CAP.CNT >8
Register Address: 423H, 1059	IA32_MC8_MISC	
MC8_MISC		If IA32_MCG_CAP.CNT >8
Register Address: 424H, 1060	IA32_MC9_CTL	
MC9_CTL		If IA32_MCG_CAP.CNT >9
Register Address: 425H, 1061	IA32_MC9_STATUS	
MC9_STATUS		If IA32_MCG_CAP.CNT >9
Register Address: 426H, 1062	IA32_MC9_ADDR ¹	
MC9_ADDR		If IA32_MCG_CAP.CNT >9
Register Address: 427H, 1063	IA32_MC9_MISC	
MC9_MISC		If IA32_MCG_CAP.CNT >9
Register Address: 428H, 1064	IA32_MC10_CTL	
MC10_CTL		If IA32_MCG_CAP.CNT >10
Register Address: 429H, 1065	IA32_MC10_STATUS	
MC10_STATUS		If IA32_MCG_CAP.CNT >10
Register Address: 42AH, 1066	IA32_MC10_ADDR ¹	
MC10_ADDR		If IA32_MCG_CAP.CNT >10
Register Address: 42BH, 1067	IA32_MC10_MISC	
MC10_MISC		If IA32_MCG_CAP.CNT >10
Register Address: 42CH, 1068	IA32_MC11_CTL	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
MC11_CTL		If IA32_MCG_CAP.CNT >11
Register Address: 42DH, 1069	IA32_MC11_STATUS	
MC11_STATUS		If IA32_MCG_CAP.CNT >11
Register Address: 42EH, 1070	IA32_MC11_ADDR ¹	
MC11_ADDR		If IA32_MCG_CAP.CNT >11
Register Address: 42FH, 1071	IA32_MC11_MISC	
MC11_MISC		If IA32_MCG_CAP.CNT >11
Register Address: 430H, 1072	IA32_MC12_CTL	
MC12_CTL		If IA32_MCG_CAP.CNT >12
Register Address: 431H, 1073	IA32_MC12_STATUS	
MC12_STATUS		If IA32_MCG_CAP.CNT >12
Register Address: 432H, 1074	IA32_MC12_ADDR ¹	
MC12_ADDR		If IA32_MCG_CAP.CNT >12
Register Address: 433H, 1075	IA32_MC12_MISC	
MC12_MISC		If IA32_MCG_CAP.CNT >12
Register Address: 434H, 1076	IA32_MC13_CTL	
MC13_CTL		If IA32_MCG_CAP.CNT >13
Register Address: 435H, 1077	IA32_MC13_STATUS	
MC13_STATUS		If IA32_MCG_CAP.CNT >13
Register Address: 436H, 1078	IA32_MC13_ADDR ¹	
MC13_ADDR		If IA32_MCG_CAP.CNT >13
Register Address: 437H, 1079	IA32_MC13_MISC	
MC13_MISC		If IA32_MCG_CAP.CNT >13
Register Address: 438H, 1080	IA32_MC14_CTL	
MC14_CTL		If IA32_MCG_CAP.CNT >14
Register Address: 439H, 1081	IA32_MC14_STATUS	
MC14_STATUS		If IA32_MCG_CAP.CNT >14
Register Address: 43AH, 1082	IA32_MC14_ADDR ¹	
MC14_ADDR		If IA32_MCG_CAP.CNT >14
Register Address: 43BH, 1083	IA32_MC14_MISC	
MC14_MISC		If IA32_MCG_CAP.CNT >14
Register Address: 43CH, 1084	IA32_MC15_CTL	
MC15_CTL		If IA32_MCG_CAP.CNT >15
Register Address: 43DH, 1085	IA32_MC15_STATUS	
MC15_STATUS		If IA32_MCG_CAP.CNT >15
Register Address: 43EH, 1086	IA32_MC15_ADDR ¹	
MC15_ADDR		If IA32_MCG_CAP.CNT >15

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 43FH, 1087	IA32_MC15_MISC	
MC15_MISC		If IA32_MCG_CAP.CNT >15
Register Address: 440H, 1088	IA32_MC16_CTL	
MC16_CTL		If IA32_MCG_CAP.CNT >16
Register Address: 441H, 1089	IA32_MC16_STATUS	
MC16_STATUS		If IA32_MCG_CAP.CNT >16
Register Address: 442H, 1090	IA32_MC16_ADDR ¹	
MC16_ADDR		If IA32_MCG_CAP.CNT >16
Register Address: 443H, 1091	IA32_MC16_MISC	
MC16_MISC		If IA32_MCG_CAP.CNT >16
Register Address: 444H, 1092	IA32_MC17_CTL	
MC17_CTL		If IA32_MCG_CAP.CNT >17
Register Address: 445H, 1093	IA32_MC17_STATUS	
MC17_STATUS		If IA32_MCG_CAP.CNT >17
Register Address: 446H, 1094	IA32_MC17_ADDR ¹	
MC17_ADDR		If IA32_MCG_CAP.CNT >17
Register Address: 447H, 1095	IA32_MC17_MISC	
MC17_MISC		If IA32_MCG_CAP.CNT >17
Register Address: 448H, 1096	IA32_MC18_CTL	
MC18_CTL		If IA32_MCG_CAP.CNT >18
Register Address: 449H, 1097	IA32_MC18_STATUS	
MC18_STATUS		If IA32_MCG_CAP.CNT >18
Register Address: 44AH, 1098	IA32_MC18_ADDR ¹	
MC18_ADDR		If IA32_MCG_CAP.CNT >18
Register Address: 44BH, 1099	IA32_MC18_MISC	
MC18_MISC		If IA32_MCG_CAP.CNT >18
Register Address: 44CH, 1100	IA32_MC19_CTL	
MC19_CTL		If IA32_MCG_CAP.CNT >19
Register Address: 44DH, 1101	IA32_MC19_STATUS	
MC19_STATUS		If IA32_MCG_CAP.CNT >19
Register Address: 44EH, 1102	IA32_MC19_ADDR ¹	
MC19_ADDR		If IA32_MCG_CAP.CNT >19
Register Address: 44FH, 1103	IA32_MC19_MISC	
MC19_MISC		If IA32_MCG_CAP.CNT >19
Register Address: 450H, 1104	IA32_MC20_CTL	
MC20_CTL		If IA32_MCG_CAP.CNT >20
Register Address: 451H, 1105	IA32_MC20_STATUS	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
MC20_STATUS		If IA32_MCG_CAP.CNT >20
Register Address: 452H, 1106		IA32_MC20_ADDR ¹
MC20_ADDR		If IA32_MCG_CAP.CNT >20
Register Address: 453H, 1107		IA32_MC20_MISC
MC20_MISC		If IA32_MCG_CAP.CNT >20
Register Address: 454H, 1108		IA32_MC21_CTL
MC21_CTL		If IA32_MCG_CAP.CNT >21
Register Address: 455H, 1109		IA32_MC21_STATUS
MC21_STATUS		If IA32_MCG_CAP.CNT >21
Register Address: 456H, 1110		IA32_MC21_ADDR ¹
MC21_ADDR		If IA32_MCG_CAP.CNT >21
Register Address: 457H, 1111		IA32_MC21_MISC
MC21_MISC		If IA32_MCG_CAP.CNT >21
Register Address: 458H, 1112		IA32_MC22_CTL
MC22_CTL		If IA32_MCG_CAP.CNT >22
Register Address: 459H, 1113		IA32_MC22_STATUS
MC22_STATUS		If IA32_MCG_CAP.CNT >22
Register Address: 45AH, 1114		IA32_MC22_ADDR ¹
MC22_ADDR		If IA32_MCG_CAP.CNT >22
Register Address: 45BH, 1115		IA32_MC22_MISC
MC22_MISC		If IA32_MCG_CAP.CNT >22
Register Address: 45CH, 1116		IA32_MC23_CTL
MC23_CTL		If IA32_MCG_CAP.CNT >23
Register Address: 45DH, 1117		IA32_MC23_STATUS
MC23_STATUS		If IA32_MCG_CAP.CNT >23
Register Address: 45EH, 1118		IA32_MC23_ADDR ¹
MC23_ADDR		If IA32_MCG_CAP.CNT >23
Register Address: 45FH, 1119		IA32_MC23_MISC
MC23_MISC		If IA32_MCG_CAP.CNT >23
Register Address: 460H, 1120		IA32_MC24_CTL
MC24_CTL		If IA32_MCG_CAP.CNT >24
Register Address: 461H, 1121		IA32_MC24_STATUS
MC24_STATUS		If IA32_MCG_CAP.CNT >24
Register Address: 462H, 1122		IA32_MC24_ADDR ¹
MC24_ADDR		If IA32_MCG_CAP.CNT >24
Register Address: 463H, 1123		IA32_MC24_MISC
MC24_MISC		If IA32_MCG_CAP.CNT >24

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 464H, 1124	IA32_MC25_CTL	
MC25_CTL		If IA32_MCG_CAP.CNT >25
Register Address: 465H, 1125	IA32_MC25_STATUS	
MC25_STATUS		If IA32_MCG_CAP.CNT >25
Register Address: 466H, 1126	IA32_MC25_ADDR ¹	
MC25_ADDR		If IA32_MCG_CAP.CNT >25
Register Address: 467H, 1127	IA32_MC25_MISC	
MC25_MISC		If IA32_MCG_CAP.CNT >25
Register Address: 468H, 1128	IA32_MC26_CTL	
MC26_CTL		If IA32_MCG_CAP.CNT >26
Register Address: 469H, 1129	IA32_MC26_STATUS	
MC26_STATUS		If IA32_MCG_CAP.CNT >26
Register Address: 46AH, 1130	IA32_MC26_ADDR ¹	
MC26_ADDR		If IA32_MCG_CAP.CNT >26
Register Address: 46BH, 1131	IA32_MC26_MISC	
MC26_MISC		If IA32_MCG_CAP.CNT >26
Register Address: 46CH, 1132	IA32_MC27_CTL	
MC27_CTL		If IA32_MCG_CAP.CNT >27
Register Address: 46DH, 1133	IA32_MC27_STATUS	
MC27_STATUS		If IA32_MCG_CAP.CNT >27
Register Address: 46EH, 1134	IA32_MC27_ADDR ¹	
MC27_ADDR		If IA32_MCG_CAP.CNT >27
Register Address: 46FH, 1135	IA32_MC27_MISC	
MC27_MISC		If IA32_MCG_CAP.CNT >27
Register Address: 470H, 1136	IA32_MC28_CTL	
MC28_CTL		If IA32_MCG_CAP.CNT >28
Register Address: 471H, 1137	IA32_MC28_STATUS	
MC28_STATUS		If IA32_MCG_CAP.CNT >28
Register Address: 472H, 1138	IA32_MC28_ADDR ¹	
MC28_ADDR		If IA32_MCG_CAP.CNT >28
Register Address: 473H, 1139	IA32_MC28_MISC	
MC28_MISC		If IA32_MCG_CAP.CNT >28
Register Address: 474H, 1140	IA32_MC29_CTL	
MC29_CTL		If IA32_MCG_CAP.CNT >29
Register Address: 475H, 1141	IA32_MC29_STATUS	
MC29_STATUS		If IA32_MCG_CAP.CNT >29
Register Address: 476H, 1142	IA32_MC29_ADDR	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
MC29_ADDR		If IA32_MCG_CAP.CNT >29
Register Address: 477H, 1143	IA32_MC29_MISC	
MC29_MISC		If IA32_MCG_CAP.CNT >29
Register Address: 478H, 1144	IA32_MC30_CTL	
MC30_CTL		If IA32_MCG_CAP.CNT >30
Register Address: 479H, 1145	IA32_MC30_STATUS	
MC30_STATUS		If IA32_MCG_CAP.CNT >30
Register Address: 47AH, 1146	IA32_MC30_ADDR	
MC30_ADDR		If IA32_MCG_CAP.CNT >30
Register Address: 47BH, 1147	IA32_MC30_MISC	
MC30_MISC		If IA32_MCG_CAP.CNT >30
Register Address: 47CH, 1148	IA32_MC31_CTL	
MC31_CTL		If IA32_MCG_CAP.CNT >31
Register Address: 47DH, 1149	IA32_MC31_STATUS	
MC31_STATUS		If IA32_MCG_CAP.CNT >31
Register Address: 47EH, 1150	IA32_MC31_ADDR	
MC31_ADDR		If IA32_MCG_CAP.CNT >31
Register Address: 47FH, 1151	IA32_MC31_MISC	
MC31_MISC		If IA32_MCG_CAP.CNT >31
Register Address: 480H, 1152	IA32_VMX_BASIC	
Reporting Register of Basic VMX Capabilities (R/O) See Appendix A.1, "Basic VMX Information."		If CPUID.01H:ECX.[5] = 1
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL	
Capability Reporting Register of Pin-Based VM-Execution Controls (R/O) See Appendix A.3.1, "Pin-Based VM-Execution Controls."		If CPUID.01H:ECX.[5] = 1
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL	
Capability Reporting Register of Primary Processor-Based VM-Execution Controls (R/O) See Appendix A.3.2, "Primary Processor-Based VM-Execution Controls."		If CPUID.01H:ECX.[5] = 1
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL	
Capability Reporting Register of Primary VM-Exit Controls (R/O) See Appendix A.4.1, "Primary VM-Exit Controls."		If CPUID.01H:ECX.[5] = 1
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTL	
Capability Reporting Register of VM-Entry Controls (R/O) See Appendix A.5, "VM-Entry Controls."		If CPUID.01H:ECX.[5] = 1
Register Address: 485H, 1157	IA32_VMX_MISC	
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Appendix A.6, "Miscellaneous Data."		If CPUID.01H:ECX.[5] = 1
Register Address: 486H, 1158	IA32_VMX_CR0_FIXED0	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Appendix A.7, "VMX-Fixed Bits in CR0."		If CPUID.01H:ECX.[5] = 1
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1	
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Appendix A.7, "VMX-Fixed Bits in CR0."		If CPUID.01H:ECX.[5] = 1
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0	
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Appendix A.8, "VMX-Fixed Bits in CR4."		If CPUID.01H:ECX.[5] = 1
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1	
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Appendix A.8, "VMX-Fixed Bits in CR4."		If CPUID.01H:ECX.[5] = 1
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM	
Capability Reporting Register of VMCS Field Enumeration (R/O) See Appendix A.9, "VMCS Enumeration."		If CPUID.01H:ECX.[5] = 1
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL2	
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Appendix A.3.3, "Secondary Processor-Based VM-Execution Controls."		If (CPUID.01H:ECX.[5] && IA32_VMX_PROCBASED_CTL2[63])
Register Address: 48CH, 1164	IA32_VMX_EPT_VPID_CAP	
Capability Reporting Register of EPT and VPID (R/O) See Appendix A.10, "VPID and EPT Capabilities."		If (CPUID.01H:ECX.[5] && IA32_VMX_PROCBASED_CTL2[63] && (IA32_VMX_PROCBASED_CTL2[3 3] IA32_VMX_PROCBASED_CTL2[3 7]))
Register Address: 48DH, 1165	IA32_VMX_TRUE_PINBASED_CTL2	
Capability Reporting Register of Pin-Based VM-Execution Flex Controls (R/O) See Appendix A.3.1, "Pin-Based VM-Execution Controls."		If (CPUID.01H:ECX.[5] && IA32_VMX_BASIC[55])
Register Address: 48EH, 1166	IA32_VMX_TRUE_PROCBASED_CTL2	
Capability Reporting Register of Primary Processor-Based VM-Execution Flex Controls (R/O) See Appendix A.3.2, "Primary Processor-Based VM-Execution Controls."		If(CPUID.01H:ECX.[5] && IA32_VMX_BASIC[55])
Register Address: 48FH, 1167	IA32_VMX_TRUE_EXIT_CTL2	
Capability Reporting Register of VM-Exit Flex Controls (R/O) See Appendix A.4, "VM-Exit Controls."		If(CPUID.01H:ECX.[5] && IA32_VMX_BASIC[55])
Register Address: 490H, 1168	IA32_VMX_TRUE_ENTRY_CTL2	
Capability Reporting Register of VM-Entry Flex Controls (R/O) See Appendix A.5, "VM-Entry Controls."		If(CPUID.01H:ECX.[5] && IA32_VMX_BASIC[55])
Register Address: 491H, 1169	IA32_VMX_VMFUNC	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Capability Reporting Register of VM-Function Controls (R/O)		If (CPUID.01H:ECX.[5] && IA32_VMX_PROCBASED_CTL[63] && IA32_VMX_PROCBASED_CTL[45])
Register Address: 492H, 1170		IA32_VMX_PROCBASED_CTL[3]
Capability Reporting Register of Tertiary Processor-Based VM-Execution Controls (R/O) See Appendix A.3.4, "Tertiary Processor-Based VM-Execution Controls."		If (CPUID.01H:ECX.[5] && IA32_VMX_PROCBASED_CTL[49])
Register Address: 493H, 1171		IA32_VMX_EXIT_CTL[2]
Capability Reporting Register of Secondary VM-Exit Controls (R/O) See Appendix A.4.2, "Secondary VM-Exit Controls."		If (CPUID.01H:ECX.[5] && IA32_VMX_EXIT_CTL[63])
Register Address: 4C1H, 1217		IA32_A_PMC0
Full Width Writable IA32_PMC0 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 0) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[0] = 1
Register Address: 4C2H, 1218		IA32_A_PMC1
Full Width Writable IA32_PMC1 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 1) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[1] = 1
Register Address: 4C3H, 1219		IA32_A_PMC2
Full Width Writable IA32_PMC2 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 2) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[2] = 1
Register Address: 4C4H, 1220		IA32_A_PMC3
Full Width Writable IA32_PMC3 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 3) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[3] = 1
Register Address: 4C5H, 1221		IA32_A_PMC4
Full Width Writable IA32_PMC4 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 4) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[4] = 1
Register Address: 4C6H, 1222		IA32_A_PMC5

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Full Width Writable IA32_PMC5 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 5) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[5] = 1
Register Address: 4C7H, 1223		IA32_A_PMC6
Full Width Writable IA32_PMC6 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 6) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[6] = 1
Register Address: 4C8H, 1224		IA32_A_PMC7
Full Width Writable IA32_PMC7 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 7) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[7] = 1
Register Address: 4C9H, 1225		IA32_A_PMC8
Full Width Writable IA32_PMC8 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 8) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[8] = 1
Register Address: 4CAH, 1226		IA32_A_PMC9
Full Width Writable IA32_PMC9 Alias (R/W)		If (CPUID.0AH:EAX[15:8] > 9) && IA32_PERF_CAPABILITIES[13] = 1) CPUID.(EAX=23H,ECX=1):EAX[9] = 1
Register Address: 4D0H, 1232		IA32_MCG_EXT_CTL
Allows software to signal some MCEs to only a single logical processor in the system. (R/W) See Section 17.3.1.4, "IA32_MCG_EXT_CTL MSR."		If IA32_MCG_CAP.LMCE_P = 1
0	LMCE_EN Enable / Disable local machine check exception.	
63:1	Reserved.	
Register Address: 500H, 1280		IA32_SGX_SVN_STATUS
Status and SVN Threshold of SGX Support for ACM (R/O)		If CPUID.(EAX=07H, ECX=0H): EBX[2] = 1
0	Lock.	See Section 40.11.3, "Interactions with Authenticated Code Modules (ACMs)."
15:1	Reserved.	
23:16	SGX_SVN_SINIT	See Section 40.11.3, "Interactions with Authenticated Code Modules (ACMs)."
63:24	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 560H, 1376		IA32_RTIT_OUTPUT_BASE
Trace Output Base Register (R/W)		If ((CPUID.(EAX=07H, ECX=0):EBX[25] = 1) && (CPUID.(EAX=14H, ECX=0):ECX[0] = 1) (CPUID.(EAX=14H, ECX=0):ECX[2] = 1))
6:0	Reserved.	
MAXPHYADDR ⁴ -1:7	Base physical address.	
63:MAXPHYADDR	Reserved.	
Register Address: 561H, 1377		IA32_RTIT_OUTPUT_MASK_PTRS
Trace Output Mask Pointers Register (R/W)		If ((CPUID.(EAX=07H, ECX=0):EBX[25] = 1) && (CPUID.(EAX=14H, ECX=0):ECX[0] = 1) (CPUID.(EAX=14H, ECX=0):ECX[2] = 1))
6:0	Reserved.	
31:7	MaskOrTableOffset.	
63:32	Output Offset.	
Register Address: 570H, 1392		IA32_RTIT_CTL
Trace Control Register (R/W)		If (CPUID.(EAX=07H, ECX=0):EBX[25] = 1)
0	TraceEn	
1	CYCEn	If (CPUID.(EAX=07H, ECX=0):EBX[1] = 1)
2	OS	
3	User	
4	PwrEvtEn	If (CPUID.(EAX=07H, ECX=1):EBX[5] = 1)
5	FUPonPTW	If (CPUID.(EAX=07H, ECX=1):EBX[4] = 1)
6	FabricEn	If (CPUID.(EAX=07H, ECX=0):ECX[3] = 1)
7	CR3Filter	If (CPUID.(EAX=14H, ECX=0):EBX[0] = 1)
8	ToPA	
9	MTCEn	If (CPUID.(EAX=07H, ECX=0):EBX[3] = 1)
10	TSCEn	
11	DisRETC	
12	PTWEn	If (CPUID.(EAX=07H, ECX=1):EBX[4] = 1)

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
13	BranchEn	
17:14	MTCFreq.	If (CPUID.(EAX=07H, ECX=0):EBX[3] = 1)
18	Reserved, must be zero.	
22:19	CycThresh	If (CPUID.(EAX=07H, ECX=0):EBX[1] = 1)
23	Reserved, must be zero.	
27:24	PSBFreq	If (CPUID.(EAX=07H, ECX=0):EBX[1] = 1)
30:28	Reserved, must be zero.	
31	EventEn	If (CPUID.(EAX=14H, ECX=0):EBX[7] = 1)
35:32	ADDR0_CFG	If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 0)
39:36	ADDR1_CFG	If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 1)
43:40	ADDR2_CFG	If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 2)
47:44	ADDR3_CFG	If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 3)
54:48	Reserved, must be zero.	
55	DisTNT	If (CPUID.(EAX=14H, ECX=0):EBX[8] = 1)
56	InjectPsbPmiOnEnable	If (CPUID.(EAX=07H, ECX=1):EBX[6] = 1)
63:57	Reserved, must be zero.	
Register Address: 571H, 1393		IA32_RTIT_STATUS
Tracing Status Register (R/W)		If (CPUID.(EAX=07H, ECX=0):EBX[25] = 1)
0	FilterEn (writes ignored).	If (CPUID.(EAX=07H, ECX=0):EBX[2] = 1)
1	ContexEn (writes ignored).	
2	TriggerEn (writes ignored).	
3	Reserved.	
4	Error	
5	Stopped	
6	PendPSB	If (CPUID.(EAX=07H, ECX=0):EBX[6] = 1)
7	PendToPAPMI	If (CPUID.(EAX=07H, ECX=0):EBX[6] = 1)
31:8	Reserved, must be zero.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
48:32	PacketByteCnt	If (CPUID.(EAX=07H, ECX=0):EBX[1] > 3)
63:49	Reserved.	
Register Address: 572H, 1394		IA32_RTIT_CR3_MATCH
Trace Filter CR3 Match Register (R/W)		If (CPUID.(EAX=07H, ECX=0):EBX[25] = 1)
4:0	Reserved.	
63:5	CR3[63:5] value to match.	
Register Address: 580H, 1408		IA32_RTIT_ADDR0_A
Region 0 Start Address (R/W)		If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 0)
47:0	Virtual Address.	
63:48	SignExt_VA	
Register Address: 581H, 1409		IA32_RTIT_ADDR0_B
Region 0 End Address (R/W)		If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 0)
47:0	Virtual Address.	
63:48	SignExt_VA	
Register Address: 582H, 1410		IA32_RTIT_ADDR1_A
Region 1 Start Address (R/W)		If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 1)
47:0	Virtual Address.	
63:48	SignExt_VA	
Register Address: 583H, 1411		IA32_RTIT_ADDR1_B
Region 1 End Address (R/W)		If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 1)
47:0	Virtual Address.	
63:48	SignExt_VA	
Register Address: 584H, 1412		IA32_RTIT_ADDR2_A
Region 2 Start Address (R/W)		If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 2)
47:0	Virtual Address.	
63:48	SignExt_VA	
Register Address: 585H, 1413		IA32_RTIT_ADDR2_B
Region 2 End Address (R/W)		If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 2)
47:0	Virtual Address.	
63:48	SignExt_VA	
Register Address: 586H, 1414		IA32_RTIT_ADDR3_A

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Region 3 Start Address (R/W)		If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 3)
47:0	Virtual Address.	
63:48	SignExt_VA	
Register Address: 587H, 1415		IA32_RTIT_ADDR3_B
Region 3 End Address (R/W)		If (CPUID.(EAX=07H, ECX=1):EAX[2:0] > 3)
47:0	Virtual Address.	
63:48	SignExt_VA	
Register Address: 600H, 1536		IA32_DS_AREA
DS Save Area (R/W) Points to the linear address of the first byte of the DS buffer management area, which is used to manage the BTS and PEBS buffers. See Section 21.6.3.4, “Debug Store (DS) Mechanism.”		If (CPUID.01H:EDX.DS[21] = 1)
63:0	The linear address of the first byte of the DS buffer management area, if IA-32e mode is active.	
31:0	The linear address of the first byte of the DS buffer management area, if not in IA-32e mode.	
63:32	Reserved if not in IA-32e mode.	
Register Address: 6A0H, 1696		IA32_U_CET
Configure User Mode CET (R/W)		Bits 1:0 are defined if CPUID.(EAX=07H, ECX=0H):ECX.CET_SS[07] = 1. Bits 5:2 and bits 63:10 are defined if CPUID.(EAX=07H, ECX=0H):EDX.CET_IBT[20] = 1.
0	SH_STK_EN: When set to 1, enable shadow stacks at CPL3.	
1	WR_SHSTK_EN: When set to 1, enables the WRSSD/WRSSQ instructions.	
2	ENDBR_EN: When set to 1, enables indirect branch tracking.	
3	LEG_IW_EN: Enable legacy compatibility treatment for indirect branch tracking.	
4	NO_TRACK_EN: When set to 1, enables use of no-track prefix for indirect branch tracking.	
5	SUPPRESS_DIS: When set to 1, disables suppression of CET indirect branch tracking on legacy compatibility.	
9:6	Reserved; must be zero.	
10	SUPPRESS: When set to 1, indirect branch tracking is suppressed. This bit can be written to 1 only if TRACKER is written as IDLE.	
11	TRACKER: Value of the indirect branch tracking state machine. Values: IDLE (0), WAIT_FOR_ENDBRANCH(1).	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)	
Bit Fields	MSR/Bit Description		Comment
63:12	EB_LEG_BITMAP_BASE: Linear address bits 63:12 of a legacy code page bitmap used for legacy compatibility when indirect branch tracking is enabled. If the processor does not support Intel 64 architecture, these fields have only 32 bits; bits 63:32 of the MSRs are reserved. On processors that support Intel 64 architecture this value cannot represent a non-canonical address. In protected mode, only 31:0 are used.		
Register Address: 6A2H, 1698		IA32_S_CET	
Configure Supervisor Mode CET (R/W)		See IA32_U_CET (6A0H) for reference; similar format.	
Register Address: 6A4H, 1700		IA32_PL0_SSP	
Linear address to be loaded into SSP on transition to privilege level 0. (R/W) If the processor does not support Intel 64 architecture, these fields have only 32 bits; bits 63:32 of the MSRs are reserved. On processors that support Intel 64 architecture this value cannot represent a non-canonical address. In protected mode, only 31:0 are loaded. Bits 1:0 of the MSR must be 0. Transitions to privilege level 0 will check that bit 2 is also 0.		If CPUID.(EAX=07H, ECX=0H):ECX.CET_SS[07] = 1	
Register Address: 6A5H, 1701		IA32_PL1_SSP	
Linear address to be loaded into SSP on transition to privilege level 1. (R/W) If the processor does not support Intel 64 architecture, these fields have only 32 bits; bits 63:32 of the MSRs are reserved. On processors that support Intel 64 architecture this value cannot represent a non-canonical address. In protected mode, only 31:0 are loaded. Bits 1:0 of the MSR must be 0. Transitions to privilege level 1 from a higher privilege level will check that bit 2 is also 0.		If CPUID.(EAX=07H, ECX=0H):ECX.CET_SS[07] = 1	
Register Address: 6A6H, 1702		IA32_PL2_SSP	
Linear address to be loaded into SSP on transition to privilege level 2. (R/W) If the processor does not support Intel 64 architecture, these fields have only 32 bits; bits 63:32 of the MSRs are reserved. On processors that support Intel 64 architecture this value cannot represent a non-canonical address. In protected mode, only 31:0 are loaded. Bits 1:0 of the MSR must be 0. Transitions to privilege level 2 from a higher privilege level will check that bit 2 is also 0.		If CPUID.(EAX=07H, ECX=0H):ECX.CET_SS[07] = 1	
Register Address: 6A7H, 1703		IA32_PL3_SSP	
Linear address to be loaded into SSP on transition to privilege level 3. (R/W) If the processor does not support Intel 64 architecture, these fields have only 32 bits; bits 63:32 of the MSRs are reserved. On processors that support Intel 64 architecture this value cannot represent a non-canonical address. In protected mode, only 31:0 are loaded. Bits 1:0 of the MSR must be 0.		If CPUID.(EAX=07H, ECX=0H):ECX.CET_SS[07] = 1	
Register Address: 6A8H, 1704		IA32_INTERRUPT_SSP_TABLE_ADDR	
Linear address of a table of seven shadow stack pointers that are selected in IA-32e mode using the IST index (when not 0) from the interrupt gate descriptor. (R/W) This MSR is not present on processors that do not support Intel 64 architecture. This field cannot represent a non-canonical address.		If CPUID.(EAX=07H, ECX=0H):ECX.CET_SS[07] = 1	
Register Address: 6E0H, 1760		IA32_TSC_DEADLINE	
TSC Target of Local APIC's TSC Deadline Mode (R/W)		If CPUID.01H:ECX.[24] = 1	
63:0	REGISTER_VALUE TSC-deadline value.		

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)	
Bit Fields	MSR/Bit Description		Comment
Register Address: 6E1H, 1761		IA32_PKRS	
Specifies the PK permissions associated with each protection domain for supervisor pages (R/W)		If CPUID.(EAX=07H, ECX=0H):ECX.PKS [31] = 1	
31:0	For domain i (i between 0 and 15), bits 2i and 2i+1 contain the AD and WD permissions, respectively.		
63:32	Reserved.		
Register Address: 770H, 1904		IA32_PM_ENABLE	
Enable/disable HWP (R/W)		If CPUID.06H:EAX.[7] = 1	
0	HWP_ENABLE (R/W) Note this bit can only be enabled once from the default value. Once set, writes to the HWP_ENABLE bit are ignored. Only RESET will clear this bit. Default = 0. See Section 16.4.2, "Enabling HWP."	If CPUID.06H:EAX.[7] = 1	
63:1	Reserved.		
Register Address: 771H, 1905		IA32_HWP_CAPABILITIES	
HWP Performance Range Enumeration (R/O)		If CPUID.06H:EAX.[7] = 1	
7:0	Highest_Performance See Section 16.4.3, "HWP Performance Range and Dynamic Capabilities."	If CPUID.06H:EAX.[7] = 1	
15:8	Guaranteed_Performance See Section 16.4.3, "HWP Performance Range and Dynamic Capabilities."	If CPUID.06H:EAX.[7] = 1	
23:16	Most_Efficient_Performance See Section 16.4.3, "HWP Performance Range and Dynamic Capabilities."	If CPUID.06H:EAX.[7] = 1	
31:24	Lowest_Performance See Section 16.4.3, "HWP Performance Range and Dynamic Capabilities."	If CPUID.06H:EAX.[7] = 1	
63:32	Reserved.		
Register Address: 772H, 1906		IA32_HWP_REQUEST_PKG	
Power Management Control Hints for All Logical Processors in a Package (R/W)		If CPUID.06H:EAX.[11] = 1	
7:0	Minimum_Performance See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[11] = 1	
15:8	Maximum_Performance See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[11] = 1	
23:16	Desired_Performance See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[11] = 1	
31:24	Energy_Performance_Preference See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[11] = 1 && CPUID.06H:EAX.[10] = 1	
41:32	Activity_Window See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[11] = 1 && CPUID.06H:EAX.[9] = 1	
63:42	Reserved.		
Register Address: 773H, 1907		IA32_HWP_INTERRUPT	
Control HWP Native Interrupts (R/W)		If CPUID.06H:EAX.[8] = 1	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
0	EN_Guaranteed_Performance_Change See Section 16.4.6, "HWP Notifications."	If CPUID.06H:EAX.[8] = 1
1	EN_Excursion_Minimum See Section 16.4.6, "HWP Notifications."	If CPUID.06H:EAX.[8] = 1
63:2	Reserved.	
Register Address: 774H, 1908		IA32_HWP_REQUEST
Power Management Control Hints to a Logical Processor (R/W)		If CPUID.06H:EAX.[7] = 1
7:0	Minimum_Performance See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[7] = 1
15:8	Maximum_Performance See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[7] = 1
23:16	Desired_Performance See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[7] = 1
31:24	Energy_Performance_Preference See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[7] = 1 && CPUID.06H:EAX.[10] = 1
41:32	Activity_Window See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[7] = 1 && CPUID.06H:EAX.[9] = 1
42	Package_Control See Section 16.4.4, "Managing HWP."	If CPUID.06H:EAX.[7] = 1 && CPUID.06H:EAX.[11] = 1
63:43	Reserved.	
Register Address: 775H, 1909		IA32_PECI_HWP_REQUEST_INFO
IA32_PECI_HWP_REQUEST_INFO		
7:0	Minimum Performance (MINIMUM_PERFORMANCE): Used by OS to read the latest value of Peci minimum performance input. Default value is 0.	
15:8	Maximum Performance (MAXIMUM_PERFORMANCE): Used by OS to read the latest value of Peci maximum performance input. Default value is 0.	
23:16	Reserved.	
31:24	Energy Performance Preference (ENERGY_PERFORMANCE_PREFERENCE): Used by OS to read the latest value of Peci Energy Performance Preference input. Default value is 0.	
59:32	Reserved.	
60	EPP Peci Override (EPP_PECI_OVERRIDE): Indicates whether Peci is currently overriding the Energy Performance Preference input. If set to '1', Peci is overriding the Energy Performance Preference input. If clear (0), OS has control over Energy Performance Preference input. Default value is 0.	
61	Reserved.	
62	Max Peci Override (MAX_PECI_OVERRIDE): Indicates whether Peci is currently overriding the Maximum Performance input. If set to '1', Peci is overriding the Maximum Performance input. If clear (0), OS has control over Maximum Performance input. Default value is 0.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
63	Min PECL Override (MIN_PECI_OVERRIDE): Indicates whether PECL is currently overriding the Minimum Performance input. If set to '1', PECL is overriding the Minimum Performance input. If clear (0), OS has control over Minimum Performance input. Default value is 0.	
Register Address: 776H, 1910		IA32_HWP_CTL
IA32_HWP_CTL		If CPUID.06H:EAX.[22] = 1
0	PKG_CTL_POLARITY Defines which HWP Request MSR is used whether Thread level or package level. When package MSR is used, the thread MSR valid bits define which thread MSR fields override the package. Default value is 0.	If CPUID.06H:EAX.[22] = 1
63:1	Reserved.	
Register Address: 777H, 1911		IA32_HWP_STATUS
Log bits indicating changes to Guaranteed & excursions to Minimum (R/W)		If CPUID.06H:EAX.[7] = 1
0	Guaranteed_Performance_Change (R/WCO) See Section 16.4.5, "HWP Feedback."	If CPUID.06H:EAX.[7] = 1
1	Reserved.	
2	Excursion_To_Minimum (R/WCO) See Section 16.4.5, "HWP Feedback."	If CPUID.06H:EAX.[7] = 1
63:3	Reserved.	
Register Address: 7A3H, 1955		IA32_MCU_EXT_SERVICE
MCU Extended Service (R/O)		If IA32_ARCH_CAPABILITIES[22] = 1
3:0	ALLOWED_PERIODS Value indicates the allowed periods for extended servicing. Value x means that all extended servicing periods are allowed till period x.	
63:4	Reserved.	
Register Address: 7A4H, 1956		IA32_MCU_ROLLBACK_MIN_ID
Minimal MCU Revision ID (R/O) Minimal MCU Revision ID that software can rollback to per boot.		If IA32_MCU_ENUMERATION[3] = 1
31:0	REVISION_ID Minimal MCU revision ID for rollback.	
63:32	Reserved for future use.	
Register Address: 7A5H, 1957		IA32_MCU_STAGING_MBOX_ADDR
IA32_MCU_STAGING_MBOX_ADDR (R/O) Reports MMIO address of MCU staging DOE mailbox.		
63:0	ADDR MMIO address base of MCU staging DOE mailbox.	
Register Address: 7B0H, 1968		IA32_ROLLBACK_SIGN_ID_0

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Rollback ID 0 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7B1H, 1969		IA32_ROLLBACK_SIGN_ID_1
Rollback ID 1 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7B2H, 1970		IA32_ROLLBACK_SIGN_ID_2
Rollback ID 2 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7B3H, 1971		IA32_ROLLBACK_SIGN_ID_3
Rollback ID 3 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7B4H, 1972		IA32_ROLLBACK_SIGN_ID_4
Rollback ID 4 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 7B5H, 1973		IA32_ROLLBACK_SIGN_ID_5
Rollback ID 5 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7B6H, 1974		IA32_ROLLBACK_SIGN_ID_6
Rollback ID 6 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7B7H, 1975		IA32_ROLLBACK_SIGN_ID_7
Rollback ID 7 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7B8H, 1976		IA32_ROLLBACK_SIGN_ID_8
Rollback ID 8 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7B9H, 1977		IA32_ROLLBACK_SIGN_ID_9
Rollback ID 9 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
63:48	Reserved.	
Register Address: 7BAH, 1978		IA32_ROLLBACK_SIGN_ID_10
Rollback ID 10 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7BBH, 1979		IA32_ROLLBACK_SIGN_ID_11
Rollback ID 11 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7BCH, 1980		IA32_ROLLBACK_SIGN_ID_12
Rollback ID 12 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7BDH, 1981		IA32_ROLLBACK_SIGN_ID_13
Rollback ID 13 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7BEH, 1982		IA32_ROLLBACK_SIGN_ID_14
Rollback ID 14 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 7BFH, 1983		IA32_ROLLBACK_SIGN_ID_15
Rollback ID 15 (R/O) Holds the Revision ID and SVN of a supported rollback target or 0 if none.		If IA32_MCU_ENUMERATION[3] = 1
31:0	MCU_ROLLBACK_ID MCU supported Rollback ID.	
47:32	ROLLBACK_MCU_SVN MCU SVN corresponding to the reported MCU Rollback ID.	
63:48	Reserved.	
Register Address: 802H, 2050		IA32_X2APIC_APICID
x2APIC ID Register (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 803H, 2051		IA32_X2APIC_VERSION
x2APIC Version Register (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 808H, 2056		IA32_X2APIC_TPR
x2APIC Task Priority Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 80AH, 2058		IA32_X2APIC_PPR
x2APIC Processor Priority Register (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 80BH, 2059		IA32_X2APIC_EOI
x2APIC EOI Register (W/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 80DH, 2061		IA32_X2APIC_LDR
x2APIC Logical Destination Register (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 80FH, 2063		IA32_X2APIC_SIVR
x2APIC Spurious Interrupt Vector Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 810H, 2064		IA32_X2APIC_ISR0
x2APIC In-Service Register Bits 31:0 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 811H, 2065		IA32_X2APIC_ISR1
x2APIC In-Service Register Bits 63:32 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 812H, 2066		IA32_X2APIC_ISR2
x2APIC In-Service Register Bits 95:64 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 813H, 2067	IA32_X2APIC_ISR3	
x2APIC In-Service Register Bits 127:96 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 814H, 2068	IA32_X2APIC_ISR4	
x2APIC In-Service Register Bits 159:128 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 815H, 2069	IA32_X2APIC_ISR5	
x2APIC In-Service Register Bits 191:160 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 816H, 2070	IA32_X2APIC_ISR6	
x2APIC In-Service Register Bits 223:192 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 817H, 2071	IA32_X2APIC_ISR7	
x2APIC In-Service Register Bits 255:224 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 818H, 2072	IA32_X2APIC_TMR0	
x2APIC Trigger Mode Register Bits 31:0 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 819H, 2073	IA32_X2APIC_TMR1	
x2APIC Trigger Mode Register Bits 63:32 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 81AH, 2074	IA32_X2APIC_TMR2	
x2APIC Trigger Mode Register Bits 95:64 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 81BH, 2075	IA32_X2APIC_TMR3	
x2APIC Trigger Mode Register Bits 127:96 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 81CH, 2076	IA32_X2APIC_TMR4	
x2APIC Trigger Mode Register Bits 159:128 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 81DH, 2077	IA32_X2APIC_TMR5	
x2APIC Trigger Mode Register Bits 191:160 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 81EH, 2078	IA32_X2APIC_TMR6	
x2APIC Trigger Mode Register Bits 223:192 (R/O)		If (CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1)
Register Address: 81FH, 2079	IA32_X2APIC_TMR7	
x2APIC Trigger Mode Register Bits 255:224 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 820H, 2080	IA32_X2APIC_IRR0	
x2APIC Interrupt Request Register Bits 31:0 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 821H, 2081	IA32_X2APIC_IRR1	
x2APIC Interrupt Request Register Bits 63:32 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 822H, 2082	IA32_X2APIC_IRR2	
x2APIC Interrupt Request Register Bits 95:64 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 823H, 2083	IA32_X2APIC_IRR3	
x2APIC Interrupt Request Register Bits 127:96 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 824H, 2084	IA32_X2APIC_IRR4	
x2APIC Interrupt Request Register Bits 159:128 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 825H, 2085	IA32_X2APIC_IRR5	
x2APIC Interrupt Request Register Bits 191:160 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 826H, 2086	IA32_X2APIC_IRR6	
x2APIC Interrupt Request Register Bits 223:192 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 827H, 2087	IA32_X2APIC_IRR7	
x2APIC Interrupt Request Register Bits 255:224 (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 828H, 2088	IA32_X2APIC_ESR	
x2APIC Error Status Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 82FH, 2095	IA32_X2APIC_LVT_CMCI	
x2APIC LVT Corrected Machine Check Interrupt Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 830H, 2096	IA32_X2APIC_ICR	
x2APIC Interrupt Command Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 832H, 2098	IA32_X2APIC_LVT_TIMER	
x2APIC LVT Timer Interrupt Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 833H, 2099	IA32_X2APIC_LVT_THERMAL	
x2APIC LVT Thermal Sensor Interrupt Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 834H, 2100	IA32_X2APIC_LVT_PMI	
x2APIC LVT Performance Monitor Interrupt Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 835H, 2101	IA32_X2APIC_LVT_LINT0	
x2APIC LVT LINT0 Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 836H, 2102		IA32_X2APIC_LVT_LINT1
x2APIC LVT LINT1 Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 837H, 2103		IA32_X2APIC_LVT_ERROR
x2APIC LVT Error Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 838H, 2104		IA32_X2APIC_INIT_COUNT
x2APIC Initial Count Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 839H, 2105		IA32_X2APIC_CUR_COUNT
x2APIC Current Count Register (R/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 83EH, 2110		IA32_X2APIC_DIV_CONF
x2APIC Divide Configuration Register (R/W)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 83FH, 2111		IA32_X2APIC_SELF_IPI
x2APIC Self IPI Register (W/O)		If CPUID.01H:ECX.[21] = 1 && IA32_APIC_BASE.[10] = 1
Register Address: 981H, 2433		IA32_TME_CAPABILITY
Memory Encryption Capability MSR		If CPUID.07H:ECX.[13] = 1
0	Support for AES-XTS 128-bit encryption algorithm. (NIST standard)	
1	Support for AES-XTS 128-bit encryption with integrity algorithm.	
2	Support for AES-XTS 256-bit encryption algorithm.	
29:3	Reserved.	
30	SUPPORT_IA32_TME_CLEAR_SAVED_KEY Support for the IA32_TME_CLEAR_SAVED_KEY MSR.	
31	TME encryption bypass supported.	
35:32	MK_TME_MAX_KEYID_BITS Number of bits which can be allocated for usage as key identifiers for multi-key memory encryption. 4 bits allow for a maximum value of 15, which could address 32K keys. Zero if TME-MK is not supported.	
50:36	MK_TME_MAX_KEYS Indicates the maximum number of keys which are available for usage. This value may not be a power of 2. KeyID 0 is specially reserved and is not accounted for in this field.	
63:51	Reserved.	
Register Address: 982H, 2434		IA32_TME_ACTIVATE

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Memory Encryption Activation MSR This MSR is used to lock the MSRs listed below. Any write to the following MSRs will be ignored after they are locked. The lock is reset when CPU is reset. ▪ IA32_TME_ACTIVATE ▪ IA32_TME_EXCLUDE_MASK ▪ IA32_TME_EXCLUDE_BASE Note that IA32_TME_EXCLUDE_MASK and IA32_TME_EXCLUDE_BASE must be configured before IA32_TME_ACTIVATE.		If CPUID.07H:ECX.[13] = 1
0	Lock R/O – Will be set upon successful WRMSR (or first SMI); written value ignored.	
1	Hardware Encryption Enable This bit also enables TME-MK; TME-MK cannot be enabled without enabling encryption hardware.	
2	Key Select 0: Create a new TME key (expected cold/warm boot). 1: Restore the TME key from storage (Expected when resume from standby).	
3	Save TME Key for Standby Save key into storage to be used when resume from standby. Note: This may not be supported in all processors.	
7:4	TME Policy/Encryption Algorithm Only algorithms enumerated in IA32_TME_CAPABILITY are allowed. For example: 0000 – AES-XTS-128. 0001 – AES-XTS-128 with integrity. 0010 – AES-XTS-256. Other values are invalid.	
30:8	Reserved.	
31	TME Encryption Bypass Enable When encryption hardware is enabled: ▪ Total Memory Encryption is enabled using a CPU generated ephemeral key based on a hardware random number generator when this bit is set to 0. ▪ Total Memory Encryption is bypassed (no encryption/decryption for KeyID0) when this bit is set to 1. Software must inspect Hardware Encryption Enable (bit 1) and TME encryption bypass Enable (bit 31) to determine if TME encryption is enabled.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
35:32	MK_TME_KEYID_BITS Reserved if TME-MK is not enumerated, otherwise: The number of key identifier bits to allocate to TME-MK usage. Similar to enumeration, this is an encoded value. Writing a value greater than MK_TME_MAX_KEYID_BITS will result in #GP. Writing a non-zero value to this field will #GP if bit 1 of EAX (Hardware Encryption Enable) is not also set to '1, as encryption hardware must be enabled to use TME-MK. Example: To support 255 keys, this field would be set to a value of 8.	
47:36	Reserved.	
63:48	MK_TME_CRYPT0_ALGS Reserved if TME-MK is not enumerated, otherwise: Bit 48: AES-XTS 128. Bit 49: AES-XTS 128 with integrity. Bit 50: AES-XTS 256. Bit 63:51: Reserved (#GP) Bitmask for BIOS to set which encryption algorithms are allowed for TME-MK, would be later enforced by the key loading ISA ('1' = allowed).	
Register Address: 983H, 2435		IA32_TME_EXCLUDE_MASK
Memory Encryption Exclude Mask		If CPUID.07H:ECX.[13] = 1
10:0	Reserved.	
11	Enable: When set to '1', then TME_EXCLUDE_BASE and TME_EXCLUDE_MASK are used to define an exclusion region for TME/TME-MK (for KeyID=0).	
MAXPHYADDR-1:12	TMEEBASE: This field indicates the bits that must match TMEEBASE in order to qualify as a TME/TME-MK (for KeyID=0) exclusion memory range access.	
63:MAXPHYADDR	Reserved; must be zero.	
Register Address: 984H, 2436		IA32_TME_EXCLUDE_BASE
Memory Encryption Exclude Base		If CPUID.07H:ECX.[13] = 1
11:0	Reserved.	
MAXPHYADDR-1:12	TMEEBASE: Base physical address to be excluded for TME/TME-MK (for KeyID=0) encryption.	
63:MAXPHYADDR	Reserved; must be zero.	
Register Address: 985H, 2437		IA32_UINTR_RR
User Interrupt Request Register (R/W)		If CPUID.07H.01H:EDX[13]=1
63:0	UIRR Bitmap of requested user interrupt vectors.	
Register Address: 986H, 2438		IA32_UINTR_HANDLER
User Interrupt Handler Address (R/W)		If CPUID.07H.01H:EDX[13]=1

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
63:0	UIHANDLER User interrupt handler linear address.	
Register Address: 987H, 2439		IA32_UINTR_STACKADJUST
User Interrupt Stack Adjustment (R/W)		If CPUID.07H.01H:EDX[13]=1
0	LOAD_RSP User interrupt stack mode.	
2:1	Reserved.	
63:3	STACK_ADJUST Stack adjust value.	
Register Address: 988H, 2440		IA32_UINTR_MISC
User-Interrupt Target-Table Size and Notification Vector (R/W)		If CPUID.07H.01H:EDX[13]=1
31:0	UITTSZ The highest index of a valid entry in the user-interrupt target table. Valid entries are indices 0..UITTSZ (inclusive).	
39:32	UINV User-interrupt notification vector.	
63:40	Reserved.	
Register Address: 989H, 2441		IA32_UINTR_PD
User Interrupt PID Address (R/W)		If CPUID.07H.01H:EDX[13]=1
5:0	Reserved.	
63:6	UPIDADDR User-interrupt notification processing accesses a UPID at this linear address.	
Register Address: 98AH, 2442		IA32_UINTR_TT
User-Interrupt Target Table (R/W)		If CPUID.07H.01H:EDX[13]=1
0	SENDUIPI_ENABLE User-interrupt target table is valid.	
3:1	Reserved.	
63:4	UITTADDR User-interrupt target table base linear address.	
Register Address: 990H, 2448		IA32_COPY_STATUS ⁵
Status of Most Recent Platform to Local or Local to Platform Copies (R/O)		If ((CPUID.19H:EBX[4] = 1) && (CPUID.(07H,0).ECX[23] = 1))
0	IWKEY_COPY_SUCCESSFUL Status of most recent copy to or from IwKeyBackup.	If ((CPUID.19H:EBX[4] = 1) && (CPUID.(07H,0).ECX[23] = 1))
63:1	Reserved.	
Register Address: 991H, 2449		IA32_IWKEYBACKUP_STATUS ⁵
Information about IwKeyBackup Register (R/O)		If ((CPUID.19H:EBX[4] = 1) && (CPUID.(07H,0).ECX[23] = 1))

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
0	Backup/Restore Valid Cleared when a write to <code>IA32_TME_CLEAR_SAVED_KEY</code> is initiated, and then set when the latest write of <code>IA32_TME_CLEAR_SAVED_KEY</code> has been written to storage that persists across S3/S4 sleep state. If S3/S4 is entered between when an <code>IA32_TME_CLEAR_SAVED_KEY</code> write occurs and when this bit is set, then <code>IA32_TME_CLEAR_SAVED_KEY</code> may not be recovered after S3/S4 exit. During S3/S4 sleep state exit (system wake up), this bit is cleared. It is set again when <code>IA32_TME_CLEAR_SAVED_KEY</code> is restored from persistent storage and thus available to be copied to <code>IA32_TME_CLEAR_SAVED_KEY</code> using <code>IA32_COPY_PLATFORM_TO_LOCAL</code> MSR. Another write to <code>IA32_TME_CLEAR_SAVED_KEY</code> (via <code>IA32_COPY_LOCAL_TO_PLATFORM</code> MSR) may fail if a previous write has not yet set this bit.	IF ((CPUID.19H:EBX[4] = 1) && (CPUID.(07H,0):ECX[23] = 1))
1	Reserved.	
2	Backup Key Storage Read/Write Error Updated prior to backup/restore valid being set. Set when an error is encountered while backing up or restoring a key to persistent storage.	IF ((CPUID.19H:EBX[4] = 1) && (CPUID.(07H,0):ECX[23] = 1))
3	<code>IA32_TME_CLEAR_SAVED_KEY</code> Consumed Set after the previous backup operation has been consumed by the platform. This does not indicate that the system is ready for a second <code>IA32_TME_CLEAR_SAVED_KEY</code> write as the previous <code>IA32_TME_CLEAR_SAVED_KEY</code> write may still need to set Backup/restore valid.	IF ((CPUID.19H:EBX[4] = 1) && (CPUID.(07H,0):ECX[23] = 1))
63:4	Reserved.	
Register Address: 9FBH, 2555		<code>IA32_TME_CLEAR_SAVED_KEY</code>
<code>IA32_TME_CLEAR_SAVED_KEY</code> (W/O)		
0	<code>TME_CLEAR_SAVED_KEY</code> Clear saved TME keys.	
63:1	Reserved.	
Register Address: C80H, 3200		<code>IA32_DEBUG_INTERFACE</code>
Silicon Debug Feature Control (R/W)		If CPUID.01H:ECX.[11] = 1
0	Enable (R/W) BIOS set 1 to enable Silicon debug features. Default is 0.	If CPUID.01H:ECX.[11] = 1
29:1	Reserved.	
30	Lock (R/W): If 1, locks any further change to the MSR. The lock bit is set automatically on the first SMI assertion even if not explicitly set by BIOS. Default is 0.	If CPUID.01H:ECX.[11] = 1
31	Debug Occurred (R/O): This “sticky bit” is set by hardware to indicate the status of bit 0. Default is 0.	If CPUID.01H:ECX.[11] = 1
63:32	Reserved.	
Register Address: C81H, 3201		<code>IA32_L3_QOS_CFG</code>
<code>L3 QOS Configuration</code> (R/W)		If (CPUID.(EAX=10H, ECX=1):ECX.[2] = 1)
0	Enable (R/W) Set 1 to enable L3 CAT masks and CLOS to operate in Code and Data Prioritization (CDP) mode.	
63:1	Reserved. Attempts to write to reserved bits result in a #GP(0).	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)	
Bit Fields	MSR/Bit Description		Comment
Register Address: C82H, 3202		IA32_L2_QOS_CFG	
L2 QOS Configuration (R/W)			If (CPUID.(EAX=10H, ECX=2):ECX.[2] = 1)
0	Enable (R/W) Set 1 to enable L2 CAT masks and CLOS to operate in Code and Data Prioritization (CDP) mode.		
63:1	Reserved. Attempts to write to reserved bits result in a #GP(0).		
Register Address: C83H, 3203		IA32_L3_IO_QOS_CFG	
L3 I/O QOS Configuration (R/W) This MSR is used to enable the I/O RDT features.			If (CPUID.(EAX=0FH, ECX=1):EAX.[10:9] = 1)
0	L3 I/O RDT Allocation Enable.		
1	L3 I/O RDT Monitoring Enable.		
63:2	Reserved.		
Register Address: C88H, 3208		IA32_RESOURCE_PRIORITY	
Thread scope Resource Priority Enable (R/W)			
0	ENABLE When set, enables model specific features that can be used to create a Resource Priority mode.		
63:1	Reserved.		
Register Address: C89H, 3209		IA32_RESOURCE_PRIORITY_PKG	
IA32_RESOURCE_PRIORITY_PKG (R/W)			
0	ENABLE Enable Resource Priority feature.		
63:1	Reserved.		
Register Address: C8DH, 3213		IA32_QM_EVTSEL	
Monitoring Event Select Register (R/W)			If (CPUID.(EAX=07H, ECX=0):EBX.[12] = 1)
7:0	Event ID: ID of a supported monitoring event to report via IA32_QM_CTR.		
31:8	Reserved.		
N+31:32	Resource Monitoring ID: ID for monitoring hardware to report monitored data via IA32_QM_CTR.		N = Ceil (Log ₂ (CPUID.(EAX= 0FH, ECX=0H).EBX[31:0] +1))
63:N+32	Reserved.		
Register Address: C8EH, 3214		IA32_QM_CTR	
Monitoring Counter Register (R/O)			If (CPUID.(EAX=07H, ECX=0):EBX.[12] = 1)
61:0	Resource Monitored Data.		
62	Unavailable: If 1, indicates data for this RMID is not available or not monitored for this resource or RMID.		
63	Error: If 1, indicates an unsupported RMID or event type was written to IA32_PQR_QM_EVTSEL.		

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: C8FH, 3215		IA32_PQR_ASSOC
Resource Association Register (R/W)		If ((CPUID.(EAX=07H, ECX=0):EBX[12] = 1) or (CPUID.(EAX=07H, ECX=0):EBX[15] = 1))
N-1:0	Resource Monitoring ID (R/W): ID for monitoring hardware to track internal operation, e.g., memory access.	N = Ceil (Log ₂ (CPUID.(EAX= 0FH, ECX=0H):EBX[31:0] +1))
31:N	Reserved.	
63:32	CLOS (R/W): The class of service (CLOS) to enforce (on writes); returns the current CLOS when read.	If (CPUID.(EAX=07H, ECX=0):EBX.[15] = 1)
Register Address: C90H–D8FH, 3216–3471		Reserved MSR Address Space for CAT Mask Registers
See Section 19.19.4.1, “Enumeration and Detection Support of Cache Allocation Technology.”		
Register Address: C90H, 3216		IA32_L3_MASK_0
L3 CAT Mask for COS0 (R/W)		If (CPUID.(EAX=10H, ECX=0H):EBX[1] != 0)
31:0	Capacity Bit Mask (R/W)	
63:32	Reserved.	
Register Address: C90H+n, 3216+n		IA32_L3_MASK_n
L3 CAT Mask for COSn (R/W)		n = CPUID.(EAX=10H, ECX=1H):EDX[15:0]
31:0	Capacity Bit Mask (R/W)	
63:32	Reserved.	
Register Address: D10H–D4FH, 3344–3407		Reserved MSR Address Space for L2 CAT Mask Registers
See Section 19.19.4.1, “Enumeration and Detection Support of Cache Allocation Technology.”		
Register Address: D10H, 3344		IA32_L2_MASK_0
L2 CAT Mask for COS0 (R/W)		If (CPUID.(EAX=10H, ECX=0H):EBX[2] != 0)
31:0	Capacity Bit Mask (R/W)	
63:32	Reserved.	
Register Address: D10H+n, 3344+n		IA32_L2_MASK_n
L2 CAT Mask for COSn (R/W)		n = CPUID.(EAX=10H, ECX=2H):EDX[15:0]
31:0	Capacity Bit Mask (R/W)	
63:32	Reserved.	
Register Address: D18H, 3352		IA32_L2_MASK_8
L2 CAT Mask for COS8 (R/W)		
15:0	WAY_MASK Capacity Bit Mask. Available ways vectors for class of service of IA core. '1' in bit indicates allocation to the way is allowed. '0' indicates allocation to the way is not allowed.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
63:16	Reserved.	
Register Address: D19H, 3353		IA32_L2_MASK_9
L2 CAT Mask for COS9 (R/W) See IA32_L2_MASK_8 (D18H) for reference; similar format.		
Register Address: D1AH, 3354		IA32_L2_MASK_10
L2 CAT Mask for COS10 (R/W) See IA32_L2_MASK_8 (D18H) for reference; similar format.		
Register Address: D1BH, 3355		IA32_L2_MASK_11
L2 CAT Mask for COS11 (R/W) See IA32_L2_MASK_8 (D18H) for reference; similar format.		
Register Address: D1CH, 3356		IA32_L2_MASK_12
L2 CAT Mask for COS12 (R/W) See IA32_L2_MASK_8 (D18H) for reference; similar format.		
Register Address: D1DH, 3357		IA32_L2_MASK_13
L2 CAT Mask for COS13 (R/W) See IA32_L2_MASK_8 (D18H) for reference; similar format.		
Register Address: D1EH, 3358		IA32_L2_MASK_14
L2 CAT Mask for COS14 (R/W) See IA32_L2_MASK_8 (D18H) for reference; similar format.		
Register Address: D1FH, 3359		IA32_L2_MASK_15
L2 CAT Mask for COS15 (R/W) See IA32_L2_MASK_8 (D18H) for reference; similar format.		
Register Address: D50H, 3408		IA32_L2_QOS_EXT_BW_THRTL_0
IA32_L2_QOS_EXT_BW_THRTL_0 (R/W) Memory Bandwidth enforcement for COS0.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 0
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D51H, 3409		IA32_L2_QOS_EXT_BW_THRTL_1
IA32_L2_QOS_EXT_BW_THRTL_1 (R/W) Memory Bandwidth enforcement for COS1.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 1
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D52H, 3410		IA32_L2_QOS_EXT_BW_THRTL_2

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA32_L2_QOS_EXT_BW_THRTL_2 (R/W) Memory Bandwidth enforcement for COS2.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 2
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D53H, 3411		IA32_L2_QOS_EXT_BW_THRTL_3
IA32_L2_QOS_EXT_BW_THRTL_3 (R/W) Memory Bandwidth enforcement for COS3.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 3
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D54H, 3412		IA32_L2_QOS_EXT_BW_THRTL_4
IA32_L2_QOS_EXT_BW_THRTL_4 (R/W) Memory Bandwidth enforcement for COS4.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 4
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D55H, 3413		IA32_L2_QOS_EXT_BW_THRTL_5
IA32_L2_QOS_EXT_BW_THRTL_5 (R/W) Memory Bandwidth enforcement for COS5.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 5
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D56H, 3414		IA32_L2_QOS_EXT_BW_THRTL_6
IA32_L2_QOS_EXT_BW_THRTL_6 (R/W) Memory Bandwidth enforcement for COS6.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 6
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D57H, 3415		IA32_L2_QOS_EXT_BW_THRTL_7
IA32_L2_QOS_EXT_BW_THRTL_7 (R/W) Memory Bandwidth enforcement for COS7.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 7

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D58H, 3416		IA32_L2_QOS_EXT_BW_THRTL_8
IA32_L2_QOS_EXT_BW_THRTL_8 (R/W) Memory Bandwidth enforcement for COS8.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 8
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D59H, 3417		IA32_L2_QOS_EXT_BW_THRTL_9
IA32_L2_QOS_EXT_BW_THRTL_9 (R/W) Memory Bandwidth enforcement for COS9.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 9
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D5AH, 3418		IA32_L2_QOS_EXT_BW_THRTL_10
IA32_L2_QOS_EXT_BW_THRTL_10 (R/W) Memory Bandwidth enforcement for COS10.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 10
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D5BH, 3419		IA32_L2_QOS_EXT_BW_THRTL_11
IA32_L2_QOS_EXT_BW_THRTL_11 (R/W) Memory Bandwidth enforcement for COS11.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 11
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D5CH, 3420		IA32_L2_QOS_EXT_BW_THRTL_12
IA32_L2_QOS_EXT_BW_THRTL_12 (R/W) Memory Bandwidth enforcement for COS12.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 12
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: D5DH, 3421		IA32_L2_QOS_EXT_BW_THRTL_13
IA32_L2_QOS_EXT_BW_THRTL_13 (R/W) Memory Bandwidth enforcement for COS13.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 13
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D5EH, 3422		IA32_L2_QOS_EXT_BW_THRTL_14
IA32_L2_QOS_EXT_BW_THRTL_14 (R/W) Memory Bandwidth enforcement for COS14.		CPUID.(EAX=10H,ECX=0H):EBX[3] and CPUID.(EAX=10H,ECX=3H):EDX ≥ 14
6:0	RBE_ENFORCEMENT_VAL Max Delay value cannot be greater than 90 percent - 0x5a.	
63:7	Reserved.	
Register Address: D90H, 3472		IA32_BNDCFGS
Supervisor State of MPX Configuration (R/W)		If (CPUID.(EAX=07H, ECX=0H):EBX[14] = 1)
0	EN: Enable Intel MPX in supervisor mode.	
1	BNDPRESERVE: Preserve the bounds registers for near branch instructions in the absence of the BND prefix.	
11:2	Reserved, must be zero.	
63:12	Base Address of Bound Directory.	
Register Address: D91H, 3473		IA32_COPY_LOCAL_TO_PLATFORM ⁵
Copy Local State to Platform State (W)		IF ((CPUID.19H:EBX[4] = 1) && (CPUID.(EAX=07H, ECX=0H):ECX[23] = 1))
0	lwKeyBackup Copy lwKey to lwKeyBackup.	IF ((CPUID.19H:EBX[4] = 1) && (CPUID.(EAX=07H, ECX=0H):ECX[23] = 1))
63:1	Reserved.	
Register Address: D92H, 3474		IA32_COPY_PLATFORM_TO_LOCAL ⁵
Copy Platform State to Local State (W)		IF ((CPUID.19H:EBX[4] = 1) && (CPUID.(EAX=07H, ECX=0H):ECX[23] = 1))
0	lwKeyBackup Copy lwKeyBackup to lwKey.	IF ((CPUID.19H:EBX[4] = 1) && (CPUID.(EAX=07H, ECX=0H):ECX[23] = 1))
63:1	Reserved.	
Register Address: D93H, 3475		IA32_PASID
Process Address Space Identifier. (R/W)		

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
19:0	Process address space identifier (PASID). Specifies the PASID of the currently running software thread.	
30:20	Reserved.	
31	Valid. Execution of ENQCMD causes a #GP if this bit is clear.	
63:32	Reserved.	
Register Address: DA0H, 3488		IA32_XSS
Extended Supervisor State Mask (R/W)		If CPUID.(0DH, 1):EAX.[3] = 1
7:0	Reserved.	
8	PT State (R/W)	
9	Reserved.	
10	PASID State (R/W)	
11	CET_U State (R/W)	
12	CET_S State (R/W)	
13	HDC State (R/W)	
14	UINTR State (R/W)	
15	LBR State (R/W)	
16	HWP State (R/W)	
63:17	Reserved.	
Register Address: DB0H, 3504		IA32_PKG_HDC_CTL
Package Level Enable/Disable HDC (R/W)		If CPUID.06H:EAX.[13] = 1
0	HDC_Pkg_Enable (R/W) Force HDC idling or wake up HDC-idled logical processors in the package. See Section 16.5.2, "Package level Enabling HDC."	If CPUID.06H:EAX.[13] = 1
63:1	Reserved.	
Register Address: DB1H, 3505		IA32_PM_CTL1
Enable/Disable the HDC Thread Level Activity (R/W)		If CPUID.06H:EAX.[13] = 1
0	SDC_ALLOWED (R/W) Set this bit to allow this thread to be forced into HDC idle state. Clearing this bit blocks HDC-enter (HW) request. Default value: 1. See Section 16.5.3.	If CPUID.06H:EAX.[13] = 1
63:1	Reserved.	
Register Address: DB2H, 3506		IA32_THREAD_STALL
Per-Logical_Processor_ID HDC Idle Residency (R/O)		If CPUID.06H:EAX.[13] = 1
63:0	Stall_Cycle_Cnt (R/W) Stalled cycles due to HDC forced idle on this logical processor. See Section 16.5.4.1.	If CPUID.06H:EAX.[13] = 1
Register Address: E00H, 3584		IA32_QOS_CORE_BW_THRTL_0
CBA Levels Based on COS for Bandwidth Throttling (R/W)		CPUID.10H.0H:EBX[5]=1
3:0	COS0_LEVEL CBA Level for COS[0]. Levels are programmed from 0 to 15.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
7:4	Reserved.	
11:8	COS1_LEVEL CBA Level for COS[1]. Levels are programmed from 0 to 15.	
15:12	Reserved.	
19:16	COS2_LEVEL CBA Level for COS[2]. Levels are programmed from 0 to 15.	
25:20	Reserved.	
27:24	COS3_LEVEL CBA Level for COS[3]. Levels are programmed from 0 to 15.	
31:28	Reserved.	
35:32	COS4_LEVEL CBA Level for COS[4]. Levels are programmed from 0 to 15.	
39:36	Reserved.	
43:40	COS5_LEVEL CBA Level for COS[5]. Levels are programmed from 0 to 15.	
47:44	Reserved.	
51:48	COS6_LEVEL CBA Level for COS[6]. Levels are programmed from 0 to 15.	
Register Address: E01H, 3585		IA32_QOS_CORE_BW_THRTL_1
CBA Levels Based on COS for Bandwidth Throttling (R/W)		CPUID.10H.0H:EBX[5]=1
3:0	COS8_LEVEL CBA Level for COS[8]. Levels are programmed from 0 to 15.	
7:4	Reserved.	
11:8	COS9_LEVEL CBA Level for COS[9]. Levels are programmed from 0 to 15.	
15:12	Reserved.	
19:16	COS10_LEVEL CBA Level for COS[10]. Levels are programmed from 0 to 15.	
25:20	Reserved.	
27:24	COS11_LEVEL CBA Level for COS[11]. Levels are programmed from 0 to 15.	
31:28	Reserved.	
35:32	COS12_LEVEL CBA Level for COS[12]. Levels are programmed from 0 to 15.	
39:36	Reserved.	
43:40	COS13_LEVEL CBA Level for COS[13]. Levels are programmed from 0 to 15.	
47:44	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
51:48	COS14_LEVEL CBA Level for COS[14]. Levels are programmed from 0 to 15.	
55:50	Reserved.	
59:56	COS15_LEVEL CBA Level for COS[15]. Levels are programmed from 0 to 15.	
63:60	Reserved	
Register Address: 1200H–121FH, 4608–4639		IA32_LBR_x_INFO
Last Branch Record Entry X Info Register (R/W) An attempt to read or write IA32_LBR_x_INFO such that $x \geq \text{IA32_LBR_DEPTH.DEPTH}$ will #GP.		
15:0	CYC_CNT The elapsed CPU cycles (saturating) since the last LBR was recorded. See Section 18.1.3.3.	Reset Value: 0
55:16	Undefined, may be zero or non-zero. Writes of non-zero values do not fault, but reads may return a different value.	Reset Value: 0
59:56	BR_TYPE The branch type recorded by this LBR. Encodings: 0000B: COND 0001B: JMP Indirect 0010B: JMP Direct 0011B: CALL Indirect 0100B: CALL Direct 0101B: RET 011xB: Reserved 1xxxB: Other Branch	Reset Value: 0
60	CYC_CNT_VALID CYC_CNT value is valid. See Section 20.1.3.3.	Reset Value: 0
61	TSX_ABORT This LBR record is a TSX abort. On processors that do not support Intel TSX (CPUID.07H.EBX.HLE[bit 4]=0 and CPUID.07H.EBX.RTM[bit 11]=0), this bit is undefined.	Reset Value: 0
62	IN_TSX This LBR record records a branch that retired during a TSX transaction. On processors that do not support Intel TSX (CPUID.07H.EBX.HLE[bit 4]=0 and CPUID.07H.EBX.RTM[bit 11]=0), this bit is undefined.	Reset Value: 0
63	MISPRED The recorded branch direction (conditional branch) or target (indirect branch) was mispredicted.	Reset Value: 0
Register Address: 1400H, 5120		IA32_SEAMRR_BASE
SEAM Memory Range Register for TDX - Base Address (R/W)		
2:0	Reserved.	
3	CONFIGURED Set to 1 by BIOS if range is configured.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)	
Bit Fields	MSR/Bit Description		Comment
24:4	Reserved.		
51:25	BASE SEAM Range Register BASE address.		
63:52	Reserved.		
Register Address: 1401H, 5121		IA32_SEAMRR_MASK	
SEAM Memory Range Register for TDX (R/W)			
9:0	Reserved.		
10	LOCK Set by BIOS to indicate range is configured and locked.		
24:11	Reserved.		
51:25	MASK Mask value for SEAMRR matching. Lowest granularity is 32M.		
63:52	Reserved.		
Register Address: 1406H, 5126		IA32_MCU_CONTROL	
MCU Control (R/W) Controls the behavior of the Microcode Update Trigger MSR, IA32_BIOS_UPDT_TRIG.			If CPUID.07H.0H:EDX[29]=1 && IA32_ARCH_CAPABILITIES.MCU_CONTROL=1
0	LOCK Once set, further writes to this MSR will cause a #GP(0) fault. Bypassed during SMM if EN_SMM_BYPASS (bit 2) is set.		
1	DIS_MCU_LOAD If this bit is set on a given logical processor, then any subsequent attempts to load a microcode update by that logical processor will be silently dropped (WRMSR 0x79 has no effect).		
2	EN_SMM_BYPASS If set, then writes to IA32_MCU_CONTROL are allowed during SMM regardless of the LOCK bit. This enables BIOS to Opt-In to the SMM Bypass functionality.		
63:3	Reserved.		
Register Address: 14CEH, 5326		IA32_LBR_CTL	
Last Branch Record Enabling and Configuration Register (R/W)			
0	LBREn When set, enables LBR recording.		Reset Value: 0
1	OS When set, allows LBR recording when CPL == 0.		Reset Value: 0
2	USR When set, allows LBR recording when CPL != 0.		Reset Value: 0
3	CALL_STACK When set, records branches in call-stack mode. See Section 20.1.2.4.		Reset Value: 0
15:4	Reserved.		Reset Value: 0

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
16	COND When set, records taken conditional branches. See Section 20.1.2.3.	
17	NEAR_REL_JMP When set, records near relative JMPs. See Section 20.1.2.3.	
18	NEAR_IND_JMP When set, records near indirect JMPs. See Section 20.1.2.3.	
19	NEAR_REL_CALL When set, records near relative CALLs. See Section 20.1.2.3.	
20	NEAR_IND_CALL When set, records near indirect CALLs. See Section 20.1.2.3.	
21	NEAR_RET When set, records near RETs. See Section 20.1.2.3.	
22	OTHER_BRANCH When set, records other branches. See Section 20.1.2.3.	
63:23	Reserved.	
Register Address: 14CFH, 5327		IA32_LBR_DEPTH
Last Branch Record Maximum Stack Depth Register (R/W)		
N:0	DEPTH The number of LBRs to be used for recording. Supported values are indicated by the bitmap in CPUID.(EAX=01CH,ECX=0):EAX[7:0]. The reset value will match the maximum supported by the CPU. Writes of unsupported values will #GP fault.	Reset Value: Varies
63:N+1	Reserved.	Reset Value: 0
Register Address: 1500H–151FH, 5376–5407		IA32_LBR_x_FROM_IP
Last Branch Record entry X source IP register (R/W). An attempt to read or write IA32_LBR_x_FROM_IP such that $x \geq \text{IA32_LBR_DEPTH.DEPTH}$ will #GP.		
63:0	FROM_IP The source IP of the recorded branch or event, in canonical form. Writes to bits above MAXLINADDR-1 are ignored.	Reset Value: 0
Register Address: 1600H–161FH, 5632–5663		IA32_LBR_x_TO_IP
Last Branch Record Entry X Destination IP Register (R/W) An attempt to read or write IA32_LBR_x_TO_IP such that $x \geq \text{IA32_LBR_DEPTH.DEPTH}$ will #GP.		
63:0	TO_IP The destination IP of the recorded branch or event, in canonical form. Writes to bits above MAXLINADDR-1 are ignored.	Reset Value: 0
Register Address: 17D0H, 6096		IA32_HW_FEEDBACK_PTR
Hardware Feedback Interface Pointer		If CPUID.06H:EAX.[19] = 1
0	Valid (R/W) When set to 1, indicates a valid pointer is programmed into the ADDR field of the MSR.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
11:1	Reserved.	
(MAXPHYADDR-1):12	ADDR (R/W) Physical address of the page frame of the first page of the hardware feedback interface structure.	
63:MAXPHYADDR	Reserved.	
Register Address: 17D1H, 6097		IA32_HW_FEEDBACK_CONFIG
Hardware Feedback Interface Configuration		If CPUID.06H:EAX.[19] = 1
0	Enable (R/W) When set to 1, enables the hardware feedback interface.	
63:1	Reserved.	
Register Address: 17D2H, 6098		IA32_THREAD_FEEDBACK_CHAR
Thread Feedback Characteristics (R/O)		If CPUID.06H:EAX.[23] = 1
7:0	Application Class ID, pointing into the Intel Thread Director structure.	
62:8	Reserved.	
63	Valid bit. When set to 1 the OS Scheduler can use the Class ID (in bits 7:0) for its scheduling decisions. If this bit is 0, the Class ID field should be ignored. It is recommended that the OS uses the last known Class ID of the software thread for its scheduling decisions.	
Register Address: 17D4H, 6100		IA32_HW_FEEDBACK_THREAD_CONFIG
Hardware Feedback Thread Configuration (R/W)		
0	Enables Intel Thread Director. When set to 1, logical processor scope Intel Thread Director is enabled. Default is 0 (disabled).	
63:1	Reserved.	
Register Address: 17DAH, 6106		IA32_HRESET_ENABLE
History Reset Enable (R/W)		
0	Enable reset of the Intel Thread Director history.	
31:1	Reserved for other capabilities that can be reset by the HRESET instruction.	
63:32	Reserved.	
Register Address: 1900H, 6400		IA32_PMC_GPO_CTR
Full Width Writable General Performance Counter 0 (R/W)		If CPUID.0AH:EAX[15:8] > 0 and IA32_PERF_CAPABILITIES[13] = 1
47:0	RELOAD_VALUE Contains the reload value to be loaded into the associated counter by Auto Counter Reload. Will be 1-extended to 48 bits.	
63:48	Reserved.	
Register Address: 1901H, 6401		IA32_PMC_GPO_CFG_A
IA32_PMC_GPO_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 0.		If CPUID.0AH:EAX[15:8] > 0

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
7:0	EVENT_SELECT Selects a performance event logic unit.	
15:8	UMASK Qualifies the microarchitectural condition to detect on the selected event logic.	
16	USR When set, events are counted only when the processor is operating at privilege levels 1, 2 or 3. This flag can be used in conjunction with the OS flag.	
17	OS When set, events are counted only when the processor is operating at privilege level 0. This flag can be used in conjunction with the USER flag.	
18	EDGE When set, enables edge detection of events.	
19	Reserved.	
20	INT When set, the processor generates an exception through its local APIC on counter overflow for this counter's thread.	
21	ANYTHREAD If CPUID.AOH.EDX[15] is 1, then this bit is deprecated. When set to 1, it enables counting the associated event conditions occurring across all logical processors sharing a processor core. When set to 0, the counter only increments the associated event conditions occurring in the logical processor which programmed the MSR.	
22	ENABLE When set, performance counting is enabled in the performance-monitoring counter; when clear, the counter is disabled.	
23	INVERT Inverts the result of the counter-mask (CMASK) comparison when set, so that both greater than equal to and less than comparisons can be made. 0: The comparison is: threshold is greater than or equal to the event 1: The comparison is inverted: threshold is less than event.	
31:24	CMASK When CMASK is not zero, the corresponding performance counter increments by 1 each cycle if the event count is $\geq$ CMASK. This mask enables counting cycles in which multiple occurrences happen (for example, two or more instructions retired per clock).	
34:32	Reserved.	
35	EN_LBR_LOG When set enables updating LBRs with that counters event occurrences, if selected event is precise.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
36	EQUAL When EQ flag is set and the INV flag is clear, the comparison evaluates to true if the selected performance monitoring event (the event) is equal to the specified Counter Mask value (CMask). When EQ flag is set and INV flag is set, the comparison evaluates to true if the event is less-than the CMask value and the event is not zero. Note if CMask is zero, the EQ flag is ignored.	
39:37	Reserved.	
47:40	UMASK2 Unit mask 2 (UMASK2) field (bits 40 through 47) - These bits qualify the condition that the selected event logic unit detects. Valid UMASK2 values for each event logic unit are specific to the unit. The new UMASK2 field may also be used in conjunction with UMASK.	
63:48	Reserved.	
Register Address: 1903H, 6403		IA32_PMC_GPO_CFG_C
IA32_PMC_GPO_CFG_C (R/W) Extended Perf event selector for GP counter 0.		
31:0	RELOAD_VALUE Contains the reload value to be loaded into the associated counter by Auto Counter Reload. Will be 1-extended to 48 bits.	
63:32	Reserved.	
Register Address: 1904H, 6404		IA32_PMC_GP1_CTR
Full Width Writable General Performance Counter 1 (R/W) See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 1 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 1905H, 6405		IA32_PMC_GP1_CFG_A
IA32_PMC_GP1_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 1. See IA32_PMC_GPO_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 1
Register Address: 1907H, 6407		IA32_PMC_GP1_CFG_C
IA32_PMC_GP1_CFG_C (R/W) Extended Perf event selector for GP counter 1. See IA32_PMC_GPO_CFG_C (1903H) for reference; similar format.		
Register Address: 1908H, 6408		IA32_PMC_GP2_CTR
Full Width Writable General Performance Counter 2 (R/W) See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 2 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 1909H, 6409		IA32_PMC_GP2_CFG_A
IA32_PMC_GP2_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 2. See IA32_PMC_GPO_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 2
Register Address: 190AH, 6410		IA32_PMC_GP2_CFG_B

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA32_PMC_GP2_CFG_B (R/W)		
GP counter reload configuration register.		
1:0	Reserved.	
2	RELOAD_PMC2 Reload GP2 when GP2 overflows.	
3	RELOAD_PMC3 Reload GP2 when GP3 overflows.	
4	RELOAD_PMC4 Reload GP2 when GP4 overflows.	
5	RELOAD_PMC5 Reload GP2 when GP5 overflows.	
6	RELOAD_PMC6 Reload GP2 when GP6 overflows.	
7	RELOAD_PMC7 Reload GP2 when GP7 overflows.	
31:8	Reserved.	
32	RELOAD_FC0 Reload GP2 when FC0 overflows.	
33	RELOAD_FC1 Reload GP2 when FC1 overflows.	
47:34	Reserved.	
48	METRICS_CLEAR Clear PERF_METRICS on overflow of GP2.	
63:49	Reserved.	
Register Address: 190BH, 6411		IA32_PMC_GP2_CFG_C
IA32_PMC_GP2_CFG_C (R/W)		
Extended Perf event selector for GP counter 2.		
See IA32_PMC_GPO_CFG_C (1903H) for reference; similar format.		
Register Address: 190CH, 6412		IA32_PMC_GP3_CTR
Full Width Writable General Performance Counter 3 (R/W)		
See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 3 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 190DH, 6413		IA32_PMC_GP3_CFG_A
IA32_PMC_GP3_CFG_A (R/W)		
Performance Event Select Register used to control the operation of the General Performance Counter 3. See IA32_PMC_GPO_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 3
Register Address: 190EH, 6414		IA32_PMC_GP3_CFG_B
IA32_PMC_GP3_CFG_B (R/W)		
GP counter reload configuration register.		
See IA32_PMC_GP2_CFG_B (190AH) for reference; similar format.		

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 190FH, 6415		IA32_PMC_GP3_CFG_C
IA32_PMC_GP3_CFG_C (R/W) Extended Perf event selector for GP counter 3. See IA32_PMC_GPO_CFG_C (1903H) for reference; similar format.		
Register Address: 1910H, 6416		IA32_PMC_GP4_CTR
Full Width Writable General Performance Counter 4 (R/W) See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 4 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 1911H, 6417		IA32_PMC_GP4_CFG_A
IA32_PMC_GP4_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 4. See IA32_PMC_GPO_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 4
Register Address: 1912H, 6418		IA32_PMC_GP4_CFG_B
IA32_PMC_GP4_CFG_B (R/W) GP counter reload configuration register. See IA32_PMC_GP2_CFG_B (190AH) for reference; similar format.		
Register Address: 1913H, 6419		IA32_PMC_GP4_CFG_C
IA32_PMC_GP4_CFG_C (R/W) Extended Perf event selector for GP counter 4. See IA32_PMC_GPO_CFG_C (1903H) for reference; similar format.		
Register Address: 1914H, 6420		IA32_PMC_GP5_CTR
Full Width Writable General Performance Counter 5 (R/W) See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 5 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 1915H, 6421		IA32_PMC_GP5_CFG_A
IA32_PMC_GP5_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 5. See IA32_PMC_GPO_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 5
Register Address: 1916H, 6422		IA32_PMC_GP5_CFG_B
IA32_PMC_GP5_CFG_B (R/W) GP counter reload configuration register. See IA32_PMC_GP2_CFG_B (190AH) for reference; similar format.		
Register Address: 1917H, 6423		IA32_PMC_GP5_CFG_C
IA32_PMC_GP5_CFG_C (R/W) Extended Perf event selector for GP counter 5. See IA32_PMC_GPO_CFG_C (1903H) for reference; similar format.		
Register Address: 1918H, 6424		IA32_PMC_GP6_CTR
Full Width Writable General Performance Counter 6 (R/W) See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 6 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 1919H, 6425		IA32_PMC_GP6_CFG_A

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA32_PMC_GP6_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 6. See IA32_PMC_GPO_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 6
Register Address: 191AH, 6426		IA32_PMC_GP6_CFG_B
IA32_PMC_GP6_CFG_B (R/W) GP counter reload configuration register. See IA32_PMC_GP2_CFG_B (190AH) for reference; similar format.		
Register Address: 191BH, 6427		IA32_PMC_GP6_CFG_C
IA32_PMC_GP6_CFG_C (R/W) Extended Perf event selector for GP counter 6. See IA32_PMC_GPO_CFG_C (1903H) for reference; similar format.		
Register Address: 191CH, 6428		IA32_PMC_GP7_CTR
Full Width Writable General Performance Counter 7 (R/W) See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 7 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 191DH, 6429		IA32_PMC_GP7_CFG_A
IA32_PMC_GP7_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 7. See IA32_PMC_GPO_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 7
Register Address: 191EH, 6430		IA32_PMC_GP7_CFG_B
IA32_PMC_GP7_CFG_B (R/W) GP counter reload configuration register. See IA32_PMC_GP2_CFG_B (190AH) for reference; similar format.		
Register Address: 191FH, 6431		IA32_PMC_GP7_CFG_C
IA32_PMC_GP7_CFG_C (R/W) Extended Perf event selector for GP counter 7. See IA32_PMC_GPO_CFG_C (1903H) for reference; similar format.		
Register Address: 1920H, 6432		IA32_PMC_GP8_CTR
Full Width Writable General Performance Counter 8 (R/W) See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 8 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 1921H, 6433		IA32_PMC_GP8_CFG_A
IA32_PMC_GP8_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 8. See IA32_PMC_GPO_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 8
Register Address: 1924H, 6436		IA32_PMC_GP9_CTR
Full Width Writable General Performance Counter 9 (R/W) See IA32_PMC_GPO_CTR (1900H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 9 and IA32_PERF_CAPABILITIES[13]=1
Register Address: 1925H, 6437		IA32_PMC_GP9_CFG_A

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA32_PMC_GP9_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 9. See IA32_PMC_GP0_CFG_A (1901H) for reference; similar format.		If CPUID.0AH:EAX[15:8] > 9
Register Address: 1980H, 6528		IA32_PMC_FX0_CTR
Fixed-Function Performance Counter 0 (R/W) Instructions retired.		If CPUID.0AH:EDX[4:0] > 0
47:0	FIXED_COUNTER Instructions Retired Counter.	
63:46	Reserved.	
Register Address: 1982H, 6530		IA32_PMC_FX0_CFG_B
Fixed-Function Counter Reload Configuration Register (R/W)		
1:0	Reserved.	
2	RELOAD_PMC2 Reload Fixed-Function Counter0 when GP2 overflows.	
3	RELOAD_PMC3 Reload Fixed-Function Counter0 when GP3 overflows.	
4	RELOAD_PMC4 Reload Fixed-Function Counter0 when GP4 overflows.	
5	RELOAD_PMC5 Reload Fixed-Function Counter0 when GP5 overflows.	
6	RELOAD_PMC6 Reload Fixed-Function Counter0 when GP6 overflows.	
7	RELOAD_PMC7 Reload Fixed-Function Counter0 when GP7 overflows.	
33:8	Reserved.	
32	RELOAD_FC0 Reload Fixed-Function Counter0 when FC0 overflows.	
33	RELOAD_FC1 Reload Fixed-Function Counter0 when FC1 overflows.	
47:34	Reserved.	
48	METRICS_CLEAR Clear PERF_METRICS on overflow of Fixed-Function Counter 0.	
63:49	Reserved.	
Register Address: 1983H, 6531		IA32_PMC_FX0_CFG_C
Extended Perf Event Selector for Fixed-Function Counter 0 (R/W)		
31:0	RELOAD_VALUE Contains the reload value to be loaded into the associated counter by Auto Counter Reload. Will be 1-extended to 48 bits.	
63:32	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 1984H, 6532		IA32_PMC_FX1_CTR
Fixed-Function Performance Counter 1 (R/W) Unhalted core clock cycles.		If CPUID.0AH:EDX[4:0] > 1
47:0	FIXED_COUNTER Unhalted core clock cycles counter.	
63:46	Reserved.	
Register Address: 1986H, 6534		IA32_PMC_FX1_CFG_B
Fixed-Function Counter Reload Configuration Register (R/W)		
1:0	Reserved.	
2	RELOAD_PMC2 Reload Fixed-Function Counter1 when GP2 overflows.	
3	RELOAD_PMC3 Reload Fixed-Function Counter1 when GP3 overflows.	
4	RELOAD_PMC4 Reload Fixed-Function Counter1 when GP4 overflows.	
5	RELOAD_PMC5 Reload Fixed-Function Counter1 when GP5 overflows.	
6	RELOAD_PMC6 Reload Fixed-Function Counter1 when GP6 overflows.	
7	RELOAD_PMC7 Reload Fixed-Function Counter1 when GP7 overflows.	
31:8	Reserved.	
32	RELOAD_FC0 Reload Fixed-Function Counter1 when FC0 overflows.	
33	RELOAD_FC1 Reload Fixed-Function Counter1 when FC1 overflows.	
47:34	Reserved.	
48	METRICS_CLEAR Clear PERF_METRICS on overflow of Fixed-Function Counter 1.	
63:49	Reserved.	
Register Address: 1987H, 6532		IA32_PMC_FX1_CFG_C
Extended Perf Event Selector for Fixed-Function Counter 1 (R/W)		
31:0	RELOAD_VALUE Contains the reload value to be loaded into the associated counter by Auto Counter Reload. Will be 1-extended to 48 bits.	
63:32	Reserved.	
Register Address: 1988H, 6536		IA32_PMC_FX2_CTR
Fixed-Function Performance Counter 2 (R/W) Unhalted core reference cycles.		If CPUID.0AH:EDX[4:0] > 2

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
47:0	FIXED_COUNTER Unhalted core reference cycles counter.	
63:48	Reserved.	
Register Address: 198BH, 6539		IA32_PMC_FX2_CFG_C
Extended Perf Event Selector for Fixed-Function Counter 2 (R/W)		
31:0	RELOAD_VALUE Contains the reload value to be loaded into the associated counter by Auto Counter Reload. Will be 1-extended to 48 bits.	
63:32	Reserved.	
Register Address: 198CH, 6540		IA32_PMC_FX3_CTR
Fixed-Function Performance Counter 3 (R/W) Top-down Microarchitecture Analysis unhalted number of available slots.		If CPUID.0AH:EDX[4:0] > 3
47:0	FIXED_COUNTER Top-down microarchitecture analysis unhalted number of available slots counter.	
63:48	Reserved.	
Register Address: 1990H, 6544		IA32_PMC_FX4_CTR
Fixed-Function Performance Counter 4 (R/W) Top-down bad speculation.		If CPUID.0AH:EDX[4:0] > 4
47:0	FIXED_COUNTER Top-down bad speculation counter.	
63:48	Reserved.	
Register Address: 1993H, 6547		IA32_PMC_FX4_CFG_C
Extended Perf Event Selector for Fixed-Function Counter 4 (R/W)		
31:0	RELOAD_VALUE Contains the reload value to be loaded into the associated counter by Auto Counter Reload. Will be 1-extended to 48 bits.	
63:32	Reserved.	
Register Address: 1994H, 6548		IA32_PMC_FX5_CTR
Fixed-Function Performance Counter 5 (R/W) Top-down frontend bound.		If CPUID.0AH:EDX[4:0] > 5
47:0	FIXED_COUNTER Top-down frontend-bound counter.	
63:48	Reserved.	
Register Address: 1997H, 6551		IA32_PMC_FX5_CFG_C
Extended Perf Event Selector for Fixed-Function Counter 5 (R/W)		
31:0	RELOAD_VALUE Contains the reload value to be loaded into the associated counter by Auto Counter Reload. Will be 1-extended to 48 bits.	
63:32	Reserved.	

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
Register Address: 1998H, 6552		IA32_PMC_FX6_CTR
Fixed-Function Performance Counter 6 (R/W) Top-down retiring.		If CPUID.0AH:EDX[4:0] > 6
47:0	FIXED_COUNTER Top-down retiring counter.	
63:48	Reserved.	
Register Address: 199BH, 6555		IA32_PMC_FX6_CFG_C
Extended Perf Event Selector for Fixed-Function Counter 6 (R/W)		
31:0	RELOAD_VALUE Contains the reload value to be loaded into the associated counter by Auto Counter Reload. Will be 1-extended to 48 bits.	
63:32	Reserved.	
Register Address: 1B01H, 6913		IA32_UARCH_MISC_CTL
IA32_UARCH_MISC_CTL (R/W)		If IA32_ARCH_CAPABILITIES[12]=1
0	Data Operand Independent Timing Mode (DOITM).	If IA32_ARCH_CAPABILITIES[12]=1
63:1	Reserved.	
Register Address: 4000_0000H–4000_00FFH		Reserved MSR Address Space
All existing and future processors will not implement MSRs in this range.		
Register Address: C000_0080H		IA32_EFER
Extended Feature Enables		If (CPUID.80000001H:EDX.[20] CPUID.80000001H:EDX.[29])
0	SYSCALL Enable: IA32_EFER.SCE (R/W) Enables SYSCALL/SYSRET instructions in 64-bit mode.	
7:1	Reserved.	
8	IA-32e Mode Enable: IA32_EFER.LME (R/W) Enables IA-32e mode operation.	
9	Reserved.	
10	IA-32e Mode Active: IA32_EFER.LMA (R) Indicates IA-32e mode is active when set.	
11	Execute Disable Bit Enable: IA32_EFER.NXE (R/W)	
63:12	Reserved.	
Register Address: C000_0081H		IA32_STAR
System Call Target Address (R/W)		If CPUID.80000001:EDX.[29] = 1
Register Address: C000_0082H		IA32_LSTAR
IA-32e Mode System Call Target Address (R/W) Target RIP for the called procedure when SYSCALL is executed in 64-bit mode.		If CPUID.80000001:EDX.[29] = 1
Register Address: C000_0083H		IA32_CSTAR

Table 2-2. IA-32 Architectural MSRs (Contd.)

Register Address: Hex, Decimal		Architectural MSR Name (Former MSR Name)
Bit Fields	MSR/Bit Description	Comment
IA-32e Mode System Call Target Address (R/W) Not used, as the SYSCALL instruction is not recognized in compatibility mode.		If CPUID.80000001:EDX.[29] = 1
Register Address: C000_0084H		IA32_FMASK
System Call Flag Mask (R/W)		If CPUID.80000001:EDX.[29] = 1
Register Address: C000_0100H		IA32_FS_BASE
Map of BASE Address of FS (R/W)		If CPUID.80000001:EDX.[29] = 1
Register Address: C000_0101H		IA32_GS_BASE
Map of BASE Address of GS (R/W)		If CPUID.80000001:EDX.[29] = 1
Register Address: C000_0102H		IA32_KERNEL_GS_BASE
Swap Target of BASE Address of GS (R/W)		If CPUID.80000001:EDX.[29] = 1
Register Address: C000_0103H		IA32_TSC_AUX
Auxiliary TSC (R/W)		If CPUID.80000001H: EDX[27] = 1 or CPUID.(EAX=7,ECX=0):ECX[bit 22] = 1
31:0	AUX: Auxiliary signature of TSC.	
63:32	Reserved.	

NOTES:

1. Some older processors may have supported this MSR as model-specific and do not enumerate it with CPUID.
2. In processors based on Intel NetBurst® microarchitecture, MSR addresses 180H-197H are supported, software must treat them as model-specific. Starting with Intel Core Duo processors, MSR addresses 180H-185H, 188H-197H are reserved.
3. The *_ADDR MSRs may or may not be present; this depends on flag settings in IA32_MCI_STATUS. See Section 17.3.2.3 and Section 17.3.2.4 for more information.
4. MAXPHYADDR is reported by CPUID.80000008H:EAX[7:0].
5. Further details on Key Locker and usage of this MSR can be found here:

<https://software.intel.com/content/www/us/en/develop/download/intel-key-locker-specification.html>.

2.2 MSRS IN THE INTEL® CORE™ 2 PROCESSOR FAMILY

Table 2-3 lists model-specific registers (MSRs) for the Intel Core 2 processor family and for Intel Xeon processors based on Intel Core microarchitecture, architectural MSR addresses are also included in Table 2-3. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_0FH, see Table 2-1.

MSRs listed in Table 2-2 and Table 2-3 are also supported by processors based on the Enhanced Intel Core microarchitecture. Processors based on the Enhanced Intel Core microarchitecture have a CPUID Signature DisplayFamily_DisplayModel value of 06_17H.

The column “Shared/Unique” applies to multi-core processors based on Intel Core microarchitecture. “Unique” means each processor core has a separate MSR, or a bit field in an MSR governs only a core independently. “Shared” means the MSR or the bit field in an MSR address governs the operation of both processor cores.

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Register Address: 0H, 0	IA32_P5_MC_ADDR	
See Section 2.23, “MSRs in Pentium Processors.”		Unique
Register Address: 1H, 1	IA32_P5_MC_TYPE	
See Section 2.23, “MSRs in Pentium Processors.”		Unique
Register Address: 6H, 6	IA32_MONITOR_FILTER_SIZE	
See Section 10.10.5, “Monitor/Mwait Address Range Determination,” and Table 2-2.		Unique
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER	
See Section 19.17, “Time-Stamp Counter,” and Table 2-2.		Unique
Register Address: 17H, 23	IA32_PLATFORM_ID	
Platform ID (R) See Table 2-2.		Shared
Register Address: 17H, 23	MSR_PLATFORM_ID	
Model Specific Platform ID (R)		Shared
7:0	Reserved.	
12:8	Maximum Qualified Ratio (R) The maximum allowed bus ratio.	
49:13	Reserved.	
52:50	See Table 2-2.	
63:53	Reserved.	
Register Address: 1BH, 27	IA32_APIC_BASE	
See Section 12.4.4, “Local APIC Status and Location,” and Table 2-2.		Unique
Register Address: 2AH, 42	MSR_EBL_CR_POWERON	
Processor Hard Power-On Configuration (R/W) Enables and disables processor features; (R) indicates current processor configuration.		Shared
0	Reserved.	
1	Data Error Checking Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processors implement R/W.	
2	Response Error Checking Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processor implements R/W.	
3	MCERR# Drive Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processors implement R/W.	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
4	Address Parity Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processors implement R/W.	
5	Reserved.	
6	Reserved.	
7	BINIT# Driver Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processors implement R/W.	
8	Output Tri-state Enabled (R/O) 1 = Enabled; 0 = Disabled.	
9	Execute BIST (R/O) 1 = Enabled; 0 = Disabled.	
10	MCERR# Observation Enabled (R/O) 1 = Enabled; 0 = Disabled.	
11	Intel TXT Capable Chipset. (R/O) 1 = Present; 0 = Not Present.	
12	BINIT# Observation Enabled (R/O) 1 = Enabled; 0 = Disabled.	
13	Reserved.	
14	1 MByte Power on Reset Vector (R/O) 1 = 1 MByte; 0 = 4 GBytes.	
15	Reserved.	
17:16	APIC Cluster ID (R/O)	
18	N/2 Non-Integer Bus Ratio (R/O) 0 = Integer ratio; 1 = Non-integer ratio.	
19	Reserved.	
21: 20	Symmetric Arbitration ID (R/O)	
26:22	Integer Bus Frequency Ratio (R/O)	
Register Address: 3AH, 58	MSR_FEATURE_CONTROL	
Control Features in Intel 64 Processor (R/W) See Table 2-2.		Unique
3	SMRR Enable (R/WL) When this bit is set and the lock bit is set, this makes the SMRR_PHYS_BASE and SMRR_PHYS_MASK registers read visible and writeable while in SMM.	Unique
Register Address: 40H, 64	MSR_LASTBRANCH_0_FROM_IP	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Last Branch Record 0 From IP (R/W) One of four pairs of last branch record registers on the last branch record stack. The From_IP part of the stack contains pointers to the source instruction. See also: ▪ Last Branch Record Stack TOS at 1C9H. ▪ Section 19.5.		Unique
Register Address: 41H, 65	MSR_LASTBRANCH_1_FROM_IP	
Last Branch Record 1 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 42H, 66	MSR_LASTBRANCH_2_FROM_IP	
Last Branch Record 2 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 43H, 67	MSR_LASTBRANCH_3_FROM_IP	
Last Branch Record 3 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 60H, 96	MSR_LASTBRANCH_0_TO_IP	
Last Branch Record 0 To IP (R/W) One of four pairs of last branch record registers on the last branch record stack. This To_IP part of the stack contains pointers to the destination instruction.		Unique
Register Address: 61H, 97	MSR_LASTBRANCH_1_TO_IP	
Last Branch Record 1 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 62H, 98	MSR_LASTBRANCH_2_TO_IP	
Last Branch Record 2 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 63H, 99	MSR_LASTBRANCH_3_TO_IP	
Last Branch Record 3 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 79H, 121	IA32_BIOS_UPDT_TRIG	
BIOS Update Trigger Register (W) See Table 2-2.		Unique
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID	
BIOS Update Signature ID (R/W) See Table 2-2.		Unique
Register Address: A0H, 160	MSR_SMRR_PHYSBASE	
System Management Mode Base Address register (WO in SMM) Model-specific implementation of SMRR-like interface, read visible and write only in SMM.		Unique
11:0	Reserved.	
31:12	PhysBase: SMRR physical Base Address.	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
63:32	Reserved.	
Register Address: A1H, 161	MSR_SMRR_PHYSMASK	
System Management Mode Physical Address Mask register (wO in SMM) Model-specific implementation of SMRR-like interface, read visible and write only in SMM.		Unique
10:0	Reserved.	
11	Valid: Physical address base and range mask are valid.	
31:12	PhysMask: SMRR physical address range mask.	
63:32	Reserved.	
Register Address: C1H, 193	IA32_PMC0	
Performance Counter Register See Table 2-2.		Unique
Register Address: C2H, 194	IA32_PMC1	
Performance Counter Register See Table 2-2.		Unique
Register Address: CDH, 205	MSR_FSB_FREQ	
Scaleable Bus Speed (R/O) This field indicates the intended scalable bus clock speed for processors based on Intel Core microarchitecture.		Shared
2:0	▪ 101B: 100 MHz (FSB 400) ▪ 001B: 133 MHz (FSB 533) ▪ 011B: 167 MHz (FSB 667) ▪ 010B: 200 MHz (FSB 800) ▪ 000B: 267 MHz (FSB 1067) ▪ 100B: 333 MHz (FSB 1333)	
	133.33 MHz should be utilized if performing calculation with System Bus Speed when encoding is 001B. 166.67 MHz should be utilized if performing calculation with System Bus Speed when encoding is 011B. 266.67 MHz should be utilized if performing calculation with System Bus Speed when encoding is 000B. 333.33 MHz should be utilized if performing calculation with System Bus Speed when encoding is 100B.	
63:3	Reserved.	
Register Address: CDH, 205	MSR_FSB_FREQ	
Scaleable Bus Speed (R/O) This field indicates the intended scalable bus clock speed for processors based on Enhanced Intel Core microarchitecture.		Shared

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
2:0	▪ 101B: 100 MHz (FSB 400) ▪ 001B: 133 MHz (FSB 533) ▪ 011B: 167 MHz (FSB 667) ▪ 010B: 200 MHz (FSB 800) ▪ 000B: 267 MHz (FSB 1067) ▪ 100B: 333 MHz (FSB 1333) ▪ 110B: 400 MHz (FSB 1600) 133.33 MHz should be utilized if performing calculation with System Bus Speed when encoding is 001B. 166.67 MHz should be utilized if performing calculation with System Bus Speed when encoding is 011B. 266.67 MHz should be utilized if performing calculation with System Bus Speed when encoding is 110B. 333.33 MHz should be utilized if performing calculation with System Bus Speed when encoding is 111B.	
63:3	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Unique
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Unique
Register Address: FEH, 254	IA32_MTRRCAP	
See Table 2-2.		Unique
11	SMRR Capability Using MSR 0A0H and 0A1H (R)	Unique
Register Address: 174H, 372	IA32_SYSENTER_CS	
See Table 2-2.		Unique
Register Address: 175H, 373	IA32_SYSENTER_ESP	
See Table 2-2.		Unique
Register Address: 176H, 374	IA32_SYSENTER_EIP	
See Table 2-2.		Unique
Register Address: 179H, 377	IA32_MCG_CAP	
See Table 2-2.		Unique
Register Address: 17AH, 378	IA32_MCG_STATUS	
Global Machine Check Status		Unique
0	RIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) can be used to restart the program. If cleared, the program cannot be reliably restarted.	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
1	EIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) is directly associated with the error.	
2	MCIP When set, bit indicates that a machine check has been generated. If a second machine check is detected while this bit is still set, the processor enters a shutdown state. Software should write this bit to 0 after processing a machine check exception.	
63:3	Reserved.	
Register Address: 186H, 390	IA32_PERFEVTSELO	
See Table 2-2.		Unique
Register Address: 187H, 391	IA32_PERFEVTSEL1	
See Table 2-2.		Unique
Register Address: 198H, 408	IA32_PERF_STATUS	
See Table 2-2.		Shared
Register Address: 198H, 408	MSR_PERF_STATUS	
Current performance status. See Section 16.1.1, "Software Interface For Initiating Performance State Transitions."		Shared
15:0	Current Performance State Value	
30:16	Reserved.	
31	XE Operation (R/O). If set, XE operation is enabled. Default is cleared.	
39:32	Reserved.	
44:40	Maximum Bus Ratio (R/O) Indicates maximum bus ratio configured for the processor.	
45	Reserved.	
46	Non-Integer Bus Ratio (R/O) Indicates non-integer bus ratio is enabled. Applies processors based on Enhanced Intel Core microarchitecture.	
63:47	Reserved.	
Register Address: 199H, 409	IA32_PERF_CTL	
See Table 2-2.		Unique
Register Address: 19AH, 410	IA32_CLOCK_MODULATION	
Clock Modulation (R/W) See Table 2-2. IA32_CLOCK_MODULATION MSR was originally named IA32_THERM_CONTROL MSR.		Unique
Register Address: 19BH, 411	IA32_THERM_INTERRUPT	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Thermal Interrupt Control (R/W) See Table 2-2.		Unique
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Unique
Register Address: 19DH, 413	MSR_THERM2_CTL	
Thermal Monitor 2 Control		Unique
15:0	Reserved.	
16	TM_SELECT (R/W) Mode of automatic thermal monitor: 0 = Thermal Monitor 1 (thermally-initiated on-die modulation of the stop-clock duty cycle). 1 = Thermal Monitor 2 (thermally-initiated frequency transitions). If bit 3 of the IA32_MISC_ENABLE register is cleared, TM_SELECT has no effect. Neither TM1 nor TM2 are enabled.	
63:16	Reserved.	
Register Address: 1A0H, 416	IA32_MISC_ENABLE	
Enable Misc. Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		
0	Fast-Strings Enable See Table 2-2.	
2:1	Reserved.	
3	Automatic Thermal Control Circuit Enable (R/W) See Table 2-2.	Unique
6:4	Reserved.	
7	Performance Monitoring Available (R) See Table 2-2.	Shared
8	Reserved.	
9	Hardware Prefetcher Disable (R/W) When set, disables the hardware prefetcher operation on streams of data. When clear (default), enables the prefetch queue. Disabling of the hardware prefetcher may impact processor performance.	
10	FERR# Multiplexing Enable (R/W) 1 = FERR# asserted by the processor to indicate a pending break event within the processor. 0 = Indicates compatible FERR# signaling behavior. This bit must be set to 1 to support XAPIC interrupt model usage.	Shared
11	Branch Trace Storage Unavailable (R/O) See Table 2-2.	Shared

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
12	Processor Event Based Sampling Unavailable (R/O) See Table 2-2.	Shared
13	TM2 Enable (R/W) When this bit is set (1) and the thermal sensor indicates that the die temperature is at the pre-determined threshold, the Thermal Monitor 2 mechanism is engaged. TM2 will reduce the bus to core ratio and voltage according to the value last written to MSR_THERM2_CTL bits 15:0. When this bit is clear (0, default), the processor does not change the VID signals or the bus to core ratio when the processor enters a thermally managed state. The BIOS must enable this feature if the TM2 feature flag (CPUID.1:ECX[8]) is set; if the TM2 feature flag is not set, this feature is not supported and BIOS must not alter the contents of the TM2 bit location. The processor is operating out of specification if both this bit and the TM1 bit are set to 0.	Shared
15:14	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W) See Table 2-2.	Shared
18	ENABLE MONITOR FSM (R/W) See Table 2-2.	Shared
19	Adjacent Cache Line Prefetch Disable (R/W) When set to 1, the processor fetches the cache line that contains data currently required by the processor. When set to 0, the processor fetches cache lines that comprise a cache line pair (128 bytes). Single processor platforms should not set this bit. Server platforms should set or clear this bit based on platform performance observed in validation and testing. BIOS may contain a setup option that controls the setting of this bit.	Shared
20	Enhanced Intel SpeedStep Technology Select Lock (R/W0) When set, this bit causes the following bits to become read-only: - Enhanced Intel SpeedStep Technology Select Lock (this bit). - Enhanced Intel SpeedStep Technology Enable bit. The bit must be set before an Enhanced Intel SpeedStep Technology transition is requested. This bit is cleared on reset.	Shared
21	Reserved.	
22	Limit CPUID Maxval (R/W) See Table 2-2.	Shared
23	xTPR Message Disable (R/W) See Table 2-2.	Shared
33:24	Reserved.	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
34	XD Bit Disable (R/W) When set to 1, the Execute Disable Bit feature (XD Bit) is disabled and the XD Bit extended feature flag will be clear (CPUID.80000001H: EDX[20]=0). When set to a 0 (default), the Execute Disable Bit feature (if available) allows the OS to enable PAE paging and take advantage of data only pages. BIOS must not alter the contents of this bit location if XD bit is not supported. Writing this bit to 1 when the XD Bit extended feature flag is set to 0 may generate a #GP exception.	Unique
36:35	Reserved.	
37	DCU Prefetcher Disable (R/W) When set to 1, the DCU L1 data cache prefetcher is disabled. The default value after reset is 0. BIOS may write '1' to disable this feature. The DCU prefetcher is an L1 data cache prefetcher. When the DCU prefetcher detects multiple loads from the same line done within a time limit, the DCU prefetcher assumes the next line will be required. The next line is prefetched in to the L1 data cache from memory or L2.	Unique
38	IDA Disable (R/W) When set to 1 on processors that support IDA, the Intel Dynamic Acceleration feature (IDA) is disabled and the IDA_Enable feature flag will be cleared (CPUID.06H: EAX[1]=0). When set to a 0 on processors that support IDA, CPUID.06H: EAX[1] reports the processor's support of IDA is enabled. Note: The power-on default value is used by BIOS to detect hardware support of IDA. If the power-on default value is 1, IDA is available in the processor. If the power-on default value is 0, IDA is not available.	Shared
39	IP Prefetcher Disable (R/W) When set to 1, the IP prefetcher is disabled. The default value after reset is 0. BIOS may write '1' to disable this feature. The IP prefetcher is an L1 data cache prefetcher. The IP prefetcher looks for sequential load history to determine whether to prefetch the next expected data into the L1 cache from memory or L2.	Unique
63:40	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-3) that points to the MSR containing the most recent branch record. See MSR_LASTBRANCH_0_FROM_IP (at 40H).		Unique
Register Address: 1D9H, 473	IA32_DEBUGCTL	
Debug Control (R/W) See Table 2-2.		Unique
Register Address: 1DDH, 477	MSR_LER_FROM_LIP	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Last Exception Record From Linear IP (R/W) Contains a pointer to the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Unique
Register Address: 1DEH, 478	MSR_LER_TO_LIP	
Last Exception Record To Linear IP (R/W) This area contains a pointer to the target of the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Unique
Register Address: 200H, 512	IA32_MTRR_PHYSBASE0	
See Table 2-2.		Unique
Register Address: 201H, 513	IA32_MTRR_PHYSMASK0	
See Table 2-2.		Unique
Register Address: 202H, 514	IA32_MTRR_PHYSBASE1	
See Table 2-2.		Unique
Register Address: 203H, 515	IA32_MTRR_PHYSMASK1	
See Table 2-2.		Unique
Register Address: 204H, 516	IA32_MTRR_PHYSBASE2	
See Table 2-2.		Unique
Register Address: 205H, 517	IA32_MTRR_PHYSMASK2	
See Table 2-2.		Unique
Register Address: 206H, 518	IA32_MTRR_PHYSBASE3	
See Table 2-2.		Unique
Register Address: 207H, 519	IA32_MTRR_PHYSMASK3	
See Table 2-2.		Unique
Register Address: 208H, 520	IA32_MTRR_PHYSBASE4	
See Table 2-2.		Unique
Register Address: 209H, 521	IA32_MTRR_PHYSMASK4	
See Table 2-2.		Unique
Register Address: 20AH, 522	IA32_MTRR_PHYSBASE5	
See Table 2-2.		Unique
Register Address: 20BH, 523	IA32_MTRR_PHYSMASK5	
See Table 2-2.		Unique
Register Address: 20CH, 524	IA32_MTRR_PHYSBASE6	
See Table 2-2.		Unique
Register Address: 20DH, 525	IA32_MTRR_PHYSMASK6	
See Table 2-2.		Unique
Register Address: 20EH, 526	IA32_MTRR_PHYSBASE7	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
See Table 2-2.		Unique
Register Address: 20FH, 527	IA32_MTRR_PHYSMASK7	
See Table 2-2.		Unique
Register Address: 250H, 592	IA32_MTRR_FIX64K_00000	
See Table 2-2.		Unique
Register Address: 258H, 600	IA32_MTRR_FIX16K_80000	
See Table 2-2.		Unique
Register Address: 259H, 601	IA32_MTRR_FIX16K_A0000	
See Table 2-2.		Unique
Register Address: 268H, 616	IA32_MTRR_FIX4K_C0000	
See Table 2-2.		Unique
Register Address: 269H, 617	IA32_MTRR_FIX4K_C8000	
See Table 2-2.		Unique
Register Address: 26AH, 618	IA32_MTRR_FIX4K_D0000	
See Table 2-2.		Unique
Register Address: 26BH, 619	IA32_MTRR_FIX4K_D8000	
See Table 2-2.		Unique
Register Address: 26CH, 620	IA32_MTRR_FIX4K_E0000	
See Table 2-2.		Unique
Register Address: 26DH, 621	IA32_MTRR_FIX4K_E8000	
See Table 2-2.		Unique
Register Address: 26EH, 622	IA32_MTRR_FIX4K_F0000	
See Table 2-2.		Unique
Register Address: 26FH, 623	IA32_MTRR_FIX4K_F8000	
See Table 2-2.		Unique
Register Address: 277H, 631	IA32_PAT	
See Table 2-2.		Unique
Register Address: 2FFH, 767	IA32_MTRR_DEF_TYPE	
Default Memory Types (R/W) See Table 2-2.		Unique
Register Address: 309H, 777	IA32_FIXED_CTR0	
Fixed-Function Performance Counter Register 0 (R/W) See Table 2-2.		Unique
Register Address: 30AH, 778	IA32_FIXED_CTR1	
Fixed-Function Performance Counter Register 1 (R/W) See Table 2-2.		Unique

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Register Address: 30BH, 779	IA32_FIXED_CTR2	
Fixed-Function Performance Counter Register 2 (R/W) See Table 2-2.		Unique
Register Address: 345H, 837	IA32_PERF_CAPABILITIES	
See Table 2-2. See Section 19.4.1, "IA32_DEBUGCTL MSR."		Unique
Register Address: 345H, 837	MSR_PERF_CAPABILITIES	
R/O. This applies to processors that do not support architectural PerfMon version 2.		Unique
5:0	LBR Format. See Table 2-2.	
6	PEBS Record Format.	
7	PEBSSaveArchRegs. See Table 2-2.	
63:8	Reserved.	
Register Address: 38DH, 909	IA32_FIXED_CTR_CTRL	
Fixed-Function-Counter Control Register (R/W) See Table 2-2.		Unique
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 38EH, 910	MSR_PERF_GLOBAL_STATUS	
See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 38FH, 911	IA32_PERF_GLOBAL_CTRL	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 38FH, 911	MSR_PERF_GLOBAL_CTRL	
See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 390H, 912	IA32_PERF_GLOBAL_OVF_CTRL	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 390H, 912	MSR_PERF_GLOBAL_OVF_CTRL	
See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
See Table 2-2. See Section 21.6.2.4, "Processor Event Based Sampling (PEBS)."		Unique
0	Enable PEBS on IA32_PMC0. (R/W)	
Register Address: 400H, 1024	IA32_MCO_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Unique
Register Address: 401H, 1025	IA32_MCO_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Unique
Register Address: 402H, 1026	IA32_MCO_ADDR	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MCO_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MCO_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 404H, 1028	IA32_MC1_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Unique
Register Address: 405H, 1029	IA32_MC1_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Unique
Register Address: 406H, 1030	IA32_MC1_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MC1_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC1_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 408H, 1032	IA32_MC2_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Unique
Register Address: 409H, 1033	IA32_MC2_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Unique
Register Address: 40AH, 1034	IA32_MC2_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 40CH, 1036	IA32_MC4_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Unique
Register Address: 40DH, 1037	IA32_MC4_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Unique
Register Address: 40EH, 1038	IA32_MC4_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The MSR_MC4_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 410H, 1040	IA32_MC3_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		
Register Address: 411H, 1041	IA32_MC3_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		
Register Address: 412H, 1042	IA32_MC3_ADDR	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The MSR_MC3_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC3_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 413H, 1043	IA32_MC3_MISC	
Machine Check Error Reporting Register: Contains additional information describing the machine-check error if the MISC_V flag in the IA32_MCI_STATUS register is set.		Unique
Register Address: 414H, 1044	IA32_MC5_CTL	
Machine Check Error Reporting Register: Controls signaling of #MC for errors produced by a particular hardware unit (or group of hardware units).		Unique
Register Address: 415H, 1045	IA32_MC5_STATUS	
Machine Check Error Reporting Register: Contains information related to a machine-check error if its VAL (valid) flag is set. Software is responsible for clearing IA32_MCI_STATUS MSRs by explicitly writing 0s to them; writing 1s to them causes a general-protection exception.		Unique
Register Address: 416H, 1046	IA32_MC5_ADDR	
Machine Check Error Reporting Register: Contains the address of the code or data memory location that produced the machine-check error if the ADDR_V flag in the IA32_MCI_STATUS register is set.		Unique
Register Address: 417H, 1047	IA32_MC5_MISC	
Machine Check Error Reporting Register: Contains additional information describing the machine-check error if the MISC_V flag in the IA32_MCI_STATUS register is set.		Unique
Register Address: 419H, 1045	IA32_MC6_STATUS	
Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 25.		Unique
Register Address: 480H, 1152	IA32_VMX_BASIC	
Reporting Register of Basic VMX Capabilities (R/O) See Table 2-2. See Appendix A.1, "Basic VMX Information."		Unique
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL	
Capability Reporting Register of Pin-Based VM-Execution Controls (R/O) See Table 2-2. See Appendix A.3, "VM-Execution Controls."		Unique
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL	
Capability Reporting Register of Primary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Unique
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL	
Capability Reporting Register of VM-Exit Controls (R/O) See Table 2-2. See Appendix A.4, "VM-Exit Controls."		Unique
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTL	
Capability Reporting Register of VM-Entry Controls (R/O) See Table 2-2. See Appendix A.5, "VM-Entry Controls."		Unique
Register Address: 485H, 1157	IA32_VMX_MISC	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Table 2-2. See Appendix A.6, "Miscellaneous Data."		Unique
Register Address: 486H, 1158	IA32_VMX_CR0_FIXED0	
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Table 2-2. See Appendix A.7, "VMX-Fixed Bits in CR0."		Unique
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1	
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Table 2-2. See Appendix A.7, "VMX-Fixed Bits in CR0."		Unique
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0	
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Table 2-2. See Appendix A.8, "VMX-Fixed Bits in CR4."		Unique
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1	
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Table 2-2. See Appendix A.8, "VMX-Fixed Bits in CR4."		Unique
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM	
Capability Reporting Register of VMCS Field Enumeration (R/O) See Table 2-2. See Appendix A.9, "VMCS Enumeration."		Unique
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL52	
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Unique
Register Address: 600H, 1536	IA32_DS_AREA	
DS Save Area (R/W) See Table 2-2. See Section 21.6.3.4, "Debug Store (DS) Mechanism."		Unique
Register Address: 107CCH, 67532	MSR_EMON_L3_CTR_CTL0	
GBUSQ Event Control/Counter Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: 107CDH, 67533	MSR_EMON_L3_CTR_CTL1	
GBUSQ Event Control/Counter Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: 107CEH, 67534	MSR_EMON_L3_CTR_CTL2	
GSPNQ Event Control/Counter Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: 107CFH, 67535	MSR_EMON_L3_CTR_CTL3	
GSPNQ Event Control/Counter Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: 107D0H, 67536	MSR_EMON_L3_CTR_CTL4	

Table 2-3. MSRs in Processors Based on Intel® Core™ Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
FSB Event Control/Counter Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: 107D1H, 67537	MSR_EMON_L3_CTR_CTL5	
FSB Event Control/Counter Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: 107D2H, 67538	MSR_EMON_L3_CTR_CTL6	
FSB Event Control/Counter Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: 107D3H, 67539	MSR_EMON_L3_CTR_CTL7	
FSB Event Control/Counter Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: 107D8H, 67544	MSR_EMON_L3_GL_CTL	
L3/FSB Common Control Register (R/W) Applies to Intel Xeon processor 7400 series (processor signature 06_1D) only. See Section 19.2.2.		Unique
Register Address: C000_0080H	IA32_EFER	
Extended Feature Enables See Table 2-2.		Unique
Register Address: C000_0081H	IA32_STAR	
System Call Target Address (R/W) See Table 2-2.		Unique
Register Address: C000_0082H	IA32_LSTAR	
IA-32e Mode System Call Target Address (R/W) See Table 2-2.		Unique
Register Address: C000_0084H	IA32_FMASK	
System Call Flag Mask (R/W) See Table 2-2.		Unique
Register Address: C000_0100H	IA32_FS_BASE	
Map of BASE Address of FS (R/W) See Table 2-2.		Unique
Register Address: C000_0101H	IA32_GS_BASE	
Map of BASE Address of GS (R/W) See Table 2-2.		Unique
Register Address: C000_0102H	IA32_KERNEL_GS_BASE	
Swap Target of BASE Address of GS (R/W) See Table 2-2.		Unique

2.3 MSRS IN THE 45 NM AND 32 NM INTEL ATOM® PROCESSOR FAMILY

Table 2-4 lists model-specific registers (MSRs) for 45 nm and 32 nm Intel Atom processors, architectural MSR addresses are also included in Table 2-4. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_1CH, 06_26H, 06_27H, 06_35H, or 06_36H; see Table 2-1.

The column “Shared/Unique” applies to logical processors sharing the same core in processors based on the Intel Atom microarchitecture. “Unique” means each logical processor has a separate MSR, or a bit field in an MSR governs only a logical processor. “Shared” means the MSR or the bit field in an MSR address governs the operation of both logical processors in the same core.

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Register Address: 0H, 0	IA32_P5_MC_ADDR	
See Section 2.23, “MSRs in Pentium Processors.”		Shared
Register Address: 1H, 1	IA32_P5_MC_TYPE	
See Section 2.23, “MSRs in Pentium Processors.”		Shared
Register Address: 6H, 6	IA32_MONITOR_FILTER_SIZE	
See Section 10.10.5, “Monitor/Mwait Address Range Determination,” and Table 2-2.		Unique
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER	
See Section 19.17, “Time-Stamp Counter,” and see Table 2-2.		Unique
Register Address: 17H, 23	IA32_PLATFORM_ID	
Platform ID (R) See Table 2-2.		Shared
Register Address: 17H, 23	MSR_PLATFORM_ID	
Model Specific Platform ID (R)		Shared
7:0	Reserved.	
12:8	Maximum Qualified Ratio (R) The maximum allowed bus ratio.	
63:13	Reserved.	
Register Address: 1BH, 27	IA32_APIC_BASE	
See Section 12.4.4, “Local APIC Status and Location,” and Table 2-2.		Unique
Register Address: 2AH, 42	MSR_EBL_CR_POWERON	
Processor Hard Power-On Configuration (R/W) Enables and disables processor features; (R) indicates current processor configuration.		Shared
0	Reserved.	
1	Data Error Checking Enable (R/W) 1 = Enabled; 0 = Disabled. Always 0.	
2	Response Error Checking Enable (R/W) 1 = Enabled; 0 = Disabled. Always 0.	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
3	AERR# Drive Enable (R/W) 1 = Enabled; 0 = Disabled. Always 0.	
4	BERR# Enable for initiator bus requests (R/W) 1 = Enabled; 0 = Disabled. Always 0.	
5	Reserved.	
6	Reserved.	
7	BINIT# Driver Enable (R/W) 1 = Enabled; 0 = Disabled. Always 0.	
8	Reserved.	
9	Execute BIST (R/O) 1 = Enabled; 0 = Disabled.	
10	AERR# Observation Enabled (R/O) 1 = Enabled; 0 = Disabled. Always 0.	
11	Reserved.	
12	BINIT# Observation Enabled (R/O) 1 = Enabled; 0 = Disabled. Always 0.	
13	Reserved.	
14	1 MByte Power on Reset Vector (R/O) 1 = 1 MByte; 0 = 4 GBytes.	
15	Reserved.	
17:16	APIC Cluster ID (R/O) Always 00B.	
19: 18	Reserved.	
21: 20	Symmetric Arbitration ID (R/O) Always 00B.	
26:22	Integer Bus Frequency Ratio (R/O)	
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64Processor (R/W) See Table 2-2.		Unique
Register Address: 40H, 64	MSR_LASTBRANCH_0_FROM_IP	
Last Branch Record 0 From IP (R/W) One of eight pairs of last branch record registers on the last branch record stack. The From_IP part of the stack contains pointers to the source instruction. See also: - Last Branch Record Stack TOS at 1C9H. - Section 19.5.		Unique

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Register Address: 41H, 65	MSR_LASTBRANCH_1_FROM_IP	
Last Branch Record 1 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 42H, 66	MSR_LASTBRANCH_2_FROM_IP	
Last Branch Record 2 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 43H, 67	MSR_LASTBRANCH_3_FROM_IP	
Last Branch Record 3 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 44H, 68	MSR_LASTBRANCH_4_FROM_IP	
Last Branch Record 4 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 45H, 69	MSR_LASTBRANCH_5_FROM_IP	
Last Branch Record 5 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 46H, 70	MSR_LASTBRANCH_6_FROM_IP	
Last Branch Record 6 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 47H, 71	MSR_LASTBRANCH_7_FROM_IP	
Last Branch Record 7 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Unique
Register Address: 60H, 96	MSR_LASTBRANCH_0_TO_IP	
Last Branch Record 0 To IP (R/W) One of eight pairs of last branch record registers on the last branch record stack. The To_IP part of the stack contains pointers to the destination instruction.		Unique
Register Address: 61H, 97	MSR_LASTBRANCH_1_TO_IP	
Last Branch Record 1 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 62H, 98	MSR_LASTBRANCH_2_TO_IP	
Last Branch Record 2 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 63H, 99	MSR_LASTBRANCH_3_TO_IP	
Last Branch Record 3 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 64H, 100	MSR_LASTBRANCH_4_TO_IP	
Last Branch Record 4 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 65H, 101	MSR_LASTBRANCH_5_TO_IP	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Last Branch Record 5 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 66H, 102	MSR_LASTBRANCH_6_TO_IP	
Last Branch Record 6 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 67H, 103	MSR_LASTBRANCH_7_TO_IP	
Last Branch Record 7 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Unique
Register Address: 79H, 121	IA32_BIOS_UPDT_TRIG	
BIOS Update Trigger Register (W) See Table 2-2.		Shared
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID	
BIOS Update Signature ID (R/W) See Table 2-2.		Unique
Register Address: C1H, 193	IA32_PMC0	
Performance counter register See Table 2-2.		Unique
Register Address: C2H, 194	IA32_PMC1	
Performance Counter Register See Table 2-2.		Unique
Register Address: CDH, 205	MSR_FSB_FREQ	
Scaleable Bus Speed (R/O) This field indicates the intended scalable bus clock speed for processors based on Intel Atom microarchitecture.		Shared
2:0	111B: 083 MHz (FSB 333) 101B: 100 MHz (FSB 400) 001B: 133 MHz (FSB 533) 011B: 167 MHz (FSB 667) 133.33 MHz should be utilized if performing calculation with System Bus Speed when encoding is 001B. 166.67 MHz should be utilized if performing calculation with System Bus Speed when encoding is 011B.	
63:3	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Unique
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Unique
Register Address: FEH, 254	IA32_MTRRCAP	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Memory Type Range Register (R) See Table 2-2.		Shared
Register Address: 11EH, 281	MSR_BBL_CR_CTL3	
Control Register 3 Used to configure the L2 Cache.		Shared
0	L2 Hardware Enabled (R/O) 1 = Indicates the L2 is hardware-enabled. 0 = Indicates the L2 is hardware-disabled.	
7:1	Reserved.	
8	L2 Enabled (R/W) 1 = L2 cache has been initialized. 0 = Disabled (default). Until this bit is set, the processor will not respond to the WBINVD instruction or the assertion of the FLUSH# input.	
22:9	Reserved.	
23	L2 Not Present (R/O) 0 = L2 Present. 1 = L2 Not Present.	
63:24	Reserved.	
Register Address: 174H, 372	IA32_SYSENTER_CS	
See Table 2-2.		Unique
Register Address: 175H, 373	IA32_SYSENTER_ESP	
See Table 2-2.		Unique
Register Address: 176H, 374	IA32_SYSENTER_EIP	
See Table 2-2.		Unique
Register Address: 179H, 377	IA32_MCG_CAP	
See Table 2-2.		Unique
Register Address: 17AH, 378	IA32_MCG_STATUS	
Global Machine Check Status		Unique
0	RIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) can be used to restart the program. If cleared, the program cannot be reliably restarted.	
1	EIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) is directly associated with the error.	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
2	MCIP When set, bit indicates that a machine check has been generated. If a second machine check is detected while this bit is still set, the processor enters a shutdown state. Software should write this bit to 0 after processing a machine check exception.	
63:3	Reserved.	
Register Address: 186H, 390	IA32_PERFEVTSELO	
See Table 2-2.		Unique
Register Address: 187H, 391	IA32_PERFEVTSEL1	
See Table 2-2.		Unique
Register Address: 198H, 408	IA32_PERF_STATUS	
See Table 2-2.		Shared
Register Address: 198H, 408	MSR_PERF_STATUS	
Performance Status		Shared
15:0	Current Performance State Value.	
39:16	Reserved.	
44:40	Maximum Bus Ratio (R/O) Indicates maximum bus ratio configured for the processor.	
63:45	Reserved.	
Register Address: 199H, 409	IA32_PERF_CTL	
See Table 2-2.		Unique
Register Address: 19AH, 410	IA32_CLOCK_MODULATION	
Clock Modulation (R/W) See Table 2-2. IA32_CLOCK_MODULATION MSR was originally named IA32_THERM_CONTROL MSR.		Unique
Register Address: 19BH, 411	IA32_THERM_INTERRUPT	
Thermal Interrupt Control (R/W) See Table 2-2.		Unique
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Unique
Register Address: 19DH, 413	MSR_THERM2_CTL	
Thermal Monitor 2 Control		Shared
15:0	Reserved.	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
16	TM_SELECT (R/W) Mode of automatic thermal monitor: 0 = Thermal Monitor 1 (thermally-initiated on-die modulation of the stop-clock duty cycle). 1 = Thermal Monitor 2 (thermally-initiated frequency transitions). If bit 3 of the IA32_MISC_ENABLE register is cleared, TM_SELECT has no effect. Neither TM1 nor TM2 are enabled.	
63:17	Reserved.	
Register Address: 1A0H, 416	IA32_MISC_ENABLE	
Enable Misc. Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		Unique
0	Fast-Strings Enable See Table 2-2.	
2:1	Reserved.	
3	Automatic Thermal Control Circuit Enable (R/W) See Table 2-2. Default value is 0.	Unique
6:4	Reserved.	
7	Performance Monitoring Available (R) See Table 2-2.	Shared
8	Reserved.	
9	Reserved.	
10	FERR# Multiplexing Enable (R/W) 1 = FERR# asserted by the processor to indicate a pending break event within the processor. 0 = Indicates compatible FERR# signaling behavior. This bit must be set to 1 to support XAPIC interrupt model usage.	Shared
11	Branch Trace Storage Unavailable (R/O) See Table 2-2.	Shared
12	Processor Event Based Sampling Unavailable (R/O) See Table 2-2.	Shared
13	TM2 Enable (R/W) When this bit is set (1) and the thermal sensor indicates that the die temperature is at the pre-determined threshold, the Thermal Monitor 2 mechanism is engaged. TM2 will reduce the bus to core ratio and voltage according to the value last written to MSR_THERM2_CTL bits 15:0. When this bit is cleared (0, default), the processor does not change the VID signals or the bus to core ratio when the processor enters a thermally managed state. The BIOS must enable this feature if the TM2 feature flag (CPUID.1:ECX[8]) is set; if the TM2 feature flag is not set, this feature is not supported and BIOS must not alter the contents of the TM2 bit location. The processor is operating out of specification if both this bit and the TM1 bit are set to 0.	Shared

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
15:14	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W) See Table 2-2.	Shared
18	ENABLE MONITOR FSM (R/W) See Table 2-2.	Shared
19	Reserved.	
20	Enhanced Intel SpeedStep Technology Select Lock (R/WO) When set, this bit causes the following bits to become read-only: - Enhanced Intel SpeedStep Technology Select Lock (this bit). - Enhanced Intel SpeedStep Technology Enable bit. The bit must be set before an Enhanced Intel SpeedStep Technology transition is requested. This bit is cleared on reset.	Shared
21	Reserved.	
22	Limit CPUID Maxval (R/W) See Table 2-2.	Unique
23	xTPR Message Disable (R/W) See Table 2-2.	Shared
33:24	Reserved.	
34	XD Bit Disable (R/W) See Table 2-3.	Unique
63:35	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-2) that points to the MSR containing the most recent branch record. See MSR_LASTBRANCH_0_FROM_IP (at 40H).		Unique
Register Address: 1D9H, 473	IA32_DEBUGCTL	
Debug Control (R/W) See Table 2-2.		Unique
Register Address: 1DDH, 477	MSR_LER_FROM_LIP	
Last Exception Record From Linear IP (R) Contains a pointer to the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Unique
Register Address: 1DEH, 478	MSR_LER_TO_LIP	
Last Exception Record To Linear IP (R) This area contains a pointer to the target of the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Unique
Register Address: 200H, 512	IA32_MTRR_PHYSBASE0	
See Table 2-2.		Shared
Register Address: 201H, 513	IA32_MTRR_PHYSMASK0	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
See Table 2-2.		Shared
Register Address: 202H, 514	IA32_MTRR_PHYSBASE1	
See Table 2-2.		Shared
Register Address: 203H, 515	IA32_MTRR_PHYSMASK1	
See Table 2-2.		Shared
Register Address: 204H, 516	IA32_MTRR_PHYSBASE2	
See Table 2-2.		Shared
Register Address: 205H, 517	IA32_MTRR_PHYSMASK2	
See Table 2-2.		Shared
Register Address: 206H, 518	IA32_MTRR_PHYSBASE3	
See Table 2-2.		Shared
Register Address: 207H, 519	IA32_MTRR_PHYSMASK3	
See Table 2-2.		Shared
Register Address: 208H, 520	IA32_MTRR_PHYSBASE4	
See Table 2-2.		Shared
Register Address: 209H, 521	IA32_MTRR_PHYSMASK4	
See Table 2-2.		Shared
Register Address: 20AH, 522	IA32_MTRR_PHYSBASE5	
See Table 2-2.		Shared
Register Address: 20BH, 523	IA32_MTRR_PHYSMASK5	
See Table 2-2.		Shared
Register Address: 20CH, 524	IA32_MTRR_PHYSBASE6	
See Table 2-2.		Shared
Register Address: 20DH, 525	IA32_MTRR_PHYSMASK6	
See Table 2-2.		Shared
Register Address: 20EH, 526	IA32_MTRR_PHYSBASE7	
See Table 2-2.		Shared
Register Address: 20FH, 527	IA32_MTRR_PHYSMASK7	
See Table 2-2.		Shared
Register Address: 250H, 592	IA32_MTRR_FIX64K_00000	
See Table 2-2.		Shared
Register Address: 258H, 600	IA32_MTRR_FIX16K_80000	
See Table 2-2.		Shared
Register Address: 259H, 601	IA32_MTRR_FIX16K_A0000	
See Table 2-2.		Shared
Register Address: 268H, 616	IA32_MTRR_FIX4K_C0000	
See Table 2-2.		Shared

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Register Address: 269H, 617	IA32_MTRR_FIX4K_C8000	
See Table 2-2.		Shared
Register Address: 26AH, 618	IA32_MTRR_FIX4K_D0000	
See Table 2-2.		Shared
Register Address: 26BH, 619	IA32_MTRR_FIX4K_D8000	
See Table 2-2.		Shared
Register Address: 26CH, 620	IA32_MTRR_FIX4K_E0000	
See Table 2-2.		Shared
Register Address: 26DH, 621	IA32_MTRR_FIX4K_E8000	
See Table 2-2.		Shared
Register Address: 26EH, 622	IA32_MTRR_FIX4K_F0000	
See Table 2-2.		Shared
Register Address: 26FH, 623	IA32_MTRR_FIX4K_F8000	
See Table 2-2.		Shared
Register Address: 277H, 631	IA32_PAT	
See Table 2-2.		Unique
Register Address: 309H, 777	IA32_FIXED_CTR0	
Fixed-Function Performance Counter Register 0 (R/W) See Table 2-2.		Unique
Register Address: 30AH, 778	IA32_FIXED_CTR1	
Fixed-Function Performance Counter Register 1 (R/W) See Table 2-2.		Unique
Register Address: 30BH, 779	IA32_FIXED_CTR2	
Fixed-Function Performance Counter Register 2 (R/W) See Table 2-2.		Unique
Register Address: 345H, 837	IA32_PERF_CAPABILITIES	
See Table 2-2. See Section 19.4.1, "IA32_DEBUGCTL MSR."		Shared
Register Address: 38DH, 909	IA32_FIXED_CTR_CTRL	
Fixed-Function-Counter Control Register (R/W) See Table 2-2.		Unique
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 38FH, 911	IA32_PERF_GLOBAL_CTRL	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 390H, 912	IA32_PERF_GLOBAL_OVF_CTRL	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Unique
Register Address: 3F1H, 1009	IA32_PEBBS_ENABLE (MSR_PEBBS_ENABLE)	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
See Table 2-2. See Section 21.6.2.4, "Processor Event Based Sampling (PEBS)."		Unique
0	Enable PEBS on IA32_PMC0 (R/W)	
Register Address: 400H, 1024	IA32_MCO_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Shared
Register Address: 401H, 1025	IA32_MCO_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Shared
Register Address: 402H, 1026	IA32_MCO_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MCO_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MCO_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Shared
Register Address: 404H, 1028	IA32_MC1_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Shared
Register Address: 405H, 1029	IA32_MC1_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Shared
Register Address: 408H, 1032	IA32_MC2_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Shared
Register Address: 409H, 1033	IA32_MC2_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Shared
Register Address: 40AH, 1034	IA32_MC2_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Shared
Register Address: 40CH, 1036	IA32_MC3_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Shared
Register Address: 40DH, 1037	IA32_MC3_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Shared
Register Address: 40EH, 1038	IA32_MC3_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The MSR_MC3_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC3_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Shared
Register Address: 410H, 1040	IA32_MC4_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Shared
Register Address: 411H, 1041	IA32_MC4_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Shared
Register Address: 412H, 1042	IA32_MC4_ADDR	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The MSR_MC4_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Shared
Register Address: 480H, 1152	IA32_VMX_BASIC	
Reporting Register of Basic VMX Capabilities (R/O) See Table 2-2. See Appendix A.1, "Basic VMX Information."		Unique
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL	
Capability Reporting Register of Pin-Based VM-Execution Controls (R/O) See Table 2-2. See Appendix A.3, "VM-Execution Controls."		Unique
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL	
Capability Reporting Register of Primary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Unique
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL	
Capability Reporting Register of VM-Exit Controls (R/O) See Table 2-2. See Appendix A.4, "VM-Exit Controls."		Unique
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTL	
Capability Reporting Register of VM-Entry Controls (R/O) See Table 2-2. See Appendix A.5, "VM-Entry Controls."		Unique
Register Address: 485H, 1157	IA32_VMX_MISC	
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Table 2-2. See Appendix A.6, "Miscellaneous Data."		Unique
Register Address: 486H, 1158	IA32_VMX_CR0_FIXED0	
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Table 2-2. See Appendix A.7, "VMX-Fixed Bits in CR0."		Unique
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1	
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Table 2-2. See Appendix A.7, "VMX-Fixed Bits in CR0."		Unique
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0	
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Table 2-2. See Appendix A.8, "VMX-Fixed Bits in CR4."		Unique
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1	
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Table 2-2. See Appendix A.8, "VMX-Fixed Bits in CR4."		Unique
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM	
Capability Reporting Register of VMCS Field Enumeration (R/O) See Table 2-2. See Appendix A.9, "VMCS Enumeration."		Unique
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL2	

Table 2-4. MSRs in the 45 nm and 32 nm Intel Atom® Processor Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Unique
Register Address: 600H, 1536	IA32_DS_AREA	
DS Save Area (R/W) See Table 2-2. See Section 21.6.3.4, "Debug Store (DS) Mechanism."		Unique
Register Address: C000_0080H	IA32_EFER	
Extended Feature Enables See Table 2-2.		Unique
Register Address: C000_0081H	IA32_STAR	
System Call Target Address (R/W) See Table 2-2.		Unique
Register Address: C000_0082H	IA32_LSTAR	
IA-32e Mode System Call Target Address (R/W) See Table 2-2.		Unique
Register Address: C000_0084H	IA32_FMASK	
System Call Flag Mask (R/W) See Table 2-2.		Unique
Register Address: C000_0100H	IA32_FS_BASE	
Map of BASE Address of FS (R/W) See Table 2-2.		Unique
Register Address: C000_0101H	IA32_GS_BASE	
Map of BASE Address of GS (R/W) See Table 2-2.		Unique
Register Address: C000_0102H	IA32_KERNEL_GS_BASE	
Swap Target of BASE Address of GS (R/W) See Table 2-2.		Unique

Table 2-5 lists model-specific registers (MSRs) that are specific to Intel Atom® processor with a CPUID Signature DisplayFamily_DisplayModel value of 06_27H.

Table 2-5. MSRs Supported by Intel Atom® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_27H

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 3F8H, 1016	MSR_PKG_C2_RESIDENCY	
Package C2 Residency Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package

Table 2-5. MSRs Supported by Intel Atom® Processors (Contd.)with a CPUID Signature DisplayFamily_DisplayModel Value of 06_27H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:0	Package C2 Residency Counter (R/O) Time that this package is in processor-specific C2 states since last reset. Counts at 1 Mhz frequency.	Package
Register Address: 3F9H, 1017	MSR_PKG_C4_RESIDENCY	
Package C4 Residency Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C4 Residency Counter. (R/O) Time that this package is in processor-specific C4 states since last reset. Counts at 1 Mhz frequency.	Package
Register Address: 3FAH, 1018	MSR_PKG_C6_RESIDENCY	
Package C6 Residency Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C6 Residency Counter. (R/O) Time that this package is in processor-specific C6 states since last reset. Counts at 1 Mhz frequency.	Package

2.4 MSRS IN INTEL PROCESSORS BASED ON SILVERMONT MICROARCHITECTURE

Table 2-6 lists model-specific registers (MSRs) common to Intel processors based on the Silvermont microarchitecture. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_37H, 06_4AH, 06_4DH, 06_5AH, or 06_5DH; see Table 2-1. The MSRs listed in Table 2-6 are also common to processors based on the Airmont microarchitecture and newer microarchitectures for next generation Intel Atom processors.

Table 2-7 lists MSRs common to processors based on the Silvermont and Airmont microarchitectures, but not newer microarchitectures.

Table 2-8, Table 2-9, and Table 2-10 lists MSRs that are model-specific across processors based on the Silvermont microarchitecture.

In the Silvermont microarchitecture, the scope column indicates the following: “Core” means each processor core has a separate MSR, or a bit field not shared with another processor core. “Module” means the MSR or the bit field is shared by a subset of the processor cores in the physical package. The number of processor cores in this subset is model specific and may differ between different processors. For all processors based on Silvermont microarchitecture, the L2 cache is also shared between cores in a module and thus CPUID leaf 04H enumeration can be used to figure out which processors are in the same module. “Package” means all processor cores in the physical package share the same MSR or bit interface.

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 0H, 0	IA32_P5_MC_ADDR	
See Section 2.23, “MSRs in Pentium Processors.”		Module
Register Address: 1H, 1	IA32_P5_MC_TYPE	

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 2.23, "MSRs in Pentium Processors."		Module
Register Address: 6H, 6	IA32_MONITOR_FILTER_SIZE	
See Section 10.10.5, "Monitor/Mwait Address Range Determination," and Table 2-2.		Core
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER	
See Section 19.17, "Time-Stamp Counter," and Table 2-2.		Core
Register Address: 1BH, 27	IA32_APIC_BASE	
See Section 12.4.4, "Local APIC Status and Location," and Table 2-2.		Core
Register Address: 2AH, 42	MSR_EBL_CR_POWERON	
Processor Hard Power-On Configuration (R/W) Writes ignored.		Module
63:0	Reserved.	
Register Address: 34H, 52	MSR_SMI_COUNT	
SMI Counter (R/O)		Core
31:0	SMI Count (R/O) Running count of SMI events since last RESET.	
63:32	Reserved.	
Register Address: 79H, 121	IA32_BIOS_UPDT_TRIG	
BIOS Update Trigger Register (W) See Table 2-2.		Core
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID	
BIOS Update Signature ID (R/W) See Table 2-2.		Core
Register Address: C1H, 193	IA32_PMC0	
Performance Counter Register See Table 2-2.		Core
Register Address: C2H, 194	IA32_PMC1	
Performance Counter Register See Table 2-2.		Core
Register Address: E4H, 228	MSR_PMG_IO_CAPTURE_BASE	
Power Management IO Redirection in C-state (R/W) See http://biosbits.org .		Module
15:0	LVL_2 Base Address (R/W) Specifies the base address visible to software for IO redirection. If IO MWAIT Redirection is enabled, reads to this address will be consumed by the power management logic and decoded to MWAIT instructions. When IO port address redirection is enabled, this is the IO port address reported to the OS/software.	

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
18:16	C-state Range (R/W) Specifies the encoding value of the maximum C-State code name to be included when IO read to MWAIT redirection is enabled by MSR_PKG_CST_CONFIG_CONTROL[bit10]: 100b - C4 is the max C-State to include 110b - C6 is the max C-State to include 111b - C7 is the max C-State to include	
63:19	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Core
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Core
Register Address: FEH, 254	IA32_MTRRCAP	
Memory Type Range Register (R) See Table 2-2.		Core
Register Address: 13CH, 316	MSR_FEATURE_CONFIG	
AES Configuration (RW-L) Privileged post-BIOS agent must provide a #GP handler to handle unsuccessful read of this MSR.		Core
1:0	AES Configuration (RW-L) Upon a successful read of this MSR, the configuration of AES instruction sets availability is as follows: 11b: AES instructions are not available until next RESET. Otherwise, AES instructions are available. Note: AES instruction set is not available if read is unsuccessful. If the configuration is not 01b, AES instructions can be mis-configured if a privileged agent unintentionally writes 11b.	
63:2	Reserved.	
Register Address: 174H, 372	IA32_SYSENTER_CS	
See Table 2-2.		Core
Register Address: 175H, 373	IA32_SYSENTER_ESP	
See Table 2-2.		Core
Register Address: 176H, 374	IA32_SYSENTER_EIP	
See Table 2-2.		Core
Register Address: 179H, 377	IA32_MCG_CAP	
See Table 2-2.		Core
Register Address: 17AH, 378	IA32_MCG_STATUS	
Global Machine Check Status		Core

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
0	RIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) can be used to restart the program. If cleared, the program cannot be reliably restarted.	
1	EIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) is directly associated with the error.	
2	MCIP When set, bit indicates that a machine check has been generated. If a second machine check is detected while this bit is still set, the processor enters a shutdown state. Software should write this bit to 0 after processing a machine check exception.	
63:3	Reserved.	
Register Address: 186H, 390	IA32_PERFEVTSELO	
See Table 2-2.		Core
7:0	Event Select	
15:8	UMask	
16	USR	
17	OS	
18	Edge	
19	PC	
20	INT	
21	Reserved.	
22	EN	
23	INV	
31:24	CMASK	
63:32	Reserved.	
Register Address: 187H, 391	IA32_PERFEVTSEL1	
See Table 2-2.		Core
Register Address: 198H, 408	IA32_PERF_STATUS	
See Table 2-2.		Module
Register Address: 199H, 409	IA32_PERF_CTL	
See Table 2-2.		Core
Register Address: 19AH, 410	IA32_CLOCK_MODULATION	
Clock Modulation (R/W) See Table 2-2. IA32_CLOCK_MODULATION MSR was originally named IA32_THERM_CONTROL MSR.		Core
Register Address: 19BH, 411	IA32_THERM_INTERRUPT	

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Thermal Interrupt Control (R/W) See Table 2-2.		Core
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Core
Register Address: 1A2H, 418	MSR_TEMPERATURE_TARGET	
Temperature Target		Package
15:0	Reserved.	
23:16	Temperature Target (R) The default thermal throttling or PROCHOT# activation temperature in degrees C. The effective temperature for thermal throttling or PROCHOT# activation is "Temperature Target" + "Target Offset".	
29:24	Target Offset (R/W) Specifies an offset in degrees C to adjust the throttling and PROCHOT# activation temperature from the default target specified in TEMPERATURE_TARGET (bits 23:16).	
63:30	Reserved.	
Register Address: 1A6H, 422	MSR_OFFCORE_RSP_0	
Offcore Response Event Select Register (R/W)		Module
Register Address: 1A7H, 423	MSR_OFFCORE_RSP_1	
Offcore Response Event Select Register (R/W)		Module
Register Address: 1B0H, 432	IA32_ENERGY_PERF_BIAS	
See Table 2-2.		Core
Register Address: 1D9H, 473	IA32_DEBUGCTL	
Debug Control (R/W) See Table 2-2.		Core
Register Address: 1DDH, 477	MSR_LER_FROM_LIP	
Last Exception Record From Linear IP (R/W) Contains a pointer to the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Core
Register Address: 1DEH, 478	MSR_LER_TO_LIP	
Last Exception Record To Linear IP (R/W) This area contains a pointer to the target of the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Core
Register Address: 1F2H, 498	IA32_SMRR_PHYSBASE	
See Table 2-2.		Core
Register Address: 1F3H, 499	IA32_SMRR_PHYSMASK	
See Table 2-2.		Core
Register Address: 200H, 512	IA32_MTRR_PHYSBASE0	
See Table 2-2.		Core

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 201H, 513	IA32_MTRR_PHYSMASK0	
See Table 2-2.		Core
Register Address: 202H, 514	IA32_MTRR_PHYSBASE1	
See Table 2-2.		Core
Register Address: 203H, 515	IA32_MTRR_PHYSMASK1	
See Table 2-2.		Core
Register Address: 204H, 516	IA32_MTRR_PHYSBASE2	
See Table 2-2.		Core
Register Address: 205H, 517	IA32_MTRR_PHYSMASK2	
See Table 2-2.		Core
Register Address: 206H, 518	IA32_MTRR_PHYSBASE3	
See Table 2-2.		Core
Register Address: 207H, 519	IA32_MTRR_PHYSMASK3	
See Table 2-2.		Core
Register Address: 208H, 520	IA32_MTRR_PHYSBASE4	
See Table 2-2.		Core
Register Address: 209H, 521	IA32_MTRR_PHYSMASK4	
See Table 2-2.		Core
Register Address: 20AH, 522	IA32_MTRR_PHYSBASE5	
See Table 2-2.		Core
Register Address: 20BH, 523	IA32_MTRR_PHYSMASK5	
See Table 2-2.		Core
Register Address: 20CH, 524	IA32_MTRR_PHYSBASE6	
See Table 2-2.		Core
Register Address: 20DH, 525	IA32_MTRR_PHYSMASK6	
See Table 2-2.		Core
Register Address: 20EH, 526	IA32_MTRR_PHYSBASE7	
See Table 2-2.		Core
Register Address: 20FH, 527	IA32_MTRR_PHYSMASK7	
See Table 2-2.		Core
Register Address: 250H, 592	IA32_MTRR_FIX64K_00000	
See Table 2-2.		Core
Register Address: 258H, 600	IA32_MTRR_FIX16K_80000	
See Table 2-2.		Core
Register Address: 259H, 601	IA32_MTRR_FIX16K_A0000	
See Table 2-2.		Core
Register Address: 268H, 616	IA32_MTRR_FIX4K_C0000	

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2.		Core
Register Address: 269H, 617	IA32_MTRR_FIX4K_C8000	
See Table 2-2.		Core
Register Address: 26AH, 618	IA32_MTRR_FIX4K_D0000	
See Table 2-2.		Core
Register Address: 26BH, 619	IA32_MTRR_FIX4K_D8000	
See Table 2-2.		Core
Register Address: 26CH, 620	IA32_MTRR_FIX4K_E0000	
See Table 2-2.		Core
Register Address: 26DH, 621	IA32_MTRR_FIX4K_E8000	
See Table 2-2.		Core
Register Address: 26EH, 622	IA32_MTRR_FIX4K_F0000	
See Table 2-2.		Core
Register Address: 26FH, 623	IA32_MTRR_FIX4K_F8000	
See Table 2-2.		Core
Register Address: 277H, 631	IA32_PAT	
See Table 2-2.		Core
Register Address: 2FFH, 767	IA32_MTRR_DEF_TYPE	
Default Memory Types (R/W) See Table 2-2.		Core
Register Address: 309H, 777	IA32_FIXED_CTR0	
Fixed-Function Performance Counter Register 0 (R/W) See Table 2-2.		Core
Register Address: 30AH, 778	IA32_FIXED_CTR1	
Fixed-Function Performance Counter Register 1 (R/W) See Table 2-2.		Core
Register Address: 30BH, 779	IA32_FIXED_CTR2	
Fixed-Function Performance Counter Register 2 (R/W) See Table 2-2.		Core
Register Address: 345H, 837	IA32_PERF_CAPABILITIES	
See Table 2-2. See Section 19.4.1, "IA32_DEBUGCTL MSR."		Core
Register Address: 38DH, 909	IA32_FIXED_CTR_CTRL	
Fixed-Function-Counter Control Register (R/W) See Table 2-2.		Core
Register Address: 38FH, 911	IA32_PERF_GLOBAL_CTRL	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Core
Register Address: 3FDH, 1021	MSR_CORE_C6_RESIDENCY	

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C6 Residency Counter (R/O) Value since last reset that this core is in processor-specific C6 states. Counts at the TSC Frequency.	
Register Address: 400H, 1024	IA32_MCO_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Module
Register Address: 401H, 1025	IA32_MCO_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Module
Register Address: 402H, 1026	IA32_MCO_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The IA32_MCO_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MCO_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Module
Register Address: 404H, 1028	IA32_MC1_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Module
Register Address: 405H, 1029	IA32_MC1_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Module
Register Address: 408H, 1032	IA32_MC2_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Module
Register Address: 409H, 1033	IA32_MC2_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Module
Register Address: 40AH, 1034	IA32_MC2_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Module
Register Address: 40CH, 1036	IA32_MC3_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 40DH, 1037	IA32_MC3_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core
Register Address: 40EH, 1038	IA32_MC3_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The MSR_MC3_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC3_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Core
Register Address: 410H, 1040	IA32_MC4_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 411H, 1041	IA32_MC4_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 412H, 1042	IA32_MC4_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The MSR_MC4_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Core
Register Address: 414H, 1044	IA32_MC5_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 415H, 1045	IA32_MC5_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Package
Register Address: 416H, 1046	IA32_MC5_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The MSR_MC4_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Package
Register Address: 480H, 1152	IA32_VMX_BASIC	
Reporting Register of Basic VMX Capabilities (R/O) See Table 2-2. See Appendix A.1, "Basic VMX Information."		Core
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL	
Capability Reporting Register of Pin-Based VM-Execution Controls (R/O) See Table 2-2. See Appendix A.3, "VM-Execution Controls."		Core
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL	
Capability Reporting Register of Primary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Core
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL	
Capability Reporting Register of VM-Exit Controls (R/O) See Table 2-2. See Appendix A.4, "VM-Exit Controls."		Core
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTL	
Capability Reporting Register of VM-Entry Controls (R/O) See Table 2-2. See Appendix A.5, "VM-Entry Controls."		Core
Register Address: 485H, 1157	IA32_VMX_MISC	
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Table 2-2. See Appendix A.6, "Miscellaneous Data."		Core
Register Address: 486H, 1158	IA32_VMX_CRO_FIXED0	

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Table 2-2. See Appendix A.7, "VMX-Fixed Bits in CR0."		Core
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1	
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Table 2-2. See Appendix A.7, "VMX-Fixed Bits in CR0."		Core
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0	
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Table 2-2. See Appendix A.8, "VMX-Fixed Bits in CR4."		Core
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1	
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Table 2-2. See Appendix A.8, "VMX-Fixed Bits in CR4."		Core
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM	
Capability Reporting Register of VMCS Field Enumeration (R/O) See Table 2-2. See Appendix A.9, "VMCS Enumeration."		Core
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL2	
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Core
Register Address: 48CH, 1164	IA32_VMX_EPT_VPID_ENUM	
Capability Reporting Register of EPT and VPID (R/O) See Table 2-2.		Core
Register Address: 48DH, 1165	IA32_VMX_TRUE_PINBASED_CTL2	
Capability Reporting Register of Pin-Based VM-Execution Flex Controls (R/O) See Table 2-2.		Core
Register Address: 48EH, 1166	IA32_VMX_TRUE_PROCBASED_CTL2	
Capability Reporting Register of Primary Processor-based VM-Execution Flex Controls (R/O) See Table 2-2.		Core
Register Address: 48FH, 1167	IA32_VMX_TRUE_EXIT_CTL2	
Capability Reporting Register of VM-Exit Flex Controls (R/O) See Table 2-2.		Core
Register Address: 490H, 1168	IA32_VMX_TRUE_ENTRY_CTL2	
Capability Reporting Register of VM-Entry Flex Controls (R/O) See Table 2-2.		Core
Register Address: 491H, 1169	IA32_VMX_FMFUNC	
Capability Reporting Register of VM-Function Controls (R/O) See Table 2-2.		Core

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 4C1H, 1217	IA32_A_PMC0	
See Table 2-2.		Core
Register Address: 4C2H, 1218	IA32_A_PMC1	
See Table 2-2.		Core
Register Address: 600H, 1536	IA32_DS_AREA	
DS Save Area (R/W) See Table 2-2 and Section 21.6.3.4, "Debug Store (DS) Mechanism."		Core
Register Address: 660H, 1632	MSR_CORE_C1_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C1 Residency Counter. (R/O) Value since last reset that this core is in processor-specific C1 states. Counts at the TSC frequency.	
Register Address: 6E0H, 1760	IA32_TSC_DEADLINE	
TSC Target of Local APIC's TSC Deadline Mode (R/W) See Table 2-2.		Core
Register Address: C000_0080H	IA32_EFER	
Extended Feature Enables See Table 2-2.		Core
Register Address: C000_0081H	IA32_STAR	
System Call Target Address (R/W) See Table 2-2.		Core
Register Address: C000_0082H	IA32_LSTAR	
IA-32e Mode System Call Target Address (R/W) See Table 2-2.		Core
Register Address: C000_0084H	IA32_FMASK	
System Call Flag Mask (R/W) See Table 2-2.		Core
Register Address: C000_0100H	IA32_FS_BASE	
Map of BASE Address of FS (R/W) See Table 2-2.		Core
Register Address: C000_0101H	IA32_GS_BASE	
Map of BASE Address of GS (R/W) See Table 2-2.		Core
Register Address: C000_0102H	IA32_KERNEL_GS_BASE	
Swap Target of BASE Address of GS (R/W) See Table 2-2.		Core
Register Address: C000_0103H	IA32_TSC_AUX	

Table 2-6. MSRs Common to Intel Atom® Processors (Silvermont and Newer Microarchitectures) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
AUXILIARY TSC Signature (R/W) See Table 2-2.		Core

Table 2-7 lists model-specific registers (MSRs) that are common to Intel Atom® processors based on the Silvermont and Airmont microarchitectures but not newer microarchitectures.

Table 2-7. MSRs Common to the Silvermont and Airmont Microarchitectures

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 17H, 23	MSR_PLATFORM_ID	
Model Specific Platform ID (R)		Module
7:0	Reserved.	
13:8	Maximum Qualified Ratio (R) The maximum allowed bus ratio.	
49:13	Reserved.	
52:50	See Table 2-2.	
63:33	Reserved.	
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64Processor (R/W) See Table 2-2.		Core
0	Lock (R/WL)	
1	Reserved.	
2	Enable VMX outside SMX operation (R/WL)	
Register Address: 40H, 64	MSR_LASTBRANCH_0_FROM_IP	
Last Branch Record 0 From IP (R/W) One of eight pairs of last branch record registers on the last branch record stack. The From_IP part of the stack contains pointers to the source instruction. See also: - Last Branch Record Stack TOS at 1C9H. - Section 19.5 and record format in Section 19.4.8.1.		Core
Register Address: 41H, 65	MSR_LASTBRANCH_1_FROM_IP	
Last Branch Record 1 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 42H, 66	MSR_LASTBRANCH_2_FROM_IP	
Last Branch Record 2 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 43H, 67	MSR_LASTBRANCH_3_FROM_IP	
Last Branch Record 3 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 44H, 68	MSR_LASTBRANCH_4_FROM_IP	
Last Branch Record 4 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core

Table 2-7. MSRs Common to the Silvermont and Airmont Microarchitectures (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 45H, 69	MSR_LASTBRANCH_5_FROM_IP	
Last Branch Record 5 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 46H, 70	MSR_LASTBRANCH_6_FROM_IP	
Last Branch Record 6 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 47H, 71	MSR_LASTBRANCH_7_FROM_IP	
Last Branch Record 7 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 60H, 96	MSR_LASTBRANCH_0_TO_IP	
Last Branch Record 0 To IP (R/W) One of eight pairs of last branch record registers on the last branch record stack. The To_IP part of the stack contains pointers to the destination instruction.		Core
Register Address: 61H, 97	MSR_LASTBRANCH_1_TO_IP	
Last Branch Record 1 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 62H, 98	MSR_LASTBRANCH_2_TO_IP	
Last Branch Record 2 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 63H, 99	MSR_LASTBRANCH_3_TO_IP	
Last Branch Record 3 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 64H, 100	MSR_LASTBRANCH_4_TO_IP	
Last Branch Record 4 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 65H, 101	MSR_LASTBRANCH_5_TO_IP	
Last Branch Record 5 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 66H, 102	MSR_LASTBRANCH_6_TO_IP	
Last Branch Record 6 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 67H, 103	MSR_LASTBRANCH_7_TO_IP	
Last Branch Record 7 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information: Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	

Table 2-7. MSRs Common to the Silvermont and Airmont Microarchitectures (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
15:8	Maximum Non-Turbo Ratio (R/O) This is the ratio of the maximum frequency that does not require turbo. Frequency = ratio * Scalable Bus Frequency.	Package
63:16	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States. See http://biosbits.org .		Module
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: C0 (no package C-state support) 001b: C1 (Behavior is the same as 000b) 100b: C4 110b: C6 111b: C7 (Silvermont only)	
9:3	Reserved.	
10	I/O MWait Redirection Enable (R/W) When set, will map IO_read instructions sent to IO register specified by MSR_PMG_IO_CAPTURE_BASE to MWait instructions.	
14:11	Reserved.	
15	CFG Lock (R/WO) When set, locks bits 15:0 of this register until next reset.	
63:16	Reserved.	
Register Address: 11EH, 281	MSR_BBL_CR_CTL3	
Control Register 3 Used to configure the L2 Cache.		Module
0	L2 Hardware Enabled (R/O) 1 = If the L2 is hardware-enabled. 0 = Indicates if the L2 is hardware-disabled.	
7:1	Reserved.	
8	L2 Enabled (R/W) 1 = L2 cache has been initialized. 0 = Disabled (default). Until this bit is set the processor will not respond to the WBINVD instruction or the assertion of the FLUSH# input.	
22:9	Reserved.	

Table 2-7. MSRs Common to the Silvermont and Airmont Microarchitectures (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
23	L2 Not Present (R/O) 0 = L2 Present. 1 = L2 Not Present.	
63:24	Reserved.	
Register Address: 1A0H, 416	IA32_MISC_ENABLE	
Enable Misc. Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		
0	Fast-Strings Enable See Table 2-2.	Core
2:1	Reserved.	
3	Automatic Thermal Control Circuit Enable (R/W) See Table 2-2. Default value is 0.	Module
6:4	Reserved.	
7	Performance Monitoring Available (R) See Table 2-2.	Core
10:8	Reserved.	
11	Branch Trace Storage Unavailable (R/O) See Table 2-2.	Core
12	Processor Event Based Sampling Unavailable (R/O) See Table 2-2.	Core
15:13	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W) See Table 2-2.	Module
18	ENABLE MONITOR FSM (R/W) See Table 2-2.	Core
21:19	Reserved.	
22	Limit CPUID Maxval (R/W) See Table 2-2.	Core
23	xTPR Message Disable (R/W) See Table 2-2.	Module
33:24	Reserved.	
34	XD Bit Disable (R/W) See Table 2-3.	Core
37:35	Reserved.	

Table 2-7. MSRs Common to the Silvermont and Airmont Microarchitectures (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
38	Turbo Mode Disable (R/W) When set to 1 on processors that support Intel Turbo Boost Technology, the turbo mode feature is disabled and the IDA_Enable feature flag will be cleared (CPUID.06H: EAX[1]=0). When set to a 0 on processors that support IDA, CPUID.06H: EAX[1] reports the processor's support of turbo mode is enabled. Note: The power-on default value is used by BIOS to detect hardware support of turbo mode. If the power-on default value is 1, turbo mode is available in the processor. If the power-on default value is 0, turbo mode is not available.	Module
63:39	Reserved.	
Register Address: 1C8H, 456	MSR_LBR_SELECT	
Last Branch Record Filtering Select Register (R/W) See Section 19.9.2, "Filtering of Last Branch Records."		Core
0	CPL_EQ_0	
1	CPL_NEQ_0	
2	JCC	
3	NEAR_REL_CALL	
4	NEAR_IND_CALL	
5	NEAR_RET	
6	NEAR_IND_JMP	
7	NEAR_REL_JMP	
8	FAR_BRANCH	
63:9	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-2) that points to the MSR containing the most recent branch record. See MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Core
Register Address: 390H, 912	IA32_PERF_GLOBAL_OVF_CTRL	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Core
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
See Table 2-2. See Section 21.6.2.4, "Processor Event Based Sampling (PEBS)."		Core
0	Enable PEBS for precise event on IA32_PMC0 (R/W)	
Register Address: 3FAH, 1018	MSR_PKG_C6_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C6 Residency Counter (R/O) Value since last reset that this package is in processor-specific C6 states. Counts at the TSC Frequency.	

Table 2-7. MSRs Common to the Silvermont and Airmont Microarchitectures (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 664H, 1636	MSR_MC6_RESIDENCY_COUNTER	
Module C6 Residency Counter (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Module
63:0	Time that this module is in module-specific C6 states since last reset. Counts at 1 Mhz frequency.	

2.4.1 MSRs with Model-Specific Behavior in the Silvermont Microarchitecture

Table 2-8 lists MSRs that are specific to the Intel Atom[®] processor E3000 Series (CUID Signature DisplayFamily_DisplayModel value of 06_37H) and Intel Atom processors (CUID Signature DisplayFamily_DisplayModel value of 06_4AH, 06_5AH, or 06_5DH).

Table 2-8. Specific MSRs Supported by Intel Atom[®] Processors with a CUID Signature DisplayFamily_DisplayModel Value of 06_37H, 06_4AH, 06_5AH, or 06_5DH

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: CDH, 205	MSR_FSB_FREQ	
Scaleable Bus Speed (R/O) This field indicates the intended scalable bus clock speed for processors based on Silvermont microarchitecture.		Module
2:0	100B: 080.0 MHz 000B: 083.3 MHz 001B: 100.0 MHz 010B: 133.3 MHz 011B: 116.7 MHz	
63:3	Reserved.	
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
	Unit Multipliers used in RAPL Interfaces (R/O) See Section 16.10.1, "RAPL Interfaces."	Package
3:0	Power Units Power related information (in milliWatts) is based on the multiplier, 2^{PU} ; where PU is an unsigned integer represented by bits 3:0. Default value is 0101b, indicating power unit is in 32 milliWatts increment.	
7:4	Reserved.	
12:8	Energy Status Units Energy related information (in microJoules) is based on the multiplier, 2^{ESU} ; where ESU is an unsigned integer represented by bits 12:8. Default value is 00101b, indicating energy unit is in 32 microJoules increment.	
15:13	Reserved.	
19:16	Time Unit The value is 0000b, indicating time unit is in one second.	
63:20	Reserved.	
Register Address: 610H, 1552	MSR_PKG_POWER_LIMIT	
PKG RAPL Power Limit Control (R/W)		Package

Table 2-8. Specific MSRs Supported by Intel Atom® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_37H, 06_4AH, 06_5AH, or 06_5DH (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
14:0	Package Power Limit #1 (R/W) See Section 16.10.3, "Package RAPL Domain," and MSR_RAPL_POWER_UNIT in Table 2-8.	
15	Enable Power Limit #1 (R/W) See Section 16.10.3, "Package RAPL Domain."	
16	Package Clamping Limitation #1 (R/W) See Section 16.10.3, "Package RAPL Domain."	
23:17	Time Window for Power Limit #1 (R/W) In unit of second. If 0 is specified in bits [23:17], defaults to 1 second window.	
63:24	Reserved.	
Register Address: 611H, 1553	MSR_PKG_ENERGY_STATUS	
PKG Energy Status (R/O) See Section 16.10.3, "Package RAPL Domain," and MSR_RAPL_POWER_UNIT in Table 2-8.		Package
Register Address: 639H, 1593	MSR_PP0_ENERGY_STATUS	
PP0 Energy Status (R/O) See Section 16.10.4, "PP0/PP1 RAPL Domains," and MSR_RAPL_POWER_UNIT in Table 2-8.		Package

Table 2-9 lists model-specific registers (MSRs) that are specific to the Intel Atom® processor E3000 Series (CPUID Signature DisplayFamily_DisplayModel value of 06_37H).

Table 2-9. Specific MSRs Supported by the Intel Atom® Processor E3000 Series with a CPUID Signature DisplayFamily_DisplayModel Value of 06_37H

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 668H, 1640	MSR_CC6_DEMOTION_POLICY_CONFIG	
Core C6 Demotion Policy Config MSR		Package
63:0	Controls per-core C6 demotion policy. Writing a value of 0 disables core level HW demotion policy.	
Register Address: 669H, 1641	MSR_MC6_DEMOTION_POLICY_CONFIG	
Module C6 Demotion Policy Config MSR		Package
63:0	Controls module (i.e., two cores sharing the second-level cache) C6 demotion policy. Writing a value of 0 disables module level HW demotion policy.	
Register Address: 664H, 1636	MSR_MC6_RESIDENCY_COUNTER	
Module C6 Residency Counter (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Module
63:0	Time that this module is in module-specific C6 states since last reset. Counts at 1 Mhz frequency.	

Table 2-10 lists model-specific registers (MSRs) that are specific to Intel Atom® processor C2000 Series (CPUID Signature DisplayFamily_DisplayModel value of 06_4DH).

Table 2-10. Specific MSRs Supported by Intel Atom® Processor C2000 Series with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4DH

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1A4H, 420	MSR_MISC_FEATURE_CONTROL	
Miscellaneous Feature Control (R/W)		
0	L2 Hardware Prefetcher Disable (R/W) If 1, disables the L2 hardware prefetcher, which fetches additional lines of code or data into the L2 cache.	Core
1	Reserved.	
2	DCU Hardware Prefetcher Disable (R/W) If 1, disables the L1 data cache prefetcher, which fetches the next cache line into L1 data cache.	Core
63:3	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode (R/W)		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 core active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 core active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 core active.	Package
39:32	Maximum Ratio Limit for 5C Maximum turbo ratio limit of 5 core active.	Package
47:40	Maximum Ratio Limit for 6C Maximum turbo ratio limit of 6 core active.	Package
55:48	Maximum Ratio Limit for 7C Maximum turbo ratio limit of 7 core active.	Package
63:56	Maximum Ratio Limit for 8C Maximum turbo ratio limit of 8 core active.	Package
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
Unit Multipliers used in RAPL Interfaces (R/O) See Section 16.10.1, "RAPL Interfaces."		Package
3:0	Power Units Power related information (in milliWatts) is based on the multiplier, 2^{PU} ; where PU is an unsigned integer represented by bits 3:0. Default value is 0101b, indicating power unit is in 32 milliWatts increment.	
7:4	Reserved.	

Table 2-10. Specific MSRs Supported by Intel Atom® Processor C2000 Series with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4DH (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
12:8	Energy Status Units. Energy related information (in microJoules) is based on the multiplier, 2^{ESU} ; where ESU is an unsigned integer represented by bits 12:8. Default value is 00101b, indicating energy unit is in 32 microJoules increment.	
15:13	Reserved.	
19:16	Time Unit The value is 0000b, indicating time unit is in one second.	
63:20	Reserved.	
Register Address: 610H, 1552	MSR_PKG_POWER_LIMIT	
PKG RAPL Power Limit Control (R/W) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 66EH, 1646	MSR_PKG_POWER_INFO	
PKG RAPL Parameter (R/O)		Package
14:0	Thermal Spec Power (R/O) The unsigned integer value is the equivalent of the thermal specification power of the package domain. The unit of this field is specified by the "Power Units" field of MSR_RAPL_POWER_UNIT.	
63:15	Reserved.	

2.4.2 MSRs in Intel Atom® Processors Based on Airmont Microarchitecture

Intel Atom processor X7-Z8000 and X5-Z8000 series are based on the Airmont microarchitecture. These processors support MSRs listed in Table 2-6, Table 2-7, Table 2-8, and Table 2-11. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_4CH; see Table 2-1.

Table 2-11. MSRs in Intel Atom® Processors Based on Airmont Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: CDH, 205	MSR_FSB_FREQ	
Scaleable Bus Speed (R/O) This field indicates the intended scalable bus clock speed for processors based on Airmont microarchitecture.		Module
3:0	▪ 0000B: 083.3 MHz ▪ 0001B: 100.0 MHz ▪ 0010B: 133.3 MHz ▪ 0011B: 116.7 MHz ▪ 0100B: 080.0 MHz ▪ 0101B: 093.3 MHz ▪ 0110B: 090.0 MHz ▪ 0111B: 088.9 MHz ▪ 1000B: 087.5 MHz	
63:5	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	

Table 2-11. MSRs in Intel Atom® Processors Based on Airmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States. See http://biosbits.org .		Module
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: No limit 001b: C1 010b: C2 110b: C6 111b: C7	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W) When set, will map IO_read instructions sent to IO register specified by MSR_PMG_IO_CAPTURE_BASE to MWAIT instructions.	
14:11	Reserved.	
15	CFG Lock (R/W0) When set, locks bits 15:0 of this register until next reset.	
63:16	Reserved.	
Register Address: E4H, 228	MSR_PMG_IO_CAPTURE_BASE	
Power Management IO Redirection in C-state (R/W) See http://biosbits.org .		Module
15:0	LVL_2 Base Address (R/W) Specifies the base address visible to software for IO redirection. If IO MWAIT Redirection is enabled, reads to this address will be consumed by the power management logic and decoded to MWAIT instructions. When IO port address redirection is enabled, this is the IO port address reported to the OS/software.	
18:16	C-state Range (R/W) Specifies the encoding value of the maximum C-State code name to be included when IO read to MWAIT redirection is enabled by MSR_PKG_CST_CONFIG_CONTROL[bit10]: 000b - C3 is the max C-State to include. 001b - Deep Power Down Technology is the max C-State. 010b - C7 is the max C-State to include.	
63:19	Reserved.	
Register Address: 638H, 1592	MSR_PPO_POWER_LIMIT	
PPO RAPL Power Limit Control (R/W)		Package

Table 2-11. MSRs in Intel Atom® Processors Based on Airmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
14:0	PP0 Power Limit #1 (R/W) See Section 16.10.4, “PP0/PP1 RAPL Domains,” and MSR_RAPL_POWER_UNIT in Table 2-8.	
15	Enable Power Limit #1 (R/W) See Section 16.10.4, “PP0/PP1 RAPL Domains.”	
16	Reserved.	
23:17	Time Window for Power Limit #1 (R/W) Specifies the time duration over which the average power must remain below PP0_POWER_LIMIT #1(14:0). Supported Encodings: 0x0: 1 second time duration. 0x1: 5 second time duration (Default). 0x2: 10 second time duration. 0x3: 15 second time duration. 0x4: 20 second time duration. 0x5: 25 second time duration. 0x6: 30 second time duration. 0x7: 35 second time duration. 0x8: 40 second time duration. 0x9: 45 second time duration. 0xA: 50 second time duration. 0xB-0x7F - reserved.	
63:24	Reserved.	

2.5 MSRS IN INTEL ATOM® PROCESSORS BASED ON GOLDMONT MICROARCHITECTURE

Intel Atom processors based on the Goldmont microarchitecture support MSRs listed in Table 2-6 and Table 2-12. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_5CH; see Table 2-1.

In the Goldmont microarchitecture, the scope column indicates the following: “Core” means each processor core has a separate MSR, or a bit field not shared with another processor core. “Module” means the MSR or the bit field is shared by a subset of the processor cores in the physical package. The number of processor cores in this subset is model specific and may differ between different processors. For all processors based on Goldmont microarchitecture, the L2 cache is also shared between cores in a module and thus CPUID leaf 04H enumeration can be used to figure out which processors are in the same module. “Package” means all processor cores in the physical package share the same MSR or bit interface.

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 17H, 23	MSR_PLATFORM_ID	
Model Specific Platform ID (R)		Module
49:0	Reserved.	
52:50	See Table 2-2.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:33	Reserved.	
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64 Processor (R/W) See Table 2-2.		Core
0	Lock (R/WL)	
1	Enable VMX inside SMX operation (R/WL)	
2	Enable VMX outside SMX operation (R/WL)	
14:8	SENTER local functions enables (R/WL)	
15	SENTER global functions enable (R/WL)	
18	SGX global functions enable (R/WL)	
63:19	Reserved.	
Register Address: 3BH, 59	IA32_TSC_ADJUST	
Per-Core TSC ADJUST (R/W) See Table 2-2.		Core
Register Address: C3H, 195	IA32_PMC2	
Performance Counter Register See Table 2-2.		Core
Register Address: C4H, 196	IA32_PMC3	
Performance Counter Register See Table 2-2.		Core
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) This is the ratio of the maximum frequency that does not require turbo. Frequency = ratio * 100 MHz.	Package
27:16	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) When set to 1, indicates that Programmable Ratio Limit for Turbo mode is enabled. When set to 0, indicates Programmable Ratio Limit for Turbo mode is disabled.	Package
29	Programmable TDP Limit for Turbo Mode (R/O) When set to 1, indicates that TDP Limit for Turbo mode is programmable. When set to 0, indicates TDP Limit for Turbo mode is not programmable.	Package
30	Programmable TJ OFFSET (R/O) When set to 1, indicates that MSR_TEMPERATURE_TARGET.[27:24] is valid and writable to specify a temperature offset.	Package
39:31	Reserved.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
47:40	Maximum Efficiency Ratio (R/O) This is the minimum ratio (maximum efficiency) that the processor can operate, in units of 100MHz.	Package
63:48	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States. See http://biosbits.org .		Core
3:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 0000b: No limit 0001b: C1 0010b: C3 0011b: C6 0100b: C7 0101b: C7S 0110b: C8 0111b: C9 1000b: C10	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W) When set, will map IO_read instructions sent to IO register specified by MSR_PMG_IO_CAPTURE_BASE to MWAIT instructions.	
14:11	Reserved.	
15	CFG Lock (R/WO) When set, locks bits 15:0 of this register until next reset.	
63:16	Reserved.	
Register Address: 17DH, 381	MSR_SMM_MCA_CAP	
Enhanced SMM Capabilities (SMM-RO) Reports SMM capability enhancement. Accessible only while in SMM.		Core
57:0	Reserved.	
58	SMM_Code_Access_Chk (SMM-RO) If set to 1 indicates that the SMM code access restriction is supported and the MSR_SMM_FEATURE_CONTROL is supported.	
59	Long_Flow_Indication (SMM-RO) If set to 1 indicates that the SMM long flow indicator is supported and the MSR_SMM_DELAYED is supported.	
63:60	Reserved.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 188H, 392	IA32_PERFEVTSEL2	
See Table 2-2.		Core
Register Address: 189H, 393	IA32_PERFEVTSEL3	
See Table 2-2.		Core
Register Address: 1A0H, 416	IA32_MISC_ENABLE	
Enable Misc. Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		
0	Fast-Strings Enable See Table 2-2.	Core
2:1	Reserved.	
3	Automatic Thermal Control Circuit Enable (R/W) See Table 2-2. Default value is 1.	Package
6:4	Reserved.	
7	Performance Monitoring Available (R) See Table 2-2.	Core
10:8	Reserved.	
11	Branch Trace Storage Unavailable (R/O) See Table 2-2.	Core
12	Processor Event Based Sampling Unavailable (R/O) See Table 2-2.	Core
15:13	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W) See Table 2-2.	Package
18	ENABLE MONITOR FSM (R/W) See Table 2-2.	Core
21:19	Reserved.	
22	Limit CPUID Maxval (R/W) See Table 2-2.	Core
23	xTPR Message Disable (R/W) See Table 2-2.	Package
33:24	Reserved.	
34	XD Bit Disable (R/W) See Table 2-3.	Core
37:35	Reserved.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
38	Turbo Mode Disable (R/W) When set to 1 on processors that support Intel Turbo Boost Technology, the turbo mode feature is disabled and the IDA_Enable feature flag will be clear (CPUID.06H: EAX[1]=0). When set to a 0 on processors that support IDA, CPUID.06H: EAX[1] reports the processor's support of turbo mode is enabled. Note: The power-on default value is used by BIOS to detect hardware support of turbo mode. If the power-on default value is 1, turbo mode is available in the processor. If the power-on default value is 0, turbo mode is not available.	Package
63:39	Reserved.	
Register Address: 1A4H, 420	MSR_MISC_FEATURE_CONTROL	
Miscellaneous Feature Control (R/W)		
0	L2 Hardware Prefetcher Disable (R/W) If 1, disables the L2 hardware prefetcher, which fetches additional lines of code or data into the L2 cache.	Core
1	Reserved.	
2	DCU Hardware Prefetcher Disable (R/W) If 1, disables the L1 data cache prefetcher, which fetches the next cache line into L1 data cache.	Core
63:3	Reserved.	
Register Address: 1AAH, 426	MSR_MISC_PWR_MGMT	
Miscellaneous Power Management Control Various model specific features enumeration. See http://biosbits.org.		Package
0	EIST Hardware Coordination Disable (R/W) When 0, enables hardware coordination of Enhanced Intel Speedstep Technology request from processor cores. When 1, disables hardware coordination of Enhanced Intel Speedstep Technology requests.	
21:1	Reserved.	
22	Thermal Interrupt Coordination Enable (R/W) If set, then thermal interrupt on one core is routed to all cores.	
63:23	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode by Core Groups (R/W) Specifies Maximum Ratio Limit for each Core Group. Max ratio for groups with more cores must decrease monotonically. For groups with less than 4 cores, the max ratio must be 32 or less. For groups with 4-5 cores, the max ratio must be 22 or less. For groups with more than 5 cores, the max ratio must be 16 or less.		Package
7:0	Maximum Ratio Limit for Active Cores in Group 0 Maximum turbo ratio limit when the number of active cores is less than or equal to the Group 0 threshold.	Package

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
15:8	Maximum Ratio Limit for Active Cores in Group 1 Maximum turbo ratio limit when the number of active cores is less than or equal to the Group 1 threshold, and greater than the Group 0 threshold.	Package
23:16	Maximum Ratio Limit for Active Cores in Group 2 Maximum turbo ratio limit when the number of active cores is less than or equal to the Group 2 threshold, and greater than the Group 1 threshold.	Package
31:24	Maximum Ratio Limit for Active Cores in Group 3 Maximum turbo ratio limit when the number of active cores is less than or equal to the Group 3 threshold, and greater than the Group 2 threshold.	Package
39:32	Maximum Ratio Limit for Active Cores in Group 4 Maximum turbo ratio limit when the number of active cores is less than or equal to the Group 4 threshold, and greater than the Group 3 threshold.	Package
47:40	Maximum Ratio Limit for Active Cores in Group 5 Maximum turbo ratio limit when the number of active cores is less than or equal to the Group 5 threshold, and greater than the Group 4 threshold.	Package
55:48	Maximum Ratio Limit for Active Cores in Group 6 Maximum turbo ratio limit when the number of active cores is less than or equal to the Group 6 threshold, and greater than the Group 5 threshold.	Package
63:56	Maximum Ratio Limit for Active Cores in Group 7 Maximum turbo ratio limit when the number of active cores is less than or equal to the Group 7 threshold, and greater than the Group 6 threshold.	Package
Register Address: 1AEH, 430	MSR_TURBO_GROUP_CORECNT	
Group Size of Active Cores for Turbo Mode Operation (R/W) Writes of 0 threshold is ignored.		Package
7:0	Group 0 Core Count Threshold Maximum number of active cores to operate under the Group 0 Max Turbo Ratio limit.	Package
15:8	Group 1 Core Count Threshold Maximum number of active cores to operate under the Group 1 Max Turbo Ratio limit. Must be greater than the Group 0 Core Count.	Package
23:16	Group 2 Core Count Threshold Maximum number of active cores to operate under the Group 2 Max Turbo Ratio limit. Must be greater than the Group 1 Core Count.	Package
31:24	Group 3 Core Count Threshold Maximum number of active cores to operate under the Group 3 Max Turbo Ratio limit. Must be greater than the Group 2 Core Count.	Package
39:32	Group 4 Core Count Threshold Maximum number of active cores to operate under the Group 4 Max Turbo Ratio limit. Must be greater than the Group 3 Core Count.	Package
47:40	Group 5 Core Count Threshold Maximum number of active cores to operate under the Group 5 Max Turbo Ratio limit. Must be greater than the Group 4 Core Count.	Package

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
55:48	Group 6 Core Count Threshold Maximum number of active cores to operate under the Group 6 Max Turbo Ratio limit. Must be greater than the Group 5 Core Count.	Package
63:56	Group 7 Core Count Threshold Maximum number of active cores to operate under the Group 7 Max Turbo Ratio limit. Must be greater than the Group 6 Core Count, and not less than the total number of processor cores in the package. E.g., specify 255.	Package
Register Address: 1C8H, 456	MSR_LBR_SELECT	
Last Branch Record Filtering Select Register (R/W) See Section 19.9.2, "Filtering of Last Branch Records."		Core
0	CPL_EQ_0	
1	CPL_NEQ_0	
2	JCC	
3	NEAR_REL_CALL	
4	NEAR_IND_CALL	
5	NEAR_RET	
6	NEAR_IND_JMP	
7	NEAR_REL_JMP	
8	FAR_BRANCH	
9	EN_CALL_STACK	
63:10	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-4) that points to the MSR containing the most recent branch record. See MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 1FCH, 508	MSR_POWER_CTL	
Power Control Register See http://biosbits.org .		Core
0	Reserved.	
1	C1E Enable (R/W) When set to '1', will enable the CPU to switch to the Minimum Enhanced Intel SpeedStep Technology operating point when all execution cores enter MWAIT (C1).	Package
63:2	Reserved.	
Register Address: 210H, 528	IA32_MTRR_PHYSBASE8	
See Table 2-2.		Core
Register Address: 211H, 529	IA32_MTRR_PHYSMASK8	
See Table 2-2.		Core
Register Address: 212H, 530	IA32_MTRR_PHYSBASE9	
See Table 2-2.		Core

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address:	IA32_MTRR_PHYSMASK9	
213H, 531	See Table 2-2.	Core
Register Address:	IA32_MC0_CTL2	
280H, 640	See Table 2-2.	Module
Register Address:	IA32_MC1_CTL2	
281H, 641	See Table 2-2.	Module
Register Address:	IA32_MC2_CTL2	
282H, 642	See Table 2-2.	Core
Register Address: 283H, 643	IA32_MC3_CTL2	
See Table 2-2.		Module
Register Address: 284H, 644	IA32_MC4_CTL2	
See Table 2-2.		Package
Register Address: 285H, 645	IA32_MC5_CTL2	
See Table 2-2.		Package
Register Address: 286H, 646	IA32_MC6_CTL2	
See Table 2-2.		Package
Register Address: 300H, 768	MSR_SGXOWNEREPPOCH0	
Lower 64 Bit CR_SGXOWNEREPPOCH (W) Writes do not update CR_SGXOWNEREPPOCH if CPUID.(EAX=12H, ECX=0):EAX.SGX1 is 1 on any thread in the package.		Package
63:0	Lower 64 bits of an 128-bit external entropy value for key derivation of an enclave.	
Register Address: 301H, 769	MSR_SGXOWNEREPPOCH1	
Upper 64 Bit CR_SGXOWNEREPPOCH (W) Writes do not update CR_SGXOWNEREPPOCH if CPUID.(EAX=12H, ECX=0):EAX.SGX1 is 1 on any thread in the package.		Package
63:0	Upper 64 bits of an 128-bit external entropy value for key derivation of an enclave.	
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	
See Table 2-2 and Section 21.2.4, "Architectural Performance Monitoring Version 4."		Core
0	Ovf_PMC0	
1	Ovf_PMC1	
2	Ovf_PMC2	
3	Ovf_PMC3	
31:4	Reserved.	
32	Ovf_FixedCtr0	
33	Ovf_FixedCtr1	
34	Ovf_FixedCtr2	
54:35	Reserved.	
55	Trace_ToPA_PMI	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
57:56	Reserved.	
58	LBR_Frz	
59	CTR_Frz	
60	ASCI	
61	Ovf_Uncore	
62	Ovf_BufDSSAVE	
63	CondChgd	
Register Address: 390H, 912	IA32_PERF_GLOBAL_STATUS_RESET	
See Table 2-2 and Section 21.2.4, "Architectural Performance Monitoring Version 4."		Core
0	Set 1 to clear Ovf_PMC0.	
1	Set 1 to clear Ovf_PMC1.	
2	Set 1 to clear Ovf_PMC2.	
3	Set 1 to clear Ovf_PMC3.	
31:4	Reserved.	
32	Set 1 to clear Ovf_FixedCtr0.	
33	Set 1 to clear Ovf_FixedCtr1.	
34	Set 1 to clear Ovf_FixedCtr2.	
54:35	Reserved.	
55	Set 1 to clear Trace_ToPA_PMI.	
57:56	Reserved.	
58	Set 1 to clear LBR_Frz.	
59	Set 1 to clear CTR_Frz.	
60	Set 1 to clear ASCI.	
61	Set 1 to clear Ovf_Uncore.	
62	Set 1 to clear Ovf_BufDSSAVE.	
63	Set 1 to clear CondChgd.	
Register Address: 391H, 913	IA32_PERF_GLOBAL_STATUS_SET	
See Table 2-2 and Section 21.2.4, "Architectural Performance Monitoring Version 4."		Core
0	Set 1 to cause Ovf_PMC0 = 1.	
1	Set 1 to cause Ovf_PMC1 = 1.	
2	Set 1 to cause Ovf_PMC2 = 1.	
3	Set 1 to cause Ovf_PMC3 = 1.	
31:4	Reserved.	
32	Set 1 to cause Ovf_FixedCtr0 = 1.	
33	Set 1 to cause Ovf_FixedCtr1 = 1.	
34	Set 1 to cause Ovf_FixedCtr2 = 1.	
54:35	Reserved.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
55	Set 1 to cause Trace_ToPA_PMI = 1.	
57:56	Reserved.	
58	Set 1 to cause LBR_Frz = 1.	
59	Set 1 to cause CTR_Frz = 1.	
60	Set 1 to cause ASCI = 1.	
61	Set 1 to cause Ovf_Uncore.	
62	Set 1 to cause Ovf_BufDSSAVE.	
63	Reserved.	
Register Address: 392H, 914	IA32_PERF_GLOBAL_INUSE	
See Table 2-2.		Core
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
See Table 2-2 and Section 21.6.2.4, "Processor Event Based Sampling (PEBS)."		Core
0	Enable PEBS trigger and recording for the programmed event (precise or otherwise) on IA32_PMC0. (R/W)	
Register Address: 3F8H, 1016	MSR_PKG_C3_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C3 Residency Counter (R/O) Value since last reset that this package is in processor-specific C3 states. Count at the same frequency as the TSC.	
Register Address: 3F9H, 1017	MSR_PKG_C6_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C6 Residency Counter (R/O) Value since last reset that this package is in processor-specific C6 states. Count at the same frequency as the TSC.	
Register Address: 3FCH, 1020	MSR_CORE_C3_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C3 Residency Counter (R/O) Value since last reset that this core is in processor-specific C3 states. Count at the same frequency as the TSC.	
Register Address: 406H, 1030	IA32_MC1_ADDR	
See Section 17.3.2.3, "IA32_MC1_ADDR MSRs." The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Module
Register Address: 418H, 1048	IA32_MC6_CTL	
See Section 17.3.2.1, "IA32_MC1_CTL MSRs."		Package
Register Address: 419H, 1049	IA32_MC6_STATUS	
See Section 17.3.2.2, "IA32_MC1_STATUS MSRs," and Chapter 18.		Package

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 41AH, 1050	IA32_MC6_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 4C3H, 1219	IA32_A_PMC2	
See Table 2-2.		Core
Register Address: 4C4H, 1220	IA32_A_PMC3	
See Table 2-2.		Core
Register Address: 4E0H, 1248	MSR_SMM_FEATURE_CONTROL	
Enhanced SMM Feature Control (SMM-RW) Reports SMM capability Enhancement. Accessible only while in SMM.		Package
0	Lock (SMM-RW) When set to '1' locks this register from further changes.	
1	Reserved.	
2	SMM_Code_Chk_En (SMM-RW) This control bit is available only if MSR_SMM_MCA_CAP[58] == 1. When set to '0' (default) none of the logical processors are prevented from executing SMM code outside the ranges defined by the SMRR. When set to '1' any logical processor in the package that attempts to execute SMM code not within the ranges defined by the SMRR will assert an unrecoverable MCE.	
63:3	Reserved.	
Register Address: 4E2H, 1250	MSR_SMM_DELAYED	
SMM Delayed (SMM-RO) Reports the interruptible state of all logical processors in the package. Available only while in SMM and MSR_SMM_MCA_CAP[LONG_FLOW_INDICATION] == 1.		Package
N-1:0	LOG_PROC_STATE (SMM-RO) Each bit represents a processor core of its state in a long flow of internal operation which delays servicing an interrupt. The corresponding bit will be set at the start of long events such as: Microcode Update Load, C6, WBINVD, Ratio Change, Throttle. The bit is automatically cleared at the end of each long event. The reset value of this field is 0. Only bit positions below N = CPUID.(EAX=0BH, ECX=PKG_LVL):EBX[15:0] can be updated.	
63:N	Reserved.	
Register Address: 4E3H, 1251	MSR_SMM_BLOCKED	
SMM Blocked (SMM-RO) Reports the blocked state of all logical processors in the package. Available only while in SMM.		Package

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
N-1:0	LOG_PROC_STATE (SMM-RO) Each bit represents a processor core of its blocked state to service an SMI. The corresponding bit will be set if the logical processor is in one of the following states: Wait For SIPI or SENTER Sleep. The reset value of this field is 0FFFH. Only bit positions below N = CPUID.(EAX=0BH, ECX=PKG_LVL):EBX[15:0] can be updated.	
63:N	Reserved.	
Register Address: 500H, 1280	IA32_SGX_SVN_STATUS	
Status and SVN Threshold of SGX Support for ACM (R/O)		Core
0	Lock See Section 40.11.3, "Interactions with Authenticated Code Modules (ACMs)."	
15:1	Reserved.	
23:16	SGX_SVN_SINIT See Section 40.11.3, "Interactions with Authenticated Code Modules (ACMs)."	
63:24	Reserved.	
Register Address: 560H, 1376	IA32_RTIT_OUTPUT_BASE	
Trace Output Base Register (R/W) See Table 2-2.		Core
Register Address: 561H, 1377	IA32_RTIT_OUTPUT_MASK_PTRS	
Trace Output Mask Pointers Register (R/W) See Table 2-2.		Core
Register Address: 570H, 1392	IA32_RTIT_CTL	
Trace Control Register (R/W)		Core
0	TraceEn	
1	CYCEn	
2	OS	
3	User	
6:4	Reserved, must be zero.	
7	CR3Filter	
8	ToPA Writing 0 will #GP if also setting TraceEn.	
9	MTCEn	
10	TSCEn	
11	DisRETC	
12	Reserved, must be zero.	
13	BranchEn	
17:14	MTCFreq	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
18	Reserved, must be zero.	
22:19	CycThresh	
23	Reserved, must be zero.	
27:24	PSBFreq	
31:28	Reserved, must be zero.	
35:32	ADDR0_CFG	
39:36	ADDR1_CFG	
63:40	Reserved, must be zero.	
Register Address: 571H, 1393	IA32_RTIT_STATUS	
Tracing Status Register (R/W)		Core
0	FilterEn Writes ignored.	
1	ContextEn Writes ignored.	
2	TriggerEn Writes ignored.	
3	Reserved	
4	Error (R/W)	
5	Stopped	
31:6	Reserved, must be zero.	
48:32	PacketByteCnt	
63:49	Reserved, must be zero.	
Register Address: 572H, 1394	IA32_RTIT_CR3_MATCH	
Trace Filter CR3 Match Register (R/W)		Core
4:0	Reserved	
63:5	CR3[63:5] value to match.	
Register Address: 580H, 1408	IA32_RTIT_ADDR0_A	
Region 0 Start Address (R/W)		Core
63:0	See Table 2-2.	
Register Address: 581H, 1409	IA32_RTIT_ADDR0_B	
Region 0 End Address (R/W)		Core
63:0	See Table 2-2.	
Register Address: 582H, 1410	IA32_RTIT_ADDR1_A	
Region 1 Start Address (R/W)		Core
63:0	See Table 2-2.	
Register Address: 583H, 1411	IA32_RTIT_ADDR1_B	
Region 1 End Address (R/W)		Core
63:0	See Table 2-2.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
Unit Multipliers used in RAPL Interfaces (R/O) See Section 16.10.1, "RAPL Interfaces."		Package
3:0	Power Units Power related information (in Watts) is in unit of $1W/2^{PU}$; where PU is an unsigned integer represented by bits 3:0. Default value is 1000b, indicating power unit is in 3.9 milliWatts increment.	
7:4	Reserved.	
12:8	Energy Status Units Energy related information (in Joules) is in unit of $1Joule/2^{ESU}$; where ESU is an unsigned integer represented by bits 12:8. Default value is 01110b, indicating energy unit is in 61 microjoules.	
15:13	Reserved.	
19:16	Time Unit Time related information (in seconds) is in unit of $1S/2^{TU}$; where TU is an unsigned integer represented by bits 19:16. Default value is 1010b, indicating power unit is in 0.977 millisecond.	
63:20	Reserved.	
Register Address: 60AH, 1546	MSR_PKG_C3_IRT_L	
Package C3 Interrupt Response Limit (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
9:0	Interrupt Response Time Limit (R/W) Specifies the limit that should be used to decide if the package should be put into a package C3 state.	
12:10	Time Unit (R/W) Specifies the encoding value of time unit of the interrupt response time limit. See Table 2-20 for supported time unit encodings.	
14:13	Reserved.	
15	Valid (R/W) Indicates whether the values in bits 12:0 are valid and can be used by the processor for package C-state management.	
63:16	Reserved.	
Register Address: 60BH, 1547	MSR_PKG_C6_IRT_L1	
Package C6/C7S Interrupt Response Limit 1 (R/W) This MSR defines the interrupt response time limit used by the processor to manage a transition to a package C6 or C7S state. Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
9:0	Interrupt Response Time Limit (R/W) Specifies the limit that should be used to decide if the package should be put into a package C6 or C7S state.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
12:10	Time Unit (R/W) Specifies the encoding value of time unit of the interrupt response time limit. See Table 2-20 for supported time unit encodings.	
14:13	Reserved.	
15	Valid (R/W) Indicates whether the values in bits 12:0 are valid and can be used by the processor for package C-state management.	
63:16	Reserved.	
Register Address: 60CH, 1548	MSR_PKGC_IRTL2	
Package C7 Interrupt Response Limit 2 (R/W) This MSR defines the interrupt response time limit used by the processor to manage a transition to a package C7 state. Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
9:0	Interrupt Response Time Limit (R/W) Specifies the limit that should be used to decide if the package should be put into a package C7 state.	
12:10	Time Unit (R/W) Specifies the encoding value of time unit of the interrupt response time limit. See Table 2-20 for supported time unit encodings.	
14:13	Reserved.	
15	Valid (R/W) Indicates whether the values in bits 12:0 are valid and can be used by the processor for package C-state management.	
63:16	Reserved.	
Register Address: 60DH, 1549	MSR_PKG_C2_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.		Package
63:0	Package C2 Residency Counter (R/O) Value since last reset that this package is in processor-specific C2 states. Count at the same frequency as the TSC.	
Register Address: 610H, 1552	MSR_PKG_POWER_LIMIT	
PKG RAPL Power Limit Control (R/W) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 611H, 1553	MSR_PKG_ENERGY_STATUS	
PKG Energy Status (R/O) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 613H, 1555	MSR_PKG_PERF_STATUS	
PKG Perf Status (R/O) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 614H, 1556	MSR_PKG_POWER_INFO	
PKG RAPL Parameters (R/W)		Package

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
14:0	Thermal Spec Power (R/W) See Section 16.10.3, "Package RAPL Domain."	
15	Reserved.	
30:16	Minimum Power (R/W) See Section 16.10.3, "Package RAPL Domain."	
31	Reserved.	
46:32	Maximum Power (R/W) See Section 16.10.3, "Package RAPL Domain."	
47	Reserved.	
54:48	Maximum Time Window (R/W) Specified by $2^Y * (1.0 + Z/4.0) * \text{Time_Unit}$, where "Y" is the unsigned integer value represented by bits 52:48, "Z" is an unsigned integer represented by bits 54:53. "Time_Unit" is specified by the "Time Units" field of MSR_RAPL_POWER_UNIT.	
63:55	Reserved.	
Register Address: 618H, 1560	MSR_DRAM_POWER_LIMIT	
DRAM RAPL Power Limit Control (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61CH, 1564	MSR_DRAM_POWER_INFO	
DRAM RAPL Parameters (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 632H, 1586	MSR_PKG_C10_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.		Package
63:0	Package C10 Residency Counter (R/O) Value since last reset that the entire SOC is in an S0i3 state. Count at the same frequency as the TSC.	
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	
PPO Energy Status (R/O) See Section 16.10.4, "PPO/PP1 RAPL Domains."		Package
Register Address: 641H, 1601	MSR_PP1_ENERGY_STATUS	
PP1 Energy Status (R/O) See Section 16.10.4, "PPO/PP1 RAPL Domains."		Package
Register Address: 64CH, 1612	MSR_TURBO_ACTIVATION_RATIO	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
ConfigTDP Control (R/W)		Package
7:0	MAX_NON_TURBO_RATIO (RW/L) System BIOS can program this field.	
30:8	Reserved.	
31	TURBO_ACTIVATION_RATIO_Lock (RW/L) When this bit is set, the content of this register is locked until a reset.	
63:32	Reserved.	
Register Address: 64FH, 1615	MSR_CORE_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in Processor Cores (R/W) (Frequency refers to processor core frequency.)		Package
0	PROCHOT Status (R0) When set, processor core frequency is reduced below the operating system request due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced below the operating system request due to a thermal event.	
2	Package-Level Power Limiting PL1 Status (R0) When set, frequency is reduced below the operating system request due to package-level power limiting PL1.	
3	Package-Level PL2 Power Limiting Status (R0) When set, frequency is reduced below the operating system request due to package-level power limiting PL2.	
8:4	Reserved.	
9	Core Power Limiting Status (R0) When set, frequency is reduced below the operating system request due to domain-level power limiting.	
10	VR Therm Alert Status (R0) When set, frequency is reduced below the operating system request due to a thermal alert from the Voltage Regulator.	
11	Max Turbo Limit Status (R0) When set, frequency is reduced below the operating system request due to multi-core turbo limits.	
12	Electrical Design Point Status (R0) When set, frequency is reduced below the operating system request due to electrical design point constraints (e.g., maximum electrical current consumption).	
13	Turbo Transition Attenuation Status (R0) When set, frequency is reduced below the operating system request due to Turbo transition attenuation. This prevents performance degradation due to frequent operating ratio changes.	
14	Maximum Efficiency Frequency Status (R0) When set, frequency is reduced below the maximum efficiency frequency.	
15	Reserved.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
18	Package-Level PL1 Power Limiting Log When set, indicates that the Package Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
19	Package-Level PL2 Power Limiting Log When set, indicates that the Package Level PL2 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
24:20	Reserved.	
25	Core Power Limiting Log When set, indicates that the Core Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
26	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
27	Max Turbo Limit Log When set, indicates that the Max Turbo Limit Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28	Electrical Design Point Log When set, indicates that the EDP Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
29	Turbo Transition Attenuation Log When set, indicates that the Turbo Transition Attenuation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
30	Maximum Efficiency Frequency Log When set, indicates that the Maximum Efficiency Frequency Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:31	Reserved.	
Register Address: 680H, 1664	MSR_LASTBRANCH_0_FROM_IP	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 0 From IP (R/W) One of 32 pairs of last branch record registers on the last branch record stack. The From_IP part of the stack contains pointers to the source instruction. See also: - Last Branch Record Stack TOS at 1C9H. - Section 19.6 and record format in Section 19.4.8.1.		Core
0:47	From Linear Address (R/W)	
62:48	Signed extension of bits 47:0.	
63	Mispred	
Register Address: 681H, 1665	MSR_LASTBRANCH_1_FROM_IP	
Last Branch Record 1 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 682H, 1666	MSR_LASTBRANCH_2_FROM_IP	
Last Branch Record 2 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 683H, 1667	MSR_LASTBRANCH_3_FROM_IP	
Last Branch Record 3 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 684H, 1668	MSR_LASTBRANCH_4_FROM_IP	
Last Branch Record 4 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 685H, 1669	MSR_LASTBRANCH_5_FROM_IP	
Last Branch Record 5 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 686H, 1670	MSR_LASTBRANCH_6_FROM_IP	
Last Branch Record 6 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 687H, 1671	MSR_LASTBRANCH_7_FROM_IP	
Last Branch Record 7 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 688H, 1672	MSR_LASTBRANCH_8_FROM_IP	
Last Branch Record 8 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 689H, 1673	MSR_LASTBRANCH_9_FROM_IP	
Last Branch Record 9 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 68AH, 1674	MSR_LASTBRANCH_10_FROM_IP	
Last Branch Record 10 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 68BH, 1675	MSR_LASTBRANCH_11_FROM_IP	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 11 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 68CH, 1676	MSR_LASTBRANCH_12_FROM_IP	
Last Branch Record 12 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 68DH, 1677	MSR_LASTBRANCH_13_FROM_IP	
Last Branch Record 13 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 68EH, 1678	MSR_LASTBRANCH_14_FROM_IP	
Last Branch Record 14 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 68FH, 1679	MSR_LASTBRANCH_15_FROM_IP	
Last Branch Record 15 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 690H, 1680	MSR_LASTBRANCH_16_FROM_IP	
Last Branch Record 16 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 691H, 1681	MSR_LASTBRANCH_17_FROM_IP	
Last Branch Record 17 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 692H, 1682	MSR_LASTBRANCH_18_FROM_IP	
Last Branch Record 18 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 693H, 1683	MSR_LASTBRANCH_19_FROM_IP	
Last Branch Record 19 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 694H, 1684	MSR_LASTBRANCH_20_FROM_IP	
Last Branch Record 20 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 695H, 1685	MSR_LASTBRANCH_21_FROM_IP	
Last Branch Record 21 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 696H, 1686	MSR_LASTBRANCH_22_FROM_IP	
Last Branch Record 22 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 697H, 1687	MSR_LASTBRANCH_23_FROM_IP	
Last Branch Record 23 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 698H, 1688	MSR_LASTBRANCH_24_FROM_IP	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 24 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 699H, 1689	MSR_LASTBRANCH_25_FROM_IP	
Last Branch Record 25 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 69AH, 1690	MSR_LASTBRANCH_26_FROM_IP	
Last Branch Record 26 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 69BH, 1691	MSR_LASTBRANCH_27_FROM_IP	
Last Branch Record 27 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 69CH, 1692	MSR_LASTBRANCH_28_FROM_IP	
Last Branch Record 28 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 69DH, 1693	MSR_LASTBRANCH_29_FROM_IP	
Last Branch Record 29 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 69EH, 1694	MSR_LASTBRANCH_30_FROM_IP	
Last Branch Record 30 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 69FH, 1695	MSR_LASTBRANCH_31_FROM_IP	
Last Branch Record 31 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Core
Register Address: 6C0H, 1728	MSR_LASTBRANCH_0_TO_IP	
Last Branch Record 0 To IP (R/W) One of 32 pairs of last branch record registers on the last branch record stack. The To_IP part of the stack contains pointers to the Destination instruction and elapsed cycles from last LBR update. See Section 19.6.		Core
0:47	Target Linear Address (R/W)	
63:48	Elapsed cycles from last update to the LBR.	
Register Address: 6C1H, 1729	MSR_LASTBRANCH_1_TO_IP	
Last Branch Record 1 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6C2H, 1730	MSR_LASTBRANCH_2_TO_IP	
Last Branch Record 2 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6C3H, 1731	MSR_LASTBRANCH_3_TO_IP	
Last Branch Record 3 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6C4H, 1732	MSR_LASTBRANCH_4_TO_IP	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 4 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6C5H, 1733	MSR_LASTBRANCH_5_TO_IP	
Last Branch Record 5 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6C6H, 1734	MSR_LASTBRANCH_6_TO_IP	
Last Branch Record 6 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6C7H, 1735	MSR_LASTBRANCH_7_TO_IP	
Last Branch Record 7 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6C8H, 1736	MSR_LASTBRANCH_8_TO_IP	
Last Branch Record 8 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6C9H, 1737	MSR_LASTBRANCH_9_TO_IP	
Last Branch Record 9 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6CAH, 1738	MSR_LASTBRANCH_10_TO_IP	
Last Branch Record 10 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6CBH, 1739	MSR_LASTBRANCH_11_TO_IP	
Last Branch Record 11 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6CCH, 1740	MSR_LASTBRANCH_12_TO_IP	
Last Branch Record 12 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6CDH, 1741	MSR_LASTBRANCH_13_TO_IP	
Last Branch Record 13 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6CEH, 1742	MSR_LASTBRANCH_14_TO_IP	
Last Branch Record 14 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6CFH, 1743	MSR_LASTBRANCH_15_TO_IP	
Last Branch Record 15 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6DOH, 1744	MSR_LASTBRANCH_16_TO_IP	
Last Branch Record 16 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D1H, 1745	MSR_LASTBRANCH_17_TO_IP	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 17 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D2H, 1746	MSR_LASTBRANCH_18_TO_IP	
Last Branch Record 18 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D3H, 1747	MSR_LASTBRANCH_19_TO_IP	
Last Branch Record 19 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D4H, 1748	MSR_LASTBRANCH_20_TO_IP	
Last Branch Record 20 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D5H, 1749	MSR_LASTBRANCH_21_TO_IP	
Last Branch Record 21 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D6H, 1750	MSR_LASTBRANCH_22_TO_IP	
Last Branch Record 22 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D7H, 1751	MSR_LASTBRANCH_23_TO_IP	
Last Branch Record 23 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D8H, 1752	MSR_LASTBRANCH_24_TO_IP	
Last Branch Record 24 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6D9H, 1753	MSR_LASTBRANCH_25_TO_IP	
Last Branch Record 25 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6DAH, 1754	MSR_LASTBRANCH_26_TO_IP	
Last Branch Record 26 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6DBH, 1755	MSR_LASTBRANCH_27_TO_IP	
Last Branch Record 27 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6DCH, 1756	MSR_LASTBRANCH_28_TO_IP	
Last Branch Record 28 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6DDH, 1757	MSR_LASTBRANCH_29_TO_IP	
Last Branch Record 29 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6DEH, 1758	MSR_LASTBRANCH_30_TO_IP	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 30 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 6DFH, 1759	MSR_LASTBRANCH_31_TO_IP	
Last Branch Record 31 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Core
Register Address: 802H, 2050	IA32_X2APIC_APICID	
x2APIC ID register (R/O)		Core
Register Address: 803H, 2051	IA32_X2APIC_VERSION	
x2APIC Version register (R/O)		Core
Register Address: 808H, 2056	IA32_X2APIC_TPR	
x2APIC Task Priority register (R/W)		Core
Register Address: 80AH, 2058	IA32_X2APIC_PPR	
x2APIC Processor Priority register (R/O)		Core
Register Address: 80BH, 2059	IA32_X2APIC_EOI	
x2APIC EOI register (W/O)		Core
Register Address: 80DH, 2061	IA32_X2APIC_LDR	
x2APIC Logical Destination register (R/O)		Core
Register Address: 80FH, 2063	IA32_X2APIC_SIVR	
x2APIC Spurious Interrupt Vector register (R/W)		Core
Register Address: 810H, 2064	IA32_X2APIC_ISR0	
x2APIC In-Service register bits [31:0] (R/O)		Core
Register Address: 811H, 2065	IA32_X2APIC_ISR1	
x2APIC In-Service register bits [63:32] (R/O)		Core
Register Address: 812H, 2066	IA32_X2APIC_ISR2	
x2APIC In-Service register bits [95:64] (R/O)		Core
Register Address: 813H, 2067	IA32_X2APIC_ISR3	
x2APIC In-Service register bits [127:96] (R/O)		Core
Register Address: 814H, 2068	IA32_X2APIC_ISR4	
x2APIC In-Service register bits [159:128] (R/O)		Core
Register Address: 815H, 2069	IA32_X2APIC_ISR5	
x2APIC In-Service register bits [191:160] (R/O)		Core
Register Address: 816H, 2070	IA32_X2APIC_ISR6	
x2APIC In-Service register bits [223:192] (R/O)		Core
Register Address: 817H, 2071	IA32_X2APIC_ISR7	
x2APIC In-Service register bits [255:224] (R/O)		Core
Register Address: 818H, 2072	IA32_X2APIC_TMR0	
x2APIC Trigger Mode register bits [31:0] (R/O)		Core
Register Address: 819H, 2073	IA32_X2APIC_TMR1	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
x2APIC Trigger Mode register bits [63:32] (R/O)		Core
Register Address: 81AH, 2074	IA32_X2APIC_TMR2	
x2APIC Trigger Mode register bits [95:64] (R/O)		Core
Register Address: 81BH, 2075	IA32_X2APIC_TMR3	
x2APIC Trigger Mode register bits [127:96] (R/O)		Core
Register Address: 81CH, 2076	IA32_X2APIC_TMR4	
x2APIC Trigger Mode register bits [159:128] (R/O)		Core
Register Address: 81DH, 2077	IA32_X2APIC_TMR5	
x2APIC Trigger Mode register bits [191:160] (R/O)		Core
Register Address: 81EH, 2078	IA32_X2APIC_TMR6	
x2APIC Trigger Mode register bits [223:192] (R/O)		Core
Register Address: 81FH, 2079	IA32_X2APIC_TMR7	
x2APIC Trigger Mode register bits [255:224] (R/O)		Core
Register Address: 820H, 2080	IA32_X2APIC_IRR0	
x2APIC Interrupt Request register bits [31:0] (R/O)		Core
Register Address: 821H, 2081	IA32_X2APIC_IRR1	
x2APIC Interrupt Request register bits [63:32] (R/O)		Core
Register Address: 822H, 2082	IA32_X2APIC_IRR2	
x2APIC Interrupt Request register bits [95:64] (R/O)		Core
Register Address: 823H, 2083	IA32_X2APIC_IRR3	
x2APIC Interrupt Request register bits [127:96] (R/O)		Core
Register Address: 824H, 2084	IA32_X2APIC_IRR4	
x2APIC Interrupt Request register bits [159:128] (R/O)		Core
Register Address: 825H, 2085	IA32_X2APIC_IRR5	
x2APIC Interrupt Request register bits [191:160] (R/O)		Core
Register Address: 826H, 2086	IA32_X2APIC_IRR6	
x2APIC Interrupt Request register bits [223:192] (R/O)		Core
Register Address: 827H, 2087	IA32_X2APIC_IRR7	
x2APIC Interrupt Request register bits [255:224] (R/O)		Core
Register Address: 828H, 2088	IA32_X2APIC_ESR	
x2APIC Error Status register (R/W)		Core
Register Address: 82FH, 2095	IA32_X2APIC_LVT_CMCI	
x2APIC LVT Corrected Machine Check Interrupt register (R/W)		Core
Register Address: 830H, 2096	IA32_X2APIC_ICR	
x2APIC Interrupt Command register (R/W)		Core
Register Address: 832H, 2098	IA32_X2APIC_LVT_TIMER	
x2APIC LVT Timer Interrupt register (R/W)		Core

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 833H, 2099	IA32_X2APIC_LVT_THERMAL	
x2APIC LVT Thermal Sensor Interrupt register (R/W)		Core
Register Address: 834H, 2100	IA32_X2APIC_LVT_PMI	
x2APIC LVT Performance Monitor register (R/W)		Core
Register Address: 835H, 2101	IA32_X2APIC_LVT_LINT0	
x2APIC LVT LINT0 register (R/W)		Core
Register Address: 836H, 2102	IA32_X2APIC_LVT_LINT1	
x2APIC LVT LINT1 register (R/W)		Core
Register Address: 837H, 2103	IA32_X2APIC_LVT_ERROR	
x2APIC LVT Error register (R/W)		Core
Register Address: 838H, 2104	IA32_X2APIC_INIT_COUNT	
x2APIC Initial Count register (R/W)		Core
Register Address: 839H, 2105	IA32_X2APIC_CUR_COUNT	
x2APIC Current Count register (R/O)		Core
Register Address: 83EH, 2110	IA32_X2APIC_DIV_CONF	
x2APIC Divide Configuration register (R/W)		Core
Register Address: 83FH, 2111	IA32_X2APIC_SELF_IPI	
x2APIC Self IPI register (W/O)		Core
Register Address: C8FH, 3215	IA32_PQR_ASSOC	
Resource Association Register (R/W)		Core
31:0	Reserved.	
33:32	CLOS (R/W)	
63: 34	Reserved.	
Register Address: D10H, 3344	IA32_L2_QOS_MASK_0	
L2 Class Of Service Mask - CLOS 0 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=0.		Module
0:7	CBM: Bit vector of available L2 ways for CLOS 0 enforcement.	
63:8	Reserved.	
Register Address: D11H, 3345	IA32_L2_QOS_MASK_1	
L2 Class Of Service Mask - CLOS 1 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=1.		Module
0:7	CBM: Bit vector of available L2 ways for CLOS 0 enforcement.	
63:8	Reserved.	
Register Address: D12H, 3346	IA32_L2_QOS_MASK_2	
L2 Class Of Service Mask - CLOS 2 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=2.		Module
0:7	CBM: Bit vector of available L2 ways for CLOS 0 enforcement.	
63:8	Reserved.	

Table 2-12. MSRs in Intel Atom® Processors Based on Goldmont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: D13H, 3347	IA32_L2_QOS_MASK_3	
L2 Class Of Service Mask - CLOS 3 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=3.		Package
0:19	CBM: Bit vector of available L2 ways for CLOS 3 enforcement.	
63:20	Reserved.	
Register Address: D90H, 3472	IA32_BNDCFGS	
See Table 2-2.		Core
Register Address: DA0H, 3488	IA32_XSS	
See Table 2-2.		Core
See Table 2-6, and Table 2-12 for MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_5CH.		

2.6 MSRS IN INTEL ATOM® PROCESSORS BASED ON GOLDMONT PLUS MICROARCHITECTURE

Intel Atom processors based on the Goldmont Plus microarchitecture support MSRs listed in Table 2-6, Table 2-12, and Table 2-13. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_7AH; see Table 2-1. For an MSR listed in Table 2-13 that also appears in the model-specific tables of prior generations, Table 2-13 supersedes prior generation tables.

In the Goldmont Plus microarchitecture, the scope column indicates the following: “Core” means each processor core has a separate MSR, or a bit field not shared with another processor core. “Module” means the MSR or the bit field is shared by a subset of the processor cores in the physical package. The number of processor cores in this subset is model specific and may differ between different processors. For all processors based on Goldmont Plus microarchitecture, the L2 cache is also shared between cores in a module and thus CPUID leaf 04H enumeration can be used to figure out which processors are in the same module. “Package” means all processor cores in the physical package share the same MSR or bit interface.

Table 2-13. MSRs in Intel Atom® Processors Based on Goldmont Plus Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64Processor (R/W) See Table 2-2.		Core
0	Lock (R/WL)	
1	Enable VMX inside SMX operation (R/WL)	
2	Enable VMX outside SMX operation (R/WL)	
14:8	SENTER local functions enables (R/WL)	
15	SENTER global functions enable (R/WL)	
17	SGX Launch Control Enable (R/WL) This bit must be set to enable runtime reconfiguration of SGX Launch Control via IA32_SGXLEPUBKEYHASHn MSR. Valid if CPUID.(EAX=07H, ECX=0H): ECX[30] = 1.	

Table 2-13. MSRs in Intel Atom® Processors Based on Goldmont Plus Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
18	SGX global functions enable (R/WL)	
63:19	Reserved.	
Register Address: 8CH, 140	IA32_SGXLEPUBKEYHASH0	
See Table 2-2.		Core
Register Address: 8DH, 141	IA32_SGXLEPUBKEYHASH1	
See Table 2-2.		Core
Register Address: 8EH, 142	IA32_SGXLEPUBKEYHASH2	
See Table 2-2.		Core
Register Address: 8FH, 143	IA32_SGXLEPUBKEYHASH3	
See Table 2-2.		Core
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
(R/W) See Table 2-2. See Section 21.6.2.4, "Processor Event Based Sampling (PEBS)."		Core
0	Enable PEBS trigger and recording for the programmed event (precise or otherwise) on IA32_PMC0.	
1	Enable PEBS trigger and recording for the programmed event (precise or otherwise) on IA32_PMC1.	
2	Enable PEBS trigger and recording for the programmed event (precise or otherwise) on IA32_PMC2.	
3	Enable PEBS trigger and recording for the programmed event (precise or otherwise) on IA32_PMC3.	
31:4	Reserved.	
32	Enable PEBS trigger and recording for IA32_FIXED_CTR0.	
33	Enable PEBS trigger and recording for IA32_FIXED_CTR1.	
34	Enable PEBS trigger and recording for IA32_FIXED_CTR2.	
63:35	Reserved.	
Register Address: 570H, 1392	IA32_RTIT_CTL	
Trace Control Register (R/W)		Core
0	TraceEn	
1	CYCEn	
2	OS	
3	User	
4	PwrEvtEn	
5	FUPonPTW	
6	FabricEn	
7	CR3Filter	
8	ToPA Writing 0 will #GP if also setting TraceEn.	
9	MTCEn	
10	TSCEn	

Table 2-13. MSRs in Intel Atom® Processors Based on Goldmont Plus Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
11	DisRETC	
12	PTWEn	
13	BranchEn	
17:14	MTCFreq	
18	Reserved, must be zero.	
22:19	CycThresh	
23	Reserved, must be zero.	
27:24	PSBFreq	
31:28	Reserved, must be zero.	
35:32	ADDR0_CFG	
39:36	ADDR1_CFG	
63:40	Reserved, must be zero.	
Register Address: 680H, 1664	MSR_LASTBRANCH_0_FROM_IP	
Last Branch Record 0 From IP (R/W) One of the three MSRs that make up the first entry of the 32-entry LBR stack. The From_IP part of the stack contains pointers to the source instruction. See also: ▪ Last Branch Record Stack TOS at 1C9H. ▪ Section 19.7, “Last Branch, Call Stack, Interrupt, and Exception Recording for Processors based on Goldmont Plus Microarchitecture.”		Core
Register Address: 681H–69FH, 1665–1695	MSR_LASTBRANCH_1_FROM_IP	
Last Branch Record 1 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP; <i>i</i> = 1-31.		Core
Register Address: 6C0H, 1728	MSR_LASTBRANCH_0_TO_IP	
Last Branch Record 0 To IP (R/W) One of the three MSRs that make up the first entry of the 32-entry LBR stack. The To_IP part of the stack contains pointers to the Destination instruction. See also: ▪ Section 19.7, “Last Branch, Call Stack, Interrupt, and Exception Recording for Processors based on Goldmont Plus Microarchitecture.”		Core
Register Address: 6C1H–6DFH, 1729–1759	MSR_LASTBRANCH_1_TO_IP	
Last Branch Record 1 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP; <i>i</i> = 1-31.		Core
Register Address: DCOH, 3520	MSR_LASTBRANCH_INFO_0	
Last Branch Record 0 Additional Information (R/W) One of the three MSRs that make up the first entry of the 32-entry LBR stack. This part of the stack contains flag and elapsed cycle information. See also: ▪ Last Branch Record Stack TOS at 1C9H. ▪ Section 19.9.1, “LBR Stack.”		Core
Register Address: DC1H, 3521	MSR_LASTBRANCH_INFO_1	
Last Branch Record 1 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core

Table 2-13. MSRs in Intel Atom® Processors Based on Goldmont Plus Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: DC2H, 3522	MSR_LASTBRANCH_INFO_2	
Last Branch Record 2 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DC3H, 3523	MSR_LASTBRANCH_INFO_3	
Last Branch Record 3 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DC4H, 3524	MSR_LASTBRANCH_INFO_4	
Last Branch Record 4 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DC5H, 3525	MSR_LASTBRANCH_INFO_5	
Last Branch Record 5 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DC6H, 3526	MSR_LASTBRANCH_INFO_6	
Last Branch Record 6 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DC7H, 3527	MSR_LASTBRANCH_INFO_7	
Last Branch Record 7 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DC8H, 3528	MSR_LASTBRANCH_INFO_8	
Last Branch Record 8 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DC9H, 3529	MSR_LASTBRANCH_INFO_9	
Last Branch Record 9 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DCAH, 3530	MSR_LASTBRANCH_INFO_10	
Last Branch Record 10 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DCBH, 3531	MSR_LASTBRANCH_INFO_11	
Last Branch Record 11 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DCCH, 3532	MSR_LASTBRANCH_INFO_12	
Last Branch Record 12 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DCDH, 3533	MSR_LASTBRANCH_INFO_13	
Last Branch Record 13 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DCEH, 3534	MSR_LASTBRANCH_INFO_14	
Last Branch Record 14 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core

Table 2-13. MSRs in Intel Atom® Processors Based on Goldmont Plus Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: DCFH, 3535	MSR_LASTBRANCH_INFO_15	
Last Branch Record 15 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DDOH, 3536	MSR_LASTBRANCH_INFO_16	
Last Branch Record 16 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD1H, 3537	MSR_LASTBRANCH_INFO_17	
Last Branch Record 17 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD2H, 3538	MSR_LASTBRANCH_INFO_18	
Last Branch Record 18 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD3H, 3539	MSR_LASTBRANCH_INFO_19	
Last Branch Record 19 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD4H, 3520	MSR_LASTBRANCH_INFO_20	
Last Branch Record 20 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD5H, 3521	MSR_LASTBRANCH_INFO_21	
Last Branch Record 21 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD6H, 3522	MSR_LASTBRANCH_INFO_22	
Last Branch Record 22 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD7H, 3523	MSR_LASTBRANCH_INFO_23	
Last Branch Record 23 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD8H, 3524	MSR_LASTBRANCH_INFO_24	
Last Branch Record 24 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DD9H, 3525	MSR_LASTBRANCH_INFO_25	
Last Branch Record 25 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DDAH, 3526	MSR_LASTBRANCH_INFO_26	
Last Branch Record 26 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DDBH, 3527	MSR_LASTBRANCH_INFO_27	
Last Branch Record 27 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core

Table 2-13. MSRs in Intel Atom® Processors Based on Goldmont Plus Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: DDCH, 3528	MSR_LASTBRANCH_INFO_28	
Last Branch Record 28 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DDDH, 3529	MSR_LASTBRANCH_INFO_29	
Last Branch Record 29 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DDEH, 3530	MSR_LASTBRANCH_INFO_30	
Last Branch Record 30 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
Register Address: DDFH, 3531	MSR_LASTBRANCH_INFO_31	
Last Branch Record 31 Additional Information (R/W) See description of MSR_LASTBRANCH_INFO_0.		Core
See Table 2-6, Table 2-12, and Table 2-13 for MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_7AH.		

2.7 MSRS IN INTEL ATOM® PROCESSORS BASED ON TREMONT MICROARCHITECTURE

Processors based on the Tremont microarchitecture support MSRs listed in Table 2-6, Table 2-12, Table 2-13, and Table 2-14. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_86H, 06_96H, or 06_9CH; see Table 2-1. For an MSR listed in Table 2-14 that also appears in the model-specific tables of prior generations, Table 2-14 supersedes prior generation tables.

In the Tremont microarchitecture, the scope column indicates the following: “Core” means each processor core has a separate MSR, or a bit field not shared with another processor core. “Module” means the MSR or the bit field is shared by a subset of the processor cores in the physical package. The number of processor cores in this subset is model specific and may differ between different processors. For all processors based on Tremont microarchitecture, the L2 cache is also shared between cores in a module and thus CPUID leaf 04H enumeration can be used to figure out which processors are in the same module. “Package” means all processor cores in the physical package share the same MSR or bit interface.

Table 2-14. MSRs in Intel Atom® Processors Based on Tremont Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 33H, 51	MSR_MEMORY_CTRL	
Memory Control Register		Core
28:0	Reserved.	
29	SPLIT_LOCK_DISABLE If set to 1, a split lock will cause an #AC(0) exception. See Section 10.1.2.3, “Features to Disable Bus Locks.”	
30	Reserved.	
31	Reserved.	
Register Address: CFH, 207	IA32_CORE_CAPABILITIES	

Table 2-14. MSRs in Intel Atom® Processors Based on Tremont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
IA32 Core Capabilities Register If CPUID.(EAX=07H, ECX=0):EDX[30] = 1.		Core
4:0	Reserved.	
5	SPLIT_LOCK_DISABLE_SUPPORTED When read as 1, software can set bit 29 of MSR_MEMORY_CTRL (MSR address 33H).	
63:6	Reserved.	
Register Address: 2A0H, 672	MSR_PRMRR_BASE_0	
Processor Reserved Memory Range Register - Physical Base Control Register (R/W)		Core
2:0	MEMTYPE: PRMRR BASE Memory Type.	
3	CONFIGURED: PRMRR BASE Configured.	
11:4	Reserved.	
51:12	BASE: PRMRR Base Address.	
63:52	Reserved.	
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
(R/W) See Table 2-2. See Section 21.6.2.4, "Processor Event Based Sampling (PEBS)."		Core
$n:0$	Enable PEBS trigger and recording for the programmed event (precise or otherwise) on IA32_PMCx. The maximum value n can be determined from CPUID.0AH:EAX[15:8].	
31: $n+1$	Reserved.	
32: $m:32$	Enable PEBS trigger and recording for IA32_FIXED_CTRx. The maximum value m can be determined from CPUID.0AH:EDX[4:0].	
59:33+m	Reserved.	
60	Pend a PerfMon Interrupt (PMI) after each PEBS event.	
62:61	Specifies PEBS output destination. Encodings: 00B: DS Save Area. 01B: Intel PT trace output. Supported if IA32_PERF_CAPABILITIES.PEBS_OUTPUT_PT_AVAIL[16] and CPUID.07H.0.EBX[25] are set. 10B: Reserved. 11B: Reserved.	
63	Reserved.	
Register Address: 1309H–130BH, 4873–4875	MSR_RELOAD_FIXED_CTRx	
Reload value for IA32_FIXED_CTRx (R/W)		
47:0	Value loaded into IA32_FIXED_CTRx when a PEBS record is generated while PEBS_EN_FIXEDx = 1 and PEBS_OUTPUT = 01B in IA32_PEBS_ENABLE, and FIXED_CTRx is overflowed.	
63:48	Reserved.	
Register Address: 14C1H–14C4H, 5313–5316	MSR_RELOAD_PMCx	

Table 2-14. MSRs in Intel Atom® Processors Based on Tremont Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Reload value for IA32_PMCx (R/W)		Core
47:0	Value loaded into IA32_PMCx when a PEBS record is generated while PEBS_EN_PMCx = 1 and PEBS_OUTPUT = 01B in IA32_PEBS_ENABLE, and PMCx is overflowed.	
63:48	Reserved.	
See Table 2-6, Table 2-12, Table 2-13, and Table 2-14 for MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_86H.		

2.8 MSRS IN PROCESSORS BASED ON NEHALEM MICROARCHITECTURE

Table 2-15 lists model-specific registers (MSRs) that are common for Nehalem microarchitecture. These include the Intel Core i7 and i5 processor family. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_1AH, 06_1EH, 06_1FH, or 06_2EH; see Table 2-1. Additional MSRs specific to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_1AH, 06_1EH, or 06_1FH are listed in Table 2-16. Some MSRs listed in these tables are used by BIOS. More information about these MSR can be found at <http://biosbits.org>.

The column “Scope” represents the package/core/thread scope of individual bit field of an MSR. “Thread” means this bit field must be programmed on each logical processor independently. “Core” means the bit field must be programmed on each processor core independently, logical processors in the same core will be affected by change of this bit on the other logical processor in the same core. “Package” means the bit field must be programmed once for each physical package. Change of a bit filed with a package scope will affect all logical processors in that physical package.

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 0H, 0	IA32_P5_MC_ADDR	
See Section 2.23, “MSRs in Pentium Processors.”		Thread
Register Address: 1H, 1	IA32_P5_MC_TYPE	
See Section 2.23, “MSRs in Pentium Processors.”		Thread
Register Address: 6H, 6	IA32_MONITOR_FILTER_SIZE	
See Section 10.10.5, “Monitor/Mwait Address Range Determination,” and Table 2-2.		Thread
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER	
See Section 19.17, “Time-Stamp Counter,” and Table 2-2.		Thread
Register Address: 17H, 23	IA32_PLATFORM_ID	
Platform ID (R) See Table 2-2.		Package
Register Address: 17H, 23	MSR_PLATFORM_ID	
Model Specific Platform ID (R)		Package
49:0	Reserved.	
52:50	See Table 2-2.	
63:53	Reserved.	
Register Address: 1BH, 27	IA32_APIC_BASE	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 12.4.4, "Local APIC Status and Location," and Table 2-2.		Thread
Register Address: 34H, 52	MSR_SMI_COUNT	
SMI Counter (R/O)		Thread
31:0	SMI Count (R/O) Running count of SMI events since last RESET.	
63:32	Reserved.	
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64Processor (R/W) See Table 2-2.		Thread
Register Address: 79H, 121	IA32_BIOS_UPDT_TRIG	
BIOS Update Trigger Register (W) See Table 2-2.		Core
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID	
BIOS Update Signature ID (R/W) See Table 2-2.		Thread
Register Address: C1H, 193	IA32_PMC0	
Performance Counter Register See Table 2-2.		Thread
Register Address: C2H, 194	IA32_PMC1	
Performance Counter Register See Table 2-2.		Thread
Register Address: C3H, 195	IA32_PMC2	
Performance Counter Register See Table 2-2.		Thread
Register Address: C4H, 196	IA32_PMC3	
Performance Counter Register See Table 2-2.		Thread
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) This is the ratio of the frequency that invariant TSC runs at. The invariant TSC frequency can be computed by multiplying this ratio by 133.33 MHz.	Package
27:16	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) When set to 1, indicates that Programmable Ratio Limit for Turbo mode is enabled. When set to 0, indicates Programmable Ratio Limit for Turbo mode is disabled.	Package

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
29	Programmable TDC-TDP Limit for Turbo Mode (R/O) When set to 1, indicates that TDC and TDP Limits for Turbo mode are programmable. When set to 0, indicates TDC and TDP Limits for Turbo mode are not programmable.	Package
39:30	Reserved.	
47:40	Maximum Efficiency Ratio (R/O) This is the minimum ratio (maximum efficiency) that the processor can operate, in units of 133.33MHz.	Package
63:48	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States. See http://biosbits.org .		Core
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: C0 (no package C-state support) 001b: C1 (Behavior is the same as 000b) 010b: C3 011b: C6 100b: C7 101b and 110b: Reserved 111: No package C-state limit. Note: This field cannot be used to limit package C-state to C3.	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W) When set, will map IO_read instructions sent to IO register specified by MSR_PMG_IO_CAPTURE_BASE to MWAIT instructions.	
14:11	Reserved.	
15	CFG Lock (R/WO) When set, locks bits 15:0 of this register until next reset.	
23:16	Reserved.	
24	Interrupt filtering enable (R/W) When set, processor cores in a deep C-State will wake only when the event message is destined for that core. When 0, all processor cores in a deep C-State will wake for an event message.	
25	C3 state auto demotion enable (R/W) When set, the processor will conditionally demote C6/C7 requests to C3 based on uncore auto-demote information.	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
26	C1 state auto demotion enable (R/W) When set, the processor will conditionally demote C3/C6/C7 requests to C1 based on uncore auto-demote information.	
27	Enable C3 Undemotion (R/W)	
28	Enable C1 Undemotion (R/W)	
29	Package C State Demotion Enable (R/W)	
30	Package C State Undemotion Enable (R/W)	
63:31	Reserved.	
Register Address: E4H, 228	MSR_PMG_IO_CAPTURE_BASE	
Power Management IO Redirection in C-state (R/W) See http://biosbits.org .		Core
15:0	LVL_2 Base Address (R/W) Specifies the base address visible to software for IO redirection. If IO MWAIT Redirection is enabled, reads to this address will be consumed by the power management logic and decoded to MWAIT instructions. When IO port address redirection is enabled, this is the IO port address reported to the OS/software.	
18:16	C-state Range (R/W) Specifies the encoding value of the maximum C-State code name to be included when IO read to MWAIT redirection is enabled by MSR_PKG_CST_CONFIG_CONTROL[bit10]: 000b - C3 is the max C-State to include. 001b - C6 is the max C-State to include. 010b - C7 is the max C-State to include.	
63:19	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: FEH, 254	IA32_MTRRCAP	
See Table 2-2.		Thread
Register Address: 174H, 372	IA32_SYSENTER_CS	
See Table 2-2.		Thread
Register Address: 175H, 373	IA32_SYSENTER_ESP	
See Table 2-2.		Thread
Register Address: 176H, 374	IA32_SYSENTER_EIP	
See Table 2-2.		Thread
Register Address: 179H, 377	IA32_MCG_CAP	
See Table 2-2.		Thread

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 17AH, 378	IA32_MCG_STATUS	
Global Machine Check Status		Thread
0	RIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) can be used to restart the program. If cleared, the program cannot be reliably restarted.	
1	EIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) is directly associated with the error.	
2	MCIP When set, bit indicates that a machine check has been generated. If a second machine check is detected while this bit is still set, the processor enters a shutdown state. Software should write this bit to 0 after processing a machine check exception.	
63:3	Reserved.	
Register Address: 186H, 390	IA32_PERFEVTSEL0	
See Table 2-2.		Thread
7:0	Event Select	
15:8	UMask	
16	USR	
17	OS	
18	Edge	
19	PC	
20	INT	
21	AnyThread	
22	EN	
23	INV	
31:24	CMASK	
63:32	Reserved.	
Register Address: 187H, 391	IA32_PERFEVTSEL1	
See Table 2-2.		Thread
Register Address: 188H, 392	IA32_PERFEVTSEL2	
See Table 2-2.		Thread
Register Address: 189H, 393	IA32_PERFEVTSEL3	
See Table 2-2.		Thread
Register Address: 198H, 408	IA32_PERF_STATUS	
See Table 2-2.		Core
15:0	Current Performance State Value.	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:16	Reserved.	
Register Address: 199H, 409	IA32_PERF_CTL	
See Table 2-2.		Thread
Register Address: 19AH, 410	IA32_CLOCK_MODULATION	
Clock Modulation (R/W) See Table 2-2. IA32_CLOCK_MODULATION MSR was originally named IA32_THERM_CONTROL MSR.		Thread
0	Reserved.	
3:1	On demand Clock Modulation Duty Cycle (R/W)	
4	On demand Clock Modulation Enable (R/W)	
63:5	Reserved.	
Register Address: 19BH, 411	IA32_THERM_INTERRUPT	
Thermal Interrupt Control (R/W) See Table 2-2.		Core
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Core
Register Address: 1A0H, 416	IA32_MISC_ENABLE	
Enable Misc. Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		
0	Fast-Strings Enable See Table 2-2.	Thread
2:1	Reserved.	
3	Automatic Thermal Control Circuit Enable (R/W) See Table 2-2. Default value is 1.	Thread
6:4	Reserved.	
7	Performance Monitoring Available (R) See Table 2-2.	Thread
10:8	Reserved.	
11	Branch Trace Storage Unavailable (R/O) See Table 2-2.	Thread
12	Processor Event Based Sampling Unavailable (R/O) See Table 2-2.	Thread
15:13	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W) See Table 2-2.	Package
18	ENABLE MONITOR FSM. (R/W) See Table 2-2.	Thread
21:19	Reserved.	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
22	Limit CPUID Maxval (R/W) See Table 2-2.	Thread
23	xTPR Message Disable (R/W) See Table 2-2.	Thread
33:24	Reserved.	
34	XD Bit Disable (R/W) See Table 2-3.	Thread
37:35	Reserved.	
38	Turbo Mode Disable (R/W) When set to 1 on processors that support Intel Turbo Boost Technology, the turbo mode feature is disabled and the IDA_Enable feature flag will be clear (CPUID.06H: EAX[1]=0). When set to a 0 on processors that support IDA, CPUID.06H: EAX[1] reports the processor's support of turbo mode is enabled. Note: The power-on default value is used by BIOS to detect hardware support of turbo mode. If the power-on default value is 1, turbo mode is available in the processor. If the power-on default value is 0, turbo mode is not available.	Package
63:39	Reserved.	
Register Address: 1A2H, 418	MSR_TEMPERATURE_TARGET	
Temperature Target		Thread
15:0	Reserved.	
23:16	Temperature Target (R) The minimum temperature at which PROCHOT# will be asserted. The value is degrees C.	
63:24	Reserved.	
Register Address: 1A4H, 420	MSR_MISC_FEATURE_CONTROL	
Miscellaneous Feature Control (R/W)		
0	L2 Hardware Prefetcher Disable (R/W) If 1, disables the L2 hardware prefetcher, which fetches additional lines of code or data into the L2 cache.	Core
1	L2 Adjacent Cache Line Prefetcher Disable (R/W) If 1, disables the adjacent cache line prefetcher, which fetches the cache line that comprises a cache line pair (128 bytes).	Core
2	DCU Hardware Prefetcher Disable (R/W) If 1, disables the L1 data cache prefetcher, which fetches the next cache line into L1 data cache.	Core
3	DCU IP Prefetcher Disable (R/W) If 1, disables the L1 data cache IP prefetcher, which uses sequential load history (based on instruction pointer of previous loads) to determine whether to prefetch additional lines.	Core
63:4	Reserved.	
Register Address: 1A6H, 422	MSR_OFFCORE_RSP_0	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Offcore Response Event Select Register (R/W)		Thread
Register Address: 1AAH, 426	MSR_MISC_PWR_MGMT	
Miscellaneous Power Management Control Various model specific features enumeration. See http://biosbits.org .		
0	EIST Hardware Coordination Disable (R/W) When 0, enables hardware coordination of Enhanced Intel Speedstep Technology request from processor cores. When 1, disables hardware coordination of Enhanced Intel Speedstep Technology requests.	Package
1	Energy/Performance Bias Enable (R/W) This bit makes the IA32_ENERGY_PERF_BIAS register (MSR 1B0h) visible to software with Ring 0 privileges. This bit's status (1 or 0) is also reflected by CPUID.(EAX=06h):ECX[3].	Thread
63:2	Reserved.	
Register Address: 1ACH, 428	MSR_TURBO_POWER_CURRENT_LIMIT	
See http://biosbits.org .		
14:0	TDP Limit (R/W) TDP limit in 1/8 Watt granularity.	Package
15	TDP Limit Override Enable (R/W) A value = 0 indicates override is not active; a value = 1 indicates override is active.	Package
30:16	TDC Limit (R/W) TDC limit in 1/8 Amp granularity.	Package
31	TDC Limit Override Enable (R/W) A value = 0 indicates override is not active; a value = 1 indicates override is active.	Package
63:32	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0. R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 core active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 core active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 core active.	Package
63:32	Reserved.	
Register Address: 1C8H, 456	MSR_LBR_SELECT	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record Filtering Select Register (R/W) See Section 19.9.2, "Filtering of Last Branch Records."		Core
0	CPL_EQ_0	
1	CPL_NEQ_0	
2	JCC	
3	NEAR_REL_CALL	
4	NEAR_IND_CALL	
5	NEAR_RET	
6	NEAR_IND_JMP	
7	NEAR_REL_JMP	
8	FAR_BRANCH	
63:9	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-3) that points to the MSR containing the most recent branch record. See MSR_LASTBRANCH_0_FROM_IP (at 680H).		Thread
Register Address: 1D9H, 473	IA32_DEBUGCTL	
Debug Control (R/W) See Table 2-2.		Thread
Register Address: 1DDH, 477	MSR_LER_FROM_LIP	
Last Exception Record From Linear IP (R) Contains a pointer to the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Thread
Register Address: 1DEH, 478	MSR_LER_TO_LIP	
Last Exception Record To Linear IP (R) This area contains a pointer to the target of the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Thread
Register Address: 1F2H, 498	IA32_SMRR_PHYSBASE	
See Table 2-2.		Core
Register Address: 1F3H, 499	IA32_SMRR_PHYSMASK	
See Table 2-2.		Core
Register Address: 1FCH, 508	MSR_POWER_CTL	
Power Control Register See http://biosbits.org .		Core
0	Reserved.	
1	C1E Enable (R/W) When set to '1', will enable the CPU to switch to the Minimum Enhanced Intel SpeedStep Technology operating point when all execution cores enter MWAIT (C1).	Package
63:2	Reserved.	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 200H, 512	IA32_MTRR_PHYSBASE0	
See Table 2-2.		Thread
Register Address: 201H, 513	IA32_MTRR_PHYSMASK0	
See Table 2-2.		Thread
Register Address: 202H, 514	IA32_MTRR_PHYSBASE1	
See Table 2-2.		Thread
Register Address: 203H, 515	IA32_MTRR_PHYSMASK1	
See Table 2-2.		Thread
Register Address: 204H, 516	IA32_MTRR_PHYSBASE2	
See Table 2-2.		Thread
Register Address: 205H, 517	IA32_MTRR_PHYSMASK2	
See Table 2-2.		Thread
Register Address: 206H, 518	IA32_MTRR_PHYSBASE3	
See Table 2-2.		Thread
Register Address: 207H, 519	IA32_MTRR_PHYSMASK3	
See Table 2-2.		Thread
Register Address: 208H, 520	IA32_MTRR_PHYSBASE4	
See Table 2-2.		Thread
Register Address: 209H, 521	IA32_MTRR_PHYSMASK4	
See Table 2-2.		Thread
Register Address: 20AH, 522	IA32_MTRR_PHYSBASE5	
See Table 2-2.		Thread
Register Address: 20BH, 523	IA32_MTRR_PHYSMASK5	
See Table 2-2.		Thread
Register Address: 20CH, 524	IA32_MTRR_PHYSBASE6	
See Table 2-2.		Thread
Register Address: 20DH, 525	IA32_MTRR_PHYSMASK6	
See Table 2-2.		Thread
Register Address: 20EH, 526	IA32_MTRR_PHYSBASE7	
See Table 2-2.		Thread
Register Address: 20FH, 527	IA32_MTRR_PHYSMASK7	
See Table 2-2.		Thread
Register Address: 210H, 528	IA32_MTRR_PHYSBASE8	
See Table 2-2.		Thread
Register Address: 211H, 529	IA32_MTRR_PHYSMASK8	
See Table 2-2.		Thread
Register Address: 212H, 530	IA32_MTRR_PHYSBASE9	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2.		Thread
Register Address: 213H, 531	IA32_MTRR_PHYSMASK9	
See Table 2-2.		Thread
Register Address: 250H, 592	IA32_MTRR_FIX64K_00000	
See Table 2-2.		Thread
Register Address: 258H, 600	IA32_MTRR_FIX16K_80000	
See Table 2-2.		Thread
Register Address: 259H, 601	IA32_MTRR_FIX16K_A0000	
See Table 2-2.		Thread
Register Address: 268H, 616	IA32_MTRR_FIX4K_C0000	
See Table 2-2.		Thread
Register Address: 269H, 617	IA32_MTRR_FIX4K_C8000	
See Table 2-2.		Thread
Register Address: 26AH, 618	IA32_MTRR_FIX4K_D0000	
See Table 2-2.		Thread
Register Address: 26BH, 619	IA32_MTRR_FIX4K_D8000	
See Table 2-2.		Thread
Register Address: 26CH, 620	IA32_MTRR_FIX4K_E0000	
See Table 2-2.		Thread
Register Address: 26DH, 621	IA32_MTRR_FIX4K_E8000	
See Table 2-2.		Thread
Register Address: 26EH, 622	IA32_MTRR_FIX4K_F0000	
See Table 2-2.		Thread
Register Address: 26FH, 623	IA32_MTRR_FIX4K_F8000	
See Table 2-2.		Thread
Register Address: 277H, 631	IA32_PAT	
See Table 2-2.		Thread
Register Address: 280H, 640	IA32_MC0_CTL2	
See Table 2-2.		Package
Register Address: 281H, 641	IA32_MC1_CTL2	
See Table 2-2.		Package
Register Address: 282H, 642	IA32_MC2_CTL2	
See Table 2-2.		Core
Register Address: 283H, 643	IA32_MC3_CTL2	
See Table 2-2.		Core
Register Address: 284H, 644	IA32_MC4_CTL2	
See Table 2-2.		Core

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 285H, 645	IA32_MC5_CTL2	
See Table 2-2.		Core
Register Address: 286H, 646	IA32_MC6_CTL2	
See Table 2-2.		Package
Register Address: 287H, 647	IA32_MC7_CTL2	
See Table 2-2.		Package
Register Address: 288H, 648	IA32_MC8_CTL2	
See Table 2-2.		Package
Register Address: 2FFH, 767	IA32_MTRR_DEF_TYPE	
Default Memory Types (R/W) See Table 2-2.		Thread
Register Address: 309H, 777	IA32_FIXED_CTR0	
Fixed-Function Performance Counter Register 0 (R/W) See Table 2-2.		Thread
Register Address: 30AH, 778	IA32_FIXED_CTR1	
Fixed-Function Performance Counter Register 1 (R/W) See Table 2-2.		Thread
Register Address: 30BH, 779	IA32_FIXED_CTR2	
Fixed-Function Performance Counter Register 2 (R/W) See Table 2-2.		Thread
Register Address: 345H, 837	IA32_PERF_CAPABILITIES	
See Table 2-2. See Section 19.4.1, "IA32_DEBUGCTL MSR."		Thread
5:0	LBR Format See Table 2-2.	
6	PEBS Record Format	
7	PEBSSaveArchRegs See Table 2-2.	
11:8	PEBS_REC_FORMAT See Table 2-2.	
12	SMM_FREEZE See Table 2-2.	
63:13	Reserved.	
Register Address: 38DH, 909	IA32_FIXED_CTR_CTRL	
Fixed-Function-Counter Control Register (R/W) See Table 2-2.		Thread
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Thread
Register Address: 38EH, 910	MSR_PERF_GLOBAL_STATUS	
Provides single-bit status used by software to query the overflow condition of each performance counter. (R/O)		Thread

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
61	UNC_Ovf Uncore overflowed if 1.	
Register Address: 38FH, 911	IA32_PERF_GLOBAL_CTRL	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities."		Thread
Register Address: 390H, 912	IA32_PERF_GLOBAL_OVF_CTRL	
See Table 2-2. See Section 21.6.2.2, "Global Counter Control Facilities." Allows software to clear counter overflow conditions on any combination of fixed-function PMCs (IA32_FIXED_CTRx) or general-purpose PMCs via a single WRMSR.		Thread
Register Address: 390H, 912	MSR_PERF_GLOBAL_OVF_CTRL	
(R/W)		Thread
61	CLR_UNC_Ovf Set 1 to clear UNC_Ovf.	
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
See Section 21.3.1.1.1, "Processor Event Based Sampling (PEBS)."		Thread
0	Enable PEBS on IA32_PMC0 (R/W)	
1	Enable PEBS on IA32_PMC1 (R/W)	
2	Enable PEBS on IA32_PMC2 (R/W)	
3	Enable PEBS on IA32_PMC3 (R/W)	
31:4	Reserved.	
32	Enable Load Latency on IA32_PMC0 (R/W)	
33	Enable Load Latency on IA32_PMC1 (R/W)	
34	Enable Load Latency on IA32_PMC2 (R/W)	
35	Enable Load Latency on IA32_PMC3 (R/W)	
63:36	Reserved.	
Register Address: 3F6H, 1014	MSR_PEBS_LD_LAT	
See Section 21.3.1.1.2, "Load Latency Performance Monitoring Facility."		Thread
15:0	Minimum threshold latency value of tagged load operation that will be counted. (R/W)	
63:36	Reserved.	
Register Address: 3F8H, 1016	MSR_PKG_C3_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C3 Residency Counter (R/O) Value since last reset that this package is in processor-specific C3 states. Count at the same frequency as the TSC.	
Register Address: 3F9H, 1017	MSR_PKG_C6_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:0	Package C6 Residency Counter (R/O) Value since last reset that this package is in processor-specific C6 states. Count at the same frequency as the TSC.	
Register Address: 3FAH, 1018	MSR_PKG_C7_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C7 Residency Counter (R/O) Value since last reset that this package is in processor-specific C7 states. Count at the same frequency as the TSC.	
Register Address: 3FCH, 1020	MSR_CORE_C3_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C3 Residency Counter (R/O) Value since last reset that this core is in processor-specific C3 states. Count at the same frequency as the TSC.	
Register Address: 3FDH, 1021	MSR_CORE_C6_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C6 Residency Counter (R/O) Value since last reset that this core is in processor-specific C6 states. Count at the same frequency as the TSC.	
Register Address: 400H, 1024	IA32_MCO_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 401H, 1025	IA32_MCO_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Package
Register Address: 402H, 1026	IA32_MCO_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MCO_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MCO_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Package
Register Address: 403H, 1027	IA32_MCO_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 404H, 1028	IA32_MC1_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 405H, 1029	IA32_MC1_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Package
Register Address: 406H, 1030	IA32_MC1_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MC1_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC1_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Package

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 407H, 1031	IA32_MC1_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 408H, 1032	IA32_MC2_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 409H, 1033	IA32_MC2_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core
Register Address: 40AH, 1034	IA32_MC2_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Core
Register Address: 40BH, 1035	IA32_MC2_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Core
Register Address: 40CH, 1036	IA32_MC3_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 40DH, 1037	IA32_MC3_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core
Register Address: 40EH, 1038	IA32_MC3_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The MSR_MC4_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Core
Register Address: 40FH, 1039	IA32_MC3_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Core
Register Address: 410H, 1040	IA32_MC4_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 411H, 1041	IA32_MC4_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core
Register Address: 412H, 1042	IA32_MC4_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The MSR_MC3_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC3_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Core
Register Address: 413H, 1043	IA32_MC4_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Core
Register Address: 414H, 1044	IA32_MC5_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 415H, 1045	IA32_MC5_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 416H, 1046	IA32_MC5_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Core
Register Address: 417H, 1047	IA32_MC5_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Core
Register Address: 418H, 1048	IA32_MC6_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 419H, 1049	IA32_MC6_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs," and Chapter 18.		Package
Register Address: 41AH, 1050	IA32_MC6_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 41BH, 1051	IA32_MC6_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 41CH, 1052	IA32_MC7_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 41DH, 1053	IA32_MC7_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs," and Chapter 18.		Package
Register Address: 41EH, 1054	IA32_MC7_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 41FH, 1055	IA32_MC7_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 420H, 1056	IA32_MC8_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 421H, 1057	IA32_MC8_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs," and Chapter 18.		Package
Register Address: 422H, 1058	IA32_MC8_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 423H, 1059	IA32_MC8_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 480H, 1152	IA32_VMX_BASIC	
Reporting Register of Basic VMX Capabilities (R/O) See Table 2-2 and Appendix A.1, "Basic VMX Information."		Thread
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL	
Capability Reporting Register of Pin-based VM-execution Controls (R/O) See Table 2-2 and Appendix A.3, "VM-Execution Controls."		Thread
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL	
Capability Reporting Register of Primary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Thread
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Capability Reporting Register of VM-Exit Controls (R/O) See Table 2-2 and Appendix A.4, "VM-Exit Controls."		Thread
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTLS	
Capability Reporting Register of VM-Entry Controls (R/O) See Table 2-2 and Appendix A.5, "VM-Entry Controls."		Thread
Register Address: 485H, 1157	IA32_VMX_MISC	
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Table 2-2 and Appendix A.6, "Miscellaneous Data."		Thread
Register Address: 486H, 1158	IA32_VMX_CR0_FIXED0	
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Table 2-2 and Appendix A.7, "VMX-Fixed Bits in CR0."		Thread
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1	
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Table 2-2 and Appendix A.7, "VMX-Fixed Bits in CR0."		Thread
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0	
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Table 2-2 and Appendix A.8, "VMX-Fixed Bits in CR4."		Thread
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1	
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Table 2-2 and Appendix A.8, "VMX-Fixed Bits in CR4."		Thread
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM	
Capability Reporting Register of VMCS Field Enumeration (R/O) See Table 2-2 and Appendix A.9, "VMCS Enumeration."		Thread
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL2	
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Thread
Register Address: 600H, 1536	IA32_DS_AREA	
DS Save Area (R/W) See Table 2-2 and Section 21.6.3.4, "Debug Store (DS) Mechanism."		Thread
Register Address: 680H, 1664	MSR_LASTBRANCH_0_FROM_IP	
Last Branch Record 0 From IP (R/W) One of sixteen pairs of last branch record registers on the last branch record stack. The From_IP part of the stack contains pointers to the source instruction. See also: - Last Branch Record Stack TOS at 1C9H. - See Section 19.9.1 and record format in Section 19.4.8.1.		Thread
Register Address: 681H, 1665	MSR_LASTBRANCH_1_FROM_IP	
Last Branch Record 1 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 682H, 1666	MSR_LASTBRANCH_2_FROM_IP	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 2 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 683H, 1667	MSR_LASTBRANCH_3_FROM_IP	
Last Branch Record 3 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 684H, 1668	MSR_LASTBRANCH_4_FROM_IP	
Last Branch Record 4 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 685H, 1669	MSR_LASTBRANCH_5_FROM_IP	
Last Branch Record 5 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 686H, 1670	MSR_LASTBRANCH_6_FROM_IP	
Last Branch Record 6 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 687H, 1671	MSR_LASTBRANCH_7_FROM_IP	
Last Branch Record 7 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 688H, 1672	MSR_LASTBRANCH_8_FROM_IP	
Last Branch Record 8 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 689H, 1673	MSR_LASTBRANCH_9_FROM_IP	
Last Branch Record 9 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 68AH, 1674	MSR_LASTBRANCH_10_FROM_IP	
Last Branch Record 10 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 68BH, 1675	MSR_LASTBRANCH_11_FROM_IP	
Last Branch Record 11 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 68CH, 1676	MSR_LASTBRANCH_12_FROM_IP	
Last Branch Record 12 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 68DH, 1677	MSR_LASTBRANCH_13_FROM_IP	
Last Branch Record 13 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 68EH, 1678	MSR_LASTBRANCH_14_FROM_IP	
Last Branch Record 14 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 68FH, 1679	MSR_LASTBRANCH_15_FROM_IP	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 15 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 6C0H, 1728	MSR_LASTBRANCH_0_TO_IP	
Last Branch Record 0 To IP (R/W) One of sixteen pairs of last branch record registers on the last branch record stack. This part of the stack contains pointers to the destination instruction.		Thread
Register Address: 6C1H, 1729	MSR_LASTBRANCH_1_TO_IP	
Last Branch Record 1 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C2H, 1730	MSR_LASTBRANCH_2_TO_IP	
Last Branch Record 2 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C3H, 1731	MSR_LASTBRANCH_3_TO_IP	
Last Branch Record 3 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C4H, 1732	MSR_LASTBRANCH_4_TO_IP	
Last Branch Record 4 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C5H, 1733	MSR_LASTBRANCH_5_TO_IP	
Last Branch Record 5 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C6H, 1734	MSR_LASTBRANCH_6_TO_IP	
Last Branch Record 6 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C7H, 1735	MSR_LASTBRANCH_7_TO_IP	
Last Branch Record 7 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C8H, 1736	MSR_LASTBRANCH_8_TO_IP	
Last Branch Record 8 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C9H, 1737	MSR_LASTBRANCH_9_TO_IP	
Last Branch Record 9 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CAH, 1738	MSR_LASTBRANCH_10_TO_IP	
Last Branch Record 10 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CBH, 1739	MSR_LASTBRANCH_11_TO_IP	
Last Branch Record 11 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 6CCH, 1740	MSR_LASTBRANCH_12_TO_IP	
Last Branch Record 12 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CDH, 1741	MSR_LASTBRANCH_13_TO_IP	
Last Branch Record 13 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CEH, 1742	MSR_LASTBRANCH_14_TO_IP	
Last Branch Record 14 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CFH, 1743	MSR_LASTBRANCH_15_TO_IP	
Last Branch Record 15 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 802H, 2050	IA32_X2APIC_APICID	
x2APIC ID Register (R/O)		Thread
Register Address: 803H, 2051	IA32_X2APIC_VERSION	
x2APIC Version Register (R/O)		Thread
Register Address: 808H, 2056	IA32_X2APIC_TPR	
x2APIC Task Priority Register (R/W)		Thread
Register Address: 80AH, 2058	IA32_X2APIC_PPR	
x2APIC Processor Priority Register (R/O)		Thread
Register Address: 80BH, 2059	IA32_X2APIC_EOI	
x2APIC EOI Register (W/O)		Thread
Register Address: 80DH, 2061	IA32_X2APIC_LDR	
x2APIC Logical Destination Register (R/O)		Thread
Register Address: 80FH, 2063	IA32_X2APIC_SIVR	
x2APIC Spurious Interrupt Vector Register (R/W)		Thread
Register Address: 810H, 2064	IA32_X2APIC_ISR0	
x2APIC In-Service Register Bits [31:0] (R/O)		Thread
Register Address: 811H, 2065	IA32_X2APIC_ISR1	
x2APIC In-Service Register Bits [63:32] (R/O)		Thread
Register Address: 812H, 2066	IA32_X2APIC_ISR2	
x2APIC In-Service Register Bits [95:64] (R/O)		Thread
Register Address: 813H, 2067	IA32_X2APIC_ISR3	
x2APIC In-Service Register Bits [127:96] (R/O)		Thread
Register Address: 814H, 2068	IA32_X2APIC_ISR4	
x2APIC In-Service Register Bits [159:128] (R/O)		Thread
Register Address: 815H, 2069	IA32_X2APIC_ISR5	
x2APIC In-Service Register Bits [191:160] (R/O)		Thread

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 816H, 2070	IA32_X2APIC_ISR6	
x2APIC In-Service Register Bits [223:192] (R/O)		Thread
Register Address: 817H, 2071	IA32_X2APIC_ISR7	
x2APIC In-Service Register Bits [255:224] (R/O)		Thread
Register Address: 818H, 2072	IA32_X2APIC_TMR0	
x2APIC Trigger Mode Register Bits [31:0] (R/O)		Thread
Register Address: 819H, 2073	IA32_X2APIC_TMR1	
x2APIC Trigger Mode Register Bits [63:32] (R/O)		Thread
Register Address: 81AH, 2074	IA32_X2APIC_TMR2	
x2APIC Trigger Mode Register Bits [95:64] (R/O)		Thread
Register Address: 81BH, 2075	IA32_X2APIC_TMR3	
x2APIC Trigger Mode Register Bits [127:96] (R/O)		Thread
Register Address: 81CH, 2076	IA32_X2APIC_TMR4	
x2APIC Trigger Mode Register Bits [159:128] (R/O)		Thread
Register Address: 81DH, 2077	IA32_X2APIC_TMR5	
x2APIC Trigger Mode Register Bits [191:160] (R/O)		Thread
Register Address: 81EH, 2078	IA32_X2APIC_TMR6	
x2APIC Trigger Mode Register Bits [223:192] (R/O)		Thread
Register Address: 81FH, 2079	IA32_X2APIC_TMR7	
x2APIC Trigger Mode Register Bits [255:224] (R/O)		Thread
Register Address: 820H, 2080	IA32_X2APIC_IRR0	
x2APIC Interrupt Request Register Bits [31:0] (R/O)		Thread
Register Address: 821H, 2081	IA32_X2APIC_IRR1	
x2APIC Interrupt Request Register Bits [63:32] (R/O)		Thread
Register Address: 822H, 2082	IA32_X2APIC_IRR2	
x2APIC Interrupt Request Register Bits [95:64] (R/O)		Thread
Register Address: 823H, 2083	IA32_X2APIC_IRR3	
x2APIC Interrupt Request Register Bits [127:96] (R/O)		Thread
Register Address: 824H, 2084	IA32_X2APIC_IRR4	
x2APIC Interrupt Request Register Bits [159:128] (R/O)		Thread
Register Address: 825H, 2085	IA32_X2APIC_IRR5	
x2APIC Interrupt Request Register Bits [191:160] (R/O)		Thread
Register Address: 826H, 2086	IA32_X2APIC_IRR6	
x2APIC Interrupt Request Register Bits [223:192] (R/O)		Thread
Register Address: 827H, 2087	IA32_X2APIC_IRR7	
x2APIC Interrupt Request Register Bits [255:224] (R/O)		Thread
Register Address: 828H, 2088	IA32_X2APIC_ESR	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
x2APIC Error Status Register (R/W)		Thread
Register Address: 82FH, 2095	IA32_X2APIC_LVT_CMCI	
x2APIC LVT Corrected Machine Check Interrupt Register (R/W)		Thread
Register Address: 830H, 2096	IA32_X2APIC_ICR	
x2APIC Interrupt Command Register (R/W)		Thread
Register Address: 832H, 2098	IA32_X2APIC_LVT_TIMER	
x2APIC LVT Timer Interrupt Register (R/W)		Thread
Register Address: 833H, 2099	IA32_X2APIC_LVT_THERMAL	
x2APIC LVT Thermal Sensor Interrupt Register (R/W)		Thread
Register Address: 834H, 2100	IA32_X2APIC_LVT_PMI	
x2APIC LVT Performance Monitor Register (R/W)		Thread
Register Address: 835H, 2101	IA32_X2APIC_LVT_LINT0	
x2APIC LVT LINT0 Register (R/W)		Thread
Register Address: 836H, 2102	IA32_X2APIC_LVT_LINT1	
x2APIC LVT LINT1 Register (R/W)		Thread
Register Address: 837H, 2103	IA32_X2APIC_LVT_ERROR	
x2APIC LVT Error Register (R/W)		Thread
Register Address: 838H, 2104	IA32_X2APIC_INIT_COUNT	
x2APIC Initial Count Register (R/W)		Thread
Register Address: 839H, 2105	IA32_X2APIC_CUR_COUNT	
x2APIC Current Count Register (R/O)		Thread
Register Address: 83EH, 2110	IA32_X2APIC_DIV_CONF	
x2APIC Divide Configuration Register (R/W)		Thread
Register Address: 83FH, 2111	IA32_X2APIC_SELF_IPI	
x2APIC Self IPI Register (W/O)		Thread
Register Address: C000_0080H	IA32_EFER	
Extended Feature Enables See Table 2-2.		Thread
Register Address: C000_0081H	IA32_STAR	
System Call Target Address (R/W) See Table 2-2.		Thread
Register Address: C000_0082H	IA32_LSTAR	
IA-32e Mode System Call Target Address (R/W) See Table 2-2.		Thread
Register Address: C000_0084H	IA32_FMASK	
System Call Flag Mask (R/W) See Table 2-2.		Thread
Register Address: C000_0100H	IA32_FS_BASE	

Table 2-15. MSRs in Processors Based on Nehalem Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Map of BASE Address of FS (R/W) See Table 2-2.		Thread
Register Address: C000_0101H	IA32_GS_BASE	
Map of BASE Address of GS (R/W) See Table 2-2.		Thread
Register Address: C000_0102H	IA32_KERNEL_GS_BASE	
Swap Target of BASE Address of GS (R/W) See Table 2-2.		Thread
Register Address: C000_0103H	IA32_TSC_AUX	
AUXILIARY TSC Signature (R/W) See Table 2-2 and Section 19.17.2, "IA32_TSC_AUX Register and RDTSCP Support."		Thread

2.8.1 Additional MSRs in the Intel® Xeon® Processor 5500 and 3400 Series

The Intel Xeon Processor 5500 and 3400 series supports additional model-specific registers listed in Table 2-16. These MSRs also apply to the Intel Core i7 and i5 processor family with a CPUID Signature DisplayFamily_DisplayModel value of 06_1AH, 06_1EH, or 06_1FH; see Table 2-1.

Table 2-16. Additional MSRs in the Intel® Xeon® Processor 5500 and 3400 Series

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Actual maximum turbo frequency is multiplied by 133.33MHz. (Not available in model 06_2EH.)		Package
7:0	Maximum Turbo Ratio Limit 1C (R/O) Maximum Turbo mode ratio limit with 1 core active.	
15:8	Maximum Turbo Ratio Limit 2C (R/O) Maximum Turbo mode ratio limit with 2 cores active.	
23:16	Maximum Turbo Ratio Limit 3C (R/O) Maximum Turbo mode ratio limit with 3 cores active.	
31:24	Maximum Turbo Ratio Limit 4C (R/O) Maximum Turbo mode ratio limit with 4 cores active.	
63:32	Reserved.	
Register Address: 301H, 769	MSR_GQ_SNOOP_MESF	
MSR_GQ_SNOOP_MESF		Package
0	From M to S (R/W)	
1	From E to S (R/W)	
2	From S to S (R/W)	
3	From F to S (R/W)	
4	From M to I (R/W)	

Table 2-16. Additional MSRs in the Intel® Xeon® Processor 5500 and 3400 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
5	From E to I (R/W)	
6	From S to I (R/W)	
7	From F to I (R/W)	
63:8	Reserved.	
Register Address: 391H, 913	MSR_UNCORE_PERF_GLOBAL_CTRL	
See Section 21.3.1.2.1, "Uncore Performance Monitoring Management Facility."		Package
Register Address: 392H, 914	MSR_UNCORE_PERF_GLOBAL_STATUS	
See Section 21.3.1.2.1, "Uncore Performance Monitoring Management Facility."		Package
Register Address: 393H, 915	MSR_UNCORE_PERF_GLOBAL_OVF_CTRL	
See Section 21.3.1.2.1, "Uncore Performance Monitoring Management Facility."		Package
Register Address: 394H, 916	MSR_UNCORE_FIXED_CTR0	
See Section 21.3.1.2.1, "Uncore Performance Monitoring Management Facility."		Package
Register Address: 395H, 917	MSR_UNCORE_FIXED_CTR_CTRL	
See Section 21.3.1.2.1, "Uncore Performance Monitoring Management Facility."		Package
Register Address: 396H, 918	MSR_UNCORE_ADDR_OPCODE_MATCH	
See Section 21.3.1.2.3, "Uncore Address/Opcode Match MSR."		Package
Register Address: 3B0H, 960	MSR_UNCORE_PMC0	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3B1H, 961	MSR_UNCORE_PMC1	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3B2H, 962	MSR_UNCORE_PMC2	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3B3H, 963	MSR_UNCORE_PMC3	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3B4H, 964	MSR_UNCORE_PMC4	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3B5H, 965	MSR_UNCORE_PMC5	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3B6H, 966	MSR_UNCORE_PMC6	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3B7H, 967	MSR_UNCORE_PMC7	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3C0H, 944	MSR_UNCORE_PERFEVTSEL0	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3C1H, 945	MSR_UNCORE_PERFEVTSEL1	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3C2H, 946	MSR_UNCORE_PERFEVTSEL2	

Table 2-16. Additional MSRs in the Intel® Xeon® Processor 5500 and 3400 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3C3H, 947	MSR_UNCORE_PERFEVTSEL3	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3C4H, 948	MSR_UNCORE_PERFEVTSEL4	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3C5H, 949	MSR_UNCORE_PERFEVTSEL5	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3C6H, 950	MSR_UNCORE_PERFEVTSEL6	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package
Register Address: 3C7H, 951	MSR_UNCORE_PERFEVTSEL7	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package

2.8.2 Additional MSRs in the Intel® Xeon® Processor 7500 Series

The Intel Xeon Processor 7500 series supports MSRs listed in Table 2-15 (except MSR address 1ADH) and additional model-specific registers listed in Table 2-17. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_2EH.

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Reserved. Attempt to read/write will cause #UD.		Package
Register Address: 289H, 649	IA32_MC9_CTL2	
See Table 2-2.		Package
Register Address: 28AH, 650	IA32_MC10_CTL2	
See Table 2-2.		Package
Register Address: 28BH, 651	IA32_MC11_CTL2	
See Table 2-2.		Package
Register Address: 28CH, 652	IA32_MC12_CTL2	
See Table 2-2.		Package
Register Address: 28DH, 653	IA32_MC13_CTL2	
See Table 2-2.		Package
Register Address: 28EH, 654	IA32_MC14_CTL2	
See Table 2-2.		Package
Register Address: 28FH, 655	IA32_MC15_CTL2	
See Table 2-2.		Package
Register Address: 290H, 656	IA32_MC16_CTL2	
See Table 2-2.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 291H, 657	IA32_MC17_CTL2	
See Table 2-2.		Package
Register Address: 292H, 658	IA32_MC18_CTL2	
See Table 2-2.		Package
Register Address: 293H, 659	IA32_MC19_CTL2	
See Table 2-2.		Package
Register Address: 294H, 660	IA32_MC20_CTL2	
See Table 2-2.		Package
Register Address: 295H, 661	IA32_MC21_CTL2	
See Table 2-2.		Package
Register Address: 394H, 816	MSR_W_PMON_FIXED_CTR	
Uncore W-box PerfMon fixed counter.		Package
Register Address: 395H, 817	MSR_W_PMON_FIXED_CTR_CTL	
Uncore U-box PerfMon fixed counter control MSR.		Package
Register Address: 424H, 1060	IA32_MC9_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 425H, 1061	IA32_MC9_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs," and Chapter 18.		Package
Register Address: 426H, 1062	IA32_MC9_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 427H, 1063	IA32_MC9_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 428H, 1064	IA32_MC10_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 429H, 1065	IA32_MC10_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs," and Chapter 18.		Package
Register Address: 42AH, 1066	IA32_MC10_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 42BH, 1067	IA32_MC10_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 42CH, 1068	IA32_MC11_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 42DH, 1069	IA32_MC11_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs," and Chapter 18.		Package
Register Address: 42EH, 1070	IA32_MC11_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 42FH, 1071	IA32_MC11_MISC	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 430H, 1072	IA32_MC12_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 431H, 1073	IA32_MC12_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Package
Register Address: 432H, 1074	IA32_MC12_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 433H, 1075	IA32_MC12_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 434H, 1076	IA32_MC13_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 435H, 1077	IA32_MC13_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Package
Register Address: 436H, 1078	IA32_MC13_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 437H, 1079	IA32_MC13_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 438H, 1080	IA32_MC14_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 439H, 1081	IA32_MC14_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Package
Register Address: 43AH, 1082	IA32_MC14_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 43BH, 1083	IA32_MC14_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 43CH, 1084	IA32_MC15_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 43DH, 1085	IA32_MC15_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Package
Register Address: 43EH, 1086	IA32_MC15_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 43FH, 1087	IA32_MC15_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 440H, 1088	IA32_MC16_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 441H, 1089	IA32_MC16_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 442H, 1090	IA32_MC16_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 443H, 1091	IA32_MC16_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 444H, 1092	IA32_MC17_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 445H, 1093	IA32_MC17_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs," and Chapter 18.		Package
Register Address: 446H, 1094	IA32_MC17_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 447H, 1095	IA32_MC17_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 448H, 1096	IA32_MC18_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 449H, 1097	IA32_MC18_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs," and Chapter 18.		Package
Register Address: 44AH, 1098	IA32_MC18_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 44BH, 1099	IA32_MC18_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 44CH, 1100	IA32_MC19_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 44DH, 1101	IA32_MC19_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs," and Chapter 18.		Package
Register Address: 44EH, 1102	IA32_MC19_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 44FH, 1103	IA32_MC19_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 450H, 1104	IA32_MC20_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 451H, 1105	IA32_MC20_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs," and Chapter 18.		Package
Register Address: 452H, 1106	IA32_MC20_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 453H, 1107	IA32_MC20_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 454H, 1108	IA32_MC21_CTL	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 455H, 1109	IA32_MC21_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs," and Chapter 18.		Package
Register Address: 456H, 1110	IA32_MC21_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 457H, 1111	IA32_MC21_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: C00H, 3072	MSR_U_PMON_GLOBAL_CTRL	
Uncore U-box PerfMon global control MSR.		Package
Register Address: C01H, 3073	MSR_U_PMON_GLOBAL_STATUS	
Uncore U-box PerfMon global status MSR.		Package
Register Address: C02H, 3074	MSR_U_PMON_GLOBAL_OVF_CTRL	
Uncore U-box PerfMon global overflow control MSR.		Package
Register Address: C10H, 3088	MSR_U_PMON_EVNT_SEL	
Uncore U-box PerfMon event select MSR.		Package
Register Address: C11H, 3089	MSR_U_PMON_CTR	
Uncore U-box PerfMon counter MSR.		Package
Register Address: C20H, 3104	MSR_B0_PMON_BOX_CTRL	
Uncore B-box 0 PerfMon local box control MSR.		Package
Register Address: C21H, 3105	MSR_B0_PMON_BOX_STATUS	
Uncore B-box 0 PerfMon local box status MSR.		Package
Register Address: C22H, 3106	MSR_B0_PMON_BOX_OVF_CTRL	
Uncore B-box 0 PerfMon local box overflow control MSR.		Package
Register Address: C30H, 3120	MSR_B0_PMON_EVNT_SELO	
Uncore B-box 0 PerfMon event select MSR.		Package
Register Address: C31H, 3121	MSR_B0_PMON_CTR0	
Uncore B-box 0 PerfMon counter MSR.		Package
Register Address: C32H, 3122	MSR_B0_PMON_EVNT_SEL1	
Uncore B-box 0 PerfMon event select MSR.		Package
Register Address: C33H, 3123	MSR_B0_PMON_CTR1	
Uncore B-box 0 PerfMon counter MSR.		Package
Register Address: C34H, 3124	MSR_B0_PMON_EVNT_SEL2	
Uncore B-box 0 PerfMon event select MSR.		Package
Register Address: C35H, 3125	MSR_B0_PMON_CTR2	
Uncore B-box 0 PerfMon counter MSR.		Package
Register Address: C36H, 3126	MSR_B0_PMON_EVNT_SEL3	
Uncore B-box 0 PerfMon event select MSR.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: C37H, 3127	MSR_B0_PMON_CTR3	
Uncore B-box 0 PerfMon counter MSR.		Package
Register Address: C40H, 3136	MSR_S0_PMON_BOX_CTRL	
Uncore S-box 0 PerfMon local box control MSR.		Package
Register Address: C41H, 3137	MSR_S0_PMON_BOX_STATUS	
Uncore S-box 0 PerfMon local box status MSR.		Package
Register Address: C42H, 3138	MSR_S0_PMON_BOX_OVF_CTRL	
Uncore S-box 0 PerfMon local box overflow control MSR.		Package
Register Address: C50H, 3152	MSR_S0_PMON_EVNT_SELO	
Uncore S-box 0 PerfMon event select MSR.		Package
Register Address: C51H, 3153	MSR_S0_PMON_CTR0	
Uncore S-box 0 PerfMon counter MSR.		Package
Register Address: C52H, 3154	MSR_S0_PMON_EVNT_SEL1	
Uncore S-box 0 PerfMon event select MSR.		Package
Register Address: C53H, 3155	MSR_S0_PMON_CTR1	
Uncore S-box 0 PerfMon counter MSR.		Package
Register Address: C54H, 3156	MSR_S0_PMON_EVNT_SEL2	
Uncore S-box 0 PerfMon event select MSR.		Package
Register Address: C55H, 3157	MSR_S0_PMON_CTR2	
Uncore S-box 0 PerfMon counter MSR.		Package
Register Address: C56H, 3158	MSR_S0_PMON_EVNT_SEL3	
Uncore S-box 0 PerfMon event select MSR.		Package
Register Address: C57H, 3159	MSR_S0_PMON_CTR3	
Uncore S-box 0 PerfMon counter MSR.		Package
Register Address: C60H, 3168	MSR_B1_PMON_BOX_CTRL	
Uncore B-box 1 PerfMon local box control MSR.		Package
Register Address: C61H, 3169	MSR_B1_PMON_BOX_STATUS	
Uncore B-box 1 PerfMon local box status MSR.		Package
Register Address: C62H, 3170	MSR_B1_PMON_BOX_OVF_CTRL	
Uncore B-box 1 PerfMon local box overflow control MSR.		Package
Register Address: C70H, 3184	MSR_B1_PMON_EVNT_SELO	
Uncore B-box 1 PerfMon event select MSR.		Package
Register Address: C71H, 3185	MSR_B1_PMON_CTR0	
Uncore B-box 1 PerfMon counter MSR.		Package
Register Address: C72H, 3186	MSR_B1_PMON_EVNT_SEL1	
Uncore B-box 1 PerfMon event select MSR.		Package
Register Address: C73H, 3187	MSR_B1_PMON_CTR1	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore B-box 1 PerfMon counter MSR.		Package
Register Address: C74H, 3188	MSR_B1_PMON_EVNT_SEL2	
Uncore B-box 1 PerfMon event select MSR.		Package
Register Address: C75H, 3189	MSR_B1_PMON_CTRL2	
Uncore B-box 1 PerfMon counter MSR.		Package
Register Address: C76H, 3190	MSR_B1_PMON_EVNT_SEL3	
Uncore B-box 1vPerfMon event select MSR.		Package
Register Address: C77H, 3191	MSR_B1_PMON_CTRL3	
Uncore B-box 1 PerfMon counter MSR.		Package
Register Address: C80H, 3120	MSR_W_PMON_BOX_CTRL	
Uncore W-box PerfMon local box control MSR.		Package
Register Address: C81H, 3121	MSR_W_PMON_BOX_STATUS	
Uncore W-box PerfMon local box status MSR.		Package
Register Address: C82H, 3122	MSR_W_PMON_BOX_OVF_CTRL	
Uncore W-box PerfMon local box overflow control MSR.		Package
Register Address: C90H, 3136	MSR_W_PMON_EVNT_SEL0	
Uncore W-box PerfMon event select MSR.		Package
Register Address: C91H, 3137	MSR_W_PMON_CTRL0	
Uncore W-box PerfMon counter MSR.		Package
Register Address: C92H, 3138	MSR_W_PMON_EVNT_SEL1	
Uncore W-box PerfMon event select MSR.		Package
Register Address: C93H, 3139	MSR_W_PMON_CTRL1	
Uncore W-box PerfMon counter MSR.		Package
Register Address: C94H, 3140	MSR_W_PMON_EVNT_SEL2	
Uncore W-box PerfMon event select MSR.		Package
Register Address: C95H, 3141	MSR_W_PMON_CTRL2	
Uncore W-box PerfMon counter MSR.		Package
Register Address: C96H, 3142	MSR_W_PMON_EVNT_SEL3	
Uncore W-box PerfMon event select MSR.		Package
Register Address: C97H, 3143	MSR_W_PMON_CTRL3	
Uncore W-box PerfMon counter MSR.		Package
Register Address: CA0H, 3232	MSR_M0_PMON_BOX_CTRL	
Uncore M-box 0 PerfMon local box control MSR.		Package
Register Address: CA1H, 3233	MSR_M0_PMON_BOX_STATUS	
Uncore M-box 0 PerfMon local box status MSR.		Package
Register Address: CA2H, 3234	MSR_M0_PMON_BOX_OVF_CTRL	
Uncore M-box 0 PerfMon local box overflow control MSR.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: CA4H, 3236	MSR_M0_PMON_TIMESTAMP	
Uncore M-box 0 PerfMon time stamp unit select MSR.		Package
Register Address: CA5H, 3237	MSR_M0_PMON_DSP	
Uncore M-box 0 PerfMon DSP unit select MSR.		Package
Register Address: CA6H, 3238	MSR_M0_PMON_ISS	
Uncore M-box 0 PerfMon ISS unit select MSR.		Package
Register Address: CA7H, 3239	MSR_M0_PMON_MAP	
Uncore M-box 0 PerfMon MAP unit select MSR.		Package
Register Address: CA8H, 3240	MSR_M0_PMON_MSC_THR	
Uncore M-box 0 PerfMon MIC THR select MSR.		Package
Register Address: CA9H, 3241	MSR_M0_PMON_PGT	
Uncore M-box 0 PerfMon PGT unit select MSR.		Package
Register Address: CAAH, 3242	MSR_M0_PMON_PLD	
Uncore M-box 0 PerfMon PLD unit select MSR.		Package
Register Address: CABH, 3243	MSR_M0_PMON_ZDP	
Uncore M-box 0 PerfMon ZDP unit select MSR.		Package
Register Address: CBOH, 3248	MSR_M0_PMON_EVNT_SELO	
Uncore M-box 0 PerfMon event select MSR.		Package
Register Address: CB1H, 3249	MSR_M0_PMON_CTR0	
Uncore M-box 0 PerfMon counter MSR.		Package
Register Address: CB2H, 3250	MSR_M0_PMON_EVNT_SEL1	
Uncore M-box 0 PerfMon event select MSR.		Package
Register Address: CB3H, 3251	MSR_M0_PMON_CTR1	
Uncore M-box 0 PerfMon counter MSR.		Package
Register Address: CB4H, 3252	MSR_M0_PMON_EVNT_SEL2	
Uncore M-box 0 PerfMon event select MSR.		Package
Register Address: CB5H, 3253	MSR_M0_PMON_CTR2	
Uncore M-box 0 PerfMon counter MSR.		Package
Register Address: CB6H, 3254	MSR_M0_PMON_EVNT_SEL3	
Uncore M-box 0 PerfMon event select MSR.		Package
Register Address: CB7H, 3255	MSR_M0_PMON_CTR3	
Uncore M-box 0 PerfMon counter MSR.		Package
Register Address: CB8H, 3256	MSR_M0_PMON_EVNT_SEL4	
Uncore M-box 0 PerfMon event select MSR.		Package
Register Address: CB9H, 3257	MSR_M0_PMON_CTR4	
Uncore M-box 0 PerfMon counter MSR.		Package
Register Address: CBAH, 3258	MSR_M0_PMON_EVNT_SEL5	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore M-box 0 PerfMon event select MSR.		Package
Register Address: CBBH, 3259	MSR_M0_PMON_CTR5	
Uncore M-box 0 PerfMon counter MSR.		Package
Register Address: CC0H, 3264	MSR_S1_PMON_BOX_CTRL	
Uncore S-box 1 PerfMon local box control MSR.		Package
Register Address: CC1H, 3265	MSR_S1_PMON_BOX_STATUS	
Uncore S-box 1 PerfMon local box status MSR.		Package
Register Address: CC2H, 3266	MSR_S1_PMON_BOX_OVF_CTRL	
Uncore S-box 1 PerfMon local box overflow control MSR.		Package
Register Address: CD0H, 3280	MSR_S1_PMON_EVNT_SELO	
Uncore S-box 1 PerfMon event select MSR.		Package
Register Address: CD1H, 3281	MSR_S1_PMON_CTR0	
Uncore S-box 1 PerfMon counter MSR.		Package
Register Address: CD2H, 3282	MSR_S1_PMON_EVNT_SEL1	
Uncore S-box 1 PerfMon event select MSR.		Package
Register Address: CD3H, 3283	MSR_S1_PMON_CTR1	
Uncore S-box 1 PerfMon counter MSR.		Package
Register Address: CD4H, 3284	MSR_S1_PMON_EVNT_SEL2	
Uncore S-box 1 PerfMon event select MSR.		Package
Register Address: CD5H, 3285	MSR_S1_PMON_CTR2	
Uncore S-box 1 PerfMon counter MSR.		Package
Register Address: CD6H, 3286	MSR_S1_PMON_EVNT_SEL3	
Uncore S-box 1 PerfMon event select MSR.		Package
Register Address: CD7H, 3287	MSR_S1_PMON_CTR3	
Uncore S-box 1 PerfMon counter MSR.		Package
Register Address: CE0H, 3296	MSR_M1_PMON_BOX_CTRL	
Uncore M-box 1 PerfMon local box control MSR.		Package
Register Address: CE1H, 3297	MSR_M1_PMON_BOX_STATUS	
Uncore M-box 1 PerfMon local box status MSR.		Package
Register Address: CE2H, 3298	MSR_M1_PMON_BOX_OVF_CTRL	
Uncore M-box 1 PerfMon local box overflow control MSR.		Package
Register Address: CE4H, 3300	MSR_M1_PMON_TIMESTAMP	
Uncore M-box 1 PerfMon time stamp unit select MSR.		Package
Register Address: CE5H, 3301	MSR_M1_PMON_DSP	
Uncore M-box 1 PerfMon DSP unit select MSR.		Package
Register Address: CE6H, 3302	MSR_M1_PMON_ISS	
Uncore M-box 1 PerfMon ISS unit select MSR.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: CE7H, 3303	MSR_M1_PMON_MAP	
Uncore M-box 1 PerfMon MAP unit select MSR.		Package
Register Address: CE8H, 3304	MSR_M1_PMON_MSC_THR	
Uncore M-box 1 PerfMon MIC THR select MSR.		Package
Register Address: CE9H, 3305	MSR_M1_PMON_PGT	
Uncore M-box 1 PerfMon PGT unit select MSR.		Package
Register Address: CEAH, 3306	MSR_M1_PMON_PLD	
Uncore M-box 1 PerfMon PLD unit select MSR.		Package
Register Address: CEBH, 3307	MSR_M1_PMON_ZDP	
Uncore M-box 1 PerfMon ZDP unit select MSR.		Package
Register Address: CF0H, 3312	MSR_M1_PMON_EVNT_SELO	
Uncore M-box 1 PerfMon event select MSR.		Package
Register Address: CF1H, 3313	MSR_M1_PMON_CTR0	
Uncore M-box 1 PerfMon counter MSR.		Package
Register Address: CF2H, 3314	MSR_M1_PMON_EVNT_SEL1	
Uncore M-box 1 PerfMon event select MSR.		Package
Register Address: CF3H, 3315	MSR_M1_PMON_CTR1	
Uncore M-box 1 PerfMon counter MSR.		Package
Register Address: CF4H, 3316	MSR_M1_PMON_EVNT_SEL2	
Uncore M-box 1 PerfMon event select MSR.		Package
Register Address: CF5H, 3317	MSR_M1_PMON_CTR2	
Uncore M-box 1 PerfMon counter MSR.		Package
Register Address: CF6H, 3318	MSR_M1_PMON_EVNT_SEL3	
Uncore M-box 1 PerfMon event select MSR.		Package
Register Address: CF7H, 3319	MSR_M1_PMON_CTR3	
Uncore M-box 1 PerfMon counter MSR.		Package
Register Address: CF8H, 3320	MSR_M1_PMON_EVNT_SEL4	
Uncore M-box 1 PerfMon event select MSR.		Package
Register Address: CF9H, 3321	MSR_M1_PMON_CTR4	
Uncore M-box 1 PerfMon counter MSR.		Package
Register Address: CFAH, 3322	MSR_M1_PMON_EVNT_SEL5	
Uncore M-box 1 PerfMon event select MSR.		Package
Register Address: CFBH, 3323	MSR_M1_PMON_CTR5	
Uncore M-box 1 PerfMon counter MSR.		Package
Register Address: D00H, 3328	MSR_CO_PMON_BOX_CTRL	
Uncore C-box 0 PerfMon local box control MSR.		Package
Register Address: D01H, 3329	MSR_CO_PMON_BOX_STATUS	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-box 0 PerfMon local box status MSR.		Package
Register Address: D02H, 3330	MSR_CO_PMON_BOX_OVF_CTRL	
Uncore C-box 0 PerfMon local box overflow control MSR.		Package
Register Address: D10H, 3344	MSR_CO_PMON_EVNT_SELO	
Uncore C-box 0 PerfMon event select MSR.		Package
Register Address: D11H, 3345	MSR_CO_PMON_CTR0	
Uncore C-box 0 PerfMon counter MSR.		Package
Register Address: D12H, 3346	MSR_CO_PMON_EVNT_SEL1	
Uncore C-box 0 PerfMon event select MSR.		Package
Register Address: D13H, 3347	MSR_CO_PMON_CTR1	
Uncore C-box 0 PerfMon counter MSR.		Package
Register Address: D14H, 3348	MSR_CO_PMON_EVNT_SEL2	
Uncore C-box 0 PerfMon event select MSR.		Package
Register Address: D15H, 3349	MSR_CO_PMON_CTR2	
Uncore C-box 0 PerfMon counter MSR.		Package
Register Address: D16H, 3350	MSR_CO_PMON_EVNT_SEL3	
Uncore C-box 0 PerfMon event select MSR.		Package
Register Address: D17H, 3351	MSR_CO_PMON_CTR3	
Uncore C-box 0 PerfMon counter MSR.		Package
Register Address: D18H, 3352	MSR_CO_PMON_EVNT_SEL4	
Uncore C-box 0 PerfMon event select MSR.		Package
Register Address: D19H, 3353	MSR_CO_PMON_CTR4	
Uncore C-box 0 PerfMon counter MSR.		Package
Register Address: D1AH, 3354	MSR_CO_PMON_EVNT_SEL5	
Uncore C-box 0 PerfMon event select MSR.		Package
Register Address: D1BH, 3355	MSR_CO_PMON_CTR5	
Uncore C-box 0 PerfMon counter MSR.		Package
Register Address: D20H, 3360	MSR_C4_PMON_BOX_CTRL	
Uncore C-box 4 PerfMon local box control MSR.		Package
Register Address: D21H, 3361	MSR_C4_PMON_BOX_STATUS	
Uncore C-box 4 PerfMon local box status MSR.		Package
Register Address: D22H, 3362	MSR_C4_PMON_BOX_OVF_CTRL	
Uncore C-box 4 PerfMon local box overflow control MSR.		Package
Register Address: D30H, 3376	MSR_C4_PMON_EVNT_SELO	
Uncore C-box 4 PerfMon event select MSR.		Package
Register Address: D31H, 3377	MSR_C4_PMON_CTR0	
Uncore C-box 4 PerfMon counter MSR.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: D32H, 3378	MSR_C4_PMON_EVNT_SEL1	
Uncore C-box 4 PerfMon event select MSR.		Package
Register Address: D33H, 3379	MSR_C4_PMON_CTR1	
Uncore C-box 4 PerfMon counter MSR.		Package
Register Address: D34H, 3380	MSR_C4_PMON_EVNT_SEL2	
Uncore C-box 4 PerfMon event select MSR.		Package
Register Address: D35H, 3381	MSR_C4_PMON_CTR2	
Uncore C-box 4 PerfMon counter MSR.		Package
Register Address: D36H, 3382	MSR_C4_PMON_EVNT_SEL3	
Uncore C-box 4 PerfMon event select MSR.		Package
Register Address: D37H, 3383	MSR_C4_PMON_CTR3	
Uncore C-box 4 PerfMon counter MSR.		Package
Register Address: D38H, 3384	MSR_C4_PMON_EVNT_SEL4	
Uncore C-box 4 PerfMon event select MSR.		Package
Register Address: D39H, 3385	MSR_C4_PMON_CTR4	
Uncore C-box 4 PerfMon counter MSR.		Package
Register Address: D3AH, 3386	MSR_C4_PMON_EVNT_SEL5	
Uncore C-box 4 PerfMon event select MSR.		Package
Register Address: D3BH, 3387	MSR_C4_PMON_CTR5	
Uncore C-box 4 PerfMon counter MSR.		Package
Register Address: D40H, 3392	MSR_C2_PMON_BOX_CTRL	
Uncore C-box 2 PerfMon local box control MSR.		Package
Register Address: D41H, 3393	MSR_C2_PMON_BOX_STATUS	
Uncore C-box 2 PerfMon local box status MSR.		Package
Register Address: D42H, 3394	MSR_C2_PMON_BOX_OVF_CTRL	
Uncore C-box 2 PerfMon local box overflow control MSR.		Package
Register Address: D50H, 3408	MSR_C2_PMON_EVNT_SELO	
Uncore C-box 2 PerfMon event select MSR.		Package
Register Address: D51H, 3409	MSR_C2_PMON_CTR0	
Uncore C-box 2 PerfMon counter MSR.		Package
Register Address: D52H, 3410	MSR_C2_PMON_EVNT_SEL1	
Uncore C-box 2 PerfMon event select MSR.		Package
Register Address: D53H, 3411	MSR_C2_PMON_CTR1	
Uncore C-box 2 PerfMon counter MSR.		Package
Register Address: D54H, 3412	MSR_C2_PMON_EVNT_SEL2	
Uncore C-box 2 PerfMon event select MSR.		Package
Register Address: D55H, 3413	MSR_C2_PMON_CTR2	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-box 2 PerfMon counter MSR.		Package
Register Address: D56H, 3414	MSR_C2_PMON_EVNT_SEL3	
Uncore C-box 2 PerfMon event select MSR.		Package
Register Address: D57H, 3415	MSR_C2_PMON_CTR3	
Uncore C-box 2 PerfMon counter MSR.		Package
Register Address: D58H, 3416	MSR_C2_PMON_EVNT_SEL4	
Uncore C-box 2 PerfMon event select MSR.		Package
Register Address: D59H, 3417	MSR_C2_PMON_CTR4	
Uncore C-box 2 PerfMon counter MSR.		Package
Register Address: D5AH, 3418	MSR_C2_PMON_EVNT_SEL5	
Uncore C-box 2 PerfMon event select MSR.		Package
Register Address: D5BH, 3419	MSR_C2_PMON_CTR5	
Uncore C-box 2 PerfMon counter MSR.		Package
Register Address: D60H, 3424	MSR_C6_PMON_BOX_CTRL	
Uncore C-box 6 PerfMon local box control MSR.		Package
Register Address: D61H, 3425	MSR_C6_PMON_BOX_STATUS	
Uncore C-box 6 PerfMon local box status MSR.		Package
Register Address: D62H, 3426	MSR_C6_PMON_BOX_OVF_CTRL	
Uncore C-box 6 PerfMon local box overflow control MSR.		Package
Register Address: D70H, 3440	MSR_C6_PMON_EVNT_SELO	
Uncore C-box 6 PerfMon event select MSR.		Package
Register Address: D71H, 3441	MSR_C6_PMON_CTR0	
Uncore C-box 6 PerfMon counter MSR.		Package
Register Address: D72H, 3442	MSR_C6_PMON_EVNT_SEL1	
Uncore C-box 6 PerfMon event select MSR.		Package
Register Address: D73H, 3443	MSR_C6_PMON_CTR1	
Uncore C-box 6 PerfMon counter MSR.		Package
Register Address: D74H, 3444	MSR_C6_PMON_EVNT_SEL2	
Uncore C-box 6 PerfMon event select MSR.		Package
Register Address: D75H, 3445	MSR_C6_PMON_CTR2	
Uncore C-box 6 PerfMon counter MSR.		Package
Register Address: D76H, 3446	MSR_C6_PMON_EVNT_SEL3	
Uncore C-box 6 PerfMon event select MSR.		Package
Register Address: D77H, 3447	MSR_C6_PMON_CTR3	
Uncore C-box 6 PerfMon counter MSR.		Package
Register Address: D78H, 3448	MSR_C6_PMON_EVNT_SEL4	
Uncore C-box 6 PerfMon event select MSR.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: D79H, 3449	MSR_C6_PMON_CTR4	
Uncore C-box 6 PerfMon counter MSR.		Package
Register Address: D7AH, 3450	MSR_C6_PMON_EVNT_SEL5	
Uncore C-box 6 PerfMon event select MSR.		Package
Register Address: D7BH, 3451	MSR_C6_PMON_CTR5	
Uncore C-box 6 PerfMon counter MSR.		Package
Register Address: D80H, 3456	MSR_C1_PMON_BOX_CTRL	
Uncore C-box 1 PerfMon local box control MSR.		Package
Register Address: D81H, 3457	MSR_C1_PMON_BOX_STATUS	
Uncore C-box 1 PerfMon local box status MSR.		Package
Register Address: D82H, 3458	MSR_C1_PMON_BOX_OVF_CTRL	
Uncore C-box 1 PerfMon local box overflow control MSR.		Package
Register Address: D90H, 3472	MSR_C1_PMON_EVNT_SELO	
Uncore C-box 1 PerfMon event select MSR.		Package
Register Address: D91H, 3473	MSR_C1_PMON_CTR0	
Uncore C-box 1 PerfMon counter MSR.		Package
Register Address: D92H, 3474	MSR_C1_PMON_EVNT_SEL1	
Uncore C-box 1 PerfMon event select MSR.		Package
Register Address: D93H, 3475	MSR_C1_PMON_CTR1	
Uncore C-box 1 PerfMon counter MSR.		Package
Register Address: D94H, 3476	MSR_C1_PMON_EVNT_SEL2	
Uncore C-box 1 PerfMon event select MSR.		Package
Register Address: D95H, 3477	MSR_C1_PMON_CTR2	
Uncore C-box 1 PerfMon counter MSR.		Package
Register Address: D96H, 3478	MSR_C1_PMON_EVNT_SEL3	
Uncore C-box 1 PerfMon event select MSR.		Package
Register Address: D97H, 3479	MSR_C1_PMON_CTR3	
Uncore C-box 1 PerfMon counter MSR.		Package
Register Address: D98H, 3480	MSR_C1_PMON_EVNT_SEL4	
Uncore C-box 1 PerfMon event select MSR.		Package
Register Address: D99H, 3481	MSR_C1_PMON_CTR4	
Uncore C-box 1 PerfMon counter MSR.		Package
Register Address: D9AH, 3482	MSR_C1_PMON_EVNT_SEL5	
Uncore C-box 1 PerfMon event select MSR.		Package
Register Address: D9BH, 3483	MSR_C1_PMON_CTR5	
Uncore C-box 1 PerfMon counter MSR.		Package
Register Address: DA0H, 3488	MSR_C5_PMON_BOX_CTRL	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-box 5 PerfMon local box control MSR.		Package
Register Address: DA1H, 3489	MSR_C5_PMON_BOX_STATUS	
Uncore C-box 5 PerfMon local box status MSR.		Package
Register Address: DA2H, 3490	MSR_C5_PMON_BOX_OVF_CTRL	
Uncore C-box 5 PerfMon local box overflow control MSR.		Package
Register Address: DB0H, 3504	MSR_C5_PMON_EVNT_SELO	
Uncore C-box 5 PerfMon event select MSR.		Package
Register Address: DB1H, 3505	MSR_C5_PMON_CTR0	
Uncore C-box 5 PerfMon counter MSR.		Package
Register Address: DB2H, 3506	MSR_C5_PMON_EVNT_SEL1	
Uncore C-box 5 PerfMon event select MSR.		Package
Register Address: DB3H, 3507	MSR_C5_PMON_CTR1	
Uncore C-box 5 PerfMon counter MSR.		Package
Register Address: DB4H, 3508	MSR_C5_PMON_EVNT_SEL2	
Uncore C-box 5 PerfMon event select MSR.		Package
Register Address: DB5H, 3509	MSR_C5_PMON_CTR2	
Uncore C-box 5 PerfMon counter MSR.		Package
Register Address: DB6H, 3510	MSR_C5_PMON_EVNT_SEL3	
Uncore C-box 5 PerfMon event select MSR.		Package
Register Address: DB7H, 3511	MSR_C5_PMON_CTR3	
Uncore C-box 5 PerfMon counter MSR.		Package
Register Address: DB8H, 3512	MSR_C5_PMON_EVNT_SEL4	
Uncore C-box 5 PerfMon event select MSR.		Package
Register Address: DB9H, 3513	MSR_C5_PMON_CTR4	
Uncore C-box 5 PerfMon counter MSR.		Package
Register Address: DBAH, 3514	MSR_C5_PMON_EVNT_SEL5	
Uncore C-box 5 PerfMon event select MSR.		Package
Register Address: DBBH, 3515	MSR_C5_PMON_CTR5	
Uncore C-box 5 PerfMon counter MSR.		Package
Register Address: DC0H, 3520	MSR_C3_PMON_BOX_CTRL	
Uncore C-box 3 PerfMon local box control MSR.		Package
Register Address: DC1H, 3521	MSR_C3_PMON_BOX_STATUS	
Uncore C-box 3 PerfMon local box status MSR.		Package
Register Address: DC2H, 3522	MSR_C3_PMON_BOX_OVF_CTRL	
Uncore C-box 3 PerfMon local box overflow control MSR.		Package
Register Address: DDOH, 3536	MSR_C3_PMON_EVNT_SELO	
Uncore C-box 3 PerfMon event select MSR.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: DD1H, 3537	MSR_C3_PMON_CTRL0	
Uncore C-box 3 PerfMon counter MSR.		Package
Register Address: DD2H, 3538	MSR_C3_PMON_EVNT_SEL1	
Uncore C-box 3 PerfMon event select MSR.		Package
Register Address: DD3H, 3539	MSR_C3_PMON_CTRL1	
Uncore C-box 3 PerfMon counter MSR.		Package
Register Address: DD4H, 3540	MSR_C3_PMON_EVNT_SEL2	
Uncore C-box 3 PerfMon event select MSR.		Package
Register Address: DD5H, 3541	MSR_C3_PMON_CTRL2	
Uncore C-box 3 PerfMon counter MSR.		Package
Register Address: DD6H, 3542	MSR_C3_PMON_EVNT_SEL3	
Uncore C-box 3 PerfMon event select MSR.		Package
Register Address: DD7H, 3543	MSR_C3_PMON_CTRL3	
Uncore C-box 3 PerfMon counter MSR.		Package
Register Address: DD8H, 3544	MSR_C3_PMON_EVNT_SEL4	
Uncore C-box 3 PerfMon event select MSR.		Package
Register Address: DD9H, 3545	MSR_C3_PMON_CTRL4	
Uncore C-box 3 PerfMon counter MSR.		Package
Register Address: DDAH, 3546	MSR_C3_PMON_EVNT_SEL5	
Uncore C-box 3 PerfMon event select MSR.		Package
Register Address: DDBH, 3547	MSR_C3_PMON_CTRL5	
Uncore C-box 3 PerfMon counter MSR.		Package
Register Address: DE0H, 3552	MSR_C7_PMON_BOX_CTRL	
Uncore C-box 7 PerfMon local box control MSR.		Package
Register Address: DE1H, 3553	MSR_C7_PMON_BOX_STATUS	
Uncore C-box 7 PerfMon local box status MSR.		Package
Register Address: DE2H, 3554	MSR_C7_PMON_BOX_OVF_CTRL	
Uncore C-box 7 PerfMon local box overflow control MSR.		Package
Register Address: DF0H, 3568	MSR_C7_PMON_EVNT_SELO	
Uncore C-box 7 PerfMon event select MSR.		Package
Register Address: DF1H, 3569	MSR_C7_PMON_CTRL0	
Uncore C-box 7 PerfMon counter MSR.		Package
Register Address: DF2H, 3570	MSR_C7_PMON_EVNT_SEL1	
Uncore C-box 7 PerfMon event select MSR.		Package
Register Address: DF3H, 3571	MSR_C7_PMON_CTRL1	
Uncore C-box 7 PerfMon counter MSR.		Package
Register Address: DF4H, 3572	MSR_C7_PMON_EVNT_SEL2	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-box 7 PerfMon event select MSR.		Package
Register Address: DF5H, 3573	MSR_C7_PMON_CTR2	
Uncore C-box 7 PerfMon counter MSR.		Package
Register Address: DF6H, 3574	MSR_C7_PMON_EVNT_SEL3	
Uncore C-box 7 PerfMon event select MSR.		Package
Register Address: DF7H, 3575	MSR_C7_PMON_CTR3	
Uncore C-box 7 PerfMon counter MSR.		Package
Register Address: DF8H, 3576	MSR_C7_PMON_EVNT_SEL4	
Uncore C-box 7 PerfMon event select MSR.		Package
Register Address: DF9H, 3577	MSR_C7_PMON_CTR4	
Uncore C-box 7 PerfMon counter MSR.		Package
Register Address: DFAH, 3578	MSR_C7_PMON_EVNT_SEL5	
Uncore C-box 7 PerfMon event select MSR.		Package
Register Address: DFBH, 3579	MSR_C7_PMON_CTR5	
Uncore C-box 7 PerfMon counter MSR.		Package
Register Address: E00H, 3584	MSR_R0_PMON_BOX_CTRL	
Uncore R-box 0 PerfMon local box control MSR.		Package
Register Address: E01H, 3585	MSR_R0_PMON_BOX_STATUS	
Uncore R-box 0 PerfMon local box status MSR.		Package
Register Address: E02H, 3586	MSR_R0_PMON_BOX_OVF_CTRL	
Uncore R-box 0 PerfMon local box overflow control MSR.		Package
Register Address: E04H, 3588	MSR_R0_PMON_IPERF0_P0	
Uncore R-box 0 PerfMon IPERF0 unit Port 0 select MSR.		Package
Register Address: E05H, 3589	MSR_R0_PMON_IPERF0_P1	
Uncore R-box 0 PerfMon IPERF0 unit Port 1 select MSR.		Package
Register Address: E06H, 3590	MSR_R0_PMON_IPERF0_P2	
Uncore R-box 0 PerfMon IPERF0 unit Port 2 select MSR.		Package
Register Address: E07H, 3591	MSR_R0_PMON_IPERF0_P3	
Uncore R-box 0 PerfMon IPERF0 unit Port 3 select MSR.		Package
Register Address: E08H, 3592	MSR_R0_PMON_IPERF0_P4	
Uncore R-box 0 PerfMon IPERF0 unit Port 4 select MSR.		Package
Register Address: E09H, 3593	MSR_R0_PMON_IPERF0_P5	
Uncore R-box 0 PerfMon IPERF0 unit Port 5 select MSR.		Package
Register Address: E0AH, 3594	MSR_R0_PMON_IPERF0_P6	
Uncore R-box 0 PerfMon IPERF0 unit Port 6 select MSR.		Package
Register Address: E0BH, 3595	MSR_R0_PMON_IPERF0_P7	
Uncore R-box 0 PerfMon IPERF0 unit Port 7 select MSR.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E0CH, 3596	MSR_R0_PMON_QLX_P0	
Uncore R-box 0 PerfMon QLX unit Port 0 select MSR.		Package
Register Address: E0DH, 3597	MSR_R0_PMON_QLX_P1	
Uncore R-box 0 PerfMon QLX unit Port 1 select MSR.		Package
Register Address: E0EH, 3598	MSR_R0_PMON_QLX_P2	
Uncore R-box 0 PerfMon QLX unit Port 2 select MSR.		Package
Register Address: E0FH, 3599	MSR_R0_PMON_QLX_P3	
Uncore R-box 0 PerfMon QLX unit Port 3 select MSR.		Package
Register Address: E10H, 3600	MSR_R0_PMON_EVNT_SELO	
Uncore R-box 0 PerfMon event select MSR.		Package
Register Address: E11H, 3601	MSR_R0_PMON_CTR0	
Uncore R-box 0 PerfMon counter MSR.		Package
Register Address: E12H, 3602	MSR_R0_PMON_EVNT_SEL1	
Uncore R-box 0 PerfMon event select MSR.		Package
Register Address: E13H, 3603	MSR_R0_PMON_CTR1	
Uncore R-box 0 PerfMon counter MSR.		Package
Register Address: E14H, 3604	MSR_R0_PMON_EVNT_SEL2	
Uncore R-box 0 PerfMon event select MSR.		Package
Register Address: E15H, 3605	MSR_R0_PMON_CTR2	
Uncore R-box 0 PerfMon counter MSR.		Package
Register Address: E16H, 3606	MSR_R0_PMON_EVNT_SEL3	
Uncore R-box 0 PerfMon event select MSR.		Package
Register Address: E17H, 3607	MSR_R0_PMON_CTR3	
Uncore R-box 0 PerfMon counter MSR.		Package
Register Address: E18H, 3608	MSR_R0_PMON_EVNT_SEL4	
Uncore R-box 0 PerfMon event select MSR.		Package
Register Address: E19H, 3609	MSR_R0_PMON_CTR4	
Uncore R-box 0 PerfMon counter MSR.		Package
Register Address: E1AH, 3610	MSR_R0_PMON_EVNT_SEL5	
Uncore R-box 0 PerfMon event select MSR.		Package
Register Address: E1BH, 3611	MSR_R0_PMON_CTR5	
Uncore R-box 0 PerfMon counter MSR.		Package
Register Address: E1CH, 3612	MSR_R0_PMON_EVNT_SEL6	
Uncore R-box 0 PerfMon event select MSR.		Package
Register Address: E1DH, 3613	MSR_R0_PMON_CTR6	
Uncore R-box 0 PerfMon counter MSR.		Package
Register Address: E1EH, 3614	MSR_R0_PMON_EVNT_SEL7	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore R-box 0 PerfMon event select MSR.		Package
Register Address: E1FH, 3615	MSR_R0_PMON_CTR7	
Uncore R-box 0 PerfMon counter MSR.		Package
Register Address: E20H, 3616	MSR_R1_PMON_BOX_CTRL	
Uncore R-box 1 PerfMon local box control MSR.		Package
Register Address: E21H, 3617	MSR_R1_PMON_BOX_STATUS	
Uncore R-box 1 PerfMon local box status MSR.		Package
Register Address: E22H, 3618	MSR_R1_PMON_BOX_OVF_CTRL	
Uncore R-box 1 PerfMon local box overflow control MSR.		Package
Register Address: E24H, 3620	MSR_R1_PMON_IPERF1_P8	
Uncore R-box 1 PerfMon IPERF1 unit Port 8 select MSR.		Package
Register Address: E25H, 3621	MSR_R1_PMON_IPERF1_P9	
Uncore R-box 1 PerfMon IPERF1 unit Port 9 select MSR.		Package
Register Address: E26H, 3622	MSR_R1_PMON_IPERF1_P10	
Uncore R-box 1 PerfMon IPERF1 unit Port 10 select MSR.		Package
Register Address: E27H, 3623	MSR_R1_PMON_IPERF1_P11	
Uncore R-box 1 PerfMon IPERF1 unit Port 11 select MSR.		Package
Register Address: E28H, 3624	MSR_R1_PMON_IPERF1_P12	
Uncore R-box 1 PerfMon IPERF1 unit Port 12 select MSR.		Package
Register Address: E29H, 3625	MSR_R1_PMON_IPERF1_P13	
Uncore R-box 1 PerfMon IPERF1 unit Port 13 select MSR.		Package
Register Address: E2AH, 3626	MSR_R1_PMON_IPERF1_P14	
Uncore R-box 1 PerfMon IPERF1 unit Port 14 select MSR.		Package
Register Address: E2BH, 3627	MSR_R1_PMON_IPERF1_P15	
Uncore R-box 1 PerfMon IPERF1 unit Port 15 select MSR.		Package
Register Address: E2CH, 3628	MSR_R1_PMON_QLX_P4	
Uncore R-box 1 PerfMon QLX unit Port 4 select MSR.		Package
Register Address: E2DH, 3629	MSR_R1_PMON_QLX_P5	
Uncore R-box 1 PerfMon QLX unit Port 5 select MSR.		Package
Register Address: E2EH, 3630	MSR_R1_PMON_QLX_P6	
Uncore R-box 1 PerfMon QLX unit Port 6 select MSR.		Package
Register Address: E2FH, 3631	MSR_R1_PMON_QLX_P7	
Uncore R-box 1 PerfMon QLX unit Port 7 select MSR.		Package
Register Address: E30H, 3632	MSR_R1_PMON_EVNT_SEL8	
Uncore R-box 1 PerfMon event select MSR.		Package
Register Address: E31H, 3633	MSR_R1_PMON_CTR8	
Uncore R-box 1 PerfMon counter MSR.		Package

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E32H, 3634	MSR_R1_PMON_EVNT_SEL9	
Uncore R-box 1 PerfMon event select MSR.		Package
Register Address: E33H, 3635	MSR_R1_PMON_CTR9	
Uncore R-box 1 PerfMon counter MSR.		Package
Register Address: E34H, 3636	MSR_R1_PMON_EVNT_SEL10	
Uncore R-box 1 PerfMon event select MSR.		Package
Register Address: E35H, 3637	MSR_R1_PMON_CTR10	
Uncore R-box 1 PerfMon counter MSR.		Package
Register Address: E36H, 3638	MSR_R1_PMON_EVNT_SEL11	
Uncore R-box 1 PerfMon event select MSR.		Package
Register Address: E37H, 3639	MSR_R1_PMON_CTR11	
Uncore R-box 1 PerfMon counter MSR.		Package
Register Address: E38H, 3640	MSR_R1_PMON_EVNT_SEL12	
Uncore R-box 1 PerfMon event select MSR.		Package
Register Address: E39H, 3641	MSR_R1_PMON_CTR12	
Uncore R-box 1 PerfMon counter MSR.		Package
Register Address: E3AH, 3642	MSR_R1_PMON_EVNT_SEL13	
Uncore R-box 1 PerfMon event select MSR.		Package
Register Address: E3BH, 3643	MSR_R1_PMON_CTR13	
Uncore R-box 1 PerfMon counter MSR.		Package
Register Address: E3CH, 3644	MSR_R1_PMON_EVNT_SEL14	
Uncore R-box 1 PerfMon event select MSR.		Package
Register Address: E3DH, 3645	MSR_R1_PMON_CTR14	
Uncore R-box 1 PerfMon counter MSR.		Package
Register Address: E3EH, 3646	MSR_R1_PMON_EVNT_SEL15	
Uncore R-box 1 PerfMon event select MSR.		Package
Register Address: E3FH, 3647	MSR_R1_PMON_CTR15	
Uncore R-box 1 PerfMon counter MSR.		Package
Register Address: E45H, 3653	MSR_B0_PMON_MATCH	
Uncore B-box 0 PerfMon local box match MSR.		Package
Register Address: E46H, 3654	MSR_B0_PMON_MASK	
Uncore B-box 0 PerfMon local box mask MSR.		Package
Register Address: E49H, 3657	MSR_S0_PMON_MATCH	
Uncore S-box 0 PerfMon local box match MSR.		Package
Register Address: E4AH, 3658	MSR_S0_PMON_MASK	
Uncore S-box 0 PerfMon local box mask MSR.		Package
Register Address: E4DH, 3661	MSR_B1_PMON_MATCH	

Table 2-17. Additional MSRs in the Intel® Xeon® Processor 7500 Series (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore B-box 1 PerfMon local box match MSR.		Package
Register Address: E4EH, 3662	MSR_B1_PMON_MASK	
Uncore B-box 1 PerfMon local box mask MSR.		Package
Register Address: E54H, 3668	MSR_M0_PMON_MM_CONFIG	
Uncore M-box 0 PerfMon local box address match/mask config MSR.		Package
Register Address: E55H, 3669	MSR_M0_PMON_ADDR_MATCH	
Uncore M-box 0 PerfMon local box address match MSR.		Package
Register Address: E56H, 3670	MSR_M0_PMON_ADDR_MASK	
Uncore M-box 0 PerfMon local box address mask MSR.		Package
Register Address: E59H, 3673	MSR_S1_PMON_MATCH	
Uncore S-box 1 PerfMon local box match MSR.		Package
Register Address: E5AH, 3674	MSR_S1_PMON_MASK	
Uncore S-box 1 PerfMon local box mask MSR.		Package
Register Address: E5CH, 3676	MSR_M1_PMON_MM_CONFIG	
Uncore M-box 1 PerfMon local box address match/mask config MSR.		Package
Register Address: E5DH, 3677	MSR_M1_PMON_ADDR_MATCH	
Uncore M-box 1 PerfMon local box address match MSR.		Package
Register Address: E5EH, 3678	MSR_M1_PMON_ADDR_MASK	
Uncore M-box 1 PerfMon local box address mask MSR.		Package
Register Address: 3B5H, 965	MSR_UNCORE_PMC5	
See Section 21.3.1.2.2, "Uncore Performance Event Configuration Facility."		Package

2.9 MSRS IN THE INTEL® XEON® PROCESSOR 5600 SERIES BASED ON WESTMERE MICROARCHITECTURE

The Intel® Xeon® Processor 5600 Series is based on Westmere microarchitecture and supports the MSR interfaces listed in Table 2-15, Table 2-16, plus additional MSRs listed in Table 2-18. These MSRs apply to the Intel Core i7, i5, and i3 processor family with a CPUID Signature DisplayFamily_DisplayModel value of 06_25H or 06_2CH; see Table 2-1.

Table 2-18. Additional MSRs Supported by Intel® Processors Based on Westmere Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 13CH, 316	MSR_FEATURE_CONFIG	
AES Configuration (RW-L) Privileged post-BIOS agent must provide a #GP handler to handle unsuccessful read of this MSR.		Core

Table 2-18. Additional MSRs Supported by Intel® Processors Based on Westmere Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
1:0	AES Configuration (RW-L) Upon a successful read of this MSR, the configuration of AES instruction set availability is as follows: 11b: AES instructions are not available until next RESET. Otherwise, AES instructions are available. Note, AES instruction set is not available if read is unsuccessful. If the configuration is not 01b, AES instructions can be mis-configured if a privileged agent unintentionally writes 11b.	
63:2	Reserved.	
Register Address: 1A7H, 423	MSR_OFFCORE_RSP_1	
Offcore Response Event Select Register (R/W)		Thread
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0. R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 core active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 core active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 core active.	Package
39:32	Maximum Ratio Limit for 5C Maximum turbo ratio limit of 5 core active.	Package
47:40	Maximum Ratio Limit for 6C Maximum turbo ratio limit of 6 core active.	Package
63:48	Reserved.	
Register Address: 1B0H, 432	IA32_ENERGY_PERF_BIAS	
See Table 2-2.		Package

2.10 MSRS IN THE INTEL® XEON® PROCESSOR E7 FAMILY BASED ON WESTMERE MICROARCHITECTURE

The Intel® Xeon® Processor E7 Family is based on the Westmere microarchitecture and supports the MSR interfaces listed in Table 2-15 (except MSR address 1ADH), Table 2-16, plus additional MSRs listed in Table 2-19. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_2FH.

Table 2-19. Additional MSRs Supported by the Intel® Xeon® Processor E7 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 13CH, 316	MSR_FEATURE_CONFIG	
AES Configuration (RW-L) Privileged post-BIOS agent must provide a #GP handler to handle unsuccessful read of this MSR.		Core
1:0	AES Configuration (RW-L) Upon a successful read of this MSR, the configuration of AES instruction set availability is as follows: 11b: AES instructions are not available until next RESET. Otherwise, AES instructions are available. Note, AES instruction set is not available if read is unsuccessful. If the configuration is not 01b, AES instructions can be mis-configured if a privileged agent unintentionally writes 11b.	
63:2	Reserved.	
Register Address: 1A7H, 423	MSR_OFFCORE_RSP_1	
Offcore Response Event Select Register (R/W)		Thread
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Reserved. Attempt to read/write will cause #UD.		Package
Register Address: 1B0H, 432	IA32_ENERGY_PERF_BIAS	
See Table 2-2.		Package
Register Address: F40H, 3904	MSR_C8_PMON_BOX_CTRL	
Uncore C-box 8 PerfMon local box control MSR.		Package
Register Address: F41H, 3905	MSR_C8_PMON_BOX_STATUS	
Uncore C-box 8 PerfMon local box status MSR.		Package
Register Address: F42H, 3906	MSR_C8_PMON_BOX_OVF_CTRL	
Uncore C-box 8 PerfMon local box overflow control MSR.		Package
Register Address: F50H, 3920	MSR_C8_PMON_EVNT_SELO	
Uncore C-box 8 PerfMon event select MSR.		Package
Register Address: F51H, 3921	MSR_C8_PMON_CTR0	
Uncore C-box 8 PerfMon counter MSR.		Package
Register Address: F52H, 3922	MSR_C8_PMON_EVNT_SEL1	
Uncore C-box 8 PerfMon event select MSR.		Package
Register Address: F53H, 3923	MSR_C8_PMON_CTR1	
Uncore C-box 8 PerfMon counter MSR.		Package
Register Address: F54H, 3924	MSR_C8_PMON_EVNT_SEL2	
Uncore C-box 8 PerfMon event select MSR.		Package
Register Address: F55H, 3925	MSR_C8_PMON_CTR2	
Uncore C-box 8 PerfMon counter MSR.		Package
Register Address: F56H, 3926	MSR_C8_PMON_EVNT_SEL3	
Uncore C-box 8 PerfMon event select MSR.		Package

Table 2-19. Additional MSRs Supported by the Intel® Xeon® Processor E7 Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: F57H, 3927	MSR_C8_PMON_CTR3	
Uncore C-box 8 PerfMon counter MSR.		Package
Register Address: F58H, 3928	MSR_C8_PMON_EVNT_SEL4	
Uncore C-box 8 PerfMon event select MSR.		Package
Register Address: F59H, 3929	MSR_C8_PMON_CTR4	
Uncore C-box 8 PerfMon counter MSR.		Package
Register Address: F5AH, 3930	MSR_C8_PMON_EVNT_SEL5	
Uncore C-box 8 PerfMon event select MSR.		Package
Register Address: F5BH, 3931	MSR_C8_PMON_CTR5	
Uncore C-box 8 PerfMon counter MSR.		Package
Register Address: FC0H, 4032	MSR_C9_PMON_BOX_CTRL	
Uncore C-box 9 PerfMon local box control MSR.		Package
Register Address: FC1H, 4033	MSR_C9_PMON_BOX_STATUS	
Uncore C-box 9 PerfMon local box status MSR.		Package
Register Address: FC2H, 4034	MSR_C9_PMON_BOX_OVF_CTRL	
Uncore C-box 9 PerfMon local box overflow control MSR.		Package
Register Address: FDOH, 4048	MSR_C9_PMON_EVNT_SELO	
Uncore C-box 9 PerfMon event select MSR.		Package
Register Address: FD1H, 4049	MSR_C9_PMON_CTR0	
Uncore C-box 9 PerfMon counter MSR.		Package
Register Address: FD2H, 4050	MSR_C9_PMON_EVNT_SEL1	
Uncore C-box 9 PerfMon event select MSR.		Package
Register Address: FD3H, 4051	MSR_C9_PMON_CTR1	
Uncore C-box 9 PerfMon counter MSR.		Package
Register Address: FD4H, 4052	MSR_C9_PMON_EVNT_SEL2	
Uncore C-box 9 PerfMon event select MSR.		Package
Register Address: FD5H, 4053	MSR_C9_PMON_CTR2	
Uncore C-box 9 PerfMon counter MSR.		Package
Register Address: FD6H, 4054	MSR_C9_PMON_EVNT_SEL3	
Uncore C-box 9 PerfMon event select MSR.		Package
Register Address: FD7H, 4055	MSR_C9_PMON_CTR3	
Uncore C-box 9 PerfMon counter MSR.		Package
Register Address: FD8H, 4056	MSR_C9_PMON_EVNT_SEL4	
Uncore C-box 9 PerfMon event select MSR.		Package
Register Address: FD9H, 4057	MSR_C9_PMON_CTR4	
Uncore C-box 9 PerfMon counter MSR.		Package
Register Address: FDAH, 4058	MSR_C9_PMON_EVNT_SEL5	

Table 2-19. Additional MSRs Supported by the Intel® Xeon® Processor E7 Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-box 9 PerfMon event select MSR.		Package
Register Address: FDBH, 4059	MSR_C9_PMON_CTR5	
Uncore C-box 9 PerfMon counter MSR.		Package

2.11 MSRS IN THE INTEL® PROCESSOR FAMILY BASED ON SANDY BRIDGE MICROARCHITECTURE

Table 2-20 lists model-specific registers (MSRs) that are common to the Intel® processor family based on Sandy Bridge microarchitecture. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_2AH or 06_2DH; see Table 2-1. Additional MSRs specific to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_2AH are listed in Table 2-21.

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 0H, 0	IA32_P5_MC_ADDR	
See Section 2.23, "MSRs in Pentium Processors."		Thread
Register Address: 1H, 1	IA32_P5_MC_TYPE	
See Section 2.23, "MSRs in Pentium Processors."		Thread
Register Address: 6H, 6	IA32_MONITOR_FILTER_SIZE	
See Section 10.10.5, "Monitor/Mwait Address Range Determination," and Table 2-2.		Thread
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER	
See Section 19.17, "Time-Stamp Counter," and see Table 2-2.		Thread
Register Address: 17H, 23	IA32_PLATFORM_ID	
Platform ID (R) See Table 2-2.		Package
Register Address: 1BH, 27	IA32_APIC_BASE	
See Section 12.4.4, "Local APIC Status and Location," and Table 2-2.		Thread
Register Address: 34H, 52	MSR_SMI_COUNT	
SMI Counter (R/O)		Thread
31:0	SMI Count (R/O) Count SMIs.	
63:32	Reserved.	
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64 Processor (R/W) See Table 2-2.		Thread
0	Lock (R/WL)	
1	Enable VMX Inside SMX Operation (R/WL)	
2	Enable VMX Outside SMX Operation (R/WL)	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
14:8	SENTER Local Functions Enables (R/WL)	
15	SENTER Global Functions Enable (R/WL)	
Register Address: 79H, 121	IA32_BIOS_UPDT_TRIG	
BIOS Update Trigger Register (W) See Table 2-2.		Core
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID	
BIOS Update Signature ID (R/W) See Table 2-2.		Thread
Register Address: C1H, 193	IA32_PMC0	
Performance Counter Register See Table 2-2.		Thread
Register Address: C2H, 194	IA32_PMC1	
Performance Counter Register See Table 2-2.		Thread
Register Address: C3H, 195	IA32_PMC2	
Performance Counter Register See Table 2-2.		Thread
Register Address: C4H, 196	IA32_PMC3	
Performance Counter Register See Table 2-2.		Thread
Register Address: C5H, 197	IA32_PMC4	
Performance Counter Register (if core not shared by threads)		Core
Register Address: C6H, 198	IA32_PMC5	
Performance Counter Register (if core not shared by threads)		Core
Register Address: C7H, 199	IA32_PMC6	
Performance Counter Register (if core not shared by threads)		Core
Register Address: C8H, 200	IA32_PMC7	
Performance Counter Register (if core not shared by threads)		Core
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) This is the ratio of the frequency that invariant TSC runs at. Frequency = ratio * 100 MHz.	Package
27:16	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) When set to 1, indicates that Programmable Ratio Limit for Turbo mode is enabled. When set to 0, indicates Programmable Ratio Limit for Turbo mode is disabled.	Package

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
29	Programmable TDP Limit for Turbo Mode (R/O) When set to 1, indicates that TDP Limit for Turbo mode is programmable. When set to 0, indicates TDP Limit for Turbo mode is not programmable.	Package
39:30	Reserved.	
47:40	Maximum Efficiency Ratio (R/O) This is the minimum ratio (maximum efficiency) that the processor can operate, in units of 100MHz.	Package
63:48	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States. See http://biosbits.org .		Core
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: C0/C1 (no package C-state support) 001b: C2 010b: C6 no retention 011b: C6 retention 100b: C7 101b: C7s 111: No package C-state limit Note: This field cannot be used to limit package C-state to C3.	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W) When set, will map IO_read instructions sent to IO register specified by MSR_PMG_IO_CAPTURE_BASE to MWAIT instructions.	
14:11	Reserved.	
15	CFG Lock (R/WO) When set, locks bits 15:0 of this register until next reset.	
24:16	Reserved.	
25	C3 State Auto Demotion Enable (R/W) When set, the processor will conditionally demote C6/C7 requests to C3 based on uncore auto-demote information.	
26	C1 State Auto Demotion Enable (R/W) When set, the processor will conditionally demote C3/C6/C7 requests to C1 based on uncore auto-demote information.	
27	Enable C3 Undemotion (R/W) When set, enables undemotion from demoted C3.	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
28	Enable C1 Undemotion (R/W) When set, enables undemotion from demoted C1.	
63:29	Reserved.	
Register Address: E4H, 228	MSR_PMG_IO_CAPTURE_BASE	
Power Management IO Redirection in C-state (R/W) See http://biosbits.org .		Core
15:0	LVL_2 Base Address (R/W) Specifies the base address visible to software for IO redirection. If IO MWAIT Redirection is enabled, reads to this address will be consumed by the power management logic and decoded to MWAIT instructions. When IO port address redirection is enabled, this is the IO port address reported to the OS/software.	
18:16	C-State Range (R/W) Specifies the encoding value of the maximum C-State code name to be included when IO read to MWAIT redirection is enabled by MSR_PKG_CST_CONFIG_CONTROL[bit10]: 000b - C3 is the max C-State to include. 001b - C6 is the max C-State to include. 010b - C7 is the max C-State to include.	
63:19	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: FEH, 254	IA32_MTRRCAP	
See Table 2-2.		Thread
Register Address: 13CH, 316	MSR_FEATURE_CONFIG	
AES Configuration (RW-L) Privileged post-BIOS agent must provide a #GP handler to handle unsuccessful read of this MSR.		Core
1:0	AES Configuration (RW-L) Upon a successful read of this MSR, the configuration of AES instruction set availability is as follows: 11b: AES instructions are not available until next RESET. Otherwise, AES instructions are available. Note, AES instruction set is not available if read is unsuccessful. If the configuration is not 01b, AES instructions can be mis-configured if a privileged agent unintentionally writes 11b.	
63:2	Reserved.	
Register Address: 174H, 372	IA32_SYSENTER_CS	
See Table 2-2.		Thread

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 175H, 373	IA32_SYSENTER_ESP	
See Table 2-2.		Thread
Register Address: 176H, 374	IA32_SYSENTER_EIP	
See Table 2-2.		Thread
Register Address: 179H, 377	IA32_MCG_CAP	
See Table 2-2.		Thread
Register Address: 17AH, 378	IA32_MCG_STATUS	
Global Machine Check Status		Thread
0	RIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) can be used to restart the program. If cleared, the program cannot be reliably restarted.	
1	EIPV When set, bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) is directly associated with the error.	
2	MCIP When set, bit indicates that a machine check has been generated. If a second machine check is detected while this bit is still set, the processor enters a shutdown state. Software should write this bit to 0 after processing a machine check exception.	
63:3	Reserved.	
Register Address: 186H, 390	IA32_PERFVTSEL0	
See Table 2-2.		Thread
Register Address: 187H, 391	IA32_PERFVTSEL1	
See Table 2-2.		Thread
Register Address: 188H, 392	IA32_PERFVTSEL2	
See Table 2-2.		Thread
Register Address: 189H, 393	IA32_PERFVTSEL3	
See Table 2-2.		Thread
Register Address: 18AH, 394	IA32_PERFVTSEL4	
See Table 2-2. If CPUID.0AH:EAX[15:8] > 4.		Core
Register Address: 18BH, 395	IA32_PERFVTSEL5	
See Table 2-2. If CPUID.0AH:EAX[15:8] > 5.		Core
Register Address: 18CH, 396	IA32_PERFVTSEL6	
See Table 2-2. If CPUID.0AH:EAX[15:8] > 6.		Core
Register Address: 18DH, 397	IA32_PERFVTSEL7	
See Table 2-2. If CPUID.0AH:EAX[15:8] > 7.		Core
Register Address: 198H, 408	IA32_PERF_STATUS	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2.		Package
15:0	Current Performance State Value	
63:16	Reserved.	
Register Address: 198H, 408	MSR_PERF_STATUS	
Performance Status		Package
47:32	Core Voltage (R/O) P-state core voltage can be computed by MSR_PERF_STATUS[37:32] * (float) 1/(2 ¹³).	
Register Address: 199H, 409	IA32_PERF_CTL	
See Table 2-2.		Thread
Register Address: 19AH, 410	IA32_CLOCK_MODULATION	
Clock Modulation (R/W) See Table 2-2. IA32_CLOCK_MODULATION MSR was originally named IA32_THERM_CONTROL MSR.		Thread
3:0	On demand Clock Modulation Duty Cycle (R/W) In 6.25% increment.	
4	On demand Clock Modulation Enable (R/W)	
63:5	Reserved.	
Register Address: 19BH, 411	IA32_THERM_INTERRUPT	
Thermal Interrupt Control (R/W) See Table 2-2.		Core
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Core
0	Thermal Status (R/O) See Table 2-2.	
1	Thermal Status Log (R/WC0) See Table 2-2.	
2	PROTCHOT # or FORCEPR# Status (R/O) See Table 2-2.	
3	PROTCHOT # or FORCEPR# Log (R/WC0) See Table 2-2.	
4	Critical Temperature Status (R/O) See Table 2-2.	
5	Critical Temperature Status Log (R/WC0) See Table 2-2.	
6	Thermal Threshold #1 Status (R/O) See Table 2-2.	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
7	Thermal Threshold #1 Log (R/WC0) See Table 2-2.	
8	Thermal Threshold #2 Status (R/O) See Table 2-2.	
9	Thermal Threshold #2 Log (R/WC0) See Table 2-2.	
10	Power Limitation Status (R/O) See Table 2-2.	
11	Power Limitation Log (R/WC0) See Table 2-2.	
15:12	Reserved.	
22:16	Digital Readout (R/O) See Table 2-2.	
26:23	Reserved.	
30:27	Resolution in Degrees Celsius (R/O) See Table 2-2.	
31	Reading Valid (R/O) See Table 2-2.	
63:32	Reserved.	
Register Address: 1A0H, 416	IA32_MISC_ENABLE	
Enable Misc. Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		
0	Fast-Strings Enable See Table 2-2.	Thread
6:1	Reserved.	
7	Performance Monitoring Available (R) See Table 2-2.	Thread
10:8	Reserved	
11	Branch Trace Storage Unavailable (R/O) See Table 2-2.	Thread
12	Processor Event Based Sampling Unavailable (R/O) See Table 2-2.	Thread
15:13	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W) See Table 2-2.	Package
18	ENABLE MONITOR FSM (R/W) See Table 2-2.	Thread
21:19	Reserved.	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
22	Limit CPUID Maxval (R/W) See Table 2-2.	Thread
23	xTPR Message Disable (R/W) See Table 2-2.	Thread
33:24	Reserved.	
34	XD Bit Disable (R/W) See Table 2-3.	Thread
37:35	Reserved.	
38	Turbo Mode Disable (R/W) When set to 1 on processors that support Intel Turbo Boost Technology, the turbo mode feature is disabled and the IDA_Enable feature flag will be clear (CPUID.06H: EAX[1]=0). When set to a 0 on processors that support IDA, CPUID.06H: EAX[1] reports the processor's support of turbo mode is enabled. Note: The power-on default value is used by BIOS to detect hardware support of turbo mode. If the power-on default value is 1, turbo mode is available in the processor. If the power-on default value is 0, turbo mode is not available.	Package
63:39	Reserved.	
Register Address: 1A2H, 418	MSR_TEMPERATURE_TARGET	
Temperature Target		Unique
15:0	Reserved.	
23:16	Temperature Target (R) The minimum temperature at which PROCHOT# will be asserted. The value is degrees C.	
63:24	Reserved.	
Register Address: 1A4H, 420	MSR_MISC_FEATURE_CONTROL	
Miscellaneous Feature Control (R/W)		
0	L2 Hardware Prefetcher Disable (R/W) If 1, disables the L2 hardware prefetcher, which fetches additional lines of code or data into the L2 cache.	Core
1	L2 Adjacent Cache Line Prefetcher Disable (R/W) If 1, disables the adjacent cache line prefetcher, which fetches the cache line that comprises a cache line pair (128 bytes).	Core
2	DCU Hardware Prefetcher Disable (R/W) If 1, disables the L1 data cache prefetcher, which fetches the next cache line into L1 data cache.	Core
3	DCU IP Prefetcher Disable (R/W) If 1, disables the L1 data cache IP prefetcher, which uses sequential load history (based on instruction pointer of previous loads) to determine whether to prefetch additional lines.	Core
63:4	Reserved.	
Register Address: 1A6H, 422	MSR_OFFCORE_RSP_0	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Offcore Response Event Select Register (R/W)		Thread
Register Address: 1A7H, 423	MSR_OFFCORE_RSP_1	
Offcore Response Event Select Register (R/W)		Thread
Register Address: 1AAH, 426	MSR_MISC_PWR_MGMT	
Miscellaneous Power Management Control Various model specific features enumeration. See http://biosbits.org .		
Register Address: 1B0H, 432	IA32_ENERGY_PERF_BIAS	
See Table 2-2.		Package
Register Address: 1B1H, 433	IA32_PACKAGE_THERM_STATUS	
See Table 2-2.		Package
Register Address: 1B2H, 434	IA32_PACKAGE_THERM_INTERRUPT	
See Table 2-2.		Package
Register Address: 1C8H, 456	MSR_LBR_SELECT	
Last Branch Record Filtering Select Register (R/W) See Section 19.9.2, "Filtering of Last Branch Records."		Thread
0	CPL_EQ_0	
1	CPL_NEQ_0	
2	JCC	
3	NEAR_REL_CALL	
4	NEAR_IND_CALL	
5	NEAR_RET	
6	NEAR_IND_JMP	
7	NEAR_REL_JMP	
8	FAR_BRANCH	
63:9	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-3) that points to the MSR containing the most recent branch record. See MSR_LASTBRANCH_0_FROM_IP (at 680H).		Thread
Register Address: 1D9H, 473	IA32_DEBUGCTL	
Debug Control (R/W) See Table 2-2.		Thread
0	LBR: Last Branch Record	
1	BTF	
5:2	Reserved.	
6	TR: Branch Trace	
7	BTS: Log Branch Trace Message to BTS buffer	
8	BTINT	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
9	BTS_OFF_OS	
10	BTS_OFF_USER	
11	FREEZE_LBR_ON_PMI	
12	FREEZE_PERFMON_ON_PMI	
13	ENABLE_UNCORE_PMI	
14	FREEZE_WHILE_SMM	
63:15	Reserved.	
Register Address: 1DDH, 477	MSR_LER_FROM_LIP	
Last Exception Record From Linear IP (R/W) Contains a pointer to the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Thread
Register Address: 1DEH, 478	MSR_LER_TO_LIP	
Last Exception Record To Linear IP (R/W) This area contains a pointer to the target of the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Thread
Register Address: 1F2H, 498	IA32_SMRR_PHYSBASE	
See Table 2-2.		Core
Register Address: 1F3H, 499	IA32_SMRR_PHYSMASK	
See Table 2-2.		Core
Register Address: 1FCH, 508	MSR_POWER_CTL	
See http://biosbits.org .		Core
Register Address: 200H, 512	IA32_MTRR_PHYSBASE0	
See Table 2-2.		Thread
Register Address: 201H, 513	IA32_MTRR_PHYSMASK0	
See Table 2-2.		Thread
Register Address: 202H, 514	IA32_MTRR_PHYSBASE1	
See Table 2-2.		Thread
Register Address: 203H, 515	IA32_MTRR_PHYSMASK1	
See Table 2-2.		Thread
Register Address: 204H, 516	IA32_MTRR_PHYSBASE2	
See Table 2-2.		Thread
Register Address: 205H, 517	IA32_MTRR_PHYSMASK2	
See Table 2-2.		Thread
Register Address: 206H, 518	IA32_MTRR_PHYSBASE3	
See Table 2-2.		Thread
Register Address: 207H, 519	IA32_MTRR_PHYSMASK3	
See Table 2-2.		Thread
Register Address: 208H, 520	IA32_MTRR_PHYSBASE4	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2.		Thread
Register Address: 209H, 521	IA32_MTRR_PHYSMASK4	
See Table 2-2.		Thread
Register Address: 20AH, 522	IA32_MTRR_PHYSBASE5	
See Table 2-2.		Thread
Register Address: 20BH, 523	IA32_MTRR_PHYSMASK5	
See Table 2-2.		Thread
Register Address: 20CH, 524	IA32_MTRR_PHYSBASE6	
See Table 2-2.		Thread
Register Address: 20DH, 525	IA32_MTRR_PHYSMASK6	
See Table 2-2.		Thread
Register Address: 20EH, 526	IA32_MTRR_PHYSBASE7	
See Table 2-2.		Thread
Register Address: 20FH, 527	IA32_MTRR_PHYSMASK7	
See Table 2-2.		Thread
Register Address: 210H, 528	IA32_MTRR_PHYSBASE8	
See Table 2-2.		Thread
Register Address: 211H, 529	IA32_MTRR_PHYSMASK8	
See Table 2-2.		Thread
Register Address: 212H, 530	IA32_MTRR_PHYSBASE9	
See Table 2-2.		Thread
Register Address: 213H, 531	IA32_MTRR_PHYSMASK9	
See Table 2-2.		Thread
Register Address: 250H, 592	IA32_MTRR_FIX64K_00000	
See Table 2-2.		Thread
Register Address: 258H, 600	IA32_MTRR_FIX16K_80000	
See Table 2-2.		Thread
Register Address: 259H, 601	IA32_MTRR_FIX16K_A0000	
See Table 2-2.		Thread
Register Address: 268H, 616	IA32_MTRR_FIX4K_C0000	
See Table 2-2.		Thread
Register Address: 269H, 617	IA32_MTRR_FIX4K_80000	
See Table 2-2.		Thread
Register Address: 26AH, 618	IA32_MTRR_FIX4K_D0000	
See Table 2-2.		Thread
Register Address: 26BH, 619	IA32_MTRR_FIX4K_D8000	
See Table 2-2.		Thread

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 26CH, 620	IA32_MTRR_FIX4K_E0000	
See Table 2-2.		Thread
Register Address: 26DH, 621	IA32_MTRR_FIX4K_E8000	
See Table 2-2.		Thread
Register Address: 26EH, 622	IA32_MTRR_FIX4K_F0000	
See Table 2-2.		Thread
Register Address: 26FH, 623	IA32_MTRR_FIX4K_F8000	
See Table 2-2.		Thread
Register Address: 277H, 631	IA32_PAT	
See Table 2-2.		Thread
Register Address: 280H, 640	IA32_MCO_CTL2	
See Table 2-2.		Core
Register Address: 281H, 641	IA32_MC1_CTL2	
See Table 2-2.		Core
Register Address: 282H, 642	IA32_MC2_CTL2	
See Table 2-2.		Core
Register Address: 283H, 643	IA32_MC3_CTL2	
See Table 2-2.		Core
Register Address: 284H, 644	IA32_MC4_CTL2	
Always 0 (CMCI not supported).		Package
Register Address: 2FFH, 767	IA32_MTRR_DEF_TYPE	
Default Memory Types (R/W) See Table 2-2.		Thread
Register Address: 309H, 777	IA32_FIXED_CTR0	
Fixed-Function Performance Counter Register 0 (R/W) See Table 2-2.		Thread
Register Address: 30AH, 778	IA32_FIXED_CTR1	
Fixed-Function Performance Counter Register 1 (R/W) See Table 2-2.		Thread
Register Address: 30BH, 779	IA32_FIXED_CTR2	
Fixed-Function Performance Counter Register 2 (R/W) See Table 2-2.		Thread
Register Address: 345H, 837	IA32_PERF_CAPABILITIES	
See Table 2-2 and Section 19.4.1, "IA32_DEBUGCTL MSR."		Thread
5:0	LBR Format See Table 2-2.	
6	PEBS Record Format.	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
7	PEBSSaveArchRegs See Table 2-2.	
11:8	PEBS_REC_FORMAT See Table 2-2.	
12	SMM_FREEZE See Table 2-2.	
63:13	Reserved.	
Register Address: 38DH, 909	IA32_FIXED_CTR_CTRL	
Fixed-Function-Counter Control Register (R/W) See Table 2-2.		Thread
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	
See Table 2-2 and Section 21.6.2.2, "Global Counter Control Facilities."		
0	Ovf_PMC0	Thread
1	Ovf_PMC1	Thread
2	Ovf_PMC2	Thread
3	Ovf_PMC3	Thread
4	Ovf_PMC4 (if CPUID.0AH:EAX[15:8] > 4)	Core
5	Ovf_PMC5 (if CPUID.0AH:EAX[15:8] > 5)	Core
6	Ovf_PMC6 (if CPUID.0AH:EAX[15:8] > 6)	Core
7	Ovf_PMC7 (if CPUID.0AH:EAX[15:8] > 7)	Core
31:8	Reserved.	
32	Ovf_FixedCtr0	Thread
33	Ovf_FixedCtr1	Thread
34	Ovf_FixedCtr2	Thread
60:35	Reserved.	
61	Ovf_Uncore	Thread
62	Ovf_BufDSSAVE	Thread
63	CondChgd	Thread
Register Address: 38FH, 911	IA32_PERF_GLOBAL_CTRL	
See Table 2-2 and Section 21.6.2.2, "Global Counter Control Facilities."		Thread
0	Set 1 to enable PMC0 to count.	Thread
1	Set 1 to enable PMC1 to count.	Thread
2	Set 1 to enable PMC2 to count.	Thread
3	Set 1 to enable PMC3 to count.	Thread
4	Set 1 to enable PMC4 to count (if CPUID.0AH:EAX[15:8] > 4).	Core
5	Set 1 to enable PMC5 to count (if CPUID.0AH:EAX[15:8] > 5).	Core
6	Set 1 to enable PMC6 to count (if CPUID.0AH:EAX[15:8] > 6).	Core
7	Set 1 to enable PMC7 to count (if CPUID.0AH:EAX[15:8] > 7).	Core

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
31:8	Reserved.	
32	Set 1 to enable FixedCtr0 to count.	Thread
33	Set 1 to enable FixedCtr1 to count.	Thread
34	Set 1 to enable FixedCtr2 to count.	Thread
63:35	Reserved.	
Register Address: 390H, 912	IA32_PERF_GLOBAL_OVF_CTRL	
See Table 2-2 and Section 21.6.2.2, "Global Counter Control Facilities."		
0	Set 1 to clear Ovf_PMC0.	Thread
1	Set 1 to clear Ovf_PMC1.	Thread
2	Set 1 to clear Ovf_PMC2.	Thread
3	Set 1 to clear Ovf_PMC3.	Thread
4	Set 1 to clear Ovf_PMC4 (if CPUID.0AH:EAX[15:8] > 4).	Core
5	Set 1 to clear Ovf_PMC5 (if CPUID.0AH:EAX[15:8] > 5).	Core
6	Set 1 to clear Ovf_PMC6 (if CPUID.0AH:EAX[15:8] > 6).	Core
7	Set 1 to clear Ovf_PMC7 (if CPUID.0AH:EAX[15:8] > 7).	Core
31:8	Reserved.	
32	Set 1 to clear Ovf_FixedCtr0.	Thread
33	Set 1 to clear Ovf_FixedCtr1.	Thread
34	Set 1 to clear Ovf_FixedCtr2.	Thread
60:35	Reserved.	
61	Set 1 to clear Ovf_Uncore.	Thread
62	Set 1 to clear Ovf_BufDSSAVE.	Thread
63	Set 1 to clear CondChgd.	Thread
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
See Section 21.3.1.1.1, "Processor Event Based Sampling (PEBS)."		Thread
0	Enable PEBS on IA32_PMC0. (R/W)	
1	Enable PEBS on IA32_PMC1. (R/W)	
2	Enable PEBS on IA32_PMC2. (R/W)	
3	Enable PEBS on IA32_PMC3. (R/W)	
31:4	Reserved.	
32	Enable Load Latency on IA32_PMC0. (R/W)	
33	Enable Load Latency on IA32_PMC1. (R/W)	
34	Enable Load Latency on IA32_PMC2. (R/W)	
35	Enable Load Latency on IA32_PMC3. (R/W)	
62:36	Reserved.	
63	Enable Precise Store (R/W)	
Register Address: 3F6H, 1014	MSR_PEBS_LD_LAT	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 21.3.1.1.2, "Load Latency Performance Monitoring Facility."		Thread
15:0	Minimum threshold latency value of tagged load operation that will be counted. (R/W)	
63:36	Reserved.	
Register Address: 3F8H, 1016	MSR_PKG_C3_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C3 Residency Counter (R/O) Value since last reset that this package is in processor-specific C3 states. Count at the same frequency as the TSC.	
Register Address: 3F9H, 1017	MSR_PKG_C6_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C6 Residency Counter. (R/O) Value since last reset that this package is in processor-specific C6 states. Count at the same frequency as the TSC.	
Register Address: 3FAH, 1018	MSR_PKG_C7_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C7 Residency Counter (R/O) Value since last reset that this package is in processor-specific C7 states. Count at the same frequency as the TSC.	
Register Address: 3FCH, 1020	MSR_CORE_C3_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C3 Residency Counter (R/O) Value since last reset that this core is in processor-specific C3 states. Count at the same frequency as the TSC.	
Register Address: 3FDH, 1021	MSR_CORE_C6_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C6 Residency Counter (R/O) Value since last reset that this core is in processor-specific C6 states. Count at the same frequency as the TSC.	
Register Address: 3FEH, 1022	MSR_CORE_C7_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C7 Residency Counter (R/O) Value since last reset that this core is in processor-specific C7 states. Count at the same frequency as the TSC.	
Register Address: 400H, 1024	IA32_MCO_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Core

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 401H, 1025	IA32_MCO_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Core
Register Address: 402H, 1026	IA32_MCO_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Core
Register Address: 403H, 1027	IA32_MCO_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Core
Register Address: 404H, 1028	IA32_MC1_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 405H, 1029	IA32_MC1_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Core
Register Address: 406H, 1030	IA32_MC1_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Core
Register Address: 407H, 1031	IA32_MC1_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Core
Register Address: 408H, 1032	IA32_MC2_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 409H, 1033	IA32_MC2_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Core
Register Address: 40AH, 1034	IA32_MC2_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Core
Register Address: 40BH, 1035	IA32_MC2_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Core
Register Address: 40CH, 1036	IA32_MC3_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 40DH, 1037	IA32_MC3_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Core
Register Address: 40EH, 1038	IA32_MC3_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Core
Register Address: 40FH, 1039	IA32_MC3_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Core
Register Address: 410H, 1040	IA32_MC4_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
0	PCU Hardware Error (R/W) When set, enables signaling of PCU hardware detected errors.	
1	PCU Controller Error (R/W) When set, enables signaling of PCU controller detected errors.	
2	PCU Firmware Error (R/W) When set, enables signaling of PCU firmware detected errors.	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:2	Reserved.	
Register Address: 411H, 1041	IA32_MC4_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRS," and Chapter 18.		Core
Register Address: 480H, 1152	IA32_VMX_BASIC	
Reporting Register of Basic VMX Capabilities (R/O) See Table 2-2 and Appendix A.1, "Basic VMX Information."		Thread
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL	
Capability Reporting Register of Pin-Based VM-Execution Controls (R/O) See Table 2-2 and Appendix A.3, "VM-Execution Controls."		Thread
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL	
Capability Reporting Register of Primary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Thread
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL	
Capability Reporting Register of VM-Exit Controls (R/O) See Table 2-2 and Appendix A.4, "VM-Exit Controls."		Thread
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTL	
Capability Reporting Register of VM-Entry Controls (R/O) See Table 2-2 and Appendix A.5, "VM-Entry Controls."		Thread
Register Address: 485H, 1157	IA32_VMX_MISC	
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Table 2-2 and Appendix A.6, "Miscellaneous Data."		Thread
Register Address: 486H, 1158	IA32_VMX_CR0_FIXED0	
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Table 2-2 and Appendix A.7, "VMX-Fixed Bits in CR0."		Thread
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1	
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Table 2-2 and Appendix A.7, "VMX-Fixed Bits in CR0."		Thread
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0	
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Table 2-2 and Appendix A.8, "VMX-Fixed Bits in CR4."		Thread
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1	
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Table 2-2 and Appendix A.8, "VMX-Fixed Bits in CR4."		Thread
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM	
Capability Reporting Register of VMCS Field Enumeration (R/O) See Table 2-2 and Appendix A.9, "VMCS Enumeration."		Thread
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL2	
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls."		Thread

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 48CH, 1164	IA32_VMX_EPT_VPID_ENUM	
Capability Reporting Register of EPT and VPID (R/O) See Table 2-2		Thread
Register Address: 48DH, 1165	IA32_VMX_TRUE_PINBASED_CTL	
Capability Reporting Register of Pin-Based VM-Execution Flex Controls (R/O) See Table 2-2		Thread
Register Address: 48EH, 1166	IA32_VMX_TRUE_PROCBASED_CTL	
Capability Reporting Register of Primary Processor-Based VM-Execution Flex Controls (R/O) See Table 2-2		Thread
Register Address: 48FH, 1167	IA32_VMX_TRUE_EXIT_CTL	
Capability Reporting Register of VM-Exit Flex Controls (R/O) See Table 2-2		Thread
Register Address: 490H, 1168	IA32_VMX_TRUE_ENTRY_CTL	
Capability Reporting Register of VM-Entry Flex Controls (R/O) See Table 2-2		Thread
Register Address: 4C1H, 1217	IA32_A_PMC0	
See Table 2-2.		Thread
Register Address: 4C2H, 1218	IA32_A_PMC1	
See Table 2-2.		Thread
Register Address: 4C3H, 1219	IA32_A_PMC2	
See Table 2-2.		Thread
Register Address: 4C4H, 1220	IA32_A_PMC3	
See Table 2-2.		Thread
Register Address: 4C5H, 1221	IA32_A_PMC4	
See Table 2-2.		Core
Register Address: 4C6H, 1222	IA32_A_PMC5	
See Table 2-2.		Core
Register Address: 4C7H, 1223	IA32_A_PMC6	
See Table 2-2.		Core
Register Address: 4C8H, 1224	IA32_A_PMC7	
See Table 2-2.		Core
Register Address: 600H, 1536	IA32_DS_AREA	
DS Save Area (R/W) See Table 2-2 and Section 21.6.3.4, "Debug Store (DS) Mechanism."		Thread
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
Unit Multipliers used in RAPL Interfaces (R/O) See Section 16.10.1, "RAPL Interfaces."		Package
Register Address: 60AH, 1546	MSR_PKG_C3_IRT	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Package C3 Interrupt Response Limit (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
9:0	Interrupt Response Time Limit (R/W) Specifies the limit that should be used to decide if the package should be put into a package C3 state.	
12:10	Time Unit (R/W) Specifies the encoding value of time unit of the interrupt response time limit. The following time unit encodings are supported: 000b: 1 ns 001b: 32 ns 010b: 1024 ns 011b: 32768 ns 100b: 1048576 ns 101b: 33554432 ns	
14:13	Reserved.	
15	Valid (R/W) Indicates whether the values in bits 12:0 are valid and can be used by the processor for package C-state management.	
63:16	Reserved.	
Register Address: 60BH, 1547	MSR_PKG_C6_IRTL	
Package C6 Interrupt Response Limit (R/W) This MSR defines the budget allocated for the package to exit from a C6 to a C0 state, where an interrupt request can be delivered to the core and serviced. Additional core-exit latency may be applicable depending on the actual C-state the core is in. Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.		Package
9:0	Interrupt Response Time Limit (R/W) Specifies the limit that should be used to decide if the package should be put into a package C6 state.	
12:10	Time Unit (R/W) Specifies the encoding value of time unit of the interrupt response time limit. The following time unit encodings are supported: 000b: 1 ns 001b: 32 ns 010b: 1024 ns 011b: 32768 ns 100b: 1048576 ns 101b: 33554432 ns	
14:13	Reserved.	
15	Valid (R/W) Indicates whether the values in bits 12:0 are valid and can be used by the processor for package C-state management.	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:16	Reserved.	
Register Address: 60DH, 1549	MSR_PKG_C2_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
63:0	Package C2 Residency Counter (R/O) Value since last reset that this package is in processor-specific C2 states. Count at the same frequency as the TSC.	
Register Address: 610H, 1552	MSR_PKG_POWER_LIMIT	
PKG RAPL Power Limit Control (R/W) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 611H, 1553	MSR_PKG_ENERGY_STATUS	
PKG Energy Status (R/O) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 614H, 1556	MSR_PKG_POWER_INFO	
PKG RAPL Parameters (R/W) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 638H, 1592	MSR_PP0_POWER_LIMIT	
PP0 RAPL Power Limit Control (R/W) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 680H, 1664	MSR_LASTBRANCH_0_FROM_IP	
Last Branch Record 0 From IP (R/W) One of sixteen pairs of last branch record registers on the last branch record stack. This part of the stack contains pointers to the source instruction. See also: - Last Branch Record Stack TOS at 1C9H. - Section 19.9.1 and record format in Section 19.4.8.1.		Thread
Register Address: 681H, 1665	MSR_LASTBRANCH_1_FROM_IP	
Last Branch Record 1 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 682H, 1666	MSR_LASTBRANCH_2_FROM_IP	
Last Branch Record 2 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 683H, 1667	MSR_LASTBRANCH_3_FROM_IP	
Last Branch Record 3 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 684H, 1668	MSR_LASTBRANCH_4_FROM_IP	
Last Branch Record 4 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 685H, 1669	MSR_LASTBRANCH_5_FROM_IP	
Last Branch Record 5 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 686H, 1670	MSR_LASTBRANCH_6_FROM_IP	
Last Branch Record 6 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 687H, 1671	MSR_LASTBRANCH_7_FROM_IP	
Last Branch Record 7 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 688H, 1672	MSR_LASTBRANCH_8_FROM_IP	
Last Branch Record 8 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 689H, 1673	MSR_LASTBRANCH_9_FROM_IP	
Last Branch Record 9 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 68AH, 1674	MSR_LASTBRANCH_10_FROM_IP	
Last Branch Record 10 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 68BH, 1675	MSR_LASTBRANCH_11_FROM_IP	
Last Branch Record 11 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 68CH, 1676	MSR_LASTBRANCH_12_FROM_IP	
Last Branch Record 12 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 68DH, 1677	MSR_LASTBRANCH_13_FROM_IP	
Last Branch Record 13 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 68EH, 1678	MSR_LASTBRANCH_14_FROM_IP	
Last Branch Record 14 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 68FH, 1679	MSR_LASTBRANCH_15_FROM_IP	
Last Branch Record 15 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.	Thread	
Register Address: 6C0H, 1728	MSR_LASTBRANCH_0_TO_IP	
Last Branch Record 0 To IP (R/W) One of sixteen pairs of last branch record registers on the last branch record stack. This part of the stack contains pointers to the destination instruction.	Thread	
Register Address: 6C1H, 1729	MSR_LASTBRANCH_1_TO_IP	
Last Branch Record 1 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.	Thread	
Register Address: 6C2H, 1730	MSR_LASTBRANCH_2_TO_IP	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 2 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C3H, 1731	MSR_LASTBRANCH_3_TO_IP	
Last Branch Record 3 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C4H, 1732	MSR_LASTBRANCH_4_TO_IP	
Last Branch Record 4 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C5H, 1733	MSR_LASTBRANCH_5_TO_IP	
Last Branch Record 5 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C6H, 1734	MSR_LASTBRANCH_6_TO_IP	
Last Branch Record 6 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C7H, 1735	MSR_LASTBRANCH_7_TO_IP	
Last Branch Record 7 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C8H, 1736	MSR_LASTBRANCH_8_TO_IP	
Last Branch Record 8 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6C9H, 1737	MSR_LASTBRANCH_9_TO_IP	
Last Branch Record 9 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CAH, 1738	MSR_LASTBRANCH_10_TO_IP	
Last Branch Record 10 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CBH, 1739	MSR_LASTBRANCH_11_TO_IP	
Last Branch Record 11 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CCH, 1740	MSR_LASTBRANCH_12_TO_IP	
Last Branch Record 12 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CDH, 1741	MSR_LASTBRANCH_13_TO_IP	
Last Branch Record 13 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CEH, 1742	MSR_LASTBRANCH_14_TO_IP	
Last Branch Record 14 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6CFH, 1743	MSR_LASTBRANCH_15_TO_IP	

Table 2-20. MSRs Supported by Intel® Processors Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 15 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6E0H, 1760	IA32_TSC_DEADLINE	
See Table 2-2.		Thread
Register Address: 802H–83FH, 2050–2111	X2APIC MSRs	
See Table 2-2.		Thread
Register Address: C000_0080H	IA32_EFER	
Extended Feature Enables See Table 2-2.		Thread
Register Address: C000_0081H	IA32_STAR	
System Call Target Address (R/W) See Table 2-2.		Thread
Register Address: C000_0082H	IA32_LSTAR	
IA-32e Mode System Call Target Address (R/W) See Table 2-2.		Thread
Register Address: C000_0084H	IA32_FMASK	
System Call Flag Mask (R/W) See Table 2-2.		Thread
Register Address: C000_0100H	IA32_FS_BASE	
Map of BASE Address of FS (R/W) See Table 2-2.		Thread
Register Address: C000_0101H	IA32_GS_BASE	
Map of BASE Address of GS (R/W) See Table 2-2.		Thread
Register Address: C000_0102H	IA32_KERNEL_GS_BASE	
Swap Target of BASE Address of GS (R/W) See Table 2-2.		Thread
Register Address: C000_0103H	IA32_TSC_AUX	
AUXILIARY TSC Signature (R/W) See Table 2-2 and Section 19.17.2, “IA32_TSC_AUX Register and RDTSCP Support.”		Thread

2.11.1 MSRs in the 2nd Generation Intel® Core™ Processor Family Based on Sandy Bridge Microarchitecture

Table 2-21 and Table 2-22 list model-specific registers (MSRs) that are specific to the 2nd generation Intel® Core™ processor family based on the Sandy Bridge microarchitecture. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_2AH; see Table 2-1.

Table 2-21. MSRs Supported by the 2nd Generation Intel® Core™ Processors (Sandy Bridge Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0. R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 core active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 core active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 core active.	Package
63:32	Reserved.	
Register Address: 60CH, 1548	MSR_PKG_C7_IRTL	
Package C7 Interrupt Response Limit (R/W) This MSR defines the budget allocated for the package to exit from a C7 to a C0 state, where interrupt request can be delivered to the core and serviced. Additional core-exit latency may be applicable depending on the actual C-state the core is in. Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.		Package
9:0	Interrupt Response Time Limit (R/W) Specifies the limit that should be used to decide if the package should be put into a package C7 state.	
12:10	Time Unit (R/W) Specifies the encoding value of time unit of the interrupt response time limit. The following time unit encodings are supported: 000b: 1 ns 001b: 32 ns 010b: 1024 ns 011b: 32768 ns 100b: 1048576 ns 101b: 33554432 ns	
14:13	Reserved.	
15	Valid (R/W) Indicates whether the values in bits 12:0 are valid and can be used by the processor for package C-state management.	
63:16	Reserved.	
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	
PPO Energy Status (R/O) See Section 16.10.4, "PPO/PP1 RAPL Domains."		Package
Register Address: 63AH, 1594	MSR_PPO_POLICY	

Table 2-21. MSRs Supported by the 2nd Generation Intel® Core™ Processors (Sandy Bridge Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
PP0 Balance Policy (R/W) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 640H, 1600	MSR_PP1_POWER_LIMIT	
PP1 RAPL Power Limit Control (R/W) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 641H, 1601	MSR_PP1_ENERGY_STATUS	
PP1 Energy Status (R/O) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 642H, 1602	MSR_PP1_POLICY	
PP1 Balance Policy (R/W) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
See Table 2-20, Table 2-21, and Table 2-22 for MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_2AH.		

Table 2-22 lists the MSRs of uncore PMU for Intel processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_2AH.

Table 2-22. Uncore PMU MSRs Supported by 2nd Generation Intel® Core™ Processors

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 391H, 913	MSR_UNC_PERF_GLOBAL_CTRL	
Uncore PMU Global Control		Package
0	Slice 0 select.	
1	Slice 1 select.	
2	Slice 2 select.	
3	Slice 3 select.	
4	Slice 4 select.	
18:5	Reserved.	
29	Enable all uncore counters.	
30	Enable wake on PMI.	
31	Enable Freezing counter when overflow.	
63:32	Reserved.	
Register Address: 392H, 914	MSR_UNC_PERF_GLOBAL_STATUS	
Uncore PMU Main Status		Package
0	Fixed counter overflowed.	
1	An ARB counter overflowed.	
2	Reserved.	
3	A CBox counter overflowed (on any slice).	
63:4	Reserved.	

Table 2-22. Uncore PMU MSRs Supported by 2nd Generation Intel® Core™ Processors (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 394H, 916	MSR_UNC_PERF_FIXED_CTRL	
Uncore Fixed Counter Control (R/W)		Package
19:0	Reserved.	
20	Enable overflow propagation.	
21	Reserved.	
22	Enable counting.	
63:23	Reserved.	
Register Address: 395H, 917	MSR_UNC_PERF_FIXED_CTR	
Uncore Fixed Counter		Package
47:0	Current count.	
63:48	Reserved.	
Register Address: 396H, 918	MSR_UNC_CBO_CONFIG	
Uncore C-Box Configuration Information (R/O)		Package
3:0	Report the number of C-Box units with performance counters, including processor cores and processor graphics.	
63:4	Reserved.	
Register Address: 3B0H, 946	MSR_UNC_ARB_PERFCTR0	
Uncore Arb Unit, Performance Counter 0		Package
Register Address: 3B1H, 947	MSR_UNC_ARB_PERFCTR1	
Uncore Arb Unit, Performance Counter 1		Package
Register Address: 3B2H, 944	MSR_UNC_ARB_PERFEVTSELO	
Uncore Arb Unit, Counter 0 Event Select MSR		Package
Register Address: 3B3H, 945	MSR_UNC_ARB_PERFEVTSEL1	
Uncore Arb unit, Counter 1 Event Select MSR		Package
Register Address: 700H, 1792	MSR_UNC_CBO_0_PERFEVTSELO	
Uncore C-Box 0, Counter 0 Event Select MSR		Package
Register Address: 701H, 1793	MSR_UNC_CBO_0_PERFEVTSEL1	
Uncore C-Box 0, Counter 1 Event Select MSR		Package
Register Address: 702H, 1794	MSR_UNC_CBO_0_PERFEVTSEL2	
Uncore C-Box 0, Counter 2 Event Select MSR		Package
Register Address: 703H, 1795	MSR_UNC_CBO_0_PERFEVTSEL3	
Uncore C-Box 0, Counter 3 Event Select MSR		Package
Register Address: 705H, 1797	MSR_UNC_CBO_0_UNIT_STATUS	
Uncore C-Box 0, Unit Status for Counter 0-3		Package
Register Address: 706H, 1798	MSR_UNC_CBO_0_PERFCTR0	
Uncore C-Box 0, Performance Counter 0		Package
Register Address: 707H, 1799	MSR_UNC_CBO_0_PERFCTR1	
Uncore C-Box 0, Performance Counter 1		Package

Table 2-22. Uncore PMU MSRs Supported by 2nd Generation Intel® Core™ Processors (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 708H, 1800	MSR_UNC_CBO_0_PERFCTR2	
Uncore C-Box 0, Performance Counter 2		Package
Register Address: 709H, 1801	MSR_UNC_CBO_0_PERFCTR3	
Uncore C-Box 0, Performance Counter 3		Package
Register Address: 710H, 1808	MSR_UNC_CBO_1_PERFEVTSELO	
Uncore C-Box 1, Counter 0 Event Select MSR		Package
Register Address: 711H, 1809	MSR_UNC_CBO_1_PERFEVTSEL1	
Uncore C-Box 1, Counter 1 Event Select MSR		Package
Register Address: 712H, 1810	MSR_UNC_CBO_1_PERFEVTSEL2	
Uncore C-Box 1, Counter 2 Event Select MSR		Package
Register Address: 713H, 1811	MSR_UNC_CBO_1_PERFEVTSEL3	
Uncore C-Box 1, Counter 3 Event Select MSR		Package
Register Address: 715H, 1813	MSR_UNC_CBO_1_UNIT_STATUS	
Uncore C-Box 1, Unit Status for Counter 0-3		Package
Register Address: 716H, 1814	MSR_UNC_CBO_1_PERFCTR0	
Uncore C-Box 1, Performance Counter 0		Package
Register Address: 717H, 1815	MSR_UNC_CBO_1_PERFCTR1	
Uncore C-Box 1, Performance Counter 1		Package
Register Address: 718H, 1816	MSR_UNC_CBO_1_PERFCTR2	
Uncore C-Box 1, Performance Counter 2		Package
Register Address: 719H, 1817	MSR_UNC_CBO_1_PERFCTR3	
Uncore C-Box 1, Performance Counter 3		Package
Register Address: 720H, 1824	MSR_UNC_CBO_2_PERFEVTSELO	
Uncore C-Box 2, Counter 0 Event Select MSR		Package
Register Address: 721H, 1825	MSR_UNC_CBO_2_PERFEVTSEL1	
Uncore C-Box 2, Counter 1 Event Select MSR		Package
Register Address: 722H, 1826	MSR_UNC_CBO_2_PERFEVTSEL2	
Uncore C-Box 2, Counter 2 Event Select MSR		Package
Register Address: 723H, 1827	MSR_UNC_CBO_2_PERFEVTSEL3	
Uncore C-Box 2, Counter 3 Event Select MSR		Package
Register Address: 725H, 1829	MSR_UNC_CBO_2_UNIT_STATUS	
Uncore C-Box 2, Unit Status for Counter 0-3		Package
Register Address: 726H, 1830	MSR_UNC_CBO_2_PERFCTR0	
Uncore C-Box 2, Performance Counter 0		Package
Register Address: 727H, 1831	MSR_UNC_CBO_2_PERFCTR1	
Uncore C-Box 2, Performance Counter 1		Package
Register Address: 728H, 1832	MSR_UNC_CBO_3_PERFCTR2	

Table 2-22. Uncore PMU MSRs Supported by 2nd Generation Intel® Core™ Processors (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 3, Performance Counter 2		Package
Register Address: 729H, 1833	MSR_UNC_CBO_3_PERFCTR3	
Uncore C-Box 3, Performance Counter 3		Package
Register Address: 730H, 1840	MSR_UNC_CBO_3_PERFEVTSELO	
Uncore C-Box 3, Counter 0 Event Select MSR		Package
Register Address: 731H, 1841	MSR_UNC_CBO_3_PERFEVTSEL1	
Uncore C-Box 3, Counter 1 Event Select MSR		Package
Register Address: 732H, 1842	MSR_UNC_CBO_3_PERFEVTSEL2	
Uncore C-Box 3, Counter 2 Event Select MSR		Package
Register Address: 733H, 1843	MSR_UNC_CBO_3_PERFEVTSEL3	
Uncore C-Box 3, counter 3 Event Select MSR		Package
Register Address: 735H, 1845	MSR_UNC_CBO_3_UNIT_STATUS	
Uncore C-Box 3, Unit Status for Counter 0-3		Package
Register Address: 736H, 1846	MSR_UNC_CBO_3_PERFCTR0	
Uncore C-Box 3, Performance Counter 0		Package
Register Address: 737H, 1847	MSR_UNC_CBO_3_PERFCTR1	
Uncore C-Box 3, Performance Counter 1		Package
Register Address: 738H, 1848	MSR_UNC_CBO_3_PERFCTR2	
Uncore C-Box 3, Performance Counter 2		Package
Register Address: 739H, 1849	MSR_UNC_CBO_3_PERFCTR3	
Uncore C-Box 3, Performance Counter 3		Package
Register Address: 740H, 1856	MSR_UNC_CBO_4_PERFEVTSELO	
Uncore C-Box 4, Counter 0 Event Select MSR		Package
Register Address: 741H, 1857	MSR_UNC_CBO_4_PERFEVTSEL1	
Uncore C-Box 4, Counter 1 Event Select MSR		Package
Register Address: 742H, 1858	MSR_UNC_CBO_4_PERFEVTSEL2	
Uncore C-Box 4, Counter 2 Event Select MSR		Package
Register Address: 743H, 1859	MSR_UNC_CBO_4_PERFEVTSEL3	
Uncore C-Box 4, Counter 3 Event Select MSR		Package
Register Address: 745H, 1861	MSR_UNC_CBO_4_UNIT_STATUS	
Uncore C-Box 4, Unit status for Counter 0-3		Package
Register Address: 746H, 1862	MSR_UNC_CBO_4_PERFCTR0	
Uncore C-Box 4, Performance Counter 0		Package
Register Address: 747H, 1863	MSR_UNC_CBO_4_PERFCTR1	
Uncore C-Box 4, Performance Counter 1		Package
Register Address: 748H, 1864	MSR_UNC_CBO_4_PERFCTR2	
Uncore C-Box 4, Performance Counter 2		Package

Table 2-22. Uncore PMU MSRs Supported by 2nd Generation Intel® Core™ Processors (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 749H, 1865	MSR_UNC_CBO_4_PERFCTR3	
Uncore C-Box 4, Performance Counter 3		Package

2.11.2 MSRs in the Intel® Xeon® Processor E5 Family Based on Sandy Bridge Microarchitecture

Table 2-23 lists additional model-specific registers (MSRs) that are specific to the Intel® Xeon® Processor E5 Family based on Sandy Bridge microarchitecture. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_2DH, and also support MSRs listed in Table 2-20 and Table 2-24.

Table 2-23. Additional MSRs Supported by the Intel® Xeon® Processors E5 Family Based on Sandy Bridge Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 17FH, 383	MSR_ERROR_CONTROL	
	MC Bank Error Configuration (R/W)	Package
0	Reserved.	
1	MemError Log Enable (R/W) When set, enables IMC status bank to log additional info in bits 36:32.	
63:2	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0. R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 cores active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 cores active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 cores active.	Package
39:32	Maximum Ratio Limit for 5C Maximum turbo ratio limit of 5 cores active.	Package
47:40	Maximum Ratio Limit for 6C Maximum turbo ratio limit of 6 cores active.	Package
55:48	Maximum Ratio Limit for 7C Maximum turbo ratio limit of 7 cores active.	Package
63:56	Maximum Ratio Limit for 8C Maximum turbo ratio limit of 8 cores active.	Package
Register Address: 285H, 645	IA32_MC5_CTL2	
See Table 2-2.		Package

Table 2-23. Additional MSRs Supported by the Intel® Xeon® Processors E5 Family Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 286H, 646	IA32_MC6_CTL2	
See Table 2-2.		Package
Register Address: 287H, 647	IA32_MC7_CTL2	
See Table 2-2.		Package
Register Address: 288H, 648	IA32_MC8_CTL2	
See Table 2-2.		Package
Register Address: 289H, 649	IA32_MC9_CTL2	
See Table 2-2.		Package
Register Address: 28AH, 650	IA32_MC10_CTL2	
See Table 2-2.		Package
Register Address: 28BH, 651	IA32_MC11_CTL2	
See Table 2-2.		Package
Register Address: 28CH, 652	IA32_MC12_CTL2	
See Table 2-2.		Package
Register Address: 28DH, 653	IA32_MC13_CTL2	
See Table 2-2.		Package
Register Address: 28EH, 654	IA32_MC14_CTL2	
See Table 2-2.		Package
Register Address: 28FH, 655	IA32_MC15_CTL2	
See Table 2-2.		Package
Register Address: 290H, 656	IA32_MC16_CTL2	
See Table 2-2.		Package
Register Address: 291H, 657	IA32_MC17_CTL2	
See Table 2-2.		Package
Register Address: 292H, 658	IA32_MC18_CTL2	
See Table 2-2.		Package
Register Address: 293H, 659	IA32_MC19_CTL2	
See Table 2-2.		Package
Register Address: 39CH, 924	MSR_PEBS_NUM_ALT	
ENABLE_PEBS_NUM_ALT (R/W)		Package
0	ENABLE_PEBS_NUM_ALT (R/W) Write 1 to enable alternate PEBS counting logic for specific events requiring additional configuration, see https://perfmon-events.intel.com/ .	
63:1	Reserved, must be zero.	
Register Address: 414H, 1044	IA32_MC5_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 415H, 1045	IA32_MC5_STATUS	

Table 2-23. Additional MSRs Supported by the Intel® Xeon® Processors E5 Family Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 416H, 1046	IA32_MC5_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 417H, 1047	IA32_MC5_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 418H, 1048	IA32_MC6_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 419H, 1049	IA32_MC6_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 41AH, 1050	IA32_MC6_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 41BH, 1051	IA32_MC6_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 41CH, 1052	IA32_MC7_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 41DH, 1053	IA32_MC7_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 41EH, 1054	IA32_MC7_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 41FH, 1055	IA32_MC7_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 420H, 1056	IA32_MC8_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 421H, 1057	IA32_MC8_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 422H, 1058	IA32_MC8_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 423H, 1059	IA32_MC8_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 424H, 1060	IA32_MC9_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 425H, 1061	IA32_MC9_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 426H, 1062	IA32_MC9_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 427H, 1063	IA32_MC9_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package

Table 2-23. Additional MSRs Supported by the Intel® Xeon® Processors E5 Family Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 428H, 1064	IA32_MC10_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 429H, 1065	IA32_MC10_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs," and Chapter 18.		Package
Register Address: 42AH, 1066	IA32_MC10_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 42BH, 1067	IA32_MC10_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 42CH, 1068	IA32_MC11_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 42DH, 1069	IA32_MC11_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs," and Chapter 18.		Package
Register Address: 42EH, 1070	IA32_MC11_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 42FH, 1071	IA32_MC11_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 430H, 1072	IA32_MC12_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 431H, 1073	IA32_MC12_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs," and Chapter 18.		Package
Register Address: 432H, 1074	IA32_MC12_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 433H, 1075	IA32_MC12_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 434H, 1076	IA32_MC13_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 435H, 1077	IA32_MC13_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs," and Chapter 18.		Package
Register Address: 436H, 1078	IA32_MC13_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Package
Register Address: 437H, 1079	IA32_MC13_MISC	
See Section 17.3.2.4, "IA32_MCi_MISC MSRs."		Package
Register Address: 438H, 1080	IA32_MC14_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Package
Register Address: 439H, 1081	IA32_MC14_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs," and Chapter 18.		Package
Register Address: 43AH, 1082	IA32_MC14_ADDR	

Table 2-23. Additional MSRs Supported by the Intel® Xeon® Processors E5 Family Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 43BH, 1083	IA32_MC14_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 43CH, 1084	IA32_MC15_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 43DH, 1085	IA32_MC15_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 43EH, 1086	IA32_MC15_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 43FH, 1087	IA32_MC15_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 440H, 1088	IA32_MC16_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 441H, 1089	IA32_MC16_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 442H, 1090	IA32_MC16_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 443H, 1091	IA32_MC16_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 444H, 1092	IA32_MC17_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 445H, 1093	IA32_MC17_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 446H, 1094	IA32_MC17_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 447H, 1095	IA32_MC17_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 448H, 1096	IA32_MC18_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 449H, 1097	IA32_MC18_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 44AH, 1098	IA32_MC18_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 44BH, 1099	IA32_MC18_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 44CH, 1100	IA32_MC19_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package

Table 2-23. Additional MSRs Supported by the Intel® Xeon® Processors E5 Family Based on Sandy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 44DH, 1101	IA32_MC19_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS," and Chapter 18.		Package
Register Address: 44EH, 1102	IA32_MC19_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 44FH, 1103	IA32_MC19_MISC	
See Section 17.3.2.4, "IA32_MCI_MISC MSRs."		Package
Register Address: 613H, 1555	MSR_PKG_PERF_STATUS	
Package RAPL Perf Status (R/O)		Package
Register Address: 618H, 1560	MSR_DRAM_POWER_LIMIT	
DRAM RAPL Power Limit Control (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61CH, 1564	MSR_DRAM_POWER_INFO	
DRAM RAPL Parameters (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 639H, 1593	MSR_PP0_ENERGY_STATUS	
PP0 Energy Status (R/O) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
See Table 2-20, Table 2-23, and Table 2-24 for MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_2DH.		

2.11.3 Additional Uncore PMU MSRs in the Intel® Xeon® Processor E5 Family

Intel Xeon Processor E5 family is based on the Sandy Bridge microarchitecture. The MSR-based uncore PMU interfaces are listed in Table 2-24. For complete details of the uncore PMU, refer to the Intel Xeon Processor E5 Product Family Uncore Performance Monitoring Guide. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_2DH.

Table 2-24. Uncore PMU MSRs in Intel® Xeon® Processor E5 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: C08H, 3080	MSR_U_PMON_UCLK_FIXED_CTL	
Uncore U-box UCLK Fixed Counter Control		Package
Register Address: C09H, 3081	MSR_U_PMON_UCLK_FIXED_CTR	

Table 2-24. Uncore PMU MSRs in Intel® Xeon® Processor E5 Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore U-box UCLK Fixed Counter		Package
Register Address: C10H, 3088	MSR_U_PMON_EVTNSELO	
Uncore U-box PerfMon Event Select for U-box Counter 0		Package
Register Address: C11H, 3089	MSR_U_PMON_EVTNSEL1	
Uncore U-box PerfMon Event Select for U-box Counter 1		Package
Register Address: C16H, 3094	MSR_U_PMON_CTR0	
Uncore U-box PerfMon Counter 0		Package
Register Address: C17H, 3095	MSR_U_PMON_CTR1	
Uncore U-box PerfMon Counter 1		Package
Register Address: C24H, 3108	MSR_PCU_PMON_BOX_CTL	
Uncore PCU PerfMon for PCU-box-wide Control		Package
Register Address: C30H, 3120	MSR_PCU_PMON_EVTNSELO	
Uncore PCU PerfMon Event Select for PCU Counter 0		Package
Register Address: C31H, 3121	MSR_PCU_PMON_EVTNSEL1	
Uncore PCU PerfMon Event Select for PCU Counter 1		Package
Register Address: C32H, 3122	MSR_PCU_PMON_EVTNSEL2	
Uncore PCU PerfMon Event Select for PCU Counter 2		Package
Register Address: C33H, 3123	MSR_PCU_PMON_EVTNSEL3	
Uncore PCU PerfMon Event Select for PCU Counter 3		Package
Register Address: C34H, 3124	MSR_PCU_PMON_BOX_FILTER	
Uncore PCU PerfMon box-wide Filter		Package
Register Address: C36H, 3126	MSR_PCU_PMON_CTR0	
Uncore PCU PerfMon Counter 0		Package
Register Address: C37H, 3127	MSR_PCU_PMON_CTR1	
Uncore PCU PerfMon Counter 1		Package
Register Address: C38H, 3128	MSR_PCU_PMON_CTR2	
Uncore PCU PerfMon Counter 2		Package
Register Address: C39H, 3129	MSR_PCU_PMON_CTR3	
Uncore PCU PerfMon Counter 3		Package
Register Address: D04H, 3332	MSR_CO_PMON_BOX_CTL	
Uncore C-box 0 PerfMon Local Box Wide Control		Package
Register Address: D10H, 3344	MSR_CO_PMON_EVTNSELO	
Uncore C-box 0 PerfMon Event Select for C-box 0 Counter 0		Package
Register Address: D11H, 3345	MSR_CO_PMON_EVTNSEL1	
Uncore C-box 0 PerfMon Event Select for C-box 0 Counter 1		Package
Register Address: D12H, 3346	MSR_CO_PMON_EVTNSEL2	
Uncore C-box 0 PerfMon Event Select for C-box 0 Counter 2		Package

Table 2-24. Uncore PMU MSRs in Intel® Xeon® Processor E5 Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: D13H, 3347	MSR_CO_PMON_EVNTSEL3	
Uncore C-box 0 PerfMon Event Select for C-box 0 Counter 3		Package
Register Address: D14H, 3348	MSR_CO_PMON_BOX_FILTER	
Uncore C-box 0 PerfMon Box Wide Filter		Package
Register Address: D16H, 3350	MSR_CO_PMON_CTR0	
Uncore C-box 0 PerfMon Counter 0		Package
Register Address: D17H, 3351	MSR_CO_PMON_CTR1	
Uncore C-box 0 PerfMon Counter 1		Package
Register Address: D18H, 3352	MSR_CO_PMON_CTR2	
Uncore C-box 0 PerfMon Counter 2		Package
Register Address: D19H, 3353	MSR_CO_PMON_CTR3	
Uncore C-box 0 PerfMon Counter 3		Package
Register Address: D24H, 3364	MSR_C1_PMON_BOX_CTL	
Uncore C-box 1 PerfMon Local Box Wide Control		Package
Register Address: D30H, 3376	MSR_C1_PMON_EVNTSEL0	
Uncore C-box 1 PerfMon Event Select for C-box 1 Counter 0		Package
Register Address: D31H, 3377	MSR_C1_PMON_EVNTSEL1	
Uncore C-box 1 PerfMon Event Select for C-box 1 Counter 1		Package
Register Address: D32H, 3378	MSR_C1_PMON_EVNTSEL2	
Uncore C-box 1 PerfMon Event Select for C-box 1 Counter 2		Package
Register Address: D33H, 3379	MSR_C1_PMON_EVNTSEL3	
Uncore C-box 1 PerfMon Event Select for C-box 1 Counter 3		Package
Register Address: D34H, 3380	MSR_C1_PMON_BOX_FILTER	
Uncore C-box 1 PerfMon Box Wide Filter		Package
Register Address: D36H, 3382	MSR_C1_PMON_CTR0	
Uncore C-box 1 PerfMon Counter 0		Package
Register Address: D37H, 3383	MSR_C1_PMON_CTR1	
Uncore C-box 1 PerfMon Counter 1		Package
Register Address: D38H, 3384	MSR_C1_PMON_CTR2	
Uncore C-box 1 PerfMon Counter 2		Package
Register Address: D39H, 3385	MSR_C1_PMON_CTR3	
Uncore C-box 1 PerfMon Counter 3		Package
Register Address: D44H, 3396	MSR_C2_PMON_BOX_CTL	
Uncore C-box 2 PerfMon Local Box Wide Control		Package
Register Address: D50H, 3408	MSR_C2_PMON_EVNTSEL0	
Uncore C-box 2 PerfMon Event Select for C-box 2 Counter 0		Package
Register Address: D51H, 3409	MSR_C2_PMON_EVNTSEL1	

Table 2-24. Uncore PMU MSRs in Intel® Xeon® Processor E5 Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-box 2 PerfMon Event Select for C-box 2 Counter 1		Package
Register Address: D52H, 3410	MSR_C2_PMON_EVNTSEL2	
Uncore C-box 2 PerfMon Event Select for C-box 2 Counter 2		Package
Register Address: D53H, 3411	MSR_C2_PMON_EVNTSEL3	
Uncore C-box 2 PerfMon Event Select for C-box 2 Counter 3		Package
Register Address: D54H, 3412	MSR_C2_PMON_BOX_FILTER	
Uncore C-box 2 PerfMon Box Wide Filter		Package
Register Address: D56H, 3414	MSR_C2_PMON_CTR0	
Uncore C-box 2 PerfMon Counter 0		Package
Register Address: D57H, 3415	MSR_C2_PMON_CTR1	
Uncore C-box 2 PerfMon Counter 1		Package
Register Address: D58H, 3416	MSR_C2_PMON_CTR2	
Uncore C-box 2 PerfMon Counter 2		Package
Register Address: D59H, 3417	MSR_C2_PMON_CTR3	
Uncore C-box 2 PerfMon Counter 3		Package
Register Address: D64H, 3428	MSR_C3_PMON_BOX_CTL	
Uncore C-box 3 PerfMon Local Box Wide Control		Package
Register Address: D70H, 3440	MSR_C3_PMON_EVNTSEL0	
Uncore C-box 3 PerfMon Event Select for C-box 3 Counter 0		Package
Register Address: D71H, 3441	MSR_C3_PMON_EVNTSEL1	
Uncore C-box 3 PerfMon Event Select for C-box 3 Counter 1		Package
Register Address: D72H, 3442	MSR_C3_PMON_EVNTSEL2	
Uncore C-box 3 PerfMon Event Select for C-box 3 Counter 2		Package
Register Address: D73H, 3443	MSR_C3_PMON_EVNTSEL3	
Uncore C-box 3 PerfMon Event Select for C-box 3 Counter 3		Package
Register Address: D74H, 3444	MSR_C3_PMON_BOX_FILTER	
Uncore C-box 3 PerfMon Box Wide Filter		Package
Register Address: D76H, 3446	MSR_C3_PMON_CTR0	
Uncore C-box 3 PerfMon Counter 0		Package
Register Address: D77H, 3447	MSR_C3_PMON_CTR1	
Uncore C-box 3 PerfMon Counter 1		Package
Register Address: D78H, 3448	MSR_C3_PMON_CTR2	
Uncore C-box 3 PerfMon Counter 2		Package
Register Address: D79H, 3449	MSR_C3_PMON_CTR3	
Uncore C-box 3 PerfMon Counter 3		Package
Register Address: D84H, 3460	MSR_C4_PMON_BOX_CTL	
Uncore C-box 4 PerfMon Local Box Wide Control		Package

Table 2-24. Uncore PMU MSRs in Intel® Xeon® Processor E5 Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: D90H, 3472	MSR_C4_PMON_EVNTSEL0	
Uncore C-box 4 PerfMon Event Select for C-box 4 Counter 0		Package
Register Address: D91H, 3473	MSR_C4_PMON_EVNTSEL1	
Uncore C-box 4 PerfMon Event Select for C-box 4 Counter 1		Package
Register Address: D92H, 3474	MSR_C4_PMON_EVNTSEL2	
Uncore C-box 4 PerfMon Event Select for C-box 4 Counter 2		Package
Register Address: D93H, 3475	MSR_C4_PMON_EVNTSEL3	
Uncore C-box 4 PerfMon Event Select for C-box 4 Counter 3		Package
Register Address: D94H, 3476	MSR_C4_PMON_BOX_FILTER	
Uncore C-box 4 PerfMon Box Wide Filter		Package
Register Address: D96H, 3478	MSR_C4_PMON_CTR0	
Uncore C-box 4 PerfMon Counter 0		Package
Register Address: D97H, 3479	MSR_C4_PMON_CTR1	
Uncore C-box 4 PerfMon Counter 1		Package
Register Address: D98H, 3480	MSR_C4_PMON_CTR2	
Uncore C-box 4 PerfMon Counter 2		Package
Register Address: D99H, 3481	MSR_C4_PMON_CTR3	
Uncore C-box 4 PerfMon Counter 3		Package
Register Address: DA4H, 3492	MSR_C5_PMON_BOX_CTL	
Uncore C-box 5 PerfMon Local Box Wide Control		Package
Register Address: DB0H, 3504	MSR_C5_PMON_EVNTSEL0	
Uncore C-box 5 PerfMon Event Select for C-box 5 Counter 0		Package
Register Address: DB1H, 3505	MSR_C5_PMON_EVNTSEL1	
Uncore C-box 5 PerfMon Event Select for C-box 5 Counter 1		Package
Register Address: DB2H, 3506	MSR_C5_PMON_EVNTSEL2	
Uncore C-box 5 PerfMon Event Select for C-box 5 Counter 2		Package
Register Address: DB3H, 3507	MSR_C5_PMON_EVNTSEL3	
Uncore C-box 5 PerfMon Event Select for C-box 5 Counter 3		Package
Register Address: DB4H, 3508	MSR_C5_PMON_BOX_FILTER	
Uncore C-box 5 PerfMon Box Wide Filter		Package
Register Address: DB6H, 3510	MSR_C5_PMON_CTR0	
Uncore C-box 5 PerfMon Counter 0		Package
Register Address: DB7H, 3511	MSR_C5_PMON_CTR1	
Uncore C-box 5 PerfMon Counter 1		Package
Register Address: DB8H, 3512	MSR_C5_PMON_CTR2	
Uncore C-box 5 PerfMon Counter 2		Package
Register Address: DB9H, 3513	MSR_C5_PMON_CTR3	

Table 2-24. Uncore PMU MSRs in Intel® Xeon® Processor E5 Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-box 5 PerfMon Counter 3		Package
Register Address: DC4H, 3524	MSR_C6_PMON_BOX_CTL	
Uncore C-box 6 PerfMon Local Box Wide Control		Package
Register Address: DD0H, 3536	MSR_C6_PMON_EVTNTSEL0	
Uncore C-box 6 PerfMon Event Select for C-box 6 Counter 0		Package
Register Address: DD1H, 3537	MSR_C6_PMON_EVTNTSEL1	
Uncore C-box 6 PerfMon Event Select for C-box 6 Counter 1		Package
Register Address: DD2H, 3538	MSR_C6_PMON_EVTNTSEL2	
Uncore C-box 6 PerfMon Event Select for C-box 6 Counter 2		Package
Register Address: DD3H, 3539	MSR_C6_PMON_EVTNTSEL3	
Uncore C-box 6 PerfMon Event Select for C-box 6 Counter 3		Package
Register Address: DD4H, 3540	MSR_C6_PMON_BOX_FILTER	
Uncore C-box 6 PerfMon Box Wide Filter		Package
Register Address: DD6H, 3542	MSR_C6_PMON_CTR0	
Uncore C-box 6 PerfMon Counter 0		Package
Register Address: DD7H, 3543	MSR_C6_PMON_CTR1	
Uncore C-box 6 PerfMon Counter 1		Package
Register Address: DD8H, 3544	MSR_C6_PMON_CTR2	
Uncore C-box 6 PerfMon Counter 2		Package
Register Address: DD9H, 3545	MSR_C6_PMON_CTR3	
Uncore C-box 6 PerfMon Counter 3		Package
Register Address: DE4H, 3556	MSR_C7_PMON_BOX_CTL	
Uncore C-box 7 PerfMon Local Box Wide Control		Package
Register Address: DF0H, 3568	MSR_C7_PMON_EVTNTSEL0	
Uncore C-box 7 PerfMon Event Select for C-box 7 Counter 0		Package
Register Address: DF1H, 3569	MSR_C7_PMON_EVTNTSEL1	
Uncore C-box 7 PerfMon Event Select for C-box 7 Counter 1		Package
Register Address: DF2H, 3570	MSR_C7_PMON_EVTNTSEL2	
Uncore C-box 7 PerfMon Event Select for C-box 7 Counter 2		Package
Register Address: DF3H, 3571	MSR_C7_PMON_EVTNTSEL3	
Uncore C-box 7 PerfMon Event Select for C-box 7 Counter 3		Package
Register Address: DF4H, 3572	MSR_C7_PMON_BOX_FILTER	
Uncore C-box 7 PerfMon Box Wide Filter		Package
Register Address: DF6H, 3574	MSR_C7_PMON_CTR0	
Uncore C-box 7 PerfMon Counter 0		Package
Register Address: DF7H, 3575	MSR_C7_PMON_CTR1	
Uncore C-box 7 PerfMon Counter 1		Package

Table 2-24. Uncore PMU MSRs in Intel® Xeon® Processor E5 Family (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: DF8H, 3576	MSR_C7_PMON_CTR2	
Uncore C-box 7 PerfMon Counter 2		Package
Register Address: DF9H, 3577	MSR_C7_PMON_CTR3	
Uncore C-box 7 PerfMon Counter 3		Package

2.12 MSRS IN THE 3RD GENERATION INTEL® CORE™ PROCESSOR FAMILY BASED ON IVY BRIDGE MICROARCHITECTURE

The 3rd generation Intel® Core™ processor family and the Intel® Xeon® processor E3-1200v2 product family based on Ivy Bridge microarchitecture support the MSR interfaces listed in Table 2-20, Table 2-21, Table 2-22, and Table 2-25. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_3AH.

Table 2-25. Additional MSRs Supported by 3rd Generation Intel® Core™ Processors Based on Ivy Bridge Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) This is the ratio of the frequency that invariant TSC runs at. Frequency = ratio * 100 MHz.	Package
27:16	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) When set to 1, indicates that Programmable Ratio Limit for Turbo mode is enabled. When set to 0, indicates Programmable Ratio Limit for Turbo mode is disabled.	Package
29	Programmable TDP Limit for Turbo Mode (R/O) When set to 1, indicates that TDP Limit for Turbo mode is programmable. When set to 0, indicates that TDP Limit for Turbo mode is not programmable.	Package
31:30	Reserved.	
32	Low Power Mode Support (LPM) (R/O) When set to 1, indicates that LPM is supported. When set to 0, indicates LPM is not supported.	Package
34:33	Number of ConfigTDP Levels (R/O) 00: Only Base TDP level available. 01: One additional TDP level available. 02: Two additional TDP level available. 03: Reserved	Package
39:35	Reserved.	

Table 2-25. Additional MSRs Supported by 3rd Generation Intel® Core™ Processors Based on Ivy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
47:40	Maximum Efficiency Ratio (R/O) This is the minimum ratio (maximum efficiency) that the processor can operate, in units of 100MHz.	Package
55:48	Minimum Operating Ratio (R/O) Contains the minimum supported operating ratio in units of 100 MHz.	Package
63:56	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
	C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States. See http://biosbits.org .	Core
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: C0/C1 (no package C-state support) 001b: C2 010b: C6 no retention 011b: C6 retention 100b: C7 101b: C7s 111: No package C-state limit. Note: This field cannot be used to limit package C-state to C3.	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W) When set, will map IO_read instructions sent to IO register specified by MSR_PMG_IO_CAPTURE_BASE to MWAIT instructions.	
14:11	Reserved.	
15	CFG Lock (R/WO) When set, locks bits 15:0 of this register until next reset.	
24:16	Reserved	
25	C3 State Auto Demotion Enable (R/W) When set, the processor will conditionally demote C6/C7 requests to C3 based on uncore auto-demote information.	
26	C1 State Auto Demotion Enable (R/W) When set, the processor will conditionally demote C3/C6/C7 requests to C1 based on uncore auto-demote information.	
27	Enable C3 Undemotion (R/W) When set, enables undemotion from demoted C3.	

Table 2-25. Additional MSRs Supported by 3rd Generation Intel® Core™ Processors Based on Ivy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
28	Enable C1 Undemotion (R/W) When set, enables undemotion from demoted C1.	
63:29	Reserved.	
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	
PP0 Energy Status (R/O) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 648H, 1608	MSR_CONFIG_TDP_NOMINAL	
Base TDP Ratio (R/O)		Package
7:0	Config_TDP_Base Base TDP level ratio to be used for this specific processor (in units of 100 MHz).	
63:8	Reserved.	
Register Address: 649H, 1609	MSR_CONFIG_TDP_LEVEL1	
	ConfigTDP Level 1 ratio and power level (R/O)	Package
14:0	PKG_TDP_LVL1 Power setting for ConfigTDP Level 1.	
15	Reserved.	
23:16	Config_TDP_LVL1_Ratio ConfigTDP level 1 ratio to be used for this specific processor.	
31:24	Reserved.	
46:32	PKG_MAX_PWR_LVL1 Max Power setting allowed for ConfigTDP Level 1.	
47	Reserved.	
62:48	PKG_MIN_PWR_LVL1 MIN Power setting allowed for ConfigTDP Level 1.	
63	Reserved.	
Register Address: 64AH, 1610	MSR_CONFIG_TDP_LEVEL2	
ConfigTDP Level 2 ratio and power level (R/O)		Package
14:0	PKG_TDP_LVL2 Power setting for ConfigTDP Level 2.	
15	Reserved.	
23:16	Config_TDP_LVL2_Ratio ConfigTDP level 2 ratio to be used for this specific processor.	
31:24	Reserved.	
46:32	PKG_MAX_PWR_LVL2 Max Power setting allowed for ConfigTDP Level 2.	
47	Reserved.	

Table 2-25. Additional MSRs Supported by 3rd Generation Intel® Core™ Processors Based on Ivy Bridge Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
62:48	PKG_MIN_PWR_LVL2 MIN Power setting allowed for ConfigTDP Level 2.	
63	Reserved.	
Register Address: 64BH, 1611	MSR_CONFIG_TDP_CONTROL	
ConfigTDP Control (R/W)		Package
1:0	TDP_LEVEL (RW/L) System BIOS can program this field.	
30:2	Reserved.	
31	Config_TDP_Lock (RW/L) When this bit is set, the content of this register is locked until a reset.	
63:32	Reserved.	
Register Address: 64CH, 1612	MSR_TURBO_ACTIVATION_RATIO	
ConfigTDP Control (R/W)		Package
7:0	MAX_NON_TURBO_RATIO (RW/L) System BIOS can program this field.	
30:8	Reserved.	
31	TURBO_ACTIVATION_RATIO_Lock (RW/L) When this bit is set, the content of this register is locked until a reset.	
63:32	Reserved.	
See Table 2-20, Table 2-21, and Table 2-22 for other MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_3AH.		

2.12.1 MSRs in the Intel® Xeon® Processor E5 v2 Product Family Based on Ivy Bridge-E Microarchitecture

Table 2-26 lists model-specific registers (MSRs) that are specific to the Intel® Xeon® Processor E5 v2 Product Family (based on Ivy Bridge-E microarchitecture). These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_3EH; see Table 2-1. These processors supports the MSR interfaces listed in Table 2-20 and Table 2-26.

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 4EH, 78	IA32_PPIN_CTL (MSR_PPIN_CTL)	
Protected Processor Inventory Number Enable Control (R/W)		Package
0	LockOut (R/WO) See Table 2-2.	
1	Enable_PPIN (R/W) See Table 2-2.	
63:2	Reserved.	

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 4FH, 79	IA32_PPIN (MSR_PPIN)	
Protected Processor Inventory Number (R/O)		Package
63:0	Protected Processor Inventory Number (R/O) See Table 2-2.	
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) This is the ratio of the frequency that invariant TSC runs at. Frequency = ratio * 100 MHz.	Package
22:16	Reserved.	
23	PPIN_CAP (R/O) When set to 1, indicates that Protected Processor Inventory Number (PPIN) capability can be enabled for a privileged system inventory agent to read PPIN from MSR_PPIN. When set to 0, PPIN capability is not supported. An attempt to access MSR_PPIN_CTL or MSR_PPIN will cause #GP.	Package
27:24	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) When set to 1, indicates that Programmable Ratio Limit for Turbo mode is enabled. When set to 0, indicates Programmable Ratio Limit for Turbo mode is disabled.	Package
29	Programmable TDP Limit for Turbo Mode (R/O) When set to 1, indicates that TDP Limit for Turbo mode is programmable. When set to 0, indicates TDP Limit for Turbo mode is not programmable.	Package
30	Programmable TJ OFFSET (R/O) When set to 1, indicates that MSR_TEMPERATURE_TARGET.[27:24] is valid and writable to specify a temperature offset.	Package
39:31	Reserved.	
47:40	Maximum Efficiency Ratio (R/O) This is the minimum ratio (maximum efficiency) that the processor can operate, in units of 100MHz.	Package
63:48	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states. See http://biosbits.org .		Core

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: C0/C1 (no package C-state support) 001b: C2 010b: C6 no retention 011b: C6 retention 100b: C7 101b: C7s 111: No package C-state limit. Note: This field cannot be used to limit package C-state to C3.	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W) When set, will map IO_read instructions sent to IO register specified by MSR_PMG_IO_CAPTURE_BASE to MWAIT instructions.	
14:11	Reserved.	
15	CFG Lock (R/WO) When set, locks bits 15:0 of this register until next reset.	
63:16	Reserved.	
Register Address: 179H, 377	IA32_MCG_CAP	
Global Machine Check Capability (R/O)		Thread
7:0	Count	
8	MCG_CTL_P	
9	MCG_EXT_P	
10	MCP_CMCI_P	
11	MCG_TES_P	
15:12	Reserved.	
23:16	MCG_EXT_CNT	
24	MCG_SER_P	
25	Reserved.	
26	MCG_ELOG_P	
63:27	Reserved.	
Register Address: 17FH, 383	MSR_ERROR_CONTROL	
MC Bank Error Configuration (R/W)		Package
0	Reserved.	
1	MemError Log Enable (R/W) When set, enables IMC status bank to log additional info in bits 36:32.	
63:2	Reserved.	

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1A2H, 418	MSR_TEMPERATURE_TARGET	
Temperature Target		Package
15:0	Reserved.	
23:16	Temperature Target (R/O) The minimum temperature at which PROCHOT# will be asserted. The value is degrees C.	
27:24	TCC Activation Offset (R/W) Specifies a temperature offset in degrees C from the temperature target (bits 23:16). PROCHOT# will assert at the offset target temperature. Write is permitted only if MSR_PLATFORM_INFO.[30] is set.	
63:28	Reserved.	
Register Address: 1AEH, 430	MSR_TURBO_RATIO_LIMIT1	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0. R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 9C Maximum turbo ratio limit of 9 core active.	Package
15:8	Maximum Ratio Limit for 10C Maximum turbo ratio limit of 10 core active.	Package
23:16	Maximum Ratio Limit for 11C Maximum turbo ratio limit of 11 core active.	Package
31:24	Maximum Ratio Limit for 12C Maximum turbo ratio limit of 12 core active.	Package
63:32	Reserved.	
Register Address: 285H, 645	IA32_MC5_CTL2	
See Table 2-2.		Package
Register Address: 286H, 646	IA32_MC6_CTL2	
See Table 2-2.		Package
Register Address: 287H, 647	IA32_MC7_CTL2	
See Table 2-2.		Package
Register Address: 288H, 648	IA32_MC8_CTL2	
See Table 2-2.		Package
Register Address: 289H, 649	IA32_MC9_CTL2	
See Table 2-2.		Package
Register Address: 28AH, 650	IA32_MC10_CTL2	
See Table 2-2.		Package
Register Address: 28BH, 651	IA32_MC11_CTL2	
See Table 2-2.		Package
Register Address: 28CH, 652	IA32_MC12_CTL2	
See Table 2-2.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 28DH, 653	IA32_MC13_CTL2	
See Table 2-2.		Package
Register Address: 28EH, 654	IA32_MC14_CTL2	
See Table 2-2.		Package
Register Address: 28FH, 655	IA32_MC15_CTL2	
See Table 2-2.		Package
Register Address: 290H, 656	IA32_MC16_CTL2	
See Table 2-2.		Package
Register Address: 291H, 657	IA32_MC17_CTL2	
See Table 2-2.		Package
Register Address: 292H, 658	IA32_MC18_CTL2	
See Table 2-2.		Package
Register Address: 293H, 659	IA32_MC19_CTL2	
See Table 2-2.		Package
Register Address: 294H, 660	IA32_MC20_CTL2	
See Table 2-2.		Package
Register Address: 295H, 661	IA32_MC21_CTL2	
See Table 2-2.		Package
Register Address: 296H, 662	IA32_MC22_CTL2	
See Table 2-2.		Package
Register Address: 297H, 663	IA32_MC23_CTL2IA32_MC23_CTL2	
See Table 2-2.		Package
Register Address: 298H, 664	IA32_MC24_CTL2	
See Table 2-2.		Package
Register Address: 299H, 665	IA32_MC25_CTL2	
See Table 2-2.		Package
Register Address: 29AH, 666	IA32_MC26_CTL2	
See Table 2-2.		Package
Register Address: 29BH, 667	IA32_MC27_CTL2	
See Table 2-2.		Package
Register Address: 29CH, 668	IA32_MC28_CTL2	
See Table 2-2.		Package
Register Address: 414H, 1044	IA32_MC5_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI module.		Package
Register Address: 415H, 1045	IA32_MC5_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI module.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 416H, 1046	IA32_MC5_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI module.		Package
Register Address: 417H, 1047	IA32_MC5_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI module.		Package
Register Address: 418H, 1048	IA32_MC6_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 419H, 1049	IA32_MC6_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41AH, 1050	IA32_MC6_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41BH, 1051	IA32_MC6_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41CH, 1052	IA32_MC7_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC7 and MC 8 report MC errors from the two home agents.		Package
Register Address: 41DH, 1053	IA32_MC7_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC7 and MC 8 report MC errors from the two home agents.		Package
Register Address: 41EH, 1054	IA32_MC7_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC7 and MC 8 report MC errors from the two home agents.		Package
Register Address: 41FH, 1055	IA32_MC7_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC7 and MC 8 report MC errors from the two home agents.		Package
Register Address: 420H, 1056	IA32_MC8_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC7 and MC 8 report MC errors from the two home agents.		Package
Register Address: 421H, 1057	IA32_MC8_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC7 and MC 8 report MC errors from the two home agents.		Package
Register Address: 422H, 1058	IA32_MC8_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC7 and MC 8 report MC errors from the two home agents.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 423H, 1059	IA32_MC8_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC7 and MC 8 report MC errors from the two home agents.		Package
Register Address: 424H, 1060	IA32_MC9_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 425H, 1061	IA32_MC9_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 426H, 1062	IA32_MC9_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 427H, 1063	IA32_MC9_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 428H, 1064	IA32_MC10_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 429H, 1065	IA32_MC10_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42AH, 1066	IA32_MC10_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42BH, 1067	IA32_MC10_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42CH, 1068	IA32_MC11_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs." Bank MC11 reports MC errors from a specific channel of the integrated memory controller.		Package
Register Address: 42DH, 1069	IA32_MC11_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs." Bank MC11 reports MC errors from a specific channel of the integrated memory controller.		Package
Register Address: 42EH, 1070	IA32_MC11_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs." Bank MC11 reports MC errors from a specific channel of the integrated memory controller.		Package
Register Address: 42FH, 1071	IA32_MC11_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs." Bank MC11 reports MC errors from a specific channel of the integrated memory controller.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 430H, 1072	IA32_MC12_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 431H, 1073	IA32_MC12_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 432H, 1074	IA32_MC12_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 433H, 1075	IA32_MC12_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 434H, 1076	IA32_MC13_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 435H, 1077	IA32_MC13_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 436H, 1078	IA32_MC13_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 437H, 1079	IA32_MC13_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 438H, 1080	IA32_MC14_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 439H, 1081	IA32_MC14_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43AH, 1082	IA32_MC14_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43BH, 1083	IA32_MC14_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43CH, 1084	IA32_MC15_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 43DH, 1085	IA32_MC15_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43EH, 1086	IA32_MC15_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43FH, 1087	IA32_MC15_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 440H, 1088	IA32_MC16_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 441H, 1089	IA32_MC16_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 442H, 1090	IA32_MC16_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 443H, 1091	IA32_MC16_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 444H, 1092	IA32_MC17_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC17 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 445H, 1093	IA32_MC17_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC17 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 446H, 1094	IA32_MC17_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC17 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 447H, 1095	IA32_MC17_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC17 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 448H, 1096	IA32_MC18_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 449H, 1097	IA32_MC18_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 44AH, 1098	IA32_MC18_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC18 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 44BH, 1099	IA32_MC18_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC18 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 44CH, 1100	IA32_MC19_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC19 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 44DH, 1101	IA32_MC19_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC19 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 44EH, 1102	IA32_MC19_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC19 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 44FH, 1103	IA32_MC19_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC19 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 450H, 1104	IA32_MC20_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs." Bank MC20 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 451H, 1105	IA32_MC20_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs." Bank MC20 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 452H, 1106	IA32_MC20_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs." Bank MC20 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 453H, 1107	IA32_MC20_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs." Bank MC20 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 454H, 1108	IA32_MC21_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 455H, 1109	IA32_MC21_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 456H, 1110	IA32_MC21_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 457H, 1111	IA32_MC21_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 458H, 1112	IA32_MC22_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC22 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 459H, 1113	IA32_MC22_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC22 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 45AH, 1114	IA32_MC22_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC22 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 45BH, 1115	IA32_MC22_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC22 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 45CH, 1116	IA32_MC23_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC23 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 45DH, 1117	IA32_MC23_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC23 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 45EH, 1118	IA32_MC23_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC23 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 45FH, 1119	IA32_MC23_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC23 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 460H, 1120	IA32_MC24_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC24 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 461H, 1121	IA32_MC24_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC24 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 462H, 1122	IA32_MC24_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC24 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 463H, 1123	IA32_MC24_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC24 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 464H, 1124	IA32_MC25_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC25 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 465H, 1125	IA32_MC25_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC25 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 466H, 1126	IA32_MC25_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC25 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 467H, 1127	IA32_MC2MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC25 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 468H, 1128	IA32_MC26_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC26 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 469H, 1129	IA32_MC26_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC26 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 46AH, 1130	IA32_MC26_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC26 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 46BH, 1131	IA32_MC26_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC26 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 46CH, 1132	IA32_MC27_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC27 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 46DH, 1133	IA32_MC27_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC27 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 46EH, 1134	IA32_MC27_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC27 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 46FH, 1135	IA32_MC27_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC27 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 470H, 1136	IA32_MC28_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC28 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package

Table 2-26. MSRs Supported by the Intel® Xeon® Processor E5 v2 Product Family (Ivy Bridge-E Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 471H, 1137	IA32_MC28_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC28 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 472H, 1138	IA32_MC28_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC28 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 473H, 1139	IA32_MC28_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC28 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 613H, 1555	MSR_PKG_PERF_STATUS	
Package RAPL Perf Status (R/O)		Package
Register Address: 618H, 1560	MSR_DRAM_POWER_LIMIT	
DRAM RAPL Power Limit Control (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61CH, 1564	MSR_DRAM_POWER_INFO	
DRAM RAPL Parameters (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	
PPO Energy Status (R/O) See Section 16.10.4, "PPO/PP1 RAPL Domains."		Package
See Table 2-20, for other MSR definitions applicable to Intel Xeon processor E5 v2 with a CPUID Signature DisplayFamily_DisplayModel value of 06_3EH.		

2.12.2 Additional MSRs Supported by the Intel® Xeon® Processor E7 v2 Family

The Intel® Xeon® processor E7 v2 family (based on Ivy Bridge-E microarchitecture) with a CPUID Signature DisplayFamily_DisplayModel value of 06_3EH supports the MSR interfaces listed in Table 2-20, Table 2-26, and Table 2-27.

Table 2-27. Additional MSRs Supported by the Intel® Xeon® Processor E7 v2 Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_3EH

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	

Table 2-27. Additional MSRs Supported by the Intel® Xeon® Processor E7 v2 Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_3EH (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Control Features in Intel 64 Processor (R/W) See Table 2-2.		Thread
0	Lock (R/WL)	
1	Enable VMX Inside SMX Operation (R/WL)	
2	Enable VMX Outside SMX Operation (R/WL)	
14:8	SENTER Local Functions Enables (R/WL)	
15	SENTER Global Functions Enable (R/WL)	
63:16	Reserved.	
Register Address: 179H, 377	IA32_MCG_CAP	
Global Machine Check Capability (R/O)		Thread
7:0	Count	
8	MCG_CTL_P	
9	MCG_EXT_P	
10	MCP_CMCI_P	
11	MCG_TES_P	
15:12	Reserved.	
23:16	MCG_EXT_CNT	
24	MCG_SER_P	
63:25	Reserved.	
Register Address: 17AH, 378	IA32_MCG_STATUS	
Global Machine Check Status (R/W)		Thread
0	RIPV	
1	EIPV	
2	MCIP	
3	LMCE Signaled	
63:4	Reserved.	
Register Address: 1AEH, 430	MSR_TURBO_RATIO_LIMIT1	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 9C Maximum turbo ratio limit of 9 core active.	Package
15:8	Maximum Ratio Limit for 10C Maximum turbo ratio limit of 10core active.	Package
23:16	Maximum Ratio Limit for 11C Maximum turbo ratio limit of 11 core active.	Package
31:24	Maximum Ratio Limit for 12C Maximum turbo ratio limit of 12 core active.	Package

Table 2-27. Additional MSRs Supported by the Intel® Xeon® Processor E7 v2 Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_3EH (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
39:32	Maximum Ratio Limit for 13C Maximum turbo ratio limit of 13 core active.	Package
47:40	Maximum Ratio Limit for 14C Maximum turbo ratio limit of 14 core active.	Package
55:48	Maximum Ratio Limit for 15C Maximum turbo ratio limit of 15 core active.	Package
62:56	Reserved.	
63	Semaphore for Turbo Ratio Limit Configuration If 1, the processor uses override configuration ¹ specified in MSR_TURBO_RATIO_LIMIT and MSR_TURBO_RATIO_LIMIT1. If 0, the processor uses factory-set configuration (Default).	Package
Register Address: 29DH, 669	IA32_MC29_CTL2	
See Table 2-2.		Package
Register Address: 29EH, 670	IA32_MC30_CTL2	
See Table 2-2.		Package
Register Address: 29FH, 671	IA32_MC31_CTL2	
See Table 2-2.		Package
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
See Section 21.3.1.1.1, "Processor Event Based Sampling (PEBS)."		Thread
$n:0$	Enable PEBS on IA32_PMCx. (R/W)	
$31:n+1$	Reserved.	
$32+m:32$	Enable Load Latency on IA32_PMCx. (R/W)	
$63:33+m$	Reserved.	
Register Address: 41BH, 1051	IA32_MC6_MISC	
Misc MAC Information of Integrated I/O (R/O) See Section 17.3.2.4.		Package
5:0	Recoverable Address LSB	
8:6	Address Mode	
15:9	Reserved.	
31:16	PCI Express Requestor ID	
39:32	PCI Express Segment Number	
63:32	Reserved.	
Register Address: 474H, 1140	IA32_MC29_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC29 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 475H, 1141	IA32_MC29_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC29 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package

Table 2-27. Additional MSRs Supported by the Intel® Xeon® Processor E7 v2 Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_3EH (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 476H, 1142	IA32_MC29_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC29 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 477H, 1143	IA32_MC29_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC29 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 478H, 1144	IA32_MC30_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC30 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 479H, 1145	IA32_MC30_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC30 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 47AH, 1146	IA32_MC30_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC30 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 47BH, 1147	IA32_MC30_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC30 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 47CH, 1148	IA32_MC31_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC31 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 47DH, 1149	IA32_MC31_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC31 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 47EH, 1150	IA32_MC31_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC31 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
Register Address: 47FH, 1147	IA32_MC31_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC31 reports MC errors from a specific CBo (core broadcast) and its corresponding slice of L3.		Package
See Table 2-20, Table 2-26 for other MSR definitions applicable to Intel Xeon processor E7 v2 with a CPUID Signature DisplayFamily_DisplayModel value of 06_3AH.		

NOTES:

1. An override configuration lower than the factory-set configuration is always supported. An override configuration higher than the factory-set configuration is dependent on features specific to the processor and the platform.

2.12.3 Additional Uncore PMU MSRs in the Intel® Xeon® Processor E5 v2 and E7 v2 Families

Intel Xeon Processor E5 v2 and E7 v2 families are based on the Ivy Bridge-E microarchitecture. The MSR-based uncore PMU interfaces are listed in Table 2-24 and Table 2-28. For complete detail of the uncore PMU, refer to Intel

Xeon Processor E5 v2 Product Family Uncore Performance Monitoring Guide. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_3EH.

Table 2-28. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v2 and E7 v2 Families

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: C00H, 3072	MSR_PMON_GLOBAL_CTL	
Uncore PerfMon Per-Socket Global Control		Package
Register Address: C01H, 3073	MSR_PMON_GLOBAL_STATUS	
Uncore PerfMon Per-Socket Global Status		Package
Register Address: C06H, 3078	MSR_PMON_GLOBAL_CONFIG	
Uncore PerfMon Per-Socket Global Configuration		Package
Register Address: C15H, 3093	MSR_U_PMON_BOX_STATUS	
Uncore U-box PerfMon U-Box Wide Status		Package
Register Address: C35H, 3125	MSR_PCU_PMON_BOX_STATUS	
Uncore PCU PerfMon Box Wide Status		Package
Register Address: D1AH, 3354	MSR_C0_PMON_BOX_FILTER1	
Uncore C-Box 0 PerfMon Box Wide Filter1		Package
Register Address: D3AH, 3386	MSR_C1_PMON_BOX_FILTER1	
Uncore C-Box 1 PerfMon Box Wide Filter1		Package
Register Address: D5AH, 3418	MSR_C2_PMON_BOX_FILTER1	
Uncore C-Box 2 PerfMon Box Wide Filter1		Package
Register Address: D7AH, 3450	MSR_C3_PMON_BOX_FILTER1	
Uncore C-Box 3 PerfMon Box Wide Filter1		Package
Register Address: D9AH, 3482	MSR_C4_PMON_BOX_FILTER1	
Uncore C-Box 4 PerfMon Box Wide Filter1		Package
Register Address: DBAH, 3514	MSR_C5_PMON_BOX_FILTER1	
Uncore C-Box 5 PerfMon Box Wide Filter1		Package
Register Address: DDAH, 3546	MSR_C6_PMON_BOX_FILTER1	
Uncore C-Box 6 PerfMon Box Wide Filter1		Package
Register Address: DFAH, 3578	MSR_C7_PMON_BOX_FILTER1	
Uncore C-Box 7 PerfMon Box Wide Filter1		Package
Register Address: E04H, 3588	MSR_C8_PMON_BOX_CTL	
Uncore C-Box 8 PerfMon Local Box Wide Control		Package
Register Address: E10H, 3600	MSR_C8_PMON_EVNTSELO	
Uncore C-Box 8 PerfMon Event Select for C-Box 8 Counter 0		Package
Register Address: E11H, 3601	MSR_C8_PMON_EVNTSEL1	
Uncore C-Box 8 PerfMon Event Select for C-Box 8 Counter 1		Package
Register Address: E12H, 3602	MSR_C8_PMON_EVNTSEL2	
Uncore C-Box 8 PerfMon Event Select for C-Box 8 Counter 2		Package
Register Address: E13H, 3603	MSR_C8_PMON_EVNTSEL3	

Table 2-28. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v2 and E7 v2 Families (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 8 PerfMon Event Select for C-Box 8 Counter 3		Package
Register Address: E14H, 3604	MSR_C8_PMON_BOX_FILTER	
Uncore C-Box 8 PerfMon Box Wide Filter		Package
Register Address: E16H, 3606	MSR_C8_PMON_CTR0	
Uncore C-Box 8 PerfMon Counter 0		Package
Register Address: E17H, 3607	MSR_C8_PMON_CTR1	
Uncore C-Box 8 PerfMon Counter 1		Package
Register Address: E18H, 3608	MSR_C8_PMON_CTR2	
Uncore C-Box 8 PerfMon Counter 2		Package
Register Address: E19H, 3609	MSR_C8_PMON_CTR3	
Uncore C-Box 8 PerfMon Counter 3		Package
Register Address: E1AH, 3610	MSR_C8_PMON_BOX_FILTER1	
Uncore C-Box 8 PerfMon Box Wide Filter1		Package
Register Address: E24H, 3620	MSR_C9_PMON_BOX_CTL	
Uncore C-Box 9 PerfMon Local Box Wide Control		Package
Register Address: E30H, 3632	MSR_C9_PMON_EVNTSEL0	
Uncore C-Box 9 PerfMon Event Select for C-box 9 Counter 0		Package
Register Address: E31H, 3633	MSR_C9_PMON_EVNTSEL1	
Uncore C-Box 9 PerfMon Event Select for C-box 9 Counter 1		Package
Register Address: E32H, 3634	MSR_C9_PMON_EVNTSEL2	
Uncore C-Box 9 PerfMon Event Select for C-box 9 Counter 2		Package
Register Address: E33H, 3635	MSR_C9_PMON_EVNTSEL3	
Uncore C-Box 9 PerfMon Event Select for C-box 9 Counter 3		Package
Register Address: E34H, 3636	MSR_C9_PMON_BOX_FILTER	
Uncore C-Box 9 PerfMon Box Wide Filter		Package
Register Address: E36H, 3638	MSR_C9_PMON_CTR0	
Uncore C-Box 9 PerfMon Counter 0		Package
Register Address: E37H, 3639	MSR_C9_PMON_CTR1	
Uncore C-Box 9 PerfMon Counter 1		Package
Register Address: E38H, 3640	MSR_C9_PMON_CTR2	
Uncore C-Box 9 PerfMon Counter 2		Package
Register Address: E39H, 3641	MSR_C9_PMON_CTR3	
Uncore C-Box 9 PerfMon Counter 3		Package
Register Address: E3AH, 3642	MSR_C9_PMON_BOX_FILTER1	
Uncore C-Box 9 PerfMon Box Wide Filter1		Package
Register Address: E44H, 3652	MSR_C10_PMON_BOX_CTL	
Uncore C-Box 10 PerfMon Local Box Wide Control		Package

Table 2-28. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v2 and E7 v2 Families (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E50H, 3664	MSR_C10_PMON_EVTNSELO	
Uncore C-Box 10 PerfMon Event Select for C-Box 10 Counter 0		Package
Register Address: E51H, 3665	MSR_C10_PMON_EVTNSEL1	
Uncore C-Box 10 PerfMon Event Select for C-Box 10 Counter 1		Package
Register Address: E52H, 3666	MSR_C10_PMON_EVTNSEL2	
Uncore C-Box 10 PerfMon Event Select for C-Box 10 Counter 2		Package
Register Address: E53H, 3667	MSR_C10_PMON_EVTNSEL3	
Uncore C-Box 10 PerfMon Event Select for C-Box 10 Counter 3		Package
Register Address: E54H, 3668	MSR_C10_PMON_BOX_FILTER	
Uncore C-Box 10 PerfMon Box Wide Filter		Package
Register Address: E56H, 3670	MSR_C10_PMON_CTR0	
Uncore C-Box 10 PerfMon Counter 0		Package
Register Address: E57H, 3671	MSR_C10_PMON_CTR1	
Uncore C-Box 10 PerfMon Counter 1		Package
Register Address: E58H, 3672	MSR_C10_PMON_CTR2	
Uncore C-Box 10 PerfMon Counter 2		Package
Register Address: E59H, 3673	MSR_C10_PMON_CTR3	
Uncore C-Box 10 PerfMon Counter 3		Package
Register Address: E5AH, 3674	MSR_C10_PMON_BOX_FILTER1	
Uncore C-Box 10 PerfMon Box Wide Filter1		Package
Register Address: E64H, 3684	MSR_C11_PMON_BOX_CTL	
Uncore C-Box 11 PerfMon Local Box Wide Control		Package
Register Address: E70H, 3696	MSR_C11_PMON_EVTNSELO	
Uncore C-Box 11 PerfMon Event Select for C-Box 11 Counter 0		Package
Register Address: E71H, 3697	MSR_C11_PMON_EVTNSEL1	
Uncore C-Box 11 PerfMon Event Select for C-Box 11 Counter 1		Package
Register Address: E72H, 3698	MSR_C11_PMON_EVTNSEL2	
Uncore C-Box 11 PerfMon Event Select for C-Box 11 Counter 2		Package
Register Address: E73H, 3699	MSR_C11_PMON_EVTNSEL3	
Uncore C-Box 11 PerfMon Event Select for C-Box 11 Counter 3		Package
Register Address: E74H, 3700	MSR_C11_PMON_BOX_FILTER	
Uncore C-Box 11 PerfMon Box Wide Filter		Package
Register Address: E76H, 3702	MSR_C11_PMON_CTR0	
Uncore C-Box 11 PerfMon Counter 0		Package
Register Address: E77H, 3703	MSR_C11_PMON_CTR1	
Uncore C-Box 11 PerfMon Counter 1		Package
Register Address: E78H, 3704	MSR_C11_PMON_CTR2	

Table 2-28. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v2 and E7 v2 Families (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 11 PerfMon Counter 2		Package
Register Address: E79H, 3705	MSR_C11_PMON_CTR3	
Uncore C-Box 11 PerfMon Counter 3		Package
Register Address: E7AH, 3706	MSR_C11_PMON_BOX_FILTER1	
Uncore C-Box 11 PerfMon Box Wide Filter1		Package
Register Address: E84H, 3716	MSR_C12_PMON_BOX_CTL	
Uncore C-Box 12 PerfMon Local Box Wide Control		Package
Register Address: E90H, 3728	MSR_C12_PMON_EVNTSELO	
Uncore C-Box 12 PerfMon Event Select for C-Box 12 Counter 0		Package
Register Address: E91H, 3729	MSR_C12_PMON_EVNTSEL1	
Uncore C-Box 12 PerfMon Event Select for C-Box 12 Counter 1		Package
Register Address: E92H, 3730	MSR_C12_PMON_EVNTSEL2	
Uncore C-Box 12 PerfMon Event Select for C-Box 12 Counter 2		Package
Register Address: E93H, 3731	MSR_C12_PMON_EVNTSEL3	
Uncore C-Box 12 PerfMon Event Select for C-Box 12 Counter 3		Package
Register Address: E94H, 3732	MSR_C12_PMON_BOX_FILTER	
Uncore C-Box 12 PerfMon Box Wide Filter		Package
Register Address: E96H, 3734	MSR_C12_PMON_CTR0	
Uncore C-Box 12 PerfMon Counter 0		Package
Register Address: E97H, 3735	MSR_C12_PMON_CTR1	
Uncore C-Box 12 PerfMon Counter 1		Package
Register Address: E98H, 3736	MSR_C12_PMON_CTR2	
Uncore C-Box 12 PerfMon Counter 2		Package
Register Address: E99H, 3737	MSR_C12_PMON_CTR3	
Uncore C-Box 12 PerfMon Counter 3		Package
Register Address: E9AH, 3738	MSR_C12_PMON_BOX_FILTER1	
Uncore C-Box 12 PerfMon Box Wide Filter1		Package
Register Address: EA4H, 3748	MSR_C13_PMON_BOX_CTL	
Uncore C-Box 13 PerfMon Local Box Wide Control		Package
Register Address: EB0H, 3760	MSR_C13_PMON_EVNTSELO	
Uncore C-Box 13 PerfMon Event Select for C-Box 13 Counter 0		Package
Register Address: EB1H, 3761	MSR_C13_PMON_EVNTSEL1	
Uncore C-Box 13 PerfMon Event Select for C-Box 13 Counter 1		Package
Register Address: EB2H, 3762	MSR_C13_PMON_EVNTSEL2	
Uncore C-Box 13 PerfMon Event Select for C-Box 13 Counter 2		Package
Register Address: EB3H, 3763	MSR_C13_PMON_EVNTSEL3	
Uncore C-Box 13 PerfMon Event Select for C-Box 13 Counter 3		Package

Table 2-28. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v2 and E7 v2 Families (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: EB4H, 3764	MSR_C13_PMON_BOX_FILTER	
Uncore C-Box 13 PerfMon Box Wide Filter		Package
Register Address: EB6H, 3766	MSR_C13_PMON_CTR0	
Uncore C-Box 13 PerfMon Counter 0		Package
Register Address: EB7H, 3767	MSR_C13_PMON_CTR1	
Uncore C-Box 13 PerfMon Counter 1		Package
Register Address: EB8H, 3768	MSR_C13_PMON_CTR2	
Uncore C-Box 13 PerfMon Counter 2		Package
Register Address: EB9H, 3769	MSR_C13_PMON_CTR3	
Uncore C-Box 13 PerfMon Counter 3		Package
Register Address: EBAH, 3770	MSR_C13_PMON_BOX_FILTER1	
Uncore C-Box 13 PerfMon Box Wide Filter1		Package
Register Address: EC4H, 3780	MSR_C14_PMON_BOX_CTL	
Uncore C-Box 14 PerfMon Local Box Wide Control		Package
Register Address: ED0H, 3792	MSR_C14_PMON_EVNTSELO	
Uncore C-Box 14 PerfMon Event Select for C-Box 14 Counter 0		Package
Register Address: ED1H, 3793	MSR_C14_PMON_EVNTSEL1	
Uncore C-Box 14 PerfMon Event Select for C-Box 14 Counter 1		Package
Register Address: ED2H, 3794	MSR_C14_PMON_EVNTSEL2	
Uncore C-Box 14 PerfMon Event Select for C-Box 14 Counter 2		Package
Register Address: ED3H, 3795	MSR_C14_PMON_EVNTSEL3	
Uncore C-Box 14 PerfMon Event Select for C-Box 14 Counter 3		Package
Register Address: ED4H, 3796	MSR_C14_PMON_BOX_FILTER	
Uncore C-Box 14 PerfMon Box Wide Filter		Package
Register Address: ED6H, 3798	MSR_C14_PMON_CTR0	
Uncore C-Box 14 PerfMon Counter 0		Package
Register Address: ED7H, 3799	MSR_C14_PMON_CTR1	
Uncore C-Box 14 PerfMon Counter 1		Package
Register Address: ED8H, 3800	MSR_C14_PMON_CTR2	
Uncore C-Box 14 PerfMon Counter 2		Package
Register Address: ED9H, 3801	MSR_C14_PMON_CTR3	
Uncore C-Box 14 PerfMon Counter 3		Package
Register Address: EDAH, 3802	MSR_C14_PMON_BOX_FILTER1	
Uncore C-Box 14 PerfMon Box Wide Filter1		Package

2.13 MSRS IN THE 4TH GENERATION INTEL® CORE™ PROCESSORS BASED ON HASWELL MICROARCHITECTURE

The 4th generation Intel® Core™ processor family and the Intel® Xeon® processor E3-1200v3 product family (based on Haswell microarchitecture), with a CPUID Signature DisplayFamily_DisplayModel value of 06_3CH, 06_45H, or 06_46H, support the MSR interfaces listed in Table 2-20, Table 2-21, Table 2-22, and Table 2-29. For an MSR listed in Table 2-20 that also appears in Table 2-29, Table 2-29 supersedes Table 2-20.

The MSRs listed in Table 2-29 also apply to processors based on Haswell-E microarchitecture (see Section 2.14).

Table 2-29. Additional MSRs Supported by Processors Based on the Haswell and Haswell-E Microarchitectures

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 3BH, 59	IA32_TSC_ADJUST	
Per-Logical-Processor TSC ADJUST (R/W) See Table 2-2.		Thread
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) This is the ratio of the frequency that invariant TSC runs at. Frequency = ratio * 100 MHz.	Package
27:16	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) When set to 1, indicates that Programmable Ratio Limit for Turbo mode is enabled. When set to 0, indicates Programmable Ratio Limit for Turbo mode is disabled.	Package
29	Programmable TDP Limit for Turbo Mode (R/O) When set to 1, indicates that TDP Limit for Turbo mode is programmable. When set to 0, indicates TDP Limit for Turbo mode is not programmable.	Package
31:30	Reserved.	
32	Low Power Mode Support (LPM) (R/O) When set to 1, indicates that LPM is supported. When set to 0, indicates LPM is not supported.	Package
34:33	Number of ConfigTDP Levels (R/O) 00: Only Base TDP level available. 01: One additional TDP level available. 02: Two additional TDP level available. 03: Reserved.	Package
39:35	Reserved.	
47:40	Maximum Efficiency Ratio (R/O) This is the minimum ratio (maximum efficiency) that the processor can operate, in units of 100MHz.	Package
55:48	Minimum Operating Ratio (R/O) Contains the minimum supported operating ratio in units of 100 MHz.	Package
63:56	Reserved.	

Table 2-29. Additional MSRs Supported by Processors Based on the Haswell and Haswell-E Microarchitectures

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 186H, 390	IA32_PERFEVTSEL0	
Performance Event Select for Counter 0 (R/W) Supports all fields described in Table 2-2 and the fields below.		Thread
32	IN_TX: See Section 21.3.6.5.1. When IN_TX (bit 32) is set, AnyThread (bit 21) should be cleared to prevent incorrect results.	
Register Address: 187H, 391	IA32_PERFEVTSEL1	
Performance Event Select for Counter 1 (R/W) Supports all fields described in Table 2-2 and the fields below.		Thread
32	IN_TX: See Section 21.3.6.5.1. When IN_TX (bit 32) is set, AnyThread (bit 21) should be cleared to prevent incorrect results.	
Register Address: 188H, 392	IA32_PERFEVTSEL2	
Performance Event Select for Counter 2 (R/W) Supports all fields described in Table 2-2 and the fields below.		Thread
32	IN_TX: See Section 21.3.6.5.1. When IN_TX (bit 32) is set, AnyThread (bit 21) should be cleared to prevent incorrect results.	
33	IN_TXCP: See Section 21.3.6.5.1. When IN_TXCP=1 & IN_TX=1 and in sampling, a spurious PMI may occur and transactions may continuously abort near overflow conditions. Software should favor using IN_TXCP for counting over sampling. If sampling, software should use large "sample-after" value after clearing the counter configured to use IN_TXCP and also always reset the counter even when no overflow condition was reported.	
Register Address: 189H, 393	IA32_PERFEVTSEL3	
Performance Event Select for Counter 3 (R/W) Supports all fields described in Table 2-2 and the fields below.		Thread
32	IN_TX: See Section 21.3.6.5.1 When IN_TX (bit 32) is set, AnyThread (bit 21) should be cleared to prevent incorrect results.	
Register Address: 1C8H, 456	MSR_LBR_SELECT	
Last Branch Record Filtering Select Register (R/W)		Thread
0	CPL_EQ_0	
1	CPL_NEQ_0	
2	JCC	
3	NEAR_REL_CALL	
4	NEAR_IND_CALL	
5	NEAR_RET	
6	NEAR_IND_JMP	
7	NEAR_REL_JMP	

Table 2-29. Additional MSRs Supported by Processors Based on the Haswell and Haswell-E Microarchitectures

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
8	FAR_BRANCH	
9	EN_CALL_STACK	
63:9	Reserved.	
Register Address: 1D9H, 473	IA32_DEBUGCTL	
Debug Control (R/W) See Table 2-2.		Thread
0	LBR: Last Branch Record	
1	BTF	
5:2	Reserved.	
6	TR: Branch Trace	
7	BTS: Log Branch Trace Message to BTS Buffer	
8	BTINT	
9	BTS_OFF_OS	
10	BTS_OFF_USER	
11	FREEZE_LBR_ON_PMI	
12	FREEZE_PERFMON_ON_PMI	
13	ENABLE_UNCORE_PMI	
14	FREEZE_WHILE_SMM	
15	RTM_DEBUG	
63:15	Reserved.	
Register Address: 491H, 1169	IA32_VMX_VMFUNC	
Capability Reporting Register of VM-Function Controls (R/O) See Table 2-2.		Thread
Register Address: 60BH, 1548	MSR_PKG_C_IRT_L1	
Package C6/C7 Interrupt Response Limit 1 (R/W) This MSR defines the interrupt response time limit used by the processor to manage a transition to a package C6 or C7 state. The latency programmed in this register is for the shorter-latency sub C-states used by an MWAIT hint to a C6 or C7 state. Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
9:0	Interrupt Response Time Limit (R/W) Specifies the limit that should be used to decide if the package should be put into a package C6 or C7 state.	
12:10	Time Unit (R/W) Specifies the encoding value of time unit of the interrupt response time limit. See Table 2-20 for supported time unit encodings.	
14:13	Reserved.	
15	Valid (R/W) Indicates whether the values in bits 12:0 are valid and can be used by the processor for package C-state management.	

Table 2-29. Additional MSRs Supported by Processors Based on the Haswell and Haswell-E Microarchitectures

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:16	Reserved.	
Register Address: 60CH, 1548	MSR_PKG_C7_INTERRUPT_RESPONSE_LIMIT_2	
Package C6/C7 Interrupt Response Limit 2 (R/W) This MSR defines the interrupt response time limit used by the processor to manage a transition to a package C6 or C7 state. The latency programmed in this register is for the longer-latency sub C-states used by an MWAIT hint to a C6 or C7 state. Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
9:0	Interrupt response time limit (R/W) Specifies the limit that should be used to decide if the package should be put into a package C6 or C7 state.	
12:10	Time Unit (R/W) Specifies the encoding value of time unit of the interrupt response time limit. See Table 2-20 for supported time unit encodings.	
14:13	Reserved.	
15	Valid (R/W) Indicates whether the values in bits 12:0 are valid and can be used by the processor for package C-state management.	
63:16	Reserved.	
Register Address: 613H, 1555	MSR_PKG_PERF_STATUS	
PKG Perf Status (R/O) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 648H, 1608	MSR_CONFIG_TDP_NOMINAL	
Base TDP Ratio (R/O)		Package
7:0	Config_TDP_Base Base TDP level ratio to be used for this specific processor (in units of 100 MHz).	
63:8	Reserved.	
Register Address: 649H, 1609	MSR_CONFIG_TDP_LEVEL1	
ConfigTDP Level 1 Ratio and Power Level (R/O)		Package
14:0	PKG_TDP_LVL1 Power setting for ConfigTDP Level 1.	
15	Reserved.	
23:16	Config_TDP_LVL1_Ratio ConfigTDP level 1 ratio to be used for this specific processor.	

Table 2-29. Additional MSRs Supported by Processors Based on the Haswell and Haswell-E Microarchitectures

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
31:24	Reserved.	
46:32	PKG_MAX_PWR_LVL1 Max Power setting allowed for ConfigTDP Level 1.	
62:47	PKG_MIN_PWR_LVL1 MIN Power setting allowed for ConfigTDP Level 1.	
63	Reserved.	
Register Address: 64AH, 1610	MSR_CONFIG_TDP_LEVEL2	
ConfigTDP Level 2 Ratio and Power Level (R/O)		Package
14:0	PKG_TDP_LVL2 Power setting for ConfigTDP Level 2.	
15	Reserved.	
23:16	Config_TDP_LVL2_Ratio ConfigTDP level 2 ratio to be used for this specific processor.	
31:24	Reserved.	
46:32	PKG_MAX_PWR_LVL2 Max Power setting allowed for ConfigTDP Level 2.	
62:47	PKG_MIN_PWR_LVL2 MIN Power setting allowed for ConfigTDP Level 2.	
63	Reserved.	
Register Address: 64BH, 1611	MSR_CONFIG_TDP_CONTROL	
ConfigTDP Control (R/W)		Package
1:0	TDP_LEVEL (RW/L) System BIOS can program this field.	
30:2	Reserved.	
31	Config_TDP_Lock (RW/L) When this bit is set, the content of this register is locked until a reset.	
63:32	Reserved.	
Register Address: 64CH, 1612	MSR_TURBO_ACTIVATION_RATIO	
ConfigTDP Control (R/W)		Package
7:0	MAX_NON_TURBO_RATIO (RW/L) System BIOS can program this field.	
30:8	Reserved.	
31	TURBO_ACTIVATION_RATIO_Lock (RW/L) When this bit is set, the content of this register is locked until a reset.	
63:32	Reserved.	
Register Address: C80H, 3200	IA32_DEBUG_INTERFACE	
Silicon Debug Feature Control (R/W) See Table 2-2.		Package

2.13.1 MSRs in the 4th Generation Intel® Core™ Processor Family Based on Haswell Microarchitecture

Table 2-30 lists model-specific registers (MSRs) that are specific to the 4th generation Intel® Core™ processor family and the Intel® Xeon® processor E3-1200 v3 product family (based on Haswell microarchitecture). These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_3CH, 06_45H, or 06_46H; see Table 2-1.

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states. See http://biosbits.org .		Core
3:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 0000b: C0/C1 (no package C-state support) 0001b: C2 0010b: C3 0011b: C6 0100b: C7 0101b: C7s Package C states C7 are not available to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_3CH.	
9:4	Reserved.	
10	I/O MWAIT Redirection Enable (R/W)	
14:11	Reserved	
15	CFG Lock (R/WO)	
24:16	Reserved.	
25	C3 State Auto Demotion Enable (R/W)	
26	C1 State Auto Demotion Enable (R/W)	
27	Enable C3 Undemotion (R/W)	
28	Enable C1 Undemotion (R/W)	
63:29	Reserved.	
Register Address: 17DH, 381	MSR_SMM_MCA_CAP	
Enhanced SMM Capabilities (SMM-RO) Reports SMM capability Enhancement. Accessible only while in SMM.		Thread
57:0	Reserved.	
58	SMM_Code_Access_Chk (SMM-RO) If set to 1, indicates that the SMM code access restriction is supported and the MSR_SMM_FEATURE_CONTROL is supported.	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
59	Long_Flow_Indication (SMM-RO) If set to 1, indicates that the SMM long flow indicator is supported and the MSR_SMM_DELAYED is supported.	
63:60	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 core active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 core active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 core active.	Package
63:32	Reserved.	
Register Address: 391H, 913	MSR_UNC_PERF_GLOBAL_CTRL	
Uncore PMU Global Control		Package
0	Core 0 select.	
1	Core 1 select.	
2	Core 2 select.	
3	Core 3 select.	
18:4	Reserved.	
29	Enable all uncore counters.	
30	Enable wake on PMI.	
31	Enable Freezing counter when overflow.	
63:32	Reserved.	
Register Address: 392H, 914	MSR_UNC_PERF_GLOBAL_STATUS	
Uncore PMU Main Status		Package
0	Fixed counter overflowed.	
1	An ARB counter overflowed.	
2	Reserved.	
3	A CBox counter overflowed (on any slice).	
63:4	Reserved.	
Register Address: 394H, 916	MSR_UNC_PERF_FIXED_CTRL	
Uncore Fixed Counter Control (R/W)		Package
19:0	Reserved.	
20	Enable overflow propagation.	
21	Reserved.	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
22	Enable counting.	
63:23	Reserved.	
Register Address: 395H, 917	MSR_UNC_PERF_FIXED_CTR	
Uncore Fixed Counter		Package
47:0	Current count.	
63:48	Reserved.	
Register Address: 396H, 918	MSR_UNC_CBO_CONFIG	
Uncore C-Box Configuration Information (R/O)		Package
3:0	Encoded number of C-Box, derive value by "-1".	
63:4	Reserved.	
Register Address: 3B0H, 946	MSR_UNC_ARB_PERFCTR0	
Uncore Arb Unit, Performance Counter 0		Package
Register Address: 3B1H, 947	MSR_UNC_ARB_PERFCTR1	
Uncore Arb Unit, Performance Counter 1		Package
Register Address: 3B2H, 944	MSR_UNC_ARB_PERFEVTSELO	
Uncore Arb Unit, Counter 0 Event Select MSR		Package
Register Address: 3B3H, 945	MSR_UNC_ARB_PERFEVTSEL1	
Uncore Arb Unit, Counter 1 Event Select MSR		Package
Register Address: 4E0H, 1248	MSR_SMM_FEATURE_CONTROL	
Enhanced SMM Feature Control (SMM-RW) Reports SMM capability Enhancement. Accessible only while in SMM.		Package
0	Lock (SMM-RW0) When set to '1' locks this register from further changes.	
1	Reserved.	
2	SMM_Code_Chk_En (SMM-RW) This control bit is available only if MSR_SMM_MCA_CAP[58] == 1. When set to '0' (default) none of the logical processors are prevented from executing SMM code outside the ranges defined by the SMRR. When set to '1' any logical processor in the package that attempts to execute SMM code not within the ranges defined by the SMRR will assert an unrecoverable MCE.	
63:3	Reserved.	
Register Address: 4E2H, 1250	MSR_SMM_DELAYED	
SMM Delayed (SMM-RO) Reports the interruptible state of all logical processors in the package. Available only while in SMM and MSR_SMM_MCA_CAP[LONG_FLOW_INDICATION] == 1.		Package

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
N-1:0	LOG_PROC_STATE (SMM-RO) Each bit represents a logical processor of its state in a long flow of internal operation which delays servicing an interrupt. The corresponding bit will be set at the start of long events such as: Microcode Update Load, C6, WBINVD, Ratio Change, Throttle. The bit is automatically cleared at the end of each long event. The reset value of this field is 0. Only bit positions below N = CPUID.(EAX=0BH, ECX=PKG_LVL):EBX[15:0] can be updated.	
63:N	Reserved.	
Register Address: 4E3H, 1251	MSR_SMM_BLOCKED	
SMM Blocked (SMM-RO) Reports the blocked state of all logical processors in the package. Available only while in SMM.		Package
N-1:0	LOG_PROC_STATE (SMM-RO) Each bit represents a logical processor of its blocked state to service an SMI. The corresponding bit will be set if the logical processor is in one of the following states: Wait For SIPI or SENTER Sleep. The reset value of this field is 0FFFH. Only bit positions below N = CPUID.(EAX=0BH, ECX=PKG_LVL):EBX[15:0] can be updated.	
63:N	Reserved.	
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
Unit Multipliers Used in RAPL Interfaces (R/O)		Package
3:0	Power Units See Section 16.10.1, "RAPL Interfaces."	Package
7:4	Reserved.	Package
12:8	Energy Status Units Energy related information (in Joules) is based on the multiplier, $1/2^{\text{ESU}}$; where ESU is an unsigned integer represented by bits 12:8. Default value is 0EH (or 61 micro-joules).	Package
15:13	Reserved.	Package
19:16	Time Units See Section 16.10.1, "RAPL Interfaces."	Package
63:20	Reserved.	
Register Address: 639H, 1593	MSR_PP0_ENERGY_STATUS	
PP0 Energy Status (R/O) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 640H, 1600	MSR_PP1_POWER_LIMIT	
PP1 RAPL Power Limit Control (R/W) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 641H, 1601	MSR_PP1_ENERGY_STATUS	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
PP1 Energy Status (R/O) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 642H, 1602	MSR_PP1_POLICY	
PP1 Balance Policy (R/W) See Section 16.10.4, "PP0/PP1 RAPL Domains."		Package
Register Address: 690H, 1680	MSR_CORE_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in Processor Cores (R/W) (Frequency refers to processor core frequency.)		Package
0	PROCHOT Status (R0) When set, processor core frequency is reduced below the operating system request due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced below the operating system request due to a thermal event.	
3:2	Reserved.	
4	Graphics Driver Status (R0) When set, frequency is reduced below the operating system request due to Processor Graphics driver override.	
5	Autonomous Utilization-Based Frequency Control Status (R0) When set, frequency is reduced below the operating system request because the processor has detected that utilization is low.	
6	VR Therm Alert Status (R0) When set, frequency is reduced below the operating system request due to a thermal alert from the Voltage Regulator.	
7	Reserved.	
8	Electrical Design Point Status (R0) When set, frequency is reduced below the operating system request due to electrical design point constraints (e.g., maximum electrical current consumption).	
9	Core Power Limiting Status (R0) When set, frequency is reduced below the operating system request due to domain-level power limiting.	
10	Package-Level Power Limiting PL1 Status (R0) When set, frequency is reduced below the operating system request due to package-level power limiting PL1.	
11	Package-Level PL2 Power Limiting Status (R0) When set, frequency is reduced below the operating system request due to package-level power limiting PL2.	
12	Max Turbo Limit Status (R0) When set, frequency is reduced below the operating system request due to multi-core turbo limits.	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
13	Turbo Transition Attenuation Status (R0) When set, frequency is reduced below the operating system request due to Turbo transition attenuation. This prevents performance degradation due to frequent operating ratio changes.	
15:14	Reserved.	
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
19:18	Reserved.	
20	Graphics Driver Log When set, indicates that the Graphics Driver Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
21	Autonomous Utilization-Based Frequency Control Log When set, indicates that the Autonomous Utilization-Based Frequency Control Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	Reserved.	
24	Electrical Design Point Log When set, indicates that the EDP Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Core Power Limiting Log When set, indicates that the Core Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
26	Package-Level PL1 Power Limiting Log When set, indicates that the Package Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
27	Package-Level PL2 Power Limiting Log When set, indicates that the Package Level PL2 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28	Max Turbo Limit Log When set, indicates that the Max Turbo Limit Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
29	Turbo Transition Attenuation Log When set, indicates that the Turbo Transition Attenuation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:30	Reserved.	
Register Address: 6B0H, 1712	MSR_GRAPHICS_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in the Processor Graphics (R/W) (Frequency refers to processor graphics frequency.)		Package
0	PROCHOT Status (R0) When set, frequency is reduced below the operating system request due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced below the operating system request due to a thermal event.	
3:2	Reserved.	
4	Graphics Driver Status (R0) When set, frequency is reduced below the operating system request due to Processor Graphics driver override.	
5	Autonomous Utilization-Based Frequency Control Status (R0) When set, frequency is reduced below the operating system request because the processor has detected that utilization is low.	
6	VR Therm Alert Status (R0) When set, frequency is reduced below the operating system request due to a thermal alert from the Voltage Regulator.	
7	Reserved.	
8	Electrical Design Point Status (R0) When set, frequency is reduced below the operating system request due to electrical design point constraints (e.g., maximum electrical current consumption).	
9	Graphics Power Limiting Status (R0) When set, frequency is reduced below the operating system request due to domain-level power limiting.	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
10	Package-Level Power Limiting PL1 Status (R0) When set, frequency is reduced below the operating system request due to package-level power limiting PL1.	
11	Package-Level PL2 Power Limiting Status (R0) When set, frequency is reduced below the operating system request due to package-level power limiting PL2.	
15:12	Reserved.	
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
19:18	Reserved.	
20	Graphics Driver Log When set, indicates that the Graphics Driver Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
21	Autonomous Utilization-Based Frequency Control Log When set, indicates that the Autonomous Utilization-Based Frequency Control Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	Reserved.	
24	Electrical Design Point Log When set, indicates that the EDP Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Core Power Limiting Log When set, indicates that the Core Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
26	Package-Level PL1 Power Limiting Log When set, indicates that the Package Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
27	Package-Level PL2 Power Limiting Log When set, indicates that the Package Level PL2 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28	Max Turbo Limit Log When set, indicates that the Max Turbo Limit Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
29	Turbo Transition Attenuation Log When set, indicates that the Turbo Transition Attenuation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:30	Reserved.	
Register Address: 6B1H, 1713	MSR_RING_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in the Ring Interconnect (R/W) (Frequency refers to ring interconnect in the uncore.)		Package
0	PROCHOT Status (R0) When set, frequency is reduced below the operating system request due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced below the operating system request due to a thermal event.	
5:2	Reserved.	
6	VR Therm Alert Status (R0) When set, frequency is reduced below the operating system request due to a thermal alert from the Voltage Regulator.	
7	Reserved.	
8	Electrical Design Point Status (R0) When set, frequency is reduced below the operating system request due to electrical design point constraints (e.g., maximum electrical current consumption).	
9	Reserved.	
10	Package-Level Power Limiting PL1 Status (R0) When set, frequency is reduced below the operating system request due to package-level power limiting PL1.	
11	Package-Level PL2 Power Limiting Status (R0) When set, frequency is reduced below the operating system request due to package-level power limiting PL2.	
15:12	Reserved.	
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
19:18	Reserved.	
20	Graphics Driver Log When set, indicates that the Graphics Driver Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
21	Autonomous Utilization-Based Frequency Control Log When set, indicates that the Autonomous Utilization-Based Frequency Control Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	Reserved.	
24	Electrical Design Point Log When set, indicates that the EDP Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Core Power Limiting Log When set, indicates that the Core Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
26	Package-Level PL1 Power Limiting Log When set, indicates that the Package Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
27	Package-Level PL2 Power Limiting Log When set, indicates that the Package Level PL2 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28	Max Turbo Limit Log When set, indicates that the Max Turbo Limit Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
29	Turbo Transition Attenuation Log When set, indicates that the Turbo Transition Attenuation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	

Table 2-30. MSRs Supported by 4th Generation Intel® Core™ Processors (Haswell Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:30	Reserved.	
Register Address: 700H, 1792	MSR_UNC_CBO_0_PERFEVTSELO	
Uncore C-Box 0, Counter 0 Event Select MSR		Package
Register Address: 701H, 1793	MSR_UNC_CBO_0_PERFEVTSEL1	
Uncore C-Box 0, Counter 1 Event Select MSR		Package
Register Address: 706H, 1798	MSR_UNC_CBO_0_PERFCTR0	
Uncore C-Box 0, Performance Counter 0		Package
Register Address: 707H, 1799	MSR_UNC_CBO_0_PERFCTR1	
Uncore C-Box 0, Performance Counter 1		Package
Register Address: 710H, 1808	MSR_UNC_CBO_1_PERFEVTSELO	
Uncore C-Box 1, Counter 0 Event Select MSR		Package
Register Address: 711H, 1809	MSR_UNC_CBO_1_PERFEVTSEL1	
Uncore C-Box 1, Counter 1 Event Select MSR		Package
Register Address: 716H, 1814	MSR_UNC_CBO_1_PERFCTR0	
Uncore C-Box 1, Performance Counter 0		Package
Register Address: 717H, 1815	MSR_UNC_CBO_1_PERFCTR1	
Uncore C-Box 1, Performance Counter 1		Package
Register Address: 720H, 1824	MSR_UNC_CBO_2_PERFEVTSELO	
Uncore C-Box 2, Counter 0 Event Select MSR		Package
Register Address: 721H, 1824	MSR_UNC_CBO_2_PERFEVTSEL1	
Uncore C-Box 2, Counter 1 Event Select MSR		Package
Register Address: 726H, 1830	MSR_UNC_CBO_2_PERFCTR0	
Uncore C-Box 2, Performance Counter 0		Package
Register Address: 727H, 1831	MSR_UNC_CBO_2_PERFCTR1	
Uncore C-Box 2, Performance Counter 1		Package
Register Address: 730H, 1840	MSR_UNC_CBO_3_PERFEVTSELO	
Uncore C-Box 3, Counter 0 Event Select MSR		Package
Register Address: 731H, 1841	MSR_UNC_CBO_3_PERFEVTSEL1	
Uncore C-Box 3, Counter 1 Event Select MSR		Package
Register Address: 736H, 1846	MSR_UNC_CBO_3_PERFCTR0	
Uncore C-Box 3, Performance Counter 0		Package
Register Address: 737H, 1847	MSR_UNC_CBO_3_PERFCTR1	
Uncore C-Box 3, Performance Counter 1		Package
See Table 2-20, Table 2-21, Table 2-22, Table 2-25, and Table 2-29 for other MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 063CH or 06_46H.		

2.13.2 Additional Residency MSRs Supported in 4th Generation Intel® Core™ Processors

The 4th generation Intel® Core™ processor family (based on Haswell microarchitecture) with a CPUID Signature DisplayFamily_DisplayModel value of 06_45H supports the MSR interfaces listed in Table 2-20, Table 2-21, Table 2-29, Table 2-30, and Table 2-31.

Table 2-31. Additional Residency MSRs Supported by 4th Generation Intel® Core™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_45H

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states. See http://biosbits.org .		Core
3:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 0000b: C0/C1 (no package C-state support) 0001b: C2 0010b: C3 0011b: C6 0100b: C7 0101b: C7s 0110b: C8 0111b: C9 1000b: C10	
9:4	Reserved.	
10	I/O MWAIT Redirection Enable (R/W)	
14:11	Reserved.	
15	CFG Lock (R/W0)	
24:16	Reserved.	
25	C3 State Auto Demotion Enable (R/W)	
26	C1 State Auto Demotion Enable (R/W)	
27	Enable C3 Undemotion (R/W)	
28	Enable C1 Undemotion (R/W)	
63:29	Reserved.	
Register Address: 630H, 1584	MSR_PKG_C8_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
59:0	Package C8 Residency Counter (R/O) Value since last reset that this package is in processor-specific C8 states. Count at the same frequency as the TSC.	
63:60	Reserved.	
Register Address: 631H, 1585	MSR_PKG_C9_RESIDENCY	

Table 2-31. Additional Residency MSRs Supported by 4th Generation Intel® Core™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_45H

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
59:0	Package C9 Residency Counter (R/O) Value since last reset that this package is in processor-specific C9 states. Count at the same frequency as the TSC.	
63:60	Reserved.	
Register Address: 632H, 1586	MSR_PKG_C10_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
59:0	Package C10 Residency Counter (R/O) Value since last reset that this package is in processor-specific C10 states. Count at the same frequency as the TSC.	
63:60	Reserved.	
See Table 2-20, Table 2-21, Table 2-22, Table 2-29, and Table 2-30 for other MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_45H.		

2.14 MSRS IN THE INTEL® XEON® PROCESSOR E5 V3 AND E7 V3 PRODUCT FAMILY

The Intel® Xeon® processor E5 v3 family and the Intel® Xeon® processor E7 v3 family are based on Haswell-E microarchitecture (CPUID Signature DisplayFamily_DisplayModel value of 06_3F). These processors support the MSR interfaces listed in Table 2-20, Table 2-29, and Table 2-32.

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 35H, 53	MSR_CORE_THREAD_COUNT	
Configured State of Enabled Processor Core Count and Logical Processor Count (R/O)		Package
- After a Power-On RESET, enumerates factory configuration of the number of processor cores and logical processors in the physical package. - Following the sequence of (i) BIOS modified a Configuration Mask which selects a subset of processor cores to be active post RESET and (ii) a RESET event after the modification, enumerates the current configuration of enabled processor core count and logical processor count in the physical package.		
15:0	THREAD_COUNT (R/O) The number of logical processors that are currently enabled (by either factory configuration or BIOS configuration) in the physical package.	
31:16	Core_COUNT (R/O) The number of processor cores that are currently enabled (by either factory configuration or BIOS configuration) in the physical package.	
63:32	Reserved.	
Register Address: 53H, 83	MSR_THREAD_ID_INFO	
A Hardware Assigned ID for the Logical Processor (R/O)		Thread

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
7:0	Logical_Processor_ID (R/O) An implementation-specific numerical value physically assigned to each logical processor. This ID is not related to Initial APIC ID or x2APIC ID, it is unique within a physical package.	
63:8	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states. See http://biosbits.org .		Core
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: C0/C1 (no package C-state support) 001b: C2 010b: C6 (non-retention) 011b: C6 (retention) 111b: No Package C state limits. All C states supported by the processor are available.	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W)	
14:11	Reserved.	
15	CFG Lock (R/WO)	
24:16	Reserved.	
25	C3 State Auto Demotion Enable (R/W)	
26	C1 State Auto Demotion Enable (R/W)	
27	Enable C3 Undemotion (R/W)	
28	Enable C1 Undemotion (R/W)	
29	Package C State Demotion Enable (R/W)	
30	Package C State Undemotion Enable (R/W)	
63:31	Reserved.	
Register Address: 179H, 377	IA32_MCG_CAP	
Global Machine Check Capability (R/O)		Thread
7:0	Count	
8	MCG_CTL_P	
9	MCG_EXT_P	
10	MCP_CMCI_P	
11	MCG_TES_P	
15:12	Reserved.	

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
23:16	MCG_EXT_CNT	
24	MCG_SER_P	
25	MCG_EM_P	
26	MCG_ELOG_P	
63:27	Reserved.	
Register Address: 17DH, 381	MSR_SMM_MCA_CAP	
Enhanced SMM Capabilities (SMM-RO) Reports SMM capability Enhancement. Accessible only while in SMM.		Thread
57:0	Reserved.	
58	SMM_Code_Access_Chk (SMM-RO) If set to 1, indicates that the SMM code access restriction is supported and a host-space interface available to SMM handler.	
59	Long_Flow_Indication (SMM-RO) If set to 1, indicates that the SMM long flow indicator is supported and a host-space interface available to SMM handler.	
63:60	Reserved.	
Register Address: 17FH, 383	MSR_ERROR_CONTROL	
MC Bank Error Configuration (R/W)		Package
0	Reserved.	
1	MemError Log Enable (R/W) When set, enables IMC status bank to log additional info in bits 36:32.	
63:2	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 core active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 core active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 core active.	Package
39:32	Maximum Ratio Limit for 5C Maximum turbo ratio limit of 5 core active.	Package
47:40	Maximum Ratio Limit for 6C Maximum turbo ratio limit of 6 core active.	Package
55:48	Maximum Ratio Limit for 7C Maximum turbo ratio limit of 7 core active.	Package

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:56	Maximum Ratio Limit for 8C Maximum turbo ratio limit of 8 core active.	Package
Register Address: 1AEH, 430	MSR_TURBO_RATIO_LIMIT1	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 9C Maximum turbo ratio limit of 9 core active.	Package
15:8	Maximum Ratio Limit for 10C Maximum turbo ratio limit of 10 core active.	Package
23:16	Maximum Ratio Limit for 11C Maximum turbo ratio limit of 11 core active.	Package
31:24	Maximum Ratio Limit for 12C Maximum turbo ratio limit of 12 core active.	Package
39:32	Maximum Ratio Limit for 13C Maximum turbo ratio limit of 13 core active.	Package
47:40	Maximum Ratio Limit for 14C Maximum turbo ratio limit of 14 core active.	Package
55:48	Maximum Ratio Limit for 15C Maximum turbo ratio limit of 15 core active.	Package
63:56	Maximum Ratio Limit for 16C Maximum turbo ratio limit of 16 core active.	Package
Register Address: 1AFH, 431	MSR_TURBO_RATIO_LIMIT2	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 17C Maximum turbo ratio limit of 17 core active.	Package
15:8	Maximum Ratio Limit for 18C Maximum turbo ratio limit of 18 core active.	Package
62:16	Reserved.	Package
63	Semaphore for Turbo Ratio Limit Configuration If 1, the processor uses override configuration ¹ specified in MSR_TURBO_RATIO_LIMIT, MSR_TURBO_RATIO_LIMIT1, and MSR_TURBO_RATIO_LIMIT2. If 0, the processor uses factory-set configuration (Default).	Package
Register Address: 414H, 1044	IA32_MC5_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI 0 module.		Package
Register Address: 415H, 1045	IA32_MC5_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI 0 module.		Package

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 416H, 1046	IA32_MC5_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI 0 module.		Package
Register Address: 417H, 1047	IA32_MC5_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI 0 module.		Package
Register Address: 418H, 1048	IA32_MC6_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 419H, 1049	IA32_MC6_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41AH, 1050	IA32_MC6_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41BH, 1051	IA32_MC6_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41CH, 1052	IA32_MC7_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41DH, 1053	IA32_MC7_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41EH, 1054	IA32_MC7_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41FH, 1055	IA32_MC7_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 420H, 1056	IA32_MC8_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC8 reports MC errors from the home agent HA 1.		Package
Register Address: 421H, 1057	IA32_MC8_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC8 reports MC errors from the home agent HA 1.		Package
Register Address: 422H, 1058	IA32_MC8_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC8 reports MC errors from the home agent HA 1.		Package

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 423H, 1059	IA32_MC8_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the home agent HA 1.		Package
Register Address: 424H, 1060	IA32_MC9_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 425H, 1061	IA32_MC9_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 426H, 1062	IA32_MC9_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 427H, 1063	IA32_MC9_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 428H, 1064	IA32_MC10_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 429H, 1065	IA32_MC10_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42AH, 1066	IA32_MC10_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42BH, 1067	IA32_MC10_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42CH, 1068	IA32_MC11_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42DH, 1069	IA32_MC11_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42EH, 1070	IA32_MC11_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42FH, 1071	IA32_MC11_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 430H, 1072	IA32_MC12_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 431H, 1073	IA32_MC12_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 432H, 1074	IA32_MC12_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 433H, 1075	IA32_MC12_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 434H, 1076	IA32_MC13_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 435H, 1077	IA32_MC13_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 436H, 1078	IA32_MC13_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 437H, 1079	IA32_MC13_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 438H, 1080	IA32_MC14_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 439H, 1081	IA32_MC14_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43AH, 1082	IA32_MC14_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43BH, 1083	IA32_MC14_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43CH, 1084	IA32_MC15_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 43DH, 1085	IA32_MC15_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 43EH, 1086	IA32_MC15_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 43FH, 1087	IA32_MC15_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 440H, 1088	IA32_MC16_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 441H, 1089	IA32_MC16_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 442H, 1090	IA32_MC16_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 443H, 1091	IA32_MC16_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 444H, 1092	IA32_MC17_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 445H, 1093	IA32_MC17_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 446H, 1094	IA32_MC17_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 447H, 1095	IA32_MC17_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 448H, 1096	IA32_MC18_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.	Package	

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 449H, 1097	IA32_MC18_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.		Package
Register Address: 44AH, 1098	IA32_MC18_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.		Package
Register Address: 44BH, 1099	IA32_MC18_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.		Package
Register Address: 44CH, 1100	IA32_MC19_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 44DH, 1101	IA32_MC19_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 44EH, 1102	IA32_MC19_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 44FH, 1103	IA32_MC19_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 450H, 1104	IA32_MC20_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC20 reports MC errors from the Intel QPI 1 module.		Package
Register Address: 451H, 1105	IA32_MC20_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC20 reports MC errors from the Intel QPI 1 module.		Package
Register Address: 452H, 1106	IA32_MC20_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC20 reports MC errors from the Intel QPI 1 module.		Package
Register Address: 453H, 1107	IA32_MC20_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC20 reports MC errors from the Intel QPI 1 module.		Package
Register Address: 454H, 1108	IA32_MC21_CTL	

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from the Intel QPI 2 module.		Package
Register Address: 455H, 1109	IA32_MC21_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from the Intel QPI 2 module.		Package
Register Address: 456H, 1110	IA32_MC21_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from the Intel QPI 2 module.		Package
Register Address: 457H, 1111	IA32_MC21_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from the Intel QPI 2 module.		Package
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
Unit Multipliers Used in RAPL Interfaces (R/O)		Package
3:0	Power Units See Section 16.10.1, "RAPL Interfaces."	Package
7:4	Reserved.	Package
12:8	Energy Status Units Energy related information (in Joules) is based on the multiplier, $1/2^{\text{ESU}}$; where ESU is an unsigned integer represented by bits 12:8. Default value is 0EH (or 61 micro-joules).	Package
15:13	Reserved.	Package
19:16	Time Units See Section 16.10.1, "RAPL Interfaces."	Package
63:20	Reserved.	
Register Address: 618H, 1560	MSR_DRAM_POWER_LIMIT	
DRAM RAPL Power Limit Control (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) Energy Consumed by DRAM devices.		Package
31:0	Energy in 15.3 micro-joules. Requires BIOS configuration to enable DRAM RAPL mode 0 (Direct VR).	
63:32	Reserved.	
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61CH, 1564	MSR_DRAM_POWER_INFO	
DRAM RAPL Parameters (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61EH, 1566	MSR_PCIE_PLL_RATIO	

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Configuration of PCIE PLL Relative to BCLK(R/W)		Package
1:0	PCIE Ratio (R/W) 00b: Use 5:5 mapping for 100MHz operation (default). 01b: Use 5:4 mapping for 125MHz operation. 10b: Use 5:3 mapping for 166MHz operation. 11b: Use 5:2 mapping for 250MHz operation.	Package
2	LPLL Select (R/W) If 1, use configured setting of PCIE Ratio.	Package
3	LONG RESET (R/W) If 1, wait an additional time-out before re-locking Gen2/Gen3 PLLs.	Package
63:4	Reserved.	
Register Address: 620H, 1568	MSR_UNCORE_RATIO_LIMIT	
Uncore Ratio Limit (R/W) Out of reset, the min_ratio and max_ratio fields represent the widest possible range of uncore frequencies. Writing to these fields allows software to control the minimum and the maximum frequency that hardware will select.		Package
6:0	MAX_RATIO This field is used to limit the max ratio of the LLC/Ring.	
7	Reserved.	
14:8	MIN_RATIO Writing to this field controls the minimum possible ratio of the LLC/Ring.	
63:15	Reserved.	
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	
Reserved (R/O) Reads return 0.		Package
Register Address: 690H, 1680	MSR_CORE_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in Processor Cores (R/W) (Frequency refers to processor core frequency.)		Package
0	PROCHOT Status (R0) When set, processor core frequency is reduced below the operating system request due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced below the operating system request due to a thermal event.	
2	Power Budget Management Status (R0) When set, frequency is reduced below the operating system request due to PBM limit	
3	Platform Configuration Services Status (R0) When set, frequency is reduced below the operating system request due to PCS limit	
4	Reserved.	

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
5	Autonomous Utilization-Based Frequency Control Status (R0) When set, frequency is reduced below the operating system request because the processor has detected that utilization is low.	
6	VR Therm Alert Status (R0) When set, frequency is reduced below the operating system request due to a thermal alert from the Voltage Regulator.	
7	Reserved.	
8	Electrical Design Point Status (R0) When set, frequency is reduced below the operating system request due to electrical design point constraints (e.g., maximum electrical current consumption).	
9	Reserved.	
10	Multi-Core Turbo Status (R0) When set, frequency is reduced below the operating system request due to Multi-Core Turbo limits.	
12:11	Reserved.	
13	Core Frequency P1 Status (R0) When set, frequency is reduced below max non-turbo P1.	
14	Core Max N-Core Turbo Frequency Limiting Status (R0) When set, frequency is reduced below max n-core turbo frequency.	
15	Core Frequency Limiting Status (R0) When set, frequency is reduced below the operating system request.	
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
18	Power Budget Management Log When set, indicates that the PBM Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
19	Platform Configuration Services Log When set, indicates that the PCS Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
20	Reserved.	
21	Autonomous Utilization-Based Frequency Control Log When set, indicates that the AUBFC Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
22	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	Reserved.	
24	Electrical Design Point Log When set, indicates that the EDP Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Reserved.	
26	Multi-Core Turbo Log When set, indicates that the Multi-Core Turbo Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28:27	Reserved.	
29	Core Frequency P1 Log When set, indicates that the Core Frequency P1 Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
30	Core Max N-Core Turbo Frequency Limiting Log When set, indicates that the Core Max n-core Turbo Frequency Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
31	Core Frequency Limiting Log When set, indicates that the Core Frequency Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:32	Reserved.	
Register Address: C8DH, 3213	IA32_QM_EVTSEL	
Monitoring Event Select Register (R/W) If CPUID.(EAX=07H, ECX=0):EBX.RDT-M[bit 12] = 1.		Thread
7:0	EventID (R/W) Event encoding: 0x0: No monitoring. 0x1: L3 occupancy monitoring. All other encoding reserved.	
31:8	Reserved.	
41:32	RMID (R/W)	
63:42	Reserved.	
Register Address: C8EH, 3214	IA32_QM_CTR	
Monitoring Counter Register (R/O) If CPUID.(EAX=07H, ECX=0):EBX.RDT-M[bit 12] = 1.		Thread

Table 2-32. Additional MSRs Supported by the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
61:0	Resource Monitored Data	
62	Unavailable: If 1, indicates data for this RMID is not available or not monitored for this resource or RMID.	
63	Error: If 1, indicates an unsupported RMID or event type was written to IA32_PQR_QM_EVTSEL.	
Register Address: C8FH, 3215	IA32_PQR_ASSOC	
Resource Association Register (R/W)		Thread
9:0	RMID	
63: 10	Reserved.	
See Table 2-20 and Table 2-29 for other MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_3FH.		

NOTES:

1. An override configuration lower than the factory-set configuration is always supported. An override configuration higher than the factory-set configuration is dependent on features specific to the processor and the platform.

2.14.1 Additional Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family

The Intel Xeon Processor E5 v3 and E7 v3 families are based on Haswell-E microarchitecture. The MSR-based uncore PMU interfaces are listed in Table 2-33. For complete details of the uncore PMU, refer to the Intel Xeon Processor E5 v3 Product Family Uncore Performance Monitoring Guide. These processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_3FH.

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 700H, 1792	MSR_PMON_GLOBAL_CTL	
Uncore PerfMon Per-Socket Global Control		Package
Register Address: 701H, 1793	MSR_PMON_GLOBAL_STATUS	
Uncore PerfMon Per-Socket Global Status		Package
Register Address: 702H, 1794	MSR_PMON_GLOBAL_CONFIG	
Uncore PerfMon Per-Socket Global Configuration		Package
Register Address: 703H, 1795	MSR_U_PMON_UCLK_FIXED_CTL	
Uncore U-Box UCLK Fixed Counter Control		Package
Register Address: 704H, 1796	MSR_U_PMON_UCLK_FIXED_CTR	
Uncore U-Box UCLK Fixed Counter		Package
Register Address: 705H, 1797	MSR_U_PMON_EVNTSELO	
Uncore U-Box PerfMon Event Select for U-Box Counter 0		Package
Register Address: 706H, 1798	MSR_U_PMON_EVNTSEL1	
Uncore U-Box PerfMon Event Select for U-Box Counter 1		Package
Register Address: 708H, 1800	MSR_U_PMON_BOX_STATUS	

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore U-Box PerfMon U-Box Wide Status		Package
Register Address: 709H, 1801	MSR_U_PMON_CTRL0	
Uncore U-Box PerfMon Counter 0		Package
Register Address: 70AH, 1802	MSR_U_PMON_CTRL1	
Uncore U-Box PerfMon Counter 1		Package
Register Address: 710H, 1808	MSR_PCU_PMON_BOX_CTL	
Uncore PCU PerfMon for PCU-Box-Wide Control		Package
Register Address: 711H, 1809	MSR_PCU_PMON_EVTSEL0	
Uncore PCU PerfMon Event Select for PCU Counter 0		Package
Register Address: 712H, 1810	MSR_PCU_PMON_EVTSEL1	
Uncore PCU PerfMon Event Select for PCU Counter 1		Package
Register Address: 713H, 1811	MSR_PCU_PMON_EVTSEL2	
Uncore PCU PerfMon Event Select for PCU Counter 2		Package
Register Address: 714H, 1812	MSR_PCU_PMON_EVTSEL3	
Uncore PCU PerfMon Event Select for PCU Counter 3		Package
Register Address: 715H, 1813	MSR_PCU_PMON_BOX_FILTER	
Uncore PCU PerfMon Box-Wide Filter		Package
Register Address: 716H, 1814	MSR_PCU_PMON_BOX_STATUS	
Uncore PCU PerfMon Box Wide Status		Package
Register Address: 717H, 1815	MSR_PCU_PMON_CTRL0	
Uncore PCU PerfMon Counter 0		Package
Register Address: 718H, 1816	MSR_PCU_PMON_CTRL1	
Uncore PCU PerfMon Counter 1		Package
Register Address: 719H, 1817	MSR_PCU_PMON_CTRL2	
Uncore PCU PerfMon Counter 2		Package
Register Address: 71AH, 1818	MSR_PCU_PMON_CTRL3	
Uncore PCU PerfMon Counter 3		Package
Register Address: 720H, 1824	MSR_S0_PMON_BOX_CTL	
Uncore SBo 0 PerfMon for SBo 0 Box-Wide Control		Package
Register Address: 721H, 1825	MSR_S0_PMON_EVTSEL0	
Uncore SBo 0 PerfMon Event Select for SBo 0 Counter 0		Package
Register Address: 722H, 1826	MSR_S0_PMON_EVTSEL1	
Uncore SBo 0 PerfMon Event Select for SBo 0 Counter 1		Package
Register Address: 723H, 1827	MSR_S0_PMON_EVTSEL2	
Uncore SBo 0 PerfMon Event Select for SBo 0 Counter 2		Package
Register Address: 724H, 1828	MSR_S0_PMON_EVTSEL3	
Uncore SBo 0 PerfMon Event Select for SBo 0 Counter 3		Package

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 725H, 1829	MSR_S0_PMON_BOX_FILTER	
Uncore SBo 0 PerfMon Box-Wide Filter		Package
Register Address: 726H, 1830	MSR_S0_PMON_CTR0	
Uncore SBo 0 PerfMon Counter 0		Package
Register Address: 727H, 1831	MSR_S0_PMON_CTR1	
Uncore SBo 0 PerfMon Counter 1		Package
Register Address: 728H, 1832	MSR_S0_PMON_CTR2	
Uncore SBo 0 PerfMon Counter 2		Package
Register Address: 729H, 1833	MSR_S0_PMON_CTR3	
Uncore SBo 0 PerfMon Counter 3		Package
Register Address: 72AH, 1834	MSR_S1_PMON_BOX_CTL	
Uncore SBo 1 PerfMon for SBo 1 Box-Wide Control		Package
Register Address: 72BH, 1835	MSR_S1_PMON_EVTNSELO	
Uncore SBo 1 PerfMon Event Select for SBo 1 Counter 0		Package
Register Address: 72CH, 1836	MSR_S1_PMON_EVTNSEL1	
Uncore SBo 1 PerfMon Event Select for SBo 1 Counter 1		Package
Register Address: 72DH, 1837	MSR_S1_PMON_EVTNSEL2	
Uncore SBo 1 PerfMon Event Select for SBo 1 Counter 2		Package
Register Address: 72EH, 1838	MSR_S1_PMON_EVTNSEL3	
Uncore SBo 1 PerfMon Event Select for SBo 1 Counter 3		Package
Register Address: 72FH, 1839	MSR_S1_PMON_BOX_FILTER	
Uncore SBo 1 PerfMon Box-Wide Filter		Package
Register Address: 730H, 1840	MSR_S1_PMON_CTR0	
Uncore SBo 1 PerfMon Counter 0		Package
Register Address: 731H, 1841	MSR_S1_PMON_CTR1	
Uncore SBo 1 PerfMon Counter 1		Package
Register Address: 732H, 1842	MSR_S1_PMON_CTR2	
Uncore SBo 1 PerfMon Counter 2		Package
Register Address: 733H, 1843	MSR_S1_PMON_CTR3	
Uncore SBo 1 PerfMon Counter 3		Package
Register Address: 734H, 1844	MSR_S2_PMON_BOX_CTL	
Uncore SBo 2 PerfMon for SBo 2 Box-Wide Control		Package
Register Address: 735H, 1845	MSR_S2_PMON_EVTNSELO	
Uncore SBo 2 PerfMon Event Select for SBo 2 Counter 0		Package
Register Address: 736H, 1846	MSR_S2_PMON_EVTNSEL1	
Uncore SBo 2 PerfMon Event Select for SBo 2 Counter 1		Package
Register Address: 737H, 1847	MSR_S2_PMON_EVTNSEL2	

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore SBo 2 PerfMon Event Select for SBo 2 Counter 2		Package
Register Address: 738H, 1848	MSR_S2_PMON_EVNTSEL3	
Uncore SBo 2 PerfMon Event Select for SBo 2 Counter 3		Package
Register Address: 739H, 1849	MSR_S2_PMON_BOX_FILTER	
Uncore SBo 2 PerfMon Box-Wide Filter		Package
Register Address: 73AH, 1850	MSR_S2_PMON_CTR0	
Uncore SBo 2 PerfMon Counter 0		Package
Register Address: 73BH, 1851	MSR_S2_PMON_CTR1	
Uncore SBo 2 PerfMon Counter 1		Package
Register Address: 73CH, 1852	MSR_S2_PMON_CTR2	
Uncore SBo 2 PerfMon Counter 2		Package
Register Address: 73DH, 1853	MSR_S2_PMON_CTR3	
Uncore SBo 2 PerfMon Counter 3		Package
Register Address: 73EH, 1854	MSR_S3_PMON_BOX_CTL	
Uncore SBo 3 PerfMon for SBo 3 Box-Wide Control		Package
Register Address: 73FH, 1855	MSR_S3_PMON_EVNTSELO	
Uncore SBo 3 PerfMon Event Select for SBo 3 Counter 0		Package
Register Address: 740H, 1856	MSR_S3_PMON_EVNTSEL1	
Uncore SBo 3 PerfMon Event Select for SBo 3 Counter 1		Package
Register Address: 741H, 1857	MSR_S3_PMON_EVNTSEL2	
Uncore SBo 3 PerfMon Event Select for SBo 3 Counter 2		Package
Register Address: 742H, 1858	MSR_S3_PMON_EVNTSEL3	
Uncore SBo 3 PerfMon Event Select for SBo 3 Counter 3		Package
Register Address: 743H, 1859	MSR_S3_PMON_BOX_FILTER	
Uncore SBo 3 PerfMon Box-Wide Filter		Package
Register Address: 744H, 1860	MSR_S3_PMON_CTR0	
Uncore SBo 3 PerfMon Counter 0		Package
Register Address: 745H, 1861	MSR_S3_PMON_CTR1	
Uncore SBo 3 PerfMon Counter 1		Package
Register Address: 746H, 1862	MSR_S3_PMON_CTR2	
Uncore SBo 3 PerfMon Counter 2		Package
Register Address: 747H, 1863	MSR_S3_PMON_CTR3	
Uncore SBo 3 PerfMon Counter 3		Package
Register Address: E00H, 3584	MSR_CO_PMON_BOX_CTL	
Uncore C-Box 0 PerfMon for Box-Wide Control		Package
Register Address: E01H, 3585	MSR_CO_PMON_EVNTSELO	
Uncore C-Box 0 PerfMon Event Select for C-Box 0 Counter 0		Package

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E02H, 3586	MSR_CO_PMON_EVTSEL1	
Uncore C-Box 0 PerfMon Event Select for C-Box 0 Counter 1		Package
Register Address: E03H, 3587	MSR_CO_PMON_EVTSEL2	
Uncore C-Box 0 PerfMon Event Select for C-Box 0 Counter 2		Package
Register Address: E04H, 3588	MSR_CO_PMON_EVTSEL3	
Uncore C-Box 0 PerfMon Event Select for C-Box 0 Counter 3		Package
Register Address: E05H, 3589	MSR_CO_PMON_BOX_FILTER0	
Uncore C-Box 0 PerfMon Box Wide Filter 0		Package
Register Address: E06H, 3590	MSR_CO_PMON_BOX_FILTER1	
Uncore C-Box 0 PerfMon Box Wide Filter 1		Package
Register Address: E07H, 3591	MSR_CO_PMON_BOX_STATUS	
Uncore C-Box 0 PerfMon Box Wide Status		Package
Register Address: E08H, 3592	MSR_CO_PMON_CTR0	
Uncore C-Box 0 PerfMon Counter 0		Package
Register Address: E09H, 3593	MSR_CO_PMON_CTR1	
Uncore C-Box 0 PerfMon Counter 1		Package
Register Address: E0AH, 3594	MSR_CO_PMON_CTR2	
Uncore C-Box 0 PerfMon Counter 2		Package
Register Address: E0BH, 3595	MSR_CO_PMON_CTR3	
Uncore C-Box 0 PerfMon Counter 3		Package
Register Address: E10H, 3600	MSR_C1_PMON_BOX_CTL	
Uncore C-Box 1 PerfMon for Box-Wide Control		Package
Register Address: E11H, 3601	MSR_C1_PMON_EVTSEL0	
Uncore C-Box 1 PerfMon Event Select for C-Box 1 Counter 0		Package
Register Address: E12H, 3602	MSR_C1_PMON_EVTSEL1	
Uncore C-Box 1 PerfMon Event Select for C-Box 1 Counter 1		Package
Register Address: E13H, 3603	MSR_C1_PMON_EVTSEL2	
Uncore C-Box 1 PerfMon Event Select for C-Box 1 Counter 2		Package
Register Address: E14H, 3604	MSR_C1_PMON_EVTSEL3	
Uncore C-Box 1 PerfMon Event Select for C-Box 1 Counter 3		Package
Register Address: E15H, 3605	MSR_C1_PMON_BOX_FILTER0	
Uncore C-Box 1 PerfMon Box Wide Filter 0		Package
Register Address: E16H, 3606	MSR_C1_PMON_BOX_FILTER1	
Uncore C-Box 1 PerfMon Box Wide Filter 1		Package
Register Address: E17H, 3607	MSR_C1_PMON_BOX_STATUS	
Uncore C-Box 1 PerfMon Box Wide Status		Package
Register Address: E18H, 3608	MSR_C1_PMON_CTR0	

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 1 PerfMon Counter 0		Package
Register Address: E19H, 3609	MSR_C1_PMON_CTR1	
Uncore C-Box 1 PerfMon Counter 1		Package
Register Address: E1AH, 3610	MSR_C1_PMON_CTR2	
Uncore C-Box 1 PerfMon Counter 2		Package
Register Address: E1BH, 3611	MSR_C1_PMON_CTR3	
Uncore C-Box 1 PerfMon Counter 3		Package
Register Address: E20H, 3616	MSR_C2_PMON_BOX_CTL	
Uncore C-Box 2 PerfMon for Box-Wide Control		Package
Register Address: E21H, 3617	MSR_C2_PMON_EVNTSELO	
Uncore C-Box 2 PerfMon Event Select for C-Box 2 Counter 0		Package
Register Address: E22H, 3618	MSR_C2_PMON_EVNTSEL1	
Uncore C-Box 2 PerfMon Event Select for C-Box 2 Counter 1		Package
Register Address: E23H, 3619	MSR_C2_PMON_EVNTSEL2	
Uncore C-Box 2 PerfMon Event Select for C-Box 2 Counter 2		Package
Register Address: E24H, 3620	MSR_C2_PMON_EVNTSEL3	
Uncore C-Box 2 PerfMon Event select for C-Box 2 Counter 3		Package
Register Address: E25H, 3621	MSR_C2_PMON_BOX_FILTER0	
Uncore C-Box 2 PerfMon Box Wide Filter 0		Package
Register Address: E26H, 3622	MSR_C2_PMON_BOX_FILTER1	
Uncore C-Box 2 PerfMon Box Wide Filter1		Package
Register Address: E27H, 3623	MSR_C2_PMON_BOX_STATUS	
Uncore C-Box 2 PerfMon Box Wide Status		Package
Register Address: E28H, 3624	MSR_C2_PMON_CTR0	
Uncore C-Box 2 PerfMon Counter 0		Package
Register Address: E29H, 3625	MSR_C2_PMON_CTR1	
Uncore C-Box 2 PerfMon Counter 1		Package
Register Address: E2AH, 3626	MSR_C2_PMON_CTR2	
Uncore C-Box 2 PerfMon Counter 2		Package
Register Address: E2BH, 3627	MSR_C2_PMON_CTR3	
Uncore C-Box 2 PerfMon Counter 3		Package
Register Address: E30H, 3632	MSR_C3_PMON_BOX_CTL	
Uncore C-Box 3 PerfMon for Box-Wide Control		Package
Register Address: E31H, 3633	MSR_C3_PMON_EVNTSELO	
Uncore C-Box 3 PerfMon Event Select for C-Box 3 Counter 0		Package
Register Address: E32H, 3634	MSR_C3_PMON_EVNTSEL1	
Uncore C-Box 3 PerfMon Event Select for C-Box 3 Counter 1		Package

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E33H, 3635	MSR_C3_PMON_EVTSEL2	
Uncore C-Box 3 PerfMon Event Select for C-Box 3 Counter 2		Package
Register Address: E34H, 3636	MSR_C3_PMON_EVTSEL3	
Uncore C-Box 3 PerfMon Event Select for C-Box 3 Counter 3		Package
Register Address: E35H, 3637	MSR_C3_PMON_BOX_FILTER0	
Uncore C-Box 3 PerfMon Box Wide Filter 0		Package
Register Address: E36H, 3638	MSR_C3_PMON_BOX_FILTER1	
Uncore C-Box 3 PerfMon Box Wide Filter1		Package
Register Address: E37H, 3639	MSR_C3_PMON_BOX_STATUS	
Uncore C-Box 3 PerfMon Box Wide Status		Package
Register Address: E38H, 3640	MSR_C3_PMON_CTR0	
Uncore C-Box 3 PerfMon Counter 0		Package
Register Address: E39H, 3641	MSR_C3_PMON_CTR1	
Uncore C-Box 3 PerfMon Counter 1		Package
Register Address: E3AH, 3642	MSR_C3_PMON_CTR2	
Uncore C-Box 3 PerfMon Counter 2		Package
Register Address: E3BH, 3643	MSR_C3_PMON_CTR3	
Uncore C-Box 3 PerfMon Counter 3		Package
Register Address: E40H, 3648	MSR_C4_PMON_BOX_CTL	
Uncore C-Box 4 PerfMon for Box-Wide Control		Package
Register Address: E41H, 3649	MSR_C4_PMON_EVTSELO	
Uncore C-Box 4 PerfMon Event Select for C-Box 4 Counter 0		Package
Register Address: E42H, 3650	MSR_C4_PMON_EVTSEL1	
Uncore C-Box 4 PerfMon Event Select for C-Box 4 Counter 1		Package
Register Address: E43H, 3651	MSR_C4_PMON_EVTSEL2	
Uncore C-Box 4 PerfMon Event Select for C-Box 4 Counter 2		Package
Register Address: E44H, 3652	MSR_C4_PMON_EVTSEL3	
Uncore C-Box 4 PerfMon Event Select for C-Box 4 Counter 3		Package
Register Address: E45H, 3653	MSR_C4_PMON_BOX_FILTER0	
Uncore C-Box 4 PerfMon Box Wide Filter 0		Package
Register Address: E46H, 3654	MSR_C4_PMON_BOX_FILTER1	
Uncore C-Box 4 PerfMon Box Wide Filter1		Package
Register Address: E47H, 3655	MSR_C4_PMON_BOX_STATUS	
Uncore C-Box 4 PerfMon Box Wide Status		Package
Register Address: E48H, 3656	MSR_C4_PMON_CTR0	
Uncore C-Box 4 PerfMon Counter 0		Package
Register Address: E49H, 3657	MSR_C4_PMON_CTR1	

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 4 PerfMon Counter 1		Package
Register Address: E4AH, 3658	MSR_C4_PMON_CTR2	
Uncore C-Box 4 PerfMon Counter 2		Package
Register Address: E4BH, 3659	MSR_C4_PMON_CTR3	
Uncore C-Box 4 PerfMon Counter 3		Package
Register Address: E50H, 3664	MSR_C5_PMON_BOX_CTL	
Uncore C-Box 5 PerfMon for Box-Wide Control		Package
Register Address: E51H, 3665	MSR_C5_PMON_EVNTSELO	
Uncore C-Box 5 PerfMon Event Select for C-Box 5 Counter 0		Package
Register Address: E52H, 3666	MSR_C5_PMON_EVNTSEL1	
Uncore C-Box 5 PerfMon Event Select for C-Box 5 Counter 1		Package
Register Address: E53H, 3667	MSR_C5_PMON_EVNTSEL2	
Uncore C-Box 5 PerfMon Event Select for C-Box 5 Counter 2		Package
Register Address: E54H, 3668	MSR_C5_PMON_EVNTSEL3	
Uncore C-Box 5 PerfMon Event Select for C-Box 5 Counter 3		Package
Register Address: E55H, 3669	MSR_C5_PMON_BOX_FILTER0	
Uncore C-Box 5 PerfMon Box Wide Filter 0		Package
Register Address: E56H, 3670	MSR_C5_PMON_BOX_FILTER1	
Uncore C-Box 5 PerfMon Box Wide Filter 1		Package
Register Address: E57H, 3671	MSR_C5_PMON_BOX_STATUS	
Uncore C-Box 5 PerfMon Box Wide Status		Package
Register Address: E58H, 3672	MSR_C5_PMON_CTR0	
Uncore C-Box 5 PerfMon Counter 0		Package
Register Address: E59H, 3673	MSR_C5_PMON_CTR1	
Uncore C-Box 5 PerfMon Counter 1		Package
Register Address: E5AH, 3674	MSR_C5_PMON_CTR2	
Uncore C-Box 5 PerfMon Counter 2		Package
Register Address: E5BH, 3675	MSR_C5_PMON_CTR3	
Uncore C-Box 5 PerfMon Counter 3		Package
Register Address: E60H, 3680	MSR_C6_PMON_BOX_CTL	
Uncore C-Box 6 PerfMon for Box-Wide Control		Package
Register Address: E61H, 3681	MSR_C6_PMON_EVNTSELO	
Uncore C-Box 6 PerfMon Event Select for C-Box 6 Counter 0		Package
Register Address: E62H, 3682	MSR_C6_PMON_EVNTSEL1	
Uncore C-Box 6 PerfMon Event Select for C-Box 6 Counter 1		Package
Register Address: E63H, 3683	MSR_C6_PMON_EVNTSEL2	
Uncore C-Box 6 PerfMon Event Select for C-Box 6 Counter 2		Package

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E64H, 3684	MSR_C6_PMON_EVTSEL3	
Uncore C-Box 6 PerfMon Event Select for C-Box 6 Counter 3		Package
Register Address: E65H, 3685	MSR_C6_PMON_BOX_FILTER0	
Uncore C-Box 6 PerfMon Box Wide Filter 0		Package
Register Address: E66H, 3686	MSR_C6_PMON_BOX_FILTER1	
Uncore C-Box 6 PerfMon Box Wide Filter 1		Package
Register Address: E67H, 3687	MSR_C6_PMON_BOX_STATUS	
Uncore C-Box 6 PerfMon Box Wide Status		Package
Register Address: E68H, 3688	MSR_C6_PMON_CTR0	
Uncore C-Box 6 PerfMon Counter 0		Package
Register Address: E69H, 3689	MSR_C6_PMON_CTR1	
Uncore C-Box 6 PerfMon Counter 1		Package
Register Address: E6AH, 3690	MSR_C6_PMON_CTR2	
Uncore C-Box 6 PerfMon Counter 2		Package
Register Address: E6BH, 3691	MSR_C6_PMON_CTR3	
Uncore C-Box 6 PerfMon Counter 3		Package
Register Address: E70H, 3696	MSR_C7_PMON_BOX_CTL	
Uncore C-Box 7 PerfMon for Box-Wide Control		Package
Register Address: E71H, 3697	MSR_C7_PMON_EVTSELO	
Uncore C-Box 7 PerfMon Event Select for C-Box 7 Counter 0		Package
Register Address: E72H, 3698	MSR_C7_PMON_EVTSEL1	
Uncore C-Box 7 PerfMon Event Select for C-Box 7 Counter 1		Package
Register Address: E73H, 3699	MSR_C7_PMON_EVTSEL2	
Uncore C-Box 7 PerfMon Event Select for C-Box 7 Counter 2		Package
Register Address: E74H, 3700	MSR_C7_PMON_EVTSEL3	
Uncore C-Box 7 PerfMon Event Select for C-Box 7 Counter 3		Package
Register Address: E75H, 3701	MSR_C7_PMON_BOX_FILTER0	
Uncore C-Box 7 PerfMon Box Wide Filter 0		Package
Register Address: E76H, 3702	MSR_C7_PMON_BOX_FILTER1	
Uncore C-Box 7 PerfMon Box Wide Filter 1		Package
Register Address: E77H, 3703	MSR_C7_PMON_BOX_STATUS	
Uncore C-Box 7 PerfMon Box Wide Status		Package
Register Address: E78H, 3704	MSR_C7_PMON_CTR0	
Uncore C-Box 7 PerfMon Counter 0		Package
Register Address: E79H, 3705	MSR_C7_PMON_CTR1	
Uncore C-Box 7 PerfMon Counter 1		Package
Register Address: E7AH, 3706	MSR_C7_PMON_CTR2	

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 7 PerfMon Counter 2		Package
Register Address: E7BH, 3707	MSR_C7_PMON_CTR3	
Uncore C-Box 7 PerfMon Counter 3		Package
Register Address: E80H, 3712	MSR_C8_PMON_BOX_CTL	
Uncore C-Box 8 PerfMon Local Box Wide Control		Package
Register Address: E81H, 3713	MSR_C8_PMON_EVNTSELO	
Uncore C-Box 8 PerfMon Event Select for C-Box 8 Counter 0		Package
Register Address: E82H, 3714	MSR_C8_PMON_EVNTSEL1	
Uncore C-Box 8 PerfMon Event Select for C-Box 8 Counter 1		Package
Register Address: E83H, 3715	MSR_C8_PMON_EVNTSEL2	
Uncore C-Box 8 PerfMon Event Select for C-Box 8 Counter 2		Package
Register Address: E84H, 3716	MSR_C8_PMON_EVNTSEL3	
Uncore C-Box 8 PerfMon Event Select for C-Box 8 Counter 3		Package
Register Address: E85H, 3717	MSR_C8_PMON_BOX_FILTER0	
Uncore C-Box 8 PerfMon Box Wide Filter 0		Package
Register Address: E86H, 3718	MSR_C8_PMON_BOX_FILTER1	
Uncore C-Box 8 PerfMon Box Wide Filter 1		Package
Register Address: E87H, 3719	MSR_C8_PMON_BOX_STATUS	
Uncore C-Box 8 PerfMon Box Wide Status		Package
Register Address: E88H, 3720	MSR_C8_PMON_CTR0	
Uncore C-Box 8 PerfMon Counter 0		Package
Register Address: E89H, 3721	MSR_C8_PMON_CTR1	
Uncore C-Box 8 PerfMon Counter 1		Package
Register Address: E8AH, 3722	MSR_C8_PMON_CTR2	
Uncore C-Box 8 PerfMon Counter 2		Package
Register Address: E8BH, 3723	MSR_C8_PMON_CTR3	
Uncore C-Box 8 PerfMon Counter 3		Package
Register Address: E90H, 3728	MSR_C9_PMON_BOX_CTL	
Uncore C-Box 9 PerfMon Local Box Wide Control		Package
Register Address: E91H, 3729	MSR_C9_PMON_EVNTSELO	
Uncore C-Box 9 PerfMon Event Select for C-Box 9 Counter 0		Package
Register Address: E92H, 3730	MSR_C9_PMON_EVNTSEL1	
Uncore C-Box 9 PerfMon Event Select for C-Box 9 Counter 1		Package
Register Address: E93H, 3731	MSR_C9_PMON_EVNTSEL2	
Uncore C-Box 9 PerfMon Event Select for C-Box 9 Counter 2		Package
Register Address: E94H, 3732	MSR_C9_PMON_EVNTSEL3	
Uncore C-Box 9 PerfMon Event Select for C-Box 9 Counter 3		Package

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E95H, 3733	MSR_C9_PMON_BOX_FILTER0	
Uncore C-Box 9 PerfMon Box Wide Filter 0		Package
Register Address: E96H, 3734	MSR_C9_PMON_BOX_FILTER1	
Uncore C-Box 9 PerfMon Box Wide Filter 1		Package
Register Address: E97H, 3735	MSR_C9_PMON_BOX_STATUS	
Uncore C-Box 9 PerfMon Box Wide Status		Package
Register Address: E98H, 3736	MSR_C9_PMON_CTR0	
Uncore C-Box 9 PerfMon Counter 0		Package
Register Address: E99H, 3737	MSR_C9_PMON_CTR1	
Uncore C-Box 9 PerfMon Counter 1		Package
Register Address: E9AH, 3738	MSR_C9_PMON_CTR2	
Uncore C-Box 9 PerfMon Counter 2		Package
Register Address: E9BH, 3739	MSR_C9_PMON_CTR3	
Uncore C-Box 9 PerfMon Counter 3		Package
Register Address: EA0H, 3744	MSR_C10_PMON_BOX_CTL	
Uncore C-Box 10 PerfMon Local Box Wide Control		Package
Register Address: EA1H, 3745	MSR_C10_PMON_EVTNSELO	
Uncore C-Box 10 PerfMon Event Select for C-Box 10 Counter 0		Package
Register Address: EA2H, 3746	MSR_C10_PMON_EVTNSEL1	
Uncore C-Box 10 PerfMon Event Select for C-Box 10 Counter 1		Package
Register Address: EA3H, 3747	MSR_C10_PMON_EVTNSEL2	
Uncore C-Box 10 PerfMon Event Select for C-Box 10 Counter 2		Package
Register Address: EA4H, 3748	MSR_C10_PMON_EVTNSEL3	
Uncore C-Box 10 PerfMon Event Select for C-Box 10 Counter 3		Package
Register Address: EA5H, 3749	MSR_C10_PMON_BOX_FILTER0	
Uncore C-Box 10 PerfMon Box Wide Filter 0		Package
Register Address: EA6H, 3750	MSR_C10_PMON_BOX_FILTER1	
Uncore C-Box 10 PerfMon Box Wide Filter 1		Package
Register Address: EA7H, 3751	MSR_C10_PMON_BOX_STATUS	
Uncore C-Box 10 PerfMon Box Wide Status		Package
Register Address: EA8H, 3752	MSR_C10_PMON_CTR0	
Uncore C-Box 10 PerfMon Counter 0		Package
Register Address: EA9H, 3753	MSR_C10_PMON_CTR1	
Uncore C-Box 10 PerfMon Counter 1		Package
Register Address: EAAH, 3754	MSR_C10_PMON_CTR2	
Uncore C-Box 10 PerfMon Counter 2		Package
Register Address: EABH, 3755	MSR_C10_PMON_CTR3	

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 10 PerfMon Counter 3		Package
Register Address: EB0H, 3760	MSR_C11_PMON_BOX_CTL	
Uncore C-Box 11 PerfMon Local Box Wide Control		Package
Register Address: EB1H, 3761	MSR_C11_PMON_EVTNSELO	
Uncore C-Box 11 PerfMon Event Select for C-Box 11 Counter 0		Package
Register Address: EB2H, 3762	MSR_C11_PMON_EVTNSEL1	
Uncore C-Box 11 PerfMon Event Select for C-Box 11 Counter 1		Package
Register Address: EB3H, 3763	MSR_C11_PMON_EVTNSEL2	
Uncore C-Box 11 PerfMon Event Select for C-Box 11 Counter 2		Package
Register Address: EB4H, 3764	MSR_C11_PMON_EVTNSEL3	
Uncore C-box 11 PerfMon Event Select for C-Box 11 Counter 3		Package
Register Address: EB5H, 3765	MSR_C11_PMON_BOX_FILTER0	
Uncore C-Box 11 PerfMon Box Wide Filter 0		Package
Register Address: EB6H, 3766	MSR_C11_PMON_BOX_FILTER1	
Uncore C-Box 11 PerfMon Box Wide Filter 1		Package
Register Address: EB7H, 3767	MSR_C11_PMON_BOX_STATUS	
Uncore C-Box 11 PerfMon Box Wide Status		Package
Register Address: EB8H, 3768	MSR_C11_PMON_CTR0	
Uncore C-Box 11 PerfMon Counter 0		Package
Register Address: EB9H, 3769	MSR_C11_PMON_CTR1	
Uncore C-Box 11 PerfMon Counter 1		Package
Register Address: EBAH, 3770	MSR_C11_PMON_CTR2	
Uncore C-Box 11 PerfMon Counter 2		Package
Register Address: EBBH, 3771	MSR_C11_PMON_CTR3	
Uncore C-Box 11 PerfMon Counter 3		Package
Register Address: EC0H, 3776	MSR_C12_PMON_BOX_CTL	
Uncore C-Box 12 PerfMon Local Box Wide Control		Package
Register Address: EC1H, 3777	MSR_C12_PMON_EVTNSELO	
Uncore C-Box 12 PerfMon Event Select for C-Box 12 Counter 0		Package
Register Address: EC2H, 3778	MSR_C12_PMON_EVTNSEL1	
Uncore C-Box 12 PerfMon Event Select for C-Box 12 Counter 1		Package
Register Address: EC3H, 3779	MSR_C12_PMON_EVTNSEL2	
Uncore C-Box 12 PerfMon Event Select for C-Box 12 Counter 2		Package
Register Address: EC4H, 3780	MSR_C12_PMON_EVTNSEL3	
Uncore C-Box 12 PerfMon Event Select for C-Box 12 Counter 3		Package
Register Address: EC5H, 3781	MSR_C12_PMON_BOX_FILTER0	
Uncore C-Box 12 PerfMon Box Wide Filter 0		Package

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: EC6H, 3782	MSR_C12_PMON_BOX_FILTER1	
Uncore C-Box 12 PerfMon Box Wide Filter 1		Package
Register Address: EC7H, 3783	MSR_C12_PMON_BOX_STATUS	
Uncore C-Box 12 PerfMon Box Wide Status		Package
Register Address: EC8H, 3784	MSR_C12_PMON_CTR0	
Uncore C-Box 12 PerfMon Counter 0		Package
Register Address: EC9H, 3785	MSR_C12_PMON_CTR1	
Uncore C-Box 12 PerfMon Counter 1		Package
Register Address: ECAH, 3786	MSR_C12_PMON_CTR2	
Uncore C-Box 12 PerfMon Counter 2		Package
Register Address: ECBH, 3787	MSR_C12_PMON_CTR3	
Uncore C-Box 12 PerfMon Counter 3		Package
Register Address: EDOH, 3792	MSR_C13_PMON_BOX_CTL	
Uncore C-Box 13 PerfMon local box wide control.		Package
Register Address: ED1H, 3793	MSR_C13_PMON_EVTSELO	
Uncore C-Box 13 PerfMon Event Select for C-Box 13 Counter 0		Package
Register Address: ED2H, 3794	MSR_C13_PMON_EVTSEL1	
Uncore C-Box 13 PerfMon Event Select for C-Box 13 Counter 1		Package
Register Address: ED3H, 3795	MSR_C13_PMON_EVTSEL2	
Uncore C-Box 13 PerfMon Event Select for C-Box 13 Counter 2		Package
Register Address: ED4H, 3796	MSR_C13_PMON_EVTSEL3	
Uncore C-Box 13 PerfMon Event Select for C-Box 13 Counter 3		Package
Register Address: ED5H, 3797	MSR_C13_PMON_BOX_FILTER0	
Uncore C-Box 13 PerfMon Box Wide Filter 0		Package
Register Address: ED6H, 3798	MSR_C13_PMON_BOX_FILTER1	
Uncore C-Box 13 PerfMon Box Wide Filter 1		Package
Register Address: ED7H, 3799	MSR_C13_PMON_BOX_STATUS	
Uncore C-Box 13 PerfMon Box Wide Status		Package
Register Address: ED8H, 3800	MSR_C13_PMON_CTR0	
Uncore C-Box 13 PerfMon Counter 0		Package
Register Address: ED9H, 3801	MSR_C13_PMON_CTR1	
Uncore C-Box 13 PerfMon Counter 1		Package
Register Address: EDAH, 3802	MSR_C13_PMON_CTR2	
Uncore C-Box 13 PerfMon Counter 2		Package
Register Address: EDBH, 3803	MSR_C13_PMON_CTR3	
Uncore C-Box 13 PerfMon Counter 3		Package
Register Address: EEOH, 3808	MSR_C14_PMON_BOX_CTL	

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 14 PerfMon Local Box Wide Control		Package
Register Address: EE1H, 3809	MSR_C14_PMON_EVNTSEL0	
Uncore C-Box 14 PerfMon Event Select for C-Box 14 Counter 0		Package
Register Address: EE2H, 3810	MSR_C14_PMON_EVNTSEL1	
Uncore C-Box 14 PerfMon Event Select for C-Box 14 Counter 1		Package
Register Address: EE3H, 3811	MSR_C14_PMON_EVNTSEL2	
Uncore C-Box 14 PerfMon Event Select for C-Box 14 Counter 2		Package
Register Address: EE4H, 3812	MSR_C14_PMON_EVNTSEL3	
Uncore C-Box 14 PerfMon Event Select for C-Box 14 Counter 3		Package
Register Address: EE5H, 3813	MSR_C14_PMON_BOX_FILTER	
Uncore C-Box 14 PerfMon Box Wide Filter 0		Package
Register Address: EE6H, 3814	MSR_C14_PMON_BOX_FILTER1	
Uncore C-Box 14 PerfMon Box Wide Filter 1		Package
Register Address: EE7H, 3815	MSR_C14_PMON_BOX_STATUS	
Uncore C-Box 14 PerfMon Box Wide Status		Package
Register Address: EE8H, 3816	MSR_C14_PMON_CTR0	
Uncore C-Box 14 PerfMon Counter 0		Package
Register Address: EE9H, 3817	MSR_C14_PMON_CTR1	
Uncore C-Box 14 PerfMon Counter 1		Package
Register Address: EEAH, 3818	MSR_C14_PMON_CTR2	
Uncore C-Box 14 PerfMon Counter 2		Package
Register Address: EEBH, 3819	MSR_C14_PMON_CTR3	
Uncore C-Box 14 PerfMon Counter 3		Package
Register Address: EF0H, 3824	MSR_C15_PMON_BOX_CTL	
Uncore C-Box 15 PerfMon Local Box Wide Control		Package
Register Address: EF1H, 3825	MSR_C15_PMON_EVNTSEL0	
Uncore C-Box 15 PerfMon Event Select for C-Box 15 Counter 0		Package
Register Address: EF2H, 3826	MSR_C15_PMON_EVNTSEL1	
Uncore C-Box 15 PerfMon Event Select for C-Box 15 Counter 1		Package
Register Address: EF3H, 3827	MSR_C15_PMON_EVNTSEL2	
Uncore C-Box 15 PerfMon Event Select for C-Box 15 Counter 2		Package
Register Address: EF4H, 3828	MSR_C15_PMON_EVNTSEL3	
Uncore C-Box 15 PerfMon Event Select for C-Box 15 Counter 3		Package
Register Address: EF5H, 3829	MSR_C15_PMON_BOX_FILTER0	
Uncore C-Box 15 PerfMon Box Wide Filter 0		Package
Register Address: EF6H, 3830	MSR_C15_PMON_BOX_FILTER1	
Uncore C-Box 15 PerfMon Box Wide Filter 1		Package

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: EF7H, 3831	MSR_C15_PMON_BOX_STATUS	
Uncore C-Box 15 PerfMon Box Wide Status		Package
Register Address: EF8H, 3832	MSR_C15_PMON_CTR0	
Uncore C-Box 15 PerfMon Counter 0		Package
Register Address: EF9H, 3833	MSR_C15_PMON_CTR1	
Uncore C-Box 15 PerfMon Counter 1		Package
Register Address: EFAH, 3834	MSR_C15_PMON_CTR2	
Uncore C-Box 15 PerfMon Counter 2		Package
Register Address: EFBH, 3835	MSR_C15_PMON_CTR3	
Uncore C-Box 15 PerfMon Counter 3		Package
Register Address: F00H, 3840	MSR_C16_PMON_BOX_CTL	
Uncore C-Box 16 PerfMon for Box-Wide Control		Package
Register Address: F01H, 3841	MSR_C16_PMON_EVTSELO	
Uncore C-Box 16 PerfMon Event Select for C-Box 16 Counter 0		Package
Register Address: F02H, 3842	MSR_C16_PMON_EVTSEL1	
Uncore C-Box 16 PerfMon Event Select for C-Box 16 Counter 1		Package
Register Address: F03H, 3843	MSR_C16_PMON_EVTSEL2	
Uncore C-Box 16 PerfMon Event Select for C-Box 16 Counter 2		Package
Register Address: F04H, 3844	MSR_C16_PMON_EVTSEL3	
Uncore C-Box 16 PerfMon Event Select for C-Box 16 Counter 3		Package
Register Address: F05H, 3845	MSR_C16_PMON_BOX_FILTER0	
Uncore C-Box 16 PerfMon Box Wide Filter 0		Package
Register Address: F06H, 3846	MSR_C16_PMON_BOX_FILTER1	
Uncore C-Box 16 PerfMon Box Wide Filter 1		Package
Register Address: F07H, 3847	MSR_C16_PMON_BOX_STATUS	
Uncore C-Box 16 PerfMon Box Wide Status		Package
Register Address: F08H, 3848	MSR_C16_PMON_CTR0	
Uncore C-Box 16 PerfMon Counter 0		Package
Register Address: F09H, 3849	MSR_C16_PMON_CTR1	
Uncore C-Box 16 PerfMon Counter 1		Package
Register Address: FOAH, 3850	MSR_C16_PMON_CTR2	
Uncore C-Box 16 PerfMon Counter 2		Package
Register Address: F0BH, 3851	MSR_C16_PMON_CTR3	
Uncore C-Box 16 PerfMon Counter 3		Package
Register Address: F10H, 3856	MSR_C17_PMON_BOX_CTL	
Uncore C-Box 17 PerfMon for Box-Wide Control		Package
Register Address: F11H, 3857	MSR_C17_PMON_EVTSELO	

Table 2-33. Uncore PMU MSRs in the Intel® Xeon® Processor E5 v3 Family (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 17 PerfMon Event Select for C-Box 17 Counter 0		Package
Register Address: F12H, 3858	MSR_C17_PMON_EVTNSEL1	
Uncore C-Box 17 PerfMon Event Select for C-Box 17 Counter 1		Package
Register Address: F13H, 3859	MSR_C17_PMON_EVTNSEL2	
Uncore C-Box 17 PerfMon Event Select for C-Box 17 Counter 2		Package
Register Address: F14H, 3860	MSR_C17_PMON_EVTNSEL3	
Uncore C-Box 17 PerfMon Event Select for C-Box 17 Counter 3		Package
Register Address: F15H, 3861	MSR_C17_PMON_BOX_FILTER0	
Uncore C-Box 17 PerfMon Box Wide Filter 0		Package
Register Address: F16H, 3862	MSR_C17_PMON_BOX_FILTER1	
Uncore C-Box 17 PerfMon Box Wide Filter 1		Package
Register Address: F17H, 3863	MSR_C17_PMON_BOX_STATUS	
Uncore C-Box 17 PerfMon Box Wide Status		Package
Register Address: F18H, 3864	MSR_C17_PMON_CTR0	
Uncore C-Box 17 PerfMon Counter 0		Package
Register Address: F19H, 3865	MSR_C17_PMON_CTR1	
Uncore C-Box 17 PerfMon Counter 1		Package
Register Address: F1AH, 3866	MSR_C17_PMON_CTR2	
Uncore C-Box 17 PerfMon Counter 2		Package
Register Address: F1BH, 3867	MSR_C17_PMON_CTR3	
Uncore C-Box 17 PerfMon Counter 3		Package

2.15 MSRS IN THE INTEL® CORE™ M PROCESSORS AND THE 5TH GENERATION INTEL® CORE™ PROCESSORS

The Intel® Core™ M-5xxx processors, 5th generation Intel® Core™ Processors, and the Intel® Xeon® Processor E3-1200 v4 family are based on Broadwell microarchitecture. The Intel® Core™ M-5xxx processors and 5th generation Intel® Core™ Processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_3DH. The Intel® Xeon® Processor E3-1200 v4 family and 5th generation Intel® Core™ Processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_47H. Processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_3DH or 06_47H support the MSR interfaces listed in Table 2-20, Table 2-21, Table 2-22, Table 2-25, Table 2-29, Table 2-30, Table 2-34, and Table 2-35. For an MSR listed in Table 2-35 that also appears in the model-specific tables of prior generations, Table 2-35 supersedes prior generation tables.

Table 2-34 lists MSRs that are common to processors based on the Broadwell microarchitectures (including CPUID Signature DisplayFamily_DisplayModel values of 06_3DH, 06_47H, 06_4FH, and 06_56H).

Table 2-34. Additional MSRs Common to Processors Based on Broadwell Microarchitectures

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	

Table 2-34. Additional MSRs Common to Processors Based on Broadwell Microarchitectures

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2 and Section 21.6.2.2, "Global Counter Control Facilities."		Thread
0	Ovf_PMC0	
1	Ovf_PMC1	
2	Ovf_PMC2	
3	Ovf_PMC3	
31:4	Reserved	
32	Ovf_FixedCtr0	
33	Ovf_FixedCtr1	
34	Ovf_FixedCtr2	
54:35	Reserved.	
55	Trace_ToPA_PMI See Section 34.2.7.2, "Table of Physical Addresses (ToPA)."	
60:56	Reserved.	
61	Ovf_Uncore	
62	Ovf_BufDSSAVE	
63	CondChgd	
Register Address: 390H, 912	IA32_PERF_GLOBAL_OVF_CTRL	
See Table 2-2 and Section 21.6.2.2, "Global Counter Control Facilities."		Thread
0	Set 1 to clear Ovf_PMC0.	
1	Set 1 to clear Ovf_PMC1.	
2	Set 1 to clear Ovf_PMC2.	
3	Set 1 to clear Ovf_PMC3.	
31:4	Reserved.	
32	Set 1 to clear Ovf_FixedCtr0.	
33	Set 1 to clear Ovf_FixedCtr1.	
34	Set 1 to clear Ovf_FixedCtr2	
54:35	Reserved.	
55	Set 1 to clear Trace_ToPA_PMI. See Section 34.2.7.2, "Table of Physical Addresses (ToPA)."	
60:56	Reserved.	
61	Set 1 to clear Ovf_Uncore.	
62	Set 1 to clear Ovf_BufDSSAVE.	
63	Set 1 to clear CondChgd.	
Register Address: 560H, 1376	IA32_RTIT_OUTPUT_BASE	
Trace Output Base Register (R/W)		Thread
6:0	Reserved.	
MAXPHYADDR ¹ -1:7	Base physical address.	
63:MAXPHYADDR	Reserved.	

Table 2-34. Additional MSRs Common to Processors Based on Broadwell Microarchitectures

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 561H, 1377	IA32_RTIT_OUTPUT_MASK_PTRS	
Trace Output Mask Pointers Register (R/W)		Thread
6:0	Reserved.	
31:7	MaskOrTableOffset	
63:32	Output Offset.	
Register Address: 570H, 1392	IA32_RTIT_CTL	
Trace Control Register (R/W)		Thread
0	TraceEn	
1	Reserved, must be zero.	
2	OS	
3	User	
6:4	Reserved, must be zero.	
7	CR3Filter	
8	ToPA Writing 0 will #GP if also setting TraceEn.	
9	Reserved, must be zero.	
10	TSCEn	
11	DisRETC	
12	Reserved, must be zero.	
13	Reserved; writing 0 will #GP if also setting TraceEn.	
63:14	Reserved, must be zero.	
Register Address: 571H, 1393	IA32_RTIT_STATUS	
Tracing Status Register (R/W)		Thread
0	Reserved, writes ignored.	
1	ContexEn, writes ignored.	
2	TriggerEn, writes ignored.	
3	Reserved	
4	Error (R/W)	
5	Stopped	
63:6	Reserved, must be zero.	
Register Address: 572H, 1394	IA32_RTIT_CR3_MATCH	
Trace Filter CR3 Match Register (R/W)		Thread
4:0	Reserved.	
63:5	CR3[63:5] value to match.	
Register Address: 620H, 1568	MSR_UNCORE_RATIO_LIMIT	
Uncore Ratio Limit (R/W) Out of reset, the min_ratio and max_ratio fields represent the widest possible range of uncore frequencies. Writing to these fields allows software to control the minimum and the maximum frequency that hardware will select.		Package

Table 2-34. Additional MSRs Common to Processors Based on Broadwell Microarchitectures

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
6:0	MAX_RATIO This field is used to limit the max ratio of the LLC/Ring.	
7	Reserved.	
14:8	MIN_RATIO Writing to this field controls the minimum possible ratio of the LLC/Ring.	
63:15	Reserved.	

NOTES:

1. MAXPHYADDR is reported by CPUID.80000008H:EAX[7:0].

Table 2-35 lists MSRs that are specific to Intel Core M processors and 5th Generation Intel Core Processors.

Table 2-35. Additional MSRs Supported by Intel® Core™ M Processors and 5th Generation Intel® Core™ Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states. See http://biosbits.org .		Core
3:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 0000b: C0/C1 (no package C-state support) 0001b: C2 0010b: C3 0011b: C6 0100b: C7 0101b: C7s 0110b: C8 0111b: C9 1000b: C10	
9:4	Reserved.	
10	I/O MWAIT Redirection Enable (R/W)	
14:11	Reserved.	
15	CFG Lock (R/WO)	
24:16	Reserved.	
25	C3 State Auto Demotion Enable (R/W)	
26	C1 State Auto Demotion Enable (R/W)	
27	Enable C3 Undemotion (R/W)	

Table 2-35. Additional MSRs Supported by Intel® Core™ M Processors and 5th Generation Intel® Core™ Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
28	Enable C1 Undemotion (R/W)	
29	Enable Package C-State Auto-Demotion (R/W)	
30	Enable Package C-State Undemotion (R/W)	
63:31	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 core active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 core active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 core active.	Package
39:32	Maximum Ratio Limit for 5C Maximum turbo ratio limit of 5core active.	Package
47:40	Maximum Ratio Limit for 6C Maximum turbo ratio limit of 6core active.	Package
63:48	Reserved.	
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	
PPO Energy Status (R/O) See Section 16.10.4, "PPO/PP1 RAPL Domains."		Package
See Table 2-20, Table 2-21, Table 2-22, Table 2-25, Table 2-29, Table 2-30, and Table 2-34 for other MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_3DH.		

2.16 MSRS IN THE INTEL® XEON® PROCESSOR E5 V4 FAMILY

The MSRs listed in Table 2-36 are available and common to the Intel® Xeon® Processor D Product Family (CPUID Signature DisplayFamily_DisplayModel value of 06_56H) and to the Intel Xeon processors E5 v4 and E7 v4 families (CPUID Signature DisplayFamily_DisplayModel value of 06_4FH). These processors are based on Broadwell microarchitecture.

See Section 2.16.1 for lists of tables of MSRs that are supported by the Intel® Xeon® Processor D Family.

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 4EH, 78	IA32_PPIN_CTL (MSR_PPIN_CTL)	
Protected Processor Inventory Number Enable Control (R/W)		Package

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
0	LockOut (R/WO) See Table 2-2.	
1	Enable_PPIN (R/W) See Table 2-2.	
63:2	Reserved	
Register Address: 4FH, 79	IA32_PPIN (MSR_PPIN)	
Protected Processor Inventory Number (R/O)		Package
63:0	Protected Processor Inventory Number (R/O) See Table 2-2.	
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) See Table 2-26.	Package
22:16	Reserved.	
23	PPIN_CAP (R/O) See Table 2-26.	Package
27:24	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) See Table 2-26.	Package
29	Programmable TDP Limit for Turbo Mode (R/O) See Table 2-26.	Package
30	Programmable TJ OFFSET (R/O) See Table 2-26.	Package
39:31	Reserved.	
47:40	Maximum Efficiency Ratio (R/O) See Table 2-26.	Package
63:48	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states. See http://biosbits.org .		Core

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: C0/C1 (no package C-state support) 001b: C2 010b: C6 (non-retention) 011b: C6 (retention) 111b: No Package C state limits. All C states supported by the processor are available.	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W)	
14:11	Reserved.	
15	CFG Lock (R/WO)	
16	Automatic C-State Conversion Enable (R/W) If 1, the processor will convert HALT or MWAIT(C1) to MWAIT(C6).	
24:17	Reserved.	
25	C3 State Auto Demotion Enable (R/W)	
26	C1 State Auto Demotion Enable (R/W)	
27	Enable C3 Undemotion (R/W)	
28	Enable C1 Undemotion (R/W)	
29	Package C State Demotion Enable (R/W)	
30	Package C State Undemotion Enable (R/W)	
63:31	Reserved.	
Register Address: 179H, 377	IA32_MCG_CAP	
Global Machine Check Capability (R/O)		Thread
7:0	Count	
8	MCG_CTL_P	
9	MCG_EXT_P	
10	MCP_CMCI_P	
11	MCG_TES_P	
15:12	Reserved	
23:16	MCG_EXT_CNT	
24	MCG_SER_P	
25	MCG_EM_P	
26	MCG_ELOG_P	
63:27	Reserved.	
Register Address: 17DH, 381	MSR_SMM_MCA_CAP	

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Enhanced SMM Capabilities (SMM-RO) Reports SMM capability Enhancement. Accessible only while in SMM.		Thread
57:0	Reserved.	
58	SMM_Code_Access_Chk (SMM-RO) If set to 1, indicates that the SMM code access restriction is supported and a host-space interface available to SMM handler.	
59	Long_Flow_Indication (SMM-RO) If set to 1, indicates that the SMM long flow indicator is supported and a host-space interface available to SMM handler.	
63:60	Reserved.	
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Core
0	Thermal Status (R/O) See Table 2-2.	
1	Thermal Status Log (R/WC0) See Table 2-2.	
2	PROTCHOT # or FORCEPR# Status (R/O) See Table 2-2.	
3	PROTCHOT # or FORCEPR# Log (R/WC0) See Table 2-2.	
4	Critical Temperature Status (R/O) See Table 2-2.	
5	Critical Temperature Status Log (R/WC0) See Table 2-2.	
6	Thermal Threshold #1 Status (R/O) See Table 2-2.	
7	Thermal Threshold #1 Log (R/WC0) See Table 2-2.	
8	Thermal Threshold #2 Status (R/O) See Table 2-2.	
9	Thermal Threshold #2 Log (R/WC0) See Table 2-2.	
10	Power Limitation Status (R/O) See Table 2-2.	
11	Power Limitation Log (R/WC0) See Table 2-2.	
12	Current Limit Status (R/O) See Table 2-2.	

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
13	Current Limit Log (R/WC0) See Table 2-2.	
14	Cross Domain Limit Status (R/O) See Table 2-2.	
15	Cross Domain Limit Log (R/WC0) See Table 2-2.	
22:16	Digital Readout (R/O) See Table 2-2.	
26:23	Reserved.	
30:27	Resolution in Degrees Celsius (R/O) See Table 2-2.	
31	Reading Valid (R/O) See Table 2-2.	
63:32	Reserved.	
Register Address: 1A2H, 418	MSR_TEMPERATURE_TARGET	
Temperature Target		Package
15:0	Reserved.	
23:16	Temperature Target (R/O) See Table 2-26.	
27:24	TCC Activation Offset (R/W) See Table 2-26.	
63:28	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 1C	Package
15:8	Maximum Ratio Limit for 2C	Package
23:16	Maximum Ratio Limit for 3C	Package
31:24	Maximum Ratio Limit for 4C	Package
39:32	Maximum Ratio Limit for 5C	Package
47:40	Maximum Ratio Limit for 6C	Package
55:48	Maximum Ratio Limit for 7C	Package
63:56	Maximum Ratio Limit for 8C	Package
Register Address: 1AEH, 430	MSR_TURBO_RATIO_LIMIT1	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
7:0	Maximum Ratio Limit for 9C	Package
15:8	Maximum Ratio Limit for 10C	Package

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
23:16	Maximum Ratio Limit for 11C	Package
31:24	Maximum Ratio Limit for 12C	Package
39:32	Maximum Ratio Limit for 13C	Package
47:40	Maximum Ratio Limit for 14C	Package
55:48	Maximum Ratio Limit for 15C	Package
63:56	Maximum Ratio Limit for 16C	Package
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
Unit Multipliers Used in RAPL Interfaces (R/O)		Package
3:0	Power Units See Section 16.10.1, "RAPL Interfaces."	Package
7:4	Reserved.	Package
12:8	Energy Status Units Energy related information (in Joules) is based on the multiplier, $1/2^{\text{ESU}}$; where ESU is an unsigned integer represented by bits 12:8. Default value is 0EH (or 61 micro-joules).	Package
15:13	Reserved.	Package
19:16	Time Units See Section 16.10.1, "RAPL Interfaces."	Package
63:20	Reserved.	
Register Address: 618H, 1560	MSR_DRAM_POWER_LIMIT	
DRAM RAPL Power Limit Control (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) Energy consumed by DRAM devices.		Package
31:0	Energy in 15.3 micro-joules. Requires BIOS configuration to enable DRAM RAPL mode 0 (Direct VR).	
63:32	Reserved.	
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61CH, 1564	MSR_DRAM_POWER_INFO	
DRAM RAPL Parameters (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 620H, 1568	MSR_UNCORE_RATIO_LIMIT	
Uncore Ratio Limit (R/W) Out of reset, the min_ratio and max_ratio fields represent the widest possible range of uncore frequencies. Writing to these fields allows software to control the minimum and the maximum frequency that hardware will select.		Package
63:15	Reserved.	

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
14:8	MIN_RATIO Writing to this field controls the minimum possible ratio of the LLC/Ring.	
7	Reserved.	
6:0	MAX_RATIO This field is used to limit the max ratio of the LLC/Ring.	
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	
Reserved (R/O) Reads return 0.		Package
Register Address: 690H, 1680	MSR_CORE_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in Processor Cores (R/W) (Frequency refers to processor core frequency.)		Package
0	PROCHOT Status (R0) When set, processor core frequency is reduced below the operating system request due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced below the operating system request due to a thermal event.	
2	Power Budget Management Status (R0) When set, frequency is reduced below the operating system request due to PBM limit.	
3	Platform Configuration Services Status (R0) When set, frequency is reduced below the operating system request due to PCS limit.	
4	Reserved.	
5	Autonomous Utilization-Based Frequency Control Status (R0) When set, frequency is reduced below the operating system request because the processor has detected that utilization is low.	
6	VR Therm Alert Status (R0) When set, frequency is reduced below the operating system request due to a thermal alert from the Voltage Regulator.	
7	Reserved.	
8	Electrical Design Point Status (R0) When set, frequency is reduced below the operating system request due to electrical design point constraints (e.g., maximum electrical current consumption).	
9	Reserved.	
10	Multi-Core Turbo Status (R0) When set, frequency is reduced below the operating system request due to Multi-Core Turbo limits.	
12:11	Reserved.	

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
13	Core Frequency P1 Status (R0) When set, frequency is reduced below max non-turbo P1.	
14	Core Max N-Core Turbo Frequency Limiting Status (R0) When set, frequency is reduced below max n-core turbo frequency.	
15	Core Frequency Limiting Status (R0) When set, frequency is reduced below the operating system request.	
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
18	Power Budget Management Log When set, indicates that the PBM Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
19	Platform Configuration Services Log When set, indicates that the PCS Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
20	Reserved.	
21	Autonomous Utilization-Based Frequency Control Log When set, indicates that the AUBFC Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	Reserved.	
24	Electrical Design Point Log When set, indicates that the EDP Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Reserved.	
26	Multi-Core Turbo Log When set, indicates that the Multi-Core Turbo Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28:27	Reserved.	

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
29	Core Frequency P1 Log When set, indicates that the Core Frequency P1 Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
30	Core Max N-Core Turbo Frequency Limiting Log When set, indicates that the Core Max n-core Turbo Frequency Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
31	Core Frequency Limiting Log When set, indicates that the Core Frequency Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:32	Reserved.	
Register Address: 770H, 1904	IA32_PM_ENABLE	
See Section 16.4.2, "Enabling HWP."		Package
Register Address: 771H, 1905	IA32_HWP_CAPABILITIES	
See Section 16.4.3, "HWP Performance Range and Dynamic Capabilities."		Thread
Register Address: 774H, 1908	IA32_HWP_REQUEST	
See Section 16.4.4, "Managing HWP."		Thread
7:0	Minimum Performance (R/W)	
15:8	Maximum Performance (R/W)	
23:16	Desired Performance (R/W)	
63:24	Reserved.	
Register Address: 777H, 1911	IA32_HWP_STATUS	
See Section 16.4.5, "HWP Feedback."		Thread
1:0	Reserved.	
2	Excursion to Minimum (R/O)	
63:3	Reserved.	
Register Address: C8DH, 3213	IA32_QM_EVTSEL	
Monitoring Event Select Register (R/W) If CPUID.(EAX=07H, ECX=0):EBX.RDT-M[bit 12] = 1.		Thread
7:0	EventID (R/W) Event encoding: 0x00: No monitoring. 0x01: L3 occupancy monitoring. 0x02: Total memory bandwidth monitoring. 0x03: Local memory bandwidth monitoring. All other encoding reserved.	
31:8	Reserved.	

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
41:32	RMID (R/W)	
63:42	Reserved.	
Register Address: C8FH, 3215	IA32_PQR_ASSOC	
Resource Association Register (R/W)		Thread
9:0	RMID	
31:10	Reserved.	
51:32	CLOS (R/W)	
63: 52	Reserved.	
Register Address: C90H, 3216	IA32_L3_QOS_MASK_0	
L3 Class Of Service Mask - CLOS 0 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=0.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 0 enforcement.	
63:20	Reserved.	
Register Address: C91H, 3217	IA32_L3_QOS_MASK_1	
L3 Class Of Service Mask - CLOS 1 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=1.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 1 enforcement.	
63:20	Reserved.	
Register Address: C92H, 3218	IA32_L3_QOS_MASK_2	
L3 Class Of Service Mask - CLOS 2 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=2.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 2 enforcement.	
63:20	Reserved.	
Register Address: C93H, 3219	IA32_L3_QOS_MASK_3	
L3 Class Of Service Mask - CLOS 3 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=3.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 3 enforcement.	
63:20	Reserved.	
Register Address: C94H, 3220	IA32_L3_QOS_MASK_4	
L3 Class Of Service Mask - CLOS 4 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=4.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 4 enforcement.	
63:20	Reserved.	
Register Address: C95H, 3221	IA32_L3_QOS_MASK_5	
L3 Class Of Service Mask - CLOS 5 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=5.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 5 enforcement.	

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:20	Reserved.	
Register Address: C96H, 3222	IA32_L3_QOS_MASK_6	
L3 Class Of Service Mask - CLOS 6 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=6.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 6 enforcement.	
63:20	Reserved.	
Register Address: C97H, 3223	IA32_L3_QOS_MASK_7	
L3 Class Of Service Mask - CLOS 7 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=7.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 7 enforcement.	
63:20	Reserved.	
Register Address: C98H, 3224	IA32_L3_QOS_MASK_8	
L3 Class Of Service Mask - CLOS 8 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=8.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 8 enforcement.	
63:20	Reserved.	
Register Address: C99H, 3225	IA32_L3_QOS_MASK_9	
L3 Class Of Service Mask - CLOS 9 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=9.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 9 enforcement.	
63:20	Reserved.	
Register Address: C9AH, 3226	IA32_L3_QOS_MASK_10	
L3 Class Of Service Mask - CLOS 10 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=10.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 10 enforcement.	
63:20	Reserved.	
Register Address: C9BH, 3227	IA32_L3_QOS_MASK_11	
L3 Class Of Service Mask - CLOS 11 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=11.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 11 enforcement.	
63:20	Reserved.	
Register Address: C9CH, 3228	IA32_L3_QOS_MASK_12	
L3 Class Of Service Mask - CLOS 12 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=12.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 12 enforcement.	
63:20	Reserved.	
Register Address: C9DH, 3229	IA32_L3_QOS_MASK_13	

Table 2-36. Additional MSRs Common to the Intel® Xeon® Processor D and the Intel® Xeon® Processor E5 v4 Family Based on Broadwell Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
L3 Class Of Service Mask - CLOS 13 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=13.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 13 enforcement.	
63:20	Reserved.	
Register Address: C9EH, 3230	IA32_L3_QOS_MASK_14	
L3 Class Of Service Mask - CLOS 14 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=14.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 14 enforcement.	
63:20	Reserved.	
Register Address: C9FH, 3231	IA32_L3_QOS_MASK_15	
L3 Class Of Service Mask - CLOS 15 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=15.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 15 enforcement.	
63:20	Reserved.	

2.16.1 Additional MSRs Supported in the Intel® Xeon® Processor D Product Family

The MSRs listed in Table 2-37 are available to Intel® Xeon® Processor D Product Family (CPUID Signature DisplayFamily_DisplayModel value of 06_56H). The Intel® Xeon® processor D product family is based on Broadwell microarchitecture and supports the MSR interfaces listed in Table 2-20, Table 2-29, Table 2-34, Table 2-36, and Table 2-37.

Table 2-37. Additional MSRs Supported by Intel® Xeon® Processor D with a CPUID Signature DisplayFamily_DisplayModel Value of 06_56H

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1ACH, 428	MSR_TURBO_RATIO_LIMIT3	
Config Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
62:0	Reserved.	Package
63	Semaphore for Turbo Ratio Limit Configuration If 1, the processor uses override configuration ¹ specified in MSR_TURBO_RATIO_LIMIT, MSR_TURBO_RATIO_LIMIT1. If 0, the processor uses factory-set configuration (Default).	Package
Register Address: 286H, 646	IA32_MC6_CTL2	
See Table 2-2.		Package
Register Address: 287H, 647	IA32_MC7_CTL2	
See Table 2-2.		Package
Register Address: 289H, 649	IA32_MC9_CTL2	
See Table 2-2.		Package

Table 2-37. Additional MSRs Supported by Intel® Xeon® Processor D with a CPUID Signature DisplayFamily_DisplayModel Value of 06_56H

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 28AH, 650	IA32_MC10_CTL2	
See Table 2-2.		Package
Register Address: 291H, 657	IA32_MC17_CTL2	
See Table 2-2.		Package
Register Address: 292H, 658	IA32_MC18_CTL2	
See Table 2-2.		Package
Register Address: 293H, 659	IA32_MC19_CTL2	
See Table 2-2.		Package
Register Address: 418H, 1048	IA32_MC6_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 419H, 1049	IA32_MC6_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41AH, 1050	IA32_MC6_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41BH, 1051	IA32_MC6_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41CH, 1052	IA32_MC7_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41DH, 1053	IA32_MC7_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41EH, 1054	IA32_MC7_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41FH, 1055	IA32_MC7_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 424H, 1060	IA32_MC9_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 10 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 425H, 1061	IA32_MC9_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 10 report MC errors from each channel of the integrated memory controllers.		Package

Table 2-37. Additional MSRs Supported by Intel® Xeon® Processor D with a CPUID Signature DisplayFamily_DisplayModel Value of 06_56H

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 426H, 1062	IA32_MC9_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 10 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 427H, 1063	IA32_MC9_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 10 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 428H, 1064	IA32_MC10_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 10 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 429H, 1065	IA32_MC10_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 10 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 42AH, 1066	IA32_MC10_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 10 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 42BH, 1067	IA32_MC10_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 10 report MC errors from each channel of the integrated memory controllers.	Package	
Register Address: 444H, 1092	IA32_MC17_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 445H, 1093	IA32_MC17_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 446H, 1094	IA32_MC17_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 447H, 1095	IA32_MC17_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 448H, 1096	IA32_MC18_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.	Package	
Register Address: 449H, 1097	IA32_MC18_STATUS	

Table 2-37. Additional MSRs Supported by Intel® Xeon® Processor D with a CPUID Signature DisplayFamily_DisplayModel Value of 06_56H

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.		Package
Register Address: 44AH, 1098	IA32_MC18_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.		Package
Register Address: 44BH, 1099	IA32_MC18_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.		Package
Register Address: 44CH, 1100	IA32_MC19_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 44DH, 1101	IA32_MC19_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 44EH, 1102	IA32_MC19_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 44FH, 1103	IA32_MC19_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
See Table 2-20, Table 2-29, Table 2-34, and Table 2-36 for other MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_56H.		

NOTES:

1. An override configuration lower than the factory-set configuration is always supported. An override configuration higher than the factory-set configuration is dependent on features specific to the processor and the platform.

2.16.2 Additional MSRs Supported in Intel® Xeon® Processors E5 v4 and E7 v4 Families

The MSRs listed in Table 2-37 are available to the Intel® Xeon® Processor E5 v4 and E7 v4 Families (CPUID Signature DisplayFamily_DisplayModel value of 06_4FH). The Intel® Xeon® processor E5 v4 family is based on Broadwell microarchitecture and supports the MSR interfaces listed in Table 2-20, Table 2-21, Table 2-29, Table 2-34, Table 2-36, and Table 2-38.

Table 2-38. Additional MSRs Supported by Intel® Xeon® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4FH

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1ACH, 428	MSR_TURBO_RATIO_LIMIT3	
Config Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1.		Package
62:0	Reserved.	Package
63	Semaphore for Turbo Ratio Limit Configuration If 1, the processor uses override configuration ¹ specified in MSR_TURBO_RATIO_LIMIT, MSR_TURBO_RATIO_LIMIT1, and MSR_TURBO_RATIO_LIMIT2. If 0, the processor uses factory-set configuration (Default).	Package
Register Address: 285H, 645	IA32_MC5_CTL2	
See Table 2-2.		Package
Register Address: 286H, 646	IA32_MC6_CTL2	
See Table 2-2.		Package
Register Address: 287H, 647	IA32_MC7_CTL2	
See Table 2-2.		Package
Register Address: 288H, 648	IA32_MC8_CTL2	
See Table 2-2.		Package
Register Address: 289H, 649	IA32_MC9_CTL2	
See Table 2-2.		Package
Register Address: 28AH, 650	IA32_MC10_CTL2	
See Table 2-2.		Package
Register Address: 28BH, 651	IA32_MC11_CTL2	
See Table 2-2.		Package
Register Address: 28CH, 652	IA32_MC12_CTL2	
See Table 2-2.		Package
Register Address: 28DH, 653	IA32_MC13_CTL2	
See Table 2-2.		Package
Register Address: 28EH, 654	IA32_MC14_CTL2	
See Table 2-2.		Package
Register Address: 28FH, 655	IA32_MC15_CTL2	
See Table 2-2.		Package
Register Address: 290H, 656	IA32_MC16_CTL2	
See Table 2-2.		Package
Register Address: 291H, 657	IA32_MC17_CTL2	
See Table 2-2.		Package
Register Address: 292H, 658	IA32_MC18_CTL2	
See Table 2-2.		Package

Table 2-38. Additional MSRs Supported by Intel® Xeon® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4FH

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 293H, 659	IA32_MC19_CTL2	
See Table 2-2.		Package
Register Address: 294H, 660	IA32_MC20_CTL2	
See Table 2-2.		Package
Register Address: 295H, 661	IA32_MC21_CTL2	
See Table 2-2.		Package
Register Address: 414H, 1044	IA32_MC5_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI 0 module.		Package
Register Address: 415H, 1045	IA32_MC5_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI 0 module.		Package
Register Address: 416H, 1046	IA32_MC5_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI 0 module.		Package
Register Address: 417H, 1047	IA32_MC5_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from the Intel QPI 0 module.		Package
Register Address: 418H, 1048	IA32_MC6_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 419H, 1049	IA32_MC6_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41AH, 1050	IA32_MC6_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41BH, 1051	IA32_MC6_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.		Package
Register Address: 41CH, 1052	IA32_MC7_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41DH, 1053	IA32_MC7_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41EH, 1054	IA32_MC7_ADDR	

Table 2-38. Additional MSRs Supported by Intel® Xeon® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4FH

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 41FH, 1055	IA32_MC7_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC7 reports MC errors from the home agent HA 0.		Package
Register Address: 420H, 1056	IA32_MC8_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the home agent HA 1.		Package
Register Address: 421H, 1057	IA32_MC8_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the home agent HA 1.		Package
Register Address: 422H, 1058	IA32_MC8_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the home agent HA 1.		Package
Register Address: 423H, 1059	IA32_MC8_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the home agent HA 1.		Package
Register Address: 424H, 1060	IA32_MC9_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 425H, 1061	IA32_MC9_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 426H, 1062	IA32_MC9_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 427H, 1063	IA32_MC9_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 428H, 1064	IA32_MC10_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 429H, 1065	IA32_MC10_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42AH, 1066	IA32_MC10_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package

Table 2-38. Additional MSRs Supported by Intel® Xeon® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4FH

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 42BH, 1067	IA32_MC10_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42CH, 1068	IA32_MC11_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42DH, 1069	IA32_MC11_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42EH, 1070	IA32_MC11_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 42FH, 1071	IA32_MC11_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 430H, 1072	IA32_MC12_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 431H, 1073	IA32_MC12_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 432H, 1074	IA32_MC12_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 433H, 1075	IA32_MC12_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 434H, 1076	IA32_MC13_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 435H, 1077	IA32_MC13_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 436H, 1078	IA32_MC13_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 437H, 1079	IA32_MC13_MISC	

Table 2-38. Additional MSRs Supported by Intel® Xeon® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4FH

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 438H, 1080	IA32_MC14_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 439H, 1081	IA32_MC14_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43AH, 1082	IA32_MC14_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43BH, 1083	IA32_MC14_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43CH, 1084	IA32_MC15_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43DH, 1085	IA32_MC15_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43EH, 1086	IA32_MC15_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 43FH, 1087	IA32_MC15_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 440H, 1088	IA32_MC16_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 441H, 1089	IA32_MC16_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 442H, 1090	IA32_MC16_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package
Register Address: 443H, 1091	IA32_MC16_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 through MC 16 report MC errors from each channel of the integrated memory controllers.		Package

Table 2-38. Additional MSRs Supported by Intel® Xeon® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4FH

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 444H, 1092	IA32_MC17_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 445H, 1093	IA32_MC17_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 446H, 1094	IA32_MC17_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 447H, 1095	IA32_MC17_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC17 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo0, CBo3, CBo6, CBo9, CBo12, CBo15.	Package	
Register Address: 448H, 1096	IA32_MC18_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.	Package	
Register Address: 449H, 1097	IA32_MC18_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.	Package	
Register Address: 44AH, 1098	IA32_MC18_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.	Package	
Register Address: 44BH, 1099	IA32_MC18_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC18 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo1, CBo4, CBo7, CBo10, CBo13, CBo16.	Package	
Register Address: 44CH, 1100	IA32_MC19_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.	Package	
Register Address: 44DH, 1101	IA32_MC19_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.	Package	
Register Address: 44EH, 1102	IA32_MC19_ADDR	

Table 2-38. Additional MSRs Supported by Intel® Xeon® Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_4FH

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 44FH, 1103	IA32_MC19_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC19 reports MC errors from the following pair of CBo/L3 Slices (if the pair is present): CBo2, CBo5, CBo8, CBo11, CBo14, CBo17.		Package
Register Address: 450H, 1104	IA32_MC20_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC20 reports MC errors from the Intel QPI 1 module.		Package
Register Address: 451H, 1105	IA32_MC20_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC20 reports MC errors from the Intel QPI 1 module.		Package
Register Address: 452H, 1106	IA32_MC20_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC20 reports MC errors from the Intel QPI 1 module.		Package
Register Address: 453H, 1107	IA32_MC20_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC20 reports MC errors from the Intel QPI 1 module.		Package
Register Address: 454H, 1108	IA32_MC21_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from the Intel QPI 2 module.		Package
Register Address: 455H, 1109	IA32_MC21_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from the Intel QPI 2 module.		Package
Register Address: 456H, 1110	IA32_MC21_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from the Intel QPI 2 module.		Package
Register Address: 457H, 1111	IA32_MC21_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC21 reports MC errors from the Intel QPI 2 module.		Package
Register Address: C81H, 3201	IA32_L3_QOS_CFG	
Cache Allocation Technology Configuration (R/W)		Package
0	CAT Enable. Set 1 to enable Cache Allocation Technology.	
63:1	Reserved.	
See Table 2-20, Table 2-21, Table 2-29, and Table 2-30 for other MSR definitions applicable to processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_45H.		

NOTES:

1. An override configuration lower than the factory-set configuration is always supported. An override configuration higher than the factory-set configuration is dependent on features specific to the processor and the platform.

2.17 MSRS IN THE 6TH—13TH GENERATION INTEL® CORE™ PROCESSORS, 1ST—5TH GENERATION INTEL® XEON® SCALABLE PROCESSOR FAMILIES, INTEL® CORE™ ULTRA 7 PROCESSORS, 8TH GENERATION INTEL® CORE™ I3 PROCESSORS, INTEL® XEON® E PROCESSORS, INTEL® XEON® 6 P-CORE PROCESSORS, INTEL® XEON® 6 E-CORE PROCESSORS, AND INTEL® SERIES 2 CORE™ ULTRA PROCESSORS

6th generation Intel® Core™ processors are based on Skylake microarchitecture and have a CPUID Signature DisplayFamily_DisplayModel value of 06_4EH or 06_5EH.

The Intel® Xeon® Scalable Processor Family based on the Skylake microarchitecture, the 2nd generation Intel® Xeon® Scalable Processor Family based on the Cascade Lake product, and the 3rd generation Intel® Xeon® Scalable Processor Family based on the Cooper Lake product all have a CPUID Signature DisplayFamily_DisplayModel value of 06_55H.

7th generation Intel® Core™ processors are based on the Kaby Lake microarchitecture, 8th generation and 9th generation Intel® Core™ processors, and Intel® Xeon® E processors are based on Coffee Lake microarchitecture; these processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_8EH or 06_9EH.

8th generation Intel® Core™ i3 processors are based on Cannon Lake microarchitecture and have a CPUID Signature DisplayFamily_DisplayModel value of 06_66H.

10th generation Intel® Core™ processors are based on Comet Lake microarchitecture (with a CPUID Signature DisplayFamily_DisplayModel value of 06_A5H or 06_A6H) and Ice Lake microarchitecture (with a CPUID Signature DisplayFamily_DisplayModel value of 06_7DH or 06_7EH).

11th generation Intel® Core™ processors are based on Tiger Lake microarchitecture and have a CPUID Signature DisplayFamily_DisplayModel value of 06_8CH or 06_8DH.

The 3rd generation Intel® Xeon® Scalable Processor Family is based on Ice Lake microarchitecture and has a CPUID Signature DisplayFamily_DisplayModel value of 06_6AH or 06_6CH.

12th generation Intel® Core™ processors supporting the Alder Lake performance hybrid architecture have a CPUID Signature DisplayFamily_DisplayModel value of 06_97H or 06_9AH.

13th generation Intel® Core™ processors supporting the Raptor Lake performance hybrid architecture have a CPUID Signature DisplayFamily_DisplayModel value of 06_BAH, 06_B7H, or 06_BFH.

The 4th generation Intel® Xeon® Scalable Processor Family is based on Sapphire Rapids microarchitecture and has a CPUID Signature DisplayFamily_DisplayModel value of 06_8FH.

The 5th generation Intel® Xeon® Scalable Processor Family is based on Emerald Rapids microarchitecture and has a CPUID Signature DisplayFamily_DisplayModel value of 06_CFH.

The Intel® Core™ Ultra 7 processors supporting the Meteor Lake hybrid architecture have a CPUID Signature DisplayFamily_DisplayModel value of 06_AAH.

The Intel® Xeon® 6 P-core processor is based on the Granite Rapids microarchitecture and has a CPUID Signature DisplayFamily_DisplayModel value of 06_ADH or 06_AEH.

The Intel® Xeon® 6 E-core processor is based on the Sierra Forest microarchitecture and has a CPUID Signature DisplayFamily_DisplayModel value of 06_AFH.

The Intel® Series 2 Core™ Ultra processors supporting the Lunar Lake performance hybrid architecture have a CPUID Signature DisplayFamily_DisplayModel value of 06_BDH.

These processors support the MSR interfaces listed in Table 2-20, Table 2-21, Table 2-25, Table 2-29, Table 2-35, and Table 2-39¹. For an MSR listed in Table 2-39 that also appears in the model-specific tables of prior generations, Table 2-39 supersedes prior generation tables.

Tables 2-40 through 2-60 list additional supported MSR interfaces introduced in specific processors; see each table for additional details.

The notation of “Platform” in the Scope column (with respect to MSR_PLATFORM_ENERGY_COUNTER and MSR_PLATFORM_POWER_LIMIT) is limited to the power-delivery domain and the specifics of the power delivery integration may vary by platform vendor’s implementation.

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64 Processor (R/W) See Table 2-2.		Thread
Register Address: FEH, 254	IA32_MTRRCAP	
MTRR Capability (R/O, Architectural) See Table 2-2		Thread
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Core
0	Thermal Status (R/O) See Table 2-2.	
1	Thermal Status Log (R/WC0) See Table 2-2.	
2	PROTCHOT # or FORCEPR# Status (R/O) See Table 2-2.	
3	PROTCHOT # or FORCEPR# Log (R/WC0) See Table 2-2.	
4	Critical Temperature Status (R/O) See Table 2-2.	
5	Critical Temperature Status Log (R/WC0) See Table 2-2.	
6	Thermal threshold #1 Status (R/O) See Table 2-2.	
7	Thermal threshold #1 Log (R/WC0) See Table 2-2.	
8	Thermal Threshold #2 Status (R/O) See Table 2-2.	

1. MSRs at the following addresses are not supported in the 12th generation Intel Core processor E-core: 3F7H. MSRs at the following addresses are not supported in the 12th generation Intel Core processor E-core or P-core: 652H, 653H, 655H, 656H, DB0H, DB1H, DB2H, and D90H.

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
9	Thermal Threshold #2 Log (R/WC0) See Table 2-2.	
10	Power Limitation Status (R/O) See Table 2-2.	
11	Power Limitation Log (R/WC0) See Table 2-2.	
12	Current Limit Status (R/O) See Table 2-2.	
13	Current Limit Log (R/WC0) See Table 2-2.	
14	Cross Domain Limit Status (R/O) See Table 2-2.	
15	Cross Domain Limit Log (R/WC0) See Table 2-2.	
22:16	Digital Readout (R/O) See Table 2-2.	
26:23	Reserved.	
30:27	Resolution in Degrees Celsius (R/O) See Table 2-2.	
31	Reading Valid (R/O) See Table 2-2.	
63:32	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode R/O if MSR_PLATFORM_INFO.[28] = 0, and R/W if MSR_PLATFORM_INFO.[28] = 1		Package
7:0	Maximum Ratio Limit for 1C Maximum turbo ratio limit of 1 core active.	Package
15:8	Maximum Ratio Limit for 2C Maximum turbo ratio limit of 2 core active.	Package
23:16	Maximum Ratio Limit for 3C Maximum turbo ratio limit of 3 core active.	Package
31:24	Maximum Ratio Limit for 4C Maximum turbo ratio limit of 4 core active.	Package
63:32	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-4) that points to the MSR containing the most recent branch record.		Thread

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1FCH, 508	MSR_POWER_CTL	
Power Control Register See http://biosbits.org .		Core
0	Reserved.	
1	C1E Enable (R/W) When set to '1', will enable the CPU to switch to the Minimum Enhanced Intel SpeedStep Technology operating point when all execution cores enter MWAIT (C1).	Package
18:2	Reserved.	
19	Disable Energy Efficiency Optimization (R/W) Setting this bit disables the P-States energy efficiency optimization. Default value is 0. Disable/enable the energy efficiency optimization in P-State legacy mode (when IA32_PM_ENABLE[HWP_ENABLE] = 0), has an effect only in the turbo range or into PERF_MIN_CTL value if it is not zero set. In HWP mode (IA32_PM_ENABLE[HWP_ENABLE] == 1), has an effect between the OS desired or OS maximize to the OS minimize performance setting.	
20	Disable Race to Halt Optimization (R/W) Setting this bit disables the Race to Halt optimization and avoids this optimization limitation to execute below the most efficient frequency ratio. Default value is 0 for processors that support Race to Halt optimization.	
63:21	Reserved.	
Register Address: 300H, 768	MSR_SGXOWNERPOCH0	
Lower 64 Bit CR_SGXOWNERPOCH (W) Writes do not update CR_SGXOWNERPOCH if CPUID.(EAX=12H, ECX=0):EAX.SGX1 is 1 on any thread in the package.		Package
63:0	Lower 64 bits of an 128-bit external entropy value for key derivation of an enclave.	
Register Address: 301H, 769	MSR_SGXOWNERPOCH1	
Upper 64 Bit CR_SGXOWNERPOCH (W) Writes do not update CR_SGXOWNERPOCH if CPUID.(EAX=12H, ECX=0):EAX.SGX1 is 1 on any thread in the package.		Package
63:0	Upper 64 bits of an 128-bit external entropy value for key derivation of an enclave.	
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	
See Table 2-2 and Section 21.2.4, "Architectural Performance Monitoring Version 4."		
0	Ovf_PMC0	Thread
1	Ovf_PMC1	Thread
2	Ovf_PMC2	Thread
3	Ovf_PMC3	Thread
4	Ovf_PMC4 (if CPUID.0AH:EAX[15:8] > 4)	Thread
5	Ovf_PMC5 (if CPUID.0AH:EAX[15:8] > 5)	Thread

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
6	Ovf_PMC6 (if CPUID.0AH:EAX[15:8] > 6)	Thread
7	Ovf_PMC7 (if CPUID.0AH:EAX[15:8] > 7)	Thread
31:8	Reserved.	
32	Ovf_FixedCtr0	Thread
33	Ovf_FixedCtr1	Thread
34	Ovf_FixedCtr2	Thread
54:35	Reserved	
55	Trace_ToPA_PMI	Thread
57:56	Reserved.	
58	LBR_Frz	Thread
59	CTR_Frz	Thread
60	ASCI	Thread
61	Ovf_Uncore	Thread
62	Ovf_BufDSSAVE	Thread
63	CondChgd	Thread
Register Address: 390H, 912	IA32_PERF_GLOBAL_STATUS_RESET	
See Table 2-2 and Section 21.2.4, “Architectural Performance Monitoring Version 4.”		
0	Set 1 to clear Ovf_PMC0.	Thread
1	Set 1 to clear Ovf_PMC1.	Thread
2	Set 1 to clear Ovf_PMC2.	Thread
3	Set 1 to clear Ovf_PMC3.	Thread
4	Set 1 to clear Ovf_PMC4 (if CPUID.0AH:EAX[15:8] > 4).	Thread
5	Set 1 to clear Ovf_PMC5 (if CPUID.0AH:EAX[15:8] > 5).	Thread
6	Set 1 to clear Ovf_PMC6 (if CPUID.0AH:EAX[15:8] > 6).	Thread
7	Set 1 to clear Ovf_PMC7 (if CPUID.0AH:EAX[15:8] > 7).	Thread
31:8	Reserved.	
32	Set 1 to clear Ovf_FixedCtr0.	Thread
33	Set 1 to clear Ovf_FixedCtr1.	Thread
34	Set 1 to clear Ovf_FixedCtr2.	Thread
54:35	Reserved.	
55	Set 1 to clear Trace_ToPA_PMI.	Thread
57:56	Reserved.	
58	Set 1 to clear LBR_Frz.	Thread
59	Set 1 to clear CTR_Frz.	Thread
60	Set 1 to clear ASCI.	Thread

Table 2-39. Additional MSRs Supported by the 6th—13th Generation Intel® Core™ Processors, 1st—5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
61	Set 1 to clear Ovf_Uncore.	Thread
62	Set 1 to clear Ovf_BufDSSAVE.	Thread
63	Set 1 to clear CondChgd.	Thread
Register Address: 391H, 913	IA32_PERF_GLOBAL_STATUS_SET	
See Table 2-2 and Section 21.2.4, “Architectural Performance Monitoring Version 4.”		
0	Set 1 to cause Ovf_PMC0 = 1.	Thread
1	Set 1 to cause Ovf_PMC1 = 1.	Thread
2	Set 1 to cause Ovf_PMC2 = 1.	Thread
3	Set 1 to cause Ovf_PMC3 = 1.	Thread
4	Set 1 to cause Ovf_PMC4=1 (if CPUID.0AH:EAX[15:8] > 4).	Thread
5	Set 1 to cause Ovf_PMC5=1 (if CPUID.0AH:EAX[15:8] > 5).	Thread
6	Set 1 to cause Ovf_PMC6=1 (if CPUID.0AH:EAX[15:8] > 6).	Thread
7	Set 1 to cause Ovf_PMC7=1 (if CPUID.0AH:EAX[15:8] > 7).	Thread
31:8	Reserved.	
32	Set 1 to cause Ovf_FixedCtr0 = 1.	Thread
33	Set 1 to cause Ovf_FixedCtr1 = 1.	Thread
34	Set 1 to cause Ovf_FixedCtr2 = 1.	Thread
54:35	Reserved.	
55	Set 1 to cause Trace_ToPA_PMI = 1.	Thread
57:56	Reserved.	
58	Set 1 to cause LBR_Frz = 1.	Thread
59	Set 1 to cause CTR_Frz = 1.	Thread
60	Set 1 to cause ASCI = 1.	Thread
61	Set 1 to cause Ovf_Uncore.	Thread
62	Set 1 to cause Ovf_BufDSSAVE.	Thread
63	Reserved.	
Register Address: 392H, 914	IA32_PERF_GLOBAL_INUSE	
See Table 2-2.		Thread
Register Address: 3F7H, 1015	MSR_PEBS_FRONTEND	
FrontEnd Precise Event Condition Select (R/W)		Thread
2:0	Event Code Select	
3	Reserved	
4	Event Code Select High	
7:5	Reserved.	
19:8	IDQ_Bubble_Length Specifier	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
22:20	IDQ_Bubble_Width Specifier	
63:23	Reserved.	
Register Address: 500H, 1280	IA32_SGX_SVN_STATUS	
Status and SVN Threshold of SGX Support for ACM (R/O)		Thread
0	Lock See Section 40.11.3, "Interactions with Authenticated Code Modules (ACMs)."	
15:1	Reserved.	
23:16	SGX_SVN_SINIT See Section 40.11.3, "Interactions with Authenticated Code Modules (ACMs)."	
63:24	Reserved.	
Register Address: 560H, 1376	IA32_RTIT_OUTPUT_BASE	
Trace Output Base Register (R/W) See Table 2-2.		Thread
Register Address: 561H, 1377	IA32_RTIT_OUTPUT_MASK_PTRS	
Trace Output Mask Pointers Register (R/W) See Table 2-2.		Thread
Register Address: 570H, 1392	IA32_RTIT_CTL	
Trace Control Register (R/W)		Thread
0	TraceEn	
1	CYCEn	
2	OS	
3	User	
6:4	Reserved, must be zero.	
7	CR3Filter	
8	ToPA Writing 0 will #GP if also setting TraceEn.	
9	MTCEn	
10	TSCEn	
11	DisRETC	
12	Reserved, must be zero.	
13	BranchEn	
17:14	MTCFreq	
18	Reserved, must be zero.	
22:19	CycThresh	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
23	Reserved, must be zero.	
27:24	PSBFreq	
31:28	Reserved, must be zero.	
35:32	ADDR0_CFG	
39:36	ADDR1_CFG	
63:40	Reserved, must be zero.	
Register Address: 571H, 1393	IA32_RTIT_STATUS	
Tracing Status Register (R/W)		Thread
0	FilterEn, writes ignored.	
1	ContexEn, writes ignored.	
2	TriggerEn, writes ignored.	
3	Reserved	
4	Error (R/W)	
5	Stopped	
31:6	Reserved, must be zero.	
48:32	PacketByteCnt	
63:49	Reserved, must be zero.	
Register Address: 572H, 1394	IA32_RTIT_CR3_MATCH	
Trace Filter CR3 Match Register (R/W)		Thread
4:0	Reserved	
63:5	CR3[63:5] value to match	
Register Address: 580H, 1408	IA32_RTIT_ADDR0_A	
Region 0 Start Address (R/W)		Thread
63:0	See Table 2-2.	
Register Address: 581H, 1409	IA32_RTIT_ADDR0_B	
Region 0 End Address (R/W)		Thread
63:0	See Table 2-2.	
Register Address: 582H, 1410	IA32_RTIT_ADDR1_A	
Region 1 Start Address (R/W)		Thread
63:0	See Table 2-2.	
Register Address: 583H, 1411	IA32_RTIT_ADDR1_B	
Region 1 End Address (R/W)		Thread
63:0	See Table 2-2.	
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
PP0 Energy Status (R/O) See Section 16.10.4, “PP0/PP1 RAPL Domains.”		Package
Register Address: 64DH, 1613	MSR_PLATFORM_ENERGY_COUNTER	
Platform Energy Counter (R/O) This MSR is valid only if both platform vendor hardware implementation and BIOS enablement support it. This MSR will read 0 if not valid.		Platform
31:0	Total energy consumed by all devices in the platform that receive power from integrated power delivery mechanism, included platform devices are processor cores, SOC, memory, add-on or peripheral devices that get powered directly from the platform power delivery means. The energy units are specified in the MSR_RAPL_POWER_UNIT.Energy_Status_Unit.	
63:32	Reserved.	
Register Address: 64EH, 1614	MSR_PPERF	
Productive Performance Count (R/O)		Thread
63:0	Hardware’s view of workload scalability. See Section 16.4.5.1.	
Register Address: 64FH, 1615	MSR_CORE_PERF_LIMIT_REASONS	
	Indicator of Frequency Clipping in Processor Cores (R/W) (Frequency refers to processor core frequency.)	Package
0	PROCHOT Status (R0) When set, frequency is reduced below the operating system request due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced below the operating system request due to a thermal event.	
3:2	Reserved.	
4	Residency State Regulation Status (R0) When set, frequency is reduced below the operating system request due to residency state regulation limit.	
5	Running Average Thermal Limit Status (R0) When set, frequency is reduced below the operating system request due to Running Average Thermal Limit (RATL).	
6	VR Therm Alert Status (R0) When set, frequency is reduced below the operating system request due to a thermal alert from a processor Voltage Regulator (VR).	
7	VR Therm Design Current Status (R0) When set, frequency is reduced below the operating system request due to VR thermal design current limit.	
8	Other Status (R0) When set, frequency is reduced below the operating system request due to electrical or other constraints.	

Table 2-39. Additional MSRs Supported by the 6th—13th Generation Intel® Core™ Processors, 1st—5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
9	Reserved.	
10	Package/Platform-Level Power Limiting PL1 Status (R0) When set, frequency is reduced below the operating system request due to package/platform-level power limiting PL1.	
11	Package/Platform-Level PL2 Power Limiting Status (R0) When set, frequency is reduced below the operating system request due to package/platform-level power limiting PL2/PL3.	
12	Max Turbo Limit Status (R0) When set, frequency is reduced below the operating system request due to multi-core turbo limits.	
13	Turbo Transition Attenuation Status (R0) When set, frequency is reduced below the operating system request due to Turbo transition attenuation. This prevents performance degradation due to frequent operating ratio changes.	
15:14	Reserved.	
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
19:18	Reserved.	
20	Residency State Regulation Log When set, indicates that the Residency State Regulation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
21	Running Average Thermal Limit Log When set, indicates that the RATL Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	VR Thermal Design Current Log When set, indicates that the VR TDC Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
24	Other Log When set, indicates that the Other Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Reserved.	
26	Package/Platform-Level PL1 Power Limiting Log When set, indicates that the Package or Platform Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
27	Package/Platform-Level PL2 Power Limiting Log When set, indicates that the Package or Platform Level PL2/PL3 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28	Max Turbo Limit Log When set, indicates that the Max Turbo Limit Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
29	Turbo Transition Attenuation Log When set, indicates that the Turbo Transition Attenuation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:30	Reserved.	
Register Address: 652H, 1618	MSR_PKG_HDC_CONFIG	
HDC Configuration (R/W)		Package
2:0	PKG_Cx_Monitor Configures Package Cx state threshold for MSR_PKG_HDC_DEEP_RESIDENCY.	
63:3	Reserved.	
Register Address: 653H, 1619	MSR_CORE_HDC_RESIDENCY	
Core HDC Idle Residency (R/O)		Core
63:0	Core_Cx_Duty_Cycle_Cnt	
Register Address: 655H, 1621	MSR_PKG_HDC_SHALLOW_RESIDENCY	
Accumulate the cycles the package was in C2 state and at least one logical processor was in forced idle (R/O)		Package
63:0	Pkg_C2_Duty_Cycle_Cnt	
Register Address: 656H, 1622	MSR_PKG_HDC_DEEP_RESIDENCY	
Package Cx HDC Idle Residency (R/O)		Package
63:0	Pkg_Cx_Duty_Cycle_Cnt	
Register Address: 658H, 1624	MSR_WEIGHTED_CORE_CO	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Core-count Weighted C0 Residency (R/O)		Package
63:0	Increment at the same rate as the TSC. The increment each cycle is weighted by the number of processor cores in the package that reside in C0. If N cores are simultaneously in C0, then each cycle the counter increments by N.	
Register Address: 659H, 1625	MSR_ANY_CORE_C0	
Any Core C0 Residency (R/O)		Package
63:0	Increment at the same rate as the TSC. The increment each cycle is one if any processor core in the package is in C0.	
Register Address: 65AH, 1626	MSR_ANY GFXE_C0	
Any Graphics Engine C0 Residency (R/O)		Package
63:0	Increment at the same rate as the TSC. The increment each cycle is one if any processor graphic device's compute engines are in C0.	
Register Address: 65BH, 1627	MSR_CORE GFXE_OVERLAP_C0	
Core and Graphics Engine Overlapped C0 Residency (R/O)		Package
63:0	Increment at the same rate as the TSC. The increment each cycle is one if at least one compute engine of the processor graphics is in C0 and at least one processor core in the package is also in C0.	
Register Address: 65CH, 1628	MSR_PLATFORM_POWER_LIMIT	
Platform Power Limit Control (R/W-L) Allows platform BIOS to limit power consumption of the platform devices to the specified values. The Long Duration power consumption is specified via Platform_Power_Limit_1 and Platform_Power_Limit_1_Time. The Short Duration power consumption limit is specified via the Platform_Power_Limit_2 with duration chosen by the processor. The processor implements an exponential-weighted algorithm in the placement of the time windows.		Platform
14:0	Platform Power Limit #1 Average Power limit value which the platform must not exceed over a time window as specified by Power_Limit_1_TIME field. The default value is the Thermal Design Power (TDP) and varies with product skus. The unit is specified in MSR_RAPLPOWER_UNIT.	
15	Enable Platform Power Limit #1 When set, enables the processor to apply control policy such that the platform power does not exceed Platform Power limit #1 over the time window specified by Power Limit #1 Time Window.	
16	Platform Clamping Limitation #1 When set, allows the processor to go below the OS requested P states in order to maintain the power below specified Platform Power Limit #1 value. This bit is writeable only when CPUID (EAX=6):EAX[4] is set.	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
23:17	Time Window for Platform Power Limit #1 Specifies the duration of the time window over which Platform Power Limit 1 value should be maintained for sustained long duration. This field is made up of two numbers from the following equation: Time Window = (float) ((1+(X/4))*(2^Y)), where: X = POWER_LIMIT_1_TIME[23:22] Y = POWER_LIMIT_1_TIME[21:17] The maximum allowed value in this field is defined in MSR_PKG_POWER_INFO[PKG_MAX_WIN]. The default value is 0DH, and the unit is specified in MSR_RAPL_POWER_UNIT[Time Unit].	
31:24	Reserved.	
46:32	Platform Power Limit #2 Average Power limit value which the platform must not exceed over the Short Duration time window chosen by the processor. The recommended default value is 1.25 times the Long Duration Power Limit (i.e., Platform Power Limit # 1).	
47	Enable Platform Power Limit #2 When set, enables the processor to apply control policy such that the platform power does not exceed Platform Power limit #2 over the Short Duration time window.	
48	Platform Clamping Limitation #2 When set, allows the processor to go below the OS requested P states in order to maintain the power below specified Platform Power Limit #2 value.	
62:49	Reserved.	
63	Lock. Setting this bit will lock all other bits of this MSR until system RESET.	
Register Address: 690H, 1680	MSR_LASTBRANCH_16_FROM_IP	
Last Branch Record 16 From IP (R/W) One of 32 triplets of last branch record registers on the last branch record stack. This part of the stack contains pointers to the source instruction. See also: - Last Branch Record Stack TOS at 1C9H. - Section 19.12.		Thread
Register Address: 691H, 1681	MSR_LASTBRANCH_17_FROM_IP	
Last Branch Record 17 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 692H, 1682	MSR_LASTBRANCH_18_FROM_IP	
Last Branch Record 18 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 693H, 1683	MSR_LASTBRANCH_19_FROM_IP	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 19 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 694H, 1684	MSR_LASTBRANCH_20_FROM_IP	
Last Branch Record 20 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 695H, 1685	MSR_LASTBRANCH_21_FROM_IP	
Last Branch Record 21 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 696H, 1686	MSR_LASTBRANCH_22_FROM_IP	
Last Branch Record 22 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 697H, 1687	MSR_LASTBRANCH_23_FROM_IP	
Last Branch Record 23 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 698H, 1688	MSR_LASTBRANCH_24_FROM_IP	
Last Branch Record 24 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 699H, 1689	MSR_LASTBRANCH_25_FROM_IP	
Last Branch Record 25 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 69AH, 1690	MSR_LASTBRANCH_26_FROM_IP	
Last Branch Record 26 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 69BH, 1691	MSR_LASTBRANCH_27_FROM_IP	
Last Branch Record 27 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 69CH, 1692	MSR_LASTBRANCH_28_FROM_IP	
Last Branch Record 28 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 69DH, 1693	MSR_LASTBRANCH_29_FROM_IP	
Last Branch Record 29 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 69EH, 1694	MSR_LASTBRANCH_30_FROM_IP	
Last Branch Record 30 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 69FH, 1695	MSR_LASTBRANCH_31_FROM_IP	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 31 From IP (R/W) See description of MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 6BOH, 1712	MSR_GRAPHICS_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in the Processor Graphics (R/W) (Frequency refers to processor graphics frequency.)		Package
0	PROCHOT Status (R0) When set, frequency is reduced due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced due to a thermal event.	
4:2	Reserved.	
5	Running Average Thermal Limit Status (R0) When set, frequency is reduced due to running average thermal limit.	
6	VR Therm Alert Status (R0) When set, frequency is reduced due to a thermal alert from a processor Voltage Regulator.	
7	VR Thermal Design Current Status (R0) When set, frequency is reduced due to VR TDC limit.	
8	Other Status (R0) When set, frequency is reduced due to electrical or other constraints.	
9	Reserved.	
10	Package/Platform-Level Power Limiting PL1 Status (R0) When set, frequency is reduced due to package/platform-level power limiting PL1.	
11	Package/Platform-Level PL2 Power Limiting Status (R0) When set, frequency is reduced due to package/platform-level power limiting PL2/PL3.	
12	Inefficient Operation Status (R0) When set, processor graphics frequency is operating below target frequency.	
15:13	Reserved.	
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
20:18	Reserved.	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
21	Running Average Thermal Limit Log When set, indicates that the RATL Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	VR Thermal Design Current Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
24	Other Log When set, indicates that the OTHER Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Reserved.	
26	Package/Platform-Level PL1 Power Limiting Log When set, indicates that the Package/Platform Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
27	Package/Platform-Level PL2 Power Limiting Log When set, indicates that the Package/Platform Level PL2 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28	Inefficient Operation Log When set, indicates that the Inefficient Operation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:29	Reserved.	
Register Address: 6B1H, 1713	MSR_RING_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in the Ring Interconnect (R/W) (Frequency refers to ring interconnect in the uncore.)		Package
0	PROCHOT Status (R0) When set, frequency is reduced due to assertion of external PROCHOT.	
1	Thermal Status (R0) When set, frequency is reduced due to a thermal event.	
4:2	Reserved.	
5	Running Average Thermal Limit Status (R0) When set, frequency is reduced due to running average thermal limit.	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
6	VR Therm Alert Status (R0) When set, frequency is reduced due to a thermal alert from a processor Voltage Regulator.	
7	VR Thermal Design Current Status (R0) When set, frequency is reduced due to VR TDC limit.	
8	Other Status (R0) When set, frequency is reduced due to electrical or other constraints.	
9	Reserved.	
10	Package/Platform-Level Power Limiting PL1 Status (R0) When set, frequency is reduced due to package/Platform-level power limiting PL1.	
11	Package/Platform-Level PL2 Power Limiting Status (R0) When set, frequency is reduced due to package/Platform-level power limiting PL2/PL3.	
15:12	Reserved	
16	PROCHOT Log When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
20:18	Reserved.	
21	Running Average Thermal Limit Log When set, indicates that the RATL Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR Therm Alert Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	VR Thermal Design Current Log When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
24	Other Log When set, indicates that the OTHER Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	

Table 2-39. Additional MSRs Supported by the 6th—13th Generation Intel® Core™ Processors, 1st—5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
25	Reserved.	
26	Package/Platform-Level PL1 Power Limiting Log When set, indicates that the Package/Platform Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
27	Package/Platform-Level PL2 Power Limiting Log When set, indicates that the Package/Platform Level PL2 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:28	Reserved.	
Register Address: 6D0H, 1744	MSR_LASTBRANCH_16_TO_IP	
Last Branch Record 16 To IP (R/W) One of 32 triplets of last branch record registers on the last branch record stack. This part of the stack contains pointers to the destination instruction. See also: - Last Branch Record Stack TOS at 1C9H. - Section 19.12.		Thread
Register Address: 6D1H, 1745	MSR_LASTBRANCH_17_TO_IP	
Last Branch Record 17 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6D2H, 1746	MSR_LASTBRANCH_18_TO_IP	
Last Branch Record 18 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6D3H, 1747	MSR_LASTBRANCH_19_TO_IP	
Last Branch Record 19 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6D4H, 1748	MSR_LASTBRANCH_20_TO_IP	
Last Branch Record 20 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6D5H, 1749	MSR_LASTBRANCH_21_TO_IP	
Last Branch Record 21 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6D6H, 1750	MSR_LASTBRANCH_22_TO_IP	
Last Branch Record 22 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6D7H, 1751	MSR_LASTBRANCH_23_TO_IP	
Last Branch Record 23 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6D8H, 1752	MSR_LASTBRANCH_24_TO_IP	

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Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 24 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6D9H, 1753	MSR_LASTBRANCH_25_TO_IP	
Last Branch Record 25 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6DAH, 1754	MSR_LASTBRANCH_26_TO_IP	
Last Branch Record 26 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6DBH, 1755	MSR_LASTBRANCH_27_TO_IP	
Last Branch Record 27 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6DCH, 1756	MSR_LASTBRANCH_28_TO_IP	
Last Branch Record 28 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6DDH, 1757	MSR_LASTBRANCH_29_TO_IP	
Last Branch Record 29 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6DEH, 1758	MSR_LASTBRANCH_30_TO_IP	
Last Branch Record 30 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 6DFH, 1759	MSR_LASTBRANCH_31_TO_IP	
Last Branch Record 31 To IP (R/W) See description of MSR_LASTBRANCH_0_TO_IP.		Thread
Register Address: 770H, 1904	IA32_PM_ENABLE	
See Section 16.4.2, “Enabling HWP.”		Package
Register Address: 771H, 1905	IA32_HWP_CAPABILITIES	
See Section 16.4.3, “HWP Performance Range and Dynamic Capabilities.”		Thread
Register Address: 772H, 1906	IA32_HWP_REQUEST_PKG	
See Section 16.4.4, “Managing HWP.”		Package
Register Address: 773H, 1907	IA32_HWP_INTERRUPT	
See Section 16.4.6, “HWP Notifications.”		Thread
Register Address: 774H, 1908	IA32_HWP_REQUEST	
See Section 16.4.4, “Managing HWP.”		Thread
7:0	Minimum Performance (R/W)	
15:8	Maximum Performance (R/W)	
23:16	Desired Performance (R/W)	

Table 2-39. Additional MSRs Supported by the 6th—13th Generation Intel® Core™ Processors, 1st—5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
31:24	Energy/Performance Preference (R/W)	
41:32	Activity Window (R/W)	
42	Package Control (R/W)	
63:43	Reserved.	
Register Address: 777H, 1911	IA32_HWP_STATUS	
See Section 16.4.5, “HWP Feedback.”		Thread
Register Address: D90H, 3472	IA32_BNDCFGS	
See Table 2-2.		Thread
Register Address: DA0H, 3488	IA32_XSS	
See Table 2-2.		Thread
Register Address: DBOH, 3504	IA32_PKG_HDC_CTL	
See Section 16.5.2, “Package level Enabling HDC.”		Package
Register Address: DB1H, 3505	IA32_PM_CTL1	
See Section 16.5.3, “Logical-Processor Level HDC Control.”		Thread
Register Address: DB2H, 3506	IA32_THREAD_STALL	
See Section 16.5.4.1, “IA32_THREAD_STALL.”		Thread
Register Address: DCOH, 3520	MSR_LBR_INFO_0	
Last Branch Record 0 Additional Information (R/W) One of 32 triplet of last branch record registers on the last branch record stack. This part of the stack contains flag, TSX-related and elapsed cycle information. See also: ▪ Last Branch Record Stack TOS at 1C9H. ▪ Section 19.9.1, “LBR Stack.”		Thread
Register Address: DC1H, 3521	MSR_LBR_INFO_1	
Last Branch Record 1 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DC2H, 3522	MSR_LBR_INFO_2	
Last Branch Record 2 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DC3H, 3523	MSR_LBR_INFO_3	
Last Branch Record 3 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DC4H, 3524	MSR_LBR_INFO_4	
Last Branch Record 4 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DC5H, 3525	MSR_LBR_INFO_5	
Last Branch Record 5 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: DC6H, 3526	MSR_LBR_INFO_6	
Last Branch Record 6 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DC7H, 3527	MSR_LBR_INFO_7	
Last Branch Record 7 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DC8H, 3528	MSR_LBR_INFO_8	
Last Branch Record 8 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DC9H, 3529	MSR_LBR_INFO_9	
Last Branch Record 9 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DCAH, 3530	MSR_LBR_INFO_10	
Last Branch Record 10 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DCBH, 3531	MSR_LBR_INFO_11	
Last Branch Record 11 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DCCH, 3532	MSR_LBR_INFO_12	
Last Branch Record 12 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DCDH, 3533	MSR_LBR_INFO_13	
Last Branch Record 13 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DCEH, 3534	MSR_LBR_INFO_14	
Last Branch Record 14 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DCFH, 3535	MSR_LBR_INFO_15	
Last Branch Record 15 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DDOH, 3536	MSR_LBR_INFO_16	
Last Branch Record 16 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD1H, 3537	MSR_LBR_INFO_17	
Last Branch Record 17 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD2H, 3538	MSR_LBR_INFO_18	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 18 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD3H, 3539	MSR_LBR_INFO_19	
Last Branch Record 19 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD4H, 3540	MSR_LBR_INFO_20	
Last Branch Record 20 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD5H, 3541	MSR_LBR_INFO_21	
Last Branch Record 21 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD6H, 3542	MSR_LBR_INFO_22	
Last Branch Record 22 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD7H, 3543	MSR_LBR_INFO_23	
Last Branch Record 23 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD8H, 3544	MSR_LBR_INFO_24	
Last Branch Record 24 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DD9H, 3545	MSR_LBR_INFO_25	
Last Branch Record 25 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DDAH, 3546	MSR_LBR_INFO_26	
Last Branch Record 26 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DDBH, 3547	MSR_LBR_INFO_27	
Last Branch Record 27 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DDCH, 3548	MSR_LBR_INFO_28	
Last Branch Record 28 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DDDH, 3549	MSR_LBR_INFO_29	
Last Branch Record 29 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DDEH, 3550	MSR_LBR_INFO_30	

Table 2-39. Additional MSRs Supported by the 6th–13th Generation Intel® Core™ Processors, 1st–5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 E-Core Processors, Intel® Xeon® 6 P-Core Processors, and Intel® Series 2 Core™ Ultra Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Last Branch Record 30 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread
Register Address: DDFH, 3551	MSR_LBR_INFO_31	
Last Branch Record 31 Additional Information (R/W) See description of MSR_LBR_INFO_0.		Thread

Table 2-40 lists the MSRs of uncore PMU for Intel processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_4EH, 06_5EH, 06_8EH, 06_9EH, or 06_66H.

Table 2-40. Uncore PMU MSRs Supported by 6th Generation, 7th Generation, and 8th Generation Intel® Core™ Processors, and 8th generation Intel® Core™ i3 Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 394H, 916	MSR_UNC_PERF_FIXED_CTRL	
Uncore Fixed Counter Control (R/W)		Package
19:0	Reserved.	
20	Enable overflow propagation.	
21	Reserved.	
22	Enable counting.	
63:23	Reserved.	
Register Address: 395H, 917	MSR_UNC_PERF_FIXED_CTR	
Uncore Fixed Counter		Package
43:0	Current count.	
63:44	Reserved.	
Register Address: 396H, 918	MSR_UNC_CBO_CONFIG	
Uncore C-Box Configuration Information (R/O)		Package
3:0	Specifies the number of C-Box units with programmable counters (including processor cores and processor graphics).	
63:4	Reserved.	
Register Address: 3B0H, 946	MSR_UNC_ARB_PERFCTR0	
Uncore Arb Unit, Performance Counter 0		Package
Register Address: 3B1H, 947	MSR_UNC_ARB_PERFCTR1	
Uncore Arb Unit, Performance Counter 1		Package
Register Address: 3B2H, 944	MSR_UNC_ARB_PERFEVTSEL0	
Uncore Arb Unit, Counter 0 Event Select MSR		Package
Register Address: 3B3H, 945	MSR_UNC_ARB_PERFEVTSEL1	
Uncore Arb Unit, Counter 1 Event Select MSR		Package

Table 2-40. Uncore PMU MSRs Supported by 6th Generation, 7th Generation, and 8th Generation Intel® Core™ Processors, and 8th generation Intel® Core™ i3 Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 700H, 1792	MSR_UNC_CBO_0_PERFEVTSEL0	
Uncore C-Box 0, Counter 0 Event Select MSR		Package
Register Address: 701H, 1793	MSR_UNC_CBO_0_PERFEVTSEL1	
Uncore C-Box 0, Counter 1 Event Select MSR		Package
Register Address: 706H, 1798	MSR_UNC_CBO_0_PERFCTR0	
Uncore C-Box 0, Performance Counter 0		Package
Register Address: 707H, 1799	MSR_UNC_CBO_0_PERFCTR1	
Uncore C-Box 0, Performance Counter 1		Package
Register Address: 710H, 1808	MSR_UNC_CBO_1_PERFEVTSEL0	
Uncore C-Box 1, Counter 0 Event Select MSR		Package
Register Address: 711H, 1809	MSR_UNC_CBO_1_PERFEVTSEL1	
Uncore C-Box 1, Counter 1 Event Select MSR		Package
Register Address: 716H, 1814	MSR_UNC_CBO_1_PERFCTR0	
Uncore C-Box 1, Performance Counter 0		Package
Register Address: 717H, 1815	MSR_UNC_CBO_1_PERFCTR1	
Uncore C-Box 1, Performance Counter 1		Package
Register Address: 720H, 1824	MSR_UNC_CBO_2_PERFEVTSEL0	
Uncore C-Box 2, Counter 0 Event Select MSR		Package
Register Address: 721H, 1825	MSR_UNC_CBO_2_PERFEVTSEL1	
Uncore C-Box 2, Counter 1 Event Select MSR		Package
Register Address: 726H, 1830	MSR_UNC_CBO_2_PERFCTR0	
Uncore C-Box 2, Performance Counter 0		Package
Register Address: 727H, 1831	MSR_UNC_CBO_2_PERFCTR1	
Uncore C-Box 2, Performance Counter 1		Package
Register Address: 730H, 1840	MSR_UNC_CBO_3_PERFEVTSEL0	
Uncore C-Box 3, Counter 0 Event Select MSR		Package
Register Address: 731H, 1841	MSR_UNC_CBO_3_PERFEVTSEL1	
Uncore C-Box 3, Counter 1 Event Select MSR		Package
Register Address: 736H, 1846	MSR_UNC_CBO_3_PERFCTR0	
Uncore C-Box 3, Performance Counter 0		Package
Register Address: 737H, 1847	MSR_UNC_CBO_3_PERFCTR1	
Uncore C-Box 3, Performance Counter 1		Package
Register Address: E01H, 3585	MSR_UNC_PERF_GLOBAL_CTRL	
Uncore PMU Global Control		Package
0	Slice 0 select.	
1	Slice 1 select.	
2	Slice 2 select.	

Table 2-40. Uncore PMU MSRs Supported by 6th Generation, 7th Generation, and 8th Generation Intel® Core™ Processors, and 8th generation Intel® Core™ i3 Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
3	Slice 3 select.	
4	Slice 4select.	
18:5	Reserved.	
29	Enable all uncore counters.	
30	Enable wake on PMI.	
31	Enable Freezing counter when overflow.	
63:32	Reserved.	
Register Address: E02H, 3586	MSR_UNC_PERF_GLOBAL_STATUS	
Uncore PMU Main Status		Package
0	Fixed counter overflowed.	
1	An ARB counter overflowed.	
2	Reserved.	
3	A CBox counter overflowed (on any slice).	
63:4	Reserved.	

2.17.1 MSRs Introduced in 7th Generation and 8th Generation Intel® Core™ Processors Based on Kaby Lake Microarchitecture and Coffee Lake Microarchitecture

Table 2-41 lists additional MSRs for 7th generation and 8th generation Intel Core processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_8EH or 06_9EH. For an MSR listed in Table 2-41 that also appears in the model-specific tables of prior generations, Table 2-41 supersedes prior generation tables.

Table 2-41. Additional MSRs Supported by the 7th Generation and 8th Generation Intel® Core™ Processors Based on Kaby Lake Microarchitecture and Coffee Lake Microarchitecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 80H, 128	MSR_TRACE_HUB_STH ACPIBAR_BASE	
NPK Address Used by AET Messages (R/W)		Package
0	Lock Bit If set, this MSR cannot be re-written anymore. Lock bit has to be set in order for the AET packets to be directed to NPK MMIO.	
17:1	Reserved.	
63:18	ACPIBAR_BASE_ADDRESS AET target address in NPK MMIO space.	
Register Address: 1F4H, 500	MSR_PRMRR_PHYS_BASE	
Processor Reserved Memory Range Register - Physical Base Control Register (R/W)		Core
2:0	MemType PRMRR BASE MemType.	
11:3	Reserved.	

Table 2-41. Additional MSRs Supported by the 7th Generation and 8th Generation Intel® Core™ Processors Based on Kaby Lake Microarchitecture and Coffee Lake Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
45:12	Base PRMRR Base Address.	
63:46	Reserved.	
Register Address: 1F5H, 501	MSR_PRMRR_PHYS_MASK	
Processor Reserved Memory Range Register - Physical Mask Control Register (R/W)		Core
9:0	Reserved.	
10	Lock Lock bit for the PRMRR.	
11	VLD Enable bit for the PRMRR.	
45:12	Mask PRMRR MASK bits.	
63:46	Reserved.	
Register Address: 1FBH, 507	MSR_PRMRR_VALID_CONFIG	
Valid PRMRR Configurations (R/W)		Core
0	1M supported MEE size.	
4:1	Reserved.	
5	32M supported MEE size.	
6	64M supported MEE size.	
7	128M supported MEE size.	
31:8	Reserved.	
Register Address: 2F4H, 756	MSR_UNCORE_PRMRR_PHYS_BASE ¹	
(R/W) The PRMRR range is used to protect the processor reserved memory from unauthorized reads and writes. Any IO access to this range is aborted. This register controls the location of the PRMRR range by indicating its starting address. It functions in tandem with the PRMRR mask register.		Package
11:0	Reserved.	
PAWIDTH-1:12	Range Base This field corresponds to bits PAWIDTH-1:12 of the base address memory range which is allocated to PRMRR memory.	
63:PAWIDTH	Reserved.	
Register Address: 2F5H, 757	MSR_UNCORE_PRMRR_PHYS_MASK ¹	
(R/W) This register controls the size of the PRMRR range by indicating which address bits must match the PRMRR base register value.		Package
9:0	Reserved.	
10	Lock Setting this bit locks all writeable settings in this register, including itself.	

Table 2-41. Additional MSRs Supported by the 7th Generation and 8th Generation Intel® Core™ Processors Based on Kaby Lake Microarchitecture and Coffee Lake Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
11	Range_En Indicates whether the PRMRR range is enabled and valid.	
38:12	Range_Mask This field indicates which address bits must match PRMRR base in order to qualify as an PRMRR access.	
63:39	Reserved.	
Register Address: 620H, 1568	MSR_RING_RATIO_LIMIT	
Ring Ratio Limit (R/W) This register provides Min/Max Ratio Limits for the LLC and Ring.		Package
6:0	MAX_Ratio This field is used to limit the max ratio of the LLC/Ring.	
7	Reserved.	
14:8	MIN_Ratio Writing to this field controls the minimum possible ratio of the LLC/Ring.	
63:15	Reserved.	

NOTES:

1. This MSR is specific to 7th generation and 8th generation Intel® Core™ processors.

2.17.2 MSRs Specific to 8th Generation Intel® Core™ i3 Processors

Table 2-42 lists additional MSRs for 8th generation Intel Core i3 processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_66H. For an MSR listed in Table 2-42 that also appears in the model-specific tables of prior generations, Table 2-42 supersedes prior generation tables.

Table 2-42. Additional MSRs Supported by the 8th Generation Intel® Core™ i3 Processors Based on Cannon Lake Microarchitecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64 Processor (R/W) See Table 2-2.		Thread
0	Lock (R/WL)	
1	Enable VMX Inside SMX Operation (R/WL)	
2	Enable VMX Outside SMX Operation (R/WL)	
14:8	SENTER Local Functions Enables (R/WL)	
15	SENTER Global Functions Enable (R/WL)	
17	SGX Launch Control Enable (R/WL) This bit must be set to enable runtime reconfiguration of SGX Launch Control via IA32_SGXLEPUBKEYHASHn MSR. Available only if CPUID.(EAX=07H, ECX=0H): ECX[30] = 1.	

**Table 2-42. Additional MSRs Supported by the 8th Generation Intel® Core™ i3 Processors
Based on Cannon Lake Microarchitecture (Contd.)**

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
18	SGX Global Functions Enable (R/WL)	
63:21	Reserved.	
Register Address: 350H, 848	MSR_BR_DETECT_CTRL	
Branch Monitoring Global Control (R/W)		
0	EnMonitoring Global enable for branch monitoring.	
1	EnExcept Enable branch monitoring event signaling on threshold trip. The branch monitoring event handler is signaled via the existing PMI signaling mechanism as programmed from the corresponding local APIC LVT entry.	
2	EnLBRFrz Enable LBR freeze on threshold trip. This will cause the LBR frozen bit 58 to be set in IA32_PERF_GLOBAL_STATUS when a triggering condition occurs and this bit is enabled.	
3	DisableInGuest When set to '1', branch monitoring, event triggering and LBR freeze actions are disabled when operating at VMX non-root operation.	
7:4	Reserved.	
17:8	WindowSize Window size defined by WindowCntSel. Values 0 - 1023 are supported. Once the Window counter reaches the WindowSize count both the Window Counter and all Branch Monitoring Counters are cleared.	
23:18	Reserved.	
25:24	WindowCntSel Window event count select: '00 = Instructions retired. '01 = Branch instructions retired '10 = Return instructions retired. '11 = Indirect branch instructions retired.	
26	CntAndMode When set to '1', the overall branch monitoring event triggering condition is true only if all enabled counters' threshold conditions are true. When '0', the threshold tripping condition is true if any enabled counters' threshold is true.	
63:27	Reserved.	
Register Address: 351H, 849	MSR_BR_DETECT_STATUS	
Branch Monitoring Global Status (R/W)		
0	Branch Monitoring Event Signaled When set to '1', Branch Monitoring event signaling is blocked until this bit is cleared by software.	

**Table 2-42. Additional MSRs Supported by the 8th Generation Intel® Core™ i3 Processors
Based on Cannon Lake Microarchitecture (Contd.)**

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
1	LBRsValid This status bit is set to '1' if the LBR state is considered valid for sampling by branch monitoring software.	
7:2	Reserved.	
8	CntrHit0 Branch monitoring counter #0 threshold hit. This status bit is sticky and once set requires clearing by software. Counter operation continues independent of the state of the bit.	
9	CntrHit1 Branch monitoring counter #1 threshold hit. This status bit is sticky and once set requires clearing by software. Counter operation continues independent of the state of the bit.	
15:10	Reserved. Reserved for additional branch monitoring counters threshold hit status.	
25:16	CountWindow The current value of the window counter. The count value is frozen on a valid branch monitoring triggering condition. This is a 10-bit unsigned value.	
31:26	Reserved. Reserved for future extension of CountWindow.	
39:32	Count0 The current value of counter 0 updated after each occurrence of the event being counted. The count value is frozen on a valid branch monitoring triggering condition (in which case CntrHit0 will also be set). This is an 8-bit signed value (2's complement). Heuristic events which only increment will saturate and freeze at maximum value 0xFF (256). RET-CALL event counter saturate at maximum value 0x7F (+127) and minimum value 0x80 (-128).	
47:40	Count1 The current value of counter 1 updated after each occurrence of the event being counted. The count value is frozen on a valid branch monitoring triggering condition (in which case CntrHit1 will also be set). This is an 8-bit signed value (2's complement). Heuristic events which only increment will saturate and freeze at maximum value 0xFF (256). RET-CALL event counter saturate at maximum value 0x7F (+127) and minimum value 0x80 (-128).	
63:48	Reserved.	
Register Address: 354H–355H, 852–853	MSR_BR_DETECT_COUNTER_CONFIG_i	
Branch Monitoring Detect Counter Configuration (R/W)		
0	CntrEn Enable counter.	

**Table 2-42. Additional MSRs Supported by the 8th Generation Intel® Core™ i3 Processors
Based on Cannon Lake Microarchitecture (Contd.)**

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
7:1	CntrEvSel Event select (other values #GP) '0000000 = RETs. '0000001 = RET-CALL bias. '0000010 = RET mispredicts. '0000011 = Branch (all) mispredicts. '0000100 = Indirect branch mispredicts. '0000101 = Far branch instructions.	
14:8	CntrThreshold Threshold (an unsigned value of 0 to 127 supported). The value 0 of counter threshold will result in event signaled after every instruction. #GP if threshold is < 2.	
15	MispredEventCnt Mispredict events counting behavior: '0 = Mispredict events are counted in a window. '1 = Mispredict events are counted based on a consecutive occurrence. CntrThreshold is treated as # of consecutive mispredicts. This control bit only applies to events specified by CntrEvSel that involve a prediction (0000010, 0000011, 0000100). Setting this bit for other events is ignored.	
63:16	Reserved.	
Register Address: 3F8H, 1016	MSR_PKG_C3_RESIDENCY	
Package C3 Residency Counter (R/O)		Package
63:0	Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.	
Register Address: 620H, 1568	MSR_RING_RATIO_LIMIT	
Ring Ratio Limit (R/W) This register provides Min/Max Ratio Limits for the LLC and Ring.		Package
6:0	MAX_Ratio This field is used to limit the max ratio of the LLC/Ring.	
7	Reserved.	
14:8	MIN_Ratio Writing to this field controls the minimum possible ratio of the LLC/Ring.	
63:15	Reserved.	
Register Address: 660H, 1632	MSR_CORE_C1_RESIDENCY	
Core C1 Residency Counter (R/O)		Core

**Table 2-42. Additional MSRs Supported by the 8th Generation Intel® Core™ i3 Processors
Based on Cannon Lake Microarchitecture (Contd.)**

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
63:0	Value since last reset for the Core C1 residency. Counter rate is the Max Non-Turbo frequency (same as TSC). This counter counts in case both of the core's threads are in an idle state and at least one of the core's thread residency is in a C1 state or in one of its sub states. The counter is updated only after a core C state exit. Note: Always reads 0 if core C1 is unsupported. A value of zero indicates that this processor does not support core C1 or never entered core C1 level state.	
Register Address: 662H, 1634	MSR_CORE_C3_RESIDENCY	
Core C3 Residency Counter (R/O)		Core
63:0	Will always return 0.	

Table 2-43 lists the MSRs of uncore PMU for Intel processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_66H.

Table 2-43. Uncore PMU MSRs Supported by Intel® Core™ Processors Based on Cannon Lake Microarchitecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 394H, 916	MSR_UNC_PERF_FIXED_CTRL	
Uncore Fixed Counter Control (R/W)		Package
19:0	Reserved.	
20	Enable overflow propagation.	
21	Reserved	
22	Enable counting.	
63:23	Reserved.	
Register Address: 395H, 917	MSR_UNC_PERF_FIXED_CTR	
Uncore Fixed Counter		Package
47:0	Current count.	
63:48	Reserved.	
Register Address: 396H, 918	MSR_UNC_CBO_CONFIG	
Uncore C-Box Configuration Information (R/O)		Package
3:0	Report the number of C-Box units with performance counters, including processor cores and processor graphics.	
63:4	Reserved.	
Register Address: 3B0H, 946	MSR_UNC_ARB_PERFCTR0	
Uncore Arb Unit, Performance Counter 0		Package
Register Address: 3B1H, 947	MSR_UNC_ARB_PERFCTR1	
Uncore Arb Unit, Performance Counter 1		Package
Register Address: 3B2H, 944	MSR_UNC_ARB_PERFEVTSELO	
Uncore Arb Unit, Counter 0 Event Select MSR		Package
Register Address: 3B3H, 945	MSR_UNC_ARB_PERFEVTSEL1	

Table 2-43. Uncore PMU MSRs Supported by Intel® Core™ Processors Based on Cannon Lake Microarchitecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore Arb unit, Counter 1 Event Select MSR		Package
Register Address: 700H, 1792	MSR_UNC_CBO_0_PERFEVTSELO	
Uncore C-Box 0, Counter 0 Event Select MSR		Package
Register Address: 701H, 1793	MSR_UNC_CBO_0_PERFEVTSEL1	
Uncore C-Box 0, Counter 1 Event Select MSR		Package
Register Address: 702H, 1794	MSR_UNC_CBO_0_PERFCTRO	
Uncore C-Box 0, Performance Counter 0		Package
Register Address: 703H, 1795	MSR_UNC_CBO_0_PERFCTR1	
Uncore C-Box 0, Performance Counter 1		Package
Register Address: 708H, 1800	MSR_UNC_CBO_1_PERFEVTSELO	
Uncore C-Box 1, Counter 0 Event Select MSR		Package
Register Address: 709H, 1801	MSR_UNC_CBO_1_PERFEVTSEL1	
Uncore C-Box 1, Counter 1 Event Select MSR		Package
Register Address: 70AH, 1802	MSR_UNC_CBO_1_PERFCTRO	
Uncore C-Box 1, Performance Counter 0		Package
Register Address: 70BH, 1803	MSR_UNC_CBO_1_PERFCTR1	
Uncore C-Box 1, Performance Counter 1		Package
Register Address: 710H, 1808	MSR_UNC_CBO_2_PERFEVTSELO	
Uncore C-Box 2, Counter 0 Event Select MSR		Package
Register Address: 711H, 1809	MSR_UNC_CBO_2_PERFEVTSEL1	
Uncore C-Box 2, Counter 1 Event Select MSR		Package
Register Address: 712H, 1810	MSR_UNC_CBO_2_PERFCTRO	
Uncore C-Box 2, Performance Counter 0		Package
Register Address: 713H, 1811	MSR_UNC_CBO_2_PERFCTR1	
Uncore C-Box 2, Performance Counter 1		Package
Register Address: 718H, 1816	MSR_UNC_CBO_3_PERFEVTSELO	
Uncore C-Box 3, Counter 0 Event Select MSR		Package
Register Address: 719H, 1817	MSR_UNC_CBO_3_PERFEVTSEL1	
Uncore C-Box 3, Counter 1 Event Select MSR		Package
Register Address: 71AH, 1818	MSR_UNC_CBO_3_PERFCTRO	
Uncore C-Box 3, Performance Counter 0		Package
Register Address: 71BH, 1819	MSR_UNC_CBO_3_PERFCTR1	
Uncore C-Box 3, Performance Counter 1		Package
Register Address: 720H, 1824	MSR_UNC_CBO_4_PERFEVTSELO	
Uncore C-Box 4, Counter 0 Event Select MSR		Package
Register Address: 721H, 1825	MSR_UNC_CBO_4_PERFEVTSEL1	
Uncore C-Box 4, Counter 1 Event Select MSR		Package

Table 2-43. Uncore PMU MSRs Supported by Intel® Core™ Processors Based on Cannon Lake Microarchitecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 722H, 1826	MSR_UNC_CBO_4_PERFCTR0	
Uncore C-Box 4, Performance Counter 0		Package
Register Address: 723H, 1827	MSR_UNC_CBO_4_PERFCTR1	
Uncore C-Box 4, Performance Counter 1		Package
Register Address: 728H, 1832	MSR_UNC_CBO_5_PERFEVTSELO	
Uncore C-Box 5, Counter 0 Event Select MSR		Package
Register Address: 729H, 1833	MSR_UNC_CBO_5_PERFEVTSEL1	
Uncore C-Box 5, Counter 1 Event Select MSR		Package
Register Address: 72AH, 1834	MSR_UNC_CBO_5_PERFCTR0	
Uncore C-Box 5, Performance Counter 0		Package
Register Address: 72BH, 1835	MSR_UNC_CBO_5_PERFCTR1	
Uncore C-Box 5, Performance Counter 1		Package
Register Address: 730H, 1840	MSR_UNC_CBO_6_PERFEVTSELO	
Uncore C-Box 6, Counter 0 Event Select MSR		Package
Register Address: 731H, 1841	MSR_UNC_CBO_6_PERFEVTSEL1	
Uncore C-Box 6, Counter 1 Event Select MSR		Package
Register Address: 732H, 1842	MSR_UNC_CBO_6_PERFCTR0	
Uncore C-Box 6, Performance Counter 0		Package
Register Address: 733H, 1843	MSR_UNC_CBO_6_PERFCTR1	
Uncore C-Box 6, Performance Counter 1		Package
Register Address: 738H, 1848	MSR_UNC_CBO_7_PERFEVTSELO	
Uncore C-Box 7, Counter 0 Event Select MSR		Package
Register Address: 739H, 1849	MSR_UNC_CBO_7_PERFEVTSEL1	
Uncore C-Box 7, Counter 1 Event Select MSR		Package
Register Address: 73AH, 1850	MSR_UNC_CBO_7_PERFCTR0	
Uncore C-Box 7, Performance Counter 0		Package
Register Address: 73BH, 1851	MSR_UNC_CBO_7_PERFCTR1	
Uncore C-Box 7, Performance Counter 1		Package
Register Address: E01H, 3585	MSR_UNC_PERF_GLOBAL_CTRL	
Uncore PMU Global Control		Package
0	Slice 0 select.	
1	Slice 1 select.	
2	Slice 2 select.	
3	Slice 3 select.	
4	Slice 4select.	
18:5	Reserved.	
29	Enable all uncore counters.	

Table 2-43. Uncore PMU MSRs Supported by Intel® Core™ Processors Based on Cannon Lake Microarchitecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
30	Enable wake on PMI.	
31	Enable Freezing counter when overflow.	
63:32	Reserved.	
Register Address: E02H, 3586	MSR_UNC_PERF_GLOBAL_STATUS	
Uncore PMU Main Status		Package
0	Fixed counter overflowed.	
1	An ARB counter overflowed.	
2	Reserved.	
3	A CBox counter overflowed (on any slice).	
63:4	Reserved.	

2.17.3 MSRs Introduced in 10th Generation Intel® Core™ Processors

Table 2-44 lists additional MSRs for 10th generation Intel Core processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_7DH or 06_7EH. For an MSR listed in Table 2-44 that also appears in the model-specific tables of prior generations, Table 2-44 supersedes prior generation tables.

Table 2-44. MSRs Supported by the 10th Generation Intel® Core™ Processors (Ice Lake Microarchitecture)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 33H, 51	MSR_MEMORY_CTRL	
Memory Control Register		Core
28:0	Reserved.	
29	SPLIT_LOCK_DISABLE If set to 1, a split lock will cause an #AC(0) exception. See Section 10.1.2.3, "Features to Disable Bus Locks."	
30	Reserved.	
31	Reserved.	
Register Address: 48H, 72	IA32_SPEC_CTRL	
See Table 2-2.		Core
Register Address: 49H, 73	IA32_PREDICT_CMD	
See Table 2-2.		Thread
Register Address: 8CH, 140	IA32_SGXLEPUBKEYHASH0	
See Table 2-2.		Thread
Register Address: 8DH, 141	IA32_SGXLEPUBKEYHASH1	
See Table 2-2.		Thread
Register Address: 8EH, 142	IA32_SGXLEPUBKEYHASH2	
See Table 2-2.		Thread
Register Address: 8FH, 143	IA32_SGXLEPUBKEYHASH3	

Table 2-44. MSRs Supported by the 10th Generation Intel® Core™ Processors (Ice Lake Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2.		Thread
Register Address: A0H, 160	MSR_BIOS_MCU_ERRORCODE	
BIOS MCU ERRORCODE (R/O) This MSR indicates if WRMSR 0x79 failed to configure PRM memory and gives a hint to debug BIOS.		Package
15:0	Error Codes (R/O)	Package
30:16	Reserved.	
31	MCU Partial Success (R/O) When set to 1, WRMSR 0x79 skipped part of the functionality during BIOS.	Thread
Register Address: A5H, 165	MSR_FIT_BIOS_ERROR	
FIT BIOS ERROR (R/W) Report error codes for debug in case the processor failed to parse the Firmware Table in BIOS. Can also be used to log BIOS information.		Thread
7:0	Error Codes (R/W) Error codes for debug.	
15:8	Entry Type (R/W) Failed FIT entry type.	
16	FIT MCU Entry (R/W) FIT contains MCU entry.	
62:17	Reserved.	
63	LOCK (R/W) When set to 1, writes to this MSR will be skipped.	
Register Address: 10BH, 267	IA32_FLUSH_CMD	
See Table 2-2.		Thread
Register Address: 151H, 337	MSR_BIOS_DONE	
BIOS Done (R/W0)		Thread
0	BIOS Done Indication (R/W0) Set by BIOS when it finishes programming the processor and wants to lock the memory configuration from changes by software that is running on this thread. Writes to the bit will be ignored if EAX[0] is 0.	Thread
1	Package BIOS Done Indication (R/O) When set to 1, all threads in the package have bit 0 of this MSR set.	Package
31:2	Reserved.	
Register Address: 1F1H, 497	MSR_CRASHLOG_CONTROL	
Write Data to a Crash Log Configuration		Thread
0	CDDIS: CrashDump_Disable If set, indicates that Crash Dump is disabled.	
63:1	Reserved.	
Register Address: 2A0H, 672	MSR_PRMRR_BASE_0	

Table 2-44. MSRs Supported by the 10th Generation Intel® Core™ Processors (Ice Lake Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Processor Reserved Memory Range Register - Physical Base Control Register (R/W)		Core
2:0	MEMTYPE: PRMRR BASE Memory Type.	
3	CONFIGURED: PRMRR BASE Configured.	
11:4	Reserved.	
51:12	BASE: PRMRR Base Address.	
63:52	Reserved.	
Register Address: 30CH, 780	IA32_FIXED_CTR3	
Fixed-Function Performance Counter Register 3 (R/W) Bit definitions are the same as found in IA32_FIXED_CTR0, offset 309H. See Table 2-2.		Thread
Register Address: 329H, 809	MSR_PERF_METRICS	
Performance Metrics (R/W) Reports metrics directly. Software can check (and/or expose to its guests) the availability of PERF_METRICS feature using IA32_PERF_CAPABILITIES.PERF_METRICS_AVAILABLE (bit 15).		Thread
7:0	Retiring. Percent of utilized slots by uops that eventually retire (commit).	
15:8	Bad Speculation. Percent of wasted slots due to incorrect speculation, covering utilized by uops that do not retire, or recovery bubbles (unutilized slots).	
23:16	Frontend Bound. Percent of unutilized slots where front-end did not deliver a uop while back-end is ready.	
31:24	Backend Bound. Percent of unutilized slots where a uop was not delivered to back-end due to lack of back-end resources.	
63:32	Reserved.	
Register Address: 3F2H, 1010	MSR_PEBS_DATA_CFG	
PEBS Data Configuration (R/W) Provides software the capability to select data groups of interest and thus reduce the record size in memory and record generation latency. Hence, a PEBS record's size and layout vary based on the selected groups. The MSR also allows software to select LBR depth for branch data records.		Thread
0	Memory Info. Setting this bit will capture memory information such as the linear address, data source and latency of the memory access in the PEBS record.	
1	GPRs. Setting this bit will capture the contents of the General Purpose registers in the PEBS record.	
2	XMMs. Setting this bit will capture the contents of the XMM registers in the PEBS record.	
3	LBRs. Setting this bit will capture LBR TO, FROM, and INFO in the PEBS record.	
23:4	Reserved.	

Table 2-44. MSRs Supported by the 10th Generation Intel® Core™ Processors (Ice Lake Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
31:24	LBR Entries. Set the field to the desired number of entries - 1. For example, if the LBR_entries field is 0, a single entry will be included in the record. To include 32 LBR entries, set the LBR_entries field to 31 (0x1F). To ensure all PEBS records are 16-byte aligned, software can use LBR_entries that is multiple of 3.	
Register Address: 541H, 1345	MSR_CORE_UARCH_CTL	
Core Microarchitecture Control MSR (R/W)		Core
0	L1 Scrubbing Enable When set to 1, enable L1 scrubbing.	
31:1	Reserved.	
Register Address: 657H, 1623	MSR_FAST_UNCORE_MSRS_CTL	
Fast WRMSR/RDMSR Control MSR (R/W)		Thread
3:0	FAST_ACCESS_ENABLE: Bit 0: When set to '1', provides a hint for the hardware to enable fast access mode for the IA32_HWP_REQUEST MSR. This bit is sticky and is cleaned by the hardware only during reset time. This bit is valid only if FAST_UNCORE_MSRS_CAPABILITY[0] is set. Setting this bit will cause CPUID[6].EAX[18] to be set.	
31:4	Reserved.	
Register Address: 65EH, 1630	MSR_FAST_UNCORE_MSRS_STATUS	
Indication of Uncore MSRs, Post Write Activates		Thread
0	Indicates whether the CPU is still in the middle of writing IA32_HWP_REQUEST MSR, even after the WRMSR instruction has retired. A value of 1 indicates the last write of IA32_HWP_REQUEST is still ongoing. A value of 0 indicates the last write of IA32_HWP_REQUEST is visible outside the logical processor. Software can use the status of this bit to avoid overwriting IA32_HWP_REQUEST.	
31:1	Reserved.	
Register Address: 65FH, 1631	MSR_FAST_UNCORE_MSRS_CAPABILITY	
Fast WRMSR/RDMSR Enumeration MSR (R/O)		Thread
3:0	MSRS_CAPABILITY: Bit 0: If set to '1', hardware supports the fast access mode for the IA32_HWP_REQUEST MSR.	
31:4	Reserved.	
Register Address: 772H, 1906	IA32_HWP_REQUEST_PKG	
See Table 2-2.		Package
Register Address: 775H, 1909	IA32_PECI_HWP_REQUEST_INFO	
See Table 2-2.		Package
Register Address: 777H, 1911	IA32_HWP_STATUS	

Table 2-44. MSRs Supported by the 10th Generation Intel® Core™ Processors (Ice Lake Microarchitecture) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2.		Thread

2.17.4 MSRs Introduced in the 11th Generation Intel® Core™ Processors based on Tiger Lake Microarchitecture

Table 2-45 lists additional MSRs for 11th generation Intel Core processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_8CH or 06_8DH. The MSRs listed in Table 2-44 are also supported by these processors. For an MSR listed in Table 2-45 that also appears in the model-specific tables of prior generations, Table 2-45 supersedes prior generation tables.

Table 2-45. Additional MSRs Supported by the 11th Generation Intel® Core™ Processors Based on Tiger Lake Microarchitecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: A0H, 160	MSR_BIOS_MCU_ERRORCODE	
BIOS MCU ERRORCODE (R/O)		Package
15:0	Error Codes	
31:16	Reserved.	
Register Address: A7H, 167	MSR_BIOS_DEBUG	
BIOS DEBUG (R/O) This MSR indicates if WRMSR 79H failed to configure PRM memory and gives a hint to debug BIOS.		Thread
30:0	Reserved.	
31	MCU Partial Success When set to 1, WRMSR 79H skipped part of the functionality during BIOS.	
63:32	Reserved.	
Register Address: CFH, 207	IA32_CORE_CAPABILITIES	
IA32 Core Capabilities Register (R/O) If CPUID.(EAX=07H, ECX=0):EDX[30] = 1. This MSR provides an architectural enumeration function for model-specific behavior.		Package
1:0	Reserved.	
2	FUSA_SUPPORTED	
3	RSM_IN_CPL0_ONLY When set to 1, the RSM instruction is only allowed in CPL0 (#GP triggered in any CPL != 0). When set to 0, then any CPL may execute the RSM instruction.	
4	Reserved.	
5	SPLIT_LOCK_DISABLE_SUPPORTED When read as 1, software can set bit 29 of MSR_MEMORY_CTRL (MSR address 33H).	
31:6	Reserved.	

Table 2-45. Additional MSRs Supported by the 11th Generation Intel® Core™ Processors Based on Tiger Lake Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 492H, 1170	IA32_VMX_PROCBASED_CTL3	
IA32_VMX_PROCBASED_CTL3 This MSR enumerates the allowed 1-settings of the third set of processor-based controls. Specifically, VM entry allows bit X of the tertiary processor-based VM-execution controls to be 1 if and only if bit X of the MSR is set to 1. If bit X of the MSR is cleared to 0, VM entry fails if control X and the “activate tertiary controls” primary processor-based VM-execution control are both 1.		Core
0	LOADIWKEY This control determines whether executions of LOADIWKEY cause VM exits.	
63:1	Reserved.	
Register Address: 601H, 1537	MSR_VR_CURRENT_CONFIG	
Power Limit 4 (PL4) Package-level maximum power limit (in Watts). It is a proactive, instantaneous limit.		Package
12:0	PL4 Value PL4 value in 0.125 A increments. This field is locked by VR_CURRENT_CONFIG[LOCK]. When the LOCK bit is set to 1b, this field becomes Read Only.	
30:13	Reserved.	
31	Lock Indication (LOCK) This bit will lock the CURRENT_LIMIT settings in this register and will also lock this setting. This means that once set to 1b, the CURRENT_LIMIT setting and this bit become Read Only until the next Warm Reset.	
62:32	Not in use.	
63	Reserved.	
Register Address: 6A0H, 1696	IA32_U_CET	
Configure User Mode CET (R/W) See Table 2-2.		
Register Address: 6A2H, 1698	IA32_S_CET	
Configure Supervisor Mode CET (R/W) See Table 2-2.		
Register Address: 6A4H, 1700	IA32_PL0_SSP	
Linear address to be loaded into SSP on transition to privilege level 0. (R/W) See Table 2-2.		
Register Address: 6A5H, 1701	IA32_PL1_SSP	
Linear address to be loaded into SSP on transition to privilege level 1. (R/W) See Table 2-2.		
Register Address: 6A6H, 1702	IA32_PL2_SSP	
Linear address to be loaded into SSP on transition to privilege level 2. (R/W) See Table 2-2.		
Register Address: 6A7H, 1703	IA32_PL3_SSP	

Table 2-45. Additional MSRs Supported by the 11th Generation Intel® Core™ Processors Based on Tiger Lake Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Linear address to be loaded into SSP on transition to privilege level 3. (R/W) See Table 2-2.		
Register Address: 6A8H, 1704	IA32_INTERRUPT_SSP_TABLE_ADDR	
Linear address of a table of seven shadow stack pointers that are selected in IA-32e mode using the IST index (when not 0) from the interrupt gate descriptor. (R/W) See Table 2-2.		
Register Address: 981H, 2433	IA32_TME_CAPABILITY	
See Table 2-2.		
Register Address: 982H, 2434	IA32_TME_ACTIVATE	
See Table 2-2.		
Register Address: 983H, 2435	IA32_TME_EXCLUDE_MASK	
See Table 2-2.		
Register Address: 984H, 2436	IA32_TME_EXCLUDE_BASE	
See Table 2-2.		
Register Address: 990H, 2448	IA32_COPY_STATUS ¹	
See Table 2-2.		Thread
Register Address: 991H, 2449	IA32_IWKEYBACKUP_STATUS ¹	
See Table 2-2.		Platform
Register Address: C82H, 3202	IA32_L2_QOS_CFG	
IA32_CR_L2_QOS_CFG This MSR provides software an enumeration of the parameters that L2 QoS (Intel RDT) support in any particular implementation.		Core
0	CDP_ENABLE When set to 1, it will enable the code and data prioritization for the L2 CAT/Intel RDT feature. When set to 0, code and data prioritization is disabled for L2 CAT/Intel RDT. See Chapter 19, “Debug, Branch Profile, TSC, and Intel® Resource Director Technology (Intel® RDT) Features,” for further details on CDP.	
31:1	Reserved.	
Register Address: D10H—D17H, 3220—3351	IA32_L2_QOS_MASK_[0-7]	
IA32_CR_L2_QOS_MASK_[0-7] Controls MLC (L2) Intel RDT allocation. For more details on CAT/RDT, see Chapter 19, “Debug, Branch Profile, TSC, and Intel® Resource Director Technology (Intel® RDT) Features.”		Package
19:0	WAYS_MASK Setting a 1 in this bit X allows threads with CLOS <n> (where N is [0-7]) to allocate to way X in the MLC. Ones are only allowed to be written to ways that physically exist in the MLC (CPUID.4.2:EBX[31:22] will indicate this). Writing a 1 to a value beyond the highest way or a non-contiguous set of 1s will cause a #GP on the WRMSR to this MSR.	
31:20	Reserved.	

Table 2-45. Additional MSRs Supported by the 11th Generation Intel® Core™ Processors Based on Tiger Lake Microarchitecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: D91H, 3473	IA32_COPY_LOCAL_TO_PLATFORM ¹	
See Table 2-2.		Thread
Register Address: D92H, 3474	IA32_COPY_PLATFORM_TO_LOCAL ¹	
See Table 2-2.		Thread

NOTES:

1. Further details on Key Locker and usage of this MSR can be found here:

<https://software.intel.com/content/www/us/en/develop/download/intel-key-locker-specification.html>.

2.17.5 MSRs Introduced in the 12th and 13th Generation Intel® Core™ Processors Supporting Performance Hybrid Architecture

Table 2-46 lists additional MSRs for 12th and 13th generation Intel Core processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_97H, 06_9AH, 06_BAH, 06_B7H, or 06_BFH. Table 2-47 lists the MSRs unique to the processor P-core. Table 2-48 lists the MSRs unique to the processor E-core.

The MSRs listed in Table 2-44¹ and Table 2-45 are also supported by these processors. For an MSR listed in Table 2-46, Table 2-47, or Table 2-48 that also appears in the model-specific tables of prior generations, Table 2-46, Table 2-47, and Table 2-48 supersede prior generation tables.

Table 2-46. Additional MSRs Supported by the 12th and 13th Generation Intel® Core™ Processors Supporting Performance Hybrid Architecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 33H, 51	MSR_MEMORY_CTRL	
Memory Control Register		Core
26:0	Reserved.	
27	UC_STORE_THROTTLE If set to 1, when enabled, the processor will only allow one in-progress UC store at a time.	
28	UC_LOCK_DISABLE If set to 1, a UC lock will cause a #GP(0) exception. See Section 10.1.2.3, "Features to Disable Bus Locks."	
29	SPLIT_LOCK_DISABLE If set to 1, a split lock will cause an #AC(0) exception. See Section 10.1.2.3, "Features to Disable Bus Locks."	
30	Reserved.	
31	Reserved.	
Register Address: BCH, 188	IA32_MISC_PACKAGE_CTL5	

1. MSRs at the following addresses are not supported in the 12th and 13th generation Intel Core processor E-core: 30CH, 329H, 541H, and 657H. The MSR at address 657H is not supported in the 12th and 13th generation Intel Core processor P-core.

Table 2-46. Additional MSRs Supported by the 12th and 13th Generation Intel® Core™ Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Power Filtering Control (R/W) IA32_ARCH_CAPABILITIES[bit 10] enumerates support for this MSR. See Table 2-2.		Package
Register Address: C7H, 199	IA32_PMC6	
General Performance Counter 6 (R/W) See Table 2-2.		Core
Register Address: C8H, 200	IA32_PMC7	
General Performance Counter 7 (R/W) See Table 2-2.		Core
Register Address: CFH, 207	IA32_CORE_CAPABILITIES	
IA32 Core Capabilities Register (R/O) If CPUID.(EAX=07H, ECX=0):EDX[30] = 1. This MSR provides an architectural enumeration function for model-specific behavior.		Package
0	STLB_QOS_SUPPORTED When set to 1, the STLB QoS feature is supported and the STLB QoS MSRs (1A8FH -1A97H) are accessible. When set to 0, access to these MSRs will #GP.	
1	Reserved.	
2	FUSA_SUPPORTED	
3	RSM_IN_CPLO_ONLY When set to 1, the RSM instruction is only allowed in CPLO (#GP triggered in any CPL != 0). When set to 0, then any CPL may execute the RSM instruction.	
4	UC_LOCK_DISABLE_SUPPORTED When read as 1, software can set bit 28 of MSR_MEMORY_CTRL (MSR address 33H).	
5	SPLIT_LOCK_DISABLE_SUPPORTED When read as 1, software can set bit 29 of MSR_MEMORY_CTRL.	
6	SNOOP_FILTER_QOS_SUPPORTED When set to 1, the Snoop Filter Qos Mask MSRs are supported. When set to 0, access to these MSRs will #GP.	
7	UC_STORE_THROTTLING_SUPPORTED When set 1, UC Store throttle capability exist through MSR_MEMORY_CTRL (33H) bit 27.	
31:8	Reserved.	
Register Address: E1H, 225	IA32_UMWAIT_CONTROL	
UMWAIT Control (R/W) See Table 2-2.		
Register Address: 10AH, 266	IA32_ARCH_CAPABILITIES	

Table 2-46. Additional MSRs Supported by the 12th and 13th Generation Intel® Core™ Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Enumeration of Architectural Features (R/O) See Table 2-2.		
Register Address: 18CH, 396	IA32_PERFEVTSEL6	
See Table 2-20.		Core
Register Address: 18DH, 397	IA32_PERFEVTSEL7	
See Table 2-20.		Core
Register Address: 195H, 405	IA32_OVERCLOCKING_STATUS	
Overclocking Status (R/O) IA32_ARCH_CAPABILITIES[bit 23] enumerates support for this MSR. See Table 2-2.		Package
Register Address: 1ADH, 429	MSR_PRIMARY_TURBO_RATIO_LIMIT	
Primary Maximum Turbo Ratio Limit (R/W) Software can configure these limits when MSR_PLATFORM_INFO[28] = 1. Specifies Maximum Ratio Limit for each group. Maximum ratio for groups with more cores must decrease monotonically.		Package
7:0	MAX_TURBO_GROUP_0: Maximum turbo ratio limit with 1 core active.	
15:8	MAX_TURBO_GROUP_1: Maximum turbo ratio limit with 2 cores active.	
23:16	MAX_TURBO_GROUP_2: Maximum turbo ratio limit with 3 cores active.	
31:24	MAX_TURBO_GROUP_3: Maximum turbo ratio limit with 4 cores active.	
39:32	MAX_TURBO_GROUP_4: Maximum turbo ratio limit with 5 cores active.	
47:40	MAX_TURBO_GROUP_5: Maximum turbo ratio limit with 6 cores active.	
55:48	MAX_TURBO_GROUP_6: Maximum turbo ratio limit with 7 cores active.	
63:56	MAX_TURBO_GROUP_7: Maximum turbo ratio limit with 8 cores active.	
Register Address: 493H, 1171	IA32_VMX_EXIT_CTL2	
See Table 2-2.		
Register Address: 4C7H, 1223	IA32_A_PMC6	
Full Width Writable IA32_PMC6 Alias (R/W) See Table 2-2.		
Register Address: 4C8H, 1224	IA32_A_PMC7	
Full Width Writable IA32_PMC7 Alias (R/W) See Table 2-2.		
Register Address: 650H, 1616	MSR_SECONDARY_TURBO_RATIO_LIMIT	

Table 2-46. Additional MSRs Supported by the 12th and 13th Generation Intel® Core™ Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Secondary Maximum Turbo Ratio Limit (R/W) Software can configure these limits when MSR_PLATFORM_INFO[28] = 1. Specifies Maximum Ratio Limit for each group. Maximum ratio for groups with more cores must decrease monotonically.		Package
7:0	MAX_TURBO_GROUP_0: Maximum turbo ratio limit with 1 core active.	
15:8	MAX_TURBO_GROUP_1: Maximum turbo ratio limit with 2 cores active.	
23:16	MAX_TURBO_GROUP_2: Maximum turbo ratio limit with 3 cores active.	
31:24	MAX_TURBO_GROUP_3: Maximum turbo ratio limit with 4 cores active.	
39:32	MAX_TURBO_GROUP_4: Maximum turbo ratio limit with 5 cores active.	
47:40	MAX_TURBO_GROUP_5: Maximum turbo ratio limit with 6 cores active.	
55:48	MAX_TURBO_GROUP_6: Maximum turbo ratio limit with 7 cores active.	
63:56	MAX_TURBO_GROUP_7: Maximum turbo ratio limit with 8 cores active.	
Register Address: 664H, 1636	MSR_MC6_RESIDENCY_COUNTER	
Module C6 Residency Counter (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Module
63:0	Time that this module is in module-specific C6 states since last reset. Counts at 1 Mhz frequency.	
Register Address: 6E1H, 1761	IA32_PKRS	
Specifies the PK permissions associated with each protection domain for supervisor pages (R/W) See Table 2-2.		
Register Address: 776H, 1910	IA32_HWP_CTL	
See Table 2-2.		
Register Address: 981H, 2433	IA32_TME_CAPABILITY	
Memory Encryption Capability MSR See Table 2-2.		
Register Address: 1200H—121FH, 4608—4639	IA32_LBR_x_INFO	
Last Branch Record Entry X Info Register (R/W) See Table 2-2.		
Register Address: 14CEH, 5326	IA32_LBR_CTL	
Last Branch Record Enabling and Configuration Register (R/W) See Table 2-2.		

Table 2-46. Additional MSRs Supported by the 12th and 13th Generation Intel® Core™ Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 14CFH, 5327	IA32_LBR_DEPTH	
Last Branch Record Maximum Stack Depth Register (R/W) See Table 2-2.		
Register Address: 1500H–151FH, 5376–5407	IA32_LBR_x_FROM_IP	
Last Branch Record Entry X Source IP Register (R/W) See Table 2-2.		
Register Address: 1600H–161FH, 5632–5663	IA32_LBR_x_TO_IP	
Last Branch Record Entry X Destination IP Register (R/W) See Table 2-2.		
Register Address: 17D2H, 6098	IA32_THREAD_FEEDBACK_CHAR	
Thread Feedback Characteristics (R/O) See Table 2-2.		
Register Address: 17D4H, 6100	IA32_HW_FEEDBACK_THREAD_CONFIG	
Hardware Feedback Thread Configuration (R/W) See Table 2-2.		
Register Address: 17DAH, 6106	IA32_HRESET_ENABLE	
History Reset Enable (R/W) See Table 2-2.		

The MSRs listed in Table 2-47 are unique to the 12th and 13th generation Intel Core processor P-core. These MSRs are not supported on the processor E-core.

Table 2-47. MSRs Supported by 12th and 13th Generation Intel® Core™ Processor P-core

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1A4H, 420	MSR_PREFETCH_CONTROL	
Prefetch Disable Bits (R/W)		
0	L2_HARDWARE_PREFETCHER_DISABLE If 1, disables the L2 hardware prefetcher, which fetches additional lines of code or data into the L2 cache.	
1	L2_ADJACENT_CACHE_LINE_PREFETCHER_DISABLE If 1, disables the adjacent cache line prefetcher, which fetches the cache line that comprises a cache line pair (128 bytes).	
2	DCU_HARDWARE_PREFETCHER_DISABLE If 1, disables the L1 data cache prefetcher, which fetches the next cache line into L1 data cache.	
3	DCU_IP_PREFETCHER_DISABLE If 1, disables the L1 data cache IP prefetcher, which uses sequential load history (based on instruction pointer of previous loads) to determine whether to prefetch additional lines.	
4	Reserved.	

Table 2-47. MSRs Supported by 12th and 13th Generation Intel® Core™ Processor P-core

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
5	AMP_PREFETCH_DISABLE If 1, disables the L2 Adaptive Multipath Probability (AMP) prefetcher.	
63:6	Reserved.	
Register Address: 3F7H, 1015	MSR_PEBS_FRONTEND	
FrontEnd Precise Event Condition Select (R/W) See Table 2-39.		Thread
Register Address: 540H, 1344	MSR_THREAD_UARCH_CTL	
Thread Microarchitectural Control (R/W)		Thread
0	WB_MEM_STRM_LD_DISABLE Disable streaming behavior for MOVNTDQA loads to WB memory type. If set, these accesses will be treated like regular cacheable loads (Data will be cached).	
63:1	Reserved.	
Register Address: 541H, 1345	MSR_CORE_UARCH_CTL	
Core Microarchitecture Control MSR (R/W) See Table 2-44.		Core
Register Address: D10H–D17H, 3220–3351	IA32_L2_QOS_MASK_[0-7]	
IA32_CR_L2_QOS_MASK_[0-7] If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] ≥ 0. Controls MLC (L2) Intel RDT allocation. For more details on CAT/RDT, see Chapter 19, “Debug, Branch Profile, TSC, and Intel® Resource Director Technology (Intel® RDT) Features.”		Core
19:0	WAYS_MASK Setting a 1 in this bit X allows threads with CLOS <n> (where N is [0-7]) to allocate to way X in the MLC. Ones are only allowed to be written to ways that physically exist in the MLC (CPUID.4.2:EBX[31:22] will indicate this). Writing a 1 to a value beyond the highest way or a non-contiguous set of 1s will cause a #GP on the WRMSR to this MSR.	
31:20	Reserved.	

The MSRs listed in Table 2-48 are unique to the 12th and 13th generation Intel Core processor E-core. These MSRs are not supported on the processor P-core.

Table 2-48. MSRs Supported by 12th and 13th Generation Intel® Core™ Processor E-core

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: D10H–D1FH, 3220–3359	IA32_L2_QOS_MASK_[0-15]	
IA32_CR_L2_QOS_MASK_[0-15] If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] ≥ 0. Controls MLC (L2) Intel RDT allocation. For more details on CAT/RDT, see Chapter 19, “Debug, Branch Profile, TSC, and Intel® Resource Director Technology (Intel® RDT) Features.”		Module

Table 2-48. MSRs Supported by 12th and 13th Generation Intel® Core™ Processor E-core

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
19:0	WAYS_MASK Setting a 1 in this bit X allows threads with CLOS <n> (where N is [0-7]) to allocate to way X in the MLC. Ones are only allowed to be written to ways that physically exist in the MLC (CPUID.4.2:EBX[31:22] will indicate this). Writing a 1 to a value beyond the highest way or a non-contiguous set of 1s will cause a #GP on the WRMSR to this MSR.	
31:20	Reserved.	
Register Address: 1309H–130BH, 4873–4875	MSR_RELOAD_FIXED_CTRx	
Reload value for IA32_FIXED_CTRx (R/W)		
47:0	Value loaded into IA32_FIXED_CTRx when a PEBS record is generated while PEBS_EN_FIXEDx = 1 and PEBS_OUTPUT = 01B in IA32_PEBS_ENABLE, and FIXED_CTRx is overflowed.	
63:48	Reserved.	
Register Address: 14C1H–14C6H, 5313–5318	MSR_RELOAD_PMCx	
Reload value for IA32_PMCx (R/W)		Core
47:0	Value loaded into IA32_PMCx when a PEBS record is generated while PEBS_EN_PMCx = 1 and PEBS_OUTPUT = 01B in IA32_PEBS_ENABLE, and PMCx is overflowed.	
63:48	Reserved.	

Table 2-49 lists the MSRs of uncore PMU for Intel processors with a CPUID Signature DisplayFamily_DisplayModel value of 06_97H, 06_9AH, 06_BAH, 06_B7H, or 06_BFH.

Table 2-49. Uncore PMU MSRs Supported by 12th and 13th Generation Intel® Core™ Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 396H, 918	MSR_UNC_CBO_CONFIG	
Uncore C-Box Configuration Information (R/O)		Package
3:0	Specifies the number of C-Box units with programmable counters (including processor cores and processor graphics).	
63:4	Reserved.	
Register Address: 2000H, 8192	MSR_UNC_CBO_0_PERFEVTSELO	
Uncore C-Box 0, Counter 0 Event Select MSR		Package
Register Address: 2001H, 8193	MSR_UNC_CBO_0_PERFEVTSEL1	
Uncore C-Box 0, Counter 1 Event Select MSR		Package
Register Address: 2002H, 8194	MSR_UNC_CBO_0_PERFCTRO	
Uncore C-Box 0, Performance Counter 0		Package
Register Address: 2003H, 8195	MSR_UNC_CBO_0_PERFCTR1	
Uncore C-Box 0, Performance Counter 1		Package
Register Address: 2008H, 8200	MSR_UNC_CBO_1_PERFEVTSELO	

Table 2-49. Uncore PMU MSRs Supported by 12th and 13th Generation Intel® Core™ Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore C-Box 1, Counter 0 Event Select MSR		Package
Register Address: 2009H, 8201	MSR_UNC_CBO_1_PERFEVTSEL1	
Uncore C-Box 1, Counter 1 Event Select MSR		Package
Register Address: 200AH, 8202	MSR_UNC_CBO_1_PERFCTRO	
Uncore C-Box 1, Performance Counter 0		Package
Register Address: 200BH, 8203	MSR_UNC_CBO_1_PERFCTR1	
Uncore C-Box 1, Performance Counter 1		Package
Register Address: 2010H, 8208	MSR_UNC_CBO_2_PERFEVTSELO	
Uncore C-Box 2, Counter 0 Event Select MSR		Package
Register Address: 2011H, 8209	MSR_UNC_CBO_2_PERFEVTSEL1	
Uncore C-Box 2, Counter 1 Event Select MSR		Package
Register Address: 2012H, 8210	MSR_UNC_CBO_2_PERFCTRO	
Uncore C-Box 2, Performance Counter 0		Package
Register Address: 2013H, 8211	MSR_UNC_CBO_2_PERFCTR1	
Uncore C-Box 2, Performance Counter 1		Package
Register Address: 2018H, 8216	MSR_UNC_CBO_3_PERFEVTSELO	
Uncore C-Box 3, Counter 0 Event Select MSR		Package
Register Address: 2019H, 8217	MSR_UNC_CBO_3_PERFEVTSEL1	
Uncore C-Box 3, Counter 1 Event Select MSR		Package
Register Address: 201AH, 8218	MSR_UNC_CBO_3_PERFCTRO	
Uncore C-Box 3, Performance Counter 0		Package
Register Address: 201BH, 8219	MSR_UNC_CBO_3_PERFCTR1	
Uncore C-Box 3, Performance Counter 1		Package
Register Address: 2020H, 8224	MSR_UNC_CBO_4_PERFEVTSELO	
Uncore C-Box 4, Counter 0 Event Select MSR		Package
Register Address: 2021H, 8225	MSR_UNC_CBO_4_PERFEVTSEL1	
Uncore C-Box 4, Counter 1 Event Select MSR		Package
Register Address: 2022H, 8226	MSR_UNC_CBO_4_PERFCTRO	
Uncore C-Box 4, Performance Counter 0		Package
Register Address: 2023H, 8227	MSR_UNC_CBO_4_PERFCTR1	
Uncore C-Box 4, Performance Counter 1		Package
Register Address: 2028H, 8232	MSR_UNC_CBO_5_PERFEVTSELO	
Uncore C-Box 5, Counter 0 Event Select MSR		Package
Register Address: 2029H, 8233	MSR_UNC_CBO_5_PERFEVTSEL1	
Uncore C-Box 5, Counter 1 Event Select MSR		Package
Register Address: 202AH, 8234	MSR_UNC_CBO_5_PERFCTRO	
Uncore C-Box 5, Performance Counter 0		Package

Table 2-49. Uncore PMU MSRs Supported by 12th and 13th Generation Intel® Core™ Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 202BH, 8235	MSR_UNC_CBO_5_PERFCTR1	
Uncore C-Box 5, Performance Counter 1		Package
Register Address: 2030H, 8240	MSR_UNC_CBO_6_PERFEVTSELO	
Uncore C-Box 6, Counter 0 Event Select MSR		Package
Register Address: 2031H, 8241	MSR_UNC_CBO_6_PERFEVTSEL1	
Uncore C-Box 6, Counter 1 Event Select MSR		Package
Register Address: 2032H, 8242	MSR_UNC_CBO_6_PERFCTRO	
Uncore C-Box 6, Performance Counter 0		Package
Register Address: 2033H, 8243	MSR_UNC_CBO_6_PERFCTR1	
Uncore C-Box 6, Performance Counter 1		Package
Register Address: 2038H, 8248	MSR_UNC_CBO_7_PERFEVTSELO	
Uncore C-Box 7, Counter 0 Event Select MSR		Package
Register Address: 2039H, 8249	MSR_UNC_CBO_7_PERFEVTSEL1	
Uncore C-Box 7, Counter 1 Event Select MSR		Package
Register Address: 203AH, 8250	MSR_UNC_CBO_7_PERFCTRO	
Uncore C-Box 7, Performance Counter 0		Package
Register Address: 203BH, 8251	MSR_UNC_CBO_7_PERFCTR1	
Uncore C-Box 7, Performance Counter 1		Package
Register Address: 2040H, 8256	MSR_UNC_CBO_8_PERFEVTSELO	
Uncore C-Box 8, Counter 0 Event Select MSR		Package
Register Address: 2041H, 8257	MSR_UNC_CBO_8_PERFEVTSEL1	
Uncore C-Box 8, Counter 1 Event Select MSR		Package
Register Address: 2042H, 8258	MSR_UNC_CBO_8_PERFCTRO	
Uncore C-Box 8, Performance Counter 0		Package
Register Address: 2043H, 8259	MSR_UNC_CBO_8_PERFCTR1	
Uncore C-Box 8, Performance Counter 1		Package
Register Address: 2048H, 8264	MSR_UNC_CBO_9_PERFEVTSELO	
Uncore C-Box 9, Counter 0 Event Select MSR		Package
Register Address: 2049H, 8265	MSR_UNC_CBO_9_PERFEVTSEL1	
Uncore C-Box 9, Counter 1 Event Select MSR		Package
Register Address: 204AH, 8266	MSR_UNC_CBO_9_PERFCTRO	
Uncore C-Box 9, Performance Counter 0		Package
Register Address: 204BH, 8267	MSR_UNC_CBO_9_PERFCTR1	
Uncore C-Box 9, Performance Counter 1		Package
Register Address: 2FD0H, 12240	MSR_UNC_ARB_0_PERFEVTSELO	
Uncore Arb Unit 0, Counter 0 Event Select MSR		Package
Register Address: 2FD1H, 12241	MSR_UNC_ARB_0_PERFEVTSEL1	

Table 2-49. Uncore PMU MSRs Supported by 12th and 13th Generation Intel® Core™ Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Uncore Arb Unit 0, Counter 1 Event Select MSR		Package
Register Address: 2FD2H, 12242	MSR_UNC_ARB_0_PERFCTR0	
Uncore Arb Unit 0, Performance Counter 0		Package
Register Address: 2FD3H, 12243	MSR_UNC_ARB_0_PERFCTR1	
Uncore Arb Unit 0, Performance Counter 1		Package
Register Address: 2FD4H, 12244	MSR_UNC_ARB_0_PERF_STATUS	
Uncore Arb Unit 0, Performance Status		Package
Register Address: 2FD5H, 12245	MSR_UNC_ARB_0_PERF_CTRL	
Uncore Arb Unit 0, Performance Control		Package
Register Address: 2FD8H, 12248	MSR_UNC_ARB_1_PERFEVTSELO	
Uncore Arb Unit 1, Counter 0 Event Select MSR		Package
Register Address: 2FD9H, 12249	MSR_UNC_ARB_1_PERFEVTSEL1	
Uncore Arb Unit 1, Counter 1 Event Select MSR		Package
Register Address: 2FDAH, 12250	MSR_UNC_ARB_1_PERFCTR0	
Uncore Arb Unit 1, Performance Counter 0		Package
Register Address: 2FDBH, 12251	MSR_UNC_ARB_1_PERFCTR1	
Uncore Arb Unit 1, Performance Counter 1		Package
Register Address: 2FDCB, 12252	MSR_UNC_ARB_1_PERF_STATUS	
Uncore Arb Unit 1, Performance Status		Package
Register Address: 2FDDH, 12253	MSR_UNC_ARB_1_PERF_CTRL	
Uncore Arb Unit 1, Performance Control		Package
Register Address: 2FDEH, 12254	MSR_UNC_PERF_FIXED_CTRL	
Uncore Fixed Counter Control (R/W)		Package
19:0	Reserved.	
20	Enable overflow propagation.	
21	Reserved.	
22	Enable counting.	
63:23	Reserved.	
Register Address: 2FDFH, 12255	MSR_UNC_PERF_FIXED_CTR	
Uncore Fixed Counter		Package
43:0	Current count.	
63:44	Reserved.	
Register Address: 2FF0H, 12272	MSR_UNC_PERF_GLOBAL_CTRL	
Uncore PMU Global Control		Package
0	Slice 0 select.	
1	Slice 1 select.	
2	Slice 2 select.	

Table 2-49. Uncore PMU MSRs Supported by 12th and 13th Generation Intel® Core™ Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
3	Slice 3 select.	
4	Slice 4 select.	
18:5	Reserved.	
29	Enable all uncore counters.	
30	Enable wake on PMI.	
31	Enable Freezing counter when overflow.	
63:32	Reserved.	
Register Address: 2FF2H, 12274	MSR_UNC_PERF_GLOBAL_STATUS	
Uncore PMU Main Status		Package
0	Fixed counter overflowed.	
1	An ARB counter overflowed.	
2	Reserved.	
3	A CBox counter overflowed (on any slice).	
63:4	Reserved.	

2.17.6 MSRs Introduced in the Intel® Xeon® Scalable Processor Family

The Intel® Xeon® Scalable Processor Family (CUID Signature DisplayFamily_DisplayModel value of 06_55H) supports the MSRs listed in Table 2-50.

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CUID Signature DisplayFamily_DisplayModel Value of 06_55H

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64 Processor (R/W) See Table 2-2.		Thread
0	Lock (R/WL)	
1	Enable VMX Inside SMX Operation (R/WL)	
2	Enable VMX Outside SMX Operation (R/WL)	
14:8	SENTER Local Functions Enables (R/WL)	
15	SENTER Global Functions Enable (R/WL)	
18	SGX Global Functions Enable (R/WL)	
20	LMCE_ENABLED (R/WL)	
63:21	Reserved.	
Register Address: 4EH, 78	IA32_PPIN_CTL (MSR_PPIN_CTL)	
Protected Processor Inventory Number Enable Control (R/W)		Package
0	LockOut (R/WO) See Table 2-2.	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
1	Enable_PPIN (R/W) See Table 2-2.	
63:2	Reserved.	
Register Address: 4FH, 79	IA32_PPIN (MSR_PPIN)	
Protected Processor Inventory Number (R/O)		Package
63:0	Protected Processor Inventory Number (R/O) See Table 2-2.	
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) See Table 2-26.	Package
22:16	Reserved.	
23	PPIN_CAP (R/O) See Table 2-26.	Package
27:24	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) See Table 2-26.	Package
29	Programmable TDP Limit for Turbo Mode (R/O) See Table 2-26.	Package
30	Programmable TJ OFFSET (R/O) See Table 2-26.	Package
39:31	Reserved.	
47:40	Maximum Efficiency Ratio (R/O) See Table 2-26.	Package
63:48	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states. See http://biosbits.org .		Core

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
2:0	Package C-State Limit (R/W) Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings are supported: 000b: C0/C1 (no package C-state support) 001b: C2 010b: C6 (non-retention) 011b: C6 (retention) 111b: No Package C state limits. All C states supported by the processor are available.	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W)	
14:11	Reserved.	
15	CFG Lock (R/W0)	
16	Automatic C-State Conversion Enable (R/W) If 1, the processor will convert HALT or MWAIT(C1) to MWAIT(C6).	
24:17	Reserved.	
25	C3 State Auto Demotion Enable (R/W)	
26	C1 State Auto Demotion Enable (R/W)	
27	Enable C3 Undemotion (R/W)	
28	Enable C1 Undemotion (R/W)	
29	Package C State Demotion Enable (R/W)	
30	Package C State Undemotion Enable (R/W)	
63:31	Reserved.	
Register Address: 179H, 377	IA32_MCG_CAP	
Global Machine Check Capability (R/O)		Thread
7:0	Count.	
8	MCG_CTL_P	
9	MCG_EXT_P	
10	MCP_CMCI_P	
11	MCG_TES_P	
15:12	Reserved.	
23:16	MCG_EXT_CNT	
24	MCG_SER_P	
25	MCG_EM_P	
26	MCG_ELOG_P	
63:27	Reserved.	
Register Address: 17DH, 381	MSR_SMM_MCA_CAP	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Enhanced SMM Capabilities (SMM-RO) Reports SMM capability Enhancement. Accessible only while in SMM.		Thread
57:0	Reserved.	
58	SMM_Code_Access_Chk (SMM-RO) If set to 1 indicates that the SMM code access restriction is supported and a host-space interface is available to SMM handler.	
59	Long_Flow_Indication (SMM-RO) If set to 1 indicates that the SMM long flow indicator is supported and a host-space interface is available to SMM handler.	
63:60	Reserved.	
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Core
0	Thermal Status (R/O) See Table 2-2.	
1	Thermal Status Log (R/WC0) See Table 2-2.	
2	PROTCHOT # or FORCEPR# Status (R/O) See Table 2-2.	
3	PROTCHOT # or FORCEPR# Log (R/WC0) See Table 2-2.	
4	Critical Temperature Status (R/O) See Table 2-2.	
5	Critical Temperature Status Log (R/WC0) See Table 2-2.	
6	Thermal Threshold #1 Status (R/O) See Table 2-2.	
7	Thermal Threshold #1 Log (R/WC0) See Table 2-2.	
8	Thermal Threshold #2 Status (R/O) See Table 2-2.	
9	Thermal Threshold #2 Log (R/WC0) See Table 2-2.	
10	Power Limitation Status (R/O) See Table 2-2.	
11	Power Limitation Log (R/WC0) See Table 2-2.	
12	Current Limit Status (R/O) See Table 2-2.	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
13	Current Limit Log (R/WC0) See Table 2-2.	
14	Cross Domain Limit Status (R/O) See Table 2-2.	
15	Cross Domain Limit Log (R/WC0) See Table 2-2.	
22:16	Digital Readout (R/O) See Table 2-2.	
26:23	Reserved.	
30:27	Resolution in Degrees Celsius (R/O) See Table 2-2.	
31	Reading Valid (R/O) See Table 2-2.	
63:32	Reserved.	
Register Address: 1A2H, 418	MSR_TEMPERATURE_TARGET	
Temperature Target		Package
15:0	Reserved.	
23:16	Temperature Target (R/O) See Table 2-26.	
27:24	TCC Activation Offset (R/W) See Table 2-26.	
63:28	Reserved.	
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
This register defines the ratio limits. RATIO[0:7] must be populated in ascending order. RATIO[i+1] must be less than or equal to RATIO[i]. Entries with RATIO[i] will be ignored. If any of the rules above are broken, the configuration is silently rejected. If the programmed ratio is: ▪ Above the fused ratio for that core count, it will be clipped to the fuse limits (assuming !OC). ▪ Below the min supported ratio, it will be clipped.		Package
7:0	RATIO_0 Defines ratio limits.	
15:8	RATIO_1 Defines ratio limits.	
23:16	RATIO_2 Defines ratio limits.	
31:24	RATIO_3 Defines ratio limits.	
39:32	RATIO_4 Defines ratio limits.	
47:40	RATIO_5 Defines ratio limits.	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
55:48	RATIO_6 Defines ratio limits.	
63:56	RATIO_7 Defines ratio limits.	
Register Address: 1AEH, 430	MSR_TURBO_RATIO_LIMIT_CORES	
This register defines the active core ranges for each frequency point. NUMCORE[0:7] must be populated in ascending order. NUMCORE[i+1] must be greater than NUMCORE[i]. Entries with NUMCORE[i] == 0 will be ignored. The last valid entry must have NUMCORE >= the number of cores in the SKU. If any of the rules above are broken, the configuration is silently rejected.		Package
7:0	NUMCORE_0 Defines the active core ranges for each frequency point.	
15:8	NUMCORE_1 Defines the active core ranges for each frequency point.	
23:16	NUMCORE_2 Defines the active core ranges for each frequency point.	
31:24	NUMCORE_3 Defines the active core ranges for each frequency point.	
39:32	NUMCORE_4 Defines the active core ranges for each frequency point.	
47:40	NUMCORE_5 Defines the active core ranges for each frequency point.	
55:48	NUMCORE_6 Defines the active core ranges for each frequency point.	
63:56	NUMCORE_7 Defines the active core ranges for each frequency point.	
Register Address: 280H, 640	IA32_MC0_CTL2	
See Table 2-2.		Core
Register Address: 281H, 641	IA32_MC1_CTL2	
See Table 2-2.		Core
Register Address: 282H, 642	IA32_MC2_CTL2	
See Table 2-2.		Core
Register Address: 283H, 643	IA32_MC3_CTL2	
See Table 2-2.		Core
Register Address: 284H, 644	IA32_MC4_CTL2	
See Table 2-2.		Package
Register Address: 285H, 645	IA32_MC5_CTL2	
See Table 2-2.		Package
Register Address: 286H, 646	IA32_MC6_CTL2	
See Table 2-2.		Package

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 287H, 647	IA32_MC7_CTL2	
See Table 2-2.		Package
Register Address: 288H, 648	IA32_MC8_CTL2	
See Table 2-2.		Package
Register Address: 289H, 649	IA32_MC9_CTL2	
See Table 2-2.		Package
Register Address: 28AH, 650	IA32_MC10_CTL2	
See Table 2-2.		Package
Register Address: 28BH, 651	IA32_MC11_CTL2	
See Table 2-2.		Package
Register Address: 28CH, 652	IA32_MC12_CTL2	
See Table 2-2.		Package
Register Address: 28DH, 653	IA32_MC13_CTL2	
See Table 2-2.		Package
Register Address: 28EH, 654	IA32_MC14_CTL2	
See Table 2-2.		Package
Register Address: 28FH, 655	IA32_MC15_CTL2	
See Table 2-2.		Package
Register Address: 290H, 656	IA32_MC16_CTL2	
See Table 2-2.		Package
Register Address: 291H, 657	IA32_MC17_CTL2	
See Table 2-2.		Package
Register Address: 292H, 658	IA32_MC18_CTL2	
See Table 2-2.		Package
Register Address: 293H, 659	IA32_MC19_CTL2	
See Table 2-2.		Package
Register Address: 400H, 1024	IA32_MCO_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MCO reports MC errors from the IFU module.		Core
Register Address: 401H, 1025	IA32_MCO_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MCO reports MC errors from the IFU module.		Core
Register Address: 402H, 1026	IA32_MCO_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MCO reports MC errors from the IFU module.		Core
Register Address: 403H, 1027	IA32_MCO_MISC	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC0 reports MC errors from the IFU module.		Core
Register Address: 404H, 1028	IA32_MC1_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC1 reports MC errors from the DCU module.		Core
Register Address: 405H, 1029	IA32_MC1_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC1 reports MC errors from the DCU module.		Core
Register Address: 406H, 1030	IA32_MC1_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC1 reports MC errors from the DCU module.		Core
Register Address: 407H, 1031	IA32_MC1_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC1 reports MC errors from the DCU module.		Core
Register Address: 408H, 1032	IA32_MC2_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC2 reports MC errors from the DTLB module.		Core
Register Address: 409H, 1033	IA32_MC2_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC2 reports MC errors from the DTLB module.		Core
Register Address: 40AH, 1034	IA32_MC2_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC2 reports MC errors from the DTLB module.		Core
Register Address: 40BH, 1035	IA32_MC2_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC2 reports MC errors from the DTLB module.		Core
Register Address: 40CH, 1036	IA32_MC3_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC3 reports MC errors from the MLC module.		Core
Register Address: 40DH, 1037	IA32_MC3_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC3 reports MC errors from the MLC module.		Core
Register Address: 40EH, 1038	IA32_MC3_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC3 reports MC errors from the MLC module.		Core
Register Address: 40FH, 1039	IA32_MC3_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC3 reports MC errors from the MLC module.		Core

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 410H, 1040	IA32_MC4_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC4 reports MC errors from the PCU module.	Package	
Register Address: 411H, 1041	IA32_MC4_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC4 reports MC errors from the PCU module.	Package	
Register Address: 412H, 1042	IA32_MC4_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC4 reports MC errors from the PCU module.	Package	
Register Address: 413H, 1043	IA32_MC4_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC4 reports MC errors from the PCU module.	Package	
Register Address: 414H, 1044	IA32_MC5_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from a link interconnect module.	Package	
Register Address: 415H, 1045	IA32_MC5_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from a link interconnect module.	Package	
Register Address: 416H, 1046	IA32_MC5_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from a link interconnect module.	Package	
Register Address: 417H, 1047	IA32_MC5_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC5 reports MC errors from a link interconnect module.	Package	
Register Address: 418H, 1048	IA32_MC6_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.	Package	
Register Address: 419H, 1049	IA32_MC6_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.	Package	
Register Address: 41AH, 1050	IA32_MC6_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.	Package	
Register Address: 41BH, 1051	IA32_MC6_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC6 reports MC errors from the integrated I/O module.	Package	
Register Address: 41CH, 1052	IA32_MC7_CTL	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC7 reports MC errors from the M2M 0.		Package
Register Address: 41DH, 1053	IA32_MC7_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC7 reports MC errors from the M2M 0.		Package
Register Address: 41EH, 1054	IA32_MC7_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC7 reports MC errors from the M2M 0.		Package
Register Address: 41FH, 1055	IA32_MC7_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC7 reports MC errors from the M2M 0.		Package
Register Address: 420H, 1056	IA32_MC8_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the M2M 1.		Package
Register Address: 421H, 1057	IA32_MC8_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the M2M 1.		Package
Register Address: 422H, 1058	IA32_MC8_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the M2M 1.		Package
Register Address: 423H, 1059	IA32_MC8_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC8 reports MC errors from the M2M 1.		Package
Register Address: 424H, 1060	IA32_MC9_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 425H, 1061	IA32_MC9_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 426H, 1062	IA32_MC9_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 427H, 1063	IA32_MC9_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 428H, 1064	IA32_MC10_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 429H, 1065	IA32_MC10_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 42AH, 1066	IA32_MC10_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 42BH, 1067	IA32_MC10_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 42CH, 1068	IA32_MC11_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 42DH, 1069	IA32_MC11_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 42EH, 1070	IA32_MC11_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 42FH, 1071	IA32_MC11_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC9 - MC11 report MC errors from the CHA.		Package
Register Address: 430H, 1072	IA32_MC12_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC12 report MC errors from each channel of a link interconnect module.		Package
Register Address: 431H, 1073	IA32_MC12_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC12 report MC errors from each channel of a link interconnect module.		Package
Register Address: 432H, 1074	IA32_MC12_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC12 report MC errors from each channel of a link interconnect module.		Package
Register Address: 433H, 1075	IA32_MC12_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC12 report MC errors from each channel of a link interconnect module.		Package
Register Address: 434H, 1076	IA32_MC13_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 435H, 1077	IA32_MC13_STATUS	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 436H, 1078	IA32_MC13_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 437H, 1079	IA32_MC13_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 438H, 1080	IA32_MC14_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 439H, 1081	IA32_MC14_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 43AH, 1082	IA32_MC14_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 43BH, 1083	IA32_MC14_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 43CH, 1084	IA32_MC15_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 43DH, 1085	IA32_MC15_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 43EH, 1086	IA32_MC15_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 43FH, 1087	IA32_MC15_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 440H, 1088	IA32_MC16_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 441H, 1089	IA32_MC16_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 442H, 1090	IA32_MC16_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 443H, 1091	IA32_MC16_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 444H, 1092	IA32_MC17_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 445H, 1093	IA32_MC17_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 446H, 1094	IA32_MC17_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 447H, 1095	IA32_MC17_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 448H, 1096	IA32_MC18_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 449H, 1097	IA32_MC18_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 44AH, 1098	IA32_MC18_ADDR	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 44BH, 1099	IA32_MC18_MISC	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Banks MC13 through MC 18 report MC errors from the integrated memory controllers.		Package
Register Address: 44CH, 1100	IA32_MC19_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from a link interconnect module.		Package
Register Address: 44DH, 1101	IA32_MC19_STATUS	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs," through Section 17.3.2.4, "IA32_MCi_MISC MSRs." Bank MC19 reports MC errors from a link interconnect module.		Package
Register Address: 44EH, 1102	IA32_MC19_ADDR	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC19 reports MC errors from a link interconnect module.		Package
Register Address: 44FH, 1103	IA32_MC19_MISC	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC19 reports MC errors from a link interconnect module.		Package
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
Unit Multipliers Used in RAPL Interfaces (R/O)		Package
3:0	Power Units See Section 16.10.1, "RAPL Interfaces."	Package
7:4	Reserved.	Package
12:8	Energy Status Units Energy related information (in Joules) is based on the multiplier, $1/2^{\text{ESU}}$; where ESU is an unsigned integer represented by bits 12:8. Default value is 0EH (or 61 micro-joules).	Package
15:13	Reserved.	Package
19:16	Time Units See Section 16.10.1, "RAPL Interfaces."	Package
63:20	Reserved.	
Register Address: 618H, 1560	MSR_DRAM_POWER_LIMIT	
DRAM RAPL Power Limit Control (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) Energy consumed by DRAM devices.		Package
31:0	Energy in 15.3 micro-joules. Requires BIOS configuration to enable DRAM RAPL mode 0 (Direct VR).	
63:32	Reserved.	
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61CH, 1564	MSR_DRAM_POWER_INFO	
DRAM RAPL Parameters (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 620H, 1568	MSR_UNCORE_RATIO_LIMIT	
Uncore Ratio Limit (R/W) Out of reset, the min_ratio and max_ratio fields represent the widest possible range of uncore frequencies. Writing to these fields allows software to control the minimum and the maximum frequency that hardware will select.		Package
63:15	Reserved.	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
14:8	MIN_RATIO Writing to this field controls the minimum possible ratio of the LLC/Ring.	
7	Reserved.	
6:0	MAX_RATIO This field is used to limit the max ratio of the LLC/Ring.	
Register Address: 639H, 1593	MSR_PP0_ENERGY_STATUS	
Reserved (R/O) Reads return 0.		Package
Register Address: C8DH, 3213	IA32_QM_EVTSEL	
Monitoring Event Select Register (R/W) If CPUID.(EAX=07H, ECX=0):EBX.RDT-M[bit 12] = 1.		Thread
7:0	EventID (R/W) Event encoding: 0x00: No monitoring. 0x01: L3 occupancy monitoring. 0x02: Total memory bandwidth monitoring. 0x03: Local memory bandwidth monitoring. All other encoding reserved.	
31:8	Reserved.	
41:32	RMID (R/W)	
63:42	Reserved.	
Register Address: C8FH, 3215	IA32_PQR_ASSOC	
Resource Association Register (R/W)		Thread
9:0	RMID	
31:10	Reserved.	
51:32	CLOS (R/W)	
63: 52	Reserved.	
Register Address: C90H, 3216	IA32_L3_QOS_MASK_0	
L3 Class Of Service Mask - CLOS 0 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=0.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 0 enforcement.	
63:20	Reserved.	
Register Address: C91H, 3217	IA32_L3_QOS_MASK_1	
L3 Class Of Service Mask - CLOS 1 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=1.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 1 enforcement.	
63:20	Reserved.	
Register Address: C92H, 3218	IA32_L3_QOS_MASK_2	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
L3 Class Of Service Mask - CLOS 2 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=2.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 2 enforcement.	
63:20	Reserved.	
Register Address: C93H, 3219	IA32_L3_QOS_MASK_3	
L3 Class Of Service Mask - CLOS 3 (R/W). If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=3.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 3 enforcement.	
63:20	Reserved.	
Register Address: C94H, 3220	IA32_L3_QOS_MASK_4	
L3 Class Of Service Mask - CLOS 4 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=4.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 4 enforcement.	
63:20	Reserved.	
Register Address: C95H, 3221	IA32_L3_QOS_MASK_5	
L3 Class Of Service Mask - CLOS 5 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=5.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 5 enforcement.	
63:20	Reserved.	
Register Address: C96H, 3222	IA32_L3_QOS_MASK_6	
L3 Class Of Service Mask - CLOS 6 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=6.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 6 enforcement.	
63:20	Reserved.	
Register Address: C97H, 3223	IA32_L3_QOS_MASK_7	
L3 Class Of Service Mask - CLOS 7 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=7.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 7 enforcement.	
63:20	Reserved.	
Register Address: C98H, 3224	IA32_L3_QOS_MASK_8	
L3 Class Of Service Mask - CLOS 8 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=8.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 8 enforcement.	
63:20	Reserved.	
Register Address: C99H, 3225	IA32_L3_QOS_MASK_9	
L3 Class Of Service Mask - CLOS 9 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=9.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 9 enforcement.	

Table 2-50. MSRs Supported by the Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_55H (Contd.)

Register Address: Hex, Decimal	Register Name (Former Register Name)	
Register Information / Bit Fields	Bit Description	Scope
63:20	Reserved.	
Register Address: C9AH, 3226	IA32_L3_QOS_MASK_10	
L3 Class Of Service Mask - CLOS 10 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=10.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 10 enforcement.	
63:20	Reserved.	
Register Address: C9BH, 3227	IA32_L3_QOS_MASK_11	
L3 Class Of Service Mask - CLOS 11 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=11.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 11 enforcement.	
63:20	Reserved.	
Register Address: C9CH, 3228	IA32_L3_QOS_MASK_12	
L3 Class Of Service Mask - CLOS 12 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=12.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 12 enforcement.	
63:20	Reserved.	
Register Address: C9DH, 3229	IA32_L3_QOS_MASK_13	
L3 Class Of Service Mask - CLOS 13 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=13.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 13 enforcement.	
63:20	Reserved.	
Register Address: C9EH, 3230	IA32_L3_QOS_MASK_14	
L3 Class Of Service Mask - CLOS 14 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=14.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 14 enforcement.	
63:20	Reserved.	
Register Address: C9FH, 3231	IA32_L3_QOS_MASK_15	
L3 Class Of Service Mask - CLOS 15 (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] >=15.		Package
0:19	CBM: Bit vector of available L3 ways for CLOS 15 enforcement.	
63:20	Reserved.	

2.17.7 MSRs Specific to the 3rd Generation Intel® Xeon® Scalable Processor Family Based on Ice Lake Microarchitecture

The 3rd generation Intel® Xeon® Scalable Processor Family based on Ice Lake microarchitecture (CPUID Signature DisplayFamily_DisplayModel value of 06_6AH or 06_6CH) support the MSRs listed in Table 2-51.

Table 2-51. MSRs Supported by the 3rd Generation Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_6AH or 06_6CH

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 612H, 1554	MSR_PACKAGE_ENERGY_TIME_STATUS	
Package energy consumed by the entire CPU (R/W)		Package
31:0	Total amount of energy consumed since last reset.	
63:32	Total time elapsed when the energy was last updated. This is a monotonic increment counter with auto wrap back to zero after overflow. Unit is 10ns.	
Register Address: 618H, 1560	MSR_DRAM_POWER_LIMIT	
Allows software to set power limits for the DRAM domain and measurement attributes associated with each limit.		Package
14:0	DRAM_PP_PWR_LIM: Power Limit[0] for DDR domain. Units = Watts, Format = 11.3, Resolution = 0.125W, Range = 0-2047.875W.	
15	PWR_LIM_CTRL_EN: Power Limit[0] enable bit for DDR domain.	
16	Reserved.	
23:17	CTRL_TIME_WIN: Power Limit[0] time window Y value, for DDR domain. Actual time_window for RAPL is: $(1/1024 \text{ seconds}) * (1 + (x/4)) * (2^y)$	
62:24	Reserved.	
63	PP_PWR_LIM_LOCK: When set, this entire register becomes read-only. This bit will typically be set by BIOS during boot.	
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
31:0	Energy in 15.3 micro-joules. Requires BIOS configuration to enable DRAM RAPL mode 0 (Direct VR).	
63:32	Reserved.	
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61CH, 1564	MSR_DRAM_POWER_INFO	
DRAM Power Parameters (R/W)		Package
14:0	Spec DRAM Power (DRAM_TDP): The Spec power allowed for DRAM. The TDP setting is typical (not guaranteed). The units for this value are defined in MSR_DRAM_POWER_INFO_UNIT[PWR_UNIT].	
15	Reserved.	

Table 2-51. MSRs Supported by the 3rd Generation Intel® Xeon® Scalable Processor Family with a CPUID Signature DisplayFamily_DisplayModel Value of 06_6AH or 06_6CH (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
30:16	Minimal DRAM Power (DRAM_MIN_PWR): The minimal power setting allowed for DRAM. Lower values will be clamped to this value. The minimum setting is typical (not guaranteed). The units for this value are defined in MSR_DRAM_POWER_INFO_UNIT[PWR_UNIT].	
31	Reserved.	
46:32	Maximal Package Power (DRAM_MAX_PWR): The maximal power setting allowed for DRAM. Higher values will be clamped to this value. The maximum setting is typical (not guaranteed). The units for this value are defined in MSR_DRAM_POWER_INFO_UNIT[PWR_UNIT].	
47	Reserved.	
54:48	Maximal Time Window (DRAM_MAX_WIN): The maximal time window allowed for the DRAM. Higher values will be clamped to this value. x = PKG_MAX_WIN[54:53] y = PKG_MAX_WIN[52:48] The timing interval window is a floating-point number given by $1.x * \text{power}(2,y)$. The unit of measurement is defined in MSR_DRAM_POWER_INFO_UNIT[TIME_UNIT].	
62:55	Reserved.	
63	LOCK: Lock bit to lock the register.	
Register Address: 981H, 2433	IA32_TME_CAPABILITY	
See Table 2-2.		
Register Address: 982H, 2434	IA32_TME_ACTIVATE	
See Table 2-2.		
Register Address: 983H, 2435	IA32_TME_EXCLUDE_MASK	
See Table 2-2.		
Register Address: 984H, 2436	IA32_TME_EXCLUDE_BASE	
See Table 2-2.		

2.17.8 MSRs Specific to the 4th and 5th Generation Intel® Xeon® Scalable Processor Families

The 4th generation Intel® Xeon® Scalable Processor Family based on Sapphire Rapids microarchitecture (CPUID Signature DisplayFamily_DisplayModel value of 06_8FH) and the 5th generation Intel® Xeon® Scalable Processor Family based on Emerald Rapids microarchitecture (CPUID Signature DisplayFamily_DisplayModel value of 06_CFH) both support the MSRs listed in Section 2.17, “MSRs In the 6th—13th Generation Intel® Core™ Proces-

sors, 1st—5th Generation Intel® Xeon® Scalable Processor Families, Intel® Core™ Ultra 7 Processors, 8th Generation Intel® Core™ i3 Processors, Intel® Xeon® E Processors, Intel® Xeon® 6 P-core processors, Intel® Xeon® 6 E-core processors, and Intel® Series 2 Core™ Ultra Processors,” including Table 2-52. For an MSR listed in Table 2-52 that also appears in the model-specific tables of prior generations, Table 2-52 supersedes prior generation tables.

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 33H, 51	MSR_MEMORY_CTRL	
Memory Control Register (R/W)		Core
27:0	Reserved.	
28	UC_LOCK_DISABLE If set to 1, a UC lock will cause a #GP(0) exception. See Section 10.1.2.3, “Features to Disable Bus Locks.”	
29	SPLIT_LOCK_DISABLE If set to 1, a split lock will cause an #AC(0) exception. See Section 10.1.2.3, “Features to Disable Bus Locks.”	
31:30	Reserved.	
Register Address: A7H, 167	MSR_BIOS_DEBUG	
BIOS DEBUG (R/O) See Table 2-45.		Thread
Register Address: BCH, 188	IA32_MISC_PACKAGE_CTL	
Power Filtering Control (R/W) IA32_ARCH_CAPABILITIES[bit 10] enumerates support for this MSR. See Table 2-2.		Package
Register Address: BFH, 191	IA32_PBOPT_CTRL	
IA32_PBOPT_OPT_CTRL If IA32_ARCH_CAPABILITIES[32] = 1 This MSR provides an architectural enumeration function for model-specific behavior.		
0	PREDICTION_CONTROL_BARRIER (R/W) When set to 0 (default), original Indirect Branch Predictor Barrier Target Array Return (IBPB TA RET) mitigation is enabled. When set to 1, alternative IBPB mitigation is enabled.	
63:1	Reserved.	
Register Address: CFH, 207	IA32_CORE_CAPABILITIES	
IA32 Core Capabilities Register (R/W) If CPUID.(EAX=07H, ECX=0):EDX[30] = 1. This MSR provides an architectural enumeration function for model-specific behavior.		Core
0	Reserved: returns zero.	
1	Reserved: returns zero.	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
2	INTEGRITY_CAPABILITIES When set to 1, the processor supports MSR_INTEGRITY_CAPABILITIES.	
3	RSM_IN_CPL0_ONLY Indicates that RSM will only be allowed in CPL0 and will #GP for all non-CPL0 privilege levels.	
4	UC_LOCK_DISABLE_SUPPORTED When read as 1, software can set bit 28 of MSR_MEMORY_CTRL (MSR address 33H).	
5	SPLIT_LOCK_DISABLE_SUPPORTED When read as 1, software can set bit 29 of MSR_MEMORY_CTRL.	
6	Reserved: returns zero.	
7	UC_STORE_THROTTLING_SUPPORTED Indicates that the snoop filter quality of service MSRs are supported on this core. This is based on the existence of a non-inclusive cache and the L2/MLC QoS feature supported.	
63:8	Reserved: returns zero.	
Register Address: E1H, 225	IA32_UMWAIT_CONTROL	
UMWAIT Control (R/W) See Table 2-2.		
Register Address: EDH, 237	MSR_RAR_CONTROL	
RAR Control (R/W)		Thread
63:32	Reserved.	
31	ENABLE RAR events are recognized. When RAR is not enabled, RARs are dropped.	
30	IGNORE_IF Allow RAR servicing at the RLP regardless of the value of RFLAGS.IF.	
29:0	Reserved.	
Register Address: EEH, 238	MSR_RAR_ACTION_VECTOR_BASE	
Pointer to RAR Action Vector (R/W)		Thread
63:MAXPHYADDR	Reserved.	
MAXPHYADDR-1:6	VECTOR_PHYSICAL_ADDRESS Pointer to the physical address of the 64B aligned RAR action vector.	
5:0	Reserved.	
Register Address: EFH, 239	MSR_RAR_PAYLOAD_TABLE_BASE	
Pointer to Base of RAR Payload Table (R/W)		Thread
63:MAXPHYADDR	Reserved.	
MAXPHYADDR-1:12	TABLE_PHYSICAL_ADDRESS Pointer to the base physical address of the 4K aligned RAR payload table.	
11:0	Reserved.	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: F0H, 240	MSR_RAR_INFO	
Read Only RAR Information (R0)		Thread
63:38	Always zero.	
37:32	Table Max Index Maximum supported payload table index.	
31:0	Supported payload type bitmap. A value of 1 in bit position [i] indicates that payload type [i] is supported.	
Register Address: 105H, 261	MSR_CORE_BIST	
Core BIST (R/W) Controls Array BIST activation and status checking as part of FUSA.		Core
31:0	BIST_ARRAY Bitmap indicating which arrays to run BIST on (WRITE). Bitmap indicating which arrays were not processed, i.e., completion mask (READ).	
39:32	BANK Array bank of the [least significant set bit] array indicated in EAX to start BIST(WRITE). Array bank interrupted or failed (READ).	
47:40	DWORD Array dword of the [least significant set bit] array indicated in EAX to start BIST (WRITE). Array dword interrupted or failed (READ).	
62:48	Reserved.	
63	CTRL_RESULT Indicates whether WRMSR should signal Machine-Check upon BIST-error (WRITE). BIST result PASS(0)/FAIL(1) of the (least significant set bit) array indicated in EAX (READ).	
Register Address: 10AH, 266	IA32_ARCH_CAPABILITIES	
Enumeration of Architectural Features (R/O) See Table 2-2.		
Register Address: 1A4H, 420	MSR_PREFETCH_CONTROL	
Prefetch Disable Bits (R/W)		
0	L2_HARDWARE_PREFETCHER_DISABLE If 1, disables the L2 hardware prefetcher, which fetches additional lines of code or data into the L2 cache.	
1	L2_ADJACENT_CACHE_LINE_PREFETCHER_DISABLE If 1, disables the adjacent cache line prefetcher, which fetches the cache line that comprises a cache line pair (128 bytes).	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
2	DCU_HARDWARE_PREFETCHER_DISABLE If 1, disables the L1 data cache prefetcher, which fetches the next cache line into L1 data cache.	
3	DCU_IP_PREFETCHER_DISABLE If 1, disables the L1 data cache IP prefetcher, which uses sequential load history (based on instruction pointer of previous loads) to determine whether to prefetch additional lines.	
4	Reserved.	
5	AMP_PREFETCH_DISABLE If 1, disables the L2 Adaptive Multipath Probability (AMP) prefetcher.	
63:6	Reserved.	
Register Address: 1ADH, 429	MSR_PRIMARY_TURBO_RATIO_LIMIT	
Primary Maximum Turbo Ratio Limit (R/W) See Table 2-46.		Package
Register Address: 1AEH, 430	MSR_TURBO_RATIO_LIMIT_CORES	
See Table 2-50.		Package
Register Address: 1C4H, 452	IA32_XFD	
Extended Feature Detect (R/W) See Table 2-2.		
Register Address: 1C5H, 453	IA32_XFD_ERR	
XFD Error Code (R/W) See Table 2-2.		
Register Address: 2C2H, 706	MSR_COPY_SCAN_HASHES	
COPY_SCAN_HASHES (W)		Die
63:0	SCAN_HASH_ADDR Contains the linear address of the SCAN Test HASH Binary loaded into memory.	
Register Address: 2C3H, 707	MSR_SCAN_HASHES_STATUS	
SCAN_HASHES_STATUS (R/O)		
15:0	CHUNK_SIZE Chunk size of the test in KB.	Die
23:16	NUM_CHUNKS Total number of chunks.	Die
31:24	Reserved: all zeros.	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
39:32	ERROR_CODE The error-code refers to the LP that runs WRMSR(2C2H). 0x0: No error reported. 0x1: Attempt to copy scan-hashes when copy already in progress. 0x2: Secure Memory not set up correctly. 0x3: Scan-image header Image_info.ProgramID doesn't match RDMSR(2D9H)[31:24], or scan-image header Processor-Signature doesn't match F/M/S, or scan-image header Processor-Flags doesn't match PlatformID. 0x4: Reserved 0x5: Integrity check failed. 0x6: Re-install of scan test image attempted when current scan test image is in use by other LPs.	Thread
50:40	Reserved: set to all zeros.	
62:51	MAX_CORE_LIMIT Maximum Number of cores that can run Intel® In-field Scan simultaneously minus 1. 0 means 1 core at a time.	Die
63	Valid Valid bit is set when COPY_SCAN_HASHES has completed successfully.	Die
Register Address: 2C4H, 708	MSR_AUTHENTICATE_AND_COPY_CHUNK	
AUTHENTICATE_AND_COPY_CHUNK (W)		Die
7:0	CHUNK_INDEX Chunk Index, should be less than the total number of chunks defined by NUM_CHUNKS (MSR_SCAN_HASHES_STATUS[23:16]).	
63:8	CHUNK_ADDR Bits 63:8 of 256B aligned Linear address of scan chunk in memory.	
Register Address: 2C5H, 709	MSR_CHUNKS_AUTHENTICATION_STATUS	
CHUNKS_AUTHENTICATION_STATUS (R/O)		
7:0	VALID_CHUNKS Total number of Valid (authenticated) chunks.	Die
15:8	TOTAL_CHUNKS Total number of chunks.	Die
31:16	Reserved: all zeros.	
39:32	ERROR_CODE The error code refers to the LP that runs WRMSR(2C4H). 0x0: No error reported. 0x1: Attempt to authenticate a CHUNK which is already marked as authentic or is currently being installed by another core. 0x2: CHUNK authentication error. HASH of chunk did not match expected value.	Thread
63:40	Reserved: set to all zeros.	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 2C6H, 710	MSR_ACTIVATE_SCAN	
ACTIVATE_SCAN (W)		Thread
7:0	CHUNK_START_INDEX Indicates chunk index to start from.	
15:8	CHUNK_STOP_INDEX Indicates what chunk index to stop at (inclusive).	
31:16	Reserved: all zeros.	
62:32	THREAD_WAIT_DELAY TSC based delay to allow threads to rendezvous.	
63	SIGNAL_MCE If 1, then on scan-error log MC in MC4_STATUS and signal MCE if machine check signaling enabled in MC4_CTL[0]. If 0, then no logging/no signaling.	
Register Address: 2C7H, 711	MSR_SCAN_STATUS	
SCAN_STATUS (R/O)		
7:0	CHUNK_NUM SCAN Chunk that was reached.	Core
15:8	CHUNK_STOP_INDEX Indicates what chunk index to stop at (inclusive). Maps to same field in WRMSR(ACTIVATE_SCAN).	Core
31:16	Reserved: return all zeros.	
39:32	ERROR_CODE 0x0: No error. 0x1: SCAN operation did not start. Other thread did not join in time. 0x2: SCAN operation did not start. Interrupt occurred prior to threads rendezvous. 0x3: SCAN operation did not start. Power Management conditions are inadequate to run Intel In-field Scan. 0x4: SCAN operation did not start. Non-valid chunks in the range CHUNK_STOP_INDEX : CHUNK_START_INDEX. 0x5: SCAN operation did not start. Mismatch in arguments between threads T0/T1. 0x6: SCAN operation did not start. Core not capable of performing SCAN currently. 0x8: SCAN operation did not start. Exceeded number of Logical Processors (LP) allowed to run Intel In-field Scan concurrently. MAX_CORE_LIMIT exceeded. 0x9: Interrupt occurred. Scan operation aborted prematurely, not all chunks requested have been executed.	Thread
61:40	Reserved: return all zeros.	
62	SCAN_CONTROL_ERROR Scan-System-Controller malfunction.	Core

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
63	SCAN_SIGNATURE_ERROR Core failed SCAN-SIGNATURE checking for this chunk.	Core
Register Address: 2C8H, 712	MSR_SCAN_MODULE_ID	
SCAN_MODULE_ID (R/O)		Module
31:0	RevID of the currently installed scan test image. Maps to Revision field in external header (offset 4).	
63:32	Reserved: return all zeros.	
Register Address: 2C9H, 713	MSR_LAST_SAF_WP	
LAST_SAF_WP (R/O)		Core
31:0	LAST_WP Provides information about the core when the last WRMSR(ACTIVATE_SCAN) was executed. Available only if enumerated in MSR_INTEGRITY_CAPABILITIES[10:9].	
63:32	Reserved: return all zeros.	
Register Address: 2D9H, 729	MSR_INTEGRITY_CAPABILITIES	
INTEGRITY_CAPABILITIES (R/O)		Module
0	STARTUP_SCAN_BIST When set, supports Intel In-field Scan.	
3:1	Reserved: return all zeros.	
4	PERIODIC_SCAN_BIST When set, supports Intel In-field Scan.	
23:5	Reserved: return all zeros.	
31:24	ID of the scan programs supported for this part. WRMSR(2C2H) verifies this value against the corresponding value in the scan-image header, i.e., Image_info.	
Register Address: 410H, 1040	IA32_MC4_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC4 reports MC errors from the PCU module. If SIGNAL_MCE is set, a Scan Status is logged in MC4_STATUS and MC4_MISC.		Package
Register Address: 411H, 1041	IA32_MC4_STATUS	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC4 reports MC errors from the PCU module. If SIGNAL_MCE is set, a Scan Status is logged in MC4_STATUS and MC4_MISC.		Package
Register Address: 412H, 1042	IA32_MC4_ADDR	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC4 reports MC errors from the PCU module. If SIGNAL_MCE is set, a Scan Status is logged in MC4_STATUS and MC4_MISC.		Package
Register Address: 413H, 1043	IA32_MC4_MISC	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Section 17.3.2.1, "IA32_MCI_CTL MSRs," through Section 17.3.2.4, "IA32_MCI_MISC MSRs." Bank MC4 reports MC errors from the PCU module. If SIGNAL_MCE is set, a Scan Status is logged in MC4_STATUS and MC4_MISC.		Package
Register Address: 492H, 1170	IA32_VMX_PROCBASED_CTL3	
Capability Reporting Register of Tertiary Processor-Based VM-Execution Controls (R/O) See Table 2-2.		
Register Address: 493H, 1171	IA32_VMX_EXIT_CTL2	
Capability Reporting Register of Secondary VM-Exit Controls (R/O) See Table 2-2.		
Register Address: 540H, 1344	MSR_THREAD_UARCH_CTL	
Thread Microarchitectural Control (R/W) See Table 2-47.		Thread
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) Energy consumed by DRAM devices.		Package
31:0	Energy in 61 micro-joules. Requires BIOS configuration to enable DRAM RAPL mode 0 (Direct VR).	
63:32	Reserved.	
Register Address: 64DH, 1613	MSR_PLATFORM_ENERGY_STATUS	
Platform Energy Status (R/O)		Package
31:0	TOTAL_ENERGY_CONSUMED Total energy consumption in J (32.0), in 10nsec units.	
63:32	TIME_STAMP Time stamp (U32.0).	
Register Address: 65CH, 1628	MSR_PLATFORM_POWER_LIMIT	
Platform Power Limit Control (R/W-L)		Package
16:0	POWER_LIMIT_1 The average power limit value that the platform must not exceed over a time window as specified by the Power_Limit_1_TIME field. The default value is the Thermal Design Power (TDP) and varies with product skus. The unit is specified in MSR_RAPL_POWER_UNIT.	
17	POWER_LIMIT_1_EN When set, the processor can apply control policies such that the platform average power does not exceed the Power_Limit_1 value over an exponential weighted moving average of the time window.	
18	CRITICAL_POWER_CLAMP_1 When set, the processor can go below the OS-requested P States to maintain the power below the specified Power_Limit_1 value.	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
25:19	POWER_LIMIT_1_TIME This indicates the time window over which the Power_Limit_1 value should be maintained. This field is made up of two numbers from the following equation: Time Window = (float) ((1+(X/4))*(2^Y)), where: X = POWER_LIMIT_1_TIME[23:22] Y = POWER_LIMIT_1_TIME[21:17] The maximum allowed value in this field is defined in MSR_PKG_POWER_INFO[PKG_MAX_WIN]. The default value is 0DH, and the unit is specified in MSR_RAPL_POWER_UNIT[Time Unit].	
31:26	Reserved.	
48:32	POWER_LIMIT_2 This is the Duration Power limit value that the platform must not exceed. The unit is specified in MSR_RAPL_POWER_UNIT.	
49	Enable Platform Power Limit #2 When set, enables the processor to apply control policy such that the platform power does not exceed Platform Power limit #2 over the Short Duration time window.	
50	Platform Clamping Limitation #2 When set, allows the processor to go below the OS requested P states in order to maintain the power below specified Platform Power Limit #2 value.	
57:51	POWER_LIMIT_2_TIME This indicates the time window over which the Power_Limit_2 value should be maintained. This field has the same format as the POWER_LIMIT_1_TIME field.	
62:58	Reserved.	
63	LOCK Setting this bit will lock all other bits of this MSR until system RESET.	
Register Address: 665H, 1637	MSR_PLATFORM_POWER_INFO	
Platform Power Information (R/W)		Package
16:0	MAX_PPL1 Maximum PP L1 value. The unit is specified in MSR_RAPL_POWER_UNIT.	
31:17	MIN_PPL1 Minimum PP L1 value. The unit is specified in MSR_RAPL_POWER_UNIT.	
48:32	MAX_PPL2 Maximum PP L2 value. The unit is specified in MSR_RAPL_POWER_UNIT.	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
55:49	MAX_TW Maximum time window. The unit is specified in MSR_RAPL_POWER_UNIT.	
62:56	Reserved.	
63	LOCK Setting this bit will lock all other bits of this MSR until system RESET.	
Register Address: 666H, 1638	MSR_PLATFORM_RAPL_SOCKET_PERF_STATUS	
Platform RAPL Socket Performance Status (R/O)		Package
31:0	Count of limited performance due to platform RAPL limit.	
Register Address: 6A0H, 1696	IA32_U_CET	
Configure User Mode CET (R/W) See Table 2-2.		
Register Address: 6A2H, 1698	IA32_S_CET	
Configure Supervisor Mode CET (R/W) See Table 2-2.		
Register Address: 6A4H, 1700	IA32_PL0_SSP	
Linear address to be loaded into SSP on transition to privilege level 0. (R/W) See Table 2-2.		
Register Address: 6A5H, 1701	IA32_PL1_SSP	
Linear address to be loaded into SSP on transition to privilege level 1. (R/W) See Table 2-2.		
Register Address: 6A6H, 1702	IA32_PL2_SSP	
Linear address to be loaded into SSP on transition to privilege level 2. (R/W) See Table 2-2.		
Register Address: 6A7H, 1703	IA32_PL3_SSP	
Linear address to be loaded into SSP on transition to privilege level 3. (R/W) See Table 2-2.		
Register Address: 6A8H, 1704	IA32_INTERRUPT_SSP_TABLE_ADDR	
Linear address of a table of seven shadow stack pointers that are selected in IA-32e mode using the IST index (when not 0) from the interrupt gate descriptor. (R/W) See Table 2-2.		
Register Address: 6E1H, 1761	IA32_PKRS	
Specifies the PK permissions associated with each protection domain for supervisor pages (R/W) See Table 2-2.		
Register Address: 776H, 1910	IA32_HWP_CTL	
See Table 2-2.		
Register Address: 981H, 2433	IA32_TME_CAPABILITY	
Memory Encryption Capability MSR See Table 2-2.		

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 985H, 2437	IA32_UINTR_RR	
User Interrupt Request Register (R/W) See Table 2-2.		
Register Address: 986H, 2438	IA32_UINTR_HANDLER	
User Interrupt Handler Address (R/W) See Table 2-2.		
Register Address: 987H, 2439	IA32_UINTR_STACKADJUST	
User Interrupt Stack Adjustment (R/W) See Table 2-2.		
Register Address: 988H, 2440	IA32_UINTR_MISC	
User-Interrupt Target-Table Size and Notification Vector (R/W) See Table 2-2.		
Register Address: 989H, 2441	IA32_UINTR_PD	
User Interrupt PID Address (R/W) See Table 2-2.		
Register Address: 98AH, 2442	IA32_UINTR_TT	
User-Interrupt Target Table (R/W) See Table 2-2.		
Register Address: C70H, 3184	MSR_B1_PMON_EVNT_SELO	
Uncore B-box 1 PerfMon event select MSR.		Package
Register Address: C71H, 3185	MSR_B1_PMON_CTR0	
Uncore B-box 1 PerfMon counter MSR.		Package
Register Address: C72H, 3186	MSR_B1_PMON_EVNT_SEL1	
Uncore B-box 1 PerfMon event select MSR.		Package
Register Address: C73H, 3187	MSR_B1_PMON_CTR1	
Uncore B-box 1 PerfMon counter MSR.		Package
Register Address: C74H, 3188	MSR_B1_PMON_EVNT_SEL2	
Uncore B-box 1 PerfMon event select MSR.		Package
Register Address: C75H, 3189	MSR_B1_PMON_CTR2	
Uncore B-box 1 PerfMon counter MSR.		Package
Register Address: C76H, 3190	MSR_B1_PMON_EVNT_SEL3	
Uncore B-box 1vPerfMon event select MSR.		Package
Register Address: C77H, 3191	MSR_B1_PMON_CTR3	
Uncore B-box 1 PerfMon counter MSR.		Package
Register Address: C82H, 3122	MSR_W_PMON_BOX_OVF_CTRL	
Uncore W-box PerfMon local box overflow control MSR.		Package
Register Address: C8FH, 3215	IA32_PQR_ASSOC	

Table 2-52. Additional MSRs Supported by the 4th and 5th Generation Intel® Xeon® Scalable Processor Families (CPUID Signature DisplayFamily_DisplayModel Values of 06_8FH and 06_CFH) (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2.		
Register Address: C90H—C9EH, 3216—3230	IA32_L3_QOS_MASK_0 through IA32_L3_QOS_MASK_14	
See Table 2-50.		Package
Register Address: D10H—D17H, 3344—3351	IA32_L2_QOS_MASK_[0-7]	
IA32_CR_L2_QOS_MASK_[0-7] If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] ≥ 0. See Table 2-2.		Core
Register Address: D93H, 3475	IA32_PASID	
See Table 2-2.		
Register Address: 1200H—121FH, 4608—4639	IA32_LBR_x_INFO	
Last Branch Record Entry X Info Register (R/W) See Table 2-2.		
Register Address: 1406H, 5126	IA32_MCU_CONTROL	
See Table 2-2.		
Register Address: 14CEH, 5326	IA32_LBR_CTL	
Last Branch Record Enabling and Configuration Register (R/W) See Table 2-2.		
Register Address: 14CFH, 5327	IA32_LBR_DEPTH	
Last Branch Record Maximum Stack Depth Register (R/W) See Table 2-2.		
Register Address: 1500H—151FH, 5376—5407	IA32_LBR_x_FROM_IP	
Last Branch Record Entry X Source IP Register (R/W) See Table 2-2.		
Register Address: 1600H—161FH, 5632—5663	IA32_LBR_x_TO_IP	
Last Branch Record Entry X Destination IP Register (R/W) See Table 2-2.		

2.17.9 MSRs Introduced in the Intel® Core™ Ultra 7 Processor Supporting Performance Hybrid Architecture

Table 2-53 lists additional MSRs for the Intel Core Ultra 7 processor with a CPUID Signature DisplayFamily_DisplayModel value of 06_AAH. Table 2-54 lists the MSRs unique to the processor P-core. Table 2-55 lists the MSRs unique to the processor E-core.

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 33H, 51	MSR_MEMORY_CTRL	
Memory Control Register		Core
26:0	Reserved.	
27	UC_STORE_THROTTLE If set to 1, when enabled, the processor will only allow one in-progress UC store at a time.	
28	UC_LOCK_DISABLE If set to 1, a UC lock will cause a #GP(0) exception. See Section 10.1.2.3, “Features to Disable Bus Locks.”	
29	SPLIT_LOCK_DISABLE If set to 1, a split lock will cause an #AC(0) exception. See Section 10.1.2.3, “Features to Disable Bus Locks.”	
63:30	Reserved.	
Register Address: 7AH, 122	IA32_FEATURE_ACTIVATION	
Feature Activation (R/W) Implements Feature Activation command. WRMSR to this address activates all ‘activatable’ features on this thread. See Table 2-2.		
Register Address: 80H, 128	MSR_TRACE_HUB_STH ACPIBAR_BASE	
MSR_TRACE_HUB_STH ACPIBAR_BASE (R/W) This register is used by BIOS to program Trace Hub STH base address that will be used by AET messages.		Thread
0	LOCK Lock bit. If set, this MSR cannot be re-written anymore. The lock bit has to be set in order for the AET packets to be directed to Trace Hub MMIO.	
17:1	Reserved.	
45:18	ADDRESS AET target address in Trace Hub MMIO space.	
63:46	Reserved.	
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration (R/W)		Core

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
3:0	PKG_C_STATE_LIMIT Specifies the lowest processor-specific C-state code name (consuming the least power) for the package. The default is set as factory-configured package C-state limit. The following C-state code name encodings may be supported: 0000b: C0/C1 (no package C-state support) 0001b: C2 0010b: C3 0011b: C6 0100b: C7 0101b: C7s 0110b: C8 0111b: C9 1000b: C10	
7:4	MAX_CORE_C_STATE Possible values are: 0000—reserved; 0001—C1; 0010—C3, 0011—C6.	
9:8	Reserved.	
10	IO_MWAIT_REDIRECTION_ENABLE When set, will map IO_read instructions sent to IO registers PMG_IO_BASE_ADDR.PMB0+0/1/2 to MWAIT(C2,3,4) instructions; applies to deepc4 too.	
14:11	Reserved.	
15	CFG_LOCK When set, locks bits 15:0 of this register for further writes, until the next reset occurs.	
24:16	Reserved.	
25	C3_STATE_AUTO_DEMOTION_ENABLE When set, processor will conditionally demote C6/C7 requests to C3 based on uncore auto-demote information.	
26	C1_STATE_AUTO_DEMOTION_ENABLE When set, processor will conditionally demote C3/C6/C7 requests to C1 based on uncore auto-demote information.	
27	ENABLE_C3_UNDEMOTION Enable Un-Demotion from Demoted C3.	
28	ENABLE_C1_UNDEMOTION Enable Un-Demotion from Demoted C1.	
29	ENABLE_PKG_C_STATE_AUTODEMOTION Enable Package C-State Auto-Demotion. It enables use of the history of past package C-state depth and residence, as a factor in determining C-State depth.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
30	ENABLE_PKG_C_UNDEMOTION Enable Package C-State Un-Demotion. It enables considering cases where demotion was the incorrect decision in determining C-State depth.	
31	TIMED_MWAIT_ENABLE When set, enables Timed MWAIT feature. MWAIT would #GP on attempts to do setup MWAIT timer if this bit is not set.	
63:32	Reserved.	
Register Address: E4H, 228	MSR_IO_CAPTURE_BASE	
IO Capture Base (R/W) Power Management IO Redirection in C-state. See http://biosbits.org .		Core
15:0	LVL_2_BASE_ADDRESS Specifies the base address visible to software for IO redirection. If MSR_PKG_CST_CONFIG_CONTROL.IO_MWAIT_REDIRECTION_ENABLE, reads to this address will be consumed by the power management logic and decoded to MWAIT instructions. When IO port address redirection is enabled, this is the IO port address reported to the OS/software.	
18:16	CST_RANGE Specifies the encoding value of the maximum C-State code name to be included when IO read to MWAIT redirection is enabled by MSR_PKG_CST_CONFIG_CONTROL.IO_MWAIT_REDIRECTION_ENABLE: 000b—C3 is the max C-State to include. 001b—C6 is the max C-State to include. 010b—C7 is the max C-State to include.	
63:19	Reserved.	
Register Address: 13CH, 316	MSR_FEATURE_CONFIG	
AES Feature Configuration (R/W)		Core
0	AESNI_LOCK Once this bit is set, writes to this register will not be allowed.	
1	AESNI_DISABLE This bit disables Advanced Encryption Standard feature on this processor core. To disable AES, BIOS will write '11 to this MSR on every core.	
63:2	Reserved.	
Register Address: 140H, 320	MSR_FEATURE_ENABLES	
Feature Enable (R/W) Miscellaneous enables for thread specific features.		Thread
0	CPUID_GP_ON_CPL_GT_0 Causes CPUID to #GP if CPL greater than 0 and not in SMM.	
63:1	Reserved.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1A2H, 418	MSR_TEMPERATURE_TARGET	
Temperature Target (R/W) Legacy register holding temperature related constants for Platform use.		Package
6:0	TCC Offset Time Window Describes the RATL averaging time window.	
7	TCC Offset Clamping Bit When enabled will allow RATL throttling below P1.	
15:8	Temperature Control Offset Fan Temperature Target Offset (a.k.a. T-Control) indicates the relative offset from the Thermal Monitor Trip Temperature at which fans should be engaged.	
23:16	TCC Activation Temperature The minimum temperature at which PROCHOT# will be asserted. The value is degrees C.	
30:24	TCC Activation Offset Specifies a temperature offset in degrees C from the temperature target (bits 23:16). PROCHOT# will assert at the offset target temperature. Write is permitted only if MSR_PLATFORM_INFO[30] is set.	
31	LOCKED When set, this entire register becomes read-only.	
63:2	Reserved.	
Register Address: 1A4H, 420	MSR_PREFETCH_CONTROL	
PREFETCH Control (R/W) Prefetch disable bits.		Thread
0	L2_HARDWARE_PREFETCHER_DISABLE If 1, disables the L2 hardware prefetcher, which fetches additional lines of code or data into the L2 cache.	
1	L2_ADJACENT_CACHE_LINE_PREFETCHER_DISABLE If 1, disables the adjacent cache line prefetcher, which fetches the cache line that comprises a cache line pair (128 bytes).	
2	DCU_HARDWARE_PREFETCHER_DISABLE If 1, disables the L1 data cache prefetcher, which fetches the next cache line into L1 data cache.	
3	DCU_IP_PREFETCHER_DISABLE If 1, disables the L1 data cache IP prefetcher, which uses sequential load history (based on instruction pointer of previous loads) to determine whether to prefetch additional lines.	
4	DCU_NEXT_PAGE_PREFETCH_DISABLE If 1, disables Next Page prefetcher.	
5	AMP_PREFETCH_DISABLE If 1, disables L2 Adaptive Multipath Probability (AMP) prefetcher.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
6	LLC_PAGE_PREFETCH_DISABLE If 1, disables the LLC Page prefetcher.	
7	AOP_PREFETCH_DISABLE	
8	STREAM_PREFETCH_CODE_FETCH_DISABLE	
63:9	Reserved.	
Register Address: 1A6H, 422	MSR_OFFCORE_RSP_0	
OFFCORE_RSP_0 (R/W) Offcore Response Event Select Register		Thread
0	TRUE_DEMAND_CACHE_LOAD Demand Data Rd = DCU reads (includes partials) that is not tagged homeless.	
1	DEMAND_RFO Demand Instruction fetch = IFU Fetches. ItoM or RFO that is not tagged homeless.	
2	DEMAND_CODE_READ Demand Instruction fetch = IFU Fetches. CRd or CRd_UC.	
3	CORE_MODIFIED_WRITEBACK WBMtoI or WBMtoE.	
4	HW_PREFETCH_MLC_LOAD L2 prefetcher requests triggered by reads from MEC (except those triggered by I-side).	
5	HW_PREFETCH_MLC_RFO L2 prefetcher requests triggered by RFOs.	
6	HW_PREFETCH_MLC_CODE L2 prefetcher requests triggered by I-side requests.	
7	HW_PREFETCH_LLC_LOAD LLC prefetch requests triggered by DRd.	
8	HW_PREFETCH_LLC_RFO LLC prefetch requests triggered by RFO.	
9	HW_PREFETCH_LLC_CODE LLC prefetch requests triggered by CRd.	
10	L1_HWPREFETCH Covers Hardware PFRFO, PFNEAR, PFMED, PFFAR, PFHW, PFNTA, PFNPP, PFIPP including the homeless versions.	
11	ALL_STREAMING_STORE Write Combining. WCiL or WCiLF.	
12	CORE_NON_MODIFIED_WB WBEFtoI or WBEFtoE.	
13	LLC_PREFETCH LLC prefetch of load/code/RFO.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
14	L1_SWPREFETCH Covers Software PFRFO, PFNEAR, PFMED, PFFAR, PFHW, PFNTA, PFNPP, PFIPP including the homeless versions.	
15	OTHER Includes CLFlush, CLFlushOPT, CLDemote, CLWB, Enqueue SetMonitor, PortIn, IntA, Lock, SplitLock, Unlock, SpCyc, ClrMonitor, PortOut, IntPriUp, IntLog, IntPhy, EOI, RdCurr, WbStol, LLCWBIInv, LLCInv, NOP, PCOMMIT.	
16	ANY_RESP Match on any response.	
17	SUPPLIER_NONE No Supplier Details. DATA_PRE [6:3] = 0.	
18	LLC_HIT_M_STATE LLC/L3, M-state, DATA_PRE [6:3] = 2.	
19	LLC_HIT_E_STATE LLC/L3, E-state, DATA_PRE [6:3] = 4.	
20	LLC_HIT_S_STATE LLC/L3, S-state, DATA_PRE [6:3] = 6.	
21	LLC_HIT_F_STATE LLC/L3, F-state, DATA_PRE [6:3] = 8.	
22	FAR_MEM_LOCAL Far Memory, Local, DATA_PRE [6:3] = 1.	
23	FAR_MEM_REMOTE_0_HOP Far Memory, Remote 0-hop, DATA_PRE [6:3] = 3.	
24	FAR_MEM_REMOTE_1_HOP Far Memory, Remote 1-hop, DATA_PRE [6:3] = 5.	
25	FAR_MEM_REMOTE_2_PLUS_HOP Far Memory, Rem 2+ hop, DATA_PRE [6:3] = 7.	
26	NEAR_MEM_MISS_LOCAL_NODE LLC Miss Local Node. Near Memory, Local DATA_PRE [6:3] = E.	
27	NEAR_MEM_REMOTE_0_HOP Near Memory, Remote 0-hop, DATA_PRE [6:3] = B	
28	NEAR_MEM_REMOTE_1_HOP Near Memory, Remote 1-hop, DATA_PRE [6:3] = D.	
29	NEAR_MEM_REMOTE_2_PLUS_HOP Near Memory, Remote 2+ hop, DATA_PRE [6:3] = F.	
30	SPL_HIT Snoop Info: SPL-hit, DATA_PRE [2:0] = 6.	
31	SNOOP_NONE No details as to Snoop-related info. Snoop Info: None, DATA_PRE [2:0] = 0.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
32	NOT_NEEDED No snoop was needed to satisfy the request. Snoop Info: Not needed, DATA_PRE [2:0] = 1.	
33	MISS No snoop was needed to satisfy the request. Snoop Info: Miss, DATA_PRE [2:0] = 2.	
34	HIT_NO_FWD A snoop was needed and it Hits in at least one snooped cache. Hit denotes a cache-line was valid before snoop effect. Snoop Info: Hit No Fwd, DATA_PRE [2:0] = 3.	
35	HIT_EF_WITH_FWD A snoop was needed and data was Forwarded from a remote socket. Snoop Info: Hit EF w/Fwd, DATA_PRE [2:0] = 4.	
36	HITM A snoop was needed and it HitMed in local or remote cache. HitM denotes a cache-line was modified before snoop effect. Snoop Info: HitM, DATA_PRE [2:0] = 5.	
37	NON_DRAM Target was non-DRAM system address. Snoop Info: HitM, DATA_PRE [2:0] = 5.	
38	GO_ERR GO-ERR, RspData[3:0] = 0100.	
39	GO_NO_GO GO-NoGO, RspData[3:0] = 0111.	
40	INPKG_MEM_LOCAL In-package Memory, Local, DATA_PRE [6:3] = 9.	
41	INPKG_MEM_NONLOCAL In-package Memory, Non-Local, DATA_PRE [6:3] = C.	
43:42	Reserved.	
44	UC_LOAD PRd or UCRdF.	
45	UC_STORE WiL.	
46	PARTIAL_STREAMING_STORES WCiL.	
47	FULL_STREAMING_STORES WCiLF.	
48	L1_MODIFIED_WB EVICTION EXTTYPE from MEC.	
49	L2_MODIFIED_WB WBMtol or WBMtoE.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
50	PSMI MemPushWr_NS (PSMI only).	
51	ITOM ItoM.	
63:52	Reserved.	
Register Address: 1A7H, 423	MSR_OFFCORE_RSP_1	
OFFCORE_RSP_1 (R/W) Offcore Response Event Select Register. See MSR_OFFCORE_RSP_0 (at 1A6H).		Thread
Register Address: 1AAH, 426	MSR_MISC_PWR_MGMT	
Miscellaneous Power Management Control (R/W) Various model-specific features enumeration. See http://biosbits.org .		Package
0	Reserved.	
1	ENABLE_HWP_VOTING_RIGHT When set (1), The CPU will take into account thread HWP requests for threads that have voting rights only (ignores thread requests if they do not have voting rights). When reset(0), The CPU will take into account all thread HWP requests, even for threads that don't have voting rights. Setting this bit will cause the HWP Base feature bit to be reported in CPUID as present; clearing will cause it to be reported as non-present.	
5:2	Reserved.	
6	ENABLE_HWP Setting this bit will cause the HWP Base feature bit to report as present in CPUID; clearing this bit will cause CPUID to report the feature as non-present.	
7	ENABLE_HWP_INTERRUPT Setting this bit will cause the HWP Interrupt feature CPUID[6].EAX[8] bit to report as present; clearing will report as non-present.	
8	ENABLE_OUT_OF_BAND_AUTONOMOUS Setting this bit will cause the HWP Autonomous feature bit to report as present; clearing will report as non-present.	
11:9	Reserved.	
12	ENABLE_HWP_EPP Enable HWP EPP. Setting this bit (1) will cause the HWP CPUID[6].EAX[10] Energy Performance Preference bit to report as present (1); clearing will report as non-present (0).	
13	LOCK Setting this bit will prevent the BIOS specific bits from changing until the next reset. i.e., only Bits [0,22] which are meant for OS use can be changed once the LOCK bit is set.	
63:14	Reserved.	
Register Address: 1ADH, 429	MSR_PRIMARY_TURBO_RATIO_LIMIT	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Primary Maximum Turbo Ratio Limit (R/W) Software can configure these limits when MSR_PLATFORM_INFO[28] = 1. Specifies Maximum Ratio Limit for each group. Maximum ratio for groups with more cores must decrease monotonically.		Package
7:0	MAX_TURBO_GROUP_0: Maximum turbo ratio limit with 1 core active.	
15:8	MAX_TURBO_GROUP_1: Maximum turbo ratio limit with 2 cores active.	
23:16	MAX_TURBO_GROUP_2: Maximum turbo ratio limit with 3 cores active.	
31:24	MAX_TURBO_GROUP_3: Maximum turbo ratio limit with 4 cores active.	
39:32	MAX_TURBO_GROUP_4: Maximum turbo ratio limit with 5 cores active.	
47:40	MAX_TURBO_GROUP_5: Maximum turbo ratio limit with 6 cores active.	
55:48	MAX_TURBO_GROUP_6: Maximum turbo ratio limit with 7 cores active.	
63:56	MAX_TURBO_GROUP_7: Maximum turbo ratio limit with 8 cores active.	
Register Address: 1F1H, 497	MSR_CRASHLOG_CONTROL	
Crash Log Control (R/W) Write data to a Crash Log configuration.		Thread
0	CDDIS CrashDump_Disable: If set, indicates that Crash Dump is disabled.	
1	EN_GPRS Collect GPRs on a crash dump. Only meaningful when CDDIS is zero.	
2	EN_GPRS_IN_SMM Collect GPRs in SMM on a crash dump. Only meaningful when CDDIS is zero. EN_GPRS will override this control,	
3	TRIPLE_FAULT_SHUTDOWN Collect a crash log on a triple fault shutdown. Only meaningful when CDDIS is zero.	
63:4	Reserved.	
Register Address: 1F5H, 501	MSR_PRMRR_PHYS_MASK	
Processor Reserved Memory Range Register - Physical Mask (R/W)		Core
9:0	Reserved.	
10	LOCK Once set, this bit prevents software from modifying the PRMRR.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
11	VALID This bit serves as the enable for the PRMRR; the PRMRR must be LOCKed before it can be enabled.	
19:12	Reserved.	
45:20	MASK PRMRR Address Mask.	
63:46	Reserved.	
Register Address: 1FCH, 508	MSR_POWER_CTL	
Power Control Register (R/W) See http://biosbits.org .		Package
0	ENABLE_BIDIR_PROCHOT Used to enable or disable the response to PROCHOT# input. When set/enabled, the platform can force the CPU to throttle to a lower power condition such as Pn/Pm by asserting prohot#. When clear/disabled (default), the CPU ignores the status of the prohot input signal.	
1	C1E_ENABLE When set to '1', will enable the CPU to switch to the Minimum Enhanced Intel SpeedStep Technology operating point when all execution cores enter MWAIT (C1).	
2	SAPM_IMC_C2_POLICY This bit determines if self-refresh activation is allowed when entering Package C2 State. If it is set to 0b, PCODE will keep the FORCE_SR_OFF bit asserted in Package C2 State and allow its negation according to the defined latency negotiations with the PCH and Display Engine in Package C3 and deeper states. Otherwise, self-refresh is allowed in Package C2 State.	
3	FAST_BRK_SNP_EN This bit controls the VID swing rate for the OTHER_SNP_WAKE events that are detected by the iMPH. This is the event that is detected by the iMPH when a non-DMI snoopable request is observed while UCLK domain is not functional. 0b: Use slow VID swing rate. 1b: Use fast VID swing rate.	
17:4	Reserved.	
18	PWR_PERF_PLTFRM_OVR Power performance platform override.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
19	EE_TURBO_DISABLE Setting this bit disables the P-States energy efficiency optimization. Default value is 0. Disable/enable the energy efficiency optimization in P-State legacy mode (when IA32_PM_ENABLE[HWP_ENABLE] = 0), has an effect only in the turbo range or into PERF_MIN_CTL value if it is not zero set. In HWP mode (IA32_PM_ENABLE[HWP_ENABLE] == 1), has an effect between the OS desired or OS maximize to the OS minimize performance setting.	
20	RTH_DISABLE Setting this bit disables the Race to Halt optimization and avoids this optimization limitation to execute below the most efficient frequency ratio. Default value is 0 for processors that support Race to Halt optimization.	
21	DIS_PROCHOT_OUT Prochot output disable.	
22	PROCHOT_RESPONSE Prochhot configurable response enable.	
23	VR_THERM_ALERT_DISABLE_LOCK When set to 1, locks PROCHOT related bits of this MSR. Once set, a reset is required to clear this bit.	
24	VR_THERM_ALERT_DISABLE When set to 1, disables the VR_THERMAL_ALERT signaling.	
25	DISABLE_RING_EE Disable Ring EE.	
26	DISABLE_SA_OPTIMIZATION Disable SA optimization.	
27	DISABLE_OOK Disable OOK.	
28	DISABLE_AUTONOMOUS Disable HWP autonomous mode.	
29	Reserved.	
30	CSTATE_PREWAKE_DISABLE C-state pre-wake disable.	
63:31	Reserved.	
Register Address: 2A0H, 672	MSR_PRMRR_BASE_0	
Processor Reserved Memory Range Register - Physical Base Control Register (R/W)		Core
2:0	MEMTYPE Memory type for PRMRR accesses.	
3	CONFIGURED PRMRR base configured.	
19:4	Reserved.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
45:20	BASE PRMRR base address.	
63:46	Reserved.	
Register Address: 474H, 1140	IA32_MC29_CTL	
MC29_CTL. See Table 2-2.		Package
Register Address: 475H, 1141	IA32_MC29_STATUS	
MC29_STATUS. See Table 2-2.		Package
Register Address: 476H, 1142	IA32_MC29_ADDR	
MC29_ADDR. See Table 2-2.		Package
Register Address: 477H, 1143	IA32_MC29_MISC	
MC29_MISC. See Table 2-2.		Package
Register Address: 478H, 1144	IA32_MC30_CTL	
MC30_CTL. See Table 2-2.		Package
Register Address: 479H, 1145	IA32_MC30_STATUS	
MC30_STATUS. See Table 2-2.		Package
Register Address: 47AH, 1146	IA32_MC30_ADDR	
MC30_ADDR. See Table 2-2.		Package
Register Address: 47BH, 1147	IA32_MC30_MISC	
MC30_MISC. See Table 2-2.		Package
Register Address: 47CH, 1148	IA32_MC31_CTL	
MC31_CTL. See Table 2-2.		Package
Register Address: 47DH, 1149	IA32_MC31_STATUS	
MC31_STATUS. See Table 2-2.		Package
Register Address: 47EH, 1150	IA32_MC31_ADDR	
MC31_ADDR. See Table 2-2.		Package
Register Address: 47FH, 1151	IA32_MC31_MISC	
MC31_MISC. See Table 2-2.		Package
Register Address: 4E0H, 1248	MSR_SMM_FEATURE_CONTROL	
Enhanced SMM Feature Control (R/W) Reports SMM capability enhancement.		Package
0	LOCK When set, locks this register from further changes.	
1	SMM_CPU_SAVE_EN If 0, SMI/RSM will save/restore state in SMRAM If 1, SMI/RSM will save/restore state from SRAM.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
2	SMM_CODE_CHK_EN When clear (default) none of the logical processors are prevented from executing SMM code outside the ranges defined by the SMRR. When set, any logical processor in the package that attempts to execute SMM code not within the ranges defined by the SMRR will assert an unrecoverable MCE.	
63:3	Reserved.	
Register Address: 601H, 1537	MSR_VR_CURRENT_CONFIG	
Power Limit 4 (PL4) (R/W) Package-level maximum power limit (in Watts). It is a proactive, instantaneous limit.		Package
15:0	CURRENT_LIMIT PL4 Value in 0.125 A increments. This field is locked by MSR_VR_CURRENT_CONFIG.LOCK. When the LOCK bit is set to 1, this field becomes Read Only.	
30:16	Reserved.	
31	LOCK This bit will lock the CURRENT_LIMIT settings in this register and will also lock this setting. This means that once set to 1, the CURRENT_LIMIT setting and this bit become Read Only until the next Warm Reset.	
63:32	Reserved.	
Register Address: 620H, 1568	MSR_UNCORE_RATIO_LIMIT	
Uncore Ratio Limit (R/W) Min/Max Ratio Limits for Uncore LLC and Ring.		Package
6:0	MAX_CLR_RATIO Maximum allowed ratio for the Ring and Last Level Cache (LLC).	
7	Reserved.	
14:8	MIN_CLR_RATIO Minimum allowed ratio for the Ring and Last Level Cache (LLC).	
63:15	Reserved.	
Register Address: 638H, 1592	MSR_PPO_POWER_LIMIT	
MSR_PPO_POWER_LIMIT (R/W) PPO RAPL power unit control.		Package
14:0	IA_PP_PWR_LIM This is the power limitation on the IA cores power plane. The unit of measurement is defined in PACKAGE_POWER_SKU_UNIT_MSR[PWR_UNIT].	
15	PWR_LIM_CTRL_EN This bit must be set in order to limit the power of the IA cores power plane. 0b: IA cores power plane power limitation is disabled. 1b: IA cores power plane power limitation is enabled.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
16	PP_CLAMP_LIM Power Plane Clamping limitation; allow going below P1. 0b: PBM is limited between P1 and P0. 1b: PBM can go below P1.	
23:17	CTRL_TIME_WIN x = CTRL_TIME_WIN[23:22] y = CTRL_TIME_WIN[21:17] The timing interval window is Floating Point number given by 1.x * power(2,y). The unit of measurement is defined in PACKAGE_POWER_SKU_UNIT_MSR[TIME_UNIT]. The maximal time window is bounded by PACKAGE_POWER_SKU_MSR[PKG_MAX_WIN]. The minimum time window is 1 unit of measurement (as defined above).	
30:24	Reserved.	
31	PP_PWR_LIM_LOCK When set, all settings in this register are locked and are treated as Read Only.	
63:32	Reserved.	
Register Address: 64FH, 1615	MSR_CORE_PERF_LIMIT_REASONS	
Core Performance Limit Reasons Indicator of Frequency Clipping in Processor Cores. (Frequency refers to processor core frequency.)		Package
0	PROCHOT (R/O) PROCHOT Status. When set, frequency is reduced below the operating system request due to assertion of external PROCHOT.	
1	THERMAL (R/O) Thermal Status. When set, frequency is reduced below the operating system request due to a thermal event.	
3:2	Reserved.	
4	RSR_LIMIT (R/O) Residency State Regulation Status. When set, frequency is reduced below the operating system request due to residency state regulation limit.	
5	RATL (R/O) Running Average Thermal Limit Status. When set, frequency is reduced below the operating system request due to Running Average Thermal Limit (RATL).	
6	VR_THERMALERT (R/O) VR Therm Alert Status. When set, frequency is reduced below the operating system request due to a thermal alert from a processor Voltage Regulator (VR).	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
7	VR_TDC (R/O) VR Therm Design Current Status. When set, frequency is reduced below the operating system request due to VR thermal design current limit.	
8	OTHER (R/O) Other Status. When set, frequency is reduced below the operating system request due to electrical or other constraints.	
9	Reserved.	
10	PBM_PL1 (R/O) Package/Platform-Level Power Limiting PL1 Status. When set, frequency is reduced below the operating system request due to package/platform-level power limiting PL1.	
11	PBM_PL2 (R/O) Package/Platform-Level PL2 Power Limiting Status. When set, frequency is reduced below the operating system request due to package/platform-level power limiting PL2/PL3.	
12	MAX_TURBO_LIMIT (R/O) Max Turbo Limit Status. When set, frequency is reduced below the operating system request due to multi-core turbo limits.	
13	TURBO_ATTEN (R/O) Turbo Transition Attenuation Status. When set, frequency is reduced below the operating system request due to Turbo transition attenuation. This prevents performance degradation due to frequent operating ratio changes.	
15:14	Reserved.	
16	PROCHOT_LOG (R/W) PROCHOT Log. When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	THERMAL_LOG (R/W) Thermal Log When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
19:18	Reserved.	
20	RSR_LIMIT_LOG (R/W) Residency State Regulation Log. When set, indicates that the Residency State Regulation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
21	RATL_LOG (R/W) Running average thermal limit Log, RW, When set by PCODE indicates that Running average thermal limit has cause IA frequency clipping. Software should write to this bit to clear the status in this bit.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
22	VR_THERMALERT_LOG (R/W) VR Therm Alert Log. When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	VR_TDC_LOG (R/W) VR Thermal Design Current Log. When set, indicates that the VR TDC Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
24	OTHER_LOG (R/W) Other Log. When set, indicates that the Other Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Reserved.	
26	PBM_PL1_LOG (R/W) Package/Platform-Level PL1 Power Limiting Log. When set, indicates that the Package or Platform Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
27	PBM_PL2_LOG (R/W) Package/Platform-Level PL2 Power Limiting Log. When set, indicates that the Package or Platform Level PL2/PL3 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28	MAX_TURBO_LIMIT_LOG (R/W) Max Turbo Limit Log. When set, indicates that the Max Turbo Limit Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
29	TURBO_ATTEN_LOG (R/W) Turbo Transition Attenuation Log. When set, indicates that the Turbo Transition Attenuation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:30	Reserved.	
Register Address: 650H, 1616	MSR_SECONDARY_TURBO_RATIO_LIMIT	
Secondary Maximum Turbo Ratio Limit (R/W) Software can configure these limits when MSR_PLATFORM_INFO[28] = 1. Specifies Maximum Ratio Limit for each group. Maximum ratio for groups with more cores must decrease monotonically.		Package
7:0	MAX_TURBO_GROUP_0: Maximum turbo ratio limit with 1 core active.	
15:8	MAX_TURBO_GROUP_1: Maximum turbo ratio limit with 2 cores active.	
23:16	MAX_TURBO_GROUP_2: Maximum turbo ratio limit with 3 cores active.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
31:24	MAX_TURBO_GROUP_3: Maximum turbo ratio limit with 4 cores active.	
39:32	MAX_TURBO_GROUP_4: Maximum turbo ratio limit with 5 cores active.	
47:40	MAX_TURBO_GROUP_5: Maximum turbo ratio limit with 6 cores active.	
55:48	MAX_TURBO_GROUP_6: Maximum turbo ratio limit with 7 cores active.	
63:56	MAX_TURBO_GROUP_7: Maximum turbo ratio limit with 8 cores active.	
Register Address: 65CH, 1628	MSR_PLATFORM_POWER_LIMIT	
Platform Power Limit Control (R/W) Allows platform BIOS to limit power consumption of the platform devices to the specified values. The Long Duration power consumption is specified via Platform_Power_Limit_1 and Platform_Power_Limit_1_Time. The Short Duration power consumption limit is specified via the Platform_Power_Limit_2 with duration chosen by the processor. The processor implements an exponential-weighted algorithm in the placement of the time windows.		Package
14:0	POWER_LIMIT_1 Average Power limit value which the platform must not exceed over a time window as specified by Power_Limit_1_TIME field. The default value is the Thermal Design Power (a.k.a TDP) and varies with product skus. The unit is specified in MSR_RAPLPOWER_UNIT.	
15	POWER_LIMIT_1_EN When set, enables the processor to apply control policy such that the platform power does not exceed Platform Power limit 1 over the time window specified by Power Limit 1 Time Window.	
16	CRITICAL_POWER_CLAMP_1 When set, allows the processor to go below the OS requested P states in order to maintain the power below specified Platform Power Limit 1 value.	
23:17	POWER_LIMIT_1_TIME Specifies the duration of the time window over which Platform Power Limit 1 value should be maintained for sustained long duration. This field is made up of two numbers from the following equation: Time Window = (float) ((1+(X/4))*(2^Y)), where: X = POWER_LIMIT_1_TIME[23:22] Y = POWER_LIMIT_1_TIME[21:17] The maximum allowed value in this field is defined in MSR_PKG_POWER_INFO[PKG_MAX_WIN]. The default value is ODH, The unit is specified in MSR_RAPLPOWER_UNIT[Time Unit]	
31:24	Reserved.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
46:32	POWER_LIMIT_2 Average Power limit value which the platform must not exceed over the Short Duration time window chosen by the processor. The recommended default value is 1.25 times the Long Duration Power Limit (i.e., Platform Power Limit 1).	
47	POWER_LIMIT_2_EN When set, enables the processor to apply control policy such that the platform power does not exceed Platform Power limit 2 over the Short Duration time window.	
48	CRITICAL_POWER_CLAMP_2 When set, allows the processor to go below the OS requested P states in order to maintain the power below specified Platform Power Limit 2 value.	
62:49	Reserved.	
63	LOCK Setting this bit will lock all other bits of this MSR until system RESET.	
Register Address: 6BOH, 1712	MSR_GRAPHICS_PERF_LIMIT_REASONS	
MSR_GRAPHICS_PERF_LIMIT_REASONS Indicator of Frequency Clipping in the Processor Graphics. (Frequency refers to processor graphics frequency.)		Package
0	PROCHOT (R/O) PROCHOT Status. When set, frequency is reduced due to assertion of external PROCHOT.	
1	THERMAL (R/O) Thermal Status. When set, frequency is reduced due to a thermal event.	
4:2	Reserved.	
5	RATL (R/O) Running Average Thermal Limit Status. When set, frequency is reduced due to running average thermal limit.	
6	VR_THERMALERT (R/O) VR Therm Alert Status. When set, frequency is reduced due to a thermal alert from a processor Voltage Regulator.	
7	VR_TDC (R/O) VR Thermal Design Current Status. When set, frequency is reduced due to VR TDC limit.	
8	OTHER (R/O) Other Status. When set, frequency is reduced due to electrical or other constraints.	
9	Reserved.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
10	PBM_PL1 (R/O) Package/Platform-Level Power Limiting PL1 Status. When set, frequency is reduced due to package/platform-level power limiting PL1.	
11	PBM_PL2 (R/O) Package/Platform-Level PL2 Power Limiting Status. When set, frequency is reduced due to package/platform-level power limiting PL2/PL3.	
12	INEFFICIENT_OPERATION (R/O) Inefficient Operation Status. When set, processor graphics frequency is operating below target frequency.	
15:13	Reserved.	
16	PROCHOT_LOG (R/W) PROCHOT Log. When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	THERMAL_LOG (R/W) Thermal Log. When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
20:18	Reserved.	
21	RATL_LOG (R/W) Running Average Thermal Limit Log. When set, indicates that the RATL Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR_THERMALERT_LOG (R/W) VR Therm Alert Log. When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	VR_TDC_LOG (R/W) VR Thermal Design Current Log. When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
24	OTHER_LOG (R/W) Other Log. When set, indicates that the OTHER Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Reserved.	
26	PBM_PL1_LOG (R/W) Package/Platform-Level PL1 Power Limiting Log. When set, indicates that the Package/Platform Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
27	PBM_PL2_LOG (R/W) Package/Platform-Level PL2 Power Limiting Log. When set, indicates that the Package/Platform Level PL2 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
28	INEFFICIENT_OPERATION_LOG (R/W) Inefficient Operation Log. When set, indicates that the Inefficient Operation Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:29	Reserved.	
Register Address: 6B1H, 1713	MSR_RING_PERF_LIMIT_REASONS	
MSR_RING_PERF_LIMIT_REASONS Indicator of Frequency Clipping in the Ring Interconnect. (Frequency refers to ring interconnect in the uncore.)		Package
0	PROCHOT (R/O) PROCHOT Status. When set, frequency is reduced due to assertion of external PROCHOT.	
1	THERMAL (R/O) Thermal Status. When set, frequency is reduced due to a thermal event.	
4:2	Reserved.	
5	RATL (R/O) Running Average Thermal Limit Status. When set, frequency is reduced due to running average thermal limit.	
6	VR_THERMALERT (R/O) VR Therm Alert Status. When set, frequency is reduced due to a thermal alert from a processor Voltage Regulator.	
7	VR_TDC (R/O) VR Thermal Design Current Status. When set, frequency is reduced due to VR TDC limit.	
8	OTHER (R/O) Other Status. When set, frequency is reduced due to electrical or other constraints.	
9	Reserved.	
10	PBM_PL1 (R/O) Package/Platform-Level Power Limiting PL1 Status. When set, frequency is reduced due to package/platform-level power limiting PL1.	
11	PBM_PL2 (R/O) Package/Platform-Level PL2 Power Limiting Status. When set, frequency is reduced due to package/platform-level power limiting PL2/PL3.	
15:12	Reserved.	

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
16	PROCHOT_LOG (R/W) PROCHOT Log. When set, indicates that the PROCHOT Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
17	THERMAL_LOG (R/W) Thermal Log. When set, indicates that the Thermal Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
20:18	Reserved.	
21	RATL_LOG (R/W) Running Average Thermal Limit Log. When set, indicates that the RATL Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
22	VR_THERMALERT_LOG (R/W) VR Therm Alert Log. When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
23	VR_TDC_LOG (R/W) VR Thermal Design Current Log. When set, indicates that the VR Therm Alert Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
24	OTHER_LOG (R/W) Other Log. When set, indicates that the OTHER Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
25	Reserved.	
26	PBM_PL1_LOG (R/W) Package/Platform-Level PL1 Power Limiting Log. When set, indicates that the Package/Platform Level PL1 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
27	PBM_PL2_LOG (R/W) Package/Platform-Level PL2 Power Limiting Log. When set, indicates that the Package/Platform Level PL2 Power Limiting Status bit has asserted since the log bit was last cleared. This log bit will remain set until cleared by software writing 0.	
63:28	Reserved.	
Register Address: 9FBH, 2555	IA32_TME_CLEAR_SAVED_KEY	
IA32_TME_CLEAR_SAVED_KEY (R/W) See Table 2-2.		Package
Register Address: 9FFH, 2559	MSR_CORE_MKTME_ACTIVATE	
MSR_CORE_MKTME_ACTIVATE (R/O) MSR to read TME_ACTIVATE[MK_TME_KEYID_BITS].		Core

Table 2-53. Additional MSRs Supported by the Intel® Core™ Ultra 7 Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
31:0	Reserved.	
35:32	READ_MK_TME_KEYID_BITS This value will be returned on a RDMSR, but must be zero on a WRMSR.	
63:36	Reserved.	

The MSRs listed in Table 2-54 are unique to the Intel Core Ultra 7 processor P-core. These MSRs are not supported on the processor E-core.

Table 2-54. MSRs Supported by the Intel® Core™ Ultra 7 Processor P-core

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 30CH, 780	IA32_FIXED_CTR3	
Fixed-Function Performance Counter 3 (R/W)		Thread
47:0	FIXED_COUNTER Top-down Microarchitecture Analysis unhalted number of available slots counter.	
63:48	Reserved.	
Register Address: 329H, 809	MSR_PERF_METRICS	
Performance Metrics (R/W) This register provides built-in support for Top-down Micro-architecture Analysis (TMA) metrics. It exposes the four TMA Level 1 metrics where the lower 32 bits are divided into four 8 bit fields, each of which is an integer percentage of the total TOPDOWN.SLOTS (as reported by fixed counter 3).		Thread
7:0	RETIRING Percent of utilized by uops that eventually retire (commit).	
15:8	BAD_SPECULATION Percent of Wasted due to incorrect speculation, covering Utilized by uops that do not retire, or Recovery Bubbles (unutilized slots).	
23:16	FRONTEND_BOUND Percent of Unutilized slots where Front-end did not deliver a uop while Back-end is ready.	
31:24	BACKEND_BOUND Percent of Unutilized slots where a uop was not delivered to Back-end due to lack of Back-end resources.	
39:32	MULTI_UOPS Frontend bound.	
47:40	BRANCH_MISPREDICTS Frontend bound.	
55:48	FRONTEND_LATENCY Frontend bound.	

Table 2-54. MSRs Supported by the Intel® Core™ Ultra 7 Processor P-core (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
63:56	MEMORY_BOUND Frontend bound.	
Register Address: 540H, 1344	MSR_THREAD_UARCH_CTL	
Thread Microarchitectural Control (R/W) See Table 2-47.		Thread
Register Address: 541H, 1345	MSR_CORE_UARCH_CTL	
Core Microarchitecture Control MSR (R/W) See Table 2-44.		Core

The MSRs listed in Table 2-48 are unique to the Intel Core Ultra 7 processor E-core. These MSRs are not supported on the processor P-core.

Table 2-55. MSRs Supported by the Intel® Core™ Ultra 7 Processor E-core

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 4F0H, 1264	MSR_SAF_CTRL	
SAF Control (W/O) Extension to SAF.		Package
0	INVALIDATE_CURRENT_STRIDE Invalidate all chunks in current stride.	
63:1	Reserved.	
Register Address: D18H–D1FH, 3352–3359	IA32_L2_MASK_[8-15]	
IA32_L2_MASK_[8-15] (R/W) If CPUID.(EAX=10H, ECX=1):EDX.COS_MAX[15:0] ≥ 0. Controls MLC (L2) Intel RDT allocation. For more details on CAT/RDT, see Chapter 19, “Debug, Branch Profile, TSC, and Intel® Resource Director Technology (Intel® RDT) Features.”		Module
15:0	WAY_MASK Capacity Bit Mask. Available ways vectors for class of service of IA core. ‘1 in bit indicates allocation to the way is allowed. ‘0 indicates allocation to the way is not allowed.	
31:16	Reserved.	
Register Address: 1309H–130BH, 4873–4875	MSR_RELOAD_FIXED_CTRx	
Reload value for IA32_FIXED_CTRx (R/W)		Thread
47:0	Value loaded into IA32_FIXED_CTRx when a PEBS record is generated while PEBS_EN_FIXEDx = 1 and PEBS_OUTPUT = 01B in IA32_PEBS_ENABLE, and FIXED_CTRx is overflowed.	
63:48	Reserved.	
Register Address: 14C1H–14C8H, 5313–5320	MSR_RELOAD_PMCx	
Reload value for IA32_PMCx (R/W)		Thread
47:0	Value loaded into IA32_PMCx when a PEBS record is generated while PEBS_EN_PMCx = 1 and PEBS_OUTPUT = 01B in IA32_PEBS_ENABLE, and PMCx is overflowed.	

Table 2-55. MSRs Supported by the Intel® Core™ Ultra 7 Processor E-core (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
63:48	Reserved.	
Register Address: 1A8EH, 6798	MSR_STLB_FILL_TRANSLATION	
STLB Fill Translation (W/O) STLB QoS MSR to fill translations into STLB.		Core
3:0	CLOS Class of service to use for the fill.	
9:4	Reserved.	
10	X Set to 1 when LA is to an executable page.	
11	RW Set to 1 when LA is to a writeable page.	
63:12	LA Logical address to use for fill.	

2.17.10 MSRs Introduced in the Intel® Xeon® 6 P-Core Processors

Table 2-56 lists additional MSRs for the Intel Xeon 6 P-core processors. Intel Xeon 6 P-core processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_ADH or 06_AEH.

For an MSR listed in Table 2-56 that also appears in the model-specific tables of prior generations, Table 2-56 supersedes prior generation tables.

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 33H, 51	MSR_MEMORY_CONTROL	
MSR_MEMORY_CONTROL (R/W) Disables split locks, which are locked instructions that split a cache line.		Core
26:0	Reserved.	
27	UC_STORE_THROTTLE If set to 1, when enabled, the processor allows one in-progress, post-retirement UC stores at a time.	
28	UC_LOCK_DISABLE If set to 1, a UC load lock will trigger a fault. If clear to 0, UC load locks proceed normally.	
29	SPLIT_LOCK_DISABLE If set to 1, a split lock will trigger an #AC fault. If clear to 0, split locks proceed normally	
63:30	Reserved.	
Register Address: 34H, 52	MSR_SMI_COUNT	
SMI Counter (R/W)		Thread

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
31:0	SMI_COUNT Running count of SMI events since the last reset.	
63:32	Reserved.	
Register Address: 39H, 57	MSR_SOCKET_ID	
Socket ID (R/W) Reassigns the package-specific portions of the APIC ID. This MSR is used on scalable DP and high-end MP platforms to resolve legacy-mode APIC ID conflicts.		Package
10:0	PACKAGE_ID: Holds package ID. This reflects the upper bits of the APIC ID.	
63:11	Reserved.	
Register Address: 7AH, 122	IA32_FEATURE_ACTIVATION	
IA32_FEATURE_ACTIVATION (R/W) Implements Feature Activation command. WRMSR to this address activates all 'activatable' features on this thread. See Table 2-2.		Thread
Register Address: 7BH, 123	IA32_MCU_ENUMERATION	
IA32_MCU_ENUMERATION (R/O) Enumeration of architectural features. See Table 2-2.		Package
Register Address: 7CH, 124	IA32_MCU_STATUS	
IA32_MCU_STATUS (R/O) Communicates results from the previous patch loads. See Table 2-2.		Package
Register Address: 82H, 130	IA32_FZM_RANGE_INDEX	
IA32_FZM_RANGE_INDEX (R/W) Index and Domain handle for a valid FZM region. Programmed by SW and used by other FRM MSRs FZM Range Index register to R/W Domain Index. See Table 2-2.		Thread
Register Address: 83H, 131	IA32_FZM_DOMAIN_CONFIG	
IA32_FZM_DOMAIN_CONFIG (R/O) Bit mask of valid regions within the domain identified by FZM_RANGE_INDEX. See Table 2-2.		Thread
Register Address: 84H, 132	IA32_FZM_RANGE_STARTADDR	
IA32_FZM_RANGE_STARTADDR (R/O) Start address of the FZM range pointed to by FZM_RANGE_INDEX. See Table 2-2.		Thread
Register Address: 85H, 133	IA32_FZM_RANGE_ENDADDR	
IA32_FZM_RANGE_ENDADDR (R/O) End address of the specified domain in FZM_RANGE_INDEX. See Table 2-2.		Thread
Register Address: 86H, 134	IA32_FZM_RANGE_WRITESTATUS	
IA32_FZM_RANGE_WRITESTATUS (R/O) Write status of the FZM range pointed to by FZM_RANGE_INDEX. See Table 2-2.		Thread
Register Address: 87H, 135	IA32_MKTME_KEYID_PARTITIONING	
MKTME KEY ID Partitioning (R/O) Enumerates the number of activated KeyIDs for Intel TME-MK and Intel TDX. See Table 2-2.		Package
Register Address: 90H, 144	IA32_SGXLEPUBKEYHASH4	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
IA32_SGXLEPUBKEYHASH4 (R/W) See Table 2-2.		Thread
Register Address: 91H, 145	IA32_SGXLEPUBKEYHASH5	
IA32_SGXLEPUBKEYHASH5 (R/W) See Table 2-2.		Thread
Register Address: 98H, 152	MSR_SEAM_WBINVDP	
SEAM WBINVDP (R/W) Allows software to WBINVD sections of the LLC.		Thread
63:0	HANDLE Caches sub-block to invalidate.	
Register Address: 99H, 153	MSR_SEAM_WBNINVDP	
SEAM WBNINVDP (R/W) Allows software to WBNINVD sections of the LLC.		Thread
63:0	HANDLE Caches sub-block to invalidate.	
Register Address: 9AH, 154	MSR_SEAM_INTR_PENDING	
SEAM Interrupt Pending (R/O) Report out some event pending bits.		Thread
0	INTR Interrupt is pending.	
1	NMI NMI is pending.	
2	SMI SMI is pending.	
4:3	OTHER_EVENTS Other events pending.	
63:5	Reserved.	
Register Address: 9BH, 155	IA32_SMM_MONITOR_CTL	
SMM Monitor Control (R/W) The SMM Monitor Configuration involves SMM code specifying the MSEG location and enabling dual-monitor treatment by writing to the corresponding MSR. See Table 2-2.		Thread
Register Address: CFH, 207	IA32_CORE_CAPABILITIES	
IA32 Core Capabilities Register (R/W) If CPUID.(EAX=07H, ECX=0):EDX[30] = 1. This MSR provides an architectural enumeration function for model-specific behavior.		Core
0	STLB_QOS When set to 1, processor supports STLB QoS.	
1	Reserved.	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
2	INTEGRITY_SUPPORTED When set to 1, processor supports Functional Safety. Specific FUSA capabilities are enumerated in MSR_FUSA_CAPABILITIES.	
3	RSM_IN_CPLO_ONLY Intel System Resources Defense: When set to 1, RSM will only be allowed in CPLO and will #GP for all non-CPLO privilege levels.	
4	UC_LOCK_DISABLE When set to 1, processor supports UC load lock disable.	
5	SPLIT_LOCK_DISABLE When set to 1, processor supports #AC on split locks.	
6	SNP_FILTER_QOS When set to 1, processor supports Snoop Filter Quality of Service MSRs.	
7	UC_STORE_THROTTLING When set to 1, processor supports UC store throttling through MSR_MEMORY_CTRL[UC_STORE_THROTTLE].	
63:8	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: FEH, 254	IA32_MTRRCAP	
Memory Type Range Register (R/O) See Table 2-2.		Core
Register Address: 105H, 261	MSR_ARRAY_BIST	
MSR_ARRAY_BIST (R/W) Triggered by writing and reading an MSR that can be written by Ring 0 software.		Core

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
31:0	ARRAY_LIST: Bit map which indicates which arrays to run MarchC- BIST ▪ Bit[0] MLC Data ▪ Bit[1] MLC Tag ▪ Bit[2] C6SRAM Data (NOP for WRMSR – used for reporting error only) ▪ Bit[3] PMA BIST (NOP for WRMSR – used for reporting error only) ▪ Bit[4] STLB Data ▪ Bit[5] IFU Data ▪ Bit[6] STLB Tag ▪ Bit[7] DCU Data ▪ Bit[8] DSB Data ▪ Bit[9] TMUL Data ▪ Bit[10] UROM pointer0 ▪ Bit[11] UROM pointer1-3 ▪ Bit[12] UROM pointer4-7 ▪ Bit[13] UROM unique0 ▪ Bit[14] UROM unique1/2 The WRMSR will run PBIST on all the arrays indicated in the bitmap, starting from the LSB. NOTE2: C6SRAM[Bit 2] and PMA[Bit 3] are only for reporting and do not execute BIST (done by EDX[15:0]uCode during Fusa-Reset).	
46:32	Reserved.	
62:47	Reserved.	
63	SIGNAL_MCE: Signal MCERR upon BIST failure.	
Register Address: 105H, 261	MSR_ARRAY_BIST_STATUS	
MSR_ARRAY_BIST_STATUS (R/O)		Core
31:0	ARRAY_COMPLETION_MASK Bitmap indicating which arrays from the ARRAY_BIST.ARRAY_LIST was not processed. 1 means not tested and 0 means tested.	
62:32	Reserved. Returns all 0s.	
63	PASS_FAIL: 0 means Pass on all arrays in the WRMSR(ARRAY_BIST.ARRAY_LIST) 1 means Fail on the LSB array in the RDMSR(ARRAY_BIST_STATUS.ARRAY_COMPLETION_MASK).	
Register Address: 122H, 290	IA32_TSX_CTRL	
IA32_TSX_CTRL (R/W) See Table 2-2.		Thread
Register Address: 140H, 320	MSR_FEATURE_ENABLES	
Miscellaneous enables for thread-specific features. (R/W)		Thread

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
0	AESNI_LOCK Once this bit is set, writes to this register will not be allowed.	
63:1	Reserved.	
Register Address: 1E0H, 480	IA32_LER_INFO	
IA32_LER_INFO (R/W) Last Event Record Destination IP Register. See Table 2-2.		Thread
Register Address: 1F9H, 505	IA32_CPU_DCA_CAP	
IA32_CPU_DCA_CAP (R/O) See Table 2-2.		Thread
Register Address: 2A1H, 673	MSR_PRMRR_BASE_1	
MSR_PRMRR_BASE_1 (R/W) Processor Reserved Memory Range Register - Physical Base Control Register.		Core
2:0	MEMTYPE Memory Type for PRMRR accesses.	
3	CONFIGURED PRMRR base configured.	
19:4	Reserved.	
51:20	BASE PRMRR Base address.	
63:52	Reserved.	
Register Address: 2A2H, 674	MSR_PRMRR_BASE_2	
MSR_PRMRR_BASE_2 (R/W) Processor Reserved Memory Range Register - Physical Base Control Register. See MSR_PRMRR_BASE_1 (2A1H) for reference; similar format.		Core
Register Address: 2A3H, 675	MSR_PRMRR_BASE_3	
MSR_PRMRR_BASE_3 (R/W) Processor Reserved Memory Range Register - Physical Base Control Register. See MSR_PRMRR_BASE_1 (2A1H) for reference; similar format.		Core
Register Address: 2A4H, 676	MSR_PRMRR_BASE_4	
MSR_PRMRR_BASE_4 (R/W) Processor Reserved Memory Range Register - Physical Base Control Register. See MSR_PRMRR_BASE_1 (2A1H) for reference; similar format.		Core
Register Address: 2A5H, 677	MSR_PRMRR_BASE_5	
MSR_PRMRR_BASE_5 (R/W) Processor Reserved Memory Range Register - Physical Base Control Register. See MSR_PRMRR_BASE_1 (2A1H) for reference; similar format.		Core
Register Address: 2A6H, 678	MSR_PRMRR_BASE_6	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
MSR_PRMRR_BASE_6 (R/W) Processor Reserved Memory Range Register - Physical Base Control Register. See MSR_PRMRR_BASE_1 (2A1H) for reference; similar format.		Core
Register Address: 2A7H, 679	MSR_PRMRR_BASE_7	
MSR_PRMRR_BASE_7 (R/W) Processor Reserved Memory Range Register - Physical Base Control Register. See MSR_PRMRR_BASE_1 (2A1H) for reference; similar format.		Core
Register Address: 2B8H, 696	MSR_COPY_SBFT_HASHES	
MSR_COPY_SBFT_HASHES (W/O)		Module
63:0	SBFT_PROGRAM_SOURCE_ADDR EDX:EAX contains the linear address base of the SBFT Binary loaded into memory.	
Register Address: 2B9H, 697	MSR_SBFT_HASHES_STATUS	
MSR_COPY_SBFT_HASHES (R/O)		Core
15:0	CHUNK_SIZE EAX[15:0] - Chunk size of the test in KB.	
31:16	TOTAL_NUM_CHUNKS EAX[31:16] - Total number of chunks.	
39:32	ERROR_CODE - EDX[7:0] The error code refers to the LP that runs WRMSR(2B8H). 0x0: Reserved. 0x1: Attempt to copy SBFT-hashes when copy already in progress. 0x2: Secure Memory not set up correctly. 0x3: Scan-Image Header Image_info.ProgramID does not match MSR_INTEGRITY_CAPABILITIES[31:24], or scan-image header Processor-Signature doesn't match F/M/S, or scan-image header Processor-Flags doesn't match PlatformID. 0x4: Reserved. 0x5: Integrity check failed. 0x6: WRMSR(0x2B8) (ACTIVATE_SBAF) Reinstall of SBFT test image attempted when current SBFT test image is in use by other LPs. 0x7: Aborted due to #PF (Page Fault). 0x8: Unable to generate a Random Value.	
48:40	NUM_CHUNKS_IN_STRIDE EDX[16:8] - Number of Chunks in stride. This is the number of chunks that are installed. 0 in this field means that the CPU does not support strides, otherwise, stride value must be >=1.	
50:49	Reserved. EDX[18:17] - Set to all zeros.	
62:51	MAX_CORE_LIMIT EDX[30:19] - Maximum Number of Cores that can run SBFTAFSBAF simultaneously -1. 0 means 1 core at a time.	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
63	Valid. EDX[31] - Valid bit is set when COPY_SBFT_HASHES completed successfully.	
Register Address: 2BAH, 698	MSR_AUTHENTICATE_AND_COPY_SBFT_CHUNK	
MSR_AUTHENTICATE_AND_COPY_SBFT_CHUNK (W/O)		Core
63:0	BASE_CHUNK_TABLE_ADDR EDX:EAX[63:0] - Linear Address pointing to the CHUNK TABLE (TABLE_BASE).	
Register Address: 2BBH, 699	MSR_SBFT_CHUNKS_AUTHENTICATION_STATUS	
MSR_SBFT_CHUNKS_AUTHENTICATION_STATUS (R/O)		Core
15:0	NUM_VALID_CHUNKS EAX[15:0] - Total number of Valid (authenticated) chunks.	
31:16	NUM_CHUNKS_IN_STRIDE EAX[31:16] - Number of Chunks in Stride.	
39:32	ERROR_CODE EDX[7:0] 0x0 - No error reported. 0x1 - Attempt to authenticate a CHUNK already marked as authentic or is currently being installed by another core. 0x2 - CHUNK authentication error. HASH of chunk did not match expected value. 0x3 - Aborted due to #PF. 0x4 - Chunk Outside the current Stride. 0x5 - Interrupted.	
47:40	Reserved. EDX[15:8] - Set to all zeros.	
63:48	CURRENT_MAX_BUNDLE_INDX EDX[31:16] - Maximum Bundle Index in current stride.	
Register Address: 2BCH, 700	MSR_ACTIVATE_SBFT	
MSR_ACTIVATE_SBFT (W/O)		Core
13:0	SBFT_BUNDLE_INDEX EAX[13:0] - Indicates SBFT Bundle Index to start from.	
15:14	SBFT_PRGM_INDEX EAX[15:14] - Indicates what SBFT Program index to run.	
31:16	Reserved. Set to all zeros.	
62:32	THREAD_WAIT_DELAY EDX[30:0] - TSC-based delay to allow threads to rendezvous.	
63	Reserved. EDX[31] - Must be set to 0. #GP fault otherwise.	
Register Address: 2BDH, 701	MSR_SBFT_STATUS	
MSR_SBFT_STATUS (R/O)		Core

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
13:0	SBFT_BUNDLE_INDEX EAX[13:0] - SBFT Bundle that was executed.	
15:14	SBFT_PGM_INDEX EAX[15:14] - Indicates what SBFT Program index that was last ran. Maps to same field in WRMSR(ACTIVATE_SBFT). On a test pass this field will be 2'b00.	
31:16	Reserved. EAX[31:16] - Return all zeros.	
39:32	ERROR_CODE EDX[7:0] 0x0 - No Error. 0x1 - SBFT operation did not start. Other thread could not join. 0x2 - SBFT operation did not start. Interrupt occurred prior to SBFT coordination. 0x3 - Reserved. 0x4 - SBFT operation did not start. Non-valid SBFT BUNDLES in the SBFT_BUNDLE_INDEX. 0x5 - SBFT operation did not start. Mismatch in arguments between threads T0/T1. 0x6 - SBFT operation did not start. Core is not capable of performing SBFT currently. 0x7 - Reserved. 0x8 - SBFT operation did not start. Exceeded number of Logical Processors (LP) allowed to run SBFT-At-Field concurrently. 0x9 - SBFT operation did not start. Interrupt occurred or timer about to expire. 0xA - SBFT operation did not start. SBFT_PGM_INDEX is not valid. 0xB - SBFT operation aborted due to corrupted chunk. 0xC - SBFT operation did not start. TAP Data error. 0xD - SBFT operation did not start. SBFT program is not valid. All other error codes are reserved.	
60:40	Reserved. EDX[28:8] - Return all zeros.	
61	TEST_FAIL EDX[29:29] - Architectural Signature failed. Last thread executed HLT and completed SBFT and EBX != 0xACED.	
63:62	SBFT_STATUS EDX[31:30] - SBFT status (result of running SBAF). 00 - PASS. 10 - INTERRUPTED. 01 - FAILED SIGNATURE CHECK. 11 - FAILED.	
Register Address: 2BEH, 702	MSR_SBFT_MODULE_ID	
MSR_SBFT_MODULE_ID (R/O)		Module
31:0	SBFT-AT-FIELD_REVID EAX[31:0] - Maps to Revision field in external header (offset 4).	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
40:32	CURRENT_STRIDE_INDEX EDX[8:0] - Stride Index.	
63:41	Reserved. EDX[31:9] - Return all zeros.	
Register Address: 2BFH, 703	MSR_SBFTAF_LAST_WP	
MSR_SBFTAF_LAST_WP (R/O)		Module
31:0	LAST_WP EAX[31:0] - Provides information about the core when the last WRMSR(ACTIVATE_SBFT) was executed. Available only if enumerated in INTEGRITY_CAPABILITIES[10:9].	
39:32	Reserved.	
63:40	Reserved. EDX[31:8] - Return all zeros.	
Register Address: 2C2H, 706	MSR_COPY_SCAN_HASHES	
MSR_COPY_SCAN_HASHES (W/O)		Module
63:0	SCAN_HASH-ADDR EDX:EAX contains the linear address of the SCAN Test HASH Binary loaded into memory	
Register Address: 2C3H, 707	MSR_SCAN_HASHES_STATUS	
MSR_SCAN_HASHES_STATUS (R/O)		Core
15:0	CHUNK_SIZE EAX[15:0] - Chunk size of the test in KB.	
31:16	TOTAL_NUM_CHUNKS EAX[31:16] - Total number of chunks.	
39:32	ERROR_CODE EDX[7:0] - The error code refers to the LP that runs WRMSR(2C2H). ▪ 0x0 - Reserved. ▪ 0x1 - Attempt to copy scan-hashes when copy already in progress. ▪ 0x2 - Secure Memory not set up correctly. ▪ 0x3 - Scan-Image Header Image_info.ProgramID does not match MSR_INTEGRITY_CAPABILITIES[31:24], or scan-image header Processor-Signature doesn't match F/M/S, or scan-image header Processor-Flags doesn't match PlatformID. ▪ 0x4 - Reserved. ▪ 0x5 - Integrity check failed. ▪ 0x6 - WRMSR(0x2C6) Re-install of scan test image attempted when current scan test image is in use by other LPs. ▪ 0x7 - Aborted due to #PF (Page Fault). ▪ 0x8 - Unable to generate a Random Value.	
48:40	NUM_CHUNKS_IN_STRIDE EDX[16:8] - Number of Chunks in stride. This is the number of chunks that are installed. 0 in this field means that the CPU does not support strides, otherwise, the stride value must be >=1.	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
50:49	Reserved. EDX[18:17] - Set to all zeros.	
62:51	NAME EDX[30:19] - Maximum Number of cores that can run Intel® In-field Scan simultaneously minus 1. 0 means 1 core at a time.	
63	VALID EDX[31] - Valid bit is set when COPY_SCAN_HASHES completed.	
Register Address: 2C4H, 708	MSR_AUTHENTICATE_AND_COPY_CHUNK	
MSR_AUTHENTICATE_AND_COPY_CHUNK (R/O)		Core
63:0	BASE_CHUNK_TABLE_ADDR EDX:EAX[63:0] - Linear Address pointing to the CHUNK TABLE (TABLE_BASE).	
Register Address: 2C5H, 709	MSR_CHUNKS_AUTHENTICATION_STATUS	
MSR_CHUNKS_AUTHENTICATION_STATUS (R/O)		Core
15:0	VALID_CHUNKS EAX[15:0] - Total number of Valid (authenticated) chunks.	
31:16	NUM_CHUNKS_IN_STRIDE EAX[31:16] - Number of Chunks in Stride.	
39:32	ERROR_CODE EDX[7:0] 0x0 - No-error reported. 0x1 - Attempt to authenticate a CHUNK which is already marked as authentic or is currently being installed by another core. 0x2 - CHUNK authentication error. HASH of chunk did not match expected value. 0x3 - Aborted due to #PF (Page Fault). 0x4 - Chunk Outside the current Stride.	
63:40	Reserved. EDX[31:8] - Set to all zeros.	
Register Address: 2C6H, 710	MSR_ACTIVATE_SCAN	
MSR_ACTIVATE_SCAN (W/O)		Core
15:0	CHUNK_START_INDEX EAX[15:0] - Indicates Chunk Index from which to start.	
31:16	CHUNK_STOP_INDEX EAX[31:16] - Indicates what chunk index to stop at (inclusive).	
62:32	THREAD_WAIT_DELAY EDX[30:0] - TSC based delay to allow threads to rendezvous.	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
63	SIGNAL_MCE EDX[31] - If 1: On scan-error log MC in MC4_STATUS and signal MCE if machine check signaling enabled in MC4_CTL[0]. - If 0: Don't no-logging/no-signaling.	
Register Address: 2C7H, 711	MSR_SCAN_STATUS	
MSR_SCAN_STATUS (R/O)		Core
15:0	CHUNK_NUM EAX[15:0] - SCAN Chunk that was reached.	
31:16	CHUNK_STOP_INDEX EAX[31:16] - Indicates what chunk index to stop at (inclusive). - Maps to same field in WRMSR(ACTIVATE_SCAN).	
39:32	ERROR_CODE EDX[7:0] 0x0 - No Error. 0x1 - SCAN operation did not start. Other thread could not join. 0x2 - SCAN operation did not start. Interrupt occurred prior to SCAN coordination. 0x3 - SCAN operation did not start. Power Management conditions are inadequate to run SAF. 0x4 - SCAN operation did not start. Non valid chunks in the range CHUNK_STOP_INDEX : CHUNK_START_INDEX. 0x5 - SCAN operation did not start. Mismatch in arguments between threads T0/T1. 0x6 - SCAN operation did not start. Core not capable of performing SCAN currently. 0x7 - Debug Mode. Scan-At-Field results not to be trusted. 0x8 - SCAN operation did not start. Exceeded number of Logical Processors (LP) allowed to run Scan-At-Field concurrently. MAX_CORE_LIMIT exceeded. 0x9 - Interrupt occurred. Scan operation aborted prematurely, not all chunks requested have been executed. 0xB - Scan operation aborted due to corrupted chunk. 0xC - Scan operation did not start. All other error codes are reserved.	
61:40	Reserved. EDX[29:8] - Return all zeros.	
62	SCAN_CONTROL_ERROR EDX[30] - SCAN error in the Scan-At-Field controller. - Non ECC error.	
63	SCAN_SIGNATURE_ERROR EDX[31] - SCAN SIGNATURE error in the SCAN pattern fetched from main memory. - Non ECC error.	
Register Address: 2C8H, 712	MSR_SCAN_MODULE_ID	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
MSR_SCAN_MODULE_ID (R/O)		Module
31:0	SCAN-AT-FIELD_REVID EAX[31:0] - Maps to Revision field in external header (offset 4).	
40:32	CURRENT_STRIDE_INDEX EDX[8:0] - Stride Index.	
63:41	Reserved. EDX[31:9] - Return all zeros.	
Register Address: 2C9H, 713		MSR_LAST_SAF_WP
MSR_LAST_SAF_WP (R/O)		Module
31:0	LAST_WP EAX[31:0] - Provides information about the core when the last WRMSR(ACTIVATE_SCAN) was executed. - Available only if enumerated in INTEGRITY_CAPABILITIES[10:9].	
39:32	Reserved. EDX[7:0]	
63:40	Reserved. EDX[31:8] - Return all zeros.	
Register Address: 2D9H, 729		MSR_INTEGRITY_CAPABILITIES
MSR_INTEGRITY_CAPABILITIES (R/O) Enumerates features supported in Functional Safety.		Thread
0	STARTUP_SCAN_BIST When set to 1, processor supports Startup SCAN BIST.	
1	STARTUP_MEM_BIST When set to 1, processor supports Startup MEM BIST.	
2	PERIODIC_MEM_BIST When set to 1, processor supports Periodic MEM BIST.	
3	LOCKSTEP When set to 1, processor supports Lock Step Mode.	
4	PERIODIC_SCAN_BIST When set to 1, processor supports Periodic SCAN BIST.	
5	PLL_LOSS_DETECT When set to 1, processor supports PLL LOSS detection.	
6	PWR_LOSS_DETECT When set to 1, processor supports Power Loss detection.	
7	PERRINJ When set to 1, processor supports FUSA PERRINJ.	
8	SBFT_AT_FIELD When set to 1, processor supports SBFT-At-Field.	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
10:9	SAF_GEN_REV 00 = REV1; 01 = REV2; 10 = REV3; 11 = REV4.	
14:11	Reserved.	
15	PRESERVE_MEMORY_NEEDED When set to 1, processor supports FUSARR_BASE/MASK MSRs.	
20:16	TID_BIT_SHIFT Number of bits to shift right on x2APICID to get a unique topology ID of all logical processors that share a scan test engine.	
21	ALL_LP_JOIN_NEEDED All logical processors that share scan test engine need to be tested together and must join using MSR_ACTIVATE_SCAN.	
23:22	Reserved.	
31:24	PATTERN_ID Processor scan pattern ID. ID of the startup and periodic scan programs supported for this part.	
63:32	Reserved.	
Register Address: 30CH, 780	IA32_FIXED_CTR3	
Fixed-Function Performance Counter 3 (R/W) See Table 2-2.		Thread
Register Address: 4D0H, 1232	IA32_MCG_EXT_CTL	
IA32_MCG_EXT_CTL (R/W) See Table 2-2.		Thread
Register Address: 4F0H, 1264	MSR_SAF_CTRL	
MSR_SAF_CTRL (W/O)		Core
0	INVALIDATE_CURRENT_STRIDE EAX[0] - Write of 1 invalidates the currently installed stride. - Clears only the VALID_CHUNKS field on a RDMSR(CHUNKS_AUTHENTICATION_STATUS).	
63:1	Reserved.	
Register Address: 4F8H, 1272	MSR_SBFT_CTRL	
MSR_SBFT_CTRL (W/O)		Module
0	INVALIDATE_CURRENT_STRIDE EAX[0] - Write of 1 invalidates the currently installed stride.	
63:1	Reserved. EDX[31:0], EAX[31:1]	
Register Address: 540H, 1344	MSR_THREAD_UARCH_CTL	
Thread Microarchitectural Control (R/W)		Thread

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
0	WB_MEM_STRM_LD_DISABLE Disable streaming behavior for MOVNTDQA loads to WB memory type. If set, these accesses will be treated like regular cacheable loads (Data will be cached).	
63:1	Reserved.	
Register Address: 541H, 1345	MSR_CORE_UARCH_CTL	
Core Microarchitecture Control MSR (R/W)		Core
0	SCRUB_DIS L1 scrubbing disable.	
63:1	Reserved.	
Register Address: 664H, 1636	MSR_MC6_RESIDENCY	
MSR_MC6_RESIDENCY (R/O) Time spent in the Module C6-State. Provided in units compatible to P1 clock frequency (Guaranteed / Maximum Core Non-Turbo Frequency).		Module
63:0	RESIDENCY Time that this module is in module-specific C6 states since last reset.	
Register Address: 6E1H, 1761	IA32_PKRS	
IA32_PKRS (R/W) Specifies the PK permissions associated with each protection domain for supervisor pages. See Table 2-2.		Thread
Register Address: 7A3H, 1955	IA32_MCU_EXT_SERVICE	
MCU Extended Service MSR (R/O) If IA32_ARCH_CAPABILITIES[22] = 1. See Table 2-2.		Module
Register Address: 7A4H, 1956	IA32_MCU_ROLLBACK_MIN_ID	
Minimal MCU Revision ID for Rollback (R/O) See Table 2-2.		Module
Register Address: 7B0H, 1968	IA32_ROLLBACK_SIGN_ID_0	
Rollback ID 0 (R/O) See Table 2-2.		Module
Register Address: 7B1H, 1969	IA32_ROLLBACK_SIGN_ID_1	
Rollback ID 1 (R/O) See Table 2-2.		Module
Register Address: 7B2H, 1970	IA32_ROLLBACK_SIGN_ID_2	
Rollback ID 2 (R/O) See Table 2-2.		Module
Register Address: 7B3H, 1971	IA32_ROLLBACK_SIGN_ID_3	
Rollback ID 3 (R/O) See Table 2-2.		Module
Register Address: 7B4H, 1972	IA32_ROLLBACK_SIGN_ID_4	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Rollback ID 4 (R/O) See Table 2-2.		Module
Register Address: 7B5H, 1973	IA32_ROLLBACK_SIGN_ID_5	
Rollback ID 5 (R/O) See Table 2-2.		Module
Register Address: 7B6H, 1974	IA32_ROLLBACK_SIGN_ID_6	
Rollback ID 6 (R/O) See Table 2-2.		Module
Register Address: 7B7H, 1975	IA32_ROLLBACK_SIGN_ID_7	
Rollback ID 7 (R/O) See Table 2-2.		Module
Register Address: 7B8H, 1976	IA32_ROLLBACK_SIGN_ID_8	
Rollback ID 8 (R/O) See Table 2-2.		Module
Register Address: 7B9H, 1977	IA32_ROLLBACK_SIGN_ID_9	
Rollback ID 9 (R/O) See Table 2-2.		Module
Register Address: 7BAH, 1978	IA32_ROLLBACK_SIGN_ID_10	
Rollback ID 10 (R/O) See Table 2-2.		Module
Register Address: 7BBH, 1979	IA32_ROLLBACK_SIGN_ID_11	
Rollback ID 11 (R/O) See Table 2-2.		Module
Register Address: 7BCH, 1980	IA32_ROLLBACK_SIGN_ID_12	
Rollback ID 12 (R/O) See Table 2-2.		Module
Register Address: 7BDH, 1981	IA32_ROLLBACK_SIGN_ID_13	
Rollback ID 13 (R/O) See Table 2-2.		Module
Register Address: 7BEH, 1982	IA32_ROLLBACK_SIGN_ID_14	
Rollback ID 14 (R/O) See Table 2-2.		Module
Register Address: 7BFH, 1983	IA32_ROLLBACK_SIGN_ID_15	
Rollback ID 15 (R/O) See Table 2-2.		Module
Register Address: 981H, 2433	IA32_TME_CAPABILITY	
IA32_TME_CAPABILITY (R/O) See Table 2-2.		Package
Register Address: 982H, 2434	IA32_TME_ACTIVATE	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
IA32_TME_ACTIVATE (R/W) See Table 2-2.		Package
Register Address: 983H, 2435	IA32_TME_EXCLUDE_MASK	
Intel TME Exclude Mask (R/W) See Table 2-2.		Package
Register Address: 984H, 2436	IA32_TME_EXCLUDE_BASE	
Intel TME Exclude Base (R/W) See Table 2-2.		Package
Register Address: 985H, 2437	IA32_UINTR_RR	
User Interrupt Request Register (R/W) See Table 2-2.		Thread
Register Address: 986H, 2438	IA32_UINTR_HANDLER	
User Interrupt Handler Address (R/W) See Table 2-2.		Thread
Register Address: 987H, 2439	IA32_UINTR_STACKADJUST	
User Interrupt Stack Adjustment (R/W) See Table 2-2.		Thread
Register Address: 988H, 2440	IA32_UINTR_NV	
User-Interrupt Size and Notification Vector (R/W) See Table 2-2.		Thread
Register Address: 989H, 2441	IA32_UINTR_PD	
User Interrupt PID Address (R/W) See Table 2-2.		Thread
Register Address: 98AH, 2442	IA32_UINTR_TT	
User-Interrupt Target Table (R/W) See Table 2-2.		Thread
Register Address: 990H, 2448	IA32_COPY_STATUS	
IA32_COPY_STATUS (R/O) See Table 2-2.		Thread
Register Address: 991H, 2449	IA32_IWKEYBACKUP_STATUS	
IA32_IWKEYBACKUP_STATUS (R/O) See Table 2-2.		Package
Register Address: 9FBH, 2555	IA32_TME_CLEAR_SAVED_KEY	
IA32_TME_CLEAR_SAVED_KEY (R/W) See Table 2-2.		Package
Register Address: 9FFH, 2559	MSR_CORE_MKTME_ACTIVATE	
MSR to read TME_ACTIVATE[MK_TME_KEYID_BITS] (R/O)		Core
31:0	Reserved.	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
35:32	READ_MK_TME_KEYID_BITS This value will be returned on a RDMSR, but must be zero on a WRMSR.	
63:36	Reserved.	
Register Address: C84H, 3204	MSR_MBA_CFG	
Memory Bandwidth Allocation (MBA) Configuration (R/W)		Package
1:0	Reserved.	
2	RAMBAE Resource Aware MBA Enable.	
63:3	Reserved.	
Register Address: CA0H, 3232	MSR_RMID_SNC_CONFIG	
RMID_SNC_CONFIG (R/W)		Package
0	RMID_LOCALIZED_DISTRIBUTION_MODE_ENABLE If set, Localized RMID distribution mode is enabled. If Clear, RMID Sharing mode is enabled.	
63:1	Reserved.	
Register Address: D50H, 3408	IA32_L2_QOS_EXT_BW_THRTL_0	
Memory Bandwidth Enforcement for COS0 (R/W) See Table 2-2.		Package
Register Address: D51H, 3409	IA32_L2_QOS_EXT_BW_THRTL_1	
Memory Bandwidth Enforcement for COS1 (R/W) See Table 2-2.		Package
Register Address: D52H, 3410	IA32_L2_QOS_EXT_BW_THRTL_2	
Memory Bandwidth Enforcement for COS2 (R/W) See Table 2-2.		Package
Register Address: D53H, 3411	IA32_L2_QOS_EXT_BW_THRTL_3	
Memory Bandwidth Enforcement for COS3 (R/W) See Table 2-2.		Package
Register Address: D54H, 3412	IA32_L2_QOS_EXT_BW_THRTL_4	
Memory Bandwidth Enforcement for COS4 (R/W) See Table 2-2.		Package
Register Address: D55H, 3413	IA32_L2_QOS_EXT_BW_THRTL_5	
Memory Bandwidth Enforcement for COS5 (R/W) See Table 2-2.		Package
Register Address: D56H, 3414	IA32_L2_QOS_EXT_BW_THRTL_6	
Memory Bandwidth Enforcement for COS6 (R/W) See Table 2-2.		Package
Register Address: D57H, 3415	IA32_L2_QOS_EXT_BW_THRTL_7	

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Memory Bandwidth Enforcement for COS7 (R/W) See Table 2-2.		Package
Register Address: D58H, 3416	IA32_L2_QOS_EXT_BW_THRTL_8	
Memory Bandwidth Enforcement for COS8 (R/W) See Table 2-2.		Package
Register Address: D59H, 3417	IA32_L2_QOS_EXT_BW_THRTL_9	
Memory Bandwidth Enforcement for COS9 (R/W) See Table 2-2.		Package
Register Address: D5AH, 3418	IA32_L2_QOS_EXT_BW_THRTL_10	
Memory Bandwidth Enforcement for COS10 (R/W) See Table 2-2.		Package
Register Address: D5BH, 3419	IA32_L2_QOS_EXT_BW_THRTL_11	
Memory Bandwidth Enforcement for COS11 (R/W) See Table 2-2.		Package
Register Address: D5CH, 3420	IA32_L2_QOS_EXT_BW_THRTL_12	
Memory Bandwidth Enforcement for COS12 (R/W) See Table 2-2.		Package
Register Address: D5DH, 3421	IA32_L2_QOS_EXT_BW_THRTL_13	
Memory Bandwidth Enforcement for COS13 (R/W) See Table 2-2.		Package
Register Address: D5EH, 3422	IA32_L2_QOS_EXT_BW_THRTL_14	
Memory Bandwidth Enforcement for COS14 (R/W) See Table 2-2.		Package
Register Address: D91H, 3473	IA32_COPY_LOCAL_TO_PLATFORM	
See Table 2-2.		Thread
Register Address: D92H, 3474	IA32_COPY_PLATFORM_TO_LOCAL	
See Table 2-2.		Thread
Register Address: D93H, 3475	IA32_PASID	
See Table 2-2.		Thread
Register Address: 1400H, 5120	IA32_SEAMRR_BASE	
SEAM Memory Range Register for TDx - Base Address (R/W) See Table 2-2.		Core
Register Address: 1401H, 5121	IA32_SEAMRR_MASK	
SEAM Memory Range Register for TDX (R/W) See Table 2-2.		Core
Register Address: 1A8FH, 6799	MSR_STLB_QOS_INFO	
STLB_QOS_INFO (R/O) STLB QoS MASK configuration.		Core

Table 2-56. Additional MSRs Supported by the Intel® Xeon® 6 P-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
5:0	NCLOS Number of CLOS supported for STLB resource using minus-1 notation.	
15:6	Reserved.	
19:16	4K_2M_CBM Length of capacity bitmask for 4K and 2M pages using minus-1 notation.	
28:20	Reserved.	
29	STLB_FILL_TRANSLATION_MSR_SUPPORTED MSR interface to fill STLB translations supported.	
30	4K_2M_ALIAS Indicates that 4K/2M pages alias into the same structure.	
63:31	Reserved.	
Register Address: 1B01H, 6913	IA32_UARCH_MISC_CTL	
IA32_UARCH_MISC_CTL (R/W) See Table 2-2.		Thread

2.17.11 MSRs Introduced in the Intel® Xeon® 6 E-Core Processors

Table 2-57 lists additional MSRs for the Intel Xeon 6 E-core processors. Intel Xeon 6 E-core processors have a CPUID Signature DisplayFamily_DisplayModel value of 06_AFH.

For an MSR listed in Table 2-57 that also appears in the model-specific tables of prior generations, Table 2-57 supersedes prior generation tables.

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 2FH, 47	IA32_BARRIER	
BARRIER (R/O) The IA32_BARRIER MSR ensures ordered execution by acting like LFENCE, controlling the sequencing of subsequent MSR reads after prior MSR reads and instructions. See Table 2-2.	Core	
Register Address: 33H, 51	MSR_MEMORY_CONTROL	
Memory Control (R/W) Disables split locks, which are locked instructions that split a cache line.	Core	
26:0	Reserved.	
27	UC_STORE_THROTTLE If set to 1, when enabled, the processor allows one in-progress, post-retirement UC stores at a time.	
28	UC_LOCK_DISABLE If set to 1, a UC load lock will trigger a fault. If clear to 0, UC load locks proceed normally.	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
29	SPLIT_LOCK_DISABLE If set to 1, a split lock will trigger an #AC fault. If clear to 0, split locks proceed normally.	
63:30	Reserved.	
Register Address: 34H, 52	MSR_SMI_COUNT	
SMI Counter (R/W)		Thread
31:0	SMI_COUNT Running count of SMI events since the last reset.	
63:32	Reserved.	
Register Address: 39H, 57	MSR_SOCKET_ID	
Socket ID (R/W) Reassigns the package-specific portions of the APIC ID. This MSR is used on scalable DP and high-end MP platforms to resolve legacy-mode APIC ID conflicts.		Package
10:0	PACKAGE_ID Holds package ID. This reflects the upper bits of the APIC ID.	
63:11	Reserved.	
Register Address: 7BH, 123	IA32_MCU_ENUMERATION	
Enumeration of Architectural Features (R/O) See Table 2-2.		Package
Register Address: 7CH, 124	IA32_MCU_STATUS	
MCU Status (R/O) Communicates results from the previous patch loads. See Table 2-2.		Package
Register Address: 87H, 135	IA32_MKTME_KEYID_PARTITIONING	
MKTME KEY ID Partitioning (R/O) Enumerates the number of activated KeyIDs for Intel TME-MK and Intel TDX. See Table 2-2.		Package
Register Address: 98H, 152	MSR_SEAM_WBINVDP	
SEAM WBINVDP (R/W) Allows software to WBINVD sections of the LLC.		Thread
63:0	HANDLE Caches sub-block to invalidate.	
Register Address: 99H, 153	MSR_SEAM_WBNINVD	
SEAM WBNINVD (R/W) Allows software to WBNINVD sections of the LLC.		Thread
63:0	HANDLE Caches sub-block to invalidate.	
Register Address: 9AH, 154	MSR_SEAM_INTR_PENDING	
SEAM Interrupt Pending (R/O) Report out some event pending bits.		Thread
0	INTR Interrupt is pending.	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
1	NMI NMI is pending.	
2	SMI SMI is pending.	
4:3	OTHER_EVENTS Other events pending.	
63:5	Reserved.	
Register Address: 9BH, 155	IA32_SMM_MONITOR_CTL	
SMM Monitor Control (R/W) The SMM Monitor Configuration involves SMM code specifying the MSEG location and enabling dual-monitor treatment by writing to the corresponding MSR. See Table 2-2.		Thread
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information (R/O) Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
15:8	MAX_NON_TURBO_LIM_RATIO This is the ratio of the frequency that invariant TSC runs at. Frequency = ratio * 100 MHz.	
25:16	Reserved.	
26	DCU_16K_MODE_AVAIL 0b: Indicates that the part does not support the 16K DCU mode. 1b: Indicates that the part supports 16K DCU mode.	
27	Reserved.	
28	PRG_TURBO_RATIO_EN Programmable Turbo Ratios per number of Active Cores. 0 = Programming Not Allowed. 1 = Programming Allowed.	
34:29	Reserved.	
35	BIOS_GUARD_ENABLE Indicates whether the BIOS Guard feature is enabled in the CPU.	
36	PEG2DMIDIS_EN 0 = PEG2DMIDIS is disabled. 1 = PEG2DMIDIS is enabled.	
39:37	Reserved.	
47:40	MAX_EFFICIENCY_RATIO Maximum Efficiency Ratio. This is given in units of 100 MHz.	
58:48	Reserved.	
59	SMM_SUPOVR_STATE_LOCK_ENABLE When set, indicates that the CPU supports MSR SMM_SUPOVR_STATE_LOCK and the Hardware Shield feature.	
63:60	Reserved.	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: CFH, 207	IA32_CORE_CAPABILITIES	
IA32 Core Capabilities Register (R/W) If CPUID.(EAX=07H, ECX=0):EDX[30] = 1. This MSR provides an architectural enumeration function for model-specific behavior.		Core
0	STLB_QOS When set to 1, processor supports STLB QoS.	
1	Reserved.	
2	INTEGRITY_SUPPORTED When set to 1, processor supports Functional Safety. Specific FUSA capabilities are enumerated in MSR_FUSA_CAPABILITIES.	
3	RSM_IN_CPLD_ONLY Intel System Resources Defense: When set to 1, RSM will only be allowed in CPLD and will #GP for all non-CPLD privilege levels.	
4	UC_LOCK_DISABLE When set to 1, processor supports UC load lock disable.	
5	SPLIT_LOCK_DISABLE When set to 1, processor supports #AC on split locks.	
6	SNP_FILTER_QOS When set to 1, processor supports Snoop Filter Quality of Service MSRs.	
7	UC_STORE_THROTTLING When set to 1, processor supports UC store throttling through MSR_MEMORY_CTRL[UC_STORE_THROTTLE].	
63:8	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: FEH, 254	IA32_MTRRCAP	
Memory Type Range Register (R/O) See Table 2-2.		Core
Register Address: 140H, 320	MSR_FEATURE_ENABLES	
Miscellaneous Enables for Thread-Specific Features (R/W)		Thread
0	AESNI_LOCK Once this bit is set, writes to this register will not be allowed.	
63:1	Reserved.	
Register Address: 1B0H, 432	IA32_ENERGY_PERF_BIAS	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
IA32_ENERGY_PERF_BIAS (R/W) See Table 2-2.		Thread
Register Address: 1B1H, 433	IA32_PACKAGE_THERM_STATUS	
IA32_PACKAGE_THERM_STATUS See Table 2-2.		Package
Register Address: 1B2H, 434	IA32_PACKAGE_THERM_INTERRUPT	
IA32_PACKAGE_THERM_INTERRUPT (R/W) See Table 2-2.		Package
Register Address: 2A1H, 673	MSR_PRMRR_BASE_1	
Processor Reserved Memory Range Register - Physical Base Control Register (R/W)		Core
2:0	MEMTYPE Memory Type for PRMRR accesses.	
3	CONFIGURED PRMRR base configured.	
11:4	Reserved.	
51:12	BASE PRMRR Base address.	
63:52	Reserved.	
Register Address: 2A2H, 674	MSR_PRMRR_BASE_2	
Processor Reserved Memory Range Register - Physical Base Control Register (R/W)		Core
2:0	MEMTYPE Memory Type for PRMRR accesses.	
3	CONFIGURED PRMRR base configured.	
11:4	Reserved.	
51:12	BASE PRMRR Base address.	
63:52	Reserved.	
Register Address: 2A3H, 675	MSR_PRMRR_BASE_3	
Processor Reserved Memory Range Register - Physical Base Control Register (R/W)		Core
2:0	MEMTYPE Memory Type for PRMRR accesses.	
3	CONFIGURED PRMRR base configured.	
11:4	Reserved.	
51:12	BASE PRMRR Base address.	
63:52	Reserved.	
Register Address: 2C2H, 706	MSR_COPY_SCAN_HASHES	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
MSR_COPY_SCAN_HASHES (W/O)		Module
63:0	SCAN_HASH-ADDR EDX:EAX contains the linear address of the SCAN Test HASH Binary loaded into memory	
Register Address: 2C3H, 707		MSR_SCAN_HASHES_STATUS
MSR_SCAN_HASHES_STATUS (R/O)		Core
15:0	CHUNK_SIZE EAX[15:0] - Chunk size of the test in KB.	
31:16	TOTAL_NUM_CHUNKS EAX[31:16] - Total number of chunks.	
39:32	ERROR_CODE EDX[7:0] - The error code refers to the LP that runs WRMSR(2C2H). 0x0 - Reserved. 0x1 - Attempt to copy scan-hashes when copy already in progress. 0x2 - Secure Memory not set up correctly. 0x3 - Scan-Image Header Image_info.ProgramID does not match MSR_INTEGRITY_CAPABILITIES[31:24], or scan-image header Processor-Signature doesn't match F/M/S, or scan-image header Processor-Flags doesn't match PlatformID. 0x4 - Reserved. 0x5 - Integrity check failed. 0x6 - WRMSR(0x2C6) Re-install of scan test image attempted when current scan test image is in use by other LPs. 0x7 - Aborted due to #PF (Page Fault). 0x8 - Unable to generate a Random Value.	
48:40	NUM_CHUNKS_IN_STRIDE EDX[16:8] - Number of Chunks in stride. This is the number of chunks that are installed. 0 in this field means that the CPU does not support strides, otherwise, the stride value must be >=1	
50:49	Reserved. EDX[18:17] - Set to all zeros.	
62:51	NAME EDX[30:19] - Maximum Number of cores that can run Intel® In-field Scan simultaneously minus 1. 0 means 1 core at a time.	
63	VALID EDX[31] - Valid bit is set when COPY_SCAN_HASHES completed.	
Register Address: 2C4H, 708		MSR_AUTHENTICATE_AND_COPY_CHUNK
MSR_AUTHENTICATE_AND_COPY_CHUNK(R/O)		Core
63:0	BASE_CHUNK_TABLE_ADDR EDX:EAX[63:0] - Linear Address pointing to the CHUNK TABLE (TABLE_BASE).	
Register Address: 2C5H, 709		MSR_CHUNKS_AUTHENTICATION_STATUS

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
MSR_CHUNKS_AUTHENTICATION_STATUS (R/O)		Core
15:0	VALID_CHUNKS EAX[15:0] - Total number of Valid (authenticated) chunks.	
31:16	NUM_CHUNKS_IN_STRIDE EAX[31:16] - Number of Chunks in Stride.	
39:32	ERROR_CODE EDX[7:0] 0x0 - No-error reported. 0x1 - Attempt to authenticate a CHUNK which is already marked as authentic or is currently being installed by another core. 0x2 - CHUNK authentication error. HASH of chunk did not match expected value. 0x3 - Aborted due to #PF (Page Fault). 0x4 - Chunk Outside the current Stride.	
63:40	Reserved. EDX[31:8] - Set to all zeros.	
Register Address: 2C6H, 710	MSR_ACTIVATE_SCAN	
MSR_ACTIVATE_SCAN (W/O)		Core
15:0	CHUNK_START_INDEX EAX[15:0] - Indicates Chunk Index from which to start.	
31:16	CHUNK_STOP_INDEX EAX[31:16] - Indicates what chunk index to stop at (inclusive).	
62:32	THREAD_WAIT_DELAY EDX[30:0] - TSC based delay to allow threads to rendezvous.	
63	SIGNAL_MCE EDX[31] - If 1: On scan-error log MC in MC4_STATUS and signal MCE if machine check signaling enabled in MC4_CTL[0]. - If 0: Don't no-logging/no-signaling.	
Register Address: 2C7H, 711	MSR_SCAN_STATUS	
MSR_SCAN_STATUS (R/O)		Core
15:0	CHUNK_NUM EAX[15:0] - SCAN Chunk that was reached.	
31:16	CHUNK_STOP_INDEX EAX[31:16] - Indicates what chunk index to stop at (inclusive). - Maps to same field in WRMSR(ACTIVATE_SCAN).	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
39:32	ERROR_CODE EDX[7:0] 0x0 - No Error. 0x1 - SCAN operation did not start. Other thread could not join. 0x2 - SCAN operation did not start. Interrupt occurred prior to SCAN coordination. 0x3 - SCAN operation did not start. Power Management conditions are inadequate to run SAF. 0x4 - SCAN operation did not start. Non valid chunks in the range CHUNK_STOP_INDEX : CHUNK_START_INDEX. 0x5 - SCAN operation did not start. Mismatch in arguments between threads T0/T1. 0x6 - SCAN operation did not start. Core not capable of performing SCAN currently. 0x7 - Debug Mode. Scan-At-Field results not to be trusted. 0x8 - SCAN operation did not start. Exceeded number of Logical Processors (LP) allowed to run Scan-At-Field concurrently. MAX_CORE_LIMIT exceeded. 0x9 - Interrupt occurred. Scan operation aborted prematurely, not all chunks requested have been executed. 0xB - Scan operation aborted due to corrupted chunk. 0xC - Scan operation did not start. All other error codes are reserved.	
61:40	Reserved. EDX[29:8] - Return all zeros.	
62	SCAN_CONTROL_ERROR EDX[30] - SCAN error in the Scan-At-Field controller. - Non ECC error.	
63	SCAN_SIGNATURE_ERROR EDX[31] - SCAN SIGNATURE error in the SCAN pattern fetched from main memory. - Non ECC error.	
Register Address: 2C8H, 712	MSR_SCAN_MODULE_ID	
MSR_SCAN_MODULE_ID (R/O)		Module
31:0	SCAN-AT-FIELD_REVID EAX[31:0] - Maps to Revision field in external header (offset 4).	
40:32	CURRENT_STRIDE_INDEX EDX[8:0] - Stride Index.	
63:41	Reserved. EDX[31:9] - Return all zeros.	
Register Address: 2C9H, 713	MSR_LAST_SAF_WP	
MSR_LAST_SAF_WP (R/O)		Module

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
31:0	LAST_WP EAX[31:0] - Provides information about the core when the last WRMSR(ACTIVATE_SCAN) was executed. - Available only if enumerated in INTEGRITY_CAPABILITIES[10:9].	
39:32	Reserved. EDX[7:0]	
63:40	Reserved. EDX[31:8] - Return all zeros.	
Register Address: 2D6H, 726	MSR_TRIGGER_PERIODIC_MEM_BIST	
MSR_TRIGGER_PERIODIC_MEM_BIST (W/O)		Core
0	SIGNAL_MCE EAX[0] - If 1, then signal MCE on fail if machine check signaling enabled in the corresponding MCI_CTL. If 0 then don't signal machine checks.	
7:1	ARRAY_BANK EAX[7:1] - Reserved.	
15:8	TST_STEP_PARAM EAX[15:8] 0: Test All Arrays, or Test Arrays in STEPs of NUM_STEPS.	
31:16	Reserved. EAX[31:16]	
63:32	Reserved. EAX[31:0]	
Register Address: 2D7H, 727	MSR_PERIODIC_MEM_BIST_STATUS	
MSR_PERIODIC_MEM_BIST_STATUS (R/O)		Core
0	MEM_BIST_STATUS 0: PASS. 1: FAIL.	
63:1	Reserved.	
Register Address: 2D9H, 729	MSR_INTEGRITY_CAPABILITIES	
MSR_INTEGRITY_CAPABILITIES (R/O) Enumerates features supported in Functional Safety.		Thread
0	STARTUP_SCAN_BIST When set to 1, processor supports Startup SCAN BIST.	
1	STARTUP_MEM_BIST When set to 1, processor supports Startup MEM BIST.	
2	PERIODIC_MEM_BIST When set to 1, processor supports Periodic MEM BIST.	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
3	LOCKSTEP When set to 1, processor supports Lock Step Mode.	
4	PERIODIC_SCAN_BIST When set to 1, processor supports Periodic SCAN BIST.	
5	PLL_LOSS_DETECT When set to 1, processor supports PLL LOSS detection.	
6	PWR_LOSS_DETECT When set to 1, processor supports Power Loss detection.	
7	PERRINJ When set to 1, processor supports FUSA PERRINJ.	
8	SBFT_AT_FIELD When set to 1, processor supports SBFT-At-Field.	
10:9	SAF_GEN_REV 00 = REV1; 01 = REV2; 10 = REV3; 11 = REV4.	
14:11	Reserved.	
15	PRESERVE_MEMORY_NEEDED When set to 1, processor supports FUSARR_BASE/MASK MSRs.	
20:16	TID_BIT_SHIFT Number of bits to shift right on x2APICID to get a unique topology ID of all logical processors that share a scan test engine.	
21	ALL_LP_JOIN_NEEDED All logical processors that share scan test engine need to be tested together and must join using MSR_ACTIVATE_SCAN.	
23:22	Reserved.	
31:24	PATTERN_ID Processor scan pattern ID. ID of the startup and periodic scan programs supported for this part.	
63:32	Reserved.	
Register Address: 2DCH, 732	IA32_INTEGRITY_STATUS	
IA32_INTEGRITY_STATUS (R/O) Provides status information for integrity features. See Table 2-2.		Thread
Register Address: 3F9H, 1017	MSR_PKG_C6_RESIDENCY	
MSR_PKG_C6_RESIDENCY (R/O)		Package
63:0	Package C6 Residency Counter	
Register Address: 3FAH, 1018	MSR_PKG_C7_RESIDENCY	
MSR_PKG_C7_RESIDENCY (R/O)		Package
63:0	Package C7 Residency Counter	
Register Address: 3FCH, 1020	MSR_CORE_C3_RESIDENCY	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
MSR_CORE_C3_RESIDENCY (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C3 Residency Counter Time spent in the Core C-State. Provided in units compatible to P1 clock frequency (Guaranteed / Maximum Core Non-Turbo Frequency).	
Register Address: 3FDH, 1021	MSR_CORE_C6_RESIDENCY	
MSR_CORE_C6_RESIDENCY (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C6 Residency Counter Time spent in the Core C-State. Provided in units compatible to P1 clock frequency (Guaranteed / Maximum Core Non-Turbo Frequency).	
Register Address: 3FEH, 1022	MSR_CORE_C7_RESIDENCY	
MSR_CORE_C7_RESIDENCY (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Core
63:0	CORE C7 Residency Counter Time spent in the Core C-State. Provided in units compatible to P1 clock frequency (Guaranteed / Maximum Core Non-Turbo Frequency).	
Register Address: 4F0H, 1264	MSR_SAF_CTRL	
MSR_SAF_CTRL (W/O)		Core
0	INVALIDATE_CURRENT_STRIDE EAX[0] - Write of 1 invalidates the currently installed stride. - Clears only the VALID_CHUNKS field on a RDMSR(CHUNKS_AUTHENTICATION_STATUS).	
63:1	Reserved.	
Register Address: 664H, 1636	MSR_MC6_RESIDENCY	
MSR_MC6_RESIDENCY (R/O) Time spent in the Module C6-State. Provided in units compatible to P1 clock frequency (Guaranteed / Maximum Core Non-Turbo Frequency).		Module
63:0	RESIDENCY Time that this module is in module-specific C6 states since last reset.	
Register Address: 6E0H, 1760	IA32_TSC_DEADLINE	
TSC Target of Local APIC's TSC Deadline Mode (R/W) See Table 2-2.		Thread
Register Address: 7A3H, 1955	IA32_MCU_EXT_SERVICE	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
MCU Extended Service (R/W) See Table 2-2.		Module
Register Address: 7A4H, 1956	IA32_MCU_ROLLBACK_MIN_ID	
Minimal MCU Revision ID (R/O) See Table 2-2.		Module
Register Address: 7A5H, 1957	IA32_MCU_STAGING_MBOX_ADDR	
IA32_MCU_STAGING_MBOX_ADDR (R/O) See Table 2-2.		Package
Register Address: 7B0H, 1968	IA32_ROLLBACK_SIGN_ID_0	
Rollback ID 0 (R/O) See Table 2-2.		Module
Register Address: 7B1H, 1969	IA32_ROLLBACK_SIGN_ID_1	
Rollback ID 1 (R/O) See Table 2-2.		Module
Register Address: 7B2H, 1970	IA32_ROLLBACK_SIGN_ID_2	
Rollback ID 2 (R/O) See Table 2-2.		Module
Register Address: 7B3H, 1971	IA32_ROLLBACK_SIGN_ID_3	
Rollback ID 3 (R/O) See Table 2-2.		Module
Register Address: 7B4H, 1972	IA32_ROLLBACK_SIGN_ID_4	
Rollback ID 4 (R/O) See Table 2-2.		Module
Register Address: 7B5H, 1973	IA32_ROLLBACK_SIGN_ID_5	
Rollback ID 5 (R/O) See Table 2-2.		Module
Register Address: 7B6H, 1974	IA32_ROLLBACK_SIGN_ID_6	
Rollback ID 6 (R/O) See Table 2-2.		Module
Register Address: 7B7H, 1975	IA32_ROLLBACK_SIGN_ID_7	
Rollback ID 7 (R/O) See Table 2-2.		Module
Register Address: 7B8H, 1976	IA32_ROLLBACK_SIGN_ID_8	
Rollback ID 8 (R/O) See Table 2-2.		Module
Register Address: 7B9H, 1977	IA32_ROLLBACK_SIGN_ID_9	
Rollback ID 9 (R/O) See Table 2-2.		Module
Register Address: 7BAH, 1978	IA32_ROLLBACK_SIGN_ID_10	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Rollback ID 10 (R/O) See Table 2-2.		Module
Register Address: 7BBH, 1979	IA32_ROLLBACK_SIGN_ID_11	
Rollback ID 11 (R/O) See Table 2-2.		Module
Register Address: 7BCH, 1980	IA32_ROLLBACK_SIGN_ID_12	
Rollback ID 12 (R/O) See Table 2-2.		Module
Register Address: 7BDH, 1981	IA32_ROLLBACK_SIGN_ID_13	
Rollback ID 13 (R/O) See Table 2-2.		Module
Register Address: 7BEH, 1982	IA32_ROLLBACK_SIGN_ID_14	
Rollback ID 14 (R/O) See Table 2-2.		Module
Register Address: 7BFH, 1983	IA32_ROLLBACK_SIGN_ID_15	
Rollback ID 15 (R/O) See Table 2-2.		Module
Register Address: 988H, 2440	IA32_UINTR_NV	
User Interrupt Size and Notification Vector (R/W) See Table 2-2.		Thread
Register Address: 9FBH, 2555	IA32_TME_CLEAR_SAVED_KEY	
IA32_TME_CLEAR_SAVED_KEY (R/W) See Table 2-2.		Package
Register Address: 9FFH, 2559	MSR_CORE_MKTME_ACTIVATE	
MSR_CORE_MKTME_ACTIVATE (R/O) MSR to read TME_ACTIVATE[MK_TME_KEYID_BITS].		Core
31:0	Reserved.	
35:32	READ_MK_TME_KEYID_BITS This value will be returned on a RDMSR, but must be zero on a WRMSR.	
63:36	Reserved.	
Register Address: C84H, 3204	MSR_MBA_CFG	
Memory Bandwidth Allocation (MBA) Configuration (R/W)		Package
1:0	Reserved.	
2	RAMBAE Resource Aware MBA Enable.	
63:3	Reserved.	
Register Address: CA0H, 3232	MSR_RMID_SNC_CONFIG	
MSR_RMID_SNC_CONFIG (R/W)		Package

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
0	RMID_LOCALIZED_DISTRIBUTION_MODE_ENABLE If set, Localized RMID distribution mode is enabled. If Clear, RMID Sharing mode is enabled.	
63:1	Reserved.	
Register Address: D50H, 3408	IA32_L2_QOS_EXT_BW_THRTL_0	
Memory Bandwidth Enforcement for COS0 (R/W) See Table 2-2.		Package
Register Address: D51H, 3409	IA32_L2_QOS_EXT_BW_THRTL_1	
Memory Bandwidth Enforcement for COS1 (R/W) See Table 2-2.		Package
Register Address: D52H, 3410	IA32_L2_QOS_EXT_BW_THRTL_2	
Memory Bandwidth Enforcement for COS2 (R/W) See Table 2-2.		Package
Register Address: D53H, 3411	IA32_L2_QOS_EXT_BW_THRTL_3	
Memory Bandwidth Enforcement for COS3 (R/W) See Table 2-2.		Package
Register Address: D54H, 3412	IA32_L2_QOS_EXT_BW_THRTL_4	
Memory Bandwidth Enforcement for COS4 (R/W) See Table 2-2.		Package
Register Address: D55H, 3413	IA32_L2_QOS_EXT_BW_THRTL_5	
Memory Bandwidth Enforcement for COS5 (R/W) See Table 2-2.		Package
Register Address: D56H, 3414	IA32_L2_QOS_EXT_BW_THRTL_6	
Memory Bandwidth Enforcement for COS6 (R/W) See Table 2-2.		Package
Register Address: D57H, 3415	IA32_L2_QOS_EXT_BW_THRTL_7	
Memory Bandwidth Enforcement for COS7 (R/W) See Table 2-2.		Package
Register Address: D58H, 3416	IA32_L2_QOS_EXT_BW_THRTL_8	
Memory Bandwidth Enforcement for COS8 (R/W) See Table 2-2.		Package
Register Address: D59H, 3417	IA32_L2_QOS_EXT_BW_THRTL_9	
Memory Bandwidth Enforcement for COS9 (R/W) See Table 2-2.		Package
Register Address: D5AH, 3418	IA32_L2_QOS_EXT_BW_THRTL_10	
Memory Bandwidth Enforcement for COS10 (R/W) See Table 2-2.		Package
Register Address: D5BH, 3419	IA32_L2_QOS_EXT_BW_THRTL_11	

Table 2-57. Additional MSRs Supported by the Intel® Xeon® 6 E-Core Processors (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Memory Bandwidth Enforcement for COS11 (R/W) See Table 2-2.		Package
Register Address: D5CH, 3420	IA32_L2_QOS_EXT_BW_THRTL_12	
Memory Bandwidth Enforcement for COS12 (R/W) See Table 2-2.		Package
Register Address: D5DH, 3421	IA32_L2_QOS_EXT_BW_THRTL_13	
Memory Bandwidth Enforcement for COS13 (R/W) See Table 2-2.		Package
Register Address: D5EH, 3422	IA32_L2_QOS_EXT_BW_THRTL_14	
Memory Bandwidth Enforcement for COS14 (R/W) See Table 2-2.		Package
Register Address: E00H, 3584	IA32_QOS_CORE_BW_THRTL_0	
CBA Levels Based on COS for Bandwidth Throttling (R/W) See Table 2-2.		Thread
Register Address: E01H, 3585	IA32_QOS_CORE_BW_THRTL_1	
CBA Levels Based on COS for Bandwidth Throttling (R/W) See Table 2-2.		Thread
Register Address: 1400H, 5120	IA32_SEAMRR_BASE	
SEAM Memory Range Register for TDX - Base Address (R/W) See Table 2-2.		Core
Register Address: 1401H, 5121	IA32_SEAMRR_MASK	
SEAM Memory Range Register for TDX (R/W) See Table 2-2.		Core
Register Address: 1A8FH, 6799	MSR_STLB_QOS_INFO	
STLB_QOS_INFO (R/O) STLB QoS MASK configuration.		Core
5:0	NCLOS Number of CLOS supported for STLB resource using minus-1 notation.	
15:6	Reserved.	
19:16	4K_2M_CBM Length of capacity bitmask for 4K and 2M pages using minus-1 notation.	
28:20	Reserved.	
29	STLB_FILL_TRANSLATION_MSR_SUPPORTED MSR interface to fill STLB translations supported.	
30	4K_2M_ALIAS Indicates that 4K/2M pages alias into the same structure.	
63:31	Reserved.	

2.17.12 MSRs Introduced in the Intel® Series 2 Core™ Ultra Processor Supporting Performance Hybrid Architecture

Table 2-58 lists additional MSRs for the Intel Series 2 Core Ultra processor with a CPUID Signature DisplayFamily_DisplayModel value of 06_BDH. Table 2-59 lists the MSRs unique to the processor P-core. Table 2-60 lists the MSRs unique to the processor E-core.

For an MSR listed in Table 2-58, Table 2-59, or Table 2-60 that also appears in the model-specific tables of prior generations, Table 2-58, Table 2-59, and Table 2-60 supersede prior generation tables.

Table 2-58. Additional MSRs Supported by the Intel® Series 2 Core™ Ultra Processors Supporting Performance Hybrid Architecture

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID	
BIOS Update Signature ID (R/W) See Table 2-2.		Thread
Register Address: 19AH, 410	IA32_CLOCK_MODULATION	
Clock Modulation (R/W) See Table 2-2.		Thread
Register Address: 601H, 1537	MSR_PKG_POWER_LIMIT_4	
Package Power Limit 4 (R/W) Package-level maximum power limit (in Watts).		Package
15:0	POWER_LIMIT_4 PL4 Value in 0.125 W increments. This field is locked by PKG_POWER_LIMIT_4.LOCK. When the LOCK bit is set to 1, this field becomes Read Only. If the value is 0, PL4 limit is disabled.	
30:16	Reserved.	
31	LOCK This bit will lock the POWER_LIMIT_4 settings in this register and will also lock this setting. This means that once set to 1, the POWER_LIMIT_4 setting and this bit become Read Only until the next Warm Reset.	
63:32	Reserved.	
Register Address: 630H, 1584	MSR_PKG_C8_RESIDENCY	
MSR_PKG_C8_RESIDENCY (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
59:0	Package C8 Residency Counter Value since last reset that this package is in processor-specific C8 states. Count at the same frequency as the TSC.	
63:60	Reserved.	
Register Address: 631H, 1585	MSR_PKG_C9_RESIDENCY	
MSR_PKG_C9_RESIDENCY (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package

Table 2-58. Additional MSRs Supported by the Intel® Series 2 Core™ Ultra Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
59:0	Package C9 Residency Counter Value since last reset that this package is in processor-specific C9 states. Count at the same frequency as the TSC.	
63:60	Reserved.	
Register Address: 632H, 1586	MSR_PKG_C10_RESIDENCY	
MSR_PKG_C10_RESIDENCY (R/O) Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-States.		Package
59:0	Package C10 Residency Counter Value since last reset that this package is in processor-specific C10 states. Count at the same frequency as the TSC.	
63:60	Reserved.	
Register Address: 651H, 1617	MSR_SECONDARY_TURBO_RATIO_LIMIT_CORES	
SECONDARY_TURBO_RATIO_LIMIT_CORES (R/W) This register defines the active core ranges for each frequency point. - NUMCORE[0:7] must be populated in ascending order. - NUMCORE[i+1] must be greater than NUMCORE[i]. - Entries with NUMCORE[i] == 0 will be ignored. - The last valid entry must have NUMCORE >= the number of cores in the SKU. If any of the rules above are broken, we will silently reject the configuration.		Package
7:0	CORE_COUNT_0 Defines the active core ranges for each frequency point.	
15:8	CORE_COUNT_1 Defines the active core ranges for each frequency point.	
23:16	CORE_COUNT_2 Defines the active core ranges for each frequency point.	
31:24	CORE_COUNT_3 Defines the active core ranges for each frequency point.	
39:32	CORE_COUNT_4 Defines the active core ranges for each frequency point.	
47:40	CORE_COUNT_5 Defines the active core ranges for each frequency point.	
55:48	CORE_COUNT_6 Defines the active core ranges for each frequency point.	
63:56	CORE_COUNT_7 Defines the active core ranges for each frequency point.	
Register Address: 658H, 1624	MSR_WEIGHTED_CORE_CO	
Core-Count Weighted C0 Residency (R/O)		Package

Table 2-58. Additional MSRs Supported by the Intel® Series 2 Core™ Ultra Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
63:0	DATA Increment at the same rate as the TSC. The increment each cycle is weighted by the number of processor cores in the package that reside in C0. If N cores are simultaneously in C0, then each cycle the counter increments by N.	
Register Address: 659H, 1625	MSR_ANY_CORE_CO	
Any Core C0 Residency (R/O)		Package
63:0	DATA Increment at the same rate as the TSC. The increment each cycle is weighted by the number of processor cores in the package that reside in C0. If N cores are simultaneously in C0, then each cycle the counter increments by N.	
Register Address: 65AH, 1626	MSR_ANY_GFXE_CO	
Any Graphics Engine C0 Residency (R/O)		Package
63:0	DATA Increment at the same rate as the TSC. The increment each cycle is one if any processor graphic device's compute engines are in C0.	
Register Address: 65BH, 1627	MSR_CORE_GFXE_OVERLAP_CO	
Core and Graphics Engine Overlapped C0 Residency (R/O)		Package
63:0	DATA Increment at the same rate as the TSC. The increment each cycle is one if at least one compute engine of the processor graphics is in C0 and at least one processor core in the package is also in C0.	
Register Address: C88H, 3208	IA32_RESOURCE_PRIORITY	
Thread scope Resource Priority Enable (R/W) See Table 2-2.		Thread
Register Address: C89H, 3209	IA32_RESOURCE_PRIORITY_PKG	
IA32_RESOURCE_PRIORITY_PKG (R/W) See Table 2-2.		Package
Register Address: 1900H, 6400	IA32_PMC_GPO_CTR	
Full Width Writable General Performance Counter 0 (R/W) See Table 2-2.		Thread
Register Address: 1901H, 6401	IA32_PMC_GPO_CFG_A	
IA32_PMC_GPO_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 0. See Table 2-2.		Thread
Register Address: 1904H, 6404	IA32_PMC_GP1_CTR	
Full Width Writable General Performance Counter 1 (R/W) See Table 2-2.		Thread
Register Address: 1905H, 6405	IA32_PMC_GP1_CFG_A	

Table 2-58. Additional MSRs Supported by the Intel® Series 2 Core™ Ultra Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
IA32_PMC_GP1_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 1. See Table 2-2.		Thread
Register Address: 1908H, 6408	IA32_PMC_GP2_CTR	
Full Width Writable General Performance Counter 2 (R/W) See Table 2-2.		Thread
Register Address: 1909H, 6409	IA32_PMC_GP2_CFG_A	
IA32_PMC_GP2_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 2. See Table 2-2.		Thread
Register Address: 190CH, 6412	IA32_PMC_GP3_CTR	
Full Width Writable General Performance Counter 3 (R/W) See Table 2-2.		Thread
Register Address: 190DH, 6413	IA32_PMC_GP3_CFG_A	
IA32_PMC_GP3_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 3. See Table 2-2.		Thread
Register Address: 1910H, 6416	IA32_PMC_GP4_CTR	
Full Width Writable General Performance Counter 4 (R/W) See Table 2-2.		Thread
Register Address: 1911H, 6417	IA32_PMC_GP4_CFG_A	
IA32_PMC_GP4_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 4. See Table 2-2.		Thread
Register Address: 1914H, 6420	IA32_PMC_GP5_CTR	
Full Width Writable General Performance Counter 5 (R/W) See Table 2-2.		Thread
Register Address: 1915H, 6421	IA32_PMC_GP5_CFG_A	
IA32_PMC_GP5_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 5. See Table 2-2.		Thread
Register Address: 1918H, 6424	IA32_PMC_GP6_CTR	
Full Width Writable General Performance Counter 6 (R/W) See Table 2-2.		Thread
Register Address: 1919H, 6425	IA32_PMC_GP6_CFG_A	
IA32_PMC_GP6_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 6. See Table 2-2.		Thread
Register Address: 191CH, 6428	IA32_PMC_GP7_CTR	

Table 2-58. Additional MSRs Supported by the Intel® Series 2 Core™ Ultra Processors Supporting Performance Hybrid Architecture (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Full Width Writable General Performance Counter 7 (R/W) See Table 2-2.		Thread
Register Address: 191DH, 6429	IA32_PMC_GP7_CFG_A	
IA32_PMC_GP7_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 7. See Table 2-2.		Thread
Register Address: 1980H, 6528	IA32_PMC_FX0_CTR	
IA32_PMC_FX0_CTR (R/W) Fixed-Function Performance Counter 0 - Instructions Retired. See Table 2-2.		Thread
Register Address: 1984H, 6532	IA32_PMC_FX1_CTR	
IA32_PMC_FX1_CTR (R/W) Fixed-Function Performance Counter 1 - Unhalted core clock cycles. See Table 2-2.		Thread
Register Address: 1988H, 6536	IA32_PMC_FX2_CTR	
IA32_PMC_FX2_CTR (R/W) Fixed-Function Performance Counter 2 - Unhalted core reference cycles. See Table 2-2.		Thread

The MSRs listed in Table 2-59 are unique to the Intel Series 2 Core Ultra processor P-core. These MSRs are not supported on the processor E-core.

Table 2-59. MSRs Supported by the Intel® Series 2 Core™ Ultra Processor P-core

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: C9H, 201	IA32_PMC8	
General Performance Counter 8 (R/W) See Table 2-2.		Thread
Register Address: CAH, 202	IA32_PMC9	
General Performance Counter 9 (R/W) See Table 2-2.		Thread
Register Address: 18EH, 398	IA32_PERFEVTSEL8	
Performance Event Select Register 8 (R/W) See Table 2-2.		Thread
Register Address: 18FH, 399	IA32_PERFEVTSEL9	
Performance Event Select Register 9 (R/W) See Table 2-2.		Thread
Register Address: 30CH, 780	IA32_FIXED_CTR3	
Fixed-Function Performance Counter 3 (R/W) See Table 2-2.		Thread
Register Address: 329H, 809	MSR_PERF_METRICS	

Table 2-59. MSRs Supported by the Intel® Series 2 Core™ Ultra Processor P-core (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
MSR_PERF_METRICS (R/W) This register provides built-in support for Top-down Micro-architecture Analysis (TMA) metrics. It exposes the four TMA Level 1 metrics where the lower 32 bits are divided into four 8 bit fields, each of which is an integer percentage of the total TOPDOWN.SLOTS (as reported by fixed-function counter 3).		Thread
7:0	RETIRING Percent of utilized by uops that eventually retire (commit).	
15:8	BAD_SPECULATION Percent of Wasted due to incorrect speculation, covering Utilized by uops that do not retire, or Recovery Bubbles (unutilized slots).	
23:16	FRONTEND_BOUND Percent of Unutilized slots where Front-end did not deliver a uop while Back-end is ready.	
31:24	BACKEND_BOUND Percent of Unutilized slots where a uop was not delivered to Back-end due to lack of Back-end resources.	
39:32	MULTI_UOPS Frontend bound.	
47:40	BRANCH_MISPREDICTS Frontend bound.	
55:48	FRONTEND_LATENCY Frontend bound.	
63:56	MEMORY_BOUND Frontend bound.	
Register Address: 4C9H, 1225	IA32_A_PMC8	
Full Width Writable IA32_PMC8 Alias (R/W) See Table 2-2.		Thread
Register Address: 4CAH, 1226	IA32_A_PMC9	
Full Width Writable IA32_PMC9 Alias (R/W) See Table 2-2.		Thread
Register Address: 540H, 1344	MSR_THREAD_UARCH_CTL	
Thread Uarch Control (R/W)		Thread
0	WB_MEM_STRM_LD_DISABLE Disable streaming behavior for MOVNTDQA loads to WB memory type. If set, these accesses will be treated like regular cacheable loads (Data will be cached).	
63:1	Reserved.	
Register Address: 540H, 1344	MSR_CORE_UARCH_CTL	
Core Uarch Control (R/W)		Core
0	SCRUB_DIS L1 scrubbing disable.	
63:1	Reserved.	

Table 2-59. MSRs Supported by the Intel® Series 2 Core™ Ultra Processor P-core (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1920H, 6432	IA32_PMC_GP8_CTR	
Full Width Writable General Performance Counter 8 (R/W) See Table 2-2.		Thread
Register Address: 1921H, 6433	IA32_PMC_GP8_CFG_A	
IA32_PMC_GP8_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 8. See Table 2-2.		Thread
Register Address: 1924H, 6436	IA32_PMC_GP9_CTR	
Full Width Writable General Performance Counter 9 (R/W) See Table 2-2.		Thread
Register Address: 1925H, 6437	IA32_PMC_GP9_CFG_A	
IA32_PMC_GP9_CFG_A (R/W) Performance Event Select Register used to control the operation of the General Performance Counter 9. See Table 2-2.		Thread
Register Address: 198CH, 6540	IA32_PMC_FX3_CTR	
IA32_PMC_FX3_CTR (R/W) See Table 2-2.		Thread

The MSRs listed in Table 2-60 are unique to the Intel Series 2 Core Ultra processor E-core. These MSRs are not supported on the processor P-core.

Table 2-60. MSRs Supported by the Intel® Series 2 Core™ Ultra Processor E-core

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 2DCH, 732	IA32_INTEGRITY_STATUS	
Status Information for Integrity Features (R/O) See Table 2-2.		Thread
Register Address: 30DH, 781	IA32_FIXED_CTR4	
Fixed-Function Performance Counter 4 - Top-down Bad Speculation (R/W) See Table 2-2.		Thread
Register Address: 30EH, 782	IA32_FIXED_CTR5	
Fixed-Function Performance Counter 5 - Top-down Frontend Bound (R/W) See Table 2-2.		Thread
Register Address: 30FH, 783	IA32_FIXED_CTR6	
Fixed-Function Performance Counter 6 - Top-down Retiring (R/W) See Table 2-2.		Thread
Register Address: D18H, 3352	IA32_L2_MASK_8	
L2 CAT Mask for COS8 (R/W) See Table 2-2.		Module
Register Address: D19H, 3353	IA32_L2_MASK_9	

Table 2-60. MSRs Supported by the Intel® Series 2 Core™ Ultra Processor E-core (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
L2 CAT Mask for COS9 (R/W) See Table 2-2.		Module
Register Address: D1AH, 3354	IA32_L2_MASK_10	
L2 CAT Mask for COS10 (R/W) See Table 2-2.		Module
Register Address: D1BH, 3355	IA32_L2_MASK_11	
L2 CAT Mask for COS11 (R/W) See Table 2-2.		Module
Register Address: D1CH, 3356	IA32_L2_MASK_12	
L2 CAT Mask for COS12 (R/W) See Table 2-2.		Module
Register Address: D1DH, 3357	IA32_L2_MASK_13	
L2 CAT Mask for COS13 (R/W) See Table 2-2.		Module
Register Address: D1EH, 3358	IA32_L2_MASK_14	
L2 CAT Mask for COS14 (R/W) See Table 2-2.		Module
Register Address: D1FH, 3359	IA32_L2_MASK_15	
L2 CAT Mask for COS15 (R/W) See Table 2-2.		Module
Register Address: 1878H, 6264	MSR_WORK_CONSERVING_CLOS	
Work Conserving CLOS (R/W)		Module
0	WC_VALID WC Valid Bit that indicates WC MSR has been setup. This bit must be set for the WC algorithm to be enabled.	
7:1	Reserved.	
11:8	CLOS_START_PRI1 Starting CLOS range for priority 1.	
15:12	CLOS_END_PRI1 Ending CLOS range for priority 1.	
19:16	CLOS_START_PRI2 Starting CLOS range for priority 2.	
23:20	CLOS_END_PRI2 Ending CLOS range for priority 2.	
27:24	CLOS_START_PRI3 Starting CLOS range for priority 3.	
31:28	CLOS_END_PRI3 Ending CLOS range for priority 3.	
63:32	Reserved.	
Register Address: 1903H, 6403	IA32_PMC_GPO_CFG_C	

Table 2-60. MSRs Supported by the Intel® Series 2 Core™ Ultra Processor E-core (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
IA32_PMC_GP0_CFG_C (R/W) See Table 2-2.		Thread
Register Address: 1907H, 6407	IA32_PMC_GP1_CFG_C	
IA32_PMC_GP1_CFG_C (R/W) See Table 2-2.		Thread
Register Address: 190AH, 6410	IA32_PMC_GP2_CFG_B	
IA32_PMC_GP2_CFG_B (R/W) See Table 2-2.		Thread
Register Address: 190BH, 6411	IA32_PMC_GP2_CFG_C	
IA32_PMC_GP2_CFG_C (R/W) See Table 2-2.		Thread
Register Address: 190EH, 6414	IA32_PMC_GP3_CFG_B	
IA32_PMC_GP3_CFG_B (R/W) See Table 2-2.		Thread
Register Address: 190FH, 6415	IA32_PMC_GP3_CFG_C	
IA32_PMC_GP3_CFG_C (R/W) See Table 2-2.		Thread
Register Address: 1912H, 6418	IA32_PMC_GP4_CFG_B	
IA32_PMC_GP4_CFG_B (R/W) See Table 2-2.		Thread
Register Address: 1913H, 6419	IA32_PMC_GP4_CFG_C	
IA32_PMC_GP4_CFG_C (R/W) See Table 2-2.		Thread
Register Address: 1916H, 6422	IA32_PMC_GP5_CFG_B	
IA32_PMC_GP5_CFG_B (R/W) See Table 2-2.		Thread
Register Address: 1917H, 6423	IA32_PMC_GP5_CFG_C	
IA32_PMC_GP5_CFG_C (R/W) See Table 2-2.		Thread
Register Address: 191AH, 6426	IA32_PMC_GP6_CFG_B	
IA32_PMC_GP6_CFG_B (R/W) See Table 2-2.		Thread
Register Address: 191BH, 6427	IA32_PMC_GP6_CFG_C	
IA32_PMC_GP6_CFG_C (R/W) See Table 2-2.		Thread
Register Address: 191EH, 6430	IA32_PMC_GP7_CFG_B	
IA32_PMC_GP7_CFG_B (R/W) See Table 2-2.		Thread
Register Address: 191FH, 6431	IA32_PMC_GP7_CFG_C	

Table 2-60. MSRs Supported by the Intel® Series 2 Core™ Ultra Processor E-core (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
IA32_PMC_GP7_CFG_C (R/W) See Table 2-2.		Thread
Register Address: 1982H, 6530	IA32_PMC_FX0_CFG_B	
Fixed-Function Counter Reload Configuration Register (R/W) See Table 2-2.		Thread
Register Address: 1983H, 6531	IA32_PMC_FX0_CFG_C	
Extended Perf Event Selector for Fixed-Function Counter 0 (R/W) See Table 2-2.		Thread
Register Address: 1986H, 6534	IA32_PMC_FX1_CFG_B	
Fixed-Function Counter Reload Configuration Register (R/W) See Table 2-2.		Thread
Register Address: 1987H, 6535	IA32_PMC_FX1_CFG_C	
Extended Perf Event Selector for Fixed-Function Counter 1 (R/W) See Table 2-2.		Thread
Register Address: 198BH, 6539	IA32_PMC_FX2_CFG_C	
Extended Perf Event Selector for Fixed-Function Counter 2 (R/W) See Table 2-2.		Thread
Register Address: 1990H, 6544	IA32_PMC_FX4_CTR	
Fixed-Function Performance Counter 4 - Top-down Bad Speculation (R/W) See Table 2-2.		Thread
Register Address: 1993H, 6547	IA32_PMC_FX4_CFG_C	
Extended Perf Event Selector for Fixed-Function Counter 4 (R/W) See Table 2-2.		Thread
Register Address: 1994H, 6548	IA32_PMC_FX5_CTR	
Fixed-Function Performance Counter 5 - Top-down Frontend Bound (R/W) See Table 2-2.		Thread
Register Address: 1997H, 6551	IA32_PMC_FX5_CFG_C	
Extended Perf Event Selector for Fixed-Function Counter 5 (R/W) See Table 2-2.		Thread
Register Address: 1998H, 6552	IA32_PMC_FX6_CTR	
Fixed-Function Performance Counter 5 - Top-down Bad Retiring (R/W) See Table 2-2.		Thread
Register Address: 199BH, 6555	IA32_PMC_FX6_CFG_C	
Extended Perf Event Selector for Fixed-Function Counter 6 (R/W) See Table 2-2.		Thread

2.18 MSRS IN THE INTEL® XEON PHI™ PROCESSOR 3200/5200/7200 SERIES AND THE INTEL® XEON PHI™ PROCESSOR 7215/7285/7295 SERIES

The Intel® Xeon Phi™ processor 3200, 5200, 7200 series, with a CPUID Signature DisplayFamily_DisplayModel value of 06_57H, supports the MSR interfaces listed in Table 2-61. These processors are based on the Knights Landing microarchitecture. The Intel® Xeon Phi™ processor 7215, 7285, 7295 series, with a CPUID Signature DisplayFamily_DisplayModel value of 06_85H, supports the MSR interfaces listed in Table 2-61 and Table 2-62. These processors are based on the Knights Mill microarchitecture. Some MSRs are shared between a pair of processor cores, and the scope is marked as module.

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 0H, 0	IA32_P5_MC_ADDR	
See Section 2.23, "MSRs in Pentium Processors."		Module
Register Address: 1H, 1	IA32_P5_MC_TYPE	
See Section 2.23, "MSRs in Pentium Processors."		Module
Register Address: 6H, 6	IA32_MONITOR_FILTER_SIZE	
See Section 10.10.5, "Monitor/Mwait Address Range Determination." See Table 2-2.		Thread
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER	
See Section 19.17, "Time-Stamp Counter," and Table 2-2.		Thread
Register Address: 17H, 23	IA32_PLATFORM_ID	
Platform ID (R) See Table 2-2.		Package
Register Address: 1BH, 27	IA32_APIC_BASE	
See Section 12.4.4, "Local APIC Status and Location," and Table 2-2.		Thread
Register Address: 34H, 52	MSR_SMI_COUNT	
SMI Counter (R/O)		Thread
31:0	SMI Count (R/O)	
63:32	Reserved.	
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in Intel 64Processor (R/W) See Table 2-2.		Thread
0	Lock. (R/WL)	
1	Reserved.	
2	Enable VMX outside SMX operation. (R/WL)	
Register Address: 3BH, 59	IA32_TSC_ADJUST	
Per-Logical-Processor TSC ADJUST (R/W) See Table 2-2.		Thread
Register Address: 4EH, 78	IA32_PPIN_CTL (MSR_PPIN_CTL)	
Protected Processor Inventory Number Enable Control (R/W)		Package
0	LockOut (R/WO) See Table 2-2.	

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
1	Enable_PPIN (R/W) See Table 2-2.	
63:2	Reserved	
Register Address: 4FH, 79	IA32_PPIN (MSR_PPIN)	
Protected Processor Inventory Number (R/O)		Package
63:0	Protected Processor Inventory Number (R/O) See Table 2-2.	
Register Address: 79H, 121	IA32_BIOS_UPDT_TRIG	
BIOS Update Trigger Register (W) See Table 2-2.		Core
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID	
BIOS Update Signature ID (R/W) See Table 2-2.		Thread
Register Address: C1H, 193	IA32_PMC0	
Performance Counter Register See Table 2-2.		Thread
Register Address: C2H, 194	IA32_PMC1	
Performance Counter Register See Table 2-2.		Thread
Register Address: CEH, 206	MSR_PLATFORM_INFO	
Platform Information Contains power management and other model specific features enumeration. See http://biosbits.org .		Package
7:0	Reserved.	
15:8	Maximum Non-Turbo Ratio (R/O) This is the ratio of the frequency that invariant TSC runs at. Frequency = ratio * 100 MHz.	Package
27:16	Reserved.	
28	Programmable Ratio Limit for Turbo Mode (R/O) When set to 1, indicates that Programmable Ratio Limit for Turbo mode is enabled. When set to 0, indicates Programmable Ratio Limit for Turbo mode is disabled.	Package
29	Programmable TDP Limit for Turbo Mode (R/O) When set to 1, indicates that TDP Limit for Turbo mode is programmable. When set to 0, indicates TDP Limit for Turbo mode is not programmable.	Package
39:30	Reserved.	
47:40	Maximum Efficiency Ratio (R/O) This is the minimum ratio (maximum efficiency) that the processor can operate, in units of 100MHz.	Package
63:48	Reserved.	

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: E2H, 226	MSR_PKG_CST_CONFIG_CONTROL	
C-State Configuration Control (R/W)		Package
2:0	Package C-State Limit (R/W) Specifies the lowest C-state for the package. This feature does not limit the processor core C-state. The power-on default value from bit[2:0] of this register reports the deepest package C-state the processor is capable to support when manufactured. It is recommended that BIOS always read the power-on default value reported from this bit field to determine the supported deepest C-state on the processor and leave it as default without changing it. 000b - C0/C1 (No package C-state support) 001b - C2 010b - C6 (non retention)* 011b - C6 (Retention)* 100b - Reserved 101b - Reserved 110b - Reserved 111b - No package C-state limit. All C-States supported by the processor are available. Note: C6 retention mode provides more power saving than C6 non-retention mode. Limiting the package to C6 non retention mode does prevent the MSR_PKG_C6_RESIDENCY counter (MSR 3F9h) from being incremented.	
9:3	Reserved.	
10	I/O MWAIT Redirection Enable (R/W) When set, will map IO_read instructions sent to IO registers at MSR_PMG_IO_CAPTURE_BASE[15:0] to MWAIT instructions.	
14:11	Reserved.	
15	CFG Lock (R/O) When set, locks bits [15:0] of this register for further writes until the next reset occurs.	
25	Reserved.	
26	C1 State Auto Demotion Enable (R/W) When set, the processor will conditionally demote C3/C6/C7 requests to C1 based on uncore auto-demote information.	
27	Reserved.	
28	C1 State Auto Undemotion Enable (R/W) When set, enables Undemotion from Demoted C1.	
29	PKG C-State Auto Demotion Enable (R/W) When set, enables Package C state demotion.	
63:30	Reserved.	
Register Address: E4H, 228	MSR_PMG_IO_CAPTURE_BASE	
Power Management IO Capture Base (R/W)		Tile

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
15:0	LVL_2 Base Address (R/W) Microcode will compare IO-read zone to this base address to determine if an MWAIT(C2/3/4) needs to be issued instead of the IO-read. Should be programmed to the chipset Plevel_2 IO address.	
22:16	C-State Range (R/W) The IO-port block size in which IO-redirection will be executed (0-127). Should be programmed based on the number of LVLx registers existing in the chipset.	
63:23	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Thread
Register Address: FEH, 254	IA32_MTRRCAP	
Memory Type Range Register (R/O) See Table 2-2.		Core
Register Address: 13CH, 316	MSR_FEATURE_CONFIG	
AES Configuration (RW-L) Privileged post-BIOS agent must provide a #GP handler to handle unsuccessful read of this MSR.		Core
1:0	AES Configuration (RW-L) Upon a successful read of this MSR, the configuration of AES instruction set availability is as follows: 11b: AES instructions are not available until next RESET. Otherwise, AES instructions are available. Note, the AES instruction set is not available if read is unsuccessful. If the configuration is not 01b, AES instructions can be mis-configured if a privileged agent unintentionally writes 11b.	
63:2	Reserved.	
Register Address: 140H, 320	MISC_FEATURE_ENABLES	
MISC_FEATURE_ENABLES		Thread
0	Reserved.	
1	User Mode MONITOR and MWAIT (R/W) If set to 1, the MONITOR and MWAIT instructions do not cause invalid-opcode exceptions when executed with CPL > 0 or in virtual-8086 mode. If MWAIT is executed when CPL > 0 or in virtual-8086 mode, and if EAX indicates a C-state other than C0 or C1, the instruction operates as if EAX indicated the C-state C1.	
63:2	Reserved.	
Register Address: 174H, 372	IA32_SYSENTER_CS	
See Table 2-2.		Thread

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 175H, 373	IA32_SYSENTER_ESP	
See Table 2-2.		Thread
Register Address: 176H, 374	IA32_SYSENTER_EIP	
See Table 2-2.		Thread
Register Address: 179H, 377	IA32_MCG_CAP	
See Table 2-2.		Thread
Register Address: 17AH, 378	IA32_MCG_STATUS	
See Table 2-2.		Thread
Register Address: 17DH, 381	MSR_SMM_MCA_CAP	
Enhanced SMM Capabilities (SMM-RO) Reports SMM capability Enhancement. Accessible only while in SMM.		Thread
31:0	Bank Support (SMM-RO) One bit per MCA bank. If the bit is set, that bank supports Enhanced MCA (Default all 0; does not support EMCA).	
55:32	Reserved.	
56	Targeted SMI (SMM-RO) Set if targeted SMI is supported.	
57	SMM_CPU_SVRSTR (SMM-RO) Set if SMM SRAM save/restore feature is supported.	
58	SMM_CODE_ACCESS_CHK (SMM-RO) Set if SMM code access check feature is supported.	
59	Long_Flow_Indication (SMM-RO) If set to 1, indicates that the SMM long flow indicator is supported and a host-space interface available to SMM handler.	
63:60	Reserved.	
Register Address: 186H, 390	IA32_PERFVTSEL0	
Performance Monitoring Event Select Register (R/W) See Table 2-2.		Thread
7:0	Event Select.	
15:8	UMask.	
16	USR.	
17	OS.	
18	Edge.	
19	PC.	
20	INT.	
21	AnyThread.	
22	EN.	
23	INV.	

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
31:24	CMASK.	
63:32	Reserved.	
Register Address: 187H, 391	IA32_PERFEVTSEL1	
See Table 2-2.		Thread
Register Address: 198H, 408	IA32_PERF_STATUS	
See Table 2-2.		Package
Register Address: 199H, 409	IA32_PERF_CTL	
See Table 2-2.		Thread
Register Address: 19AH, 410	IA32_CLOCK_MODULATION	
Clock Modulation (R/W) See Table 2-2.		Thread
Register Address: 19BH, 411	IA32_THERM_INTERRUPT	
Thermal Interrupt Control (R/W) See Table 2-2.		Module
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2.		Module
0	Thermal Status (R/O)	
1	Thermal Status Log (R/WC0)	
2	PROTCHOT # or FORCEPR# Status (R/O)	
3	PROTCHOT # or FORCEPR# Log (R/WC0)	
4	Critical Temperature Status (R/O)	
5	Critical Temperature Status Log (R/WC0)	
6	Thermal Threshold #1 Status (R/O)	
7	Thermal Threshold #1 Log (R/WC0)	
8	Thermal Threshold #2 Status (R/O)	
9	Thermal Threshold #2 Log (R/WC0)	
10	Power Limitation Status (R/O)	
11	Power Limitation Log (R/WC0)	
15:12	Reserved.	
22:16	Digital Readout (R/O)	
26:23	Reserved.	
30:27	Resolution in Degrees Celsius (R/O)	
31	Reading Valid (R/O)	
63:32	Reserved.	
Register Address: 1A0H, 416	IA32_MISC_ENABLE	

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Enable Misc. Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		Thread
0	Fast-Strings Enable	
2:1	Reserved.	
3	Automatic Thermal Control Circuit Enable (R/W)	
6:4	Reserved.	
7	Performance Monitoring Available (R)	
10:8	Reserved.	
11	Branch Trace Storage Unavailable (R/O)	
12	Processor Event Based Sampling Unavailable (R/O)	
15:13	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W)	
18	ENABLE MONITOR FSM (R/W)	
21:19	Reserved.	
22	Limit CPUID Maxval (R/W)	
23	xTPR Message Disable (R/W)	
33:24	Reserved.	
34	XD Bit Disable (R/W) See Table 2-3.	
37:35	Reserved.	
38	Turbo Mode Disable (R/W)	
63:39	Reserved.	
Register Address: 1A2H, 418	MSR_TEMPERATURE_TARGET	
Temperature Target		Package
15:0	Reserved.	
23:16	Temperature Target (R)	
29:24	Target Offset (R/W)	
63:30	Reserved.	
Register Address: 1A4H, 420	MSR_MISC_FEATURE_CONTROL	
Miscellaneous Feature Control (R/W)		
0	DCU Hardware Prefetcher Disable (R/W) If 1, disables the L1 data cache prefetcher.	Core
1	L2 Hardware Prefetcher Disable (R/W) If 1, disables the L2 hardware prefetcher.	Core
63:2	Reserved.	
Register Address: 1A6H, 422	MSR_OFFCORE_RSP_0	
Offcore Response Event Select Register (R/W)		Shared

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 1A7H, 423	MSR_OFFCORE_RSP_1	
Offcore Response Event Select Register (R/W)		Shared
Register Address: 1ADH, 429	MSR_TURBO_RATIO_LIMIT	
Maximum Ratio Limit of Turbo Mode for Groups of Cores (R/W)		Package
0	Reserved.	
7:1	Maximum Number of Cores in Group 0 Number active processor cores which operates under the maximum ratio limit for group 0.	Package
15:8	Maximum Ratio Limit for Group 0 Maximum turbo ratio limit when the number of active cores are not more than the group 0 maximum core count.	Package
20:16	Number of Incremental Cores Added to Group 1 Group 1, which includes the specified number of additional cores plus the cores in group 0, operates under the group 1 turbo max ratio limit = "group 0 Max ratio limit" - "group ratio delta for group 1".	Package
23:21	Group Ratio Delta for Group 1 An unsigned integer specifying the ratio decrement relative to the Max ratio limit to Group 0.	Package
28:24	Number of Incremental Cores Added to Group 2 Group 2, which includes the specified number of additional cores plus all the cores in group 1, operates under the group 2 turbo max ratio limit = "group 1 Max ratio limit" - "group ratio delta for group 2".	Package
31:29	Group Ratio Delta for Group 2 An unsigned integer specifying the ratio decrement relative to the Max ratio limit for Group 1.	Package
36:32	Number of Incremental Cores Added to Group 3 Group 3, which includes the specified number of additional cores plus all the cores in group 2, operates under the group 3 turbo max ratio limit = "group 2 Max ratio limit" - "group ratio delta for group 3".	Package
39:37	Group Ratio Delta for Group 3 An unsigned integer specifying the ratio decrement relative to the Max ratio limit for Group 2.	Package
44:40	Number of Incremental Cores Added to Group 4 Group 4, which includes the specified number of additional cores plus all the cores in group 3, operates under the group 4 turbo max ratio limit = "group 3 Max ratio limit" - "group ratio delta for group 4".	Package
47:45	Group Ratio Delta for Group 4 An unsigned integer specifying the ratio decrement relative to the Max ratio limit for Group 3.	Package
52:48	Number of Incremental Cores Added to Group 5 Group 5, which includes the specified number of additional cores plus all the cores in group 4, operates under the group 5 turbo max ratio limit = "group 4 Max ratio limit" - "group ratio delta for group 5".	Package

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
55:53	Group Ratio Delta for Group 5 An unsigned integer specifying the ratio decrement relative to the Max ratio limit for Group 4.	Package
60:56	Number of Incremental Cores Added to Group 6 Group 6, which includes the specified number of additional cores plus all the cores in group 5, operates under the group 6 turbo max ratio limit = "group 5 Max ratio limit" - "group ratio delta for group 6".	Package
63:61	Group Ratio Delta for Group 6 An unsigned integer specifying the ratio decrement relative to the Max ratio limit for Group 5.	Package
Register Address: 1B0H, 432	IA32_ENERGY_PERF_BIAS	
See Table 2-2.		Thread
Register Address: 1B1H, 433	IA32_PACKAGE_THERM_STATUS	
See Table 2-2.		Package
Register Address: 1B2H, 434	IA32_PACKAGE_THERM_INTERRUPT	
See Table 2-2.		Package
Register Address: 1C8H, 456	MSR_LBR_SELECT	
Last Branch Record Filtering Select Register (R/W) See Section 19.9.2, "Filtering of Last Branch Records."		Thread
0	CPL_EQ_0	
1	CPL_NEQ_0	
2	JCC	
3	NEAR_REL_CALL	
4	NEAR_IND_CALL	
5	NEAR_RET	
6	NEAR_IND_JMP	
7	NEAR_REL_JMP	
8	FAR_BRANCH	
63:9	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-2) that points to the MSR containing the most recent branch record. See MSR_LASTBRANCH_0_FROM_IP.		Thread
Register Address: 1D9H, 473	IA32_DEBUGCTL	
Debug Control (R/W)		Thread
0	LBR Setting this bit to 1 enables the processor to record a running trace of the most recent branches taken by the processor in the LBR stack.	

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
1	BTF Setting this bit to 1 enables the processor to treat EFLAGS.TF as single-step on branches instead of single-step on instructions.	
5:2	Reserved.	
6	TR Setting this bit to 1 enables branch trace messages to be sent.	
7	BTS Setting this bit enables branch trace messages (BTMs) to be logged in a BTS buffer.	
8	BTINT When clear, BTMs are logged in a BTS buffer in circular fashion. When this bit is set, an interrupt is generated by the BTS facility when the BTS buffer is full.	
9	BTS_OFF_OS When set, BTS or BTM is skipped if CPL = 0.	
10	BTS_OFF_USR When set, BTS or BTM is skipped if CPL > 0.	
11	FREEZE_LBRS_ON_PMI When set, the LBR stack is frozen on a PMI request.	
12	FREEZE_PERFMON_ON_PMI When set, each ENABLE bit of the global counter control MSR are frozen (address 3BFH) on a PMI request.	
13	Reserved.	
14	FREEZE_WHILE_SMM When set, freezes PerfMon and trace messages while in SMM.	
31:15	Reserved.	
Register Address: 1DDH, 477	MSR_LER_FROM_LIP	
Last Exception Record from Linear IP (R)		Thread
Register Address: 1DEH, 478	MSR_LER_TO_LIP	
Last Exception Record to Linear IP (R)		Thread
Register Address: 1F2H, 498	IA32_SMRR_PHYSBASE	
See Table 2-2.		Core
Register Address: 1F3H, 499	IA32_SMRR_PHYSMASK	
See Table 2-2.		Core
Register Address: 200H, 512	IA32_MTRR_PHYSBASE0	
See Table 2-2.		Core
Register Address: 201H, 513	IA32_MTRR_PHYSMASK0	
See Table 2-2.		Core
Register Address: 202H, 514	IA32_MTRR_PHYSBASE1	

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
See Table 2-2.		Core
Register Address: 203H, 515	IA32_MTRR_PHYSMASK1	
See Table 2-2.		Core
Register Address: 204H, 516	IA32_MTRR_PHYSBASE2	
See Table 2-2.		Core
Register Address: 205H, 517	IA32_MTRR_PHYSMASK2	
See Table 2-2.		Core
Register Address: 206H, 518	IA32_MTRR_PHYSBASE3	
See Table 2-2.		Core
Register Address: 207H, 519	IA32_MTRR_PHYSMASK3	
See Table 2-2.		Core
Register Address: 208H, 520	IA32_MTRR_PHYSBASE4	
See Table 2-2.		Core
Register Address: 209H, 521	IA32_MTRR_PHYSMASK4	
See Table 2-2.		Core
Register Address: 20AH, 522	IA32_MTRR_PHYSBASE5	
See Table 2-2.		Core
Register Address: 20BH, 523	IA32_MTRR_PHYSMASK5	
See Table 2-2.		Core
Register Address: 20CH, 524	IA32_MTRR_PHYSBASE6	
See Table 2-2.		Core
Register Address: 20DH, 525	IA32_MTRR_PHYSMASK6	
See Table 2-2.		Core
Register Address: 20EH, 526	IA32_MTRR_PHYSBASE7	
See Table 2-2.		Core
Register Address: 20FH, 527	IA32_MTRR_PHYSMASK7	
See Table 2-2.		Core
Register Address: 250H, 592	IA32_MTRR_FIX64K_00000	
See Table 2-2.		Core
Register Address: 258H, 600	IA32_MTRR_FIX16K_80000	
See Table 2-2.		Core
Register Address: 259H, 601	IA32_MTRR_FIX16K_A0000	
See Table 2-2.		Core
Register Address: 268H, 616	IA32_MTRR_FIX4K_C0000	
See Table 2-2.		Core
Register Address: 269H, 617	IA32_MTRR_FIX4K_C8000	
See Table 2-2.		Core

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 26AH, 618	IA32_MTRR_FIX4K_D0000	
See Table 2-2.		Core
Register Address: 26BH, 619	IA32_MTRR_FIX4K_D8000	
See Table 2-2.		Core
Register Address: 26CH, 620	IA32_MTRR_FIX4K_E0000	
See Table 2-2.		Core
Register Address: 26DH, 621	IA32_MTRR_FIX4K_E8000	
See Table 2-2.		Core
Register Address: 26EH, 622	IA32_MTRR_FIX4K_F0000	
See Table 2-2.		Core
Register Address: 26FH, 623	IA32_MTRR_FIX4K_F8000	
See Table 2-2.		Core
Register Address: 277H, 631	IA32_PAT	
See Table 2-2.		Core
Register Address: 2FFH, 767	IA32_MTRR_DEF_TYPE	
Default Memory Types (R/W) See Table 2-2.		Core
Register Address: 309H, 777	IA32_FIXED_CTR0	
Fixed-Function Performance Counter Register 0 (R/W) See Table 2-2.		Thread
Register Address: 30AH, 778	IA32_FIXED_CTR1	
Fixed-Function Performance Counter Register 1 (R/W) See Table 2-2.		Thread
Register Address: 30BH, 779	IA32_FIXED_CTR2	
Fixed-Function Performance Counter Register 2 (R/W) See Table 2-2.		Thread
Register Address: 345H, 837	IA32_PERF_CAPABILITIES	
See Table 2-2. See Section 19.4.1, "IA32_DEBUGCTL MSR."		Package
Register Address: 38DH, 909	IA32_FIXED_CTR_CTRL	
Fixed-Function-Counter Control Register (R/W) See Table 2-2.		Thread
Register Address: 38EH, 910	IA32_PERF_GLOBAL_STATUS	
See Table 2-2.		Thread
Register Address: 38FH, 911	IA32_PERF_GLOBAL_CTRL	
See Table 2-2.		Thread
Register Address: 390H, 912	IA32_PERF_GLOBAL_OVF_CTRL	
See Table 2-2.		Thread

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)	
See Table 2-2.		Thread
Register Address: 3F8H, 1016	MSR_PKG_C3_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.		Package
63:0	Package C3 Residency Counter (R/O)	
Register Address: 3F9H, 1017	MSR_PKG_C6_RESIDENCY	
63:0	Package C6 Residency Counter (R/O)	Package
Register Address: 3FAH, 1018	MSR_PKG_C7_RESIDENCY	
63:0	Package C7 Residency Counter (R/O)	Package
Register Address: 3FCH, 1020	MSR_MCO_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.		Module
63:0	Module C0 Residency Counter (R/O)	
Register Address: 3FDH, 1021	MSR_MC6_RESIDENCY	
63:0	Module C6 Residency Counter (R/O)	Module
Register Address: 3FFH, 1023	MSR_CORE_C6_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.		Core
63:0	CORE C6 Residency Counter (R/O)	
Register Address: 400H, 1024	IA32_MCO_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 401H, 1025	IA32_MCO_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core
Register Address: 402H, 1026	IA32_MCO_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Core
Register Address: 404H, 1028	IA32_MC1_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 405H, 1029	IA32_MC1_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core
Register Address: 408H, 1032	IA32_MC2_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core
Register Address: 409H, 1033	IA32_MC2_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Core
Register Address: 40AH, 1034	IA32_MC2_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs."		Core
Register Address: 40CH, 1036	IA32_MC3_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Core

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 40DH, 1037	IA32_MC3_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS."		Core
Register Address: 40EH, 1038	IA32_MC3_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Core
Register Address: 410H, 1040	IA32_MC4_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Core
Register Address: 411H, 1041	IA32_MC4_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS."		Core
Register Address: 412H, 1042	IA32_MC4_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The MSR_MC4_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Core
Register Address: 414H, 1044	IA32_MC5_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Package
Register Address: 415H, 1045	IA32_MC5_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRS."		Package
Register Address: 416H, 1046	IA32_MC5_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs."		Package
Register Address: 4C1H, 1217	IA32_A_PMC0	
See Table 2-2.		Thread
Register Address: 4C2H, 1218	IA32_A_PMC1	
See Table 2-2.		Thread
Register Address: 600H, 1536	IA32_DS_AREA	
DS Save Area (R/W) See Table 2-2.		Thread
Register Address: 606H, 1542	MSR_RAPL_POWER_UNIT	
Unit Multipliers Used in RAPL Interfaces (R/O)		Package
3:0	Power Units See Section 16.10.1, "RAPL Interfaces."	Package
7:4	Reserved.	Package
12:8	Energy Status Units Energy related information (in Joules) is based on the multiplier, $1/2^{\text{ESU}}$; where ESU is an unsigned integer represented by bits 12:8. Default value is 0EH (or 61 micro-joules).	Package
15:13	Reserved.	Package
19:16	Time Units See Section 16.10.1, "RAPL Interfaces."	Package

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
63:20	Reserved.	
Register Address: 60DH, 1549	MSR_PKG_C2_RESIDENCY	
Note: C-state values are processor specific C-state code names, unrelated to MWAIT extension C-state parameters or ACPI C-states.		Package
63:0	Package C2 Residency Counter (R/O)	
Register Address: 610H, 1552	MSR_PKG_POWER_LIMIT	
PKG RAPL Power Limit Control (R/W) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 611H, 1553	MSR_PKG_ENERGY_STATUS	
PKG Energy Status (R/O) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 613H, 1555	MSR_PKG_PERF_STATUS	
PKG Perf Status (R/O) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 614H, 1556	MSR_PKG_POWER_INFO	
PKG RAPL Parameters (R/W) See Section 16.10.3, "Package RAPL Domain."		Package
Register Address: 618H, 1560	MSR_DRAM_POWER_LIMIT	
DRAM RAPL Power Limit Control (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 619H, 1561	MSR_DRAM_ENERGY_STATUS	
DRAM Energy Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61BH, 1563	MSR_DRAM_PERF_STATUS	
DRAM Performance Throttling Status (R/O) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 61CH, 1564	MSR_DRAM_POWER_INFO	
DRAM RAPL Parameters (R/W) See Section 16.10.5, "DRAM RAPL Domain."		Package
Register Address: 638H, 1592	MSR_PPO_POWER_LIMIT	
PPO RAPL Power Limit Control (R/W) See Section 16.10.4, "PPO/PP1 RAPL Domains."		Package
Register Address: 639H, 1593	MSR_PPO_ENERGY_STATUS	
PPO Energy Status (R/O) See Section 16.10.4, "PPO/PP1 RAPL Domains."		Package
Register Address: 648H, 1608	MSR_CONFIG_TDP_NOMINAL	
Base TDP Ratio (R/O) See Table 2-25.		Package

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 649H, 1609	MSR_CONFIG_TDP_LEVEL1	
ConfigTDP Level 1 ratio and power level (R/O) See Table 2-25.		Package
Register Address: 64AH, 1610	MSR_CONFIG_TDP_LEVEL2	
ConfigTDP Level 2 ratio and power level (R/O) See Table 2-25.		Package
Register Address: 64BH, 1611	MSR_CONFIG_TDP_CONTROL	
ConfigTDP Control (R/W) See Table 2-25.		Package
Register Address: 64CH, 1612	MSR_TURBO_ACTIVATION_RATIO	
ConfigTDP Control (R/W) See Table 2-25.		Package
Register Address: 690H, 1680	MSR_CORE_PERF_LIMIT_REASONS	
Indicator of Frequency Clipping in Processor Cores (R/W) (Frequency refers to processor core frequency.)		Package
0	PROCHOT Status (R0)	
1	Thermal Status (R0)	
5:2	Reserved.	
6	VR Therm Alert Status (R0)	
7	Reserved.	
8	Electrical Design Point Status (R0)	
63:9	Reserved.	
Register Address: 6E0H, 1760	IA32_TSC_DEADLINE	
TSC Target of Local APIC's TSC Deadline Mode (R/W) See Table 2-2.		Core
Register Address: 802H, 2050	IA32_X2APIC_APICID	
x2APIC ID Register (R/O)		Thread
Register Address: 803H, 2051	IA32_X2APIC_VERSION	
x2APIC Version Register (R/O)		Thread
Register Address: 808H, 2056	IA32_X2APIC_TPR	
x2APIC Task Priority Register (R/W)		Thread
Register Address: 80AH, 2058	IA32_X2APIC_PPR	
x2APIC Processor Priority Register (R/O)		Thread
Register Address: 80BH, 2059	IA32_X2APIC_EOI	
x2APIC EOI Register (W/O)		Thread
Register Address: 80DH, 2061	IA32_X2APIC_LDR	
x2APIC Logical Destination Register (R/O)		Thread

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 80FH, 2063	IA32_X2APIC_SIVR	
x2APIC Spurious Interrupt Vector Register (R/W)		Thread
Register Address: 810H, 2064	IA32_X2APIC_ISR0	
x2APIC In-Service Register Bits [31:0] (R/O)		Thread
Register Address: 811H, 2065	IA32_X2APIC_ISR1	
x2APIC In-Service Register Bits [63:32] (R/O)		Thread
Register Address: 812H, 2066	IA32_X2APIC_ISR2	
x2APIC In-Service Register Bits [95:64] (R/O)		Thread
Register Address: 813H, 2067	IA32_X2APIC_ISR3	
x2APIC In-Service Register Bits [127:96] (R/O)		Thread
Register Address: 814H, 2068	IA32_X2APIC_ISR4	
x2APIC In-Service Register Bits [159:128] (R/O)		Thread
Register Address: 815H, 2069	IA32_X2APIC_ISR5	
x2APIC In-Service Register Bits [191:160] (R/O)		Thread
Register Address: 816H, 2070	IA32_X2APIC_ISR6	
x2APIC In-Service Register Bits [223:192] (R/O)		Thread
Register Address: 817H, 2071	IA32_X2APIC_ISR7	
x2APIC In-Service Register Bits [255:224] (R/O)		Thread
Register Address: 818H, 2072	IA32_X2APIC_TMR0	
x2APIC Trigger Mode Register Bits [31:0] (R/O)		Thread
Register Address: 819H, 2073	IA32_X2APIC_TMR1	
x2APIC Trigger Mode Register Bits [63:32] (R/O)		Thread
Register Address: 81AH, 2074	IA32_X2APIC_TMR2	
x2APIC Trigger Mode Register Bits [95:64] (R/O)		Thread
Register Address: 81BH, 2075	IA32_X2APIC_TMR3	
x2APIC Trigger Mode Register Bits [127:96] (R/O)		Thread
Register Address: 81CH, 2076	IA32_X2APIC_TMR4	
x2APIC Trigger Mode Register Bits [159:128] (R/O)		Thread
Register Address: 81DH, 2077	IA32_X2APIC_TMR5	
x2APIC Trigger Mode Register Bits [191:160] (R/O)		Thread
Register Address: 81EH, 2078	IA32_X2APIC_TMR6	
x2APIC Trigger Mode Register Bits [223:192] (R/O)		Thread
Register Address: 81FH, 2079	IA32_X2APIC_TMR7	
x2APIC Trigger Mode Register Bits [255:224] (R/O)		Thread
Register Address: 820H, 2080	IA32_X2APIC_IRR0	
x2APIC Interrupt Request Register Bits [31:0] (R/O)		Thread
Register Address: 821H, 2081	IA32_X2APIC_IRR1	

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
x2APIC Interrupt Request Register Bits [63:32] (R/O)		Thread
Register Address: 822H, 2082	IA32_X2APIC_IRR2	
x2APIC Interrupt Request Register Bits [95:64] (R/O)		Thread
Register Address: 823H, 2083	IA32_X2APIC_IRR3	
x2APIC Interrupt Request Register Bits [127:96] (R/O)		Thread
Register Address: 824H, 2084	IA32_X2APIC_IRR4	
x2APIC Interrupt Request Register Bits [159:128] (R/O)		Thread
Register Address: 825H, 2085	IA32_X2APIC_IRR5	
x2APIC Interrupt Request Register Bits [191:160] (R/O)		Thread
Register Address: 826H, 2086	IA32_X2APIC_IRR6	
x2APIC Interrupt Request Register Bits [223:192] (R/O)		Thread
Register Address: 827H, 2087	IA32_X2APIC_IRR7	
x2APIC Interrupt Request Register Bits [255:224] (R/O)		Thread
Register Address: 828H, 2088	IA32_X2APIC_ESR	
x2APIC Error Status Register (R/W)		Thread
Register Address: 82FH, 2095	IA32_X2APIC_LVT_CMCI	
x2APIC LVT Corrected Machine Check Interrupt Register (R/W)		Thread
Register Address: 830H, 2096	IA32_X2APIC_ICR	
x2APIC Interrupt Command Register (R/W)		Thread
Register Address: 832H, 2098	IA32_X2APIC_LVT_TIMER	
x2APIC LVT Timer Interrupt Register (R/W)		Thread
Register Address: 833H, 2099	IA32_X2APIC_LVT_THERMAL	
x2APIC LVT Thermal Sensor Interrupt Register (R/W)		Thread
Register Address: 834H, 2100	IA32_X2APIC_LVT_PMI	
x2APIC LVT Performance Monitor Register (R/W)		Thread
Register Address: 835H, 2101	IA32_X2APIC_LVT_LINT0	
x2APIC LVT LINT0 Register (R/W)		Thread
Register Address: 836H, 2102	IA32_X2APIC_LVT_LINT1	
x2APIC LVT LINT1 Register (R/W)		Thread
Register Address: 837H, 2103	IA32_X2APIC_LVT_ERROR	
x2APIC LVT Error Register (R/W)		Thread
Register Address: 838H, 2104	IA32_X2APIC_INIT_COUNT	
x2APIC Initial Count Register (R/W)		Thread
Register Address: 839H, 2105	IA32_X2APIC_CUR_COUNT	
x2APIC Current Count Register (R/O)		Thread
Register Address: 83EH, 2110	IA32_X2APIC_DIV_CONF	
x2APIC Divide Configuration Register (R/W)		Thread

Table 2-61. Selected MSRs Supported by Intel® Xeon Phi™ Processors with a CPUID Signature DisplayFamily_DisplayModel Value of 06_57H or 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 83FH, 2111	IA32_X2APIC_SELF_IPI	
x2APIC Self IPI Register (W/O)		Thread
Register Address: C000_0080H	IA32_EFER	
Extended Feature Enables See Table 2-2.		Thread
Register Address: C000_0081H	IA32_STAR	
System Call Target Address (R/W) See Table 2-2.		Thread
Register Address: C000_0082H	IA32_LSTAR	
IA-32e Mode System Call Target Address (R/W) See Table 2-2.		Thread
Register Address: C000_0084H	IA32_FMASK	
System Call Flag Mask (R/W) See Table 2-2.		Thread
Register Address: C000_0100H	IA32_FS_BASE	
Map of BASE Address of FS (R/W) See Table 2-2.		Thread
Register Address: C000_0101H	IA32_GS_BASE	
Map of BASE Address of GS (R/W) See Table 2-2.		Thread
Register Address: C000_0102H	IA32_KERNEL_GS_BASE	
Swap Target of BASE Address of GS (R/W) See Table 2-2.		Thread
Register Address: C000_0103H	IA32_TSC_AUX	
AUXILIARY TSC Signature (R/W) See Table 2-2		Thread

Table 2-62 lists model-specific registers that are supported by the Intel® Xeon Phi™ processor 7215, 7285, 7295 series based on the Knights Mill microarchitecture.

Table 2-62. Additional MSRs Supported by the Intel® Xeon Phi™ Processor 7215, 7285, 7295 Series with a CPUID Signature DisplayFamily_DisplayModel Value of 06_85H

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Register Address: 9BH, 155	IA32_SMM_MONITOR_CTL	
SMM Monitor Configuration (R/W) This MSR is readable only if VMX is enabled, and writeable only if VMX is enabled and in SMM mode, and is used to configure the VMX MSEG base address. See Table 2-2.		Core
Register Address: 480H, 1152	IA32_VMX_BASIC	

Table 2-62. Additional MSRs Supported by the Intel® Xeon Phi™ Processor 7215, 7285, 7295 Series with a CPUID Signature DisplayFamily_DisplayModel Value of 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Reporting Register of Basic VMX Capabilities (R/O) See Table 2-2.		Core
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL5	
Capability Reporting Register of Pin-based VM-execution Controls (R/O) See Table 2-2.		Core
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL5	
Capability Reporting Register of Primary Processor-based VM-execution Controls (R/O)		Core
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL5	
Capability Reporting Register of VM-exit Controls (R/O) See Table 2-2.		Core
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTL5	
Capability Reporting Register of VM-entry Controls (R/O) See Table 2-2.		Core
Register Address: 485H, 1157	IA32_VMX_MISC	
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Table 2-2.		Core
Register Address: 486H, 1158	IA32_VMX_CR0_FIXED0	
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Table 2-2.		Core
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1	
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Table 2-2.		Core
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0	
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Table 2-2.		Core
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1	
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Table 2-2.		Core
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM	
Capability Reporting Register of VMCS Field Enumeration (R/O) See Table 2-2.		Core
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL52	
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Table 2-2.		Core
Register Address: 48CH, 1164	IA32_VMX_EPT_VPID_ENUM	
Capability Reporting Register of EPT and VPID (R/O) See Table 2-2.		Core
Register Address: 48DH, 1165	IA32_VMX_TRUE_PINBASED_CTL5	

Table 2-62. Additional MSRs Supported by the Intel® Xeon Phi™ Processor 7215, 7285, 7295 Series with a CPUID Signature DisplayFamily_DisplayModel Value of 06_85H (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Scope
Capability Reporting Register of Pin-Based VM-Execution Flex Controls (R/O) See Table 2-2.		Core
Register Address: 48EH, 1166	IA32_VMX_TRUE_PROCBASED_CTL5	
Capability Reporting Register of Primary Processor-Based VM-Execution Flex Controls (R/O) See Table 2-2.		Core
Register Address: 48FH, 1167	IA32_VMX_TRUE_EXIT_CTL5	
Capability Reporting Register of VM-Exit Flex Controls (R/O) See Table 2-2.		Core
Register Address: 490H, 1168	IA32_VMX_TRUE_ENTRY_CTL5	
Capability Reporting Register of VM-Entry Flex Controls (R/O) See Table 2-2.		Core
Register Address: 491H, 1169	IA32_VMX_FMFUNC	
Capability Reporting Register of VM-Function Controls (R/O) See Table 2-2.		Core

2.19 MSRS IN THE PENTIUM® 4 AND INTEL® XEON® PROCESSORS

Table 2-63 lists MSRs (architectural and model-specific) that are defined across processor generations based on Intel NetBurst microarchitecture. The processor can be identified by its CPUID signatures of DisplayFamily encoding of 0FH, see Table 2-1.

- MSRs with an "IA32_" prefix are designated as "architectural." This means that the functions of these MSRs and their addresses remain the same for succeeding families of IA-32 processors.
- MSRs with an "MSR_" prefix are model specific with respect to address functionalities. The column "Model Availability" lists the model encoding value(s) within the Pentium 4 and Intel Xeon processor family at the specified register address. The model encoding value of a processor can be queried using CPUID. See "CPUID—CPU Identification" in Chapter 3 of the Intel® 64 and IA-32 Architectures Software Developer's Manual, Volume 2A.

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Register Address: 0H, 0	IA32_P5_MC_ADDR		
See Section 2.23, "MSRs in Pentium Processors."		0, 1, 2, 3, 4, 6	Shared
Register Address: 1H, 1	IA32_P5_MC_TYPE		
See Section 2.23, "MSRs in Pentium Processors."		0, 1, 2, 3, 4, 6	Shared
Register Address: 6H, 6	IA32_MONITOR_FILTER_LINE_SIZE		
See Section 10.10.5, "Monitor/Mwait Address Range Determination."		3, 4, 6	Shared
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER		
Time Stamp Counter See Table 2-2.		0, 1, 2, 3, 4, 6	Unique

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
On earlier processors, only the lower 32 bits are writable. On any write to the lower 32 bits, the upper 32 bits are cleared. For processor family 0FH, models 3 and 4: all 64 bits are writable.			
Register Address: 17H, 23	IA32_PLATFORM_ID		
Platform ID (R) See Table 2-2. The operating system can use this MSR to determine “slot” information for the processor and the proper microcode update to load.		0, 1, 2, 3, 4, 6	Shared
Register Address: 1BH, 27	IA32_APIC_BASE		
APIC Location and Status (R/W) See Table 2-2. See Section 12.4.4, “Local APIC Status and Location.”		0, 1, 2, 3, 4, 6	Unique
Register Address: 2AH, 42	MSR_EBC_HARD_POWERON		
Processor Hard Power-On Configuration (R/W) Enables and disables processor features. (R) Indicates current processor configuration.		0, 1, 2, 3, 4, 6	Shared
0	Output Tri-state Enabled (R) Indicates whether tri-state output is enabled (1) or disabled (0) as set by the strapping of SMI#. The value in this bit is written on the deassertion of RESET#; the bit is set to 1 when the address bus signal is asserted.		
1	Execute BIST (R) Indicates whether the execution of the BIST is enabled (1) or disabled (0) as set by the strapping of INIT#. The value in this bit is written on the deassertion of RESET#; the bit is set to 1 when the address bus signal is asserted.		
2	In Order Queue Depth (R) Indicates whether the in order queue depth for the system bus is 1 (1) or up to 12 (0) as set by the strapping of A7#. The value in this bit is written on the deassertion of RESET#; the bit is set to 1 when the address bus signal is asserted.		
3	MCERR# Observation Disabled (R) Indicates whether MCERR# observation is enabled (0) or disabled (1) as determined by the strapping of A9#. The value in this bit is written on the deassertion of RESET#; the bit is set to 1 when the address bus signal is asserted.		
4	BINIT# Observation Enabled (R) Indicates whether BINIT# observation is enabled (0) or disabled (1) as determined by the strapping of A10#. The value in this bit is written on the deassertion of RESET#; the bit is set to 1 when the address bus signal is asserted.		
6:5	APIC Cluster ID (R) Contains the logical APIC cluster ID value as set by the strapping of A12# and A11#. The logical cluster ID value is written into the field on the deassertion of RESET#; the field is set to 1 when the address bus signal is asserted.		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
7	Bus Park Disable (R) Indicates whether bus park is enabled (0) or disabled (1) as set by the strapping of A15#. The value in this bit is written on the deassertion of RESET#; the bit is set to 1 when the address bus signal is asserted.		
11:8	Reserved.		
13:12	Agent ID (R) Contains the logical agent ID value as set by the strapping of BR[3:0]. The logical ID value is written into the field on the deassertion of RESET#; the field is set to 1 when the address bus signal is asserted.		
63:14	Reserved.		
Register Address: 2BH, 43	MSR_EBC_SOFT_POWERON		
Processor Soft Power-On Configuration (R/W) Enables and disables processor features.		0, 1, 2, 3, 4, 6	Shared
0	RCNT/SCNT On Request Encoding Enable (R/W) Controls the driving of RCNT/SCNT on the request encoding. Set to enable (1); clear to disabled (0, default).		
1	Data Error Checking Disable (R/W) Set to disable system data bus parity checking; clear to enable parity checking.		
2	Response Error Checking Disable (R/W) Set to disable (default); clear to enable.		
3	Address/Request Error Checking Disable (R/W) Set to disable (default); clear to enable.		
4	Initiator MCERR# Disable (R/W) Set to disable MCERR# driving for initiator bus requests (default); clear to enable.		
5	Internal MCERR# Disable (R/W) Set to disable MCERR# driving for initiator internal errors (default); clear to enable.		
6	BINIT# Driver Disable (R/W) Set to disable BINIT# driver (default); clear to enable driver.		
63:7	Reserved.		
Register Address: 2CH, 44	MSR_EBC_FREQUENCY_ID		
Processor Frequency Configuration The bit field layout of this MSR varies according to the MODEL value in the CPUID version information. The following bit field layout applies to Pentium 4 and Xeon Processors with MODEL encoding equal or greater than 2. (R) The field Indicates the current processor frequency configuration.		2,3, 4, 6	Shared
15:0	Reserved.		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
18:16	Scalable Bus Speed (R/W) Indicates the intended scalable bus speed: <u>Encoding Scalable Bus Speed</u> 000B 100 MHz (Model 2) 000B 266 MHz (Model 3 or 4) 001B 133 MHz 010B 200 MHz 011B 166 MHz 100B 333 MHz (Model 6)		
	133.33 MHz should be utilized if performing calculation with System Bus Speed when encoding is 001B. 166.67 MHz should be utilized if performing calculation with System Bus Speed when encoding is 011B.		
	266.67 MHz should be utilized if performing calculation with System Bus Speed when encoding is 000B and model encoding = 3 or 4. 333.33 MHz should be utilized if performing calculation with System Bus Speed when encoding is 100B and model encoding = 6. All other values are reserved.		
23:19	Reserved.		
31:24	Core Clock Frequency to System Bus Frequency Ratio (R) The processor core clock frequency to system bus frequency ratio observed at the deassertion of the reset pin.		
63:32	Reserved.		
Register Address: 2CH, 44	MSR_EBC_FREQUENCY_ID		
Processor Frequency Configuration (R) The bit field layout of this MSR varies according to the MODEL value of the CPUID version information. This bit field layout applies to Pentium 4 and Xeon Processors with MODEL encoding less than 2. Indicates current processor frequency configuration.		0, 1	Shared
20:0	Reserved.		
23:21	Scalable Bus Speed (R/W) Indicates the intended scalable bus speed: <u>Encoding Scalable Bus Speed</u> 000B 100 MHz All others values reserved.		
63:24	Reserved.		
Register Address: 3AH, 58	IA32_FEATURE_CONTROL		
Control Features in IA-32 Processor (R/W) See Table 2-2. (If CPUID.01H:ECX.[bit 5])		3, 4, 6	Unique
Register Address: 79H, 121	IA32_BIOS_UPDT_TRIG		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
BIOS Update Trigger Register (W) See Table 2-2.		0, 1, 2, 3, 4, 6	Shared
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID		
BIOS Update Signature ID (R/W) See Table 2-2.		0, 1, 2, 3, 4, 6	Unique
Register Address: 9BH, 155	IA32_SMM_MONITOR_CTL		
SMM Monitor Configuration (R/W) See Table 2-2.		3, 4, 6	Unique
Register Address: FEH, 254	IA32_MTRRCAP		
MTRR Information See Section 13.11.1, "MTRR Feature Identification."		0, 1, 2, 3, 4, 6	Unique
Register Address: 174H, 372	IA32_SYSENTER_CS		
CS Register Target for CPL 0 Code (R/W) See Table 2-2 and Section 6.8.7, "Performing Fast Calls to System Procedures with the SYSENTER and SYSEXIT Instructions."		0, 1, 2, 3, 4, 6	Unique
Register Address: 175H, 373	IA32_SYSENTER_ESP		
Stack Pointer for CPL 0 Stack (R/W) See Table 2-2 and Section 6.8.7, "Performing Fast Calls to System Procedures with the SYSENTER and SYSEXIT Instructions."		0, 1, 2, 3, 4, 6	Unique
Register Address: 176H, 374	IA32_SYSENTER_EIP		
CPL 0 Code Entry Point (R/W) See Table 2-2 and Section 6.8.7, "Performing Fast Calls to System Procedures with the SYSENTER and SYSEXIT Instructions."		0, 1, 2, 3, 4, 6	Unique
Register Address: 179H, 377	IA32_MCG_CAP		
Machine Check Capabilities (R) See Table 2-2 and Section 17.3.1.1, "IA32_MCG_CAP MSR."		0, 1, 2, 3, 4, 6	Unique
Register Address: 17AH, 378	IA32_MCG_STATUS		
Machine Check Status (R) See Table 2-2 and Section 17.3.1.2, "IA32_MCG_STATUS MSR."		0, 1, 2, 3, 4, 6	Unique
Register Address: 17BH, 379	IA32_MCG_CTL		
Machine Check Feature Enable (R/W) See Table 2-2 and Section 17.3.1.3, "IA32_MCG_CTL MSR."			
Register Address: 180H, 384	MSR_MCG_RAX		
Machine Check EAX/RAX Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 181H, 385	MSR_MCG_RBX		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Machine Check EBX/RBX Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 182H, 386	MSR_MCG_RCX		
Machine Check ECX/RX Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 183H, 387	MSR_MCG_RDX		
Machine Check EDX/RDX Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 184H, 388	MSR_MCG_RSI		
Machine Check ESI/RSI Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 185H, 389	MSR_MCG_RDI		
Machine Check EDI/RDI Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 186H, 390	MSR_MCG_RBP		
Machine Check EBP/RBP Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 187H, 391	MSR_MCG_RSP		
Machine Check ESP/RSP Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 188H, 392	MSR_MCG_RFLAGS		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Machine Check EFLAGS/RFLAG Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 189H, 393	MSR_MCG_RIP		
Machine Check EIP/RIP Save State See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Contains register state at time of machine check error. When in non-64-bit modes at the time of the error, bits 63-32 do not contain valid data.		
Register Address: 18AH, 394	MSR_MCG_MISC		
Machine Check Miscellaneous See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
0	DS When set, the bit indicates that a page assist or page fault occurred during DS normal operation. The processors response is to shut down. The bit is used as an aid for debugging DS handling code. It is the responsibility of the user (BIOS or operating system) to clear this bit for normal operation.		
63:1	Reserved.		
Register Address: 18BH–18FH, 395–399	MSR_MCG_RESERVED1–MSR_MCG_RESERVED5		
Reserved.			
Register Address: 190H, 400	MSR_MCG_R8		
Machine Check R8 See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Registers R8-15 (and the associated state-save MSRs) exist only in Intel 64 processors. These registers contain valid information only when the processor is operating in 64-bit mode at the time of the error.		
Register Address: 191H, 401	MSR_MCG_R9		
Machine Check R9D/R9 See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Registers R8-15 (and the associated state-save MSRs) exist only in Intel 64 processors. These registers contain valid information only when the processor is operating in 64-bit mode at the time of the error.		
Register Address: 192H, 402	MSR_MCG_R10		
Machine Check R10 See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
63:0	Registers R8-15 (and the associated state-save MSRs) exist only in Intel 64 processors. These registers contain valid information only when the processor is operating in 64-bit mode at the time of the error.		
Register Address: 193H, 403	MSR_MCG_R11		
Machine Check R11 See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Registers R8-15 (and the associated state-save MSRs) exist only in Intel 64 processors. These registers contain valid information only when the processor is operating in 64-bit mode at the time of the error.		
Register Address: 194H, 404	MSR_MCG_R12		
Machine Check R12 See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Registers R8-15 (and the associated state-save MSRs) exist only in Intel 64 processors. These registers contain valid information only when the processor is operating in 64-bit mode at the time of the error.		
Register Address: 195H, 405	MSR_MCG_R13		
Machine Check R13 See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Registers R8-15 (and the associated state-save MSRs) exist only in Intel 64 processors. These registers contain valid information only when the processor is operating in 64-bit mode at the time of the error.		
Register Address: 196H, 406	MSR_MCG_R14		
Machine Check R14 See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Registers R8-15 (and the associated state-save MSRs) exist only in Intel 64 processors. These registers contain valid information only when the processor is operating in 64-bit mode at the time of the error.		
Register Address: 197H, 407	MSR_MCG_R15		
Machine Check R15 See Section 17.3.2.6, "IA32_MCG Extended Machine Check State MSRs."		0, 1, 2, 3, 4, 6	Unique
63:0	Registers R8-15 (and the associated state-save MSRs) exist only in Intel 64 processors. These registers contain valid information only when the processor is operating in 64-bit mode at the time of the error.		
Register Address: 198H, 408	IA32_PERF_STATUS		
See Table 2-2. See Section 16.1, "Enhanced Intel Speedstep® Technology."		3, 4, 6	Unique
Register Address: 199H, 409	IA32_PERF_CTL		
See Table 2-2. See Section 16.1, "Enhanced Intel Speedstep® Technology."		3, 4, 6	Unique

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Register Address: 19AH, 410	IA32_CLOCK_MODULATION		
Thermal Monitor Control (R/W) See Table 2-2 and Section 16.8.3, “Software Controlled Clock Modulation.”		0, 1, 2, 3, 4, 6	Unique
Register Address: 19BH, 411	IA32_THERM_INTERRUPT		
Thermal Interrupt Control (R/W) See Section 16.8.2, “Thermal Monitor,” and Table 2-2.		0, 1, 2, 3, 4, 6	Unique
Register Address: 19CH, 412	IA32_THERM_STATUS		
Thermal Monitor Status (R/W) See Section 16.8.2, “Thermal Monitor,” and Table 2-2.		0, 1, 2, 3, 4, 6	Shared
Register Address: 19DH, 413	MSR_THERM2_CTL		
Thermal Monitor 2 Control			
For Family F, Model 3 processors: When read, specifies the value of the target TM2 transition last written. When set, it sets the next target value for TM2 transition.		3	Shared
For Family F, Model 4 and Model 6 processors: When read, specifies the value of the target TM2 transition last written. Writes may cause #GP exceptions.		4, 6	Shared
Register Address: 1A0H, 416	IA32_MISC_ENABLE		
Enable Miscellaneous Processor Features (R/W)		0, 1, 2, 3, 4, 6	Shared
0	Fast-Strings Enable. See Table 2-2.		
1	Reserved.		
2	x87 FPU Fopcode Compatibility Mode Enable		
3	Thermal Monitor 1 Enable See Section 16.8.2, “Thermal Monitor,” and Table 2-2.		
4	Split-Lock Disable When set, the bit causes an #AC exception to be issued instead of a split-lock cycle. Operating systems that set this bit must align system structures to avoid split-lock scenarios. When the bit is clear (default), normal split-locks are issued to the bus. This debug feature is specific to the Pentium 4 processor.		
5	Reserved.		
6	Third-Level Cache Disable (R/W) When set, the third-level cache is disabled; when clear (default) the third-level cache is enabled. This flag is reserved for processors that do not have a third-level cache. Note that the bit controls only the third-level cache; and only if overall caching is enabled through the CD flag of control register CR0, the page-level cache controls, and/or the MTRRs. See Section 13.5.4, “Disabling and Enabling the L3 Cache.”		
7	Performance Monitoring Available (R) See Table 2-2.		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
8	Suppress Lock Enable When set, assertion of LOCK on the bus is suppressed during a Split Lock access. When clear (default), LOCK is not suppressed.		
9	Prefetch Queue Disable When set, disables the prefetch queue. When clear (default), enables the prefetch queue.		
10	FERR# Interrupt Reporting Enable (R/W) When set, interrupt reporting through the FERR# pin is enabled; when clear, this interrupt reporting function is disabled. When this flag is set and the processor is in the stop-clock state (STPCLK# is asserted), asserting the FERR# pin signals to the processor that an interrupt (such as, INIT#, BINIT#, INTR, NMI, SMI#, or RESET#) is pending and that the processor should return to normal operation to handle the interrupt. This flag does not affect the normal operation of the FERR# pin (to indicate an unmasked floating-point error) when the STPCLK# pin is not asserted.		
11	Branch Trace Storage Unavailable (BTS_UNAVAILABLE) (R) See Table 2-2. When set, the processor does not support branch trace storage (BTS); when clear, BTS is supported.		
12	PEBS_UNAVAILABLE: Processor Event Based Sampling Unavailable (R) See Table 2-2. When set, the processor does not support processor event-based sampling (PEBS); when clear, PEBS is supported.		
13	TM2 Enable (R/W) When this bit is set (1) and the thermal sensor indicates that the die temperature is at the pre-determined threshold, the Thermal Monitor 2 mechanism is engaged. TM2 will reduce the bus to core ratio and voltage according to the value last written to MSR_THERM2_CTL bits 15:0. When this bit is clear (0, default), the processor does not change the VID signals or the bus to core ratio when the processor enters a thermal managed state. If the TM2 feature flag (ECX[8]) is not set to 1 after executing CPUID with EAX = 1, then this feature is not supported and BIOS must not alter the contents of this bit location. The processor is operating out of spec if both this bit and the TM1 bit are set to disabled states.	3	
17:14	Reserved.		
18	ENABLE MONITOR FSM (R/W) See Table 2-2.	3, 4, 6	

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
19	Adjacent Cache Line Prefetch Disable (R/W) When set to 1, the processor fetches the cache line of the 128-byte sector containing currently required data. When set to 0, the processor fetches both cache lines in the sector. Single processor platforms should not set this bit. Server platforms should set or clear this bit based on platform performance observed in validation and testing. BIOS may contain a setup option that controls the setting of this bit.		
21:20	Reserved.		
22	Limit CPUID MAXVAL (R/W) See Table 2-2. Setting this can cause unexpected behavior to software that depends on the availability of CPUID leaves greater than 3.	3, 4, 6	
23	xTPR Message Disable (R/W) See Table 2-2.		Shared
24	L1 Data Cache Context Mode (R/W) When set, the L1 data cache is placed in shared mode; when clear (default), the cache is placed in adaptive mode. This bit is only enabled for IA-32 processors that support Intel Hyper-Threading Technology. See Section 13.5.6, "L1 Data Cache Context Mode." When L1 is running in adaptive mode and CR3s are identical, data in L1 is shared across logical processors. Otherwise, L1 is not shared and cache use is competitive. If the Context ID feature flag (ECX[10]) is set to 0 after executing CPUID with EAX = 1, the ability to switch modes is not supported. BIOS must not alter the contents of IA32_MISC_ENABLE[24].		
33:25	Reserved.		
34	XD Bit Disable (R/W) See Table 2-3.		Unique
63:35	Reserved.		
Register Address: 1A1H, 417	MSR_PLATFORM_BRV		
Platform Feature Requirements (R)		3, 4, 6	Shared
17:0	Reserved.		
18	PLATFORM Requirements When set to 1, indicates the processor has specific platform requirements. The details of the platform requirements are listed in the respective data sheets of the processor.		
63:19	Reserved.		
Register Address: 1D7H, 471	MSR_LER_FROM_LIP		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Last Exception Record From Linear IP (R) Contains a pointer to the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled. See Section 19.13.3, "Last Exception Records."		0, 1, 2, 3, 4, 6	Unique
31:0	From Linear IP Linear address of the last branch instruction.		
63:32	Reserved.		
Register Address: 1D7H, 471	MSR_LER_FROM_LIP		
63:0	From Linear IP Linear address of the last branch instruction (If IA-32e mode is active).		Unique
Register Address: 1D8H, 472	MSR_LER_TO_LIP		
Last Exception Record To Linear IP (R) This area contains a pointer to the target of the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled. See Section 19.13.3, "Last Exception Records."		0, 1, 2, 3, 4, 6	Unique
31:0	From Linear IP Linear address of the target of the last branch instruction.		
63:32	Reserved.		
Register Address: 1D8H, 472	MSR_LER_TO_LIP		
63:0	From Linear IP Linear address of the target of the last branch instruction (If IA-32e mode is active).		Unique
Register Address: 1D9H, 473	MSR_DEBUGCTLA		
Debug Control (R/W) Controls how several debug features are used. Bit definitions are discussed in the referenced section. See Section 19.13.1, "MSR_DEBUGCTLA MSR."		0, 1, 2, 3, 4, 6	Unique
Register Address: 1DAH, 474	MSR_LASTBRANCH_TOS		
Last Branch Record Stack TOS (R/O) Contains an index (0-3 or 0-15) that points to the top of the last branch record stack (that is, that points the index of the MSR containing the most recent branch record). See Section 19.13.2, "LBR Stack for Processors Based on Intel NetBurst® Microarchitecture," and addresses 1DBH-1DEH and 680H-68FH.		0, 1, 2, 3, 4, 6	Unique
Register Address: 1DBH, 475	MSR_LASTBRANCH_0		
Last Branch Record 0 (R/O) One of four last branch record registers on the last branch record stack. It contains pointers to the source and destination instruction for one of the last four branches, exceptions, or interrupts that the processor took. MSR_LASTBRANCH_0 through MSR_LASTBRANCH_3 at 1DBH-1DEH are available only on family 0FH, models 0H-02H. They have been replaced by the MSRs at 680H-68FH and 6C0H-6CFH. See Section 19.12, "Last Branch, Call Stack, Interrupt, and Exception Recording for Processors based on Skylake Microarchitecture."		0, 1, 2	Unique

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Register Address: 1DCH, 476	MSR_LASTBRANCH_1		
Last Branch Record 1 See description of the MSR_LASTBRANCH_0 MSR at 1DBH.		0, 1, 2	Unique
Register Address: 1DDH, 477	MSR_LASTBRANCH_2		
Last Branch Record 2 See description of the MSR_LASTBRANCH_0 MSR at 1DBH.		0, 1, 2	Unique
Register Address: 1DEH, 478	MSR_LASTBRANCH_3		
Last Branch Record 3 See description of the MSR_LASTBRANCH_0 MSR at 1DBH.		0, 1, 2	Unique
Register Address: 200H, 512	IA32_MTRR_PHYSBASE0		
Variable Range Base MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 201H, 513	IA32_MTRR_PHYSMASK0		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 202H, 514	IA32_MTRR_PHYSBASE1		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 203H, 515	IA32_MTRR_PHYSMASK1		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 204H, 516	IA32_MTRR_PHYSBASE2		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 205H, 517	IA32_MTRR_PHYSMASK2		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 206H, 518	IA32_MTRR_PHYSBASE3		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 207H, 519	IA32_MTRR_PHYSMASK3		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 208H, 520	IA32_MTRR_PHYSBASE4		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 209H, 521	IA32_MTRR_PHYSMASK4		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 20AH, 522	IA32_MTRR_PHYSBASE5		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 20BH, 523	IA32_MTRR_PHYSMASK5		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 20CH, 524	IA32_MTRR_PHYSBASE6		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 20DH, 525	IA32_MTRR_PHYSMASK6		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 20EH, 526	IA32_MTRR_PHYSBASE7		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 20FH, 527	IA32_MTRR_PHYSMASK7		
Variable Range Mask MTRR See Section 13.11.2.3, "Variable Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 250H, 592	IA32_MTRR_FIX64K_00000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 258H, 600	IA32_MTRR_FIX16K_80000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 259H, 601	IA32_MTRR_FIX16K_A0000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 268H, 616	IA32_MTRR_FIX4K_C0000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 269H, 617	IA32_MTRR_FIX4K_C8000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 26AH, 618	IA32_MTRR_FIX4K_D0000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Register Address: 26BH, 619	IA32_MTRR_FIX4K_D8000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 26CH, 620	IA32_MTRR_FIX4K_E0000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 26DH, 621	IA32_MTRR_FIX4K_E8000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 26EH, 622	IA32_MTRR_FIX4K_F0000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 26FH, 623	IA32_MTRR_FIX4K_F8000		
Fixed Range MTRR See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 277H, 631	IA32_PAT		
Page Attribute Table See Section 13.11.2.2, "Fixed Range MTRRs."		0, 1, 2, 3, 4, 6	Unique
Register Address: 2FFH, 767	IA32_MTRR_DEF_TYPE		
Default Memory Types (R/W) See Table 2-2 and Section 13.11.2.1, "IA32_MTRR_DEF_TYPE MSR."		0, 1, 2, 3, 4, 6	Shared
Register Address: 300H, 768	MSR_BPU_COUNTER0		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 301H, 769	MSR_BPU_COUNTER1		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 302H, 770	MSR_BPU_COUNTER2		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 303H, 771	MSR_BPU_COUNTER3		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 304H, 772	MSR_MS_COUNTER0		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 305H, 773	MSR_MS_COUNTER1		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 306H, 774	MSR_MS_COUNTER2		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 307H, 775	MSR_MS_COUNTER3		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 308H, 776	MSR_FLAME_COUNTER0		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 309H, 777	MSR_FLAME_COUNTER1		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 30AH, 778	MSR_FLAME_COUNTER2		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 30BH, 779	MSR_FLAME_COUNTER3		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 30CH, 780	MSR_IQ_COUNTER0		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 30DH, 781	MSR_IQ_COUNTER1		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 30EH, 782	MSR_IQ_COUNTER2		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 30FH, 783	MSR_IQ_COUNTER3		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 310H, 784	MSR_IQ_COUNTER4		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 311H, 785	MSR_IQ_COUNTER5		
See Section 21.6.3.2, "Performance Counters."		0, 1, 2, 3, 4, 6	Shared
Register Address: 360H, 864	MSR_BPU_CCCR0		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 361H, 865	MSR_BPU_CCCR1		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 362H, 866	MSR_BPU_CCCR2		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 363H, 867	MSR_BPU_CCCR3		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 364H, 868	MSR_MS_CCCR0		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 365H, 869	MSR_MS_CCCR1		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 366H, 870	MSR_MS_CCCR2		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 367H, 871	MSR_MS_CCCR3		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 368H, 872	MSR_FLAME_CCCR0		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Register Address: 369H, 873	MSR_FLAME_CCCR1		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 36AH, 874	MSR_FLAME_CCCR2		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 36BH, 875	MSR_FLAME_CCCR3		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 36CH, 876	MSR_IQ_CCCR0		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 36DH, 877	MSR_IQ_CCCR1		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 36EH, 878	MSR_IQ_CCCR2		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 36FH, 879	MSR_IQ_CCCR3		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 370H, 880	MSR_IQ_CCCR4		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 371H, 881	MSR_IQ_CCCR5		
See Section 21.6.3.3, "CCCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A0H, 928	MSR_BSU_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A1H, 929	MSR_BSU_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A2H, 930	MSR_FSB_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A3H, 931	MSR_FSB_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A4H, 932	MSR_FIRM_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A5H, 933	MSR_FIRM_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A6H, 934	MSR_FLAME_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A7H, 935	MSR_FLAME_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A8H, 936	MSR_DAC_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3A9H, 937	MSR_DAC_ESCR1		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3AAH, 938	MSR_MOB_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3ABH, 939	MSR_MOB_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3ACH, 940	MSR_PMH_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3ADH, 941	MSR_PMH_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3AEH, 942	MSR_SAAT_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3AFH, 943	MSR_SAAT_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3BOH, 944	MSR_U2L_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B1H, 945	MSR_U2L_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B2H, 946	MSR_BPU_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B3H, 947	MSR_BPU_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B4H, 948	MSR_IS_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B5H, 949	MSR_IS_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B6H, 950	MSR_ITLB_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B7H, 951	MSR_ITLB_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B8H, 952	MSR_CRU_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3B9H, 953	MSR_CRU_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3BAH, 954	MSR_IQ_ESCR0		
See Section 21.6.3.1, "ESCR MSRs." This MSR is not available on later processors. It is only available on processor family 0FH, models 01H-02H.		0, 1, 2	Shared

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Register Address: 3BBH, 955	MSR_IQ_ESCR1		
See Section 21.6.3.1, "ESCR MSRs." This MSR is not available on later processors. It is only available on processor family 0FH, models 01H-02H.		0, 1, 2	Shared
Register Address: 3BCH, 956	MSR_RAT_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3BDH, 957	MSR_RAT_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3BEH, 958	MSR_SSU_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3C0H, 960	MSR_MS_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3C1H, 961	MSR_MS_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3C2H, 962	MSR_TBPU_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3C3H, 963	MSR_TBPU_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3C4H, 964	MSR_TC_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3C5H, 965	MSR_TC_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3C8H, 968	MSR_IX_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3C9H, 969	MSR_IX_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3CAH, 970	MSR_ALF_ESCR0		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3CBH, 971	MSR_ALF_ESCR1		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3CCH, 972	MSR_CRU_ESCR2		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3CDH, 973	MSR_CRU_ESCR3		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3E0H, 992	MSR_CRU_ESCR4		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3E1H, 993	MSR_CRU_ESCR5		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3F0H, 1008	MSR_TC_PRECISE_EVENT		
See Section 21.6.3.1, "ESCR MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 3F1H, 1009	IA32_PEBS_ENABLE (MSR_PEBS_ENABLE)		
Processor Event Based Sampling (PEBS) (R/W) Controls the enabling of processor event sampling and replay tagging.		0, 1, 2, 3, 4, 6	Shared
12:0	See https://perfmon-events.intel.com/ .		
23:13	Reserved.		
24	UOP Tag Enables replay tagging when set.		
25	ENABLE_PEBS_MY_THR (R/W) Enables PEBS for the target logical processor when set; disables PEBS when clear (default). See Section 21.6.4.3, "IA32_PEBS_ENABLE MSR," for an explanation of the target logical processor. This bit is called ENABLE_PEBS in IA-32 processors that do not support Intel Hyper-Threading Technology.		
26	ENABLE_PEBS_OTH_THR (R/W) Enables PEBS for the target logical processor when set; disables PEBS when clear (default). See Section 21.6.4.3, "IA32_PEBS_ENABLE MSR," for an explanation of the target logical processor. This bit is reserved for IA-32 processors that do not support Intel Hyper-Threading Technology.		
63:27	Reserved.		
Register Address: 3F2H, 1010	MSR_PEBS_MATRIX_VERT		
See https://perfmon-events.intel.com/ .		0, 1, 2, 3, 4, 6	Shared
Register Address: 400H, 1024	IA32_MCO_CTL		
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 401H, 1025	IA32_MCO_STATUS		
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 402H, 1026	IA32_MCO_ADDR		
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MCO_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MCO_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		0, 1, 2, 3, 4, 6	Shared
Register Address: 403H, 1027	IA32_MCO_MISC		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
See Section 17.3.2.4, "IA32_MCi_MISC MSRs." The IA32_MC0_MISC MSR is either not implemented or does not contain additional information if the MISCV flag in the IA32_MC0_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		0, 1, 2, 3, 4, 6	Shared
Register Address: 404H, 1028	IA32_MC1_CTL		
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 405H, 1029	IA32_MC1_STATUS		
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 406H, 1030	IA32_MC1_ADDR		
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The IA32_MC1_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC1_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		0, 1, 2, 3, 4, 6	Shared
Register Address: 407H, 1031	IA32_MC1_MISC		
See Section 17.3.2.4, "IA32_MCi_MISC MSRs." The IA32_MC1_MISC MSR is either not implemented or does not contain additional information if the MISCV flag in the IA32_MC1_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.			Shared
Register Address: 408H, 1032	IA32_MC2_CTL		
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 409H, 1033	IA32_MC2_STATUS		
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 40AH, 1034	IA32_MC2_ADDR		
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.			
Register Address: 40BH, 1035	IA32_MC2_MISC		
See Section 17.3.2.4, "IA32_MCi_MISC MSRs." The IA32_MC2_MISC MSR is either not implemented or does not contain additional information if the MISCV flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.			
Register Address: 40CH, 1036	IA32_MC3_CTL		
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 40DH, 1037	IA32_MC3_STATUS		
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 40EH, 1038	IA32_MC3_ADDR		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The IA32_MC3_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC3_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		0, 1, 2, 3, 4, 6	Shared
Register Address: 40FH, 1039	IA32_MC3_MISC		
See Section 17.3.2.4, "IA32_MCi_MISC MSRs." The IA32_MC3_MISC MSR is either not implemented or does not contain additional information if the MISCV flag in the IA32_MC3_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		0, 1, 2, 3, 4, 6	Shared
Register Address: 410H, 1040	IA32_MC4_CTL		
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 411H, 1041	IA32_MC4_STATUS		
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		0, 1, 2, 3, 4, 6	Shared
Register Address: 412H, 1042	IA32_MC4_ADDR		
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.			
Register Address: 413H, 1043	IA32_MC4_MISC		
See Section 17.3.2.4, "IA32_MCi_MISC MSRs." The IA32_MC2_MISC MSR is either not implemented or does not contain additional information if the MISCV flag in the IA32_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.			
Register Address: 480H, 1152	IA32_VMX_BASIC		
Reporting Register of Basic VMX Capabilities (R/O) See Table 2-2 and Appendix A.1, "Basic VMX Information."		3, 4, 6	Unique
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL		
Capability Reporting Register of Pin-Based VM-Execution Controls (R/O) See Table 2-2 and Appendix A.3, "VM-Execution Controls."		3, 4, 6	Unique
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL		
Capability Reporting Register of Primary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls," and Table 2-2.		3, 4, 6	Unique
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL		
Capability Reporting Register of VM-Exit Controls (R/O) See Appendix A.4, "VM-Exit Controls," and Table 2-2.		3, 4, 6	Unique
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTL		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Capability Reporting Register of VM-Entry Controls (R/O) See Appendix A.5, "VM-Entry Controls," and Table 2-2.		3, 4, 6	Unique
Register Address: 485H, 1157	IA32_VMX_MISC		
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Appendix A.6, "Miscellaneous Data," and Table 2-2.		3, 4, 6	Unique
Register Address: 486H, 1158	IA32_VMX_CR0_FIXED0		
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Appendix A.7, "VMX-Fixed Bits in CR0," and Table 2-2.		3, 4, 6	Unique
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1		
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Appendix A.7, "VMX-Fixed Bits in CR0," and Table 2-2.		3, 4, 6	Unique
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0		
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Appendix A.8, "VMX-Fixed Bits in CR4," and Table 2-2.		3, 4, 6	Unique
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1		
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Appendix A.8, "VMX-Fixed Bits in CR4," and Table 2-2.		3, 4, 6	Unique
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM		
Capability Reporting Register of VMCS Field Enumeration (R/O) See Appendix A.9, "VMCS Enumeration," and Table 2-2.		3, 4, 6	Unique
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL2		
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls," and Table 2-2.		3, 4, 6	Unique
Register Address: 600H, 1536	IA32_DS_AREA		
DS Save Area (R/W) See Table 2-2 and Section 21.6.3.4, "Debug Store (DS) Mechanism."		0, 1, 2, 3, 4, 6	Unique
Register Address: 680H, 1664	MSR_LASTBRANCH_0_FROM_IP		
Last Branch Record 0 (R/W) One of 16 pairs of last branch record registers on the last branch record stack (680H-68FH). This part of the stack contains pointers to the source instruction for one of the last 16 branches, exceptions, or interrupts taken by the processor. The MSRs at 680H-68FH, 6C0H-6CfH are not available in processor releases before family 0FH, model 03H. These MSRs replace MSRs previously located at 1DBH-1DEH, which performed the same function for early releases. See Section 19.12, "Last Branch, Call Stack, Interrupt, and Exception Recording for Processors based on Skylake Microarchitecture."		3, 4, 6	Unique
Register Address: 681H, 1665	MSR_LASTBRANCH_1_FROM_IP		
Last Branch Record 1 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 682H, 1666	MSR_LASTBRANCH_2_FROM_IP		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Last Branch Record 2 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 683H, 1667	MSR_LASTBRANCH_3_FROM_IP		
Last Branch Record 3 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 684H, 1668	MSR_LASTBRANCH_4_FROM_IP		
Last Branch Record 4 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 685H, 1669	MSR_LASTBRANCH_5_FROM_IP		
Last Branch Record 5 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 686H, 1670	MSR_LASTBRANCH_6_FROM_IP		
Last Branch Record 6 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 687H, 1671	MSR_LASTBRANCH_7_FROM_IP		
Last Branch Record 7 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 688H, 1672	MSR_LASTBRANCH_8_FROM_IP		
Last Branch Record 8 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 689H, 1673	MSR_LASTBRANCH_9_FROM_IP		
Last Branch Record 9 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 68AH, 1674	MSR_LASTBRANCH_10_FROM_IP		
Last Branch Record 10 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 68BH, 1675	MSR_LASTBRANCH_11_FROM_IP		
Last Branch Record 11 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 68CH, 1676	MSR_LASTBRANCH_12_FROM_IP		
Last Branch Record 12 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 68DH, 1677	MSR_LASTBRANCH_13_FROM_IP		
Last Branch Record 13 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 68EH, 1678	MSR_LASTBRANCH_14_FROM_IP		
Last Branch Record 14 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Register Address: 68FH, 1679	MSR_LASTBRANCH_15_FROM_IP		
Last Branch Record 15 See description of MSR_LASTBRANCH_0 at 680H.		3, 4, 6	Unique
Register Address: 6C0H, 1728	MSR_LASTBRANCH_0_TO_IP		
Last Branch Record 0 (R/W) One of 16 pairs of last branch record registers on the last branch record stack (6C0H-6CFH). This part of the stack contains pointers to the destination instruction for one of the last 16 branches, exceptions, or interrupts that the processor took. See Section 19.12, "Last Branch, Call Stack, Interrupt, and Exception Recording for Processors based on Skylake Microarchitecture."		3, 4, 6	Unique
Register Address: 6C1H, 1729	MSR_LASTBRANCH_1_TO_IP		
Last Branch Record 1 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6C2H, 1730	MSR_LASTBRANCH_2_TO_IP		
Last Branch Record 2 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6C3H, 1731	MSR_LASTBRANCH_3_TO_IP		
Last Branch Record 3 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6C4H, 1732	MSR_LASTBRANCH_4_TO_IP		
Last Branch Record 4 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6C5H, 1733	MSR_LASTBRANCH_5_TO_IP		
Last Branch Record 5 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6C6H, 1734	MSR_LASTBRANCH_6_TO_IP		
Last Branch Record 6 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6C7H, 1735	MSR_LASTBRANCH_7_TO_IP		
Last Branch Record 7 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6C8H, 1736	MSR_LASTBRANCH_8_TO_IP		
Last Branch Record 8 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6C9H, 1737	MSR_LASTBRANCH_9_TO_IP		
Last Branch Record 9 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6CAH, 1738	MSR_LASTBRANCH_10_TO_IP		

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
Last Branch Record 10 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6CBH, 1739	MSR_LASTBRANCH_11_TO_IP		
Last Branch Record 11 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6CCH, 1740	MSR_LASTBRANCH_12_TO_IP		
Last Branch Record 12 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6CDH, 1741	MSR_LASTBRANCH_13_TO_IP		
Last Branch Record 13 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6CEH, 1742	MSR_LASTBRANCH_14_TO_IP		
Last Branch Record 14 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: 6CFH, 1743	MSR_LASTBRANCH_15_TO_IP		
Last Branch Record 15 See description of MSR_LASTBRANCH_0 at 6C0H.		3, 4, 6	Unique
Register Address: C000_0080H	IA32_EFER		
Extended Feature Enables See Table 2-2.		3, 4, 6	Unique
Register Address: C000_0081H	IA32_STAR		
System Call Target Address (R/W) See Table 2-2.		3, 4, 6	Unique
Register Address: C000_0082H	IA32_LSTAR		
IA-32e Mode System Call Target Address (R/W) See Table 2-2.		3, 4, 6	Unique
Register Address: C000_0084H	IA32_FMASK		
System Call Flag Mask (R/W) See Table 2-2.		3, 4, 6	Unique
Register Address: C000_0100H	IA32_FS_BASE		
Map of BASE Address of FS (R/W) See Table 2-2.		3, 4, 6	Unique
Register Address: C000_0101H	IA32_GS_BASE		
Map of BASE Address of GS (R/W) See Table 2-2.		3, 4, 6	Unique
Register Address: C000_0102H	IA32_KERNEL_GS_BASE		
Swap Target of BASE Address of GS (R/W) See Table 2-2.		3, 4, 6	Unique

Table 2-63. MSRs in the Pentium® 4 and Intel® Xeon® Processors (Contd.)

Register Address: Hex, Decimal	Register Name		
Register Information / Bit Fields	Bit Description	Model Availability	Shared/ Unique ¹
NOTES 1. For HT-enabled processors, there may be more than one logical processors per physical unit. If an MSR is Shared, this means that one MSR is shared between logical processors. If an MSR is unique, this means that each logical processor has its own MSR.			

2.19.1 MSRs Unique to Intel® Xeon® Processor MP with L3 Cache

The MSRs listed in Table 2-64 apply to Intel® Xeon® Processor MP with up to 8MB level three cache. These processors can be detected by enumerating the deterministic cache parameter leaf of CPUID instruction (with EAX = 4 as input) to detect the presence of the third level cache, and with CPUID reporting family encoding 0FH, model encoding 3 or 4 (see CPUID instruction for more details).

Table 2-64. MSRs Unique to 64-bit Intel® Xeon® Processor MP with Up to an 8 MB L3 Cache

Register Address: Hex	Register Name		
Register Information		Model Availability	Shared/ Unique
Register Address: 107CCH	MSR_IFSB_BUSQ0		
IFSB BUSQ Event Control and Counter Register (R/W) See Section 21.6.6, "Performance Monitoring on 64-bit Intel® Xeon® Processor MP with Up to 8-MByte L3 Cache."		3, 4	Shared
Register Address: 107CDH	MSR_IFSB_BUSQ1		
IFSB BUSQ Event Control and Counter Register (R/W)		3, 4	Shared
Register Address: 107CEH	MSR_IFSB_SNPQ0		
IFSB SNPQ Event Control and Counter Register (R/W) See Section 21.6.6, "Performance Monitoring on 64-bit Intel® Xeon® Processor MP with Up to 8-MByte L3 Cache."		3, 4	Shared
Register Address: 107CFH	MSR_IFSB_SNPQ1		
IFSB SNPQ Event Control and Counter Register (R/W)		3, 4	Shared
Register Address: 107D0H	MSR_EFSB_DRDY0		
EFSB DRDY Event Control and Counter Register (R/W) See Section 21.6.6, "Performance Monitoring on 64-bit Intel® Xeon® Processor MP with Up to 8-MByte L3 Cache."		3, 4	Shared
Register Address: 107D1H	MSR_EFSB_DRDY1		
EFSB DRDY Event Control and Counter Register (R/W)		3, 4	Shared
Register Address: 107D2H	MSR_IFSB_CTL6		
IFSB Latency Event Control Register (R/W) See Section 21.6.6, "Performance Monitoring on 64-bit Intel® Xeon® Processor MP with Up to 8-MByte L3 Cache."		3, 4	Shared
Register Address: 107D3H	MSR_IFSB_CNTR7		
IFSB Latency Event Counter Register (R/W) See Section 21.6.6, "Performance Monitoring on 64-bit Intel® Xeon® Processor MP with Up to 8-MByte L3 Cache."		3, 4	Shared

The MSRs listed in Table 2-65 apply to Intel® Xeon® Processor 7100 series. These processors can be detected by enumerating the deterministic cache parameter leaf of CPUID instruction (with EAX = 4 as input) to detect the

presence of the third level cache, and with CPUID reporting family encoding 0FH, model encoding 6 (See CPUID instruction for more details.). The performance monitoring MSRs listed in Table 2-65 are shared between logical processors in the same core, but are replicated for each core.

Table 2-65. MSRs Unique to Intel® Xeon® Processor 7100 Series

Register Address: Hex	Register Name		
Register Information		Model Availability	Shared/Unique
Register Address: 107CCH	MSR_EMON_L3_CTR_CTL0		
GBUSQ Event Control and Counter Register (R/W) See Section 21.6.6, "Performance Monitoring on 64-bit Intel® Xeon® Processor MP with Up to 8-MByte L3 Cache."		6	Shared
Register Address: 107CDH	MSR_EMON_L3_CTR_CTL1		
GBUSQ Event Control and Counter Register (R/W)		6	Shared
Register Address: 107CEH	MSR_EMON_L3_CTR_CTL2		
GSNPQ Event Control and Counter Register (R/W) See Section 21.6.6, "Performance Monitoring on 64-bit Intel® Xeon® Processor MP with Up to 8-MByte L3 Cache."		6	Shared
Register Address: 107CFH	MSR_EMON_L3_CTR_CTL3		
GSNPQ Event Control and Counter Register (R/W)		6	Shared
Register Address: 107D0H	MSR_EMON_L3_CTR_CTL4		
FSB Event Control and Counter Register (R/W) See Section 21.6.6, "Performance Monitoring on 64-bit Intel® Xeon® Processor MP with Up to 8-MByte L3 Cache."		6	Shared
Register Address: 107D1H	MSR_EMON_L3_CTR_CTL5		
FSB Event Control and Counter Register (R/W)		6	Shared
Register Address: 107D2H	MSR_EMON_L3_CTR_CTL6		
FSB Event Control and Counter Register (R/W)		6	Shared
Register Address: 107D3H	MSR_EMON_L3_CTR_CTL7		
FSB Event Control and Counter Register (R/W)		6	Shared

2.20 MSRS IN INTEL® CORE™ SOLO AND INTEL® CORE™ DUO PROCESSORS

Model-specific registers (MSRs) for Intel Core Solo, Intel Core Duo processors, and Dual-core Intel Xeon processor LV are listed in Table 2-66. The column "Shared/Unique" applies to Intel Core Duo processor. "Unique" means each processor core has a separate MSR, or a bit field in an MSR governs only a core independently. "Shared" means the MSR or the bit field in an MSR address governs the operation of both processor cores.

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/Unique
Register Address: 0H, 0	P5_MC_ADDR	
See Section 2.23, "MSRs in Pentium Processors," and Table 2-2.		Unique
Register Address: 1H, 1	P5_MC_TYPE	
See Section 2.23, "MSRs in Pentium Processors," and Table 2-2.		Unique

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Register Address: 6H, 6	IA32_MONITOR_FILTER_SIZE	
See Section 10.10.5, "Monitor/Mwait Address Range Determination," and Table 2-2.		Unique
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER	
See Section 19.17, "Time-Stamp Counter," and Table 2-2.		Unique
Register Address: 17H, 23	IA32_PLATFORM_ID	
Platform ID (R) See Table 2-2. The operating system can use this MSR to determine "slot" information for the processor and the proper microcode update to load.		Shared
Register Address: 1BH, 27	IA32_APIC_BASE	
See Section 12.4.4, "Local APIC Status and Location," and Table 2-2.		Unique
Register Address: 2AH, 42	MSR_EBL_CR_POWERON	
Processor Hard Power-On Configuration (R/W) Enables and disables processor features; (R) indicates current processor configuration.		Shared
0	Reserved.	
1	Data Error Checking Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processor implements R/W.	
2	Response Error Checking Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processor implements R/W.	
3	MCERR# Drive Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processor implements R/W.	
4	Address Parity Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processor implements R/W.	
6: 5	Reserved.	
7	BINIT# Driver Enable (R/W) 1 = Enabled; 0 = Disabled. Note: Not all processor implements R/W.	
8	Output Tri-state Enabled (R/O) 1 = Enabled; 0 = Disabled.	
9	Execute BIST (R/O) 1 = Enabled; 0 = Disabled.	
10	MCERR# Observation Enabled (R/O) 1 = Enabled; 0 = Disabled.	
11	Reserved.	
12	BINIT# Observation Enabled (R/O) 1 = Enabled; 0 = Disabled.	

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
13	Reserved	
14	1 MByte Power on Reset Vector (R/O) 1 = 1 MByte; 0 = 4 GBytes	
15	Reserved.	
17:16	APIC Cluster ID (R/O)	
18	System Bus Frequency (R/O) 0 = 100 MHz. 1 = Reserved.	
19	Reserved.	
21:20	Symmetric Arbitration ID (R/O)	
26:22	Clock Frequency Ratio (R/O)	
Register Address: 3AH, 58	IA32_FEATURE_CONTROL	
Control Features in IA-32 Processor (R/W) See Table 2-2.		Unique
Register Address: 40H, 64	MSR_LASTBRANCH_0	
Last Branch Record 0 (R/W) One of 8 last branch record registers on the last branch record stack: bits 31-0 hold the 'from' address and bits 63-32 hold the 'to' address. See also: - Last Branch Record Stack TOS at 1C9H. - Section 19.15, "Last Branch, Interrupt, and Exception Recording (Pentium M Processors)."		Unique
Register Address: 41H, 65	MSR_LASTBRANCH_1	
Last Branch Record 1 (R/W) See description of MSR_LASTBRANCH_0.		Unique
Register Address: 42H, 66	MSR_LASTBRANCH_2	
Last Branch Record 2 (R/W) See description of MSR_LASTBRANCH_0.		Unique
Register Address: 43H, 67	MSR_LASTBRANCH_3	
Last Branch Record 3 (R/W) See description of MSR_LASTBRANCH_0.		Unique
Register Address: 44H, 68	MSR_LASTBRANCH_4	
Last Branch Record 4 (R/W) See description of MSR_LASTBRANCH_0.		Unique
Register Address: 45H, 69	MSR_LASTBRANCH_5	
Last Branch Record 5 (R/W) See description of MSR_LASTBRANCH_0.		Unique
Register Address: 46H, 70	MSR_LASTBRANCH_6	
Last Branch Record 6 (R/W) See description of MSR_LASTBRANCH_0.		Unique
Register Address: 47H, 71	MSR_LASTBRANCH_7	

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Last Branch Record 7 (R/W) See description of MSR_LASTBRANCH_0.		Unique
Register Address: 79H, 121	IA32_BIOS_UPDT_TRIG	
BIOS Update Trigger Register (W) See Table 2-2.		Unique
Register Address: 8BH, 139	IA32_BIOS_SIGN_ID	
BIOS Update Signature ID (R/W) See Table 2-2.		Unique
Register Address: C1H, 193	IA32_PMC0	
Performance Counter Register See Table 2-2.		Unique
Register Address: C2H, 194	IA32_PMC1	
Performance Counter Register See Table 2-2.		Unique
Register Address: CDH, 205	MSR_FSB_FREQ	
Scaleable Bus Speed (R/O) This field indicates the scalable bus clock speed.		Shared
2:0	101B: 100 MHz (FSB 400) 001B: 133 MHz (FSB 533) 011B: 167 MHz (FSB 667) 133.33 MHz should be utilized if performing calculation with System Bus Speed when encoding is 101B. 166.67 MHz should be utilized if performing calculation with System Bus Speed when encoding is 001B.	
63:3	Reserved.	
Register Address: E7H, 231	IA32_MPERF	
Maximum Performance Frequency Clock Count (R/W) See Table 2-2.		Unique
Register Address: E8H, 232	IA32_APERF	
Actual Performance Frequency Clock Count (R/W) See Table 2-2.		Unique
Register Address: FEH, 254	IA32_MTRRCAP	
See Table 2-2.		Unique
Register Address: 11EH, 281	MSR_BBL_CR_CTL3	
Control Register 3 Used to configure the L2 Cache.		Shared
0	L2 Hardware Enabled (R/O) 1 = If the L2 is hardware-enabled. 0 = Indicates if the L2 is hardware-disabled.	

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
7:1	Reserved.	
8	L2 Enabled (R/W) 1 = L2 cache has been initialized. 0 = Disabled (default). Until this bit is set the processor will not respond to the WBINVD instruction or the assertion of the FLUSH# input.	
22:9	Reserved.	
23	L2 Not Present (R/O) 0 = L2 Present. 1 = L2 Not Present.	
63:24	Reserved.	
Register Address: 174H, 372	IA32_SYSENTER_CS	
See Table 2-2.		Unique
Register Address: 175H, 373	IA32_SYSENTER_ESP	
See Table 2-2.		Unique
Register Address: 176H, 374	IA32_SYSENTER_EIP	
See Table 2-2.		Unique
Register Address: 179H, 377	IA32_MCG_CAP	
See Table 2-2.		Unique
Register Address: 17AH, 378	IA32_MCG_STATUS	
Global Machine Check Status		Unique
0	RIPV When set, this bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) can be used to restart the program. If this bit is cleared, the program cannot be reliably restarted.	
1	EIPV When set, this bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) is directly associated with the error.	
2	MCIP When set, this bit indicates that a machine check has been generated. If a second machine check is detected while this bit is still set, the processor enters a shutdown state. Software should write this bit to 0 after processing a machine check exception.	
63:3	Reserved	
Register Address: 186H, 390	IA32_PERFVTSELO	
See Table 2-2.		Unique
Register Address: 187H, 391	IA32_PERFVTSEL1	
See Table 2-2.		Unique
Register Address: 198H, 408	IA32_PERF_STATUS	

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
See Table 2-2.		Shared
Register Address: 199H, 409	IA32_PERF_CTL	
See Table 2-2.		Unique
Register Address: 19AH, 410	IA32_CLOCK_MODULATION	
Clock Modulation (R/W) See Table 2-2.		Unique
Register Address: 19BH, 411	IA32_THERM_INTERRUPT	
Thermal Interrupt Control (R/W) See Table 2-2 and Section 16.8.2, "Thermal Monitor."		Unique
Register Address: 19CH, 412	IA32_THERM_STATUS	
Thermal Monitor Status (R/W) See Table 2-2 and Section 16.8.2, "Thermal Monitor".		Unique
Register Address: 19DH, 413	MSR_THERM2_CTL	
Thermal Monitor 2 Control		Unique
15:0	Reserved.	
16	TM_SELECT (R/W) Mode of automatic thermal monitor: 0 = Thermal Monitor 1 (thermally-initiated on-die modulation of the stop-clock duty cycle) 1 = Thermal Monitor 2 (thermally-initiated frequency transitions) If bit 3 of the IA32_MISC_ENABLE register is cleared, TM_SELECT has no effect. Neither TM1 nor TM2 will be enabled.	
63:16	Reserved.	
Register Address: 1A0H, 416	IA32_MISC_ENABLE	
Enable Miscellaneous Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.		
2:0	Reserved.	
3	Automatic Thermal Control Circuit Enable (R/W) See Table 2-2.	Unique
6:4	Reserved.	
7	Performance Monitoring Available (R) See Table 2-2.	Shared
9:8	Reserved.	
10	FERR# Multiplexing Enable (R/W) 1 = FERR# asserted by the processor to indicate a pending break event within the processor 0 = Indicates compatible FERR# signaling behavior This bit must be set to 1 to support XAPIC interrupt model usage.	Shared
11	Branch Trace Storage Unavailable (R/O) See Table 2-2.	Shared

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
12	Reserved.	
13	TM2 Enable (R/W) When this bit is set (1) and the thermal sensor indicates that the die temperature is at the pre-determined threshold, the Thermal Monitor 2 mechanism is engaged. TM2 will reduce the bus to core ratio and voltage according to the value last written to MSR_THERM2_CTL bits 15:0. When this bit is clear (0, default), the processor does not change the VID signals or the bus to core ratio when the processor enters a thermal managed state. If the TM2 feature flag (ECX[8]) is not set to 1 after executing CPUID with EAX = 1, then this feature is not supported and BIOS must not alter the contents of this bit location. The processor is operating out of spec if both this bit and the TM1 bit are set to disabled states.	Shared
15:14	Reserved.	
16	Enhanced Intel SpeedStep Technology Enable (R/W) 1 = Enhanced Intel SpeedStep Technology enabled	Shared
18	ENABLE MONITOR FSM (R/W) See Table 2-2.	Shared
19	Reserved.	
22	Limit CPUID Maxval (R/W) See Table 2-2. Setting this bit may cause behavior in software that depends on the availability of CPUID leaves greater than 2.	Shared
33:23	Reserved.	
34	XD Bit Disable (R/W) See Table 2-3.	Shared
63:35	Reserved.	
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS	
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-3) that points to the MSR containing the most recent branch record. See MSR_LASTBRANCH_0_FROM_IP (at 40H).		Unique
Register Address: 1D9H, 473	IA32_DEBUGCTL	
Debug Control (R/W) Controls how several debug features are used. Bit definitions are discussed in Table 2-2.		Unique
Register Address: 1DDH, 477	MSR_LER_FROM_LIP	
Last Exception Record From Linear IP (R) Contains a pointer to the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Unique
Register Address: 1DEH, 478	MSR_LER_TO_LIP	
Last Exception Record To Linear IP (R) This area contains a pointer to the target of the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled.		Unique
Register Address: 200H, 512	MTRRphysBase0	

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Memory Type Range Registers		Unique
Register Address: 201H, 513	MTRRphysMask0	
Memory Type Range Registers		Unique
Register Address: 202H, 514	MTRRphysBase1	
Memory Type Range Registers		Unique
Register Address: 203H, 515	MTRRphysMask1	
Memory Type Range Registers		Unique
Register Address: 204H, 516	MTRRphysBase2	
Memory Type Range Registers		Unique
Register Address: 205H, 517	MTRRphysMask2	
Memory Type Range Registers		Unique
Register Address: 206H, 518	MTRRphysBase3	
Memory Type Range Registers		Unique
Register Address: 207H, 519	MTRRphysMask3	
Memory Type Range Registers		Unique
Register Address: 208H, 520	MTRRphysBase4	
Memory Type Range Registers		Unique
Register Address: 209H, 521	MTRRphysMask4	
Memory Type Range Registers		Unique
Register Address: 20AH, 522	MTRRphysBase5	
Memory Type Range Registers		Unique
Register Address: 20BH, 523	MTRRphysMask5	
Memory Type Range Registers		Unique
Register Address: 20CH, 524	MTRRphysBase6	
Memory Type Range Registers		Unique
Register Address: 20DH, 525	MTRRphysMask6	
Memory Type Range Registers		Unique
Register Address: 20EH, 526	MTRRphysBase7	
Memory Type Range Registers		Unique
Register Address: 20FH, 527	MTRRphysMask7	
Memory Type Range Registers		Unique
Register Address: 250H, 592	MTRRfix64K_00000	
Memory Type Range Registers		Unique
Register Address: 258H, 600	MTRRfix16K_80000	
Memory Type Range Registers		Unique
Register Address: 259H, 601	MTRRfix16K_A0000	
Memory Type Range Registers		Unique

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Register Address: 268H, 616	MTRRfix4K_C0000	
Memory Type Range Registers		Unique
Register Address: 269H, 617	MTRRfix4K_C8000	
Memory Type Range Registers		Unique
Register Address: 26AH, 618	MTRRfix4K_D0000	
Memory Type Range Registers		Unique
Register Address: 26BH, 619	MTRRfix4K_D8000	
Memory Type Range Registers		Unique
Register Address: 26CH, 620	MTRRfix4K_E0000	
Memory Type Range Registers		Unique
Register Address: 26DH, 621	MTRRfix4K_E8000	
Memory Type Range Registers		Unique
Register Address: 26EH, 622	MTRRfix4K_F0000	
Memory Type Range Registers		Unique
Register Address: 26FH, 623	MTRRfix4K_F8000	
Memory Type Range Registers		Unique
Register Address: 2FFH, 767	IA32_MTRR_DEF_TYPE	
Default Memory Types (R/W) See Table 2-2 and Section 13.11.2.1, "IA32_MTRR_DEF_TYPE MSR."		Unique
Register Address: 400H, 1024	IA32_MCO_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Unique
Register Address: 401H, 1025	IA32_MCO_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Unique
Register Address: 402H, 1026	IA32_MCO_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MCO_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MCO_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 404H, 1028	IA32_MC1_CTL	
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."		Unique
Register Address: 405H, 1029	IA32_MC1_STATUS	
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."		Unique
Register Address: 406H, 1030	IA32_MC1_ADDR	
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The IA32_MC1_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC1_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 408H, 1032	IA32_MC2_CTL	

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Unique
Register Address: 409H, 1033	IA32_MC2_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Unique
Register Address: 40AH, 1034	IA32_MC2_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 40CH, 1036	MSR_MC4_CTL	
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."		Unique
Register Address: 40DH, 1037	MSR_MC4_STATUS	
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."		Unique
Register Address: 40EH, 1038	MSR_MC4_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The MSR_MC4_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 410H, 1040	IA32_MC3_CTL	
IA32_MC3_CTL	See Section 17.3.2.1, "IA32_MCi_CTL MSRs."	
Register Address: 411H, 1041	IA32_MC3_STATUS	
IA32_MC3_STATUS	See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."	
Register Address: 412H, 1042	MSR_MC3_ADDR	
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs." The MSR_MC3_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC3_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.		Unique
Register Address: 413H, 1043	MSR_MC3_MISC	
Machine Check Error Reporting Register - contains additional information describing the machine-check error if the MISC_V flag in the IA32_MCi_STATUS register is set.		Unique
Register Address: 414H, 1044	MSR_MC5_CTL	
Machine Check Error Reporting Register - controls signaling of #MC for errors produced by a particular hardware unit (or group of hardware units).		Unique
Register Address: 415H, 1045	MSR_MC5_STATUS	
Machine Check Error Reporting Register - contains information related to a machine-check error if its VAL (valid) flag is set. Software is responsible for clearing IA32_MCi_STATUS MSRs by explicitly writing 0s to them; writing 1s to them causes a general-protection exception.		Unique
Register Address: 416H, 1046	MSR_MC5_ADDR	
Machine Check Error Reporting Register - contains the address of the code or data memory location that produced the machine-check error if the ADDR_V flag in the IA32_MCi_STATUS register is set.		Unique
Register Address: 417H, 1047	MSR_MC5_MISC	

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
Machine Check Error Reporting Register - contains additional information describing the machine-check error if the MISCV flag in the IA32_MCI_STATUS register is set.		Unique
Register Address: 480H, 1152	IA32_VMX_BASIC	
Reporting Register of Basic VMX Capabilities (R/O) See Table 2-2 and Appendix A.1, "Basic VMX Information." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 481H, 1153	IA32_VMX_PINBASED_CTL	
Capability Reporting Register of Pin-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 482H, 1154	IA32_VMX_PROCBASED_CTL	
Capability Reporting Register of Primary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 483H, 1155	IA32_VMX_EXIT_CTL	
Capability Reporting Register of VM-Exit Controls (R/O) See Appendix A.4, "VM-Exit Controls." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 484H, 1156	IA32_VMX_ENTRY_CTL	
Capability Reporting Register of VM-Entry Controls (R/O) See Appendix A.5, "VM-Entry Controls." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 485H, 1157	IA32_VMX_MISC	
Reporting Register of Miscellaneous VMX Capabilities (R/O) See Appendix A.6, "Miscellaneous Data." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 486H, 1158	IA32_VMX_CR0_FIXED0	
Capability Reporting Register of CR0 Bits Fixed to 0 (R/O) See Appendix A.7, "VMX-Fixed Bits in CR0." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 487H, 1159	IA32_VMX_CR0_FIXED1	
Capability Reporting Register of CR0 Bits Fixed to 1 (R/O) See Appendix A.7, "VMX-Fixed Bits in CR0." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 488H, 1160	IA32_VMX_CR4_FIXED0	
Capability Reporting Register of CR4 Bits Fixed to 0 (R/O) See Appendix A.8, "VMX-Fixed Bits in CR4." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 489H, 1161	IA32_VMX_CR4_FIXED1	
Capability Reporting Register of CR4 Bits Fixed to 1 (R/O) See Appendix A.8, "VMX-Fixed Bits in CR4." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 48AH, 1162	IA32_VMX_VMCS_ENUM	
Capability Reporting Register of VMCS Field Enumeration (R/O) See Appendix A.9, "VMCS Enumeration." (If CPUID.01H:ECX.[bit 5])		Unique
Register Address: 48BH, 1163	IA32_VMX_PROCBASED_CTL2	
Capability Reporting Register of Secondary Processor-Based VM-Execution Controls (R/O) See Appendix A.3, "VM-Execution Controls." (If CPUID.01H:ECX.[bit 5] and IA32_VMX_PROCBASED_CTL[bit 63])		Unique
Register Address: 600H, 1536	IA32_DS_AREA	

Table 2-66. MSRs in Intel® Core™ Solo, Intel® Core™ Duo Processors, and Dual-Core Intel® Xeon® Processor LV (Contd.)

Register Address: Hex, Decimal	Register Name	
Register Information / Bit Fields	Bit Description	Shared/ Unique
DS Save Area (R/W) See Table 2-2 and Section 21.6.3.4, "Debug Store (DS) Mechanism."		Unique
31:0	DS Buffer Management Area Linear address of the first byte of the DS buffer management area.	
63:32	Reserved.	
Register Address: C000_0080H	IA32_EFER	
See Table 2-2.		Unique
10:0	Reserved.	
11	Execute Disable Bit Enable	
63:12	Reserved.	

2.21 MSRS IN THE PENTIUM M PROCESSOR

Model-specific registers (MSRs) for the Pentium M processor are similar to those described in Section 2.22 for P6 family processors. The following table describes new MSRs and MSRs whose behavior has changed on the Pentium M processor.

Table 2-67. MSRs in Pentium M Processors

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Register Address: 0H, 0	P5_MC_ADDR
See Section 2.23, "MSRs in Pentium Processors."	
Register Address: 1H, 1	P5_MC_TYPE
See Section 2.23, "MSRs in Pentium Processors."	
Register Address: 10H, 16	IA32_TIME_STAMP_COUNTER
See Section 19.17, "Time-Stamp Counter," and see Table 2-2.	
Register Address: 17H, 23	IA32_PLATFORM_ID
Platform ID (R) See Table 2-2. The operating system can use this MSR to determine "slot" information for the processor and the proper microcode update to load.	
Register Address: 2AH, 42	MSR_EBL_CR_POWERON
Processor Hard Power-On Configuration (R/W) Enables and disables processor features. (R) Indicates current processor configuration.	
0	Reserved.
1	Data Error Checking Enable (R) 0 = Disabled. Always 0 on the Pentium M processor.

Table 2-67. MSRs in Pentium M Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
2	Response Error Checking Enable (R) 0 = Disabled. Always 0 on the Pentium M processor.
3	MCERR# Drive Enable (R) 0 = Disabled. Always 0 on the Pentium M processor.
4	Address Parity Enable (R) 0 = Disabled. Always 0 on the Pentium M processor.
6:5	Reserved.
7	BINIT# Driver Enable (R) 1 = Enabled; 0 = Disabled. Always 0 on the Pentium M processor.
8	Output Tri-state Enabled (R/O) 1 = Enabled; 0 = Disabled.
9	Execute BIST (R/O) 1 = Enabled; 0 = Disabled.
10	MCERR# Observation Enabled (R/O) 1 = Enabled; 0 = Disabled. Always 0 on the Pentium M processor.
11	Reserved.
12	BINIT# Observation Enabled (R/O) 1 = Enabled; 0 = Disabled. Always 0 on the Pentium M processor.
13	Reserved.
14	1 MByte Power on Reset Vector (R/O) 1 = 1 MByte; 0 = 4 GBytes. Always 0 on the Pentium M processor.
15	Reserved.
17:16	APIC Cluster ID (R/O) Always 00B on the Pentium M processor.
18	System Bus Frequency (R/O) 0 = 100 MHz. 1 = Reserved. Always 0 on the Pentium M processor.
19	Reserved.
21:20	Symmetric Arbitration ID (R/O) Always 00B on the Pentium M processor.
26:22	Clock Frequency Ratio (R/O)
Register Address: 40H, 64	MSR_LASTBRANCH_0

Table 2-67. MSRs in Pentium M Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Last Branch Record 0 (R/W) One of 8 last branch record registers on the last branch record stack: bits 31-0 hold the 'from' address and bits 63-32 hold the to address. See also: ▪ Last Branch Record Stack TOS at 1C9H. ▪ Section 19.15, "Last Branch, Interrupt, and Exception Recording (Pentium M Processors)."	
Register Address: 41H, 65	MSR_LASTBRANCH_1
Last Branch Record 1 (R/W) See description of MSR_LASTBRANCH_0.	
Register Address: 42H, 66	MSR_LASTBRANCH_2
Last Branch Record 2 (R/W) See description of MSR_LASTBRANCH_0.	
Register Address: 43H, 67	MSR_LASTBRANCH_3
Last Branch Record 3 (R/W) See description of MSR_LASTBRANCH_0.	
Register Address: 44H, 68	MSR_LASTBRANCH_4
Last Branch Record 4 (R/W) See description of MSR_LASTBRANCH_0.	
Register Address: 45H, 69	MSR_LASTBRANCH_5
Last Branch Record 5 (R/W) See description of MSR_LASTBRANCH_0.	
Register Address: 46H, 70	MSR_LASTBRANCH_6
Last Branch Record 6 (R/W) See description of MSR_LASTBRANCH_0.	
Register Address: 47H, 71	MSR_LASTBRANCH_7
Last Branch Record 7 (R/W) See description of MSR_LASTBRANCH_0.	
Register Address: 119H, 281	MSR_BBL_CR_CTL
Control Register Used to program L2 commands to be issued via cache configuration accesses mechanism. Also receives L2 lookup response.	
63:0	Reserved.
Register Address: 11EH, 281	MSR_BBL_CR_CTL3
Control Register 3 Used to configure the L2 Cache.	
0	L2 Hardware Enabled (R/O) 1 = If the L2 is hardware-enabled. 0 = Indicates if the L2 is hardware-disabled.
4:1	Reserved.

Table 2-67. MSRs in Pentium M Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
5	ECC Check Enable (R/O) This bit enables ECC checking on the cache data bus. ECC is always generated on write cycles. 0 = Disabled (default). 1 = Enabled. For the Pentium M processor, ECC checking on the cache data bus is always enabled.
7:6	Reserved.
8	L2 Enabled (R/W) 1 = L2 cache has been initialized. 0 = Disabled (default). Until this bit is set the processor will not respond to the WBINVD instruction or the assertion of the FLUSH# input.
22:9	Reserved.
23	L2 Not Present (R/O) 0 = L2 Present. 1 = L2 Not Present.
63:24	Reserved.
Register Address: 179H, 377	IA32_MCG_CAP
Read-only register that provides information about the machine-check architecture of the processor.	
7:0	Count (R/O) Indicates the number of hardware unit error reporting banks available in the processor.
8	IA32_MCG_CTL Present (R/O) 1 = Indicates that the processor implements the MSR_MCG_CTL register found at MSR 17BH. 0 = Not supported.
63:9	Reserved.
Register Address: 17AH, 378	IA32_MCG_STATUS
Global Machine Check Status	
0	RIPV When set, this bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) can be used to restart the program. If this bit is cleared, the program cannot be reliably restarted.
1	EIPV When set, this bit indicates that the instruction addressed by the instruction pointer pushed on the stack (when the machine check was generated) is directly associated with the error.
2	MCIP When set, this bit indicates that a machine check has been generated. If a second machine check is detected while this bit is still set, the processor enters a shutdown state. Software should write this bit to 0 after processing a machine check exception.
63:3	Reserved.
Register Address: 198H, 408	IA32_PERF_STATUS

Table 2-67. MSRs in Pentium M Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
See Table 2-2.	
Register Address: 199H, 409	IA32_PERF_CTL
See Table 2-2.	
Register Address: 19AH, 410	IA32_CLOCK_MODULATION
Clock Modulation (R/W). See Table 2-2 and Section 16.8.3, "Software Controlled Clock Modulation."	
Register Address: 19BH, 411	IA32_THERM_INTERRUPT
Thermal Interrupt Control (R/W) See Table 2-2 and Section 16.8.2, "Thermal Monitor."	
Register Address: 19CH, 412	IA32_THERM_STATUS
Thermal Monitor Status (R/W) See Table 2-2 and Section 16.8.2, "Thermal Monitor."	
Register Address: 19DH, 413	MSR_THERM2_CTL
Thermal Monitor 2 Control	
15:0	Reserved.
16	TM_SELECT (R/W) Mode of automatic thermal monitor: 0 = Thermal Monitor 1 (thermally-initiated on-die modulation of the stop-clock duty cycle) 1 = Thermal Monitor 2 (thermally-initiated frequency transitions) If bit 3 of the IA32_MISC_ENABLE register is cleared, TM_SELECT has no effect. Neither TM1 nor TM2 will be enabled.
63:16	Reserved.
Register Address: 1A0H, 416	IA32_MISC_ENABLE
Enable Miscellaneous Processor Features (R/W) Allows a variety of processor functions to be enabled and disabled.	
2:0	Reserved.
3	Automatic Thermal Control Circuit Enable (R/W) 1 = Setting this bit enables the thermal control circuit (TCC) portion of the Intel Thermal Monitor feature. This allows processor clocks to be automatically modulated based on the processor's thermal sensor operation. 0 = Disabled (default). The automatic thermal control circuit enable bit determines if the thermal control circuit (TCC) will be activated when the processor's internal thermal sensor determines the processor is about to exceed its maximum operating temperature. When the TCC is activated and TM1 is enabled, the processors clocks will be forced to a 50% duty cycle. BIOS must enable this feature. The bit should not be confused with the on-demand thermal control circuit enable bit.
6:4	Reserved.
7	Performance Monitoring Available (R) 1 = Performance monitoring enabled. 0 = Performance monitoring disabled.

Table 2-67. MSRs in Pentium M Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
9:8	Reserved.
10	FERR# Multiplexing Enable (R/W) 1 = FERR# asserted by the processor to indicate a pending break event within the processor. 0 = Indicates compatible FERR# signaling behavior. This bit must be set to 1 to support XAPIC interrupt model usage.
	Branch Trace Storage Unavailable (R/O) 1 = Processor doesn't support branch trace storage (BTS) 0 = BTS is supported
12	Processor Event Based Sampling Unavailable (R/O) 1 = Processor does not support processor event based sampling (PEBS); 0 = PEBS is supported. The Pentium M processor does not support PEBS.
15:13	Reserved.
16	Enhanced Intel SpeedStep Technology Enable (R/W) 1 = Enhanced Intel SpeedStep Technology enabled. On the Pentium M processor, this bit may be configured to be read-only.
22:17	Reserved.
23	xTPR Message Disable (R/W) When set to 1, xTPR messages are disabled. xTPR messages are optional messages that allow the processor to inform the chipset of its priority. The default is processor specific.
63:24	Reserved.
Register Address: 1C9H, 457	MSR_LASTBRANCH_TOS
Last Branch Record Stack TOS (R/W) Contains an index (bits 0-3) that points to the MSR containing the most recent branch record. See also: - MSR_LASTBRANCH_0_FROM_IP (at 40H). - Section 19.15, "Last Branch, Interrupt, and Exception Recording (Pentium M Processors)."	
Register Address: 1D9H, 473	MSR_DEBUGCTLB
Debug Control (R/W) Controls how several debug features are used. Bit definitions are discussed in the referenced section. See Section 19.15, "Last Branch, Interrupt, and Exception Recording (Pentium M Processors)."	
Register Address: 1DDH, 477	MSR_LER_TO_LIP
Last Exception Record To Linear IP (R) This area contains a pointer to the target of the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled. See Section 19.15, "Last Branch, Interrupt, and Exception Recording (Pentium M Processors)," and Section 19.16.2, "Last Branch and Last Exception MSRs."	
Register Address: 1DEH, 478	MSR_LER_FROM_LIP

Table 2-67. MSRs in Pentium M Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Last Exception Record From Linear IP (R) Contains a pointer to the last branch instruction that the processor executed prior to the last exception that was generated or the last interrupt that was handled. See Section 19.15, "Last Branch, Interrupt, and Exception Recording (Pentium M Processors)," and Section 19.16.2, "Last Branch and Last Exception MSRs."	
Register Address: 2FFH, 767	IA32_MTRR_DEF_TYPE
Default Memory Types (R/W) Sets the memory type for the regions of physical memory that are not mapped by the MTRRs. See Section 13.11.2.1, "IA32_MTRR_DEF_TYPE MSR."	
Register Address: 400H, 1024	IA32_MCO_CTL
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."	
Register Address: 401H, 1025	IA32_MCO_STATUS
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."	
Register Address: 402H, 1026	IA32_MCO_ADDR
See Section 14.3.2.3, "IA32_MCi_ADDR MSRs". The IA32_MCO_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MCO_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.	
Register Address: 404H, 1028	IA32_MC1_CTL
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."	
Register Address: 405H, 1029	IA32_MC1_STATUS
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."	
Register Address: 406H, 1030	IA32_MC1_ADDR
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs". The IA32_MC1_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC1_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.	
Register Address: 408H, 1032	IA32_MC2_CTL
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."	
Register Address: 409H, 1033	IA32_MC2_STATUS
See Chapter 17.3.2.2, "IA32_MCi_STATUS MSRs."	
Register Address: 40AH, 1034	IA32_MC2_ADDR
See Section 17.3.2.3, "IA32_MCi_ADDR MSRs". The IA32_MC2_ADDR register is either not implemented or contains no address if the ADDR_V flag in the IA32_MC2_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.	
Register Address: 40CH, 1036	MSR_MC4_CTL
See Section 17.3.2.1, "IA32_MCi_CTL MSRs."	
Register Address: 40DH, 1037	MSR_MC4_STATUS
See Section 17.3.2.2, "IA32_MCi_STATUS MSRs."	
Register Address: 40EH, 1038	MSR_MC4_ADDR

Table 2-67. MSRs in Pentium M Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The MSR_MC4_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC4_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.	
Register Address: 410H, 1040	MSR_MC3_CTL
See Section 17.3.2.1, "IA32_MCI_CTL MSRs."	
Register Address: 411H, 1041	MSR_MC3_STATUS
See Section 17.3.2.2, "IA32_MCI_STATUS MSRs."	
Register Address: 412H, 1042	MSR_MC3_ADDR
See Section 17.3.2.3, "IA32_MCI_ADDR MSRs." The MSR_MC3_ADDR register is either not implemented or contains no address if the ADDR_V flag in the MSR_MC3_STATUS register is clear. When not implemented in the processor, all reads and writes to this MSR will cause a general-protection exception.	
Register Address: 600H, 1536	IA32_DS_AREA
DS Save Area (R/W) See Table 2-2. Points to the DS buffer management area, which is used to manage the BTS and PEBS buffers. See Section 21.6.3.4, "Debug Store (DS) Mechanism."	
31:0	DS Buffer Management Area Linear address of the first byte of the DS buffer management area.
63:32	Reserved.

2.22 MSRS IN THE P6 FAMILY PROCESSORS

The following MSRs are defined for the P6 family processors. The MSRs in this table that are shaded are available only in the Pentium II and Pentium III processors. Beginning with the Pentium 4 processor, some of the MSRs in this list have been designated as "architectural" and have had their names changed. See Table 2-2 for a list of the architectural MSRs.

Table 2-68. MSRs in the P6 Family Processors

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Register Address: 0H, 0	P5_MC_ADDR
See Section 2.23, "MSRs in Pentium Processors."	
Register Address: 1H, 1	P5_MC_TYPE
See Section 2.23, "MSRs in Pentium Processors."	
Register Address: 10H, 16	TSC
See Section 19.17, "Time-Stamp Counter."	
Register Address: 17H, 23	IA32_PLATFORM_ID
Platform ID (R) The operating system can use this MSR to determine "slot" information for the processor and the proper microcode update to load.	
49:0	Reserved.

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
52:50	Platform Id (R) Contains information concerning the intended platform for the processor. 52 51 50 0 0 0 Processor Flag 0 0 0 1 Processor Flag 1 0 1 0 Processor Flag 2 0 1 1 Processor Flag 3 1 0 0 Processor Flag 4 1 0 1 Processor Flag 5 1 1 0 Processor Flag 6 1 1 1 Processor Flag 7
56:53	L2 Cache Latency Read.
59:57	Reserved.
60	Clock Frequency Ratio Read.
63:61	Reserved.
Register Address: 1BH, 27	APIC_BASE
Section 12.4.4, "Local APIC Status and Location."	
7:0	Reserved.
8	Boot Strap Processor Indicator Bit 1 = BSP
10:9	Reserved.
11	APIC Global Enable Bit - Permanent till reset 1 = Enabled. 0 = Disabled.
31:12	APIC Base Address.
63:32	Reserved.
Register Address: 2AH, 42	EBL_CR_POWERON
Processor Hard Power-On Configuration (R/W) Enables and disables processor features, and (R) indicates current processor configuration.	
0	Reserved ¹
1	Data Error Checking Enable (R/W) 1 = Enabled. 0 = Disabled.
2	Response Error Checking Enable FRCERR Observation Enable (R/W) 1 = Enabled. 0 = Disabled.
3	AERR# Drive Enable (R/W) 1 = Enabled. 0 = Disabled.
4	BERR# Enable for Initiator Bus Requests (R/W) 1 = Enabled. 0 = Disabled.

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
5	Reserved.
6	BERR# Driver Enable for Initiator Internal Errors (R/W) 1 = Enabled. 0 = Disabled.
7	BINIT# Driver Enable (R/W) 1 = Enabled. 0 = Disabled.
8	Output Tri-state Enabled (R) 1 = Enabled. 0 = Disabled.
9	Execute BIST (R) 1 = Enabled. 0 = Disabled.
10	AERR# Observation Enabled (R) 1 = Enabled. 0 = Disabled.
11	Reserved.
12	BINIT# Observation Enabled (R) 1 = Enabled. 0 = Disabled.
13	In Order Queue Depth (R) 1 = 1. 0 = 8.
14	1-MByte Power on Reset Vector (R) 1 = 1MByte. 0 = 4GBytes.
15	FRC Mode Enable (R) 1 = Enabled. 0 = Disabled.
17:16	APIC Cluster ID (R)
19:18	System Bus Frequency (R) 00 = 66MHz. 10 = 100Mhz. 01 = 133MHz. 11 = Reserved.
21:20	Symmetric Arbitration ID (R)
25:22	Clock Frequency Ratio (R)
26	Low Power Mode Enable (R/W)
27	Clock Frequency Ratio
63:28	Reserved. ¹
Register Address: 33H, 51	MSR_TEST_CTRL

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Test Control Register	
29:0	Reserved.
30	Streaming Buffer Disable
31	Disable LOCK# Assertion for split locked access.
Register Address: 79H, 121	BIOS_UPDT_TRIG
BIOS Update Trigger Register.	
Register Address: 88H, 136	BBL_CR_D0[63:0]
Chunk 0 data register D[63:0]: used to write to and read from the L2.	
Register Address: 89H, 137	BBL_CR_D1
Chunk 1 data register D[63:0]: used to write to and read from the L2.	
Register Address: 8AH, 138	BBL_CR_D2
Chunk 2 data register D[63:0]: used to write to and read from the L2.	
Register Address: 8BH, 139	BIOS_SIGN/BBL_CR_D3
BIOS Update Signature Register or Chunk 3 data register D[63:0]. Used to write to and read from the L2 depending on the usage model.	
Register Address: C1H, 193	PerfCtr0 (PERFCTR0)
Performance Counter Register See Table 2-2.	
Register Address: C2H, 194	PerfCtr1 (PERFCTR1)
Performance Counter Register See Table 2-2.	
Register Address: FEH, 254	MTRRcap
Memory Type Range Registers	
Register Address: 116H, 278	BBL_CR_ADDR
Address register: used to send specified address (A31-A3) to L2 during cache initialization accesses.	
2:0	Reserved; set to 0.
31:3	Address bits [35:3].
63:32	Reserved.
Register Address: 118H, 280	BBL_CR_DECC
Data ECC register D[7:0]: used to write ECC and read ECC to/from L2.	
Register Address: 119H, 281	BBL_CR_CTL
Control register: used to program L2 commands to be issued via cache configuration accesses mechanism. Also receives L2 lookup response.	

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
4:0	L2 Command: 01100 = Data Read w/ LRU update (RLU). 01110 = Tag Read w/ Data Read (TRR). 01111 = Tag Inquire (TI). 00010 = L2 Control Register Read (CR). 00011 = L2 Control Register Write (CW). 010 + MESI encode = Tag Write w/ Data Read (TWR). 111 + MESI encode = Tag Write w/ Data Write (TWW). 100 + MESI encode = Tag Write (TW).
6:5	
7	State to L2
9:8	Reserved.
11:10	Way 0 - 00, Way 1 - 01, Way 2 - 10, Way 3 - 11 Way to L2
13:12	Modified - 11, Exclusive - 10, Shared - 01, Invalid - 00 Way from L2
15:14	State from L2.
16	Reserved.
17	L2 Hit.
18	Reserved.
20:19	User supplied ECC.
21	Processor number: ² Disable = 1. Enable = 0. Reserved.
63:22	Reserved.
Register Address: 11AH, 282	BBL_CR_TRIG
Trigger register: used to initiate a cache configuration accesses access, Write only with Data = 0.	
Register Address: 11BH, 283	BBL_CR_BUSY
Busy register: indicates when a cache configuration accesses L2 command is in progress. D[0] = 1 = BUSY.	
Register Address: 11EH, 286	BBL_CR_CTL3
Control register 3: used to configure the L2 Cache.	
0	L2 Configured (read/write).
4:1	L2 Cache Latency (read/write).
5	ECC Check Enable (read/write).
6	Address Parity Check Enable (read/write).
7	CRTN Parity Check Enable (read/write).
8	L2 Enabled (read/write).

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
10:9	L2 Associativity (read only): 00 = Direct Mapped. 01 = 2 Way. 10 = 4 Way. 11 = Reserved.
12:11	Number of L2 banks (read only).
17:13	Cache size per bank (read/write): 00001 = 256 KBytes. 00010 = 512 KBytes. 00100 = 1 MByte. 01000 = 2 MBytes. 10000 = 4 MBytes.
18	Cache State error checking enable (read/write).
19	Reserved.
22:20	L2 Physical Address Range support: 111 = 64 GBytes. 110 = 32 GBytes. 101 = 16 GBytes. 100 = 8 GBytes. 011 = 4 GBytes. 010 = 2 GBytes. 001 = 1 GByte. 000 = 512 MBytes.
23	L2 Hardware Disable (read only).
24	Reserved.
25	Cache bus fraction (read only).
63:26	Reserved.
Register Address: 174H, 372	SYSENTER_CS_MSR
CS register target for CPL 0 code	
Register Address: 175H, 373	SYSENTER_ESP_MSR
Stack pointer for CPL 0 stack	
Register Address: 176H, 374	SYSENTER_EIP_MSR
CPL 0 code entry point	
Register Address: 179H, 377	MCG_CAP
Machine Check Global Control Register	
Register Address: 17AH, 378	MCG_STATUS
Machine Check Error Reporting Register - contains information related to a machine-check error if its VAL (valid) flag is set. Software is responsible for clearing IA32_MCI_STATUS MSRs by explicitly writing 0s to them; writing 1s to them causes a general-protection exception.	
Register Address: 17BH, 379	MCG_CTL
Machine Check Error Reporting Register - controls signaling of #MC for errors produced by a particular hardware unit (or group of hardware units).	
Register Address: 186H, 390	PerfEvtSel0 (EVNTSEL0)

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Performance Event Select Register 0 (R/W)	
7:0	Event Select Refer to Performance Counter section for a list of event encodings.
15:8	UMASK (Unit Mask) Unit mask register set to 0 to enable all count options.
16	USER Controls the counting of events at Privilege levels of 1, 2, and 3.
17	OS Controls the counting of events at Privilege level of 0.
18	E Occurrence/Duration Mode Select: 1 = Occurrence. 0 = Duration.
19	PC Enabled the signaling of performance counter overflow via BPO pin.
20	INT Enables the signaling of counter overflow via input to APIC: 1 = Enable. 0 = Disable.
22	ENABLE Enables the counting of performance events in both counters: 1 = Enable. 0 = Disable.
23	INV Inverts the result of the CMASK condition: 1 = Inverted. 0 = Non-Inverted.
31:24	CMASK (Counter Mask)
Register Address: 187H, 391	PerfEvtSel1 (EVNTSEL1)
Performance Event Select for Counter 1 (R/W)	
7:0	Event Select Refer to Performance Counter section for a list of event encodings.
15:8	UMASK (Unit Mask) Unit mask register set to 0 to enable all count options.
16	USER Controls the counting of events at Privilege levels of 1, 2, and 3.
17	OS Controls the counting of events at Privilege level of 0.

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
18	E Occurrence/Duration Mode Select: 1 = Occurrence. 0 = Duration.
19	PC Enabled the signaling of performance counter overflow via BP0 pin.
20	INT Enables the signaling of counter overflow via input to APIC. 1 = Enable. 0 = Disable.
23	INV Inverts the result of the CMASK condition. 1 = Inverted. 0 = Non-Inverted.
31:24	CMASK (Counter Mask)
Register Address: 1D9H, 473	DEBUGCTLMR
Enables last branch, interrupt, and exception recording; taken branch breakpoints; the breakpoint reporting pins; and trace messages. This register can be written to using the WRMSR instruction, when operating at privilege level 0 or when in real-address mode.	
0	Enable/Disable Last Branch Records
1	Branch Trap Flag
2	Performance Monitoring/Break Point Pins
3	Performance Monitoring/Break Point Pins
4	Performance Monitoring/Break Point Pins
5	Performance Monitoring/Break Point Pins
6	Enable/Disable Execution Trace Messages
31:7	Reserved.
Register Address: 1DBH, 475	LASTBRANCHFROMIP
32-bit register for recording the instruction pointers for the last branch, interrupt, or exception that the processor took prior to a debug exception being generated.	
Register Address: 1DCH, 476	LASTBRANCHTOIP
32-bit register for recording the instruction pointers for the last branch, interrupt, or exception that the processor took prior to a debug exception being generated.	
Register Address: 1DDH, 477	LASTINTFROMIP
Last INT from IP	
Register Address: 1DEH, 478	LASTINTTOIP
Last INT to IP	
Register Address: 200H, 512	MTRRphysBase0
Memory Type Range Registers	
Register Address: 201H, 513	MTRRphysMask0
Memory Type Range Registers	

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Register Address: 202H, 514	MTRRphysBase1
Memory Type Range Registers	
Register Address: 203H, 515	MTRRphysMask1
Memory Type Range Registers	
Register Address: 204H, 516	MTRRphysBase2
Memory Type Range Registers	
Register Address: 205H, 517	MTRRphysMask2
Memory Type Range Registers	
Register Address: 206H, 518	MTRRphysBase3
Memory Type Range Registers	
Register Address: 207H, 519	MTRRphysMask3
Memory Type Range Registers	
Register Address: 208H, 520	MTRRphysBase4
Memory Type Range Registers	
Register Address: 209H, 521	MTRRphysMask4
Memory Type Range Registers	
Register Address: 20AH, 522	MTRRphysBase5
Memory Type Range Registers	
Register Address: 20BH, 523	MTRRphysMask5
Memory Type Range Registers	
Register Address: 20CH, 524	MTRRphysBase6
Memory Type Range Registers	
Register Address: 20DH, 525	MTRRphysMask6
Memory Type Range Registers	
Register Address: 20EH, 526	MTRRphysBase7
Memory Type Range Registers	
Register Address: 20FH, 527	MTRRphysMask7
Memory Type Range Registers	
Register Address: 250H, 592	MTRRfix64K_00000
Memory Type Range Registers	
Register Address: 258H, 600	MTRRfix16K_80000
Memory Type Range Registers	
Register Address: 259H, 601	MTRRfix16K_A0000
Memory Type Range Registers	
Register Address: 268H, 616	MTRRfix4K_C0000
Memory Type Range Registers	
Register Address: 269H, 617	MTRRfix4K_C8000

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Memory Type Range Registers	
Register Address: 26AH, 618	MTRRfix4K_D0000
Memory Type Range Registers	
Register Address: 26BH, 619	MTRRfix4K_D8000
Memory Type Range Registers	
Register Address: 26CH, 620	MTRRfix4K_E0000
Memory Type Range Registers	
Register Address: 26DH, 621	MTRRfix4K_E8000
Memory Type Range Registers	
Register Address: 26EH, 622	MTRRfix4K_F0000
Memory Type Range Registers	
Register Address: 26FH, 623	MTRRfix4K_F8000
Memory Type Range Registers	
Register Address: 2FFH, 767	MTRRdefType
Memory Type Range Registers	
2:0	Default memory type
10	Fixed MTRR enable
11	MTRR Enable
Register Address: 400H, 1024	MCO_CTL
Machine Check Error Reporting Register - controls signaling of #MC for errors produced by a particular hardware unit (or group of hardware units).	
Register Address: 401H, 1025	MCO_STATUS
Machine Check Error Reporting Register - contains information related to a machine-check error if its VAL (valid) flag is set. Software is responsible for clearing IA32_MCI_STATUS MSRs by explicitly writing 0s to them; writing 1s to them causes a general-protection exception.	
15:0	MC_STATUS_MCACOD
31:16	MC_STATUS_MSCOD
57	MC_STATUS_DAM
58	MC_STATUS_ADDRV
59	MC_STATUS_MISCV
60	MC_STATUS_EN. (Note: For MCO_STATUS only, this bit is hardcoded to 1.)
61	MC_STATUS_UC
62	MC_STATUS_O
63	MC_STATUS_V
Register Address: 402H, 1026	MCO_ADDR
Register Address: 403H, 1027	MCO_MISC
Defined in MCA architecture but not implemented in the P6 family processors.	
Register Address: 404H, 1028	MC1_CTL

Table 2-68. MSRs in the P6 Family Processors (Contd.)

Register Address: Hex, Decimal	Register Name
Register Information / Bit Fields	Bit Description
Register Address: 405H, 1029	MC1_STATUS
Bit definitions same as MC0_STATUS.	
Register Address: 406H, 1030	MC1_ADDR
Register Address: 407H, 1031	MC1_MISC
Defined in MCA architecture but not implemented in the P6 family processors.	
Register Address: 408H, 1032	MC2_CTL
Register Address: 409H, 1033	MC2_STATUS
Bit definitions same as MC0_STATUS.	
Register Address: 40AH, 1034	MC2_ADDR
Register Address: 40BH, 1035	MC2_MISC
Defined in MCA architecture but not implemented in the P6 family processors.	
Register Address: 40CH, 1036	MC4_CTL
Register Address: 40DH, 1037	MC4_STATUS
Bit definitions same as MC0_STATUS, except bits 0, 4, 57, and 61 are hardcoded to 1.	
Register Address: 40EH, 1038	MC4_ADDR
Defined in MCA architecture but not implemented in P6 Family processors.	
Register Address: 40FH, 1039	MC4_MISC
Defined in MCA architecture but not implemented in the P6 family processors.	
Register Address: 410H, 1040	MC3_CTL
Register Address: 411H, 1041	MC3_STATUS
Bit definitions same as MC0_STATUS.	
Register Address: 412H, 1042	MC3_ADDR
Register Address: 413H, 1043	MC3_MISC
Defined in MCA architecture but not implemented in the P6 family processors.	
NOTES 1. Bit 0 of this register has been redefined several times, and is no longer used in P6 family processors. 2. The processor number feature may be disabled by setting bit 21 of the BBL_CR_CTL MSR (model-specific register address 119h) to "1". Once set, bit 21 of the BBL_CR_CTL may not be cleared. This bit is write-once. The processor number feature will be disabled until the processor is reset. 3. The Pentium III processor will prevent FSB frequency overclocking with a new shutdown mechanism. If the FSB frequency selected is greater than the internal FSB frequency the processor will shutdown. If the FSB selected is less than the internal FSB frequency the BIOS may choose to use bit 11 to implement its own shutdown policy.	

2.23 MSRS IN PENTIUM PROCESSORS

The following MSRs are defined for the Pentium processors. The P5_MC_ADDR, P5_MC_TYPE, and TSC MSRs (named IA32_P5_MC_ADDR, IA32_P5_MC_TYPE, and IA32_TIME_STAMP_COUNTER in the Pentium 4 processor) are architectural; that is, code that accesses these registers will run on Pentium 4 and P6 family processors without generating exceptions (see Section 2.1, "Architectural MSRs"). The CESR, CTR0, and CTR1 MSRs are unique to Pentium processors; code that accesses these registers will generate exceptions on Pentium 4 and P6 family processors.

Table 2-69. MSRs in the Pentium Processor

Register Address: Hex, Decimal	Register Name
Register Information	
Register Address: 0H, 0	P5_MC_ADDR
See Section 17.11.2, "Pentium Processor Machine-Check Exception Handling."	
Register Address: 1H, 1	P5_MC_TYPE
See Section 17.11.2, "Pentium Processor Machine-Check Exception Handling."	
Register Address: 10H, 16	TSC
See Section 19.17, "Time-Stamp Counter."	
Register Address: 11H, 17	CESR
See Section 21.6.9.1, "Control and Event Select Register (CESR)."	
Register Address: 12H, 18	CTR0
Section 21.6.9.3, "Events Counted."	
Register Address: 13H, 19	CTR1
Section 21.6.9.3, "Events Counted."	

A

Advanced programmable interrupt controller (see I/O APIC or Local APIC)
APIC (see I/O APIC or Local APIC)

B

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Branch trace message (see BTM)
Branch trace store (see BTS)
BTS (branch trace store) facilities
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